

Ping-Qi PAN

Linear Programming Computation

Second Edition



Springer

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*Dedicated to my parents, Chao Pan and
Yinyun Yan; my wife, Minghua Jiang; my son,
Yunpeng; my granddaughter, Heidi*

Preface to First Edition

Linear programming (LP) founded by Dantzig (1948–1951b) might be one of the most well-known and widely used mathematical tools in the world. As a branch of optimization, it serves as the most important cornerstone of operations research, decision science, and management science.

This branch emerged when the American mathematician George B. Dantzig created the LP model and the simplex method in 1947. The computer, emerging around the same period, propelled the development of LP and the simplex method toward practical applications. As a basic branch, LP orchestrated the birth of a number of new branches, such as nonlinear programming, network flow and combinatorial optimization, stochastic programming, integer programming, and complementary theory, etc., and activated the whole field of operations research.

A prominent feature of LP is given by its broad applications. Closely related to LP, many individuals have made pioneering contributions in their respective fields. In the field of economics, in particular, Russian-American economist Wassily Leontief took the 1973 Nobel Economic Prize for his epoch-making contribution to quantitative analysis of economic activities. The academician L. V. Kantorovich of the former Soviet Academy of Science and American economist Professor T. C. Koopmans won the 1975 Nobel Prize for their optimal allocation theory of resources using LP. The same prize was also given to Professors K. Arrow, P. Samuelson, H. Simon, and L. Herwicz, several decades later when they paid close attention to LP at the starting days of their professional careers.

The simplex method has also achieved great success in practice. As it is known, its applications to many fields, such as economy, commerce, production, science and technology, and defense and military affairs, etc, have brought about astonishing economic and social benefits. It is recognized as one of The Ten Algorithms in the Twenty Century (IEEE2002; see Cipra (2000)).

After more than 70 years since its birth, LP is now a relatively mature but still developing branch. Nevertheless, there exist great challenges. The importance of large-scale sparse LP models is nowadays enhanced further by globalization. The everyday practice calls upon the research community to provide more powerful solution tools just to keep up with the ever-increasing problem sizes. Therefore, this

book does not only present fundamental materials but also attempts to reflect the state of the art of LP. It has been my long-lasting belief that research, in operations research/management science, in particular, should be of practical value, at least potentially. The author, therefore, focuses on theories, methods, and implementation techniques that are closely related to LP computation.

This book consists of two parts. Part I mainly covers fundamental and conventional materials, such as the geometry of the feasible region, the simplex method, duality principle, and dual simplex method, implementation of the simplex method, sensitivity analysis, parametric LP, variants of the simplex method, decomposition method, and interior-point method. In addition, integer linear programming (ILP), differing from LP in nature, is also discussed not only because ILP models can be handled by solving a sequence of LP models but also because they are so rich in practice and form a major application area of LP computations.

Part II presents published or unpublished new results achieved by the author himself, such as pivot rule, dual pivot rule, simplex phase-I method, dual simplex phase-I method, reduced simplex method, generalized reduced simplex method, deficient-basis method, dual deficient-basis method, face method, dual face method, and pivotal interior-point method. The last chapter contains miscellaneous topics, such as some special forms of the LP problem, approaches to intercepting for primal and dual optimal sets, practical pricing schemes, relaxation principle, local duality, decomposition principle, and ILP method based on the generalized reduced simplex framework.

To make the material easier to follow and understand, the algorithms in this book were formulated and illustrative examples were worked out wherever possible. If the book is used as a textbook for upper-level undergraduate/graduate course, (Part I) may be suitable to be as basic course material.

Acknowledgments

At this very moment, I deeply cherish the memory of Professor Xuchu He, my former mentor at Nanjing University, aroused my interest in optimization. I honor the Father of linear programming and the simplex method, Professor George. B. Dantzig for the encouragement given during the 16th International Symposium on Mathematical Programming, Lausanne EPFL., in August 1997. I am very grateful to Professor Michael Saunders of Stanford University for his thoughtful comments and suggestions given in the past. He selflessly offered the MINOS 5.51 package (the latest version of MINOS at that time) and related materials, from which my work and this book benefited greatly. MINOS has been the benchmark and platform of our computational experiments. I thank Professor R. Tyrrell Rockafellar of the University of Washington, Professor Thomas F. Coleman of Cornell University, and Professor Weide Zheng of Nanjing University for their support and assistance. I would also like to thank the following colleagues for their support and assistance: Professor James V. Burke of the University of Washington, Professor Liqun

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Nanjing, China
December 2012

Ping-Qi Pan

Preface to Second Edition

Since its birth, linear programming (LP) has achieved great success in many fields, such as economy, commerce, production, science, and technology, and brought about amazing economic and social benefits. After 70 years of development, LP now becomes a relatively mature branch in OR/MS. Nevertheless, the academic community faces a major challenge of the growing demand, which needs more powerful and robust tools to deal with large-scale and difficult problems.

To meet the challenge, this book draws materials from a practical point of view, focusing on theories, methods, and implementation techniques that are closely related to LP computation. Its first edition consists of two parts. Roughly speaking, Part I covers fundamental materials of LP, and Part II includes published/unpublished new results achieved by the author.

Little attention was received via the author's ResearchGate Web page until Shi, Zhang, and Zhu published their book review in the European Journal of Operational Research (June 2018). Since then, the book quickly attracted considerable attention from academic communities around the world. As of November 6, 2022, the top ten chapter reads are as follows:

Chapter	Reads
1. Duality Principle and Dual Simplex Method (Part I)	15419
2. Simplex Feasible-Pointx Method (Part II)	7861
3. Integer Linear Programming (ILP) (Part I)	7340
4. Implementation of the Simplex Method (Part I)	3284
5. Dual Simplex Phase-I Method (Part II)	2613
6. Simplex Method (Part I)	2576
7. Variants of the Simplex Method (Part I)	2328
8. Geometry of the Feasible Region (Part I)	1951
9. Pivot Rule (Part II)	1844
10. Simplex Phase-I Method (Part I)	1506

The preceding data are consistent with corresponding downloads from the Springer website.

Surprisingly enough, the chapter "Dual Principle and Dual Simplex Method" not only received the maximum number of reads as a whole but also almost weekly, even if it is devoted to such a classical topic. This might be because the discussions are comprehensive, compared with popular literature.

Proposed for the first time, the Feasible-Point Simplex Method, which appeared as the final section of the chapter "Pivotal Interior-Point Method," received the second-highest reads as a whole. It achieved this in a short period of 10 weeks. Indeed, supported by the solid computational results, the method itself might exceed all expectations.

The final chapter of Part I, "Integer Linear Programming (ILP)," usually received the second-highest reads weekly. This happened, I guess, because many researchers are interested in the newly proposed *controlled-branch* method and *controlled-cutting* method, which have potential applications for developing new ILP solvers.

The author's main contributions seem to have not been fully valued by now, including most pivot rules, reduced and D-reduced simplex methods, deficient-basis and dual deficient-basis methods, face and dual face methods, and the decomposition principle, included in Part II. This is somewhat surprising since these methods are supported by solid computational results, except for the newly introduced reduced and D-reduced simplex methods, and the decomposition principle. In the author's view, the reduced simplex method is particularly noteworthy. It is the first simplex method that searches along the *objective-edge*. In contrast to existing simplex methods, it first determines pivot row and then pivot column, and does so without any selection essentially. As a dual version of it, the D-reduced simplex method shares similar attractive features, through determining pivot column first and pivot row later. As for the decomposition principle, it allows for solving arbitrarily large-scale (even dense) LP problems, in essence, giving a glimmer of light on some other types of separable large-scale problems. Time will likely clarify the value of these methods vs other LP solvers.

We are also optimistic about the face and dual face methods. Without exploiting sparsity, the original methods were implemented using orthogonal transformation with favorable computational results reported. In the second edition, we added two chapters devoted to new methods with LU factorization for sparse computing. Indeed, this is a very natural development which the author cannot help himself from breaking into.

Some changes on chapters and sections were made with corrections and improvements, and the whole second edition was organized into two parts. Roughly speaking, Part I (Foundations) contains Part I of the first edition, and Part II (Advances) includes Part II. The Simplex Feasible-Point Algorithm was improved, and removed from the chapter "Pivotal Interior-Point Method" to form an independent chapter with its new title "Simplex Interior-Point Method," since it actually represents a new class of interior-point algorithms, which can be transformed from the traditional simplex algorithms. The title of the original chapter was changed to "Facial Interior-Point Method," since the remaining algorithms represent another new class of

interior-point algorithms, which can be transformed from the normal interior-point algorithms. In particular, the chapter “Integer LP” was rewritten in great gains of the introduction of the objective cutting. Another exciting improvement was the reduced simplex method. The original derivation of its prototype was presented in a chapter with the same title, and then transformed into the so-called “improved” one in another chapter. Fortunately, inspired by the bisection simplex method, we recently found a quite concise new derivation, so we can introduce it in a single chapter now.

Finally, I am grateful to many individuals for their valuable comments and suggestions on the first edition of this book. In particular, I appreciate comments and encouragement given by Dr. Y.-Y Shi, Professor L.-H Zhang, and Professor W.-X Zhu.

Nanjing, China
November 2022

Ping-Qi Pan

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About the Book

This book represents a real breakthrough in the field of linear programming (LP). Being both thoughtful and informative, it focuses on reflecting and promoting the state of the art by highlighting new achievements in LP. This new edition was organized into two parts. The first part addresses foundations of LP, including geometry of feasible region, simplex method, implementation of simplex method, duality and dual simplex method, primal-dual simplex method, sensitivity analysis and parametric LP, generalized simplex method, decomposition method, interior-point method, and integer LP method. The second part mainly introduces contributions of the author himself, such as efficient primal/dual pivot rules, primal/dual Phase-I methods, reduced/D-reduced simplex methods, generalized reduced simplex method, primal/dual deficient-basis methods, primal/dual face methods, and new decomposition principle.

Many important improvements were made in this edition. The first part includes new results, such as mixed two-phase simplex algorithm, dual elimination, fresh pricing scheme for reduced cost, and bilevel LP model and intercepting of optimal solution set. In particular, the chapter "Integer LP Method" was rewritten in great gains of the objective cutting for new ILP solvers *controlled-cutting/branch* methods, as well as an attractive implementation of the controlled-branch method. In the second part, the "simplex feasible-point algorithm" was rewritten and removed from the chapter "Pivotal Interior-Point Method" to form an independent chapter with the new title "Simplex Interior-Point Method," as it represents a class of new interior-point algorithms transformed from traditional simplex algorithms. The title of the original chapter was then changed to "Facial Interior-Point Method," as the remaining algorithms represent another class of new interior-point algorithms transformed from normal interior-point algorithms. Without exploiting sparsity, the original primal/dual face methods were implemented using Cholesky factorization. To cope sparse computation, two new chapters for ones with LU factorization were added to the second part. Another exciting improvement was the reduced simplex method. In the first edition, the derivation of its prototype was presented in a chapter with the same title, and then converted into the so-called "improved" one in another chapter. Fortunately, the author recently found a quite concise new derivation, so

he can now introduce the distinctive and very promising new simplex method in a single chapter.

This book is a rare work in LP. With a focus on computation, it contains many novel ideas and methods, supported by sufficient numerical results. Being clear and succinct, its content deduces in a fresh manner, from simple to profound. In particular, a larger number of examples were worked out to demonstrate algorithms. So, it is an indispensable tool for undergraduate/graduate students, teachers, practitioners, and researchers in LP and related fields.

About the Author

Ping-Qi Pan is a professor and doctoral supervisor of the Department of Mathematics at Southeast University, Nanjing, China. He was a visiting scholar at the University of Washington (1986–1987) and visiting scientist at Cornell University (1987–1988). His research interest focuses on mathematical programming and operations research, especially large-scale linear optimization. He was standing council member of the Mathematical Programming Society of China and standing council member of the Operation Research Society of China. Professor Pan has received the honorary title of Outstanding Scientific-Technical Worker of Jiangsu Province of China. He was nominated as Top 100 scientist of 2012 by the International Biographical Centre, Cambridge, England. He won the Lifetime Achievement Award by Who's Who in the World in 2017.

Notation

In this book, in general, uppercase English letters are used to denote matrices, lowercase English letters used to denote vectors, and lowercase Greek letters do denote reals. Set is designated by uppercase English or Greek letters. Unless indicated otherwise, all vectors are column ones.

LP	Linear programming
ILP	Integer linear programming
$\mathcal{R}^n$	Euclidean space of dimension n
$\mathcal{R}$	Euclidean space of dimension 1
0	Origin of $\mathcal{R}^n$ (or the zero vector)
e_i	Unit vector with the i th component 1 (dimension will be clear from the context; the same below)
e	Vector of all ones
I	Unit matrix
A	Coefficient matrix of the linear program; also stands for index set of A 's columns, i.e., $A = \{1, \dots, n\}$
m	Number of rows of A
n	Number of columns of A
b	Right-hand side of the equality constraint
c	Cost vector, i.e., coefficient vector of the objective function
B	Basis (matrix). Also the set of basic indices
N	Nonbasic columns of A . Also the set of nonbasic indices
a_j	The j th column of A
a_{ij}	The entry of the i th row and j th column of A
v_j	The j th component of vector v
$\ v\ $	Euclidean norm of vector v
$\max(v)$	The largest component of vector v
A_J	Submatrix consisting of columns corresponding to index set J
v_I	Subvector consisting of components corresponding to row index set I
A^T	Transpose of A
X	Diagonal matrix whose diagonal entries are components of vector x

$\nabla f(x)$	Gradient of function $f(x)$
$\Pi \subset \Gamma$	Π is a subset of Γ
$\Pi \cup \Gamma$	Union set, i.e., $\{\tau \mid \tau \in \Pi \text{ or } \tau \in \Gamma\}$
$\Pi \cap \Gamma$	Intersection set, i.e., $\{\tau \mid \tau \in \Pi, \tau \in \Gamma\}$
$\Pi \setminus \Gamma$	Complementary set, i.e., $\{\tau \mid \tau \in \Pi, \tau \notin \Gamma\}$
$\emptyset$	Empty set
$ $	Such that. For example, $\{x \mid Ax = b\}$ means the set of all x such that $Ax = b$ holds
$\ll$	Far less than
$\gg$	Far greater than
$O(\alpha)$	Implies a number less than and equal to $k\alpha$, where k is a fixed integer constant independent of α
$ \tau $	Absolute value of τ if τ is a real, or cardinality of τ if τ is a set
$\text{sign}(t)$	Sign of real t
$\text{range } H$	Column space of matrix H
$\text{null } H$	Null space of matrix H
$\dim \Gamma$	Dimension of set Γ
C_n^m	Number of combinations of taking m from n elements
$\text{int } P$	Interior of set P
$\lfloor \alpha \rfloor$	The largest integer no more than α
$\lceil \alpha \rceil$	The smallest integer no less than α

This book involves the following two simplex FORTRAN codes:

- MINOS** The sophisticated smooth optimization package developed by Murtagh and Saunders (1998) at Department of Management Science and Engineering of Stanford University. Based on the simplex method, the LP option of this sparsity-exploiting code was used as the benchmark and platform for empirical evaluation of new LP methods
- RSA** The author's private code based on the revised two-phase simplex algorithm without exploiting sparsity. It uses the Harris two-pass row rule

Part I

Foundations

Chapter 1

Introduction



As intelligent creatures, human beings plan and carry out activities with pre-set objectives. Early human ancestors relied on their experience only, whereas their modern-day descendants make their decisions based on scientific tools.

The word *optimization* means to achieve the best results with minimum effort. For example, how would one allocate limited resources (labor, money, materials, etc.) to maximize returns or minimize costs while meeting a specific demand. People collect information and data and express real-life problems with appropriate forms mathematically using numbers, functions, and equations, etc., through a so-called *mathematical modeling* process. Then, optimization methods/software are applied to deal with these models, consequently offering quantitative bases for sound decision making. This process has been revolutionized by the advent of the electronic computer.

The linear programming (LP) branch emerged when the American mathematician George B. Dantzig founded the LP model and the simplex method in 1947 (for the birth of LP, see Appendix A). The LP model has a simple structure, involving linear functions and equalities/inequalities; somewhat surprisingly, it turns out to be one of the most important and widely used models in the world. This seems to be in response to the words of Lao Tzu:¹

The Great Road Is the Simplest!

Of course, “simplest” does not necessarily mean “easy.” The challenge is the huge scale of LP models that are difficult to handle. After 70 years of development, LP methods/software are now capable to solve models with considerably large scales. Even 30 years ago, for example, an LP model with 12,753,313 variables was solved successfully to provide an optimal scheme for an airline crew scheduling application (Bixby et al. 1992). Nevertheless, the academic community still faces major challenges to deal with large-scale and difficult models. Therefore, it might

¹ Lao Tzu (about 571–471 BC) was an ancient Chinese philosopher.

be time to review the foundations as well as advances in LP and to provide a book for undergraduate/graduate students, teachers, researchers, and practitioners in LP and the related fields.

In this chapter, we first introduce primitive notation and concepts of numerical algebra that are closely related to LP, including errors of floating-point arithmetic, the mathematical model of LP, illustrative applications, the standard LP problem, the basis matrix, and the basic solution.

1.1 Error of Floating-Point Arithmetic

As a basic tool for modern numerical computation, a computer is indispensable. LP without it is not imaginable. In using a computer, however, numerical errors are unavoidable because it can only represent a proper subset F of real numbers, and as a result, data can only be stored and calculated with finite digits. On one hand, primary data may be out of F and expressed approximately; on the other hand, numerical results yielded from subsequent operations of arithmetic may have to be rounded. Errors arising therefrom are called *rounding errors*.

The set F of the floating-point values represented by a computer is characterized by 4 positive integers: the *base* β , *precision* (mantissa digits) t , and *exponent range* $[L, U]$ (Forsythe et al. 1977, pp 10–29). More precisely, a computer can represent all nonzero numbers of the following form:

$$fl(x) = \pm \alpha_1 \dots \alpha_t \beta^e, \quad 0 \leq \alpha_i < \beta, \quad i = 1, \dots, t, \quad \alpha_1 \neq 0; \quad L \leq e \leq U, \quad (1.1)$$

which together with 0 constitute F , and where $\alpha_1 \dots \alpha_t$ is called *mantissa part* and β^e is the *exponent part*.

We use the notation

$$m = \beta^{L-1}, \quad M = \beta^U (1 - \beta^{-t}).$$

It is clear that for any floating-point number $fl(x) \in F$, it holds that $m \leq |x| \leq M$. Thus, F is a set of non-uniformly distributed rational numbers. Define the set

$$S = \{x \in \mathcal{R} \mid m \leq |x| \leq M\}.$$

Then, for any $0 \neq x \in S$, the approximate number $fl(x)$, yielded from the computer, only holds mantissa of t digits, via the following two methods:

- (i) Rounding method: It is based on the $(t + 1)$ th digit of the mantissa of x . The digits at this position and after are all taken off if this digit is less than a half of β ; in the other case, the digits are taken off after a unit is added to the previous digit. The associated absolute error bound is then

$$|fl(x) - x| \leq \frac{1}{2}\beta^{e-t}. \quad (1.2)$$

- (ii) Truncation method: The $(t + 1)$ th figure and after are all taken off. Thereby, $fl(x)$ is the floating-point number nearest to x in F , whose absolute values are no more than $|x|$, and hence, the associated absolute error bound is

$$|fl(x) - x| \leq \beta^{e-t}. \quad (1.3)$$

The t is a key quantity to numerical computations. To see this, define

$$\epsilon = \begin{cases} \frac{1}{2}\beta^{1-t} & \text{rounding} \\ \beta^{1-t} & \text{truncating} \end{cases}. \quad (1.4)$$

Then, from either (1.2) or (1.3) it follows that

$$|fl(x) - x| \leq \epsilon\beta^{e-1}.$$

Since

$$|x| \geq .\alpha_1\beta^e > 0,$$

it holds that

$$\left| \frac{fl(x) - x}{x} \right| \leq \frac{\epsilon\beta^{e-1}}{.\alpha_1\beta^e} = \epsilon/\alpha_1 \leq \epsilon,$$

that is

$$fl(x) = x(1 + \delta), \quad |\delta| \leq \epsilon.$$

Therefore, ϵ is just a relative error bound of the approximate number $fl(x)$. The so-called *machine precision*, ϵ , is the smallest positive value in F , satisfying

$$fl(1 + \epsilon) \neq 1.$$

Thereby, one may get to know the precision of a computer.

Assume $a, b \in F$, and represent any of the four arithmetic operations $(+, -, *, /)$ by $\square$. It is clear that if $a \square b \notin S$, then the computer cannot handle and display *overflow* (when $|a \square b| > M$) or *underflow* (when $0 < |a \square b| < m$) and terminate the computation (some computers and compilers place value 0 in the case of underflow). In normal cases, for the computed value $fl(a \square b)$ of $a \square b$, it is clear that

$$fl(a \square b) = (a \square b)(1 + \delta), \quad |\delta| < \epsilon.$$

Therefore, the relative error bound for a single arithmetic operation is ϵ , a small number. Unfortunately, the situation may be entirely contrary when executing an algorithm or a series of operations.

Usually, the algorithm's performance may be affected by rounding or truncation errors greatly. Due to error accumulations, final results could be entirely meaningless, as they could be far from the theoretical value. It is therefore important to take care of numerical errors in computations. In this aspect, the following simple rules should be followed:

1. Avoid using a number close to 0 as the denominator or divisor.
2. Avoid subtracting a number from another that is close to it.
3. Avoid operating numbers with a very different order of magnitude, so that "large swallows small whole."
4. Avoid results close to m or M .
5. Reduce the number of operations, especially multiplications or divisions.

When $\delta \ll 1$, for instance, calculating from the right-hand side of each equation below gives a more accurate result than from the left-hand side.

$$\begin{aligned} \sqrt{x + \delta} - \sqrt{x} &= \delta / (\sqrt{x + \delta} + \sqrt{x}), \\ \cos(x + \delta) - \cos x &= -2 \sin(x + \delta/2) \sin(\delta/2). \end{aligned}$$

Due to different errors introduced in computations, algorithms, which are equivalent theoretically, are often not equivalent numerically. An algorithm is said to be of good (*numerical*) *stability* if the resulting numerical error is under control and has little impact on the results. Unstable algorithms are unreliable.

Usually, it is not an easy task to accurately evaluate the error bound of results computed by algorithms. Instead, the so-called *backward error analysis* is applied. Interested readers are referred to Wilkinson (1971).

1.2 From Real-Life Issue to LP Model

In practice, there are usually many schemes to be chosen by a decision maker, and different schemes may lead to widely diverging outcomes, as is crucial in fierce competition. This situation provides a wide stage for the decision maker to play.

Making decisions just based on experience cannot be mentioned in the same breath with mathematical tools.

The formation of a mathematical model is the first step toward the application of LP. Based on full and reliable data gathered, a good model comes from one's knowledge, skill, experience, and wisdom. This topic is beyond the scope of this book, and this section only gives the reader a taste of simple examples.

Example 1.2.1 A manufacturer, who has 1100 tons of log, is committed by contract to supply 470 tons of lath to a company. There is a 6% loss in log when it is processed into laths. In addition, the production costs have risen since the signing of the contract, while the selling price of lath remains the same. In fact, he can make more profit by selling the log as raw material to another firm. What can the manufacturer do to have the maximum profit while still honoring the contract?

This problem can be solved algebraically. Let x be the number of tons of logs the manufacturer can sell as raw material, and let y be the number of logs available to produce lath. It is clear that

$$x + y = 1100,$$

where x should be maximized to gain the maximum profit. On the other hand, since 6% out of y is lost during the manufacturing process, the actual amount of lath produced is

$$y - 0.06y,$$

which, associated with the contract, must be equal to 470 tons, i.e.,

$$y - 0.06y = 470.$$

Consequently, we have the following system of linear equations:

$$\begin{cases} y - 0.06y = 470 \\ 1100 - y = x, \end{cases} \quad (1.5)$$

which has a unique solution

$$x = 600, \quad y = 500.$$

Therefore, the manufacturer should sell 600 tons of logs as raw material and use the rest 500 tons for the production of lath.

In the preceding, the manufacturer only has a unique scheme to choose. Usually, decision makers face multiple or even unlimited choices, as shown in the following example.

Example 1.2.2 A manufacturer produces lath and sheet piles. Profit per lath is 2 dollars, and profit per sheet pile is 5 dollars. The equipment capability of the manufacturer can produce at most 6000 lath or 4000 sheet piles per day. According to the sales contract, he can sell 3000 sheet piles at most, and the sales volume of laths and sheet piles cannot exceed 5000 per day. How should he arrange to produce lath and sheet piles to gain a maximum profit?

Let x be the amount of lath, and let y be that of sheet piles to be produced per day (in thousand as the unit). For values of each pair of variables (x, y) corresponding to a decision, the associated profit is $f(x, y) = 2x + 5y$ thousand dollars. So, what the manufacturer wants to do is to maximize the function value of $f(x, y)$.

It is clear that the yielded profit would be infinitely large if there was no restriction on the production. It is not the case, of course. According to the equipment capability, the average rate of producing lath and sheet piles is 6 thousand and 4 thousand per day, respectively. Therefore, variables x and y should satisfy inequality $x/6 + y/4 \leq 1$, or equivalently,

$$2x + 3y \leq 12.$$

According to the selling limitation, in addition, there is another inequality

$$x + y \leq 5,$$

and finally,

$$y \leq 3.$$

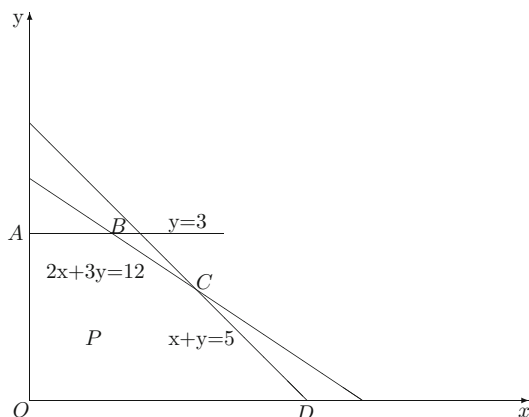
In addition, the amount of output should be nonnegative. As the amount is large, we use the restriction $x, y \geq 0$ instead for simplicity and finally obtain integer results by rounding.

The preceding analysis can be put into the following so-called *mathematical model*:

$$\begin{aligned} \max \quad & f(x, y) = 2x + 5y \\ \text{s.t.} \quad & -2x - 3y \geq -12 \\ & -x - y \geq -5 \\ & -y \geq -3 \\ & x, y \geq 0, \end{aligned} \tag{1.6}$$

where x and y are called *decision variables*, and $f(x, y)$ is called *objective function*; the inequalities are called *constraints*, (x, y) satisfying constraints is a *feasible solution*, and the set of all feasible solutions is the *feasible region*. Solving a model is trying to find its *optimal solution*, that is, a feasible solution maximizing the objective function over the feasible region.

Fig. 1.1 The feasible region of Example 1.2.2



In contrast with Example 1.2.1, which has a single feasible solution, Example 1.2.2 involves infinitely many feasible solutions. In fact, the problem has the following feasible region:

$$P = \{(x, y) \mid 2x + 3y \leq 12, x + y \leq 5, y \leq 3, x \geq 0, y \geq 0\}.$$

Since there is a 1-to-1 correspondence between pairs of real numbers and points in a rectangular coordinate system, the feasible region P can be represented geometrically by Fig. 1.1, where x (thousand) represents the number of laths and y (thousand) that of sheep piles, and where each boundary is labeled by an equation corresponding to the closed half-plane. The feasible region P is just the intersection of these half-planes, within which each point corresponds to a feasible solution.

Therefore, the problem now is how to find a point in P corresponding to the maximum possible value of function $2x + 5y$. Since there are infinitely many points in P , it is impossible to pick up and compare them one by one, not to mention much more challenging large-scale problems with hundreds of thousands of variables and constraints. Although this simple instance can be solved by a graphical approach (Sect. 2.6), dealing with general LP problems requires sophisticated mathematical tools.

1.3 Illustrative Applications

LP has a very wide range of applications, touching upon almost all fields related to decision making and management. This section only brings up a few illustrative instances. Strictly speaking, some of these instances involve variables of nonnegative integer values and hence belong to the so-called *integer/mixed-integer* LP models, although the latter requirement is ignored here for simplicity.

Example 1.3.1 (Production Planning) A factory produces furniture A, B, C. Each furniture production goes through 3 procedures: component processing, electroplating, and assembling. The production capacity of each procedure per day is converted into effective working hours. Below are the required effective working hours and getatable profit for each piece of the furniture.

Procedure	Hours per piece			Available hours per day
	Product A	Product B	Product C	
Component processing	0.025	0.05	0.3	400
Electrofacing	0.20	0.05	0.1	900
Assembling	0.04	0.02	0.20	600
Profit (dollars/piece)	1.25	1.5	2.25	

Now how can the factory achieve the highest profit?

Answer Let x_1 , x_2 , and x_3 be yields of furniture A,B,C, respectively, and let f be the total profit. To gain the highest profit, construct the following model:

$$\begin{aligned}
 \max \quad & f = 1.25x_1 + 1.5x_2 + 2.25x_3 \\
 \text{s.t.} \quad & 0.025x_1 + 0.05x_2 + 0.3x_3 \leq 400 \\
 & 0.20x_1 + 0.05x_2 + 0.1x_3 \leq 900 \\
 & 0.04x_1 + 0.02x_2 + 0.20x_3 \leq 600 \\
 & x_1, x_2, x_3 \geq 0.
 \end{aligned}$$

An optimal solution obtained by the simplex method is

$$x_1 = 2860, x_2 = 6570, x_3 = 0,$$

and the associated objective function value is $f = 13430$. That is to say, the factory should produce 2860 pieces of product A, 6570 pieces of product B, and no product C, achieving the highest profit 13430 dollars.

Example 1.3.2 (Transportation) A company has 8800, 7200, and 5700 containers at ports A, B, C, respectively. These containers should be transported to plants 1, 2, 3, 4, 5, whose working ability is 3200, 5300, 4100, 6200, and 2900 containers,

respectively. The following table lists the freight rate (dollars/container) of transport service from the ports to the plants:

Port	1	2	3	4	5
A	55	150	78	38	90
B	25	60	25	43	35
C	102	80	52	74	60

How should the company arrange to achieve the least total freight charges?

Answer This is a *balanced* transportation problem, as the total number of containers is equal to the total number of containers, i.e.,

$$8800 + 7200 + 5700 = 3200 + 5300 + 4100 + 6200 + 2900 = 21700.$$

Let x_{ij} , $i = 1, 2, 3$; $j = 1, \dots, 5$, be the number of containers that are transported from ports A, B, C to plants 1–5, and let f be the total freight charges. The model is as follows:

$$\begin{aligned}
 \min \quad & f = 55x_{11} + 150x_{12} + 78x_{13} + 38x_{14} + 90x_{15} \\
 & + 25x_{21} + 60x_{22} + 25x_{23} + 43x_{24} + 35x_{25} \\
 & + 102x_{31} + 80x_{32} + 52x_{33} + 74x_{34} + 60x_{35} \\
 \text{s.t.} \quad & x_{11} + x_{12} + x_{13} + x_{14} + x_{15} = 8800 \\
 & x_{21} + x_{22} + x_{23} + x_{24} + x_{25} = 7200 \\
 & x_{31} + x_{32} + x_{33} + x_{34} + x_{35} = 5700 \\
 & x_{11} + x_{21} + x_{31} = 3200 \\
 & x_{12} + x_{22} + x_{32} = 5300 \\
 & x_{13} + x_{23} + x_{33} = 4100 \\
 & x_{14} + x_{24} + x_{34} = 6200 \\
 & x_{15} + x_{25} + x_{35} = 2900 \\
 & x_{ij} \geq 0, \quad i = 1, 2, 3; \quad j = 1, \dots, 5.
 \end{aligned}$$

An optimal solution obtained by the simplex method is

$$x_{11} = 2600, \quad x_{14} = 6200, \quad x_{21} = 600, \quad x_{23} = 4100, \quad x_{25} = 2500, \quad x_{32} = 5300, \quad x_{35} = 400,$$

and all the other variables take value 0. Then the company pays the minimum total freight 1031600 dollars.

Example 1.3.3 (Burdening) To feed animals, someone needs 30 grams of mineral, 700 grams of protein, and 100 milligrams of vitamin per day, at least. There are 5 kinds of feed to buy. The nutrition and price per kilogram for each feed are as follows:

Fodder	Mineral (g.)	Protein (g.)	Vitamin (mg.)	Price (dollars/kg.)
A	0.5	18.0	0.8	0.8
B	2.0	6.0	2.0	0.3
C	0.2	1.0	0.2	0.4
D	0.5	2.0	1.0	0.7
E	1.0	3.0	0.5	0.2

Find a buying scheme to have a minimum cost to serve animals' needs.

Answer Let x_j (kg.), $j = 1, \dots, 5$, be the purchase quantities of the 5 kinds of feed, respectively, and let f be the total cost. The goal is to determine the values of these decision variables to achieve the minimum cost. The model is as follows:

$$\begin{aligned}
 \min \quad & f = 0.8x_1 + 0.3x_2 + 0.4x_3 + 0.7x_4 + 0.2x_5 \\
 \text{s.t.} \quad & 0.5x_1 + 2.0x_2 + 0.2x_3 + 0.5x_4 + x_5 \geq 30 \\
 & 18.0x_1 + 6.0x_2 + x_3 + 2.0x_4 + 0.3x_5 \geq 700 \\
 & 0.8x_1 + 2.0x_2 + 0.2x_3 + x_4 + 0.5x_5 \geq 100 \\
 & x_j \geq 0, \quad j = 1, \dots, 5.
 \end{aligned}$$

An optimal solution is obtained by the simplex method, i.e.,

$$x_1 = 25.6, \quad x_2 = 39.7, \quad x_3 = x_4 = x_5 = 0,$$

with minimum objective function value $f = 32.28$. So, he should buy 25.6 kg. of feed A, 39.7 kg. of feed B, and none of the other 3 feeds with the minimum cost 32.28 dollars.

Example 1.3.4 (Human Resource Arrangement) A round-the-clock supermarket arranges salespersons. Every day falls into 6 time intervals, 4 h each. Every salesperson starts work at the beginning of the intervals and then continuously works for 8 h. The required numbers of salesperson for each of the intervals are as follows:

Interval	1	2	3	4	5	6
	2:00–6:00	6:00–10:00	10:00–14:00	14:00–18:00	18:00–22:00	22:00–2:00
Number	7	15	25	20	30	7

How to arrange to achieve the least number of salespersons per day?

Answer Let x_j , $j = 1, \dots, 6$, be the number of salespersons who start work at the beginning of interval j , and let f be the total number of salespersons. The goal is to determine the variable values to achieve the minimum number of salespersons working per day. The model is as follows:

$$\begin{aligned}
 \min \quad & f = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 \\
 \text{s.t.} \quad & x_1 + x_6 \geq 7 \\
 & x_1 + x_2 \geq 15 \\
 & x_2 + x_3 \geq 25 \\
 & x_3 + x_4 \geq 20 \\
 & x_4 + x_5 \geq 30 \\
 & x_5 + x_6 \geq 7 \\
 & x_j \geq 0, \quad j = 1, \dots, 6.
 \end{aligned}$$

An optimal solution obtained by the simplex method is

$$x_2 = 25, \quad x_4 = 30, \quad x_6 = 7, \quad x_1 = x_3 = x_5 = 0,$$

and the associated objective function value is $f = 62$. So, the business should arrange 25 salespersons working from 6:00, 30 ones working from 14:00, and 7 from 22:00. This scheme requires only 62 salespersons.

Example 1.3.5 (Laying-off) A shop is asked to make 450 steel frames. Each frame needs 1.5, 2.1, and 2.9 meters long angle steel. Each raw material is 7.4 meters long. Find the optimum laying-off way.

Answer A direct way is to make a frame using raw material to get the 3 kinds of angle steel. If so, there would be a need for 450 raw materials, each of which would leave an 0.9 meters long offcut. To obtain the optimum scheme, consider the following ways, each of which would leave an offcut no more than 0.9 meters long.

Length (meters)	Way					
	A	B	C	D	E	F
2.9	1	2	0	1	0	1
2.1	0	0	2	2	1	1
1.5	3	1	2	0	3	1
Total	7.4	7.3	7.2	7.1	6.6	6.5
Offcut	0.0	0.1	0.2	0.3	0.8	0.9

These ways should be mixed and matched to gain the minimum total length of offcuts. Let x_j , $j = 1, \dots, 6$, be numbers of raw materials taken, corresponding to ways A, B, C, D, E, F , and let f be the total length of offcuts. Construct the following model:

$$\begin{aligned}
 \min \quad & f = 0.1x_2 + 0.2x_3 + 0.3x_4 + 0.8x_5 + 0.9x_6 \\
 \text{s.t.} \quad & x_1 + 2x_2 \quad \quad \quad + x_4 \quad \quad \quad + x_6 = 450 \\
 & \quad \quad \quad 2x_3 + 2x_4 + x_5 + x_6 = 450 \\
 & 3x_1 + x_2 + 2x_3 \quad \quad \quad + 3x_5 + x_6 = 450 \\
 & x_j \geq 0, \quad j = 1, \dots, 6.
 \end{aligned}$$

The optimal solution given by the simplex method is

$$x_1 = 135, \quad x_2 = 45, \quad x_4 = 225, \quad x_3 = x_5 = x_6 = 0,$$

with objective function value $f = 72$. Therefore, the shop should use 135 raw materials for way A, 45 for way B, and 225 for way D, with a total length of 72 meters of offcuts. The minimum total number of used raw materials is 405.

The sizes of the preceding problems are very small, which are just for the reader to get a taste of LP applications. Nowadays, real-world problems are usually very large. Even so, small problems still help to convey ideas and illustrate algorithms.

1.4 Standard LP Problem

Some of LP problems in the previous section seek for maximizing the objective function, while others seek for minimizing it. Besides, some of their constraints are equalities, while others are inequalities of “ $\geq$ ” or “ $\leq$.” For convenience, we introduce the following problem in a so-called *standard form*:

$$\begin{aligned}
 \min \quad & f = c_1x_1 + c_2x_2 + \cdots + c_nx_n \\
 \text{s.t.} \quad & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\
 & a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\
 & \quad \quad \quad \vdots \\
 & a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \\
 & x_j \geq 0, \quad j = 1, \dots, n.
 \end{aligned} \tag{1.7}$$

All kinds of LP problems can be transformed into standard forms:

1. Any maximization problem can be transformed into a minimization problem by introducing the negative objective function instead.

Using $f = -f'$, e.g., the objective

$$\max f' = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

can be replaced by

$$\min f = -c_1x_1 - c_2x_2 - \cdots - c_nx_n.$$

It is clear that this does not matter to the optimal solution but changes the sign of the optimal objective value only.

2. Any inequality can be transformed into equality by introducing an extra nonnegative variable.

Introducing a so-called *slack variable* $x_{k+1} \geq 0$, e.g., the “ $\leq$ ” type of inequality

$$\alpha_1x_1 + \alpha_2x_2 + \cdots + \alpha_kx_k \leq \beta$$

can be transformed into equality

$$\alpha_1x_1 + \alpha_2x_2 + \cdots + \alpha_kx_k + x_{k+1} = \beta,$$

and the “ $\geq$ ” type of inequality

$$\alpha_1x_1 + \alpha_2x_2 + \cdots + \alpha_kx_k \geq \beta$$

can be transformed into the equality

$$\alpha_1x_1 + \alpha_2x_2 + \cdots + \alpha_kx_k - x_{k+1} = \beta.$$

3. Nonnegative constraints on variables can be obtained by introducing auxiliary variables.

Using variable transformation $x'_k = -x_k$, e.g., nonpositive constraint $x'_k \leq 0$, can be converted into $x_k \geq 0$. A free variable, say x'_k , can be eliminated by the following transformation:

$$x'_k = x_k - x_{k+1}, \quad x_k, x_{k+1} \geq 0.$$

Example 1.4.1 Transform the following LP problem into the standard form:

$$\begin{aligned}
 \max \quad & f' = -2x_1 + x_2 - 5x'_3 \\
 \text{s.t.} \quad & -3x_1 + x_2 + x'_3 \leq 2 \\
 & x_1 - 7x_2 + 4x'_3 \geq -3 \\
 & x_1 - 3x_2 - 2x'_3 \leq 1 \\
 & x_1, x_2 \geq 0.
 \end{aligned}$$

Answer Maximizing f' is transformed into minimizing $f = -f'$. The free variable x'_3 is substituted by $x'_3 = x_3 - x_4$, $x_3, x_4 \geq 0$. The first 3 inequalities are transformed into equalities by introducing slack variables x_5, x_6, x_7 , respectively. The resulting standard problem is as follows:

$$\begin{aligned}
 \min \quad & f = 2x_1 - x_2 + 5x_3 - 5x_4 \\
 \text{s.t.} \quad & -3x_1 + x_2 + x_3 - x_4 + x_5 = 2 \\
 & x_1 - 7x_2 + 4x_3 - 4x_4 - x_6 = -3 \\
 & x_1 - 3x_2 - 2x_3 + 2x_4 + x_7 = 1 \\
 & x_j \geq 0, \quad j = 1, \dots, 7.
 \end{aligned}$$

Using vector and matrix notation

$$A = (a_{ij}), \quad x = (x_1, \dots, x_n)^T, \quad c = (c_1, \dots, c_n)^T, \quad b = (b_1, \dots, b_m)^T,$$

we put the standard problem (1.7) into the compact form below:

$$\begin{aligned}
 \min \quad & f = c^T x \\
 \text{s.t.} \quad & Ax = b, \quad x \geq 0,
 \end{aligned} \tag{1.8}$$

where $A \in \mathcal{R}^{m \times n}$, $c \in \mathcal{R}^n$, $b \in \mathcal{R}^m$, $m < n$. It is assumed that $\text{rank}(A) = m$, except indicated otherwise.

It goes without saying that each row or column of coefficient matrix A does not vanish. $f = c^T x$ is called *objective function*, and c is *price (cost)*. The system $Ax = b$ is called *equality constraint*, and $x \geq 0$ *nonnegative constraint*.

Usually, theories/methods are developed to deal with standard problems, although they are essentially applicable to general LP problems.

Definition 1.4.1 A solution fulfilling the constraints is a *feasible solution*, and the set of all feasible solutions is the *feasible region*, i.e.,

$$P = \{x \in \mathcal{R}^n \mid Ax = b, x \geq 0\}. \tag{1.9}$$

A feasible solution with the minimum objective function value over the feasible region is an *optimal solution*.

The minimization problem is called (lower) unbounded if its objective value in the feasible region can be arbitrarily small.

1.5 Basis and Feasible Basic Solution

The standard LP problem (1.8) involves the linear system of equations, $Ax = b$, as equality constraint. Note that it is a so-called *underdetermined* system, as the number of equations is less than that of unknowns.

The set of all solutions is called *solution set* of the system. Systems are equivalent if they have the same solution set. There are two basic types of equivalent transformations into systems. In this respect, the validity of the following propositions is evident.

Proposition 1.5.1 *A system, resulting from multiplying any equation of it by a nonzero, is equivalent to the original.*

Proposition 1.5.2 *A system, resulting from adding a multiple of any equation to another, is equivalent to the original.*

Any of the preceding operations is called *elementary (row) transformation*. The second type of elementary transformation is especially important since it can eliminate a nonzero entry of A . Using a series of such kinds of transformations, e.g., the *Gauss–Jordan elimination*, converts a linear system into a so-called *canonical form* that is readily solvable.

Let us bring up an example of 3×5 standard LP problem:

$$\begin{aligned}
 \min \quad & f = x_1 + 2x_2 - x_4 + x_5 \\
 \text{s.t.} \quad & 2x_1 + x_2 + 3x_3 + 2x_4 = 5 \\
 & x_1 - x_2 + 2x_3 - x_4 + 3x_5 = 1 \\
 & x_1 - 2x_3 - 2x_5 = -1 \\
 & x_j \geq 0, \quad j = 1, \dots, 5.
 \end{aligned} \tag{1.10}$$

For convenience, we strip the augmented coefficient matrix from the linear system and put it into the following tableau:

x_1	x_2	x_3	x_4	x_5	RHS
2	1	3	2		5
1	-1	2	-1	3	1
1		-2		-2	-1

where empty cells stand for value 0 (the same below).

The preceding tableau and the system itself are regarded as the same. Carrying out elementary transformations with the tableau amounts to doing the same thing with the system itself.

A so-called *pivot* is a nonzero entry in the tableau, which determines a Gauss–Jordan elimination process. As an example, take the first entry 2 in the first column as pivot. Multiply by $1/2$ the first row to turn the pivot to 1, leading to the following tableau:

x_1	x_2	x_3	x_4	x_5	RHS
1	$1/2$	$3/2$	1		$5/2$
1	-1	2	-1	3	1
1		-2		-2	-1

Then add -1 times of the first row to the second and third row, respectively, to eliminate entries in the first column except the pivot, giving the following tableau:

x_1	x_2	x_3	x_4	x_5	RHS
1	$1/2$	$3/2$	1		$5/2$
	$-3/2$	$1/2$	-2	3	$-3/2$
	$-1/2$	$-7/2$	-1	-2	$-7/2$

Take the second entry in the second column as the next pivot. Multiply by $-2/3$ the second row to turn the pivot to 1, leading to the following tableau:

x_1	x_2	x_3	x_4	x_5	RHS
1	$1/2$	$3/2$	1		$5/2$
	1	$-1/3$	$4/3$	-2	1
	$-1/2$	$-7/2$	-1	-2	$-7/2$

Add $-1/2$ and $1/2$ times of the second row to the first and third rows, respectively, to eliminate entries in the second column except the pivot, giving the following tableau:

x_1	x_2	x_3	x_4	x_5	RHS
1		$5/3$	$1/3$	1	2
	1	$-1/3$	$4/3$	-2	1
		$-11/3$	$-1/3$	-3	-3

Take the third entry of the third column as pivot. Multiply by $-3/11$ the third row to turn the pivot to 1, leading to the following tableau:

x_1	x_2	x_3	x_4	x_5	RHS
1		5/3	1/3	1	2
	1	-1/3	4/3	-2	1
		1	1/11	9/11	9/11

Add $-5/3$ and $1/3$ times of the third row to the first and second rows, respectively, to eliminate entries in the third column except the pivot, giving the following tableau:

x_1	x_2	x_3	x_4	x_5	RHS
1			2/11	-4/11	7/11
	1		15/11	-19/11	14/11
		1	1/11	9/11	9/11

Consequently, the preceding tableau includes a 3×3 unit matrix, and Gauss–Jordan elimination is completed. The corresponding so-called *canonical* system can be written as

$$\begin{cases} x_1 = 7/11 - (2/11)x_4 + (4/11)x_5 \\ x_2 = 14/11 - (15/11)x_4 + (19/11)x_5 \\ x_3 = 9/11 - (1/11)x_4 - (9/11)x_5, \end{cases} \quad (1.11)$$

which is equivalent to the original system (Propositions 1.5.1 and 1.5.2). However, the preceding is so simple that itself exhibits general solutions of the system if x_4 and x_5 are viewed as independent variables.

In particular, setting zero to both x_4 and x_5 leads to

$$x_1 = 7/11, \quad x_2 = 14/11, \quad x_3 = 9/11, \quad x_4 = x_5 = 0,$$

which is said to be *basic solution*. In fact, this solution can be directly read from the RHS side of the last tableau.

Nevertheless, Gauss–Jordan elimination does not always go smoothly if pivots are determined in the natural order. The process breaks down if some pivot happens to be zero. Even if all pivots are nonzero, the outcome may still be useless if some pivot is too close to zero (Sect. 1.1).

Table 1.5.1 Canonical tableau

x_{j_1}	x_{j_2}	$\cdots$	x_{j_r}	$x_{j_{r+1}}$	$\cdots$	x_{j_n}	<i>RHS</i>
1				$\bar{a}_{i_1 j_{r+1}}$	$\cdots$	$\bar{a}_{i_1 j_n}$	$\bar{b}_{i_1}$
	1			$\bar{a}_{i_2 j_{r+1}}$	$\cdots$	$\bar{a}_{i_2 j_n}$	$\bar{b}_{i_2}$
		$\ddots$		$\vdots$	$\vdots$	$\vdots$	$\vdots$
			1	$\bar{a}_{i_r j_{r+1}}$	$\cdots$	$\bar{a}_{i_r j_n}$	$\bar{b}_{i_r}$
				$\bar{a}_{i_{r+1} j_{r+1}}$	$\cdots$	$\bar{a}_{i_{r+1} j_n}$	$\bar{b}_{i_{r+1}}$
				$\vdots$	$\vdots$	$\vdots$	$\vdots$
				$\bar{a}_{i_m j_{r+1}}$	$\cdots$	$\bar{a}_{i_m j_n}$	$\bar{b}_{i_m}$

Let us turn to the so-called Gauss–Jordan elimination with *complete pivoting*. Put the augmented matrix $(A|b)$ to the following tableau:

x_1	x_2	$\cdots$	x_n	<i>RHS</i>
a_{11}	a_{12}	$\cdots$	a_{1n}	b_1
a_{21}	a_{22}	$\cdots$	a_{2n}	b_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
a_{m1}	a_{m2}	$\cdots$	a_{mn}	b_m

Without loss of generality, suppose that the first r pivots are in the natural order and eliminations are performed such that the $r \times r$ northwest corner submatrix of A becomes a unit one, as shown in Tableau 1.5.1.

At the next step, select $\bar{a}_{pq}$ from entries within the non-pivot rows and columns such that

$$|\bar{a}_{pq}| = \max\{|\bar{a}_{ij}| \mid i = i_{r+1}, \dots, i_m; j = j_{r+1}, \dots, j_n\}. \quad (1.12)$$

Assume that it is nonzero. Taking it as the $(r + 1)$ -th pivot, we multiply row p by $1/\bar{a}_{pq}$ and add $-\bar{a}_{iq}$, ($i = i_{r+1}, \dots, i_m; i \neq p$) times of the row to corresponding rows, respectively. To complete this step, we perform row and column permutations such that the $(r + 1) \times (r + 1)$ northwest submatrix becomes unit.

The process terminates if $r = m$, or $r < m$ but $\bar{a}_{pq} = 0$, that is, the $(m - r) \times (n - r)$ southeast submatrix vanishes. The final tableau is called *canonical tableau*. It is clear that if $r = m$, or $r < m$ but $\bar{b}_{i_t}, \forall t = r + 1, \dots, m$, vanish, and hence all non-pivot rows can be dropped from the tableau, system $Ax = b$ is consistent.

According to linear algebra, such a key positive integer r equals the rank of A . Thus, the following can be claimed.

Theorem 1.5.1 *The system $Ax = b$ ($m < n$) has solutions if and only if $r = m$, or $r < m$ but all entries in the non-pivot rows of the right-hand side vanish. In the latter case, it has infinitely many solutions.*

The index set $B = \{j_1, \dots, j_r\}$ of pivot columns is said *basic*, and $N = A \setminus B = \{j_{r+1}, \dots, j_n\}$ said *nonbasic*. The associated variables are said *basic* and *nonbasic* correspondingly.

Recall that for the standard LP problem, it is usually assumed that $\text{rank } A = m$. That is to say, $Ax = b$ is always consistent, and the final canonical tableau offers a set of solutions to $Ax = b$, i.e.,

$$\begin{cases} x_B = \bar{b} - \bar{N}x_N, \\ x_N = x_N, \end{cases}$$

where the nonbasic variables of x_N on the right-hand side are viewed as independent.

Definition 1.5.1 The solution resulting from setting the nonbasic variables to zero is a basic solution. If its basic components are all nonnegative, it is a feasible basic solution.

Thus, the canonical tableau corresponds to the basic solution

$$\bar{x}_B = \bar{b}, \quad \bar{x}_N = 0,$$

which is a feasible basic solution if $\bar{x}_B \geq 0$.

Theorem 1.5.2 A feasible solution is a feasible basic solution if and only if the columns of A corresponding to its positive components are linearly independent.

Proof Necessity. It is clear that all positive components of a feasible basic solution are basic ones and hence correspond to linearly independent columns.

Sufficiency. Assume that a feasible solution has s positive components corresponding to linearly independent columns. Since $\text{rank } A = m$, it holds that $s \leq m$. If $s = m$, then the feasible solution is clearly a basic solution. If, otherwise, $s < m$, there exist additional $m - s$ columns of A , which and the s columns together constitute a linearly independent set of columns. Clearly, the solution is just the basic one associated with the canonical form, corresponding to the set taken as basic columns. $\square$

Definition 1.5.2 A nonsingular $m \times m$ submatrix of A is a basis. It is a *feasible/optimal* basis if the corresponding basic solution is feasible/optimal.

There exists a basis for the standard LP problem since A has full row rank $r = m$. Therefore, $Ax = b$ has an infinite number of solutions. The concept of basis plays a central role in the simplex methodology.

By Gauss–Jordan elimination with column/row permutations, any basis can be converted into $m \times m$ unit matrix, and hence, $(A|b)$ converted into a canonical tableau. It is to say, a basis corresponds to a canonical tableau, and hence a unique basic solution. But it is not a 1-to-1 correspondence: a basic solution could correspond to multiple bases, which is closely related to the so-called *degeneracy* (Sect. 3.4).

Based on the fact that the objective function is continuous over a closed feasible region, we conclude this section with the following claim:

Proposition 1.5.3 *The answer to any LP problem must be one of the following three cases:*

- (i) *Infeasible problem: There exists no feasible solution.*
- (ii) *Unbounded problem: There exists a feasible solution, but the feasible value is lower unbounded over the feasible region.*
- (iii) *There exists an optimal basic solution.*

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Chapter 2

Geometry of Feasible Region



The feasible region P , defined by Definition 1.4.1, is of great importance to the LP problem. Theories and methods of LP are closely related to P , without exception. In this chapter, we introduce its special structure in geometry, including P as a polyhedral convex set, interior point, relative interior point, face, vertex, extreme direction, representation of P , optimal face and optimal vertex, graphic approach to LP, heuristic characteristic of an optimal solution, and feasible direction and active constraint.

In general, P can be degenerate or even empty. The following lemma (Farkas 1902) renders a sufficient and necessary condition for nonempty P (for the proof, see the end of Sect. 5.2). Readers can refer to the literature on the basic concepts and operations of $\mathcal{R}^n$ space. $\mathcal{R}^n$ space.

Lemma 2.0.3 (Farkas) Assume $A \in \mathcal{R}^{m \times n}$ and $b \in \mathcal{R}^m$. The feasible region P is nonempty if and only if

$$b^T y \geq 0, \quad \forall y \in \{y \in \mathcal{R}^m \mid A^T y \geq 0\}.$$

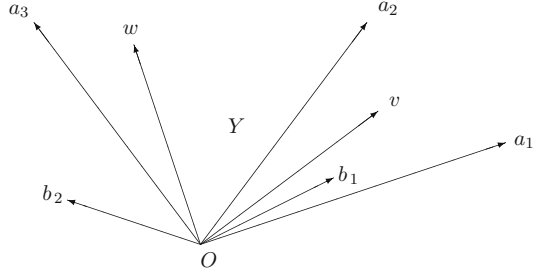
Figure 2.1 serves as a geometric explanation for the preceding Lemma, where a_1, a_2, a_3 are the columns of $A \in \mathcal{R}^{2 \times 3}$.

The area Y between the vectors v and w corresponds to all vectors that form an angle no more than $\pi/2$ with each column vector of A . It is seen that each $y \in Y$ forms an angle no more than $\pi/2$ with vector b_1 , but not with b_2 . Therefore, P is nonempty when $b = b_1$ but empty when $b = b_2$.

Thereafter in this book, discussions will be based on the following:

Assumption 1 The feasible region P is nonempty and infinite.

Fig. 2.1 A geometrical explanation for Farkas' Lemma



2.1 Feasible Region as Polyhedral Convex Set

For any given two points $x, y \in \mathcal{R}^n$, the set

$$S = \{\alpha x + (1 - \alpha)y \mid \alpha \in \mathcal{R}\}$$

is a (*straight*) *line* through them. If $0 < \alpha < 1$, it is an *open segment* with endpoints x and y , denoted by (x, y) ; if $0 \leq \alpha \leq 1$, it is a *closed segment*, denoted by $[x, y]$. Hereafter, a so-called *segment* will be a closed one, unless indicated otherwise.

Definition 2.1.1 Π is an affine set if it includes the whole line through any two points whenever it includes them. The affine hull of a set is the smallest affine set that includes the set.

Lines in $\mathcal{R}^2$ and planes in $\mathcal{R}^3$ are instances of affine sets. The whole space $\mathcal{R}^n$ is an affine set. An empty set and a single point set are viewed as affine sets. It is clear that the intersection of several affine sets is an affine set.

For any given α_i , $i = 1, \dots, k$ satisfying $\sum_{i=1}^k \alpha_i = 1$, the point

$$x = \sum_{i=1}^k \alpha_i x^i$$

is called an *affine combination* of $x^1, \dots, x^k$. It is easy to show that the set of all such affine combinations, i.e.,

$$\left\{ \sum_{i=1}^k \alpha_i x^i \mid \sum_{i=1}^k \alpha_i = 1, \alpha_i \in \mathcal{R}, i = 1, \dots, k \right\},$$

is an affine set, called *affine hull* of these points. The two points in Definition 2.1.1 can be replaced by multiple points: it is easy to show that Π is an affine set if and only if the affine hull of any finitely many points within Π belongs to Π .

The set L is a *subspace* of $\mathcal{R}^n$ if it is closed for all linear operations, that is, for any $x, y \in L$ and $\alpha, \beta \in \mathcal{R}$, it holds that $\alpha x + \beta y \in L$.

Theorem 2.1.1 *An affine set is an affine subspace if and only if it includes the origin.*

Proof The necessity is clear. Sufficiency. Let Π be an affine set including the origin. Then for any $x \in \Pi$ and $\alpha \in \mathcal{R}$, it holds that

$$\alpha x = \alpha x + (1 - \alpha)0 \in \Pi.$$

On the other hand, it holds for any $x, y \in \Pi$ that

$$(x + y)/2 = (1/2)x + (1 - 1/2)y \in \Pi.$$

Therefore, Π is closed for linear operations and is hence an affine subspace. $\square$

Theorem 2.1.2 *For any nonempty affine set Π , there exists vector p so that*

$$L = \{x + p \mid x \in \Pi\}$$

is an affine subspace, and such subspace is unique.

Proof According to Theorem 2.1.1, Π is an affine subspace if $0 \in \Pi$. Note that $L = \Pi$ corresponds to $p = 0$. Assume that $0 \notin \Pi$. Since $\Pi \neq \emptyset$, there exists $0 \neq y \in \Pi$. Letting $p = -y$, it is clear that

$$L = \{x + p \mid x \in \Pi\}$$

is an affine set including the origin and is hence an affine subspace.

Now let us show the uniqueness. Assume that L_1, L_2 are affine subspaces such that

$$L_1 = \{y + p_1 \mid y \in \Pi\}, \quad L_2 = \{y + p_2 \mid y \in \Pi\}.$$

It is clear that

$$\Pi = \{x - p_1 \mid x \in L_1\}.$$

If $p = -p_1 + p_2$, therefore, it holds that

$$L_2 = \{x + p \mid x \in L_1\},$$

from which it follows that $x + p \in L_2$ for any $x \in L_1$. Since $0 \in L_1$, $p \in L_2$ holds. Further, since L_2 is a subspace, $x = (x + p) - p \in L_2$ holds. Therefore, $L_1 \subset L_2$. $L_2 \subset L_1$ can be similarly derived. So it can be asserted that $L_1 = L_2$. $\square$

Geometrically, the affine subspace L along the vector p is parallel to the affine set Π . It is therefore called a *parallel subspace* of Π . The dimension of L is said to be that of Π , which is equal to the number of independent components (coordinates) of elements in Π . It is clear that an affine set is unbounded.

Let a be a nonzero vector, and let η be a real number. The set

$$H = \{x \in \mathcal{R}^n \mid a^T x = \eta\} \quad (2.1)$$

is called *hyperplane*, whose normal vector is a ($a \perp H$); in fact, for any two points $x, y \in H$, it holds that

$$a^T(x - y) = a^T x - a^T y = \eta - \eta = 0.$$

It is clear that any hyperplane is an affine set.

Any straight line in $\mathcal{R}^2$ and any plane in $\mathcal{R}^3$ are instances of hyperplanes.

Definition 2.1.2 The signed distance from point $\bar{x}$ to hyperplane H given by (2.1) is

$$(a^T \bar{x} - \eta) / \|a\|.$$

If $r = 0$, the point $\bar{x}$ is within H . It might be well to assume $\eta > 0$. Then, if $r < 0$, the origin and $\bar{x}$ are both on the same side of H ; if $r > 0$, the two points are on different sides of H .

The hyperplane associated with the objective function is of significance to LP. Regarding objective value f as a parameter, the sets

$$H(f) = \{x \mid c^T x = f\}$$

are a family of isosurfaces of the objective function. The gradient $\nabla f = c$ of the objective function is the common normal vector of all objective isosurfaces, pointing to the increasing side of the objective value.

The following gives a general expression of affine set.

Theorem 2.1.3 The set Π is an affine set if and only if there exists $W \in \mathcal{R}^{k \times n}$ and $h \in \mathcal{R}^k$ such that

$$\Pi = \{x \mid Wx = h\}. \quad (2.2)$$

Proof We might as well rule out the trivial case when Π is empty or the whole space.

Sufficiency. Let Π be defined by (2.2). For any $x, y \in \Pi$ and real α , it holds that

$$W(\alpha x + (1 - \alpha)y) = \alpha Wx + (1 - \alpha)Wy = \alpha h + (1 - \alpha)h = h,$$

leading to $\alpha x + (1 - \alpha)y \in \Pi$. Therefore, Π is an affine set.

Necessity. Let Π be an affine set. Assume that L is a parallel affine subspace, and $w^1, \dots, w^k$ are basis of the orthogonal complementary space of it. Then, it follows that

$$L = \{y \mid (w^i)^T y = 0, i = 1, \dots, k\} = \{y \mid Wy = 0\},$$

where rows of $W \in \mathcal{R}^{k \times n}$ are $(w^1)^T, \dots, (w^k)^T$. Introduce $h = Wp$. Since L is a parallel subspace of Π , there exists a vector p such that

$$L = \{x - p \mid x \in \Pi\}.$$

Thus, for any $x \in \Pi$, it holds that $x - p \in L$; and $W(x - p) = 0$ means that $x \in \{x \mid Wx = h\}$. If $x \in \{x \mid Wx = h\}$, conversely, then $Wx - Wp = W(x - p) = 0$; hence, from $x - p \in L$, it follows that $x \in \Pi$. So, Π has expression (2.2). $\square$

The preceding says that a set is an affine set if and only if it is the intersection of finitely many hyperplanes. It is clear that the dimension of the affine set is equal to $n - \text{rank}(W)$. In particular, the solution set

$$\Delta = \{x \mid Ax = b\} \quad (2.3)$$

of the constraint system of the standard LP problem is an affine set, in which $\text{rank } A = m$, $\Delta \neq \emptyset$ and $\dim \Delta = n - m$.

Definition 2.1.3 C is a convex set if it includes the whole segment whenever it includes its two endpoints. The smallest convex set including a set is the convex hull of the latter.

Quadrants and disks in $\mathcal{R}^2$ and octants and spheres in $\mathcal{R}^3$ are instances for convex sets; segments and the whole space are also convex sets. An empty set and a single point set are regarded as convex sets; intersections of convex sets are convex.

It is clear that any affine set is convex, but a convex set is not an affine set in general, e.g., disks and spheres are not affine sets. Note that any convex set has an affine hull. The dimension of the latter is said to be the dimension of the former. We might as well assume that the convex sets are all closed ones.

For any $\alpha_i \geq 0$, $i = 1, \dots, k$ satisfying $\sum_{i=1}^k \alpha_i = 1$, the point

$$x = \sum_{i=1}^k \alpha_i x^i$$

is called a *convex combination* of $x^1, \dots, x^k$. It is obvious that the set of all such convex combinations, i.e.,

$$\left\{ \sum_{i=1}^k \alpha_i x^i \mid \sum_{i=1}^k \alpha_i = 1; \alpha_i \geq 0, i = 1, \dots, k \right\}$$

is the convex hull of these points. Any set C is a convex set if and only if C includes the convex hull of any finite number of points within C .

Any hyperplane H divides the whole space into two *closed half-spaces*, i.e.,

$$H_L = \{x \mid a^T x \leq \eta\}, \quad \text{and} \quad H_R = \{x \mid a^T x \geq \eta\}. \quad (2.4)$$

The intersection of an infinite number of closed half-spaces is called *polyhedral*, and bounded polyhedral called *polyhedron*. It could degenerate to a segment or point, or even an empty set.

It is clear that any polyhedral/polyhedron is a convex set.

A convex set C is called *polyhedral convex cone* if $\alpha x \in C$ holds for any $x \in C$ and $\alpha \geq 0$. It is easy to show that a set is a polyhedral convex cone if and only if it is the intersection of finitely many closed half-spaces passing through the origin, as expressed by $\{x \in \mathcal{R}^n \mid Ax \geq 0\}$.

The constraint $x \geq 0$ in the standard problem corresponds to the positive octant, which is the intersection of the n closed half-spaces with the coordinate planes as its boundary. Therefore, the feasible region P is the intersection of the affine set Δ and the positive octant. As any hyperplane $a^T x = \eta$ may be viewed as the intersection of two closed half-spaces (2.4), P may also be viewed as the intersection of a finite number of closed half-spaces. Therefore, we make the following statement.

Proposition 2.1.1 *The feasible region P is a polyhedral convex set.*

2.2 Interior Point and Relative Interior Point

The feasible region itself could be degenerate, as it is closely related to the real dimension of it. This section is devoted to this topic.

Introduce the following index set:

$$J = \{j \in A \mid x_j = 0, x \in P\}, \quad (2.5)$$

which is said to be the *index set of zero components*. It is noted that if $x \in P$ and $x_j > 0$, then $j \in \bar{J}$.

Proposition 2.2.1 *Denote by I_J the coefficient matrix of the system $x_j = 0, j \in J'$. Then it holds that*

$$n - \min\{m + |J|, n\} \leq \dim P = n - r \leq n - \max\{m, |J|\}, \quad (2.6)$$

where

$$r = \text{rank} \begin{pmatrix} A \\ I_J \end{pmatrix}.$$

Proof Note that $\text{rank } A = m$ and P is nonempty infinite, according to the basic assumption.

It is clear that $|J| \leq r$ and

$$P = \{x \in \mathcal{R}^n \mid Ax = b, x \geq 0; x_j = 0, j \in J\}.$$

IF $|J| = r$, then $x_j = 0$, $j \in J$ is a canonical form of system

$$Ax = b; \quad x_j = 0, \quad j \in J.$$

If $|J| < r$, besides x_j $j \in J$, there are additional $r - |J|$ basic variables, and hence a canonical form of the preceding system. It is to say that there exists a canonical form, whose $n - r$ nonbasic variables do not belong to J . Therefore, it holds that $\dim P = n - r$, and (2.6) follows from $\min\{m + |J|, n\} \geq r \geq \max\{m, |J|\}$. $\square$

In particular, $\dim P = n - m$ if $J = \emptyset$.

Usually, if $x \in P$ and $x > 0$, it is called an interior point. If $J \neq \emptyset$, therefore, the feasible region P has no interior point in the normal sense, as is often the case for real problems. For the sake of application, we introduce the following concept.

Definition 2.2.1 Let Δ be defined by (2.3), and let J be defined by (2.5). Assume that $J \neq \emptyset$ and $\bar{x} \in P$. If there exists $\delta > 0$ such that a neighborhood of $\bar{x}$ is included in P , i.e.,

$$\Omega(\delta) = \{x \in \Delta \mid x_j = 0, \quad j \in J; \quad \|x - \bar{x}\| < \delta\} \subset P, \quad (2.7)$$

then $\bar{x}$ is a *relative interior point* of P ; otherwise, it is a *boundary point*.

The set of all relative interior points of P is called its *relative interior*, denoted by $\text{int } P$.

Theorem 2.2.1 Assume $\bar{x} \in P$. Then $\bar{x} \in \text{int } P$ if and only if

$$\bar{x}_j > 0, \quad j \in \bar{J}. \quad (2.8)$$

Proof Sufficiency. Let the point $\bar{x} \in P$ satisfy (2.8). Using

$$\delta = \min_{j \in \bar{J}} \bar{x}_j > 0 \quad (2.9)$$

and

$$\Omega(\delta) = \{x \in \Delta \mid x_j = 0, \quad j \in J; \quad \|x - \bar{x}\| < \delta\},$$

then for any $x \in \Omega(\delta)$, it holds that

$$x \in \Delta \quad \text{and} \quad x_j = 0, \quad j \in J.$$

Moreover, since

$$\|x - \bar{x}\| = \sqrt{\sum_{j=1}^n (x_j - \bar{x}_j)^2} < \delta,$$

it holds that

$$|\bar{x}_j - x_j| \leq \|x - \bar{x}\| < \delta, \quad j \in \bar{J},$$

which together with (2.9) gives

$$x_j > \bar{x}_j - \delta \geq 0, \quad j \in \bar{J}.$$

Thus $x \in P$. Therefore $\Omega(\delta) \subset P$, and $\bar{x}$ is an interior point of P .

Necessity. Assuming $\bar{x} \in \text{int } P$, then there is $\delta > 0$ such that $\Omega(\delta) \subset P$; also there is $p \in \bar{J}$ such that $\bar{x}_p = 0$. According to the definition of $\bar{J}$, there is $x' \in P$ such that $x'_p > 0$. It is clear that for any $\alpha > 0$ there exists

$$x = -\alpha x' + (1 + \alpha)\bar{x} = \bar{x} + \alpha(\bar{x} - x') \in \Delta. \quad (2.10)$$

Hence, when α is sufficiently small, it holds that

$$\|x - \bar{x}\| = \alpha \|\bar{x} - x'\| < \delta.$$

In addition, it is clear that $x'_j, \bar{x}_j = 0, j \in J$, and hence $x_j = 0, j \in J$. Therefore, $x \in \Omega(\delta)$. On the other hand, from (2.10), $\bar{x}_p = 0$, and $\alpha > 0, x'_p > 0$, it follows that

$$x_p = -\alpha x'_p + (1 + \alpha)\bar{x}_p = -\alpha x'_p < 0,$$

which contradicts $\Omega(\delta) \subset P$. Therefore, (2.8) holds if $\bar{x} \in \text{int } P$. $\square$

Proposition 2.2.2 *If $\dim P \geq 1$, then P has a relative interior point.*

Proof The assumption of the proposition implies $\bar{J} \neq \emptyset$, because otherwise it holds that $J = A$ and hence $\dim P = 0$, leading to contradiction. On the other hand, for any $j \in \bar{J}$, there exists $x \in P$ such that $x_j > 0$; hence from the convexity of P , it follows that there is $x \in P$ satisfying $x_j > 0, j \in \bar{J}$. According to Theorem 2.2.1, it is known that $x \in \text{int } P \neq \emptyset$. $\square$

Example 2.2.1 Investigate the interior of the feasible region of the following problem:

$$\begin{aligned} \min \quad & x_1 + x_2 + x_3 + x_4 + x_5, \\ \text{s.t.} \quad & x_1 - x_3 + x_4 + x_5 = 6, \\ & x_2 - x_3 - x_4 - x_5 = 0, \\ & -x_2 + 3x_3 + x_4 + x_5 = 0, \\ & x_j \geq 0, \quad j = 1, \dots, 5. \end{aligned}$$

Answer Adding the second constraint equality to the third gives

$$2x_3 = 0.$$

It is clear that the feasible region is nonempty, and the x_3 component of all feasible points equals 0. Eliminating x_3 from the problem leads to

$$\begin{aligned} \min \quad & x_1 + x_2 + x_4 + x_5, \\ \text{s.t.} \quad & x_1 + x_4 + x_5 = 6, \\ & x_2 - x_4 - x_5 = 0, \\ & x_j \geq 0, \quad j = 1, 2, 4, 5, \end{aligned}$$

the interior of the feasible region of which is clearly nonempty, corresponding to the relative interior of the feasible region of the original problem.

2.3 Face, Vertex, and Extreme Direction

In this section, we further investigate the geometric structure of the feasible region.

Definition 2.3.1 Let P' be a nonempty convex subset of P . It is a face of P if for any $x \in P'$, satisfying $x \in (y, z) \subset P$, it holds that $y, z \in P'$, where (y, z) denotes the open segment with endpoints y and z .

The preceding means that a face includes the whole segment if it includes an interior point of any segment of P .

The following renders a general expression of P .

Theorem 2.3.1 *The nonempty convex subset P' is a face of P if and only if there exists index set J' such that*

$$P' = \{x \in P \mid x_j = 0, \quad j \in J'\}. \quad (2.11)$$

Proof Sufficiency. Let $P' \subset P$ be defined by (2.11). If $v \in P'$ is an interior point of segment (y, z) and $y, z \in P$, then there exists $0 < \alpha < 1$ such that

$$v = \alpha y + (1 - \alpha)z.$$

Hence from $\alpha > 0$, $1 - \alpha > 0$, $y, z \geq 0$, and

$$v_j = \alpha y_j + (1 - \alpha)z_j = 0, \quad j \in J',$$

it follows that

$$y_j, z_j = 0, \quad j \in J'.$$

Therefore, $y, z \in P'$, and hence, P' is a face of P .

Necessity. Let $P' \neq \emptyset$ be a face of P . Introduce

$$J' = \{j \in A \mid x_j = 0, x \in P'\}. \quad (2.12)$$

Now we show that P' is equivalent to

$$P'' = \{x \in P \mid x_j = 0, j \in J'\}. \quad (2.13)$$

It is clear that $P' \subset P'' \subset P$.

If P' includes the origin O only, it follows that $b = 0$ and $J' = A$. And P'' clearly includes 0 , hence $P' = P''$. Now assuming that P' does not include 0 uniquely, we show $P'' \subset P'$.

Assume $x \in P''$. If $x = 0$ ($b = 0$), then for any $0 \neq v \in P'$ and

$$y = 2v,$$

it holds that

$$v = \frac{y}{2} + \frac{0}{2},$$

which means that $v \in (y, 0)$. Since P' is a face and $y, 0 \in P$, we have $x = 0 \in P'$. On the other hand, if $x \neq 0$, i.e.,

$$S = \{j \in A \mid x_j > 0\}$$

is nonempty, then from $x \in P''$ and (2.13), it follows that $j \notin J'$ for any $j \in S$. Therefore, there exists $u \in P'$ such that $u_j > 0$. If S includes two or more indices, then $u, w \in P'$ satisfy $u_i, w_j > 0$ for any $i, j \in S$; hence, $z = u/2 + w/2 \in P'$ satisfies $z_i, z_j > 0$. This means that there exists $v \in P'$ such that $v_j > 0$ for all $j \in S$. As for the relation between x and v , there are the following two cases only:

(i) $\{j \in S \mid x_j > v_j\} = \emptyset$. It is clear that

$$z = 2v - x = v + (v - x) \in P.$$

Since P' is a face of P , and $v = z/2 + x/2 \in (x, z)$, it holds that $x \in P'$.

(ii) $\{j \in S \mid x_j > v_j\} \neq \emptyset$. Define

$$z = x + \beta(v - x), \quad \beta = \min\{x_j/(x_j - v_j) \mid x_j - v_j > 0, j \in S\} > 1.$$

It is easy to verify that $z \in P$, and

$$v = \alpha z + (1 - \alpha)x \in (x, z), \quad 0 < \alpha = 1/\beta < 1.$$

In addition, P' is a face of P , hence $x \in P'$. Therefore, $P'' \subset P'$.

We state the following result without proof.

Proposition 2.3.1 *A subset of feasible region P is a face if and only if it is in the relative interior of P .*

Definition 2.3.2 A point x is a vertex of P if $x \in [y, z]$ leads to $x = y$ or $x = z$ for any $y, z \in P$.

Vertex is also termed extreme point, as it is not an interior point of any segment of P . It is clear that the origin is a vertex if it belongs to P , and a vertex of face is a vertex of P .

Lemma 2.3.2 *A point $x \in P$ is a vertex if and only if columns of A corresponding to its positive components are linearly independent.*

Proof It might be well to let the first s components of x be great than 0, and let the rest be 0. Assume that $\bar{x}$ is the subvector consisting of the first s components of x and $\bar{A}$ is the submatrix consisting of the first s columns of A . Then $\bar{A}\bar{x} = b$.

Necessity. Let x be a vertex of P . If columns of $\bar{A}$ are linearly dependent, there is a nonzero vector $\bar{v}$ such that $\bar{A}\bar{v} = 0$. Introduce

$$\bar{y} = \bar{x} + \alpha \bar{v}, \quad \bar{z} = \bar{x} - \alpha \bar{v}.$$

It is clear that for any real α it holds that

$$\bar{A}\bar{y} = \bar{A}\bar{z} = b.$$

Take sufficiently small $\alpha > 0$, such that $\bar{y}, \bar{z} \geq 0$. Construct vectors y and z , so that the first s components of them, respectively, constitute $\bar{y}$ and $\bar{z}$, and the others are 0. Then it is clear that $y, z \in P$ and $x = y/2 + z/2$. Thus x is not a vertex of P , as leads to a contradiction. Therefore, columns corresponding to all positive components of x are linearly independent if x is a vertex of P .

Sufficiency. Assume that columns of $\bar{A}$ are linearly independent. If $x \in P$ is not a vertex, there are two points $y, z \in P$ and a real $\alpha \in (0, 1)$ such that

$$x = \alpha y + (1 - \alpha)z,$$

from which it is known that the last $n - s$ components of y and z are both 0. Therefore, $v = x - y \neq 0$ and

$$\bar{A}\bar{v} = Av = Ax - Ay = b - b = 0,$$

which means that columns of $\bar{A}$ are linearly dependent, as a contradiction. Therefore x is a vertex if columns of $\bar{A}$ are linearly independent. $\square$

In view of $\text{rank}(A) = m$, the preceding theorem implies that the number of positive components of a vertex of P is no more than m .

Lemma 2.3.3 *A point $\bar{x}$ is a vertex of feasible region if and only if it is the feasible basic solution.*

Proof It is from Theorem 1.5.2 and Lemma 2.3.2. $\square$

Therefore, the vertex and feasible basic solution are, respectively, geometrical and algebraical names of the same thing. It is usually difficult to determine whether a point is a vertex but can determine a feasible basic solution conveniently. Recall that every canonical form of $Ax = b$ corresponds to a basic solution, which is a feasible basic solution if it is nonnegative (Sect. 1.5).

Lemma 2.3.4 *A nonempty feasible region has a vertex.*

Proof It is clearly the case when P is a single point set. Let P be an infinite set, and $\bar{x} \in P$. If $\bar{x}$ is not a vertex, there are two distinct points $y, z \in P$ and a real $\bar{\alpha} \in (0, 1)$ such that

$$\bar{x} = \bar{\alpha}y + (1 - \bar{\alpha})z = z + \bar{\alpha}(y - z).$$

Thus, a component of $\bar{x}$ is 0 if and only if the corresponding components of both y, z are 0. Introduce

$$T = \{j \in A \mid \bar{x}_j > 0\}.$$

Then $T \neq \emptyset$, since $T = \emptyset$ implies that $\bar{x} = 0$ is a vertex. It might be well to assume that for some $i \in \{1, \dots, n\}$ it holds that $z_i > y_i$, and hence $i \in T$. This means that

$$\{j \in T \mid z_j - y_j > 0\} \neq \emptyset.$$

Assuming

$$\bar{x} = \alpha_1 y + (1 - \alpha_1)z, \quad \alpha_1 = z_q / (z_q - y_q) = \min \{z_j / (z_j - y_j) \mid z_j - y_j > 0, j \in T\},$$

it is easy to show that $\bar{x} \in P$ and $\bar{x}_q = 0$. Thus $|T|$ for the new $\bar{x}$ is less than that for the old by 1, at least. Repeating no more than n times, therefore, the preceding process terminates at a vertex. $\square$

It is clear that feasible region P is the maximum face. The above proof produces a series of feasible points, corresponding to faces, each of which is a proper face of its predecessor until reaching a vertex, viewed as 0-dimensional face. Such a technique for shifting to faces of lower dimensions is useful.

The feasible region P has a finite number of faces. If $\dim P' = (\dim P) - 1$, the P' is called *facet*. An 1-dimensional face is also called *edge*. There exist at most C_n^{m+1} edges and $C_n^{n-m} = C_n^m$ vertices.

For any given point x and vector $d \neq 0$, the set $\{x + \alpha d \mid \alpha \geq 0\}$ is said to be a *ray (or half-line)*, emanating from x along the direction of d . It is clear that a ray is an infinite set.

Definition 2.3.3 A nonzero vector d is an unbounded direction if P includes rays, emanating from all $x \in P$ along d .

Two unbounded directions having the same direction are regarded as the same.

Theorem 2.3.5 A vector d is an unbounded direction of the nonempty feasible region if and only if

$$Ad = 0, \quad d \neq 0, \quad d \geq 0. \quad (2.14)$$

Proof Sufficiency. With the assumptions, it is easy to verify that for any given $x \in P$ and $\alpha \geq 0$ it holds that $x + \alpha d \in P$; therefore $d \neq 0$ is an unbounded direction.

Necessity. Let $d \neq 0$ be an unbounded direction. Thus, there is $x \in P$, satisfying $x + \alpha d \in P$ for any $\alpha \geq 0$. Hence, from $Ax = b$ and $A(x + \alpha d) = b$, it follows that $Ad = 0$. In addition, $d \geq 0$ holds, because, otherwise, d has a negative component, and hence, the corresponding component of $x + \alpha d$ is negative whenever $\alpha > 0$ becomes sufficiently large, which contradicts $x + \alpha d \in P$. $\square$

Corollary 2.3.6 A nonzero vector d is an unbounded direction if P includes the ray, emanating from some $x \in P$ along d .

It is clear that any nonnegative linear combination of finitely many unbounded directions is an unbounded direction if the combination coefficients are not all zero. Note that the “unbounded direction” is meaningless to an empty P .

Theorem 2.3.7 The feasible region is unbounded if and only if it has an unbounded direction.

Proof Sufficiency is clear, and it is only needed to show necessity.

If $v \in P$, then the translation of P , i.e.,

$$C = \{x - v \mid x \in P\}$$

clearly includes the origin, and P is unbounded if and only if C is unbounded. It might be well to assume that $0 \in P$.

Let $S' = \{x^k\} \in P$ be an unbounded sequence of points. Without loss of generality, assume that

$$\|x^k\| \rightarrow \infty \quad \text{as} \quad k \rightarrow \infty.$$

Then the sequence

$$S'' = \{x^k / \|x^k\|\}$$

on the unit sphere is bounded and hence has a cluster point. Letting x be its cluster point, then S'' includes a subsequence converging to x . We might as well assume

$$x^k / \|x^k\| \rightarrow x.$$

Now it should be shown that x is an unbounded direction of P . Let M be a given positive number. Since $\|x^k\| \rightarrow \infty$, there is a positive integer K such that $\|x^k\| > M$ or $M/\|x^k\| < 1$ when $k \geq K$. Introduce

$$y^k = (M/\|x^k\|)x^k, \quad k = K, K+1, \dots$$

Since P is convex, we have $0, x^k \in P$ and $y^k \in (0, x^k)$, and hence, $y^k \in P$ and $y^k \rightarrow Mx$ when $k \geq K$. As P is closed, it can be asserted that $Mx \in P$. $\square$

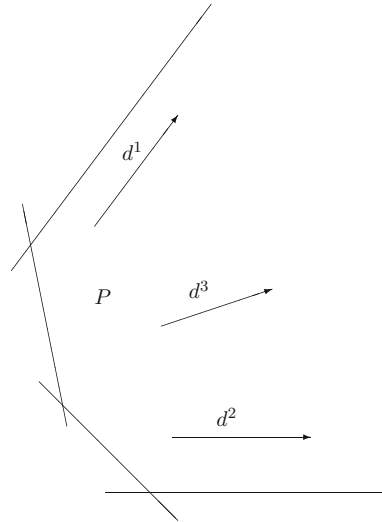
Definition 2.3.4 An extreme (unbounded) direction is not a positive linear combination of two distinct unbounded directions.

According to the preceding, that d is an extreme direction means that if there are unbounded directions d', d'' and positive numbers $\sigma_1, \sigma_2 > 0$, satisfying $d = \sigma_1 d' + \sigma_2 d''$, then there must be $d' = \sigma d'$, where $\sigma > 0$. Two extreme directions having the same direction are regarded as the same.

In Fig. 2.2, d^1, d^2, d^3 are unbounded directions of P . d^1 and d^2 are extreme directions, but d^3 is not.

Theorem 2.3.8 An unbounded direction is an extreme direction if and only if the rank of columns, corresponding to its positive components, is less than the number of columns by 1.

Fig. 2.2 d^1, d^2, d^3 are unbounded directions. d^1, d^2 are extreme directions, but d^3 not



Proof We might as well assume that k positive components of unbounded direction d correspond to the set of columns $a_1, \dots, a_k$. Satisfying $Ad = 0$ implies that the set is linearly dependent. Denote the rank of the set by r ; then it is clear that $r < k$. Without loss of generality, assume that the first r columns are linear independent. Introduce $B = (a_1, \dots, a_r)$. It is clear that

$$\text{rank } B = r \leq \text{rank } A = m.$$

Note that $k \geq 2$ because if $k = 1$, it follows from $Ad = 0$ that $a_1 = 0$, a contradiction.

Necessity. Assume that d is an extreme direction, but $r \neq k - 1$, that is, $a_1, \dots, a_{k-1}$ are linearly dependent. Thus, there exists

$$y = (y_1, \dots, y_{k-1}, 0, \dots, 0)^T \neq 0$$

such that

$$Ay = \sum_{j=1}^{k-1} y_j a_j = 0.$$

Clearly, for sufficiently small $\delta > 0$, it holds that

$$0 \neq d' = d + \delta y \geq 0, \quad 0 \neq d'' = d - \delta y \geq 0,$$

and $Ad' = Ad'' = 0$. Therefore, d', d'' are unbounded directions of P and hence not of the same direction. But $d = (d' + d'')/2$, contradicting that d is an extreme direction. It therefore holds that $r = k - 1$.

Sufficiency. Assume $r = k - 1$. If there exist unbounded directions d', d'' and $\sigma_1, \sigma_2 > 0$ such that

$$d = \sigma_1 d' + \sigma_2 d'',$$

then zero components of d clearly correspond to zero components of d' and d'' . So, the last $n - k$ components of d' and d'' are all zero. In addition, since d', d'' are unbounded directions, it holds that $Ad' = Ad'' = 0$ (Theorem 2.3.5), and hence that

$$Bd'_B + d'_k a_k = 0, \quad Bd''_B + d''_k a_k = 0.$$

It is clear that $d'_k, d''_k > 0$. Premultiplying the two sides of the preceding two equalities by B^T gives

$$B^T B d'_B + d'_k B^T a_k = 0, \quad B^T B d''_B + d''_k B^T a_k = 0,$$

from which it follows that

$$d'_B = -d'_k(B^T B)^{-1} B^T a_k, \quad d''_B = -d''_k(B^T B)^{-1} B^T a_k.$$

Therefore, it holds that

$$d'' = (d''_k/d'_k)d',$$

which means that unbounded directions d' and d'' have the same direction, and hence d is an extreme direction. $\square$

The following reveals the relationship between the extreme direction and 1-dimensional face (edge).

Theorem 2.3.9 *A vector is an extreme direction if and only if it is the unbounded direction of an edge.*

Proof Necessity. Assume that d is an extreme direction. Based on Theorem 2.3.8, assume that columns corresponding to its positive components are

$$a_1, \dots, a_r, a_{m+1},$$

where the first $r \leq m$ columns are linearly independent. Thus

$$d_1, \dots, d_r, d_{m+1} > 0, \quad d_{r+1}, \dots, d_m, d_{m+2}, \dots, d_n = 0. \quad (2.15)$$

Since the rank of A is m , there are $m - r$ columns that together with $a_1, \dots, a_r$ form a basis when $r < m$. Without loss of generality, assume that the first m columns of A constitute a basis, i.e.,

$$B = (a_1, \dots, a_r, a_{r+1}, \dots, a_m), \quad N = \{a_{m+1}, \dots, a_n\}. \quad (2.16)$$

From (2.15) and $Ad = 0$, it follows that

$$\sum_{i=1}^r d_i a_i + d_{m+1} a_{m+1} = 0,$$

hence

$$a_{m+1} = -\left(\sum_{i=1}^r (d_i/d_{m+1}) a_i\right).$$

Assume that the basis B corresponds to the canonical form below:

$$x_B = \bar{b} - \bar{N}x_N. \quad (2.17)$$

Since its augmented matrix $(I \ \bar{N} \mid \bar{b})$ comes from $(A \mid b) = (B \ N \mid b)$ by elementary transformations, and a_{m+1} is a linear combination of $a_1, \dots, a_r$, the last $m - r$ components of column $\bar{a}_{m+1}$ in the canonical form are all zero.

Let $\bar{x}$ belong to P . If $\bar{x}_N = 0$, then $\bar{x}$ is the feasible basic solution corresponding to basis B . Now assume, otherwise, that $\bar{x}_N \neq 0$. We will show that the resulting B corresponds to a feasible basic solution if properly exchanging a_{m+1} and one of $a_1, \dots, a_r$, as well as $a_{r+1}, \dots, a_m$ and $a_{m+2}, \dots, a_n$.

Assume that $\bar{x}_j > 0$ holds for some $j \in N$, $j \neq m+1$. Reducing $\bar{x}_j$ and keeping the other nonbasic components unchanged, we determine the corresponding value of $\bar{x}_B$ such that $\bar{x}$ satisfies (2.17), resulting in a new solution $\bar{x}$. For the column corresponding to $\bar{x}_j$, of the canonical form (2.17), there are only two cases arising:

(i) $\bar{a}_j \geq 0$.

It is clear that $\bar{x}_{m+1}$ may decrease to 0 and associated $\bar{x}_B$ remains nonnegative, and thus we set $\bar{x}_j = 0$.

(ii) $\bar{a}_j \not\geq 0$.

If the first r components of $\bar{a}_j$ are nonnegative, one of the last $m - r$ components of $\bar{x}_B$ decreases to 0 the earliest (the so-called blocking) as $\bar{x}_j$ decreases. Assume that it is component i ($r+1 \leq i \leq m$). Set $\bar{x}_j$ to the associated value, and interchange a_j and a_i to update B and N . By exchanging their indices at the same time, the nonbasic component $\bar{x}_j$ of the new solution $\bar{x}$ becomes 0.

If some of the first r components of $\bar{a}_j$ are negative, then we can determine a $\sigma > 0$ such that the first r basic components of

$$\bar{x} := \bar{x} + \sigma d$$

are sufficiently large and all the nonbasic components remain unchanged, except for $\bar{x}_{m+1}$, such that no broking happens to the first r components of $\bar{x}_B$ as $\bar{x}_j$ decreases. If no broking happens to the last $m - r$ components either, we set $\bar{x}_j = 0$. If broking happens to component $r+1 \leq i \leq m$, we set $\bar{x}_j$ to the according value and interchange a_j and a_i to update B and N . Consequently, by exchanging their indices, the nonbasic component $\bar{x}_j$ of the new solution $\bar{x}$ is now equal to 0.

As such, we can transform all $\bar{x}_j$, $j \in N$, $j \neq m+1$ to 0, without affecting the first r indices of B . Then, if $\bar{x}_{m+1} = 0$, we are done.

If $\bar{x}_{m+1} > 0$, we deduce it and keep the other nonbasic components unchanged. Since the last $m - r$ components of $\bar{a}_{m+1}$ are zero, the corresponding components of $\bar{x}_B$ remain unchanged, and there will be only two cases arising:

(i) The first r components of $\bar{a}_{m+1}$ are all nonnegative. Then it is clear that $\bar{x}_{m+1}$ can decrease to 0 and according $\bar{x}_B$ remain nonnegative; thus we set $\bar{x}_{m+1} = 0$.

(ii) Some of the first r components of $\bar{a}_{m+1}$ are negative. If the according $\bar{x}_B$ remains nonnegative as $\bar{x}_{m+1}$ decreases to 0, we set $\bar{x}_{m+1} = 0$; otherwise, if broking happens for component $1 \leq i \leq r$ of $\bar{x}_B$, we set $\bar{x}_{m+1}$ to the associated value and interchange a_{m+1} and a_i to update B and N . Then, by exchanging their indices, the nonbasic component $\bar{x}_{m+1}$ of the new $\bar{x}$ is equal to zero.

Therefore, it can be asserted that the basis B , defined by (2.16), corresponds to the feasible basic solution $\bar{x}$.

Consider the following face

$$\begin{aligned} P' &= \{x \in \mathcal{R}^n \mid Ax = b, x \geq 0, x_j = 0, j = m+2, \dots, n\} \\ &= \{x \in \mathcal{R}^n \mid Bx_B + x_{m+1}a_{m+1} = b, x_B, x_{m+1} \geq 0; x_j = 0, j = m+2, \dots, n\} \\ &= \{x \in \mathcal{R}^n \mid x_B + x_{m+1}\bar{a}_{m+1} = \bar{b}, x_B, x_{m+1} \geq 0; x_j = 0, j = m+2, \dots, n\}. \end{aligned} \quad (2.18)$$

It is clear that $\bar{x} \in P'$. Hence, from (2.15) and

$$Bd_B + d_{m+1}a_{m+1} = 0 \quad (2.19)$$

it is known that

$$\bar{x} + \alpha d \in P', \quad \forall \alpha \geq 0.$$

Therefore, d is an unbounded direction of P' . It is now only needed to show $\dim P' = 1$.

For any $x' \in P'$ and $x' \neq \bar{x}$, introduce $d' = x' - \bar{x}$. It is clear that

$$d'_1, \dots, d'_r, d'_{m+1} > 0, \quad d'_{r+1}, \dots, d'_m, d'_{m+2}, \dots, d'_n = 0 \quad (2.20)$$

and

$$Bd'_B + d'_{m+1}a_{m+1} = 0. \quad (2.21)$$

From (2.19) and (2.21) it follows, respectively, that

$$d_B = -d_{m+1}B^{-1}a_{m+1}, \quad d'_B = -d'_{m+1}B^{-1}a_{m+1}.$$

Therefore, $d' = (d'_{m+1}/d_{m+1})d$, where $d'_{m+1}/d_{m+1} > 0$. This implies that $\dim P' = 1$. Therefore, P' is a 1-dimensional face or edge, and d is an unbounded direction of it.

Sufficiency. Assume that d is an unbounded direction of edge P' (2.18) and hence satisfies (2.19). If d is a positive linear combination of unbounded directions d', d'' of P , then there exists a correspondence between zero components of d and of d', d'' , and hence d', d'' are also unbounded directions of P' . Since $\dim P' = 1$, in addition, d' and d'' have the same direction. Therefore, d is an extreme direction. $\square$

Lemma 2.3.10 *An unbounded direction that is not an extreme one is a positive linear combination of two unparallel unbounded directions.*

Proof Without loss of generality, assume that positive columns $a_1, \dots, a_k$ correspond to positive components of an unbounded direction d , and the first r columns are linearly independent. As d is not an extreme direction, it holds by Theorem 2.3.8 that $r < k - 1$, or

$$k - r \geq 2. \quad (2.22)$$

Introduce $B_1 = (a_1, \dots, a_r)$. Within the set of $a_{k+1}, \dots, a_n$, determine $m - r$ columns, corresponding to $B_1 = (a_{k+1}, \dots, a_{k+m-r})$, to construct basis $B = (B_1, B_1)$ (it is clear that $m - r \leq n - k$). Then the nonbasic matrix is $N = (N_1, N_2)$, where $N_1 = (a_{r+1}, \dots, a_k)$, $N_2 = (a_{k+m-r+1}, \dots, a_n)$. Assume that B corresponds to the canonical form $x_B = \bar{b} - \bar{N}x_N$. Then d satisfies

$$B_1 d_{B_1} = -\bar{N}_1 d_{N_1}, \quad d_{N_2} = 0,$$

or equivalently,

$$d_{B_1} = -(B_1^T B_1)^{-1} B_1^T \bar{N}_1 d_{N_1}, \quad d_{N_2} = 0.$$

Introduce $e \in \mathcal{R}^{k-r}$ and

$$d'_{B_1} = -(B_1^T B_1)^{-1} B_1^T \bar{N}_1 (d_{N_1} + \epsilon e), \quad d''_{B_1} = -(B_1^T B_1)^{-1} B_1^T \bar{N}_1 (d_{N_1} - \epsilon e).$$

Letting $\epsilon = (\epsilon_1, \dots, \epsilon_{k-r})^T > 0$, then

$$d' = \begin{pmatrix} d'_{B_1} \\ 0 \\ d_{N_1} + \epsilon \\ 0 \end{pmatrix} \quad \text{and} \quad d'' = \begin{pmatrix} d''_{B_1} \\ 0 \\ d_{N_1} - \epsilon \\ 0 \end{pmatrix}$$

satisfy $d = d'/2 + d''/2$. It is clear that $d', d'' \neq 0$ and $Ad' = 0, Ad'' = 0$. As (2.22) holds, it is known that there exists a sufficiently small ϵ such that $d', d'' \geq 0$ are unparallel unbounded directions. $\square$

Theorem 2.3.11 *If P has an unbounded direction, it has an extreme direction.*

Proof Let d be an unbounded direction. Assume that it has k positive components and the rank of the corresponding columns is $r \leq k - 1$. If $r = k - 1$, then d is an extreme direction (Theorem 2.3.8). Otherwise, by Lemma 2.3.10, d can be expressed as

$$d = \sigma_1 d' + \sigma_2 d'',$$

where d', d'' are unparallel unbounded directions, and $\sigma_1, \sigma_2 > 0$. Without loss of generality, assume that d'' has at least one component greater than the corresponding

component of d' . Thus, the following vector

$$\hat{d} = d'' - \alpha(d'' - d'), \quad \alpha = \min\{d_j''/(d_j'' - d_j') \mid d_j'' - d_j' > 0, j = 1, \dots, n\} > 0$$

is well-defined. Consider

$$A\tilde{d} = 0, \quad \tilde{d} \geq 0, \quad \tilde{d} \neq 0,$$

where the first two equalities clearly hold. If the third does not hold, then $d'' - \alpha(d'' - d') = 0$, as leads to

$$d' = \frac{\alpha - 1}{\alpha} d'',$$

implies that d', d'' are parallel, as a contradiction. Therefore, the third equality also holds, and hence, $\tilde{d}$ is an unbounded direction. In addition, it is known that the number of zero components of $\tilde{d}$ is less than that of d by 1, at least. Then set $d = \tilde{d}$ and repeat the preceding steps. It is clear that such a process can only be repeated finitely many times, and it terminates at some extreme direction. $\square$

According to Theorem 2.3.9, extreme directions and unbounded edges of P are 1-to-1 correspondent (extreme directions having the same direction are viewed as the same). As there are finitely many edges, the number of extreme directions is finite.

2.4 Representation of Feasible Region

It is now time to lay a theoretical basis for the Dantzig–Wolfe decomposition method for solving large-scale LP problems (Chap. 9).

Theorem 2.4.1 (Representation Theorem of Feasible Region) *Let P be nonempty. Assume that $\{u^1, \dots, u^s\}$ is the vertex set and $\{v^1, \dots, v^t\}$ is the extreme direction set. Point x belongs to feasible region if and only if*

$$\begin{aligned} x &= \sum_{i=1}^s \alpha_i u^i + \sum_{j=1}^t \beta_j v^j, \\ \sum_{i=1}^s \alpha_i &= 1, \quad \alpha_i \geq 0, \quad i = 1, \dots, s, \\ \beta_j &\geq 0, \quad j = 1, \dots, t. \end{aligned} \tag{2.23}$$

Proof When (2.23) holds, it is clear that $x \in P$. So it is only needed to show necessity. We use an inductive method to dimensions.

If $\dim P = 0$, P is a single point set, including a vertex. The conclusion holds clearly. Assume that it holds for $\dim P < k$. We will show that it holds for $\dim P = k \geq 1$.

From Proposition 2.2.2, it follows that $\text{int } P \neq \emptyset$. Assume $x \in \text{int } P$, and consider

$$x' = x - \lambda(u^1 - x). \quad (2.24)$$

Note that $u^1 \neq x$. There are the following two cases arising:

(i) $u^1 - x \not\leq 0$.

Determine λ such that

$$\lambda = x_q / (u_q^1 - x_q) = \min\{x_j / (u_j^1 - x_j) \mid u_j^1 - x_j > 0, j = 1, \dots, n\} > 0.$$

Then, it is easy to verify that x' , defined by (2.24), satisfies $x' \in P$ and $x'_q = 0$. Therefore, x' belongs to some proper face with its dimension less than k . According to the assumption of induction, x' can be expressed as the sum of a convex combination of vertices and a nonnegative combination of extreme directions of the proper face. Since vertices and extreme directions of a face are also that of P , therefore, x' can be expressed as the sum of a convex combination of vertices and a nonnegative combination of extreme directions of P , i.e.,

$$\begin{aligned} x' &= \sum_{i=1}^{s_1} \alpha'_i u^i + \sum_{j=1}^{t_1} \beta'_j v^j, \\ \sum_{i=1}^{s_1} \alpha'_i &= 1, \quad \alpha'_i \geq 0, \quad i = 1, \dots, s_1, \\ \beta'_j &\geq 0, \quad j = 1, \dots, t_1, \end{aligned}$$

where u^i and v^j are vertices and extreme directions of P , respectively. Substituting the preceding into

$$x = \frac{1}{1+\lambda} x' + \left(1 - \frac{1}{1+\lambda}\right) u^1,$$

which is equivalent to (2.24), leads to the expression of the form of (2.23), i.e.,

$$x = \sum_{i=1}^{s_1} \frac{1}{1+\lambda} \alpha'_i u^i + \left(1 - \frac{1}{1+\lambda}\right) u^1 + \sum_{j=1}^{t_1} \frac{1}{1+\lambda} \beta'_j v^j.$$

(ii) $u^1 - x \leq 0$.

Then the point x' , defined by (2.24), is the feasible point for any $\lambda \geq 0$; hence, $-(u^1 - x)$ is an unbounded direction. According to Lemma 2.3.11, there is an extreme direction, say v^1 . Now take a sufficiently large μ such that

$$\tilde{x} = u^1 + \mu v^1$$

has at least one component greater than the corresponding component of x . Consequently, the point defined by

$$x' = x - \lambda(\tilde{x} - x), \quad \lambda = x_q/(\tilde{x}_q - x_q) = \min\{x_j/(\tilde{x}_j - x_j) \mid \tilde{x}_j - x_j > 0, j = 1, \dots, n\} > 0$$

is a feasible point, satisfying $x'_q = 0$. Therefore, this point belongs to some proper face with dimension less than k and hence can be expressed as the sum of a convex combination of vertices and a nonnegative combination of extreme directions of P . As a result, an expression of the form (2.23) can be obtained in an analogous manner as case (i). $\square$

The preceding theorem indicates that a feasible point is the sum of a convex combination of vertices and a nonnegative combination of extreme directions and vice versa. In particular, the following is a direct corollary.

Corollary 2.4.2 *Let the feasible region be bounded. A point is feasible if and only if it is a convex combination of vertices.*

2.5 Optimal Face and Optimal Vertex

To begin with, we present a basic result without the proof (see, e.g., Rockafellar 1997).

Theorem 2.5.1 *Let $\bar{x}$ be a boundary point of convex set S . Then there exists a hyperplane including $\bar{x}$ and partitioning the total space into two half-spaces, one of which includes S .*

The hyperplane involved in the preceding theorem is said to be *supporting hyperplane* of S . That is to say, there is a supporting hyperplane through every boundary point of a convex set. Although the result applies to any convex set, we are only concerned with the feasible region P , in particular.

Supporting hyperplane of P is closely related to its face, as the following reveals.

Lemma 2.5.2 *The intersection of P and a supporting hyperplane is a face.*

Proof Assume that $H = \{x \in \mathcal{R}^n \mid a^T x = \eta\}$ is a supporting hyperplane of P , and $P' = P \cap H$. Without loss of generality, let $a^T x \leq \eta$ hold for all $x \in P$.

Assume that $v \in P'$ is an interior point of segment (y, z) , and $y, z \in P$, i.e.,

$$v = \alpha y + (1 - \alpha)z, \quad (2.25)$$

where $0 < \alpha < 1$. Note that P' is a nonempty convex set.

It is only needed to show $y, z \in H$, as leads to $y, z \in P'$; hence, P' is a face of P .

Assume $y, z \notin H$. Then it holds that

$$a^T y < \eta, \quad a^T z < \eta.$$

Multiplying the preceding two formulas, respectively, by $\alpha > 0$ and $1 - \alpha > 0$ and then adding the results give

$$a^T(\alpha y + (1 - \alpha)z) < \eta\alpha + \eta(1 - \alpha) = \eta,$$

combining which and (2.25) leads to $v \notin H$, and hence $v \notin P'$. This contradicts the assumption $v \in P'$. Therefore, at least one of y and z belongs to H .

Without loss of generality, assume $z \in H$. Then, it follows from (2.25) that

$$y = (1/\alpha)v + (1 - 1/\alpha)z,$$

implying that y is in the straight line through v and z . Moreover, $z, v \in H$ and H is a supporting hyperplane, therefore $y \in H$. $\square$

Definition 2.5.1 The optimal (solution) set is the set of all optimal solutions.

Lemma 2.5.3 Let f^* be the optimal value of the standard LP problem. The set F is the optimal set if and only if it is the intersection of P and the optimal objective isosurface, i.e.,

$$\bar{H} = \{x \in \mathcal{R}^n \mid c^T x = f^*\}. \quad (2.26)$$

Proof Assume $F = P \cap \bar{H}$.

It is clear that any optimal solution $\bar{x} \in P$ satisfies $c^T \bar{x} = f^*$, implying $\bar{x} \in \bar{H}$, hence $\bar{x} \in F$. Therefore, F is the set of optimal solutions. If F is the set of optimal solutions, conversely, then $c^T \bar{x} = f^*$ holds for any $\bar{x} \in F \subset P$. Therefore, $\bar{x} \in \bar{H}$, and hence, $\bar{x} \in P \cap \bar{H}$. $\square$

The $\bar{H}$ is clearly a supporting hyperplane of P , referred to as the *objective supporting hyperplane*.

A face is optimal if its elements are all optimal solutions. A vertex is optimal if it is an optimal solution.

Lemma 2.5.4 If there exists an optimal solution, there exists an optimal face.

Proof According to Lemma 2.5.3, a nonempty set of optimal solutions is the intersection of feasible region P and the objective isosurface $\bar{H}$. Therefore, it is an optimal face, associated to Lemma 2.5.2 and the definition of an optimal face. $\square$

Theorem 2.5.5 *If there is a feasible solution, there is a feasible basic solution. If there is an optimal solution, there is an optimal basic solution (vertex).*

Proof By Lemmas 2.3.3 and 2.3.4, it is known that nonempty feasible region has a feasible basic solution. By Lemma 2.5.4, if there is an optimal solution, there is an optimal face, which is a nonempty polyhedral convex set and hence having an optimal convex or optimal basic solution. $\square$

A vertex is a 0-dimensional optimal face. In general, there exist optimal faces of higher dimensions. The *optimal set* is just the largest optimal face, i.e., the optimal face of the highest dimension. The following is obvious.

Proposition 2.5.1 *If there exists an optimal solution, there exists the largest optimal face.*

2.6 Graphic Approach

An LP problem of 2-dimension can be solved via a graphic approach. To do so, let us return to Example 1.2.2. The area of polygon OABCD in Fig. 2.3 is the feasible region (ignore the straight line $x + 2y = 10$, at the moment). It is required to

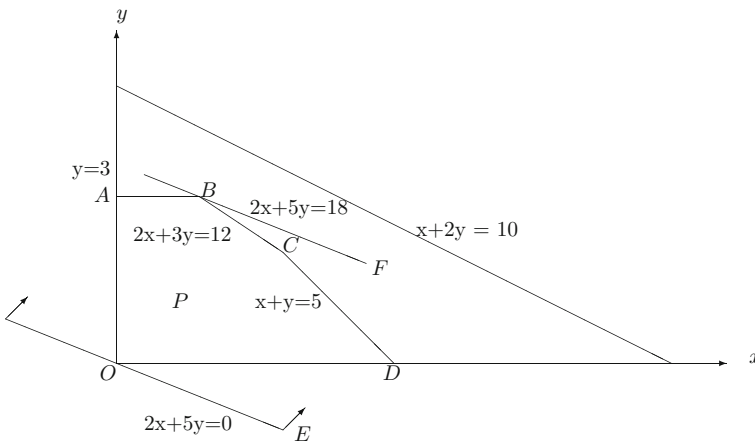


Fig. 2.3 Graphic solution to Example 1.2.2

determine a point over the area such that the objective function reaches the highest value at the point.

In the figure, equation $2x + 5y = 0$ of the isoline of the objection function corresponds to the straight line OE , going through the origin, all points on which correspond to the same objective value 0. The line's slope, the tangent of the angle between it and the x axis, is $-2/5 = -0.4$. Therefore, the corresponding isoline parallel shifts up to the upper-right side as the objective value increases from 0. The points on the intersection of the line and the area $OABCD$ are all feasible ones. The parallel shifting should be carried out as far as the intersection remains nonempty while achieving the largest possible objective value. It is seen that the farthest shifting line is BF though vertex B . The optimal set, i.e., the intersection of the line and the feasible region, includes a single vertex B , corresponding to the optimal basic solution. Consequently, problem (1.6) is solved after measuring coordinates of point B and calculating the associated objective value.

If the figure or measurement is not accurate, however, the outcome may involve unacceptable errors. For a graphic approach, therefore, it might be desirable to use coordinate paper together with simple calculating. Once B is known to be the optimal vertex, its coordinates can be obtained by solving the following system of equations:

$$\begin{cases} 2x + 3y = 12 \\ y = 3 \end{cases} \quad (2.27)$$

from which the optimal basic solution $\bar{x} = 1.5$, $\bar{y} = 3$ follows with optimal value $f = 18$. That is to say, the manufacturer should arrange production of 1.5 thousand laths and 3 thousand sheep piles daily, gaining 18 thousand dollars profit.

Although the graphic approach does not apply to the case of $n \geq 3$, it may give us some useful enlightenment that is the theme of the next section.

2.7 Heuristic Characteristic of Optimal Solution

From the graphic approach, it is seen that the optimal solution to the LP problem is attained at a vertex. It is imaginable that if the feasible region has an edge passing through a vertex, which is parallel to objective isolines, then the whole edge corresponds to the optimal set with the same optimal value.

Therefore, the key is how to determine lines intersecting at the optimal vertex. In other words, it is only needed to know which inequalities are binding at the optimal solution. Once these inequalities are known, what left to do is just to solve a linear system.

It is observed that normal directions (pointing to the interior of the feasible region) of lines AB and BC , intersecting at the optimal vertex B , form the *maximum angles* with the parallel shifting direction of the isoline BF among all lines.

Now turn to the more general minimization problem

$$\begin{aligned} \min f &= c^T x \\ \text{s.t. } Ax &\geq b, \end{aligned} \quad (2.28)$$

where $A \in \mathcal{R}^{m \times n}$, $c \in \mathcal{R}^n$, $b \in \mathcal{R}^m$, $m > 1$.

We can make a similar image in R^n space. Now examine angles between normal directions and parallel shifting direction of the isosurface. For minimization, the shifting direction is the negative gradient of the objective function. This leads to the following plausible statement (Pan 1990):

Proposition 2.7.1 (Heuristic Characteristic of Optimal Solution 1) *Gradients of constraints binding at optimal solution tend to form the maximum angles with the negative objective gradient.*

Thus, we need to quantify sizes of angles. Denote the i th row vector of A by $\bar{a}_i^T$. The cosine of the angle between the i th constraint gradient and the negative objective gradient is then

$$\cos \langle \bar{a}_i, c \rangle = -\bar{a}_i^T c / (\|\bar{a}_i\| \|c\|).$$

For simplicity, we ignore factor $1/\|c\|$ and introduce the following:

Definition 2.7.1 The pivoting-index of the i th constraint is defined as

$$\alpha_i = -\bar{a}_i^T c / \|\bar{a}_i\|. \quad (2.29)$$

Then, we are able to compare angles with pivoting-indices, and Proposition 11.1.2 can be transformed into:

Proposition 2.7.2 (Heuristic Characteristic of Optimal Solution 2) *The gradients of constraints binding at an optimal solution tend to have the minimum pivoting-indices.*

Example 2.7.1 Investigate problem (1.6) via pivoting-indices:

$$\begin{aligned} \min f &= -2x - 5y \\ \text{s.t. } \quad &-2x - 3y \geq -12 \\ &-x - y \geq -5 \\ &-y \geq -3 \\ &x, y \geq 0. \end{aligned} \quad (2.30)$$

Answer Calculate pivoting-indices of the constraints, and put them in the following table in the order of increasing pivoting-indices:

Constraints	α_i
$-2x - 3y \geq -12$	-5.26
$-y \geq -3$	-5.00
$-x - y \geq -5$	-4.95
$x \geq 0$	2.00
$y \geq 0$	5.00

From the preceding table, it is seen that $-2x - 3y \geq -12$ and $-y \geq -3$ are the two smallest pivoting-indexed constraints. Thus, the two are binding constraints at the optimal solution. This leads to the system (2.27) again, coinciding with the outcome of the graphic approach.

If Proposition 11.1.2 were true in general, solving an LP problem amounts to solving a system of linear equations by $O(m^3)$ basic arithmetics (!) But the characteristic of the optimal solution is *heuristic* only. Counterexamples can be easily constructed. If adding a constraint $x + 2y \leq 10$ to (1.6) (see Fig. 2.3), e.g., the added constraint is not binding at the optimal solution, although its gradient forms the maximum angle with the objective gradient (with pivoting-index -5.37), since adding is a redundant constraint.

Someday in the Fall of 1986 when the author was visiting the Mathematics Department of University of Washington, after he talked with Professor Rockafellar about his idea on the characteristic of an optimal solution, the latter quickly showed a counterexample by sketching on a piece of paper.

For all that, Proposition 11.1.2 might still serve as a clue to the optimal solution. Such a trick will be referred to as *the most-obtuse-angle heuristic* in this book.

In fact, in some cases, it is possible to detect the unboundedness of a problem simply by signs of pivoting-indices.

Theorem 2.7.1 *Assume that the feasible region is nonempty. If the pivoting-indices of constraints are all nonnegative, then the problem (2.28) is unbounded.*

Proof By contradiction. Assume that the problem is bounded under the assumptions. It follows that

$$c^T v \leq 0, \quad \forall v \in \{v \in \mathcal{R}^n \mid Av \geq 0\}. \quad (2.31)$$

In fact, there is v such that

$$Av \geq 0, \quad c^T v > 0. \quad (2.32)$$

Thus, for any $\alpha \geq 0$ and feasible solution $\bar{x}$, vector $x = \bar{x} + \alpha v$ satisfies $Ax \geq b$, that is, v is an unbounded direction. Further, it is known from (2.32) that

$$c^T x = c^T \bar{x} + \alpha c^T v \rightarrow \infty, \quad (\text{as } \alpha \rightarrow \infty),$$

which contradicts that (2.28) is upper bounded. Therefore, (2.31) holds.

Thus, according to Farkas' Lemma 2.0.3, there is $y \geq 0$ such that

$$-c = A^T y,$$

premultiplying which by $-c^T$ gives

$$0 < c^T c = -y^T A c.$$

Thus $A c \geq 0$ follows from the non-negativeness of pivoting-indices. This implies that the right-hand side of the preceding is less than or equal to 0, as is a contradiction. Therefore, the problem is unbounded. $\square$

An alternative proof of the preceding theorem is via showing c to be an unbounded direction. The following corollary gives a necessary condition for the existence of an optimal solution.

Corollary 2.7.2 *If there is an optimal solution, then there is at least one constraint bearing a negative pivoting-index.*

Example 2.7.2 Investigate the following LP problem by pivoting-indices:

min	$f = -x - y$	α_i
s.t.	$2x - y \geq -3$	0.71
	$-3x + 4y \geq 4$	0.20
	$2x + 2.5y \geq 5$	1.41
	$-2x + 4y \geq -8$	0.45
	$y \geq 1.2$	1.00
	$x \geq 0$	1.00
	$y \geq 0$	1.00

(2.33)

Answer Calculate indices of the constraints, and fill in the right-hand side of the preceding table. According to Theorem 2.7.1, the problem is unbounded since all indices are nonnegative (see Fig. 2.4).

2.8 Feasible Direction and Active Constraint

Methods for solving mathematical problems fall into two categories: direct and iterative methods. The latter produces a sequence of points by iterations, offering an exact or approximate solution. Methods presented in this book belong to the iterative category.

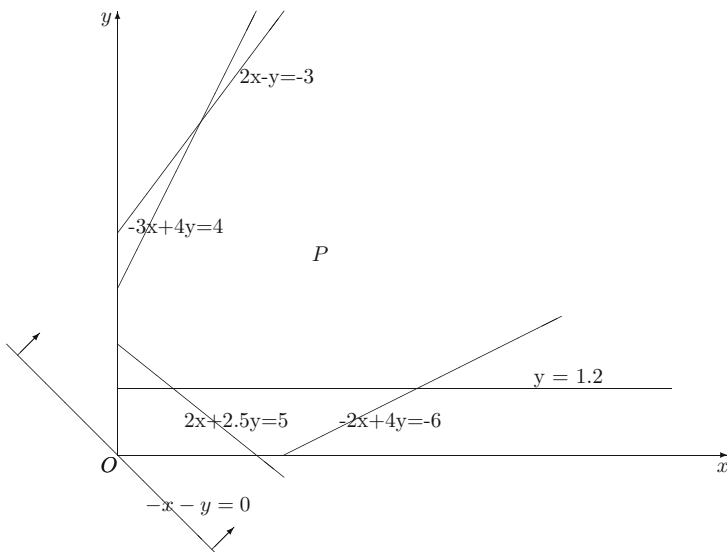


Fig. 2.4 Unbounded problem

Line search is the mostly used iterative approach in optimization. At a current point $\bar{x}$, in each iteration, a new point $\hat{x}$ is determined along a ray starting from $\bar{x}$ along a nonzero vector d , i.e.,

$$\hat{x} = \bar{x} + \alpha d, \quad (2.34)$$

where d is referred to as *search direction*, $\alpha > 0$ as *stepsize*. Once the two are available, $\hat{x}$ can be calculated and then one iteration is complete. Repeating this process yields a sequence of points, until a solution is reached. Formula (2.34) is referred to as *line search* or *iterative scheme*.

The determination of a search direction is crucial. In the presence of constraints, d should be such that the intersection of the ray (2.34) and the feasible region is nonempty. To be precise, we introduce the following concept:

Definition 2.8.1 Let P be the feasible region. Assume that $\bar{x} \in P$, $d \neq 0$. If there exists $\bar{\alpha} > 0$ such that

$$\bar{x} + \alpha d \in P, \quad \forall \alpha \in [0, \bar{\alpha}],$$

d is a feasible direction at $\bar{x}$.

The preceding is relevant to general constrained optimization problems, including LP problems. The following are some instances, in conjunction with feasible region P .

Example 2.8.1 In Fig. 2.5, $\bar{x}$ is an interior point of P , and hence any direction is feasible at it.

Fig. 2.5 Any direction is feasible at $\bar{x}$

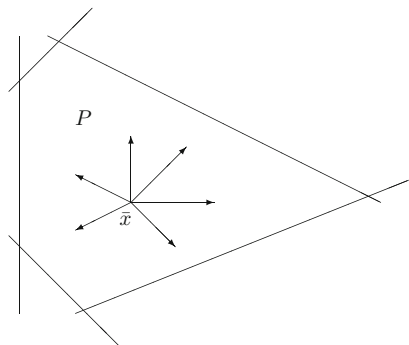


Fig. 2.6 d^3, d^4 are feasible at $\bar{x}$, but d^1, d^2 not

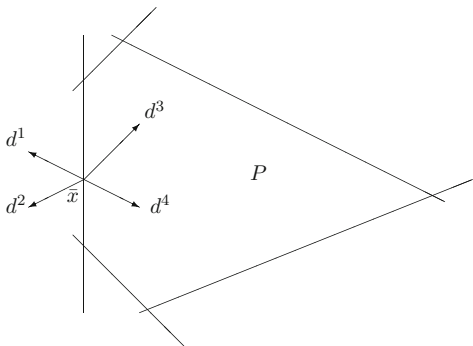
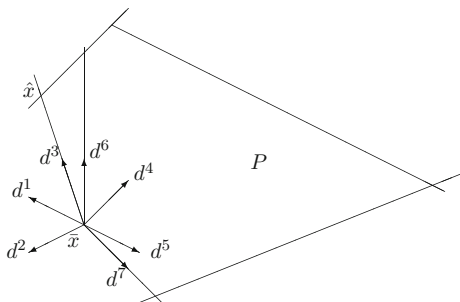


Fig. 2.7 d^4, d^5, d^6, d^7 are feasible at $\bar{x}$, but d^1, d^2, d^3 not



Example 2.8.2 In Fig. 2.6, $\bar{x}$ is a boundary point of P . It is seen that d^3, d^4 are feasible directions at $\bar{x}$, but d^1, d^2 are not.

Example 2.8.3 In Fig. 2.7, $\bar{x}$ is a vertex of P . Any direction, e.g., d^4 or d^5 , within the angle area between d^6 and d^7 (which are along the two edges of P , respectively) is feasible at $\bar{x}$. The vectors d^1, d^2, d^3 are not feasible (Fig. 2.4).

Let d be a feasible search direction. It is possible to maintain feasibility of new iterate $\hat{x}$. In order for $\hat{x}$ to be close to an optimal solution, it is needed to take into account the objective function. To this end, consider the following problem:

$$\begin{aligned} \min \quad & f = c^T x \\ \text{s.t.} \quad & a_i^T x \geq b_i, \quad i = 1, \dots, m, \end{aligned} \quad (2.35)$$

where $m > n$, and whose feasible region is

$$P = \{x \in \mathcal{R}^n \mid a_i^T x \geq b_i, \quad i = 1, \dots, m\}.$$

Definition 2.8.2 The vector d satisfying condition $c^T d < 0$ is a descent direction. If d is also a feasible direction at $\bar{x} \in P$, it is a feasible descent direction at $\bar{x}$.

It is clear that d is a feasible descent direction at $\bar{x}$ if and only if it is a feasible direction and forms an obtuse angle with the objective gradient c . Once such a direction d is available, a stepsize $\alpha > 0$ can be determined, and hence a new iterate $\hat{x} \in P$, obtained by (2.34), corresponding to a smaller objective value. Then, one iteration is complete.

It is noticeable that not all constraints affect the determination of a feasible descent direction at a current point.

Definition 2.8.3 A constraint that is violated, or satisfied as an equation, by the current point is an active constraint.

Aimed at a *feasible* point, the *active* constraint is usually defined as one satisfied as equality at the point. But it seems to be useful to include an infeasible point by regarding a constraint violated as active.

Let us bring up (2.35) as an example. If $a_i^T \bar{x} = b$, then $a_i^T x \geq b$ is active at $\bar{x}$; if, otherwise, $a_i^T \bar{x} > b$, then $a_i^T x \geq b$ is not active at that point. So, the current point is on a boundary of an active constraint. From the simple instances given in the preceding figures, it is seen that a feasible descent direction can be determined by taking into account active constraints only.

In practice, the preceding definition for an active constraint is not very satisfactory since a point close to some boundary would lead to a too small stepsize. Therefore, Powell (1989) proposes the so-called ϵ -active constraint, where ϵ is a small positive number. For example, if $a_i^T \bar{x} - b \leq \epsilon$, then $a_i^T x \geq b$ is an ϵ -active constraint at $\bar{x}$; whereas it is not if $a_i^T \bar{x} - b > \epsilon$.

The LP problem can be solved by the so-called *active set method*, which is often used for solving nonlinear programming problems though some scholars prefer it to be an LP problem solver (e.g., Fletcher 1981 and Hager 2002). We outline the method in conjunction with the problem (2.35) in the remainder of this section.

Assume that the current vertex $\bar{x}$ is a unique solution to the linear system below:

$$a_i^T x = b_i, \quad i \in \mathcal{A},$$

where $\mathcal{A}$ is called *the active set*, consisting of n indices of (total or part) active constraints, with linearly independent gradients a_i . If $\bar{x}$ is judged to be optimal under some criterion, we are done; otherwise, some index $p \in \mathcal{A}$ is selected such that the $n - 1$ equalities

$$a_i^T x = b_i, \quad i \in \mathcal{A} \setminus \{p\}, \quad (2.36)$$

determine a descent edge (1-dimensional face).

In fact, since the rank of the coefficient matrix of (2.36) is $n - 1$, the associated homogeneous system

$$a_i^T d = 0, \quad i \in \mathcal{A} \setminus \{p\},$$

has a solution d such that

$$d \neq 0, \quad c^T d < 0.$$

It can be verified that for $\alpha \geq 0$, all points on the ray

$$\hat{x} = \bar{x} + \alpha d$$

satisfy (2.36). Under the condition that the objective value is lower bounded, the following stepsize is well-defined:

$$\alpha = (b_q - a_q^T \bar{x}) / a_q^T d = \min\{(b_i - a_i^T \bar{x}) / a_i^T d \mid a_i^T d < 0, i \notin \mathcal{A}\} \geq 0.$$

In fact, such an α is the maximum stepsize possible to maintain the feasibility of $\hat{x}$. Since $a_q^T \hat{x} = b_q$, the $a_q^T x \geq b_q$ is an active constraint at $\hat{x}$.

Consequently, setting $\bar{x} = \hat{x}$ and redefining $\mathcal{A} = \mathcal{A} \setminus \{p\} \cup \{q\}$ complete an iteration of the active set method. If $\alpha > 0$, the new vertex corresponds to a lower objective value. Note however that if there are multiple active constraints at the current vertex $\bar{x}$, then α defined by (2.36) would vanish so that the descent edge degenerates to a vertex. That is to say, the “new vertex” is just the old, though set $\mathcal{A}$ changes. Such a vertex is called *degenerate*. In Fig. 2.7, e.g., $\bar{x}$ is a degenerate vertex, at which the three edges along d^3, d^6, d^7 intersect.

The simplex method can be viewed as a special scheme of the active set method, as will be discussed in Sect. 3.6. In history, the former emerged before the latter from another path by fully taking advantage of the linear structure.

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Chapter 3

Simplex Method



The simplex method is the most famous and widely used method for solving LP problems. Since created by George B. Dantzig in 1947, it has been dominating this field for more than 70 years. This chapter presents this method, including tableau as well as revised simplex algorithm, starting-up, degeneracy, cycling, and finite pivot rule. We conclude with some remarks in the final section.

The basic idea behind the simplex method seems to be simple. It moves from a vertex to an adjacent vertex, while improving the objective value, until reaching an optimal vertex. This is based on Theorem 2.5.5, guaranteeing the existence of an optimal basic solution if any. Since the feasible region has a finite number of vertices, it seems to be natural to hunt among vertices.

The idea may be traced back to as early as Fourier (1823). It was materialized algebraically by Dantzig (1951a). In this chapter, the simplex method will be presented in a tableau form first, and then it is revised to a more applicable version. Related topics are also discussed, such as how to get the method started, the finiteness, computational complexity, etc. The final section will make remarks on the simplex method.

3.1 Simplex Algorithm: Tableau Form

We begin by introducing the so-called *simplex tableau* for problem (1.10). In Sect. 1.5, we already obtained the canonical form 1.5.1 of its constraint system, without touching the objective function at all. Now we move the objective function

$$x_1 + 2x_2 - x_4 + x_5 - f = 0,$$

where f is called *objective variable*, to the bottom of the constraint system, so that the tableau becomes

x_1	x_2	x_3	x_4	x_5	f	RHS
1			2/11	-4/11		7/11
	1		15/11	-19/11		14/11
		1	1/11	9/11		9/11
1	2		-1	1	-1	

Then we eliminate the nonzero entries, corresponding to pivot columns, associated with x_1, x_2, x_3 , in the bottom (objective) row. To do so, add -1 times of the first row to the bottom row at first:

x_1	x_2	x_3	x_4	x_5	f	RHS
1			2/11	-4/11		7/11
	1		15/11	-19/11		14/11
		1	1/11	9/11		9/11
	2		-13/11	15/11	-1	-7/11

Add -2 times of row 2 to the bottom row:

x_1	x_2	x_3	x_4	x_5	f	RHS
1			2/11	-4/11		7/11
	1		15/11	-19/11		14/11
		1	1/11	9/11		9/11
			-43/11	53/11	-1	-35/11

where the northwest corner is the unit matrix, corresponding to zero entries in the bottom row. Thereby, the tableau offers not only a basic solution

$$x_1 = 7/11, \quad x_2 = 14/11, \quad x_3 = 9/11, \quad x_4 = x_5 = 0, \quad (3.1)$$

but also a reduced form of the objective function over the feasible region. Clearly, the solution is a feasible basic solution with objective value $35/11$, which is equal to the negative southeast corner entry.

It cannot be asserted that the feasible basic solution given by (3.1) is optimal because the reduced cost with x_4 is negative. In fact, as the value of x_4 increases from 0, while that of x_5 is fixed at 0, the objective value can reduce further.

The new value of x_4 should be as large as possible, so that the corresponding objective value becomes as low as possible, subject to

$$\begin{cases} x_1 = 7/11 - (2/11)x_4 \geq 0, \\ x_2 = 14/11 - (15/11)x_4 \geq 0, \\ x_3 = 9/11 - (1/11)x_4 \geq 0, \end{cases} \quad (3.2)$$

and keeping nonnegativity of values of x_1, x_2, x_3 . The preceding inequalities are equivalent to

$$\begin{cases} x_4 \leq 7/2 \\ x_4 \leq 14/15 \\ x_4 \leq 9. \end{cases}$$

The preceding determines the maximum possible value taken by x_4 , i.e.,

$$x_4 \leq \min\{7/2, 14/15, 9\} = 14/15,$$

substituting which into (3.2) gives the new feasible solution

$$\bar{x} = (7/15, 0, 11/15, 14/15, 0)^T. \quad (3.3)$$

The associated new objective value $\bar{f} = -7/15$ is less than $35/11$.

The related new simplex tableau can be obtained by taking entry $15/11$ at row 2 and column 4 as the pivot. First, multiply row 2 by $11/15$ to turn the pivot to 1, leading to

x_1	x_2	x_3	x_4	x_5	f	RHS
1			2/11	-4/11		7/11
	11/15		1	-19/15		14/15
		1	1/11	9/11		9/11
			-43/11	53/11	-1	-35/11

Then add $-2/11$, $-1/11$, and $43/11$ times of row 2 to rows 1, 3, and 4, respectively, giving the new simplex tableau

x_1	x_2	x_3	x_4	x_5	f	RHS
1	-2/15			-2/15		7/15
	11/15		1	-19/15		14/15
	-1/15	1		14/15		11/15
	43/15			-2/15	-1	7/15

which clearly corresponds to the new feasible basic solution (3.3).

As the reduced cost with x_5 in the objective line is still negative, it cannot be asserted that the new solution is optimal. Similarly as in the previous step, we consider the following set of inequalities:

$$\begin{cases} x_1 = 7/15 + (2/15)x_5 \geq 0 \\ x_4 = 14/15 + (19/15)x_5 \geq 0 \\ x_3 = 11/15 - (14/15)x_5 \geq 0. \end{cases}$$

Since coefficients of x_5 in the first two inequalities are positive, the values of x_1 and x_4 remain nonnegative as x_5 increases from 0, while x_2 is fixed on zero. It is therefore only needed to consider the third inequality associated with the negative coefficient of x_5 . Setting $x_3 = 0$ in the third equation gives $x_5 = 11/14$, leading to the feasible basic solution

$$\bar{x} = (4/7, 0, 0, 27/14, 11/14)^T, \quad (3.4)$$

with objective value $\bar{f} = -4/7$, which is lower than $-7/15$.

Taking the entry $14/15$ at row 3 and column 5 as the pivot, the corresponding simplex tableau can be obtained by entering x_5 to and dropping x_3 from the basis. To do so, multiplying row 3 by $15/14$ gives:

x_1	x_2	x_3	x_4	x_5	f	RHS
1	$-2/15$			$-2/15$		$7/15$
	$11/15$		1	$-19/15$		$14/15$
	$-1/14$	$15/14$		1		$11/14$
	$43/15$			$-2/15$	-1	$7/15$

Then adding $2/15$, $19/15$, and $2/15$ times of row 3 to rows 1, 2, and 4, respectively, leads to

x_1	x_2	x_3	x_4	x_5	f	RHS
1	$-1/7$	$1/7$				$4/7$
	$9/14$	$19/14$	1			$27/14$
	$-1/14$	$15/14$		1		$11/14$
	$20/7$	$1/7$			-1	$4/7$

with reduced costs in the bottom row all nonnegative. As will be shown a little later, it is now can be asserted that the final feasible basic solution is optimal, as shown by (3.4), and we are done.

Table 3.1 Simplex tableau

x_{j_1}	x_{j_2}	$\cdots$	x_{j_m}	$x_{j_{m+1}}$	$\cdots$	x_{j_n}	f	RHS
1				$\bar{a}_{1\ j_{m+1}}$	$\cdots$	$\bar{a}_{1\ j_n}$		$\bar{b}_1$
	1			$\bar{a}_{2\ j_{m+1}}$	$\cdots$	$\bar{a}_{2\ j_n}$		$\bar{b}_2$
		$\ddots$		$\vdots$	$\vdots$	$\vdots$		$\vdots$
			1	$\bar{a}_{m\ j_{m+1}}$	$\cdots$	$\bar{a}_{m\ j_n}$		$\bar{b}_m$
				$\bar{z}_{j_{m+1}}$	$\cdots$	$\bar{z}_{j_n}$	-1	$-\bar{f}$

In the preceding, we transformed the tableau iteration by iteration until reaching an optimal solution. Such types of tableaus are called *simplex tableaus*, each of which involves a unit matrix. Table 3.1 gives a general form of the simplex tableau:¹

Let us return to the standard LP problem (1.8) and give the formal definition of a basis with respect to the LP problem.

Definition 3.1.1 Given a partition $A = (B, N)$, where $B \in R^{m \times m}$ has full rank m . The submatrix B is termed basis, and N termed nonbasis.

Without confusion, B and N also denote the corresponding index sets of columns of the basis and nonbasis, respectively.

Then, the problem (1.8) can be put into the following tableau:

x_B	x_N	f	RHS
B	N		b
c_B^T	c_N^T	-1	

It is usually assumed that the basic and nonbasic index sets are as follows:

$$B = \{j_1, \dots, j_m\}, \quad N = A \setminus B, \quad (3.5)$$

where j_i , $i = 1, \dots, m$ is the index of the basic variable in the i th row of the tableau.

Assume that through some elementary transformations, the preceding tableau becomes the simplex tableau 3.1, which can be concisely expressed as

$$\begin{array}{c|c|c|c}
 x_B & x_N & f & \text{RHS} \\
 \hline
 I & \bar{N} & & \bar{b} \\
 \hline
 & \bar{z}_N^T & -1 & -\bar{f}
 \end{array} \quad (3.6)$$

¹ It is always possible to arrange the $m \times m$ unit matrix at the northwest corner of the tableau by column and row permutations. Practically, however, it is not necessary, and the submatrix corresponding to basic variables is usually a permutation matrix.

The preceding simplex tableau is frequently used thereafter, exhibiting the corresponding basic solution

$$\bar{x}_B = \bar{b}, \quad \bar{x}_N = 0,$$

and the negative of the end entry in the bottom line is equal to the associated objective value. The $\bar{z}_N$ in the bottom line is termed *reduced costs/gradient*.

If $\bar{x}_B \geq 0$, the solution/tableau is said to be *feasible basic*. If the objective function attains the minimum value over the feasible region, the solution/tableau is said to be *optimal basic*.

Lemma 3.1.1 *If the reduced costs are all nonnegative, the feasible simplex tableau is optimal, giving an optimal basic solution.*

Proof The simplex tableau results from a series of elementary transformations and hence is equivalent to the original problem. Its bottom row represents the equation

$$f = \bar{f} + \bar{z}_N^T x_N. \quad (3.7)$$

Assume that $\tilde{x}$ is any feasible solution with objective value $\tilde{f}$. Substituting it into (3.7) together with $\bar{z}_N \geq 0$ and $\tilde{x} \geq 0$ leads to

$$\tilde{f} = \bar{f} + \bar{z}_N \tilde{x}_N \geq \bar{f},$$

which implies that $\bar{x}$ is an optimal basic solution. $\square$

The reduced costs are often called *check numbers*, as their sign can be used to judge the optimality of a feasible simplex tableau. Usually, there are negative check numbers in the tableau.

Lemma 3.1.2 *Assuming that $\bar{z}_q < 0$ holds for some $q \in N$ and that*

$$\bar{a}_{i,q} \leq 0, \quad i = 1, \dots, m, \quad (3.8)$$

then the LP problem is (lower) unbounded.

Proof The simplex tableau is associated with the constraint system

$$x_{ji} = \bar{b}_i - \sum_{j \in N} \bar{a}_{ij} x_j, \quad i = 1, \dots, m.$$

Setting $x_j = 0$, $j \in N$, $j \neq q$ in the preceding and combining the result with the nonnegative constraints give

$$x_{ji} = \bar{b}_i - \bar{a}_{iq} x_q \geq 0, \quad i = 1, \dots, m. \quad (3.9)$$

It is known from (3.8) that the set (3.9) of inequalities hold for all $x_q = \alpha \geq 0$, associated with feasible value

$$\hat{f} = \bar{f} + \alpha \bar{z}_q, \quad (3.10)$$

which, since $\bar{z}_q < 0$, can be arbitrarily low as α increases. Therefore, the problem is lower unbounded. $\square$

If (3.8) does not hold, then the value that the nonbasic variable x_q takes on will be restricted by set (3.9) of inequalities. It is verified that the following rule gives the maximum possible value α .

Rule 3.1.1 (Row Pivot Rule) *Determine a row index p and stepsize α such that*

$$\alpha = \bar{b}_p / \bar{a}_{pq} = \min\{\bar{b}_i / \bar{a}_{iq} \mid \bar{a}_{iq} > 0, i = 1, \dots, m\} \geq 0, \quad (3.11)$$

which is often called the *minimum ratio*.

Setting $x_q = \alpha$ in the equation of (3.9) gives a new feasible basic solution, i.e.,

$$\hat{x}_{j_i} = \bar{b}_i - \alpha \bar{a}_{iq}, \quad i = 1, \dots, m; \quad \hat{x}_j = 0, \quad j \in N, j \neq q; \quad \hat{x}_q = \alpha. \quad (3.12)$$

Taking $\bar{a}_{pq}$ as pivot, the associated simplex tableau is obtained by multiplying row p by $1/\bar{a}_{pq}$ to convert the pivot into 1, adding $-\bar{a}_{iq}$ times of row p to rows $i = 1, \dots, m, i \neq p$, and $-\bar{z}_q$ times of it to objective row. The iteration ends with (B, N) updated by exchanging j_p and q .

It is seen from (3.10) that the objective value decreases strictly if $\alpha > 0$, and no actual decrease is possible if $\alpha = 0$.

Definition 3.1.2 If some components of $\bar{b}$ are equal to zero, then the associated feasible basic solution/tableau is degenerate.

In degeneracy case, the stepsize α defined by (3.11) would vanish, and hence, the objective function remains unchanged (see (3.10)). That is to say, the associated “new solution” is actually the same as the old, though the basis is changed.

In general, there are multiple choices for q , as any column index j with negative $\bar{z}_j$ is eligible to be chosen. Dantzig’s original rule (1947) is as follows.

Rule 3.1.2 (Dantzig’s Column Rule) *Select a column index q such that*

$$q \in \arg \min_{j \in N} \bar{z}_j. \quad (3.13)$$

Thus, this rule selects the column with the most negative reduced cost as the pivot column.² For a unit increment of the nonbasic variable x_q , this choice leads to the maximum possible amount of decrease in the objective value.

The overall steps are summarized into the following algorithm (Dantzig 1948, 1963).

Algorithm 3.1.1 (Simplex Algorithm: Tableau Form) *Initial: a feasible simplex tableau of form (3.6). This algorithm solves the standard LP problem (1.8).*

1. Determine a pivot column index $q \in \arg \min_{j \in N} \bar{z}_j$.
2. Stop if $\bar{z}_q \geq 0$.
3. Stop if $I = \{i = 1, \dots, m \mid \bar{a}_{iq} > 0\} = \emptyset$.
4. Determine a pivot row index $p \in \arg \min_{i \in I} \bar{b}_i / \bar{a}_{iq}$.
5. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
6. Go to Step 1.

Theorem 3.1.3 *Under nondegeneracy assumption, the simplex algorithm terminates either at:*

- (i) Step 2, generating an optimal basic solution; or at
- (ii) Step 3, detecting lower unboundedness of the problem

Proof Under nondegeneracy assumption, stepsizes are all positive in the solution process, and hence, Algorithm 3.1.1 generates a sequence of feasible basic solutions with the objective value decreases strictly from iteration to iteration. This means that any basic solution can only appear once at most. Thus, the infiniteness of the process implies that there are infinitely many feasible basic solutions, which is a contradiction. Therefore, Algorithm 3.1.1 terminates.

The meanings of the exits of Algorithm 3.1.1 come from Lemmas 3.1.1 and 3.1.2. □

Proposition 3.1.1 *A leaving column does not enter the basis in the next iteration.*

Proof Since an entering column corresponds to a negative reduced cost and the pivot determined is positive (see (3.11)), a leaving column must correspond to a positive reduced cost after the elementary transformations performed and hence never enters the basis in the next iteration. □

On the other hand, however, it is not difficult to construct an instance such that a column just entering basis leaves it immediately. Moreover, since practical problems are always degenerate, termination of the simplex algorithm 3.1.1 is not guaranteed, that is, the possibility is not ruled out that indices enter and leave basis infinitely many times. There are a few instances that cannot be solved by the simplex method (see Sects. 3.4 and 3.5) although this hardly happens in practice.

² Any choice is eligible if there is a tie when there are multiple most negative reduced costs.

The simplex tableau is just a concise expression of the LP problem. Representing the same original problem, all resulting tableaus in the solution process can be viewed as equivalent. Recursive formulas between a tableau and its predecessor are listed below:

1. The objective row

$$\begin{aligned}\beta &= -\bar{z}_q/\bar{a}_{pq}, \\ \hat{f} &= \bar{f} - \beta\bar{b}_p, \\ \hat{z}_j &= \bar{z}_j + \beta\bar{a}_{pj}, \quad j \in N, \\ \hat{z}_{ji} &= \begin{cases} \beta & i = p, \\ 0 & i = 1, \dots, m, i \neq p. \end{cases}\end{aligned}\tag{3.14}$$

2. The right-hand side

$$\begin{aligned}\alpha &= \bar{b}_p/\bar{a}_{pq}, \\ \hat{b}_i &= \begin{cases} \bar{b}_i - \alpha\bar{a}_{iq} & i = 1, \dots, m, i \neq p, \\ \alpha & i = p. \end{cases}\end{aligned}\tag{3.15}$$

3. Entries of the constraint matrix

$$\begin{aligned}\hat{a}_{t,j} &= \begin{cases} 0, & t = 1, \dots, m, t \neq p; j = q. \\ 1, & t = p; j = q, \\ \bar{a}_{tj} - (\bar{a}_{pj}/\bar{a}_{pq})\bar{a}_{tq}, & t = 1, \dots, m, t \neq p; j \in N, j \neq q, \\ \bar{a}_{pj}/\bar{a}_{pq}, & t = p; j \in N, j \neq q. \end{cases} \\ \hat{a}_{tj_i} &= \begin{cases} 0, & t = 1, \dots, m; i = 1, \dots, m, i \neq p; i \neq t, \\ 1, & t = i = 1, \dots, m; i \neq p, \\ -\bar{a}_{tq}/\bar{a}_{pq}, & t = 1, \dots, m, t \neq p; i = p, \\ 1/\bar{a}_{pq}, & t = i = p. \end{cases}\end{aligned}\tag{3.16}$$

Example 3.1.1 Solve the following problem by Algorithm 3.1.1:

$$\begin{aligned}\min f &= -4x_1 - 3x_2 - 5x_3 \\ \text{s.t. } 2x_1 + x_2 + 3x_3 &+ x_5 = 15 \\ x_1 + x_2 + x_3 + x_4 &= 12 \\ -2x_1 + x_2 - 3x_3 &+ x_7 = 3 \\ 2x_1 + x_2 &+ x_6 = 9 \\ x_j &\geq 0, \quad j = 1, \dots, 7.\end{aligned}$$

Answer Initial: The following feasible simplex tableau can be directly obtained from the problem:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
2	1	3*		1			15
1	1	1	1				12
-2	1	-3				1	3
2	1				1		9
-4	-3	-5					

Iteration 1:

1. $\min\{-4, -3, -5\} = -5 < 0$, $q = 3$.
3. $I = \{1, 2\} \neq \emptyset$.
4. $\min\{15/3, 12/1\} = 15/3$, $p = 1$.
5. Take 3 in row 1 and column 3 as the pivot (marked by “*,” the same below).

Multiply row 1 by $1/3$, and then add -1 ; 3 ; 5 times of row 1 to rows 2, 3, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$2/3$	$1/3$	1		$1/3$			5
$1/3$	$2/3$		1	$-1/3$			7
	2^*			1		1	18
2	1				1		9
$-2/3$	$-4/3$			$5/3$			25

Iteration 2:

1. $\min\{-2/3, -4/3, 5/3\} = -4/3 < 0$, $q = 2$.
3. $I = \{1, 2, 3, 4\} \neq \emptyset$.
4. $\min\{5/(1/3), 7/(2/3), 18/2, 9/1\} = 9/1$, $p = 3$.
5. Multiply row 3 by $1/2$, and then add $-1/3$, $-2/3$, -1 , $4/3$ times of row 3 to rows 1, 2, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$2/3$		1		$1/6$		$-1/6$	2
$1/3$			1	$-2/3$		$-1/3$	1
	1			$1/2$		$1/2$	9
2^*				$-1/2$	1	$-1/2$	0
$-2/3$				$7/3$		$2/3$	37

Iteration 3:

1. $\min\{-2/3, 7/3, 2/3\} = -2/3 < 0$, $q = 1$.
3. $I = \{1, 2, 4\} \neq \emptyset$.
4. $\min\{2/(2/3), 1/(1/3), 0/2\} = 0$, $p = 4$.
5. Multiply row 4 by $1/2$, and then add $-2/3$; $-1/3$; $2/3$ times of row 4 to rows 1, 2, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		1		$1/3$	$-1/3$		2
			1	$-7/12$	$-1/6$	$-1/4$	1
	1			$1/2$		$1/2$	9
1				$-1/4$	$1/2$	$-1/4$	
				$13/6$	$1/3$	$1/2$	37

Now all reduced costs in the preceding tableau are nonnegative, and hence, we have following optimal basic solution with optimal objective value:

$$x^* = (0, 9, 2, 1, 0, 0, 0)^T, \quad f^* = -37.$$

It is seen from the preceding example that the tableau in the second iteration already attained the basic optimal solution, but an additional iteration was performed, due to the existence of a negative reduced cost. This occurred because of degeneracy, leading to a zero stepsize. Therefore, the optimality condition that the reduced costs are all nonnegative is only sufficient, not necessary.

The following indicates that the condition is necessary only if the nondegeneracy is assumed.

Proposition 3.1.2 *If an optimal basic solution is nondegenerate, the reduced costs in the associated simplex tableau are all nonnegative.*

Proof Assume that there are some negative reduced costs in tableau 3.1. Without loss of generality, assume $\bar{z}_q < 0$. If (3.8) holds, then unboundedness of the problem follows from Theorem 3.1.2, which contradicts the presence of the optimal solution; if otherwise, (3.8) does not hold, it is known from the nondegeneracy assumption that $\alpha > 0$, and hence, there is a feasible value strictly less than the optimal value, as is a contradiction. Therefore, the claim is valid. $\square$

The following result concerns the case of multiple optimal solutions.

Proposition 3.1.3 *If the reduced costs are all positive, there is a unique optimal solution to the LP problem. If an optimal basic solution is nondegenerate and there exists a zero-valued reduced cost, then there are an infinite number of optimal solutions; in the case when the feasible region is bounded, there exist multiple basic optimal solutions.*

Proof We prove the first half first. Assume that the reduced costs in an optimal tableau are all positive, corresponding to optimal basic solution $\bar{x}$. For any feasible solution $\hat{x} \geq 0$ different from $\bar{x}$, there is an index $s \in N$ such that $\hat{x}_s > 0$ (otherwise, the two are the same). Therefore, substituting $\hat{x}$ to (3.7) leads to

$$\hat{f} = \bar{f} + \sum_{j \in N} \bar{z}_j \hat{x}_j > \bar{f},$$

which implies that $\hat{x}$ is not optimal, as is a contradiction. Therefore, there is a unique optimal solution.

To prove the second half, assume that a tableau, say 3.1.1, gives a nondegenerate optimal basic solution and has some zero reduced costs. Without loss of generality, assume $\bar{z}_q = 0$. If (3.8) holds, then inequalities of the right-hand side of (3.9) hold for any $x_q = \alpha > 0$, that is, there are infinitely many feasible solutions, corresponding to the same optimal value $\bar{f}$ (see (3.10)); if the feasible region is bounded, then (3.8) does not hold; hence, it is known from $\bar{b}_p > 0$ that stepsize α defined by (3.11) is positive. Thus, for any value of x_q in $[0, \alpha]$, a feasible solution can be determined by the equalities of (3.9), which corresponds to optimal value $\bar{f}$. Therefore, there are an infinite number of optimal solutions. It is clear that entering x_q to and dropping x_{j_p} from basis give a different optimal basic solution. $\square$

3.2 Getting Started

Algorithm 3.1.1 must start from a feasible simplex tableau. In Example 3.1.1, there is a feasible simplex tableau available, which is not the case in general. LP problems are usually solved by some so-called *two-phase simplex algorithm*. At first, a *Phase-I* procedure is carried out to achieve feasibility, and then Algorithm 3.1.1 is used as Phase-II to achieve optimality. In this section, a classical Phase-I using artificial variables will be presented, followed by another Phase-I, termed *big M method*.

To begin with, assume that all entries of the right-hand side are nonnegative, i.e.,

$$b_i \geq 0, i = 1, \dots, m.$$

If it is not the case, multiply by -1 each equation with a negative right-hand side. Then construct an auxiliary problem as follows.

For each $i = 1, \dots, m$, introduce a nonnegative *artificial variable* x_{n+i} to the i th equation, and take the sum of all artificial variables as objective function, i.e.,

$$f' = \sum_{i=1}^m x_{n+i}.$$

Then, using the constraint system, we eliminate all artificial variables in the objective, resulting in the following auxiliary problem:

$$\begin{aligned}
 \min \quad & f' = \sum_{i=1}^m b_i - (\sum_{i=1}^m a_{i1})x_1 - \cdots - (\sum_{i=1}^m a_{in})x_n \\
 \text{s.t.} \quad & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n + x_{n+1} = b_1 \\
 & a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n + x_{n+2} = b_2 \\
 & \vdots \\
 & a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n + x_{n+m} = b_m \\
 & x_j \geq 0, \quad j = 1, \dots, n+m,
 \end{aligned} \tag{3.17}$$

which is associated with a feasible simplex tableau, corresponding to feasible basic solution

$$x' = (\underbrace{0, \dots, 0}_n, \underbrace{b_1, \dots, b_m}_m)^T \geq 0.$$

Therefore, Algorithm 3.1.1 can get itself started with the auxiliary problem.

Regarding the outcome, we have the following.

Theorem 3.2.1 *The auxiliary problem (3.17) has an optimal solution. There are the following two cases:*

- (i) *If the optimal value is strictly greater than zero, the original problem is infeasible.*
- (ii) *If the optimal value is equal to zero, the first n components of the optimal solution form a feasible solution to the original problem.*

Proof Note that the feasible region of the auxiliary problem is nonempty, and the feasible objective value is bounded below by zero. According to Proposition 1.5.3, the auxiliary problem has an optimal basic solution.

- (i) In case when the optimal value is strictly greater than zero, the original problem must be infeasible, since if there was a feasible solution, say $\bar{x}_j \geq 0$, $j = 1, \dots, n$, then

$$\bar{x}_1, \dots, \bar{x}_n; \bar{x}_{n+1} = \cdots = \bar{x}_{n+m} = 0$$

forms a feasible solution to the auxiliary problem, which is associated with zero auxiliary objective value. This means that there is an auxiliary objective value that is feasible, and less than the optimal value, which is a contradiction.

- (ii) In case when the optimal value is zero, artificial components of the optimal solution are all zero. Then substituting the optimal solution to the auxiliary constraints exhibits that the solution resulting from dropping the auxiliary components is feasible to the original problem.

Corollary 3.2.2 *The original problem is feasible if and only if the optimal value of the auxiliary problem vanishes.*

Note that if a constraint matrix includes some columns of the unit matrix, such columns should be employed to reduce the number of artificial variables. The preceding discussions are still valid, though the auxiliary objective function involves artificial variables only.

Once a feasible solution to the original problem is obtained, a feasible simplex tableau can be yielded by steps based on the relationship between the auxiliary problem and the original problem.

Following-up steps:

- (A) Delete columns, associated to all nonbasic artificial variables (which can be deleted once the corresponding artificial variable leaves the basis).
- (B) Go to step D if there is no artificial basic variable.
- (C) Delete the row, associated with an artificial basic variable, if all its nonbasic components are zero (see the Note below); otherwise, take a nonzero component of it as the pivot to let the artificial variable become nonbasic, and then delete the associated column. This is repeated until no artificial variable is basic.
- (D) Overwrite the auxiliary objective row by the original costs, and then eliminate all basic components of this row, giving a feasible simplex tableau to the original problem.

Note In step C, the row is deleted because substituting 0 to the associated artificial variable turns the corresponding equation into an identity, as it reflects the dependence of the original constraint equations. So, the method can get rid of such dependency.

Overall steps can be put into the following algorithm.

Algorithm 3.2.1 (Phase 1: Artificial Variable) *This algorithm finds a feasible tableau:*

1. Introduce artificial variables, and construct auxiliary problem of form (3.17).
2. Call the simplex Algorithm 3.1.1.
3. If the optimal value of the auxiliary problem is zero, create a feasible tableau via “Following-up steps.”
4. The original problem is infeasible if the optimal value of the auxiliary problem is strictly greater than zero.

Example 3.2.1 Find a feasible simplex tableau to the following problem:

$$\begin{aligned}
 \min \quad & f = -x_1 + x_2 - 2x_3 \\
 \text{s.t.} \quad & x_1 - 3x_2 - 2x_3 + x_4 = -4 \\
 & x_1 - x_2 + 4x_3 - x_5 = 2 \\
 & -3x_1 + x_2 + x_3 + x_6 = 8 \\
 & x_j \geq 0, \quad j = 1, \dots, 6.
 \end{aligned}$$

Answer Construct auxiliary problem: the first constraint equation is multiplied by -1 to turn its right-hand side to nonnegative; as the coefficients of x_6 give a unit vector $(0, 0, 0, 1)^T$, only two artificial variables x_7, x_8 are introduced.

$$\begin{aligned}
 \min \quad & f' = x_7 + x_8 \\
 \text{s.t.} \quad & -x_1 + 3x_2 + 2x_3 - x_4 \qquad \qquad \qquad + x_7 = 4 \\
 & x_1 - x_2 + 4x_3 \qquad \qquad - x_5 \qquad \qquad \qquad + x_8 = 2 \\
 & -3x_1 + x_2 + x_3 \qquad \qquad \qquad + x_6 \qquad \qquad = 8 \\
 & x_j \geq 0, \quad j = 1, \dots, 8.
 \end{aligned}$$

Put the preceding auxiliary problem into the following tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
-1	3	2	-1			1		4
1	-1	4		-1			1	2
-3	1	1			1			8
							1	1

Turn the preceding to a simplex tableau: eliminate nonzeros in x_7 and x_8 columns at the bottom (objective) row by adding -1 times of row 1 and of row 2 to that row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
-1	3	2	-1			1		4
1	-1	4*		-1			1	2
-3	1	1			1			8
		-2	-6	1	1			-6

which is a feasible simplex tableau to the auxiliary problem. Call Algorithm 3.1.1 to solve it:

Iteration 1:

1. $\min\{0, -2, -6, 1, 1\} = -6 < 0, \quad q = 3.$
3. $I = \{1, 2, 3\} \neq \emptyset.$
4. $\min\{4/2, 2/4, 8/1\} = 1/2, \quad p = 2.$
5. Multiply row 2 by $1/4$, and then add $-2, -1, 6$ times of row 2 to rows 1,3,4, respectively.

(Erase x_8 column after artificial variable x_8 becomes nonbasic):

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-3/2$	$7/2^*$		-1	$1/2$		1	3
$1/4$	$-1/4$	1		$-1/4$			$1/2$
$-13/4$	$5/4$			$1/4$	1		$15/2$
$3/2$	$-7/2$		1	$-1/2$			-3

Iteration 2:

1. $\min\{3/2, -7/2, 1, -1/2\} = -7/2 < 0$, $q = 2$.
3. $I = \{1, 3\} \neq \emptyset$.
4. $\min\{3/(7/2), (15/2)/(5/4)\} = 6/7$, $p = 1$.
5. Multiply row 1 by $2/7$, and then add $1/4$; $-5/4$; $7/2$ times of row 1 to rows 2, 3, 4, respectively.

(Erase x_7 column after artificial variable x_7 becomes nonbasic):

Now, all the artificial variables become nonbasic; hence, reached is the optimal objective value 0 of the auxiliary problem.

Overwriting the bottom row by original costs leads to

x_1	x_2	x_3	x_4	x_5	x_6	RHS
$-3/7$	1		$-2/7$	$1/7$		$6/7$
$1/7$		1	$-1/14$	$-3/14$		$5/7$
$-19/7$			$5/14$	$1/14$	1	$45/7$
-1	1	-2				

Adding -1 times of row 1 and 2 times of row 2 to the bottom row gives a feasible tableau of the original problem, i.e.,

x_1	x_2	x_3	x_4	x_5	x_6	RHS
$-3/7$	1		$-2/7$	$1/7$		$6/7$
$1/7$		1	$-1/14$	$-3/14$		$5/7$
$-19/7$			$5/14$	$1/14$	1	$45/7$
$-2/7$			$1/7$	$-4/7$		$4/7$

Thus, the preceding can be taken as an initial feasible tableau to get Algorithm 3.1.1 started to solve the original problem. Solving LP problems usually requires two phases, both of which carry out the simplex algorithm.

On the other hand, it seems to be attractive to solve LP problems in a single phase, which leads to the following so-called *big-M method*.

The auxiliary problem shares the same constraints with (3.17), while its objective function is the sum of the original objective function and M times of the sum of all

artificial variables, i.e.,

$$\begin{aligned}
 \min f' &= c_1x_1 + c_2x_2 + \cdots + c_nx_n + M(x_{n+1} + x_{n+2} \cdots + x_{n+m}) \\
 \text{s.t. } a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n + x_{n+1} &= b_1 \\
 a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n + x_{n+2} &= b_2 \\
 &\vdots \\
 a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n + x_{n+m} &= b_m \\
 x_j &\geq 0, \quad j = 1, \dots, n + m.
 \end{aligned}$$

Artificial variables in the objective function are eliminated using the constraint system. As a result, there will be a feasible simplex tableau to the auxiliary problem, which can be taken as an initial one to get the simplex algorithm started.

The reason for using such an auxiliary objective function is as follows. Its artificial variable part may be regarded as a *penalty function*, in which the M , serving as *penalty factor*, is a sufficiently large positive number (far larger than the absolute value of any number involved in the computation). Such a big M imposes a penalty on the possible increase of values of artificial variables, consequently forcing artificial variables minimized before the original objective.

It is difficult however to determine a suitable M in advance. A too large M could lead to bad numerical stability, while a too small M degrades the method's effectiveness. It depends not only on the problem to be solved but also on the computer used. A practical way is to take the M as a parameter in the solution process.

To demonstrate, we again bring up Example 3.2.1. Its auxiliary problem is of the following form:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
-1	3	2	-1			1		4
1	-1	4		-1			1	2
-3	1	1			1			8
-1	1	-2				M	M	

Adding M times of row 1 and row 2 to the objective row gives

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
-1	3	2	-1			1		4
1	-1	4*		-1			1	2
-3	1	1			1			8
-1	$1 - 2M$	$-2 - 6M$	M	M				$-6M$

Thereby, the simplex algorithm can get itself stated from the preceding tableau. In selection of pivot column index q , however, it should be noted that the M is so

large that the signs of the reduced costs depend upon coefficients of the M only. In the preceding tableau, e.g., x_3 column is selected because M 's coefficients are 0, -2 , -6 , 1, 1 in costs of nonbasic variables x_1 through x_5 , respectively, and

$$\min\{0, -2, -6, 1, 1\} = -6, \quad q = 3.$$

Row 2 is selected as pivot row by the minimum-ratio test below:

$$\min\{4/2, 2/4, 8/1\} = 1/2, \quad p = 2.$$

Then elementary transformations are performed to make the associated basis change, completing the first iteration.

If the process continued, it can be found that sequences of iterates created by the Big M method and the two-phase simplex method are the same. This is not surprising because the big *penalty factor* M forces values of artificial variables to vanish first before pursuing optimality of the original problem—the two methods are essentially the same.

Practically, however, the two-phase method is certainly preferable to the big M method since the former involves no parameter and is easier to realize. In principle, it can be used to solve any LP problem, achieving an optimal basic solution, if any, or detecting infeasibility or unboundedness, otherwise.

3.3 Simplex Algorithm

The simplex tableau 3.1 or (3.6) is expressed at the scalar level. It is possible to derive a so-called *revised simplex tableau* at the vector/matrix level instead.

Premultiplying the system $Ax = b$ by B^{-1} with transposition of terms gives

$$x_B + B^{-1}Nx_N = B^{-1}b.$$

Substituting the preceding into

$$c_B^T x_B + c_N^T x_N - f = 0$$

leads to

$$(c_N^T - c_B^T B^{-1}N)x_N - f = -c_B^T B^{-1}b,$$

Table 3.2 Equivalence between the associated quantities

Quantity	Simplex tableau	Relation	Revised simplex tableau
Objective row	$\bar{z}_N^T$	=	$c_N^T - c_B^T B^{-1} N$
Pivot column	$\bar{a}_q$	=	$B^{-1} a_q$
Right-hand side	$\bar{b}$	=	$B^{-1} b$
Pivot row	$e_p^T \bar{N}$	=	$e_p^T B^{-1} N$

which can be put into

$$\begin{array}{ccc|c}
 x_B & x_N & f & \text{RHS} \\
 \hline
 I & B^{-1} N & & B^{-1} b \\
 \hline
 & c_N^T - c_B^T B^{-1} N & -1 & -c_B^T B^{-1} b
 \end{array} \quad (3.18)$$

corresponding to basic solution $\bar{x}_B = B^{-1}b$, $\bar{x}_N = 0$. The preceding, representing a problem equivalent to the original, is called *revised simplex tableau*.

For simplicity, x_B^T and f columns in the preceding two tableaus may be omitted, as they remain unchanged as the basis changes.

Proposition 3.3.1 *Any simplex tableau and revised simplex tableau, corresponding to the same basis, are equivalent.*

Proof Denote by (3.6) and (3.18) the two tableaus, having the same basis B . Since problems represented by them are equivalent, the corresponding entries of the two tableaus are equal. $\square$

Based on the preceding Proposition, Table 3.2 gives equivalence between quantities of tableaus (3.6) and (3.18), involved in simplex steps.

In simplex context, each iteration corresponds to a basis B (or the inverse B^{-1}), with which entries in a simplex tableau can be calculated from the original data (A, b, c) . Thereby, Table 3.2 will be used as a tool to derive simplex variants.

It is noted that although the simplex tableau is calculated by Algorithm 3.1.1, only a part of its $(m+1) \times (n+1)$ entries are useful, that is, the objective row, the pivot column, and right-hand side, which are used for determining pivot. Using B^{-1} , therefore, a variant can be derived by calculating the first 3 rows in Table 3.2.

Let us consider updating of B^{-1} . Assume that column index q and row index p are already determined. Replacing the p th column of B with nonbasic column a_q gives the new basis below:

$$\widehat{B} = (a_{j_1}, \dots, a_{j_{p-1}}, a_q, a_{j_{p+1}}, \dots, a_{j_m}). \quad (3.19)$$

Then, compute $\widehat{B}^{-1}$ as follows.

Note that $\bar{a}_q = B^{-1}a_q$. Taking $\bar{a}_{pq}$ as pivot, the associated elementary transformations amount to premultiplying the first m rows of the tableau by the

following $m \times m$ elementary matrix:

$$E_p = \begin{pmatrix} 1 & & -\bar{a}_{1q}/\bar{a}_{pq} & & \\ & \ddots & \vdots & & \\ & & -\bar{a}_{p-1,q}/\bar{a}_{pq} & & \\ & & 1/\bar{a}_{pq} & & \\ & & -\bar{a}_{p+1,q}/\bar{a}_{pq} & & \\ & & \vdots & \ddots & \\ & & -\bar{a}_{mq}/\bar{a}_{pq} & & 1 \end{pmatrix} \quad p, \quad (3.20)$$

p

which may also be obtained by executing the same elementary transformations on the unit matrix. Such a matrix, which is the same as the unit matrix except for the p th column, is determined only by $\bar{a}_q$. Combining (3.19) and (3.20) gives

$$\begin{aligned} E_p B^{-1} \hat{B} &= E_p (B^{-1} a_{j_1}, \dots, B^{-1} a_{j_{p-1}}, B^{-1} a_q, B^{-1} a_{j_{p+1}}, \dots, B^{-1} a_{j_m}) \\ &= E_p (e_1, \dots, e_{p-1}, \bar{a}_q, e_{p+1}, \dots, e_m) \\ &= (E_p e_1, \dots, E_p e_{p-1}, E_p \bar{a}_q, E_p e_{p+1}, \dots, E_p e_m) = I, \end{aligned}$$

from which the recursion formula follows, i.e.,

$$\hat{B}^{-1} = E_p B^{-1}. \quad (3.21)$$

Based on the preceding discussions and the equivalence between the simplex tableau and revised simplex tableau, we can turn Algorithm 3.1.1 into the following revision (Dantzig and Orchard-Hays 1953):

Algorithm 3.3.1 (Simplex Algorithm 1) *Initial:* (B, N) , B^{-1} , $\bar{x}_B = B^{-1}b \geq 0$, and $\bar{f} = c_B^T \bar{x}_B$. This algorithm solves the standard LP problem (1.8).

1. Compute $\bar{z}_N = c_N - N^T \bar{y}$, $\bar{y} = B^{-T} c_B$.
2. Determine pivot column index $q \in \arg \min_{j \in N} \bar{z}_j$.
3. Stop if $\bar{z}_q \geq 0$ (optimality achieved).
4. Compute $\bar{a}_q = B^{-1} a_q$.
5. Stop if $\bar{a}_q \leq 0$ (unbounded problem).
6. Determine stepsize α and pivot row index p such that:
 $\alpha = \bar{x}_{j_p} / \bar{a}_{pq} = \min \{ \bar{x}_{j_i} / \bar{a}_{iq} \mid \bar{a}_{iq} > 0; i = 1, \dots, m \}.$
7. Set $\bar{x}_q = \alpha$, and update $\bar{x}_B = \bar{x}_B - \alpha \bar{a}_q$, $\bar{f} = \bar{f} + \alpha \bar{z}_q$ if $\alpha \neq 0$.
8. Update B^{-1} by (3.21).
9. Update (B, N) by exchanging j_p and q .
10. Go to step 1.

In step 1, vector $\bar{y}$ is calculated first, and it is then used to compute the reduced costs $\bar{z}_N$, which is referred to as *pricing*. $\bar{y}$ is called *simplex multipliers (vector)*, whose additional meanings will be clear later.

Example 3.3.1 Solve the following problem by Algorithm 3.3.1:

$$\begin{aligned} \min f &= -4x_1 - 3x_2 - 5x_3 \\ \text{s.t. } 2x_1 + x_2 + 3x_3 &+ x_5 &= 15 \\ x_1 + x_2 + x_3 + x_4 & &= 12 \\ x_2 - 3x_3 &+ x_7 &= 3 \\ 2x_1 + x_2 &+ x_6 &= 9 \\ x_j &\geq 0, \quad j = 1, \dots, 7. \end{aligned}$$

Answer Initial: $B = \{5, 4, 7, 6\}$, $N = \{1, 2, 3\}$, $B^{-1} = I$, $\bar{x}_B = (15, 12, 3, 9)^T$, $f = 0$.

Iteration 1:

1. $\bar{y} = B^{-T}c_B = (0, 0, 0, 0)^T$, $\bar{z}_N = c_N - N^T\bar{y} = (-4, -3, -5)^T$.
2. $\min\{-4, -3, -5\} = -5 < 0$, $q = 3$.
4. $\bar{a}_3 = B^{-1}a_3 = (3, 1, -3, 0)^T$.
6. $\alpha = \min\{15/3, 12/1\} = 12/1 = 12$, $p = 1$.
7. $\bar{x}_B = (15, 12, 3, 9)^T - 12 \times (3, 1, -3, 0)^T = (3, 0, 39, 9)^T$, $x_3 = 12$, $f = 5 \times (-5) = -25$.
8. $B^{-1} = \begin{pmatrix} 1/3 & & & \\ -1/3 & 1 & & \\ 1 & & 1 & \\ 0 & & & 1 \end{pmatrix}$.
9. $B = \{3, 4, 7, 6\}$, $N = \{1, 2, 5\}$, $\bar{x}_B = (3, 0, 39, 9)^T$.

Iteration 2:

1. $\bar{y} = (-5/3, 0, 0, 0)^T$, $\bar{z}_N = (-4, -3, 0)^T - (-10/3, -5/3, -5/3)^T = (-2/3, -4/3, 5/3)^T$.
2. $\min\{-2/3, -4/3, 5/3\} = -4/3 < 0$, $q = 2$.
4. $\bar{a}_2 = (1/3, 2/3, 2, 1)^T$.
6. $\alpha = \min\{15, 21/2, 9, 9\} = 9$, $p = 3$.
7. $\bar{x}_B = (3, 0, 39, 9)^T - 9 \times (1/3, 2/3, 2, 1)^T = (2, 1, 0, 0)^T$, $x_2 = 9$, $f = -25 + 9 \times (-4/3) = -37$.
8. $B^{-1} = \begin{pmatrix} 1 & -1/6 & & \\ 0 & 1 & -1/3 & \\ 0 & 1/2 & & \\ 0 & -1/2 & 1 & \end{pmatrix} \begin{pmatrix} 1/3 & & & \\ -1/3 & 1 & & \\ 1 & & 1 & \\ 0 & & & 1 \end{pmatrix} = \begin{pmatrix} 1/6 & -1/6 & & \\ -2/3 & 1 & -1/3 & \\ 1/2 & 1/2 & & \\ -1/2 & -1/2 & 1 & \end{pmatrix}$.
9. $B = \{3, 4, 2, 6\}$, $N = \{1, 7, 5\}$, $\bar{x}_B = (2, 1, 9, 0)^T$.

Iteration 3:

1. $\bar{y} = (-7/3, 0, -2/3, 0)^T$, $\bar{z}_N = (-4, 0, 0)^T - (-10/3, -2/3, -7/3)^T = (-2/3, 2/3, 7/3)^T$.
2. $\min\{-2/3, 2/3, 7/3\} = -2/3$, $q = 1$.
4. $\bar{a}_1 = (2/3, 1/3, 0, 2)^T$.
6. $\alpha = \min\{2/(2/3), 1/(1/3), 0/2\} = 0$, $p = 4$.
7. $\bar{x}_B = (2, 1, 9, 0)^T$, $x_6 = 0$, $f = -37$.
8. $B^{-1} = \begin{pmatrix} 1 & -1/3 \\ 0 & 1 & -1/6 \\ 0 & 1 & 0 \\ 0 & & 1/2 \end{pmatrix} \begin{pmatrix} 1/6 & -1/6 \\ -2/3 & 1 & -1/3 \\ 1/2 & 1/2 \\ -1/2 & -1/2 & 1 \end{pmatrix} = \begin{pmatrix} 1/3 & 0 & -1/3 \\ -7/12 & 1 & -1/4 & -1/6 \\ 1/2 & 1/2 & 0 \\ -1/4 & -1/4 & 1/2 \end{pmatrix}$.
8. $B = \{3, 4, 2, 1\}$, $N = \{6, 7, 5\}$, $\bar{x}_B = (2, 1, 9, 0)^T$.
9. Omitted.

Iteration 4:

1. $\bar{y} = (-13/6, 0, -1/2, -1/3)^T$, $\bar{z}_N = (0, 0, 0)^T - (-1/3, -1/2, -13/6)^T = (1/3, 1/2, 13/6)^T \geq 0$.
2. The optimal basic solution and optimal value:

$$x^* = (0, 9, 2, 1, 0, 0, 0)^T, \quad f^* = -37.$$

It is possible to derive a variant, in which row p is computed instead of column q , and objective row is computed by recursion (see (3.14)).

Algorithm 3.3.2 (Simplex Algorithm 2) *Initial: (B, N) , B^{-1} , $\bar{x}_B = B^{-1}b \geq 0$, $\bar{z}_N = c_N - N^T B^{-T} c_B$, and $\bar{f} = c_B^T \bar{x}_B$. This algorithm solves the standard LP problem (1.8).*

1. Determine pivot column index $q \in \arg \min_{j \in N} \bar{z}_j$.
2. Stop if $\bar{z}_q \geq 0$ (optimality achieved).
3. Compute $\bar{a}_q = B^{-1} a_q$.
4. Stop if $\bar{a}_q \leq 0$ (unbounded).
5. Determine stepsize α and pivot row index p such that $\alpha = \bar{x}_{j_p} / \bar{a}_{p q} = \min\{\bar{x}_{j_i} / \bar{a}_{i q} \mid \bar{a}_{i q} > 0; i = 1, \dots, m\}$.
6. Set $\bar{x}_q = \alpha$, and update $\bar{x}_B = \bar{x}_B - \alpha \bar{a}_q$, $\bar{f} = \bar{f} + \alpha \bar{z}_q$ if $\alpha \neq 0$.
7. Compute $\sigma_N = N^T v$, where $v = B^{-T} e_p$.
8. Update by: $\bar{z}_N = \bar{z}_N + \beta \sigma_N$, $\bar{z}_{j_p} = \beta$, where $\beta = -\bar{z}_q / \bar{a}_{p q}$.
9. Update B^{-1} by (3.21).
10. Update (B, N) by exchanging j_p and q .
11. Go to step 1.

Algorithm 3.3.1 was derived based on the equivalence of the simplex tableau and revised simplex tableau. Alternatively, it can be derived by calculating a descent edge, emanating from the current vertex, as the search direction.

Without loss of generality, let $B = \{1, \dots, m\}$ and $N = \{m+1, \dots, n\}$ be, respectively, the basis and nonbasis, associated with feasible basic solution $\bar{x}$. Assume that pivot column index $q \in N$ has been determined such that

$$\bar{z}_q = c_q - a_q^T B^{-T} c_B < 0.$$

Introduce the vector

$$\Delta x = \begin{pmatrix} B^{-1} a_q \\ -e_{q-m} \end{pmatrix}, \quad (3.22)$$

where e_{q-m} is the $(n-m)$ -dimensional unit vector with the $(q-m)$ th component 1. It is clear that

$$-c^T \Delta x = c_q - a_q^T B^{-T} c_B = \bar{z}_q < 0. \quad (3.23)$$

Therefore, $-\Delta x$ is descent with respect to $c^T x$. Taking it as search direction gives the following line search scheme:

$$\hat{x} = \bar{x} - \alpha \Delta x, \quad (3.24)$$

where $\alpha \geq 0$ is a stepsize to be determined.

Since $\bar{x}$ is feasible, it holds for any $\alpha \geq 0$ that

$$A\hat{x} = A\bar{x} - \alpha[B, N]\Delta x = A\bar{x} = b.$$

Therefore, maximizing α subject to $\hat{x}_B \geq 0$ is reasonable. When $B^{-1}a_q \not\leq 0$, such doing results in α and p such that

$$\alpha = \bar{x}_{j_p} / (B^{-1}a_{pq}) = \min\{\bar{x}_{j_i} / (B^{-1}a_{iq}) \mid B^{-1}a_{iq} > 0, i = 1, \dots, m\}. \quad (3.25)$$

It is clear that the associated new solution $\hat{x}$ is still feasible. In fact, it is verified that $\hat{x}$ is just feasible basic solution, associated with the new basis, resulting from the old by exchanging j_p and q .

The relation between the new and old basis matrices is

$$\hat{B} = B + (a_q - a_{j_p})e_p^T.$$

In view of that a_{j_p} is the p th column of B and that $B^{-1}a_{j_p} = e_p$ and $B^{-1}a_q = \bar{a}_q$ hold, it is not difficult to derive the following result from Sherman–Morrison

formula (Golub and Van Loan 1989):

$$\widehat{B}^{-1} = B^{-1} - \frac{B^{-1}(a_q - a_{j_p})e_p^T B^{-1}}{1 + e_p^T B^{-1}(a_q - a_{j_p})} = \left(I - \frac{(\bar{a}_q - e_p)e_p^T}{\bar{a}_{pq}} \right) B^{-1}, \quad (3.26)$$

which may serve as an update of B^{-1} . In fact, it is verified that the preceding and (3.21) are actually equivalent.

The search direction $-\Delta x$, defined by (3.22), can be further investigated geometrically. Concerning the set

$$\begin{aligned} E &= \{x \in \mathcal{R}^n \mid Ax = b, x \geq 0, x_j = 0, q \neq j \in N\} \\ &= \{x \in \mathcal{R}^n \mid x_B = B^{-1}b - x_q(B^{-1}a_q) \geq 0, x_q \geq 0, x_j = 0, q \neq j \in N\}, \end{aligned} \quad (3.27)$$

we have the following result.

Proposition 3.3.2 *The set E is a descent edge, emanating from the current vertex $\bar{x}$, and $-\Delta x$ is its direction. If $B^{-1}a_q \leq 0$, then $-\Delta x$ is an (unbounded) extreme direction.*

Proof It is clear that E is a half-line or edge, emanating from $\bar{x}$. By (3.27), (3.25), and (3.22), for any $x \in E \subset P$, it holds that

$$\begin{aligned} E &= \{x \in \mathcal{R}^n \mid x_B = B^{-1}b - x_q(B^{-1}a_q), 0 \leq x_q \leq \alpha, x_j = 0, q \neq j \in N\} \\ &= \{x \in \mathcal{R}^n \mid x = \bar{x} - x_q \Delta x, x_q \in [0, \alpha]\}. \end{aligned}$$

By (3.23), it is known that the associated objective value satisfies

$$f = c^T x = c^T \bar{x} - x_q c^T \Delta x = c^T \bar{x} + x_q \bar{z}_q \leq c^T \bar{x}.$$

Note that (3.25) is well-defined when $B^{-1}a_q > 0$. If, in addition, $\bar{x}_{j_p} = 0$, then $\alpha = 0$, and hence E degenerates to vertex $\bar{x}$. If $\bar{x}_{j_p} > 0$, hence $\alpha > 0$, then the associated objective value strictly decreases with $x_q \in [0, \alpha]$. Therefore, $-\Delta x$ is a direction of the descent edge E . When $B^{-1}a_q \leq 0$, it is clear that $\alpha = +\infty$ corresponds to the edge $E \in P$, and hence, $-\Delta x$ is an extreme direction. $\square$

3.4 Degeneracy and Cycling

It was seen that a zero stepsize leads to the same feasible basic solution. Therefore, the finiteness of the simple method is questionable. Soon after its emerging, it is found that the simplex method does not terminate in a few cases. E.M.L. Beale (1955) and A.J. Hoffman (1953) offered such instances independently.

The following is due to Beale.

Example 3.4.1 Solve the following problem by Algorithm 3.1.1:

$$\begin{aligned}
 \min f &= -3/4x_4 + 20x_5 - 1/2x_6 + 6x_7 \\
 \text{s.t. } x_1 &+ 1/4x_4 - 8x_5 - x_6 + 9x_7 = 0 \\
 x_2 &+ 1/2x_4 - 12x_5 - 1/2x_6 + 3x_7 = 0 \\
 x_3 &+ x_6 = 1 \\
 x_j &\geq 0, \quad j = 1, \dots, 7.
 \end{aligned}$$

Answer Initial: The following feasible tableau is available from the preceding:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1			1/4*	-8	-1	9	
	1		1/2	-12	-1/2	3	
		1			1		1
			-3/4	20	-1/2	6	

Iteration 1:

1. $\min\{-3/4, 20, 1, -1/2, 6\} = -3/4 < 0$, $q = 4$.
3. $I = \{1, 2\} \neq \emptyset$.
4. $\min\{0/(1/4), 0/(1/2)\} = 0$, $p = 1$.
5. Multiply row 1 by 4, and then add $-1/2, 3/4$ times of row 1 to rows 2,4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
4			1	-32	-4	36	
-2	1			4*	3/2	-15	
		1			1		1
3				-4	-7/2	33	

Iteration 2:

1. $\min\{3, -4, -7/2, 33\} = -4 < 0$, $q = 5$.
3. $I = \{2\} \neq \emptyset$.
4. $\min\{0/4\} = 0$, $p = 2$.

5. Multiply row 2 by $1/4$, and then add 32, 4 times of row 2 to rows 1, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
-12	8		1		8*	-84	
-1/2	1/4			1	3/8	-15/4	
		1			1		1
1	1				-2	18	

Iteration 3:

- $\min\{1, 1, -2, 18\} = -2 < 0$, $q = 6$.
- $I = \{1, 2, 3\} \neq \emptyset$.
- $\min\{0/8, 0/(3/8), 1/1\} = 0$, $p = 1$.
- Multiply row 1 by $1/8$, and then add $-3/8$, -1 , 2 times of row 1 to rows 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
-3/2	1		1/8		1	-21/2	
1/16	-1/8		-3/64	1		3/16*	
3/2	-1	1	-1/8			21/2	1
-2	3		1/4			-3	

Iteration 4:

- $\min\{-2, 3, 1/4, -3\} = -3 < 0$, $q = 7$.
- $I = \{2, 3\} \neq \emptyset$.
- $\min\{0/(3/16), 1/(21/2)\} = 0$, $p = 2$.
- Multiply row 2 by $16/3$, and then add $21/2$, $-21/2$, 3 times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
2*	-6		-5/2	56	1		
1/3	-2/3		-1/4	16/3		1	
-2	6	1	5/2	-56			1
-1	1		-1/2	16			

Iteration 5:

- $\min\{-1, 1, -1/2, 16\} = -1 < 0$, $q = 1$.
- $I = \{1, 2\} \neq \emptyset$.
- $\min\{0/2, 0/(1/3)\} = 0$, $p = 1$.

5. Multiply row 1 by $1/2$, and then add $-1/3, 2, 1$ times of row 1 to rows 2,3,4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1	-3		-5/4	28	1/2		
	1/3*		1/6	-4	-1/6	1	
		1			1		1
	-2		-7/4	44	1/2		

Iteration 6:

1. $\min\{-2, -7/4, 44, 1/2\} = -2 < 0$, $q = 2$.
3. $I = \{2\} \neq \emptyset$.
4. $\min\{0/(1/3)\} = 0$, $p = 2$.
5. Multiply row 2 by 3, and then add 3, 2 times of row 2 to rows 1, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1			1/4	-8	-1		
	1		1/2	-12	-1/2	3	
		1			1		1
			-3/4	20	-1/2	6	

It is seen that stepsizes are equally zero in all the 6 iterations, and the last tableau is the same as the first. Thus, continuing the process always generates the same sequence of tableaus, which is called *cycling*. Consequently, the simplex algorithm failed to solve Beale's problem. It is clear that cycling could occur only in case of degeneracy.

As was addressed in Sect. 2.8, the simplex method can be explained in terms of the active set method. In each iteration, a vertex is actually determined by $Ax = b$ and $x_j = 0$, $j \in N$ corresponding to n active constraints. Since it has zero basic components, a degenerate vertex is the intersection of hyperplanes, the number of which is strictly greater than n . At first glance, this situation seems rarely to occur in practice. Surprisingly, however, it is just the opposite: problems from practice are almost all degenerate.

The real problem caused by degeneracy seems to be *stalling*, which degrades the method's performance seriously since a large number of iterations stay at a vertex for too long a time before exiting it. It is especially a headache for highly degenerate problems, in which the vanished basic components of the solution occupy a very high proportion, leading to a huge number of iterations required.

As a preliminary countermeasure, we put forward the following would be favorable variant of the classical row rule. Some newly proposed pivot rules will be presented in Part II.

Rule 3.4.1 (Row Rule) Define $I = \{i \mid \bar{a}_{iq} > 0, i = 1, \dots, m\}$, $I_1 = \{i \mid \bar{b}_i = 0, i \in I\}$. Determine pivot row index p and stepsize α by

$$p \in \begin{cases} \arg \max\{\bar{a}_{iq} \mid i \in I_1\}, & \alpha = 0, & \text{if } I_1 \neq \emptyset, \\ \arg \min\{\bar{b}_i/\bar{a}_{iq} \mid i \in I\}, & \alpha = \bar{b}_p/\bar{a}_{pq}, & \text{otherwise.} \end{cases} \quad (3.28)$$

3.5 Finite Pivot Rule

As was shown in the previous section, the finiteness of the simplex method is not guaranteed. An approach or pivot rule that turns the method to finite one is called *finite*.

Is there any finite approach/rule? The answer is positive. This is the theme of this section.

For this purpose, Charnes (1952) added a perturbation term to the right-hand side, that is,

$$w = (\epsilon, \epsilon^2, \dots, \epsilon^m)^T,$$

where $\epsilon > 0$ is a sufficiently small parameter while still using classical pivot rule.

Theorem 3.5.1 Charnes' perturbation approach is finite.

Proof Put the perturbation term in the form $w = Iw$ so that in any iteration, the right-hand side can be written as

$$v \triangleq \bar{b} + Uw,$$

where U is a permutation matrix, resulting from performing related elementary transformations on I . Note that U and I have the same rank m , and hence, each row of U is nonzero. First, it holds that $\bar{b} \geq 0$, because if, otherwise, $\bar{b}_i < 0$ for some $i \in \{1, \dots, m\}$, then it follows that $v_i < 0$, as it contradicts to feasibility. Further, it is clear that $v_i > 0$ holds for all row indices i , satisfying $\bar{b}_i > 0$; on the other hand, $v_i > 0$ also holds for all row index i with $\bar{b}_i = 0$, since the first nonzero of the i th row of U is positive (otherwise, it contradicts feasibility). Therefore, $v > 0$ holds. Since each tableau corresponds to a nondegenerate feasible basic solution, there is no possibility of cycling; hence, the process terminates within a finite number of iterations. Consequently, eliminating all parametric terms in the final optimal tableau gives an optimal one to the original problem. $\square$

Once a pivot column index q is determined, Charnes' approach amounts to selecting pivot row index p such that

$$p \in \arg \min\{(\bar{b}_i + u_{i1}\epsilon + u_{i2}\epsilon^2 + \dots + u_{im}\epsilon^m)/\bar{a}_{iq} \mid \bar{a}_{iq} > 0; i = 1, \dots, m\}. \quad (3.29)$$

Dantzig et al. (1955) proposed a pivot rule based on the so-called *lexicographic* order. This order compares vectors by their first different components. Normal equal vectors are also regarded as equal. $(\lambda_1, \dots, \lambda_t)^T < (\mu_1, \dots, \mu_t)^T$ means that the former is less than the latter in lexicographic order, that is, for the smallest subscript i , satisfying $\lambda_i \neq \mu_i$, it holds that $\lambda_i < \mu_i$. Similarly, the “ $>$ ” denotes “greater than” in lexicographic order.

Rule 3.5.1 (Lexicographic Rule) *Given pivot column index q , the row index p is determined such that*

$$p \in \arg \min \{(\bar{b}_i, u_{i1}, u_{i2}, \dots, u_{im})/\bar{a}_{iq} \mid \bar{a}_{iq} > 0; i = 1, \dots, m\}, \quad (3.30)$$

where u_{ij} denotes the entry of U , and “ $\min$ ” means minimizing the objective function in the sense of lexicographic order.

The preceding is equivalent to (3.29), in which ϵ is sufficiently small. Therefore, it is a finite rule.

Among the existing finite rules, Bland rule (1977) draws considerable attention due to its simplicity.

Rule 3.5.2 (Bland Rule) *Among nonbasic variables, corresponding to the negative reduced costs, select the smallest-indexed one to enter the basis. When there are multiple rows, attaining the same minimum ratio, select the basic variable with the smallest index to leave the basis.*

Theorem 3.5.2 *The Bland rule is finite.*

Proof Assume that cycling occurs with the simplex algorithm using Bland rule. If some variable leaves the basis in a circle, it must enter the basis gain. Denote by T the index set of these shuttling variables, and define

$$t = \max j \in T.$$

Note that the stepsize is always equal to 0 in each iteration in the circle and hence leads to the same feasible basic solution; besides, the h -indexed component of the feasible basic solution is 0 for any $h \in T$.

Assume that x_t is selected to enter the basis for simplex tableau

$$\begin{array}{cc|c} \bar{A} & 0 & \bar{b} \\ \hline \bar{z}^T & -1 & \bar{\beta} \end{array}, \quad (3.31)$$

thus $\bar{z}_t < 0$, and $\bar{z}_j \geq 0$ for any reduced cost's index $j < t$.

Assume that at another simplex tableau

$$\begin{array}{cc|c} \hat{A} & 0 & \hat{b} \\ \hline \hat{c}^T & -1 & \hat{\beta} \end{array}, \quad (3.32)$$

basic variable x_t in row p leaves and nonbasic variable x_s enters the basis. Let $x_{j_1}, \dots, x_{j_m}$ be basic variables ($x_{j_p} \equiv x_t$). It follows that $\hat{c}_s < 0$, and $\hat{c}_j \geq 0$ for any reduced cost's index $j < s$. Note that the pivot is positive, i.e., $\hat{a}_{p_s} > 0$; since $s \in T$, therefore, it holds that $s < t$.

Define v_k , $k = 1, \dots, n, n+1$ as follows:

$$v_k = \begin{cases} 1, & k = s, \\ -\hat{a}_{is}, & k = j_i, \quad i = 1, \dots, m, \\ \hat{c}_s, & k = n+1, \\ 0, & \text{otherwise.} \end{cases} \quad (3.33)$$

Note that basic columns of $\hat{A}$ constitute a permutation. Nonbasic components of the vector

$$v = (v_1, \dots, v_n, v_{n+1})^T$$

are all 0, except for $v_s = 1$. For $i = 1, \dots, m$, on the other hand, the basic entries in row i of $\hat{A}$, except for $\hat{a}_{ij_i} = 1$, are zero; basic entries of $\hat{c}$ are zero as well. Therefore, it holds that

$$\begin{pmatrix} \hat{A} & 0 \\ \hat{c}^T & -1 \end{pmatrix} v = \begin{pmatrix} \hat{a}_{1s} - \hat{a}_{1s} \\ \vdots \\ \hat{a}_{ms} - \hat{a}_{ms} \\ \hat{c}_s - \hat{c}_s \end{pmatrix} = 0. \quad (3.34)$$

Since (3.31) can be obtained from (3.32) by premultiplying a series of elementary matrices, it follows that

$$\begin{pmatrix} \bar{A} & 0 \\ \bar{z}^T & -1 \end{pmatrix} v = 0, \quad (3.35)$$

where the last equality is

$$\sum_{k=1}^n \bar{z}_k v_k - v_{n+1} = 0,$$

or

$$\sum_{k=1}^n \bar{z}_k v_k = v_{n+1} = \hat{c}_s < 0.$$

Therefore, there exists some index $h < n + 1$ such that

$$\bar{z}_h v_h < 0, \quad (3.36)$$

giving $\bar{z}_h \neq 0$ and $v_h \neq 0$.

On the other hand, it is known by $v_h \neq 0$ and the definition of v that $h \in \{j_1, \dots, j_m, s\}$. Thus, there are only the following three cases:

- (i) $h = s$. $v_h = 1$ in this case. Since x_t is an entering variable for simplex tableau (3.31) and $h = s < t$, hence $\bar{z}_h > 0$, it follows that $\bar{z}_h v_h = \bar{z}_h > 0$, contradicting (3.36).
- (ii) $h = j_p = t$. In this case, from $\bar{z}_h = \bar{z}_t < 0$ and $v_h = -\hat{a}_{ps} < 0$, it follows that $\bar{z}_h v_h > 0$, contradicting (3.36).
- (iii) $h = j_i \neq j_p$ or $h \neq t$. Now x_h is a nonbasic variable of simplex tableau (3.31) (otherwise, $\bar{z}_h = 0$); it is also a basic index of simplex tableau (3.32), thus $h \in T$. It follows that

$$\hat{b}_i = 0, \quad h < t, \quad (3.37)$$

and hence $\bar{z}_h > 0$. Further, it holds that

$$v_h = -\hat{a}_{i,s} > 0,$$

since, otherwise, $v_h \neq 0$ gives $\hat{a}_{i,s} > 0$, from which and (3.37) it follows that x_h , rather than x_t , were selected to leave the basis for tableau (3.32), as a contradiction to (3.36). Therefore, Bland rule is finite. $\square$

Terlaky (1985), and Wang (1987) independently proposed a *criss-cross* finite variant of Bland rule, which is embedded in a somehow different context, compared with the simplex method.

Unfortunately, it turns out that these finite rules are very slow in practice and cannot be mentioned in the same breath as the classical rule. This is not surprising, however. For example, Rule 3.5.2 gives nonbasic variables with small index priority to enter the basis, while we all know that basic variables of an optimal solution are not necessarily small indexed.

Rule 3.5.2 actually uses a priority order, coinciding with decreasing indices, for the selection of the entering variable. It is clear that the “ideal” order, if any, should enter a basic variable of an optimal solution to basis. According to the heuristic Proposition 11.1.2, inequality constraints with small pivoting-indices should be satisfied as equations, or the corresponding variables would be better to be nonbasic. Therefore, variables with larger pivoting-indices have priority to enter the basis.

Thus, we have the following variant of Bland rule (Pan 1990).

Rule 3.5.3 *Among nonbasic variables, corresponding to the negative reduced costs, select the maximum pivoting-indexed to enter the basis. When there are*

multiple rows, attaining the same minimum ratio, select the maximum pivoting-indexed basic variable to leave basis. When there is a tie, take the maximum indexed.

Theorem 3.5.3 Rule 3.5.3 is finite.

Proof This rule is equivalent to Rule 3.5.2 if variables are re-given indices in accordance with their pivoting-indices. $\square$

Preliminary computational experiments with small problems show that performance of Rule 3.5.3 is much better than Bland Rule. It might be the best among known finite rules. However, it is still inferior to the conventional rule, requiring more iterations in general.

The Bland Rule can be generalized to the following finite rule.

Rule 3.5.4 Assume we are given an order for the variables. Among nonbasic variables, corresponding to the negative reduced costs, select one the smallest in this order to enter the basis. When there are multiple rows attaining the same minimum ratio, select the basic variable smallest in the order to leave basis.

In Example 3.4.1, we have seen that cycling occurred with the simplex algorithm with the conventional rule. The situation is different if Rule 3.5.3 instead.

Example 3.5.1 Solve the Beale problem by Algorithm 3.1.1 using Rule 3.5.3:

$$\begin{aligned}
 \min f &= -3/4x_4 + 20x_5 - 1/2x_6 + 6x_7 \\
 \text{s.t. } x_1 &+ 1/4x_4 - 8x_5 - x_6 + 9x_7 = 0 \\
 x_2 &+ 1/2x_4 - 12x_5 - 1/2x_6 + 3x_7 = 0 \\
 x_3 &+ x_6 = 1 \\
 x_j &\geq 0, \quad j = 1, \dots, 7.
 \end{aligned}$$

Answer As the coefficient matrix includes a unit matrix, it is easy to transform the constraints to the “ $\geq$ ” type of inequalities:

$$\begin{aligned}
 \min f &= -3/4x_4 + 20x_5 - 1/2x_6 + 6x_7 \\
 \text{s.t. } -1/4x_4 + 8x_5 + x_6 - 9x_7 &\geq 0 \\
 -1/2x_4 + 12x_5 + 1/2x_6 - 3x_7 &\geq 0 \\
 -x_6 &\geq -1 \\
 x_j &\geq 0, \quad j = 4, \dots, 7.
 \end{aligned}$$

Note that the first 3 constraints correspond to the original variables x_1, x_2, x_3 , respectively; pivoting-indices of constraints may be regarded as those for the associated variables.

It is noted that the gradient of the objective function is $c = (-3/4, 20, -1/2, 6)^T$. The gradient of the first constraint is $a_1 = (-1/4, 8, 1, -9)^T$, and the calculated pivoting-index of this constraint (or corresponding variable x_1) is $\alpha_1 =$

$-a_1^T c / \|a_1\| = -8.74$. Similarly, we calculate all pivoting-indices and put them into the following table in decreasing order:

Variable	Constraint	α_i
x_4	$x_4 \geq 0$	0.75
x_6	$x_6 \geq 0$	0.50
x_3	$-x_6 \geq -1$	-0.50
x_7	$x_7 \geq 0$	-6.00
x_1	$-1/4x_4 + 8x_5 + x_6 - 9x_7 \geq 0$	-8.74
x_5	$x_5 \geq 0$	-20.00
x_2	$-1/2x_4 + 12x_5 + 1/2x_6 - 3x_7 \geq 0$	-114.78

Now call Algorithm 3.1.1 with Rule 3.5.3.

Initial: The following feasible simplex tableau is obtained directly from the preceding problem:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1			1/4*	-8	-1	9	
	1		1/2	-12	-1/2	3	
		1			1		1
			-3/4	20	-1/2	6	

Iteration 1:

1. Among nonbasic variables x_4 ($\alpha_4 = 0.75$) and x_6 ($\alpha_6 = 0.50$) corresponding to negative reduced costs, select the maximum pivoting-indexed x_4 to enter the basis, $q = 4$.
3. $I = \{1, 2\} \neq \emptyset$.
4. $\min\{0/(1/4), 0/(1/2)\} = 0$. Among basic variables in rows 1 and 2, select the maximum pivoting-indexed x_1 ($\alpha_1 = -8.74 > -114.78 = \alpha_2$) to leave the basis, $p = 1$.
5. Multiply row 1 by 4, and then add $-1/2, 3/4$ times of row 1 to rows 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
4			1	-32	-4	36	
-2	1			4	3/2*	-15	
		1			1		1
3				-4	-7/2	33	

Iteration 2:

1. Among nonbasic variables x_5 ($\alpha_5 = -20.00$) and x_6 ($\alpha_6 = 0.50$) with negative reduced costs, select the maximum pivoting-indexed x_6 to enter the basis, $q = 6$.
3. $I = \{2, 3\} \neq \emptyset$.
4. $\min\{0/(3/2), 1/1\} = 0$, only x_2 is eligible for leaving the basis, $p = 2$.
5. Multiply row 2 by $2/3$, and then add 4, -1 , $7/2$ times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-4/3$	$8/3$		1	$-64/3$		-4	
$-4/3$	$2/3$			$8/3$	1	-10	
$4/3$	$-2/3$	1		$-8/3$		10^*	1
$-5/3$	$7/3$			$16/3$		-2	

Iteration 3:

1. Among nonbasic variables x_1 ($\alpha_1 = -8.74$) and x_7 ($\alpha_7 = -6.00$) with negative reduced costs, select the maximum pivoting-indexed x_7 to enter the basis, $q = 7$.
3. $I = \{3\} \neq \emptyset$.
4. Only x_3 is eligible for leaving the basis, $p = 3$.
5. Multiply row 3 by $1/10$, and then add 4, 10, 2 times of row 3 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-4/5$	$12/5$	$2/5$	1	$-112/5$			$2/5$
		1			1		1
$2/15^*$	$-1/15$	$1/10$		$-4/15$		1	$1/10$
$-7/5$	$11/5$	$1/5$		$24/5$			$1/5$

Iteration 4:

1. Only nonbasic variable x_1 is eligible for entering the basis, $q = 1$.
3. $I = \{3\} \neq \emptyset$.
4. Only x_7 is eligible for leaving the basis, $p = 3$.

5. Multiply row 3 by $15/2$, and then add $4/5$; $7/5$ times of row 3 to rows 1, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	2	$2/5$	1	-24		6	1
		1			1		1
1	$-1/2$	$3/4$		-2		$15/2$	$3/4$
	$3/2$	$5/4$		2		$21/2$	$5/4$

All the reduced costs are now nonnegative. The optimal solution and optimal value are

$$x^* = (3/4, 0, 0, 1, 0, 1, 0)^T, \quad f^* = -5/4.$$

Thus, the Beale problem is solved without cycling.

3.6 Notes on Simplex Method

Related to the linear structure of the LP model, the simplex method is pivotal in nature, which progresses through a series of pivot operations (for the origin of the simplex method, see Appendix A). Its iteration is characterized by a basis: once a basis is known, so is the corresponding feasible basic solution. It selects a pivot and makes the corresponding basis change to improve the objective value until reaching the optimality unless detecting the unboundedness. Therefore, the pivot selection is crucial to the simplex method.

The past research on the simplex method has been on the two aspects: on one side, new pivot rules were suggested from time to time even if Dantzig's original rule has been widely used in practice for many years. On the other hand, efforts have been made to improve its framework.

The evaluation of an algorithm involves the number of required arithmetics and storage, numerical stability, and difficulty for programming. Its time complexity and storage complexity depend on the size of the problem handled. For the standard LP problem, its size may be measured by (m, n) roughly. However, the true size depends on values of (A, b, c) , as can be determined by the total number L , called *input length* of binary digits of input data. If the number of arithmetics is bounded above by some function $\tau f(m, n, L)$, where $\tau > 0$ is some constant, the time complexity of the algorithm is said $O(f(m, n, L))$. If $f(m, n, L)$ is polynomial in m, n, L , it is of polynomial-time complexity, as is commonly regarded as “good.” If $f(m, n, L)$ is exponential in m, n, L , it is of exponential time complexity, as regarded as “bad” since it can fail to solve large-scale problems by consuming too much time. Note, however, that this is just the worst-case complexity since $\tau f(m, n, L)$ is an *upper bound* of the required arithmetics.

The following table lists the numbers of arithmetics per iteration and storage locations required by tableau simplex Algorithm 3.3.1 vs. the revised simplex Algorithm 3.1.1:

	Algorithm 3.3.1	Algorithm 3.1.1
Multiplications	$m(n + 2m) + 2m + 1$	$(m + 1)(n - m + 1)$
Additions	$m(n + 2m) - 2m + 1$	$m(n - m + 1)$
Storages	$(m + 1)(m + 2)$	$(m + 1)(n + 1)$

In the preceding, the number of storage locations required by Algorithm 3.1.1 excludes that for the original data (A, b, c) , though these data should be stored, in computer. Therefore, the storage requirement of the tableau simplex algorithm is significantly high, compared with the revised simplex algorithm, especially when $n \gg m$.

The two algorithms require the same iterations in solving any standard LP problem, in principle. Therefore, we only compare the arithmetics, mainly multiplications, in a single iteration. The table shows that the revised simplex algorithm is better than the tableau simplex algorithm, especially when $n \gg m$. For sparse computations, the former is even far superior. It is not surprising that the latter appears only in textbooks.

The time complexity of the simplex method mainly depends on the required number of iterations. Klee and Minty (1972) offered an example indicating that the simplex method using the classical pivot rule passes through all the 2^m vertices of the feasible region. Therefore, the rule is not polynomial-time, not to mention the instances in which the simplex method fails to solve because of cycling. It turns out that many other rules, such as the *most improvement* rule, the *steepest-edge* rule, even Bland's finite rule, etc., are all not polynomial-time. It has not been clear whether there exists a polynomial-time rule though such possibility seems to be very low if any. Indeed, the simplex method is unsatisfactory for solving a certain type of LP problems, e.g., hard large-scale problems or problems with combinatorial constraints, such as those with zero/one coefficients or those from *Krawchuk polynomials*. Schrijver (1986, p.141) wrote:

The main problem seems to be that the simplex method is “myopic,” cannot see “cut-off paths.” It does not find a path which is locally bad but globally good.

Nevertheless, the evaluation of methods is essentially a practical issue. The simplex method performs very well and has been very successful in practice. For solving small/medium LP problems, in particular, it usually requires iterations no more than $4m$ to $6m$ (including Phase-1). Therefore, the worst-case complexity has a limited value and is even misleading.

On the other hand, the complexity under some probability sense seems to be closer to reality. Borgwardt (1982a,b) showed that an average complexity of the

simplex method is polynomial. Specifically, for the LP problem

$$\begin{array}{ll} \min & c^T x \\ \text{s.t.} & Ax \leq b, \end{array}$$

where $A \in \mathcal{R}^{m \times n}$, $b > 0$, and components of c are random variables under certain assumptions, he proved that the mathematical expectation of iterations, required by the simplex method using a special pivot rule, is

$$O(n^3 m^{1/(n-1)}).$$

Using another probability model and pivot rule, Smale (1983a,b) proved that the average complexity of the simplex method is bounded above by

$$O((\log n)^{m^2+m}),$$

which is not polynomial, but still better than Borgwardt's result when $n \gg m$. These results coincide with reality.

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Chapter 4

Implementation of Simplex Method



All algorithms formulated in this book, such as the simplex algorithm and the dual simplex algorithm, are theoretical or conceptual and cannot be put into use directly. Software, resulting by the following algorithms, step by step would solve textbook instances only. Implementation techniques are crucial to the success of optimization methods.

This chapter will highlight implementation issues for the simplex method. There are two aspects: one is to reduce involved computational effort and the other to improve numerical stability. Only limited sparse techniques will be considered, as a *full* treatment of sparsity is beyond the scope of this book. Associated contents include the miscellaneous, including feasibility/optimality tolerance, machine 0, etc., and scaling, LU factorization as well as sparse LU factorization of basis, updating LU factors, crashing for initial basis, Harris pivot rule, tolerance expending, and pricing scheme of the reduced costs.

4.1 Miscellaneous

As rounding errors affect computational procedures and results (see Sect. 1.1), restricting such effects to improve the numerical stability is important to the implementation of the simplex algorithm.

First, theoretical 0 or ∞ involved in the algorithm should be treated properly by giving tolerances, corresponding to the computer precision used. Based on MINOS¹ (Murtagh and Saunders 1998), for instance, main parameters are listed below, where

¹ MINOS is an optimization package developed by Murtagh and Saunders at Dept of Management Science and Engineering of Stanford University.

the computer precision is assumed to be

$$\epsilon = 2^{-52} \approx 2.22 \times 10^{-16}.$$

Introduce

$$\begin{aligned}\epsilon_0 &= \epsilon^{0.8} \approx 3.00 \times 10^{-13}, \\ \epsilon_1 &= \epsilon^{0.67} \approx 3.25 \times 10^{-11}, \\ \text{featol} &= 10^{-6}, \\ \text{plinfy} &= 10^{20},\end{aligned}$$

where:

- Quantity in the solution process, whose absolute value is no more than ϵ_0 , is treated as machine 0.
- Quantity, whose absolute value is more than plinfy, is treated as ∞ .
- $\text{tolpiv} = \epsilon_1$ is used to avoid a too small pivot; e.g., the minimum-ratio test (3.11) is actually replaced by

$$\alpha = \bar{b}_p / \bar{a}_{pq} = \min\{\bar{b}_i / \bar{a}_{iq} \mid \bar{a}_{iq} > \text{tolpiv}, i = 1, \dots, m\} \geq 0. \quad (4.1)$$

Thereby, any pivot chosen satisfies $\bar{a}_{pq} > \text{tolpiv}$. The problem is regarded as unbounded if no such a pivot is found.

The magnitude of a pivot seriously affects the numerical stability of the algorithm. The determination of the numerical nonzero pivot not only depends on the computer precision but also on the problem itself to be solved. It is difficult to give the “best” tolpiv . Some value between ϵ_1 and 10^{-7} seems to be acceptable, though still far from satisfactory. This theme will be discussed further in Sect. 4.7.

- featol is a primal or dual feasibility tolerance. For $j = 1, \dots, n$, $x_j \geq -\text{featol}$ is used to replace the nonnegativity condition $x_j \geq 0$ (for primal feasibility). For $j \in N$, in optimality test, $\bar{z}_j \geq -\text{featol}$ is used to replace $\bar{z}_j \geq 0$ (for dual feasibility). Therefore, the column rule (3.13) is replaced by

$$q \in \arg \min\{\bar{z}_j \mid \bar{z}_j < -\text{featol}, j \in N\}. \quad (4.2)$$

If there is no such an q found, i.e.,

$$\bar{z}_j \geq -\text{featol}, \forall j \in N,$$

then optimality is declared.

In the solution process, however, it is not easy to ensure all solutions to be within the feasibility tolerance, let alone degeneracy being intertwined with this. Solving large-scale LP problems usually involves a large number of iterations. After many iterations, due to accumulated errors, intermediate results could be too inaccurate to be used. It is, therefore, necessary to perform a so-called

restarting strategy: rebooting the process from scratch, by recomputing from the original data, inverse B^{-1} or its LU factorization of the current basis. It might be well to restart every 100 iterations. In addition, residuals of equality constraints at iterates should be monitored in the solution process: once residuals are found to exceed some tolerance, carry out the restarting strategy, it is also necessary to predetermine an upper bound on the number of iterations to limit running time. Once iterations exceed such a bound, terminate the solution process after recording the end basis for a possible later *warm start*.

Real-world demand is a powerful pushing force for implementation techniques for solving large-scale LP problems. Nowadays, LP problems yielding from practice become larger and larger. Huge problems are quite common. The solution to such types of problems is very challenging. Storage required would be too large to meet; even if it is met, the running time consumed would still be too high to afford.

Like many other models, fortunately, most large-scale LP problems are sparse, i.e., data A, b, c involve a large number of zeros. Usually, the proportion of nonzeros of A is no more than 1%, hardly 5%. This is so because variables, involved in large systems, usually belong to different groups corresponding to different regions, departments, or properties, and so on), and constraints mainly reflect relations within the interior of a group, so that each variable would appear in a few constraints. With the help of sparse matrix techniques, which are quite mature nowadays, it is possible to proceed without storing and manipulating zeros at all. Therefore, the key to solving large-scale sparse problems is to take advantage of sparsity to reduce storage and arithmetic requirements. In professional optimization packages, the original nonzero data are usually expressed via a so-called *MPS format* in an input data file (Appendix B).

4.2 Scaling

The large difference within the original data affects calculated results considerably, which is related to the accumulation of rounding errors. To reduce such differences, the original data should be preprocessed by some so-called *scaling* technique (Hamming 1971).

Adjusting the unit of measurement is a common scaling method. It amounts to performing variable transformation $x = D_1 x'$, where D_1 , called *scaling factor*, is a diagonal square matrix with positive diagonals. For instance, simply taking

$$D_1 = \begin{pmatrix} 1/\|a_1\| & & & \\ & 1/\|a_2\| & & \\ & & \ddots & \\ & & & 1/\|a_n\| \end{pmatrix} \quad (4.3)$$

converts the standard LP problem into

$$\begin{aligned} \min f &= (c^T D_1)x', \\ \text{s.t. } (AD_1)x' &= b, \quad x' \geq 0, \end{aligned} \quad (4.4)$$

where each column of AD_1 has norm 1, and hence, differences between columns are balanced. Then, once an optimal solution $\bar{x}'$ to (4.4) is obtained, the corresponding optimal solution to the original is readily available, i.e., $\bar{x} = D_1 \bar{x}'$.

Usually, the preceding scaled problem is amenable to be solved, compared to the original. It will be seen that such doing enhances pivot column selection, consequently reducing the number of iterations required.

An analogous scaling is to balance differences among rows. Denote row i of A by d_i^T and introduce

$$D_2 = \begin{pmatrix} 1/\|d_1\| & & & \\ & 1/\|d_2\| & & \\ & & \ddots & \\ & & & 1/\|d_m\| \end{pmatrix}. \quad (4.5)$$

Then the standard LP problem can be converted into

$$\begin{aligned} \min f &= c^T x, \\ \text{s.t. } (D_2 A)x &= D_2 b, \quad x \geq 0, \end{aligned} \quad (4.6)$$

where each row of the coefficient matrix $D_2 A$ has norm 1.

Professional codes for handling large-scale sparse problems generally use more sophisticated scaling methods. MINOS applies an iterative procedure based on Fourer (1979), which alternately balances nonzeros of rows and columns to obtain scaling factors by geometric mean. After the scaled problem is solved, relevant transformations are carried out to recover an optimal solution of the original. More precisely, the main steps are put in the following algorithm.

Algorithm 4.2.1 (Scaling) Given a scaling tolerance $sctol < 1$. In the following, $a_{i,j}$ is the nonzero entry of A .

1. Compute $aratio = \max_j \{\max_i |a_{i,j}| / \min_i |a_{i,j}|\}$.
2. Divide rows $i = 1, \dots, m$ of $Ax = b$ by $\sqrt{\min_j |a_{i,j}| \times \max_j |a_{i,j}|}$.
3. Divide columns $i = 1, \dots, n$ of $\left(\frac{A}{c^T}\right)$ by $\sqrt{\min_i |a_{i,j}| \times \max_i |a_{i,j}|}$.
4. Compute $sratio$ by formula in step 1.
5. Return if $sratio \geq sctol * aratio$; else, go to step 1.

As for more general LP problems, scaling factors for bounded variables, free or fixed variables should be determined differently, as omitted here.

The simplex method is not invariant to different scaling methods. The effect of scaling is problem-dependent and is difficult to make a theoretical analysis, which

is not clear what kind of scaling is the best. At any event, scaling is indispensable in real applications. Even simple scaling would improve software's performance significantly. Without scaling, only a few real-world problems can be handled.

4.3 LU Factorization of Basis

For large-scale sparse computation, it is important to maintain the sparsity of matrices and vectors in the solution process. To this end, this section presents a product form of the inverse of basis first and then turns to LU factorization of basis and its sparsity version, in particular.

The simplex algorithm 3.3.1 and others utilize the inverse of basis explicitly and update it iteration by iteration. This is just for the convenience of expression, not suitable for sparse computation, since the inverse would be very dense even though the basis itself is sparse, as is shown by the following instance (Orchard-Hays 1971).

Example 4.3.1 It is verified that the inverse of matrix

$$B = \begin{pmatrix} 1 & & & & 1 \\ & 1 & & & \\ & & 1 & & \\ & & & 1 & \\ & & & & 1 \end{pmatrix}$$

is

$$B^{-1} = \begin{pmatrix} 1 & 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 & -1 \\ -1 & 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 \end{pmatrix},$$

all of its entries being nonzero.

That an original zero entry becomes nonzero is said to be *fill-in*. In the preceding instance, a large proportion of fill-ins lead to the inverse losing sparsity completely. An approach to bypassing this difficulty is to express inverse in product form of elementary matrices.

As it is known, Gauss–Jordan elimination turns basis B to unite matrix. Such doing amounts to premultiplying B by a sequence of Gauss transformation (elementary) matrices $E_1, \dots, E_m$ such that

$$E_m \cdots E_1 B = I.$$

Therefore, the product form of B^{-1} follows, i.e.,

$$B^{-1} = E_m \cdots E_1.$$

Then, all operations involving B^{-1} can be carried out through the factors. Instead of B^{-1} itself, consequently, there is only a need for storing elementary matrix factors, which is usually sparse, relative to B^{-1} itself. It is noted that only a single vector needs to be stored for each factor since the elementary matrix differs from a unit matrix by a column only.

The preceding approach was considered in the early days of the simplex method. We will not go into detail but turn to a more desirable and commonly used alternative, i.e., the LU factorization.

Concerning the computation in the simplex method, the inverse itself is not a goal. What one needs are vectors

$$\bar{a}_q = B^{-1}a_q, \quad \text{and} \quad \bar{y} = B^{-T}c_B,$$

which can be obtained by solving the following two systems, respectively:

$$B\bar{a}_q = a_q, \quad \text{and} \quad B^T\bar{y} = c_B. \quad (4.7)$$

And a linear system can be solved through LU factorization of the coefficient matrix efficiently. In particular, there are sparse matrix techniques that solve very large sparse systems.

Assume that the LU factorization $B = LU$ is available, where L is a unit lower triangular matrix (with all diagonals 1), and U is an upper triangular matrix. Then, the first system of (4.7) is converted into the following two triangular systems:

$$Lu = a_q, \quad U\bar{a}_q = u, \quad (4.8)$$

and the second to the two triangular systems below:

$$U^T v = c_B, \quad L^T \bar{y} = v. \quad (4.9)$$

It is an easy task to solve such a system, as only $m^2/2$ multiplications are required.

Matrix factorization is closely related to elimination. The LU factors of a matrix can be obtained by Gauss elimination. In each step, Gauss elimination annihilates nonzeros below the diagonal in a column, which amounts to premultiplying the matrix by a Gauss transformation.

Assume that at the end of the $(k - 1)$ th step ($k < m$), there are Gauss transformations $G_1, \dots, G_{k-1}$ such that the entries below the diagonals of the first $k - 1$ columns are zero, i.e.,

$$B^{(k-1)} = G_{k-1} \cdots G_1 B = \begin{pmatrix} B_{11}^{(k-1)} & B_{12}^{(k-1)} \\ & B_{22}^{(k-1)} \end{pmatrix}, \quad (4.10)$$

where $B_{11}^{(k-1)}$ is upper triangular of order $k-1$. Assume that

$$B_{22}^{(k-1)} = \begin{pmatrix} b_{kk}^{(k-1)} & \cdots & b_{km}^{(k-1)} \\ \vdots & & \vdots \\ b_{mk}^{(k-1)} & \cdots & b_{mm}^{(k-1)} \end{pmatrix}, \quad (4.11)$$

and that $b_{kk}^{(k-1)} \neq 0$. Thereby, the following factors are well defined:

$$l_{ik} = b_{ik}^{(k-1)} / b_{kk}^{(k-1)}, \quad i = k+1, \dots, m. \quad (4.12)$$

It is clear that the Gauss transformation

$$G_k = I - h^{(k)} e_k^T, \quad h^{(k)} = (\underbrace{0, \dots, 0}_k, l_{k+1,k}, \dots, l_{m,k})^T \quad (4.13)$$

satisfies

$$B^{(k)} = G_k B^{(k-1)} = \begin{pmatrix} B_{11}^{(k)} & B_{12}^{(k)} \\ & B_{22}^{(k)} \end{pmatrix},$$

where $B_{11}^{(k)}$ is upper triangular of order k . Thus, the k th step of Gauss elimination is complete, provided the k th diagonal (pivot) is nonzero. In this manner, the sequence $G_1, \dots, G_{m-1}$ of Gauss transformations can be determined such that

$$B^{(m-1)} = G_{m-1} \cdots G_1 B \stackrel{\Delta}{=} U$$

is an upper triangular with nonzero diagonals.

It is verified that $G_k^{-1} = I + h^{(k)} e_k^T$. Then, the following LU factorization follows:

$$B = LU,$$

where

$$L = G_1^{-1} \cdots G_{m-1}^{-1} = \prod_{k=1}^{m-1} (I + h^{(k)} e_k^T) = I + \sum_{k=1}^{m-1} h^{(k)} e_k^T = I + (h^{(1)}, \dots, h^{(m-1)}, 0)$$

is unit lower triangular. The preceding LU factorization has a practicable form, i.e.,

$$L^{-1} B = U, \quad L^{-1} = G_{m-1} \cdots G_1. \quad (4.14)$$

Therefore, (4.8) becomes

$$U\bar{a}_q = G_{m-1} \cdots G_1 a_q, \quad (4.15)$$

and (4.9) does

$$U^T v = c_B, \quad \bar{y} = G_1^T \cdots G_{m-1}^T v. \quad (4.16)$$

Consequently, there is no need for an explicit expression of L^{-1} , but holding Gauss factors $G_k, k = 1, \dots, m-1$, is enough.

It is often the case when L^{-1} is dense even though the Gauss factors are sparse. Expressing L^{-1} in a product form is hence suitable for sparse computation. The required storage of G_k is not high, since only the nonzero components of $h^{(k)}$ are stored. In dense computations, these nonzeros are usually stored in the lower triangular part of the matrix, while U is stored in the upper triangular part.

It is noticeable that the LU factorization of a matrix is not always a clear road. The existence condition is that the principal submatrices are all nonsingular, guaranteeing a nonzero pivot in each step (Golub and Van Loan 1989); e.g., there is no LU factorization for the following simple nonsingular matrix:

$$\begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$$

because the first diagonal is 0, not eligible to be a pivot.

Fortunately, there exists LU factorization for a permuted matrix, resulting from properly rearranging rows and columns, and such doing does not matter essentially but amounts to exchanging the order of equations/variables.

4.4 Sparse LU Factorization of Basis

On the other hand, rearranging rows and columns does affect the sparsity of Gauss transformations significantly. Let us look at a simple example.

Example 4.4.1 Use Gauss elimination to convert the matrix

$$B = \begin{pmatrix} 2 & 4 & 2 \\ 3 & & \\ -1 & & 1 \end{pmatrix}$$

into upper triangular.

Answer

$$G_1 = \begin{pmatrix} 1 & & \\ -3/2 & 1 & \\ 1/2 & & 1 \end{pmatrix} \longrightarrow G_1 B = \begin{pmatrix} 2 & 4 & 2 \\ -6 & -3 & \\ 2 & 2 & \end{pmatrix},$$

$$G_2 = \begin{pmatrix} 1 & & \\ & 1 & \\ & 1/3 & 1 \end{pmatrix} \longrightarrow G_2 G_1 B = \begin{pmatrix} 2 & 4 & 2 \\ -6 & -2 & \\ & & 1 \end{pmatrix} \triangleq U.$$

The right-hand side U is already upper triangular. Vectors h_1, h_2 , related to the Gauss transformations, may be stored in the lower triangular part of U :

$$\begin{pmatrix} 2 & 4 & 2 \\ -3/2 & -6 & -2 \\ 1/2 & 1/3 & 1 \end{pmatrix}.$$

It is seen that the resulting matrix is filled with nonzeros. Let us now rearrange rows and columns, according to the order of row indices 2, 3, 1 and of column indices 1, 3, 2 as follows in advance:

$$\bar{B} = \begin{pmatrix} 3 & & \\ -1 & 1 & \\ 2 & 2 & 4 \end{pmatrix}.$$

Then carry normal eliminations in order:

$$\bar{G}_1 = \begin{pmatrix} 1 & & \\ 1/3 & 1 & \\ -2/3 & & 1 \end{pmatrix} \longrightarrow \bar{G}_1 \bar{B} = \begin{pmatrix} 3 & & \\ 1 & & \\ 2 & 4 & \end{pmatrix},$$

$$\bar{G}_2 = \begin{pmatrix} 1 & & \\ & 1 & \\ & -2 & 4 \end{pmatrix} \longrightarrow \bar{G}_2 \bar{G}_1 \bar{B} = \begin{pmatrix} 3 & & \\ 1 & & \\ & & 4 \end{pmatrix} \triangleq \bar{U}.$$

Vectors $\bar{h}_1, \bar{h}_2$ related to the Gauss transformations are stored in the lower triangular part of $\bar{U}$, i.e.,

$$\begin{pmatrix} 3 & & \\ 1/3 & 1 & \\ -2/3 & -2 & 4 \end{pmatrix}.$$

It is seen that the resulting matrix involves 3 zeros, just as the original B , and hence no fill-in is introduced. This is the case because the rearranged matrix is lower triangular, which is conducive to Gauss elimination.

For the instance below, the left-hand side is the original matrix, where the symbol “ $\times$ ” denotes a nonzeros entry. If classical Gauss elimination is carried out, its sparsity is completely lost. But if rows and columns are rearranged as the matrix on the right-hand side, no fill-ins will occur at all.

$$\begin{pmatrix} \times & \times & \times & \times & \times & \times \\ \times & \times & & & & \\ \times & & \times & & & \\ \times & & & \times & & \\ \times & & & & \times & \\ \times & & & & & \times \end{pmatrix} \longrightarrow \begin{pmatrix} \times & & & & & \times \\ & \times & & & & \times \\ & & \times & & & \times \\ & & & \times & & \times \\ & & & & \times & \times \\ \times & \times & \times & \times & \times & \times \end{pmatrix}.$$

Note that the right-hand side matrix is close to lower triangular, with the bratty bump in the right-most column.

The so-called *preassigned* pivot procedure (P^3) and *partitioned preassigned* pivot procedure (P^4) describe how to rearrange rows and columns for triangularization (Hellerman and Rarick 1971, 1972). Later, a stable revision, $P5$, of $P4$ was proposed (Duff et al. 1986). All such types of approaches provide the LU factorization of some permuted version of a matrix, i.e.,

$$L^{-1}PBQ = U, \quad L^{-1} = G_{m-1} \cdots G_1, \quad (4.17)$$

where P and Q are the permutation matrices (resulting from the unit matrix by permutation). By (4.17), it is easy to convert the two systems in (4.7) into the following equivalent forms

$$Uw = G_{m-1} \cdots G_1 Pa_q, \quad \bar{a}_q = Qw \quad (4.18)$$

and

$$U^T v = Q^T c_B, \quad \bar{y} = P^T G_1^T \cdots G_{m-1}^T v. \quad (4.19)$$

A combination of such approaches and the classical Gauss elimination perform very well in practice. We will not go into detail; though, interested readers are referred to the related literature.

We prefer another approach to select pivots in the elimination process, rather than via preassigning order. In this aspect, Markowitz's rule (1957) is especially successful in practice.

Assume that at the end of step $k - 1$, Gauss elimination generates (4.10)~(4.11). Currently, only the $(m - k + 1) \times (m - k + 1)$ southeast corner submatrix is *active*, and hence, the elimination proceeds only within it without touching the other part.

At step k , the standard Gauss elimination would take $b_{kk}^{(k-1)}$ as pivot, provided it is nonzero.

Denote by r_i^k the number of nonzeros of row i of $B_{22}^{(k-1)}$, and by c_j^k the number of nonzeros of column j . Markowitz selects a pivot, within nonzeros of $B_{22}^{(k-1)}$ such that $(r_i^k - 1) \times (c_j^k - 1)$ is the minimum. The pivot chosen is then moved to the k th diagonal's position by row and column permutations, and step k is complete.

It is noted that $(r_p^k - 1) \times (c_q^k - 1)$ rather than $r_p^k \times c_q^k$ is used; therefore, a row or column involving only a single nonzero is selected, if any, and hence, no fill-in occurs at step k .

However, the so-called *zero* or *nonzero* only has conceptual meaning, since a pivot too close to zero would lead to computational failure. Considering numerical stability, the pivot chosen should not be too small, compared with other nonzeros. Therefore, there is a contradiction between sparsity and stability. In the following rule, the two aspects are balanced to some extent.

Rule 4.4.1 (Markowitz Pivot Rule) *Given a real number α such that $0 < \sigma \leq 1$. Select pivot $b_{pq}^{(k-1)}$ such that*

$$(r_p^k - 1) \times (c_q^k - 1) = \min\{(r_i^k - 1) \times (c_j^k - 1) \mid |b_{ij}^{(k-1)}| \geq \sigma\theta, i, j = k, \dots, m\}, \quad (4.20)$$

where

$$\theta = \max\{|b_{ij}^{(k-1)}| \mid i, j = k, \dots, m\}. \quad (4.21)$$

It is clear that a large σ value contributes to stability, while a small one does to sparsity. Value $\sigma = 0.5$ or so seems to be acceptable. To avoid going through every nonzero of the active submatrix, on the other hand, some variants of the preceding rule are designed, in which only a part of rows/columns is involved.

Such kind of approaches lead to the following LU factorization:

$$G_{m-1}P_{m-1} \cdots G_1P_1BQ_1 \cdots Q_{m-1} = U, \quad (4.22)$$

where $P_i, Q_i, i = 1, \dots, m-1$, are permutations. Setting

$$P = P_{m-1} \cdots P_1, \quad Q = Q_1 \cdots Q_{m-1},$$

then from (4.22), it is known that the pivot determinations and row/column permutations, carried out alternately, have the same effects as the corresponding prepermutations followed by the standard factorization, i.e.,

$$L^{-1}PBQ = U, \quad L^{-1} = G_{m-1}P_{m-1} \cdots G_1P_1P^T.$$

Therefore, the two systems in (4.7) can still be, respectively, converted into

$$Uw = G_{m-1}P_{m-1} \cdots G_1P_1a_q, \quad \bar{a}_q = Qw$$

and

$$U^T v = Q^T c_B, \quad \bar{y} = P_1^T \cdots G_1^T \cdots P_{m-1}^T G_{m-1}^T v.$$

It is favorable that there is no need for moving around related entries for permutations in the process; two integer arrays are enough to record the positions of pivot rows and columns.

4.5 Updating LU Factors

The LU factorization of a basis from scratch usually requires a large number of operations. Fortunately, it is unnecessary to do so in each iteration. Once an initial basis is factorized, subsequent LU factors can be updated iteration by iteration. A basis is refactorized only when the number of iterations exceeds some predetermined bound, and hence, the LU factors are unacceptably inaccurate or dense, or the sequence of the factors is too long to store (Sect. 4.1).

Only one column of basis is different from that of its predecessor; therefore, it is possible to obtain LU factors from the old ones cheaply. In this section, we present updating approaches commonly used in practice.

Assume that the LU factorization of the current basis is

$$L^{-1}B = U, \quad L^{-1} = G_s P_s \cdots G_1 P_1 \quad (4.23)$$

and that a pivot column index q and row index p have been determined.

The first updating approach, proposed by Bartels and Golub (1969), drops the p th column a_{j_p} from B , then moves the $p+1$ through m th columns forward by one column distance, and adds a_q at the end as the m th column, resulting in the new basis

$$\widehat{B} = (a_{j_1}, \dots, a_{j_{p-1}}, a_{j_{p+1}}, \dots, a_{j_m}, a_q).$$

Consider the matrix

$$H \triangleq L^{-1}\widehat{B} = (L^{-1}a_{j_1}, \dots, L^{-1}a_{j_{p-1}}, L^{-1}a_{j_{p+1}}, \dots, L^{-1}a_{j_m}, L^{-1}a_q).$$

Note that the last column $\tilde{a}_q = L^{-1}a_q$ of the new basis is available since it was computed as an intermediate result when solving $B\tilde{a}_q = a_q$. It is also seen that H may result from dropping the p th column from the upper triangular matrix U ,

moving forward the $p + 1$ through m th columns by one column position, and putting $\tilde{a}_q$ as the last column. Therefore, H is an upper Hessenberg matrix with nonzero subdiagonals in the p through $m - 1$ th columns, the pattern of which looks like:

$$H = \begin{pmatrix} \times & \times & \times & \times & \times & \times & \times & \times & \times & \times \\ & \times & \times & \times & \times & \times & \times & \times & \times & \times \\ & & \times & \times & \times & \times & \times & \times & \times & \times \\ & & & \times & \times & \times & \times & \times & \times & \times \\ & & & & * & * & * & * & * & * \\ & & & & \times & \times & \times & \times & \times & \times \\ & & & & & \times & \times & \times & \times & \times \\ & & & & & & \times & \times & \times & \times \\ & & & & & & & \times & \times & \times \\ & & & & & & & & \times & \times \end{pmatrix}.$$

Then eliminate H 's nonzero subdiagonals in the p through $m - 1$ th columns, in turn, by elementary transformations to lead to upper triangular.

However, the preceding process is impossible when some diagonals are zero. Bartels–Golub's approach overcomes this difficulty by exchanging rows: in the elimination process it takes a diagonal as pivot whenever its absolute value is large enough, compared with the corresponding subdiagonal; otherwise, it interchanges the two rows where the diagonal and subdiagonal are located. Such doing leads to better numerical stability, though exchanging rows frequently could degrade efficiency.

As a variant of Bartels–Golub approach, Forrest and Tomlin (1972) moves the p th row of H (marked by $*$) to the bottom (the p through $m - 1$ th rows is moved up by one row position). The resulting matrix differs from upper triangular only in p through $(m - 1)$ -indexed possible nonzeros in row m :

$$\tilde{H} = \tilde{P}H = \begin{pmatrix} \times & \times & \times & \times & \times & \times & \times & \times & \times & \times \\ & \times & \times & \times & \times & \times & \times & \times & \times & \times \\ & & \times & \times & \times & \times & \times & \times & \times & \times \\ & & & \times & \times & \times & \times & \times & \times & \times \\ & & & & \times & \times & \times & \times & \times & \times \\ & & & & & \times & \times & \times & \times & \times \\ & & & & & & \times & \times & \times & \times \\ & & & & & & & \times & \times & \times \\ & & & & & & & & \times & \times \\ & & & & & & & & & * & * & * & * & * & * \end{pmatrix}.$$

Doing so amounts to premultiplying H by a permutation, say $\tilde{P}$. Assume that the resulting matrix is $\tilde{H} = \tilde{P}H$, in which the first nonzero in row m is p -indexed, i.e., $\tilde{h}_{mp} \neq 0$. Then, taking diagonals in the p through $(m - 1)$ th columns as pivots,

After that, remove the p th row, then move the $p + 1$ through t th rows up a row distance, and put the original p th row as the new t th row, i.e.,

$$H = \begin{pmatrix} \times & \times & \times & \times & \times & \times & \times & \times & \times & \times \\ & \times & \times & \times & \times & \times & \times & \times & \times & \times \\ & & \times & \times & \times & \times & \times & \times & \times & \times \\ & & & \times & \times & \times & \times & \times & \times & \times \\ & & & & \times & \times & \times & \times & \times & \times \\ & & & & & \times & \times & \times & \times & \times \\ & & & & & & \times & \times & \times & \times \\ & & & & & & & \times & \times & \times \\ & & & & & & & & \times & \times \\ & & & & & & & & & \times \end{pmatrix} \begin{matrix} p \\ t \end{matrix}$$

Finally, eliminating nonzeros in the p through $t - 1$ th columns of the t th row leads to the wanted upper triangular matrix.

At present, Forrest–Tomlin’s approach and its variants, such as Reid tactics, are commonly used for updating LU factors in practice. Suhl and Suhl (1993) give very detailed considerations on the implementation.

4.6 Crash Procedure for Initial Basis

The simplex method and its variants require an initial basis to get them started. There is no doubt, quality of the initial basis greatly affects the required number of iterations and storage.

There are several so-called *crash* procedures for determination of initial basis, for which being sparser, as well as tentatively reducing subsequent iterations required.

Let us bring up the crash procedure of MINOS, designed for solving general linear and nonlinear programming problems. Once the original data is entered, it introduces a so-called *logical variable* to each equality of the constraint system to convert the problem into a bounded-variable one with a zero right-hand side. As a result, the coefficient matrix of logical variables forms a unit matrix, which is taken by a crash option of MINOS as an initial basis.

Considering that an initial basis will be LU-factorized, some crash options of MINOS select basic columns to form a nearly lower triangular matrix to limit the number of fill-ins. A standard option of crash includes 5 stages. At the first stage, all free logical variables enter the basis, the number of which is usually less than m . At the second stage, other free variables (or those with very wide bounds) enter. At the third stage, columns having only a single nonzero in the active submatrix enter. At

the fourth stage, columns having two nonzeros do. If the number of selected basic columns is still less than m , some logical variables are taken to form the end basis, at the fourth stage.

It is “ideal” that the generated initial basis, say B , happens to be optimal, and what to do is to solve system $Bx_B = b$ only. This is of course impossible in general. But it would be possible to be close to optimal to some extent. In this respect, it seems to be favorable to determine basis based on the *most-obtuse-angle heuristics*: according to Proposition 2.7.1, lower pivoting-indexed inequalities tend to be satisfied as equalities at an optimal solution, that is, the associated slack variables should be taken as nonbasic variables or larger pivoting-indexed variables taken as basic ones. Let us illustrate by the instance below.

Example 4.6.1 Determine an initial basis by the most-obtuse-angle heuristics:

$$\begin{aligned} \min f &= 3x_1 - 2x_2 + x_3, \\ \text{s.t. } x_1 + 2x_2 - x_3 - x_4 &= 1, \\ x_1 + x_2 - x_3 + x_5 &= 3, \\ -x_1 + x_2 + x_3 + x_6 &= 2, \\ x_j &\geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Answer The problem can be converted into

$$\begin{aligned} \min f &= 3x_1 - 2x_2 + x_3, \\ \text{s.t. } x_4 &= -1 + x_1 + 2x_2 - x_3 \geq 0, \\ x_5 &= 3 - x_1 - x_2 + x_3 \geq 0, \\ x_6 &= 2 + x_1 - x_2 - x_3 \geq 0, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

The following table lists pivoting-indices α_i for variables:

Variable	Constraints	α_j
x_1	$x_1 \geq 0$	-3.00
x_6	$x_1 - x_2 - x_3 \geq -2$	-2.31
x_3	$x_3 \geq 0$	-1.00
x_5	$-x_1 - x_2 + x_3 \geq -3$	0.00
x_4	$x_1 + 2x_2 - x_3 \geq 1$	1.63
x_2	$x_2 \geq 0$	2.00

The first 3 pivoting-indices are the smallest, corresponding to constraints whose gradients form the most-obtuse angles with the negative objective gradient.

Therefore, correspondingly x_1, x_3, x_6 are taken as nonbasic variables. This leads to

$$B = \{2, 4, 5\}, \quad N = \{1, 3, 6\}. \quad B^{-1} = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & 0 & 1 \\ -1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}.$$

Therefore, if one solves $Bx_B = b$, the basic solution follows, that is,

$$\bar{x}_B = B^{-1}b = \begin{pmatrix} 0 & 0 & 1 \\ -1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \geq 0, \quad \bar{x}_N = (0, 0, 0)^T,$$

the optimality of which can be verified by calculating the associated dual solution, i.e.,

$$\bar{y} = B^{-T}c_B = \begin{pmatrix} 0 & 0 & 1 \\ -1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}^T \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix},$$

$$\bar{z}_N = c_N - N^T \bar{y} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{pmatrix}^T \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \geq 0.$$

Therefore, there is no need for any iteration.

It is interesting that for small textbook instances, an optimal basis can often be obtained in this way; even if it is not the case, only a few iterations are required subsequently to achieve optimality. It is noted however that the approach has little effect if differences between pivoting-indices are insignificant. For large-scale LP problems, there is a serious conflict between basis' sparsity and the most-obtuse-angle heuristics that are not easy to balance. Interested readers are referred to the related literature, e.g., (Li 2004; Hu and Pan 2006, 2008).

4.7 Harris Rule and Tolerance Expending

For the success of the simplex algorithm, controlling the size of the pivot is not only important but also cumbersome. A code based on the standard row rule (see step 6 of Algorithm 3.3.1) is not practical. It can solve some very small textbook problems because such a pivot could be too close to zero to ensure numerical stability.

It is necessary to impose restriction on pivot size. At least, the parameter $tolpiv$ should be such that

$$\epsilon_1 \approx 10^{-11} < tolpiv < 10^{-7} \quad (4.26)$$

(see Sect. 4.1). It is still unsatisfactory though. In fact, $tolpiv = 10^{-11}$ is too small for many problems, while $tolpiv = 10^{-7}$ is often too large. It is difficult to find a $tolpiv$ value that is suitable for all LP problems. In this aspect, Harris (1973) rule with $tolpiv$ satisfying (4.26) seems to be feasible overall.

She designed a so-called *two-pass procedure*, which passes m rows twice in each iteration. In the first pass, a prospective stepsize α_1 is determined by allowing the bounds violated by a small tolerance δ ($0 < \delta \ll 1$), which is equal to the current feasibility tolerance (this will be clear a little later). In the second pass, a pivot row is selected from rows, corresponding to real stepsizes no more than α_1 . The rule can be formulated as follows.

Rule 4.7.1 (Harris Two-Pass Row Rule) *First, determine*

$$\alpha_1 = \min\{(\bar{b}_i + \delta)/\bar{a}_{iq} \mid \bar{a}_{iq} > tolpiv; i = 1, \dots, m\}, \quad (4.27)$$

and then select pivot row index

$$p \in \arg \max\{\bar{a}_{iq} \mid \bar{b}_i/\bar{a}_{iq} \leq \alpha_1, \bar{a}_{iq} > tolpiv; i = 1, \dots, m\}. \quad (4.28)$$

The preceding rule widens the scope of pivot row candidates so that a pivot with a larger size is chosen to improve numerical stability. In addition, geometrically the gradient of the nonnegative constraint $x_{jp} \geq 0$ and the descent edge indexed by q form a larger angle, which is conformable with the heuristic characteristic of the optimal solution. Harris' two-pass rule performs well in practice and is at present an indispensable part of simplex LP codes. However, the stability difficulty still occurs occasionally.

On the other hand, the conceptual stepsize $\alpha = \bar{b}_p/\bar{a}_{pq}$ seems to be another source of troubles. Even if pivot $\bar{a}_{pq}$ has a normal size, there is still a somewhat sticky problem when $\bar{b}_p$ is slightly less than zero, leading to a little increase of objective value. Essentially, this is an effect of degeneracy or near degeneracy. To guarantee a positive stepsize, and hence a strict decrease in objective value, Gill et al. (1989) suggest a tactic termed *tolerance expanding*.

In each iteration, the current feasible tolerance δ is increased by a tiny amount, say τ , i.e.,

$$\delta = \delta + \tau.$$

Assuming that upper bound *featol* of δ is set to 10^{-6} (see 4.1), and the initial value is set to

$$\delta_0 = 0.5 \times 10^{-6},$$

then the increment

$$\tau = \frac{(0.99 - 0.5) \times 10^{-6}}{10000} = 0.49 \times 10^{-10}$$

seems to be acceptable. A positive stepsize is then set by

$$\alpha = \max\{\bar{b}_p/\bar{a}_{pq}, \tau/\bar{a}_{pq}\}. \quad (4.29)$$

After every 10000 of iterations, $\delta = \delta_0$ is reset, the basis matrix is refactorized, and the associated solution is recalculated. Of course, a phase-1 procedure will be carried out again if the solution becomes infeasible. After optimality is achieved, in addition, the solution process has to be restarted by setting $\delta = \delta_0$ to ensure the quality of the final optimal solution.

4.8 Pricing for Reduced Cost

All pivot rules are closely related to the reduced costs. Availability of all or a part of the reduced costs is a prerequisite for the determination of the pivot. This section is devoted to approaches to calculating them or pricing.

We might as well assume that the current basis is $B = \{1, \dots, m\}$, and nonbasis is $N = \{m + 1, \dots, n\}$. The classical pricing method is based on the following formulas:

$$B^T y = c_B, \quad z_N = c_N - N^T y. \quad (4.30)$$

We will consider several alternative schemes, some of which are being used in practice widely.

Assume that a column index $q \in N$ and row index $p \in B$ have been determined. Let the *prime* mark the quantity associated with new basis, compared with those associated with old basis, e.g.,

$$B' = (B \setminus p) \cup q, \quad N' = (N \setminus q) \cup p. \quad (4.31)$$

Then, the following relation holds:

$$B' = B + (a_q - a_p)e_p^T, \quad (4.32)$$

and y' satisfies

$$B'^T y' = c_{B'}. \quad (4.33)$$

If h denotes the solution to the following system:

$$B^T h = e_p, \quad (4.34)$$

then h' satisfies

$$B'^T h' = e_p. \quad (4.35)$$

Thereby, the following recurrence formulas can be shown:

$$y' = y + z_q h', \quad (4.36)$$

$$h = \bar{a}_{p,q} h', \quad (4.37)$$

$$z'_{N'} = z_{N'} - z_q N'^T h', \quad (4.38)$$

$$= z_{N'} - (z_q / \bar{a}_{p,q}) N'^T h, \quad (4.39)$$

where $\bar{a}_{p,q}$ is the pivot, i.e., the p th component of the solution to $B\bar{a}_q = a_q$, which can be calculated independently of pricing. Note that $z_p = 0$ in the preceding formulas.

As an alternative scheme, the reduced costs can be calculated based on (4.38) or (4.39). Zoutendijk (1960) uses formulas (4.38) and (4.35). Bixby (1994) uses (4.39) and (4.34). An advantage of the latter is that solution h/h' to (4.34)/(4.35) is usually much sparser than y and especially suitable with the steepest-edge rule (Part II), which also needs to solve (4.34)/(4.35).

Nevertheless, recurrence is not suitable for partial pricing. In contrast to the preceding schemes, instead, Tomlin uses (4.36) and (4.35) to compute the simplex multipliers in recurrence manner, which is amenable to partial pricing.

As in the classical pricing, the preceding schemes need to solve two triangular systems. Assume that LU factorization with row and column exchanges of B is

$$PBQ = LU, \quad (4.40)$$

where L is the unit lower triangular, U is the nonsingular upper triangular, and P and Q are permutations. Recall that each system involved in simplex iterations can be converted into two triangular systems. For $B\bar{a}_q = a_q$, e.g., one may solve

$$Lv = Pa_q, \quad Uu = v, \quad (4.41)$$

and set $\bar{a}_q = Qu$. It is similar to solve (4.34)/(4.35).

Hu and Pan (2008) modified Tomlin's scheme so that only a single triangular system needs to be solved for pricing. It is found later that Goldfarb (1977) proposed the same approach along another line earlier without offering any numerical results.

In the following, we will focus on this scheme and report encouraging results, obtained in extensive computational experiments. To this end, first, we derive the Bartels–Golub update in a way, slightly different from that in Sect. 4.5 (Forrest–Tomlin's variant).

From (4.32) and (4.40), it follows that

$$\begin{aligned} PB'Q &= LU + P(a_q - a_p)e_p^T Q, \\ &= L(U + (L^{-1}Pa_q - L^{-1}Pa_p)e_p^T Q). \end{aligned}$$

By (4.40) and $Be_p = a_p$, it holds that

$$Pa_p = PBe_p = LUQ^Te_p.$$

Besides, it is clear that there is an integer $1 \leq r \leq m$ such that

$$e_r = Q^Te_p. \quad (4.42)$$

Combining the preceding 3 expressions leads to

$$PB'Q = L(U + (L^{-1}Pa_q - UQ^Te_p)e_p^T Q) = LR, \quad (4.43)$$

where v satisfies the first triangular system of (4.41), and the matrix

$$R = (U + (v - Ue_r)e_r^T$$

is upper triangular, except for the r th column, which results from U by replacing the r th column by v . Move the r th column backward to the end position and the $(r + 1)$ through m th columns forward one column. Then put the r th row to the bottom and the $(r + 1)$ to m th rows up one row. If $\hat{Q}$ denotes the associated permutation, then the resulting matrix $\hat{Q}^TR\hat{Q}$ is upper triangular, except for entries in the r through $m - 1$ th columns in the m th row perhaps being nonzero ($m \geq 2$). These entries can be eliminated by a series of Gauss transformations; while proceeding, exchange the m th row and the row at which the diagonal locates whenever a diagonal is too small in magnitude, compared with the corresponding nonzero entry. Thus, there are permutations P_i and lower triangular matrices L_i , $i = 1, \dots, s$ ($1 \leq s \leq m - r$) such that

$$L_s^{-1}P_s \cdots L_1^{-1}P_1\hat{Q}^TR\hat{Q} \triangleq U' \quad (4.44)$$

is nonsingular upper triangular. On the other hand, it is clear that

$$L' = L \hat{Q} P_1^T L_1 \cdots P_s^T L_s \quad (4.45)$$

is lower triangular with row exchanges. From (4.44), (4.45), and (4.43), it is verified that the new basis B' has the following LU factorization with row and column exchanges:

$$P' B' Q' = L' U', \quad (4.46)$$

where

$$P' = P, \quad Q' = Q \hat{Q}. \quad (4.47)$$

This process simplifies computation of simplex multipliers.

Theorem 4.8.1 *Let y be a solution to $B^T y = c_B$ and z_q be reduced cost related to column index q . If w' is the solution to*

$$L'^T P w' = (1/u'_{mm}) e_m, \quad (4.48)$$

where u'_{mm} is the m th diagonal of U' , then $y' = y + z_q w'$ is the solution to (4.33).

Proof Based on relations between quantities, related to the old and new bases, it is only required to show that w' is the solution to (4.35).

Premultiplying (4.48) by U'^T gives

$$U'^T L'^T P w' = (1/u'_{mm}) U'^T e_m,$$

which, combined with (4.46), (4.47), and $U'^T e_m = u'_{mm} e_m$, leads to

$$\hat{Q}^T Q^T B'^T w' = e_m.$$

Then by (4.42) and the definition of permutation $\hat{Q}$, it is known that

$$e_p^T Q \hat{Q} = e_r^T \hat{Q} = e_m^T.$$

Finally, it follows from the preceding two expressions that

$$B'^T w' = Q \hat{Q} e_m = Q \hat{Q} \hat{Q}^T Q^T e_p^T = e_p.$$

□

This theorem says that after new LU factors are obtained by Forrest–Tomlin's update, one can solve the triangular system (4.48) for w' , calculate the new simplex multipliers y' , and then calculate the new reduced costs by (4.38).

Table 4.1 Iteration and time ratios of MINOS 5.51 to NEW

Problem	Small (38)	Medium (41)	Large (17)	Ken. (16)	BPMPD (17)	Total average (129)
Iterations	1.00	1.01	1.12	1.05	1.12	1.10
Time	1.03	1.17	1.28	1.28	1.23	1.24

Our numerical experiments involve the following two codes (Hu and Pan 2008):

1. MINOS 5.51.
2. NEW: MINOS 5.51 with the presented approach to pricing.

The preceding codes both adopted default sectional pricing options (partial price 10).

There were 96 Netlib, 16 Kennington, and 17 BPMPD test problems involved (Appendix C: Tables C.1–C.3). All the Netlib problems fall into three sets: Small (38), Medium (41), and Large (17) (in the order of increasing $m + n$). Table 4.1 lists the total iteration and time ratios of MINOS 5.51 to NEW.

From the preceding table, it is seen that the presented pricing scheme outperformed the standard scheme with each set of problems (except for small Netlib problems), in terms of either iteration or running time. The overall iteration and average time ratios attain 1.10 and 1.24, respectively.

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Chapter 5

Duality Principle and Dual Simplex Method



Duality is an essential part of LP theory. It is related to the special relationship between one LP problem and another, both of which involve the same data (A, b, c) , but positioned differently (except for the self-duality, see below). The former is referred to as *primal problem*, and the latter as *dual problem*. Based on duality, once a primal problem is solved, so is the dual problem, and vice versa. This chapter is devoted to such topics as the relationship between their feasible regions as well as optimal solutions, optimality condition, dual simplex algorithm in tableau form and in revised form, shadow price, and dual elimination. Before making notes on the duality in the final section, we introduce a special bilevel LP problem together with a related approach to intercepting the optimal set.

5.1 Dual LP Problem

Now call the standard LP problem (1.8), i.e.,

$$(P) \quad \begin{aligned} \min f &= c^T x \\ \text{s.t. } Ax &= b, \quad x \geq 0 \end{aligned} \quad (5.1)$$

primal problem, and the following problem

$$(D) \quad \begin{aligned} \max g &= b^T y \\ \text{s.t. } A^T y + z &= c, \quad z \geq 0, \end{aligned} \quad (5.2)$$

dual problem. There is 1-1 correspondence between their variables/constraints.

(y, z) satisfying $A^T y + z = c$ is called *dual feasible solution*. The set

$$D = \{(y, z) \in \mathcal{R}^m \times \mathcal{R}^n \mid A^T y + z = c, z \geq 0\}$$

is called *dual feasible region*, which includes all dual feasible solutions.

Given basis B , setting $z_B = 0$ in $B^T y + z_B = c_B$ gives

$$\bar{y} = B^{-T} c_B, \quad \bar{z}_B = 0, \quad \bar{z}_N = c_N - N^T \bar{y}, \quad (5.3)$$

called *dual basic solution*. $\bar{z}$ is just the reduced costs; and $\bar{y}$ the simplex multiplier. If $\bar{z}_N \geq 0$, $(\bar{y}, \bar{z})$ is a dual feasible basic solution, corresponding to a vertex in D . For simplicity, thereafter $\bar{z}_N$ alone is often said to be dual basic solution. In particular, $(\bar{y} = 0, \bar{z} = c)$ is a dual feasible solution if $c \geq 0$.

The following equivalent form of dual problem (5.2)

$$(D) \quad \begin{aligned} \max \quad & g = b^T y, \\ \text{s.t.} \quad & A^T y \leq c \end{aligned} \quad (5.4)$$

is useful. Problems (5.2) and (5.4) will be regarded as the same.

As it can be converted into a standard one, any LP problem corresponds to a dual problem. By introducing slack variables $u \geq 0$, e.g., the problem

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b, \quad x \geq 0 \end{aligned} \quad (5.5)$$

can be turned to the standard problem

$$\begin{aligned} \min \quad & -c^T x \\ \text{s.t.} \quad & Ax + u = b, \quad x, u \geq 0 \end{aligned}$$

and the dual problem of which is

$$\begin{aligned} \max \quad & b^T y' \\ \text{s.t.} \quad & \begin{pmatrix} A^T \\ I \end{pmatrix} y' \leq \begin{pmatrix} -c \\ 0 \end{pmatrix}. \end{aligned}$$

Then, setting $y = -y'$ turns the preceding to the dual problem of (5.5), i.e.,

$$\begin{aligned} \min \quad & b^T y \\ \text{s.t.} \quad & A^T y \geq c, \quad y \geq 0. \end{aligned}$$

The correspondence between the primal and dual problems is summarized into the table below:

Primal problem		Dual problem	
Objective function	min	Objective function	max
Variables	Nonnegative Nonpositive Free	Constraints	$\leq$ $\geq$ $=$
Constraints	$\geq$ $\leq$ $=$	Variables	Nonnegative Nonpositive Free

Note In applications of the preceding table, sign restriction is not handled as a constraint but attributed to the associated variable.

For example, the so-called bounded-variable LP problem

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b, \quad l \leq x \leq u : \end{aligned} \quad (5.6)$$

can be transformed into

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b, \\ & x + s = u, \\ & -x + t = -l, \\ & s, t \geq 0. \end{aligned} \quad (5.7)$$

According to the preceding table, the dual problem of (5.6) is

$$\begin{aligned} \max \quad & b^T y - u^T v + l^T w \\ \text{s.t.} \quad & A^T y - v + w = c, \quad v, w \geq 0. \end{aligned} \quad (5.8)$$

The so-called *self-duality* refers to a special case when the dual problem of an LP problem is just itself. Combining primal problem (5.1) and dual problem (5.4), we construct the following problem:

$$\begin{aligned} \min \quad & c^T x - b^T y \\ \text{s.t.} \quad & Ax = b, \quad x \geq 0, \\ & A^T y \leq c. \end{aligned} \quad (5.9)$$

According to the preceding table, the dual problem of it is

$$\begin{aligned} \max \quad & b^T y + c^T v \\ \text{s.t.} \quad & A^T y \leq c, \\ & Av = -b, \quad v \leq 0. \end{aligned}$$

By setting $v = -x$ and handling the objective function properly, the preceding can be transformed into the original problem (5.9). Therefore, (5.9) is a self-dual problem.

5.2 Duality Theorem

This section only focuses on the nonsymmetric duality of (P) and (D), as obtained results are valid for the general case.

Theorem 5.2.1 (Symmetry) *The dual problem of the dual problem is the primal problem.*

Proof Introduce the slack variable vector $u \geq 0$ to the dual problem (D), and make the variable transformation $y = y_1 - y_2$ to convert it into

$$\begin{aligned} \max \quad & b^T (y_1 - y_2) \\ \text{s.t.} \quad & A^T (y_1 - y_2) + u = c, \quad y_1, y_2; u \geq 0 \end{aligned}$$

or equivalently,

$$\begin{aligned} \min \quad & (-b^T, b^T, 0)(y_1^T, y_2^T, u^T)^T \\ \text{s.t.} \quad & (A^T \mid -A^T \mid I)(y_1^T, y_2^T, u^T)^T = c, \quad y_1, y_2, u \geq 0. \end{aligned}$$

The dual problem of the preceding is

$$\begin{aligned} \max \quad & c^T x' \\ \text{s.t.} \quad & \begin{pmatrix} A \\ -A \\ I \end{pmatrix} x' \leq \begin{pmatrix} -b \\ b \\ 0 \end{pmatrix}, \end{aligned}$$

that is,

$$\begin{aligned} \max \quad & c^T x' \\ \text{s.t.} \quad & Ax' = -b, \quad x' \leq 0, \end{aligned}$$

which becomes (P) by setting $x' = -x$. □

The preceding says that any of the primal and dual problems is the dual problem of the other. So, the two are symmetric in position. This is why any fact, holding for one of them, has its counterpart for the other. It is important that there is a special relationship between feasible/optimal solutions to the two problems.

Theorem 5.2.2 (Weak Duality) *If x and y are feasible solutions to primal and dual problems, respectively, then $c^T x \geq b^T y$.*

Proof Premultiplying $c \geq A^T y$ by $x \geq 0$ gives $c^T x \geq y^T A x$, substituting $b = A x$ to which leads to $c^T x \geq b^T y$. $\square$

According to the preceding, if there are feasible solutions to both primal and dual problems, any feasible value of the former is an upper bound of all feasible values of the latter; on the other hand, any feasible value of the latter is a lower bound of all feasible values of the former.

Corollary 5.2.3 *If any of the primal and dual problems are unbounded, there exists no feasible solution to the other.*

Proof By contradiction. Assume that there is a feasible solution to the dual problem. Then it follows from Theorem 5.2.2 that feasible values of the primal problem are bounded below. Analogously, if the primal problem is feasible, the dual problem is bounded above. Therefore, the claim is valid. $\square$

Corollary 5.2.4 *Let $\bar{x}$ and $\bar{y}$ be primal and dual feasible solutions, respectively. If $c^T \bar{x} = b^T \bar{y}$ holds, they are optimal solutions.*

Proof According to Theorem 5.2.2, for any primal feasible solution x , it holds that $c^T x \geq b^T \bar{y} = c^T \bar{x}$; therefore, $\bar{x}$ is a primal optimal solution. Similarly, $\bar{y}$ is a dual optimal solution. $\square$

Theorem 5.2.5 [Strong duality] *If there exists an optimal solution to any of the primal and dual problems, then there exists an optimal one to the other, and the associated optimal values are equal.*

Proof Assume that there is an optimal solution to the primal problem. According to Theorem 2.5.5, there is a basic optimal solution. Let B and N be optimal basis and nonbasis, respectively. Then

$$c_N^T - c_B^T B^{-1} N \geq 0, \quad B^{-1} b \geq 0.$$

Thus, setting

$$\bar{y} = B^{-T} c_B \tag{5.10}$$

leads to

$$A^T \bar{y} - c = \begin{pmatrix} B^T \\ N^T \end{pmatrix} \bar{y} - \begin{pmatrix} c_B \\ c_N \end{pmatrix} \leq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Therefore $\bar{y}$ is a feasible solution to the dual problem. By (5.10), on the other hand, the feasible basic solution $(\bar{x}_B = B^{-1}b, \bar{x}_N = 0)$ satisfies

$$c^T \bar{x} = c_B^T \bar{x}_B = c_B^T B^{-1}b = b^T \bar{y}.$$

By Corollary 5.2.4, therefore, $\bar{x}$ and $\bar{y}$ are the respective optimal solutions to the primal and dual problems with the same optimal value. Moreover, it is known by Theorem 5.2.1 that if there is an optimal solution to the dual problem, so is to the primal problem, with the same optimal value. $\square$

Based on the strong duality, thereafter the primal and dual optimal values will not be distinguished.

It is clear that if there is an optimal solution to one of the pairs of (5.1) and (5.4), so is the self-dual problem (5.9). Moreover, the optimal value of the latter is equal to zero, and the optimal solution of the latter gives the primal and dual optimal solutions to the pair. A variation of it will be used to derive the so-called *homogeneous and self-dual interior-point method* (10.4.4).

In case when any of the primal and dual problems are infeasible, it can be asserted that the other problem is infeasible or unbounded. The computation would be finished then. In some applications, however, it would be needed to distinguish between the two cases. This can be resolved via the duality as follows.

Assume now that the primal problem (5.1) is infeasible. To determine whether the dual problem (5.2) is infeasible or unbounded, consider

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = 0, \quad x \geq 0, \end{aligned} \tag{5.11}$$

which has a feasible solution $x = 0$. Solve the preceding problem by the simplex algorithm. If (5.11) is unbounded, then the problem

$$\begin{aligned} \min \quad & 0 \\ \text{s.t.} \quad & A^T y + z = c, \quad z \geq 0 \end{aligned}$$

is infeasible (Corollary 5.2.3). Therefore, (5.2) is infeasible either. If an optimal solution to (5.11) is reached, then there exists an optimal solution to the preceding problem (Theorem 5.2.5), which indicates that (5.2) is feasible; therefore, it can be further asserted that (5.2) is unbounded.

Using the duality, now we are able to prove Farkas Lemma 2.0.3 concisely.

Lemma 5.2.6 (Farkas) Assume $A \in \mathcal{R}^{m \times n}$ and $b \in \mathcal{R}^m$. The feasible region P is nonempty if and only if

$$b^T y \geq 0, \quad \forall y \in \{y \in \mathcal{R}^m \mid A^T y \geq 0\}.$$

Proof Consider the following LP problem

$$\begin{aligned} \min & 0 \\ \text{s.t. } & Ax = b, \quad x \geq 0, \end{aligned} \quad (5.12)$$

the dual problem of which is

$$\begin{aligned} \max & b^T y' \\ \text{s.t. } & A^T y' \leq 0. \end{aligned} \quad (5.13)$$

Note that $y' = 0$ is a feasible solution to it, with the feasible value 0.

Necessity. Assume that the feasible region P of (5.12) is nonempty; hence, all feasible solutions correspond to the same objective value 0. According to Theorem 5.2.2, for any $y' \in \{y' \in \mathcal{R}^m \mid A^T y' \leq 0\}$, it holds that $b^T y' \leq 0$. By setting $y = -y'$, it is known that $b^T y \geq 0$ holds for $y \in \{y \in \mathcal{R}^m \mid A^T y \geq 0\}$.

Sufficiency. Assume that for any $y \in \{y \in \mathcal{R}^m \mid A^T y \geq 0\}$ it holds that $b^T y \geq 0$. Then, for $y' \in \{y' \in \mathcal{R}^m \mid A^T y' \leq 0\}$, we have $b^T y' \leq 0$; hence, there is an optimal solution to (5.13). According to Theorem 5.2.5, therefore, there is an optimal solution to (5.12), which implies P being nonempty. $\square$

5.3 Optimality Condition

From duality theorems presented in the previous section, it is possible to derive a set of conditions for primal and dual solutions to be optimal, which serve as a theoretical basis for LP. We consider the standard LP problem first and more general LP problems then.

Assume that x and (y, z) are the primal and dual (not necessarily feasible) solutions, respectively.

Definition 5.3.1 The difference $c^T x - b^T y$ between the primal and dual objective values is the *duality gap* between x and (y, z) .

Definition 5.3.2 If $x^T z = 0$, x and (y, z) are *complementary*; if $x + z > 0$, in addition, the two are strictly complementary.

The quantity $x^T z$ is termed *complementarity residual*.

Lemma 5.3.1 The duality gap and complementarity residual of x and (y, z) are equal; x and (y, z) are complementary if and only if their duality gap equals zero.

Proof Since x and (y, z) satisfy the equality constraints, it is verified that

$$c^T x - b^T y = x^T c - (Ax)^T y = x^T (c - A^T y) = x^T z.$$

$\square$

A “ $\geq$ ” or “ $\leq$ ” type of inequality is said to be tightly satisfied if the “ $=$ ” holds; it is said to be slackly satisfied if the “ $>$ ” or “ $<$ ” holds.

In case when the nonnegative constraints are satisfied, the complementarity of x and (y, z) is equivalent to the satisfaction of

$$x_j z_j = 0, \quad \forall j = 1, \dots, n. \quad (5.14)$$

It is clear that for $j = 1, \dots, n$, it holds that $x_j = 0/z_j = 0$ if $z_j > 0/x_j > 0$. If a component of x/z slackly satisfies the associated nonnegativity constraint, therefore, the corresponding component of z/x must satisfy the associated nonnegativity constraint tightly.

Theorem 5.3.2 (Optimality Condition for the Standard LP Problem) *x is an optimal solution of the standard LP problem if and only if there exist y, z such that:*

$$\begin{aligned} (i) \quad & Ax = b, \quad x \geq 0, \text{ (primal feasible)} \\ (ii) \quad & A^T y + z = c, \quad z \geq 0, \text{ (dual feasible)} \\ (iii) \quad & x^T z = 0. \quad \text{(complementary)} \end{aligned} \quad (5.15)$$

Proof Note that for x and (y, z) , zero duality gap is equivalent to complementarity (Lemma 5.3.1).

Sufficiency. By Corollary 5.2.4 and the equivalence of zero duality gap and complementarity, it follows from (5.15) that x and (y, z) are primal and dual optimal solutions, respectively.

Necessity. If x is a primal optimal solution, it satisfies condition (i). By Theorem 5.2.5, in addition, there is a dual optimal solution (y, z) such that the duality gap is zero; hence, conditions (ii) and (iii) are satisfied. $\square$

The following result is stated without proof (Goldman and Tucker 1956b).

Theorem 5.3.3 (Strict Complementarity) *If there exists a pair of primal and dual optimal solutions, then there exists a strictly complementary pair of such solutions.*

The significance of the optimal conditions speaks for itself. As for method research, any type of optimality criteria in various contexts must coincide with these conditions. From them, it is understandable that LP methods always solve the pair of problems at the same time. For instance, once the simplex method reaches the primal optimal solution

$$\bar{x}_B = B^{-1}b, \quad \bar{x}_N = 0,$$

it also gives a dual optimal solution

$$\bar{y} = B^{-T}c_B, \quad \bar{z}_B = 0, \quad \bar{z}_N = c_N - N^T \bar{y}.$$

In view of the symmetry between primal and dual problems, moreover, it is not surprising why LP methods often present in pair: if a method solves the primal problem, there is another method that solves the dual problem, and vice versa. For example, the dual method matches the simplex method.

The interior-point methods presented later often judge the degree of approaching optimality through the complementarity residual: the smaller the residual is, the closer to optimality; when it vanishes, the primal and dual optimal solutions are attained respectively. It is noticeable moreover that direct dealing with the optimal conditions as a system of equalities and inequalities can lead to some interior-point algorithms (10.4). Such algorithms usually generate a strictly complementary pair of optimal solutions in the limit, which is important for asymptotic analysis (Güler and Ye 1993).

For the bounded-variable LP problem (5.6), we have the following result.

Theorem 5.3.4 (Optimal Condition for the Bounded-Variable Problem) x is an optimal solution of the bounded-variable LP problem if and only if there exist y, v, w such that:

$$\begin{aligned} (i) \quad & Ax = b, \quad l \leq x \leq u, \quad (\text{primal feasible}) \\ (ii) \quad & A^T y - v + w = c, \quad v, w, \geq 0, \quad (\text{dual feasible}) \\ (iii) \quad & (x - l)^T w = 0, \quad (u - x)^T v = 0. \quad (\text{complementary}) \end{aligned} \quad (5.16)$$

Proof It follows from (5.6), (5.7), (5.8), and Theorem 5.3.2. $\square$

Consider the following more general minimization problem:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & x \in \Omega, \end{aligned} \quad (5.17)$$

where $\Omega \subset \mathcal{R}^n$ is a convex set.

Lemma 5.3.5 $x^* \in \Omega$ is an optimal solution to (5.17) if and only if it satisfies

$$c^T(x - x^*) \geq 0, \quad \forall \quad x \in \Omega. \quad (5.18)$$

Proof Assume that x^* is an optimal solution. If (5.18) does not hold, i.e., there is a point $\bar{x} \in \Omega$ such that

$$c^T(\bar{x} - x^*) < 0, \quad (5.19)$$

then it holds that

$$c^T \bar{x} < c^T x^*, \quad (5.20)$$

which contradicts that x^* is optimal. Conversely, assume that (5.18) holds. If x^* is not optimal to (5.17), then there exists $\bar{x} \in \Omega$ satisfying (5.20), which implies (5.19), contradicting that (5.18) holds. $\square$

The preceding lemma says that a sufficient and necessary condition for x^* to be an optimal solution is that there exists no feasible descent direction at x^* with respect to the objective function; or geometrically, the angle between any feasible direction and the gradient of the objective is not obtuse. Note that, the lemma is actually valid for an arbitrary set Ω , including the feasible region of the LP problem.

Definition 5.3.3 Let x^* be a feasible solution to an LP problem. Another LP problem generated by dropping all or part of inactive constraints at x^* is said to be a relaxed LP problem.

Theorem 5.3.6 x^* is optimal to relaxed problem if and only if it is optimal to the original problem.

Proof Denote by P the feasible region of the original problem and by P' the feasible region of the relaxed problem. Note that both P and P' are convex sets.

Necessity is clear.

Sufficiency. Assume that x^* is optimal to the original problem, but not optimal to the relaxed problem. According to Theorem 5.3.5, there is solution $x' \in P'$ such that

$$c^T(x' - x^*) < 0. \quad (5.21)$$

It is clear that $x' \notin P$, and there is some $\bar{x} \in (x^*, x')$ that belongs to P . Hence, there is some $0 < \sigma < 1$ such that

$$\bar{x} - x^* = \sigma(x' - x^*), \quad (5.22)$$

from which and (5.21), it follows that

$$c^T(\bar{x} - x^*) = \sigma c^T(x' - x^*) < 0. \quad (5.23)$$

According to Lemma 5.3.5, the preceding implies that x^* is not optimal to the original problem, which is a contradiction, and the claim is therefore valid. $\square$

The preceding says that dropping or adding inactive constraints at an optimal solution does not matter with its optimality. In addition, from its proof and Definition 2.8.2, we claim the following.

Corollary 5.3.7 A vector is a feasible descent direction at x^* to the relaxed problem if and only if so is it to the original problem.

5.4 Dual Simplex Algorithm: Tableau Form

The dual simplex method is very important in optimization, not only because it is an efficient method for solving LP problems, but also because it serves as a basic tool for solving ILP problems. This section derives its tableau version first.

Starting from a feasible simplex tableau, in which the right-hand side is nonnegative, the tableau simplex algorithm turns negative entries of the objective row to nonnegative ones while maintaining non-negativeness of the right-hand side, in each iteration. It proceeds, iteration by iteration until all entries of the objective row become nonnegative, and then declares an optimal simplex tableau obtained.

In contrast, the dual simplex method presented in this section starts from the so-called *dual feasible tableau*, in which entries of the objective row are nonnegative, and proceeds to achieve non-negativeness of the right-hand side, while maintaining non-negativeness of the objective row.

Consider the standard LP problem (5.1). Let (3.6) be a current dual feasible simplex tableau, satisfying $\bar{z}_N \geq 0$ but $\bar{b} \not\geq 0$.

This time, we use determine pivot row first by the following rule.

Rule 5.4.1 (Dual Row Rule) *Select row index by*

$$p \in \arg \min \{\bar{b}_i \mid i = 1, \dots, m\}.$$

Thus, the preceding will drop the basic variable x_{j_p} from basis.

Lemma 5.4.1 *Assume that $\bar{z}_N \geq 0$ and $\bar{b}_p < 0$. If column index set*

$$J = \{j \in N \mid \bar{a}_{pj} < 0\} \quad (5.24)$$

is empty, then the problem is infeasible.

Proof $\bar{z}_N \geq 0$ indicates dual feasibility. Assume that the dual problem is bounded; hence, there is an optimal dual solution. According to Theorem 5.2.5, there is an optimal primal solution. Say $\hat{x} \geq 0$, which satisfies the equality, corresponding to the p th row of the simplex tableau, i.e.,

$$\hat{x}_{j_p} + \sum_{j \in N} \bar{a}_{pj} \hat{x}_j = \bar{b}_p.$$

From (5.24) and $\hat{x} \geq 0$, it follows that the left-side of the preceding is nonnegative, which contradicts $\bar{b}_p < 0$. Therefore, the problem is dual unbounded, and hence infeasible. $\square$

Assume that p has been determined and (星期三, 十一月 10, 2021 21:06 (5.24)) does not hold. Then the following is well-defined.

Rule 5.4.2 (Dual Column Rule) Determine β and column index q such that

$$\beta = -\bar{z}_q / \bar{a}_{pq} = \min_{j \in J} -\bar{z}_j / \bar{a}_{pj} \geq 0. \quad (5.25)$$

The β is a *dual stepsize* to be determined.

Once a pivot is determined, perform elementary transformations to turn $\bar{a}_{pq}$ to 1 and eliminate all other nonzeros in the q -indexed column. Then adding β times of the p th row to the objective row results in a new simplex tableau. It is not difficult to show that the new tableau remains dual feasible, that is, its objective row is still nonnegative.

The objective value of the resulting tableau is then

$$-\hat{f} = -\bar{f} + \beta \bar{b}_p \leq -\bar{f}. \quad (5.26)$$

If $\bar{z}_N$ has zero components, it is called *dual degenerate*. In this case, β could vanish, and hence, the corresponding objective value remains unchanged.

Overall steps can be put in the following algorithm (Lemke 1954; Beale 1954).

Algorithm 5.4.1 (Dual Simplex Algorithm: Tableau Form) *Initial: a dual feasible simplex tableau in form (3.6). This algorithm solves the standard LP problem (1.8).*

1. Select pivot row index $p \in \arg \min \{\bar{b}_i \mid i = 1, \dots, m\}$.
2. Stop if $\bar{b}_p \geq 0$.
3. Stop if $J = \{j \in N \mid \bar{a}_{pj} < 0\} = \emptyset$.
4. Determine pivot column $q \in \arg \min_{j \in J} -\bar{z}_j / \bar{a}_{pj}$.
5. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
6. Go to step 1.

Theorem 5.4.2 *Under the dual nondegeneracy, Algorithm 5.4.1 terminates either at:*

- (i) Step 2, achieving a pair of primal and dual optimal solutions; or at
- (ii) Step 3, detecting infeasibility of the problem

Proof The proof on the termination is similar to the simplex algorithm. The meanings of its exits come from Lemmas 3.1.1 and 5.4.1 and discussions preceding the algorithm. $\square$

If dual degeneracy presents, the algorithm could stall at the iteration for a time in the solution process, even fail to solve a problem due to cycling (Beale 1955). Nevertheless, despite the dual degeneracy always occurs, the dual simplex method performs satisfactorily in practice, as is just in the primal simplex context.

Algorithm 5.4.1 starts from a dual feasible simplex tableau. In general, there is a need for a dual Phase-I procedure for this purpose (see Sect. 6.1).

Example 5.4.1 Solve the following LP problem by Algorithm 5.4.1:

$$\begin{aligned}
 \min f &= x_1 + 2x_2 + x_3 \\
 \text{s.t. } 2x_1 + x_2 + x_3 - x_4 &= 1 \\
 -x_1 + 4x_2 + x_3 &\geq 2 \\
 x_1 + 3x_2 &\leq 4 \\
 x_j &\geq 0, \quad j = 1, \dots, 4.
 \end{aligned}$$

Answer Initial: turn the problem to the standard form by introducing slack variables $x_5, x_6 \geq 0$ in constraints. Then, premultiply the first two constraints by -1 , respectively:

$$\begin{aligned}
 \min f &= x_1 + 2x_2 + x_3 \\
 \text{s.t. } -2x_1 - x_2 - x_3 + x_4 &= -1 \\
 x_1 - 4x_2 - x_3 + x_5 &= -2 \\
 x_1 + 3x_2 + x_6 &= 4 \\
 x_j &\geq 0, \quad j = 1, \dots, 6,
 \end{aligned}$$

which corresponds to an available dual feasible simplex tableau, i.e.,

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-2	-1	-1	1			-1
1	-4*	-1		1		-2
1	3				1	4
1	2	1				

Iteration 1:

1. $\min\{-1, -2, 4\} = -2 < 0$, $p = 2$.
3. $J = \{2, 3\}$.
4. $\min\{-2/(-4), -1/(-1)\} = 1/2$, $q = 2$.
5. Multiply row 2 by $-1/4$, and then add 1, -3 , -2 times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-9/4*		-3/4	1	-1/4		-1/2
-1/4	1	1/4		-1/4		1/2
7/4		-3/4		3/4	1	5/2
3/2		1/2		1/2		-1

Iteration 2:

1. $\min\{-1/2, 1/2, 5/2\} = -1/2 < 0$, $p = 1$.
3. $J = \{1, 3, 5\}$.
4. $\min\{-(3/2)/(-9/4), -(1/2)/(-3/4), -(1/2)/(-1/4)\} = 2/3$, $q = 1$.
5. Multiply row 1 by $-4/9$, and then add $1/4$; $-7/4$; $-3/2$ times of row 1 to rows 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		1/3	-4/9	1/9		2/9
	1	1/3	-1/9	-2/9		5/9
		-4/3	7/9	5/9	1	19/9
			2/3	1/3		-4/3

The right-hand side of the preceding is now nonnegative, hence obtained an optimal simplex tableau. The optimal solution and associated objective value are

$$x^* = (2/9, 5/9, 0, 0)^T, \quad f^* = 4/3.$$

5.5 Dual Simplex Algorithm

In this section, we first derive the revised version of Algorithm 5.4.1 based on the equivalence between the simplex tableau and the revised simplex tableau, just as what we have done for deriving the simplex method from its tableau version. Then, we derive it alternatively to reveal the fact that it essentially solves the dual problem.

Like the simplex algorithm 3.3.2, in each iteration, the dual simplex algorithm computes the objective row, the right-hand side, the pivot column, and row. The objective row and/or the right-hand side can be computed in a recurrence manner (see (3.14) and (3.15)). The pivot column and row can be computed through B^{-1} and the original data, just as in the simplex method.

If nonbasic entries in the pivot row are all nonnegative, i.e.,

$$\sigma_N^T \triangleq e_p^T \bar{N} = e_p^T B^{-1} N \geq 0,$$

the dual problem is unbounded; hence, the original problem is infeasible. B^{-1} will be updated in the same way as in the simplex method.

Based on table 3.6.1, therefore, Algorithm 5.4.1 can be revised to the following algorithm.

Algorithm 5.5.1 (Dual Simplex Algorithm) *Initial:* (B, N) , B^{-1} , $\bar{z}_N \geq 0$, $\bar{x}_B = B^{-1}b$, and $\bar{f} = c_B^T \bar{x}_B$. This algorithm solves the standard LP problem (1.8):

1. Select row index $p \in \arg \min\{\bar{x}_{j_i} \mid i = 1, \dots, m\}$.
2. Stop if $\bar{x}_{j_p} \geq 0$ (optimality achieved).
3. Compute $\sigma_N = N^T B^{-T} e_p$.
4. Stop if $J = \{j \mid \sigma_j < 0, j \in N\} = \emptyset$ (infeasible problem).
5. Determine β and column index q such that $\beta = -\bar{z}_q / \sigma_q = \min_{j \in J} -\bar{z}_j / \sigma_j$.
6. Set $\bar{z}_{j_p} = \beta$, and update $\bar{z}_N = \bar{z}_N + \beta \sigma_N$, $\bar{f} = \bar{f} - \beta \bar{x}_{j_p}$ if $\beta \neq 0$.
7. Compute $\bar{a}_q = B^{-1} a_q$.
8. Update by $\bar{x}_B = \bar{x}_B - \alpha \bar{a}_q$, $\bar{x}_q = \alpha$, where $\alpha = \bar{x}_{j_p} / \sigma_q$.
9. Update B^{-1} by (3.21).
10. Update (B, N) by exchanging j_p and q .
11. Go to step 1.

Alternatively, Algorithm 5.5.1 can be derived by solving the dual problem itself as follows.

Consider the dual problem

$$(D) \quad \begin{aligned} \max \quad & g = b^T y \\ \text{s.t.} \quad & A^T y \leq c. \end{aligned}$$

Given (B, N) , B^{-1} . It is easy to verify that $\bar{y} = B^{-T} c_B$ satisfies the dual constraints, i.e.,

$$\bar{z}_B = c_B - B^T \bar{y} = 0, \quad \bar{z}_N = c_N - N^T \bar{y} \geq 0. \quad (5.27)$$

$\bar{y}$ is a dual feasible basic solution, or geometrically a vertex in the dual feasible region

$$D = \{y \mid A^T y \leq c\}.$$

In the primal simplex context, $\bar{y}$ is usually called *simplex multipliers*.

Consider the associated primal basic solution

$$\bar{x}_B = B^{-1}b, \quad \bar{x}_N = 0.$$

If $\bar{x}_B \geq 0$, then $\bar{x}$ and $(\bar{y}, \bar{z})$ satisfy the optimality condition and are therefore a pair of primal and dual basic optimal solutions.

Now assume that $\bar{x}_B = B^{-1}b \not\geq 0$. Determine row index p such that

$$\bar{x}_{j_p} = \min\{\bar{x}_{j_i} \mid i = 1, \dots, m\} < 0. \quad (5.28)$$

Introduce vector

$$h = B^{-T} e_p, \quad (5.29)$$

which with (5.28) gives

$$-b^T h = -b^T B^{-T} e_p = -e_p^T (B^{-1} b) = -\bar{x}_{j_p} > 0, \quad (5.30)$$

implying that $-h$ is an ascent direction, with respect to the dual objective function g .

Now consider the line search scheme below:

$$\hat{y} = \bar{y} - \beta h, \quad (5.31)$$

where β is a dual stepsize to be determined. From (5.31), (5.29), and (5.27), it follows that

$$\hat{z}_B = c_B - B^T \hat{y} = c_B - B^T (\bar{y} - \beta h) = \beta e_p \geq 0, \quad (5.32)$$

$$\hat{z}_N = c_N - N^T \hat{y} = c_N - N^T (\bar{y} - \beta h) = \bar{z}_N + \beta N^T h. \quad (5.33)$$

If $\sigma_N = N^T h \not\geq 0$, then it is seen from (5.33) that a too large $\beta > 0$ will lead to $\hat{z}_N \not\geq 0$, as violates the dual feasibility. It is easy to determine the maximum possible β and the corresponding column index q , subject to $\hat{z}_N \geq 0$ (see step 5 of Algorithm 5.4.1). Then, drop j_p from and enter q to the basis. It is easy to verify that $\hat{y}$ is just the dual feasible basic solution, corresponding to the resulting basis.

The following is valid in the other case.

Proposition 5.5.1 *If $\sigma_N = N^T h \geq 0$, the dual problem is unbounded, and $-h$ is an ascent extreme direction in the dual feasible region D .*

Proof If $\sigma_N \geq 0$, it is seen from (5.27) and (5.32), (5.33) that

$$\hat{z} = c - A^T \hat{y} \geq 0, \quad \forall \beta \geq 0,$$

implying feasibility of the new solution $\hat{y}$ given by (5.31). On the other hand, it is known from (5.31) and (5.30) that the associated new objective value is

$$b^T \hat{y} = b^T \bar{y} - \beta \bar{x}_{j_p}, \quad (5.34)$$

which goes to ∞ , as β tends to ∞ . Thus the dual problem is unbounded. This means that $-h$ is an ascent unbounded direction of D . In fact, it is seen that $-h$ is the direction of 1-dimensional face or edge

$$\{y \in \mathcal{R}^m \mid A^T y \leq c; a_{ji}^T y = c_{j_i}, i = 1, \dots, m, i \neq p\}$$

and therefore an ascent extreme direction. $\square$

The preceding analysis is actually valid for any given negative $\bar{x}_{j_p}$. Though far from the best, Rule (5.28) is simple and easy to use, and the corresponding objective increment is the maximum possible for a unit stepsize. More efficient dual pivot rules will be presented in Part II.

Since the beginning of the nineties of the last century, successful applications of the dual steepest-edge rule (Forrest and Goldfarb 1992), some of its approximations and bound-flipping (Kirillova et al. 1979) have injected fresh vigor to the dual simplex method, so that it becomes one of the most powerful methods for solving LP problems (Bixby 2002).

Example 5.5.1 Solve the following problem by Algorithm 5.5.1:

$$\begin{aligned} \min \quad & f = x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & -2x_1 - x_2 - x_3 + x_4 = -1 \\ & x_1 - 4x_2 - x_3 + x_5 = -2 \\ & x_1 + 3x_2 + x_6 = 4 \\ & x_j \geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Answer Initial: $B = \{4, 5, 6\}$, $N = \{1, 2, 3\}$, $B^{-1} = I$, $\bar{z}_N = (1, 2, 1)^T$, $\bar{x}_B = (-1, -2, 4)^T$, $f = 0$.

Iteration 1:

1. $\min\{-1, -2, 4\} = -2 < 0$, $p = 2$, x_5 leaves the basis.
3. $\sigma_N = (1, -4, -1)^T$.
5. $\beta = \min\{2/4, 1/1\} = 1/2$, $q = 2$, x_2 enters the basis.
6. $\bar{z}_N = (3/2, 0, 1/2)^T$, $\bar{z}_{j_p} = 1/2$, $\bar{f} = 0 - (1/2)(-2) = 1$.
7. $\bar{a}_q = (-1, -4, 3)^T$.
8. $\alpha = -2/-4 = 1/2$, $\bar{x}_B = (-1, -2, 4)^T - (1/2)(-1, -4, 3)^T = (-1/2, 0, 5/2)^T$, $\bar{x}_2 = \alpha = 1/2$.
9. $B^{-1} = \begin{pmatrix} 1 & 1/2 & -4 \\ 1/2 & -4 & 1 \\ 3/4 & 1 & 1 \end{pmatrix}$.
10. $B = \{4, 2, 6\}$, $N = \{1, 5, 3\}$, $\bar{z}_N = (3/2, 1/2, 1/2)^T \geq 0$, $\bar{x}_B = (-1/2, 1/2, 5/2)^T$.

Iteration 2:

1. $\min\{-1/2, 1/2, 5/2\} = -1/2 < 0$, $p = 1$, x_4 leaves the basis.
3. $\sigma_N = (-9/4, -1/4, -3/4)^T$.
5. $\beta = \min\{(3/2)/(9/4), (1/2)/(1/4), (1/2)/(3/4)\} = 2/3$, $q = 1$, x_1 enters the basis.
6. $\bar{z}_N = (3/2, 1/2, 1/2)^T + (2/3)(-9/4, -1/4, -3/4)^T = (0, 1/3, 0)^T$. $\bar{z}_{j_p} = 2/3$, $\bar{f} = 1 - (2/3)(-1/2) = 4/3$.
7. $\bar{a}_q = (-9/4, -1/4, 7/4)^T$.

$$8. \alpha = (-1/2)/(-9/4) = 2/9, \quad \bar{x}_B = (-1/2, 1/2, 5/2)^T - (2/9)(-9/4, -1/4, 7/4)^T = (0, 5/9, 19/9)^T, \quad \bar{x}_1 = 2/9.$$

$$9. B^{-1} = \begin{pmatrix} -4/9 & 1 \\ -1/9 & 1 \\ 7/9 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1/4 \\ 1/4 & 1 \\ 3/4 & 1 \end{pmatrix} = \begin{pmatrix} -4/9 & 1/9 \\ -1/9 & -2/9 \\ 7/9 & 5/9 \end{pmatrix}.$$

$$10. B = \{1, 2, 6\}, \quad N = \{4, 5, 3\}, \quad \bar{z}_N = (2/3, 1/3, 0)^T \geq 0, \quad \bar{x}_B = (2/9, 5/9, 19/9)^T \geq 0.$$

The optimal solution and objective value:

$$x^* = (2/9, 5/9, 0, 0, 0, 19/9)^T, \quad f^* = 4/3.$$

5.6 Economic Interpretation of Duality: Shadow Price

The dual problem is of an interesting economic interpretation. Assume that the primal problem

$$\begin{aligned} \max f &= c^T x \\ \text{s.t. } Ax &\leq b, \quad x \geq 0 \end{aligned}$$

is a plan model for a manufacturer to produce n products using m resources. The available amount of resource i is b_i , $i = 1, \dots, m$, units; producing a unit of product j consumes a_{ij} units of resource i . The profit of a unit of product j is c_j , $j = 1, \dots, n$. The goal is to achieve the highest profit with the limited resources.

The dual problem is

$$\begin{aligned} \min b^T y \\ \text{s.t. } A^T y &\geq c, \quad y \geq 0. \end{aligned}$$

Let x^* and y^* be the primal and dual optimal solutions, respectively. According to the strong duality theorem, the associated primal and dual optimal values are equal, i.e.,

$$v = c^T x^* = b^T y^*.$$

Optimal value's partial derivative with respect to b_i is

$$\frac{\partial v}{\partial b_i} = y_i^*.$$

Therefore, y_i^* is equal to the increment of the highest profit, created by adding one unit of resource i , and can be taken as manufacturer's assessment for resource i ,

as is named *shadow price* by Paul Samuelson.¹ Shadow price $\bar{y}_i$ is the upper price limit that the manufacturer can afford to buy resource i . When the market price of resource i is lower than shadow price y_i^* , the manufacturer should consider buying it to expand the production scale, whereas he should consider selling it to reduce the production scale in the other case. The manufacturer will not buy resource i anymore, no matter how low its price is, whenever the optimal solution x^* satisfies the i th primal inequality constraint slackly, as it implies that resource i is not fully used. In fact, the shadow price y_i^* vanishes in this case.

Let x and y be any primal and dual feasible solutions, respectively; hence, $c^T x \leq b^T y$ holds according to the weak duality. The inequality

$$c^T x < b^T y$$

implies that the total profit (output) of the plan is less than the available value (input) of the resources. In economic terms, the input–output system is said *instable (non-optimal)* in this case. It is a stable (optimal) system only when the output is equal to the input.

Consider the economic implication of the dual constraints. The manufacturer negotiates with the supplier at price y_i for resource i , which is calculated to purchase resources b_i , $i = 1, \dots, m$ by overall payment $b^T y$. For $j = 1, \dots, n$, on the other hand, the supplier asks for resource prices to produce a unit product j being no less than the profit of a unit of product j , as it satisfies the j th dual constraint

$$\sum_{i=1}^m a_{i,j} y_i \geq c_j.$$

If the supplier asks for too high prices, that is, the dual optimal solution y^* satisfies the j th dual constraint slackly, then $\bar{x}_j$ vanishes, as it implies that the manufacturer should not arrange for producing product j at all, no matter how high the profit of a unit of the product is.

5.7 Dual Elimination

Either the primal or dual simplex algorithm solves the two $m \times m$ systems of equations in each iteration, which consumes the major cost, especially when the number m of the rows of the LP problem is very large. On the other hand, the decomposition approach introduced in Chap. 16 of Part II can deal with large-scale LP problems, especially with a very large number n of columns.

¹ Paul Samuelson (1915–2009), American economist, the winner of The Nobel Economics Prize (1970), the first American to win this prize.

In this section, we introduce the so-called *dual elimination* technique, transforming the *dual* problem into an $(n - m) \times m$ standard LP problem. Therefore, one can solve two $(n - m) \times (n - m)$ system in each iteration instead. It is especially favorable if $m \gg n - m$. It is interesting that the objective row of the optimal tableau of the transformed problem exhibits the primal optimal solution, compared with the objective row of the normal optimal tableau, giving the dual optimal solution.

Consider the tableau of the dual problem (5.2), with $\max g = b^T y$ replaced by $\min -g = -b^T y$:

$$\begin{array}{c|c|c}
 y & z & g \\
 \hline
 A^T & I & c \\
 \hline
 -b^T & 1 & \\
 \hline
 \end{array} \quad \begin{array}{c} \text{RHS} \\ n \\ 1 \end{array} \quad (5.35)$$

Note that A has full row rank. Through Gauss elimination with row exchanges, the preceding can be converted into a tableau in the form below:

$$\begin{array}{c|c|c|c}
 y & z & g & \text{rhs} \\
 \hline
 U & G_1 & d_1 & m \\
 & G_2 & d_2 & n - m \\
 \hline
 \bar{x}^T & 1 & \bar{g} & 1 \\
 \hline
 \end{array} \quad (5.36)$$

where $U \in \mathcal{R}^{m \times m}$ is nonsingular upper triangular, $G_1 \in \mathcal{R}^{m \times n}$, $G_2 \in \mathcal{R}^{(n-m) \times n}$. The preceding corresponds to an equivalent form of the dual problem, i.e.,

$$\begin{aligned}
 \min -g &= -\bar{g} + \bar{x}^T z, \\
 \text{s.t. } G_2 z &= d_2, \quad z \geq 0, \\
 Uy + G_1 z &= d_1,
 \end{aligned} \quad (5.37)$$

where $\text{rank } G_2 = n - m$. Therefore, it is only needed to solve an $(n - m) \times n$ standard problem with respect to variable z , corresponding to the last $n - m + 1$ rows of tableau (5.36), i.e.,

$$\begin{array}{c|c|c}
 z & g & \text{RHS} \\
 \hline
 G_2 & d_2 & \\
 \hline
 \bar{x}^T & 1 & \bar{g} \\
 \hline
 \end{array} \quad (5.38)$$

Starting from the preceding tableau, problem (5.37) (hence the original problem) can be solved by a two-phase primal or dual tableau simplex algorithm. In case when $n - m \ll m$, handling the problem with $n - m$ rows is much cheaper than handling the original problem with m rows. When an optimal dual solution z^* is obtained, solving triangular system

$$Uy = d_1 - G_1 z^*$$

gives the associated solution y^* . Consequently, the simplex algorithm will involve systems of small order $(n - m) \times (n - m)$ rather than the original order $m \times m$.

It is favorable that the associated primal optimal basic solution is readily available. To explain, let us give the following result first.

Lemma 5.7.1 *If there is a nonsingular matrix $G \in \mathcal{R}^{n \times n}$ such that*

$$GA^T = \begin{pmatrix} U \\ 0 \end{pmatrix}, \quad (5.39)$$

where $U \in \mathcal{R}^{m \times m}$ is nonsingular upper triangular, then the range space of G_2^T is just the null space of A , where G_2 consists of the last $n - m$ rows of G .

Proof Assume that G_1 consists of the first m rows of G . It follows from (5.39) that

$$\left(AG_1^T \mid AG_2^T \right) = A \left(G_1^T \mid G_2^T \right) = AG^T = \begin{pmatrix} U^T \mid 0 \end{pmatrix}, \quad (5.40)$$

hence leading to $AG_2^T = 0$. Since G is nonsingular, it holds that $\text{rank } G_2 = n - m$. Therefore, the range space of G_2^T is the null space of A . $\square$

It is not difficult to generalize Lemma 5.7.1 to allow the case when A is not of full row rank. In addition, we stress that the lemma is valid for any $n \times n$ nonsingular matrix G . In particular, performing Gauss elimination with row exchanges on (5.35) amounts to premultiplying the first n rows by

$$G = \tilde{G}_m P_m \cdots \tilde{G}_1 P_1,$$

where $\tilde{G}_i \in \mathcal{R}^{n \times n}$, $i = 1, \dots, m$, are Gauss transformations and $P_i \in \mathcal{R}^{n \times n}$, $i = 1, \dots, m$, are permutations.

Theorem 5.7.2 *If there is an optimal tableau to (5.38), then its bottom row gives a basic optimal solution to the original problem.*

Proof Let (5.38) be an optimal simplex tableau. It is known that it is equivalent to

z	g	RHS
G_2		$G_2 c$
$b^T U^{-1} G_1$	1	$b^T U^{-1} d_1$

(5.41)

from which it is obtained that

$$\bar{x} = G_1^T U^{-T} b.$$

Thus, it holds by (5.40) that

$$A\bar{x} = (AG_1^T)U^{-T}b = U^T U^{-T}b = b.$$

Therefore, the bottom row of (5.36) gives an original solution. By Lemma 5.7.1, in addition, the range space of G_2^T gives the null space of A . This is true for all G_2^T 's in the successive tableaus, generated in the process, and each bottom row comes from its predecessor plus a multiple of some row of a G_2^T . Therefore, the bottom rows of all these simplex tableaus give primal solutions. Since the bottom row of the optimal simplex tableau is nonnegative and exhibits complementarity with the dual basic optimal solution, it gives an optimal basic solution to the original problem. $\square$

The dual tableau (5.35) is of $(n+1) \times (m+n+1)$ order, which is larger than $(m+1) \times (n+1)$, the order of the primal tableau. At a first glance, the approach described above seems to be impracticable. It is different if one takes a closer look at it. In fact, the triangularization of the former amounts to performing LU factorization of

$$\begin{pmatrix} A^T \\ -b^T \end{pmatrix}$$

(with row exchanges, except for the bottom row). Contents in Sect. 4.3 are still valid in this context. When $n - m \ll m$, the extra cost is not as much, but the return is considerable, since the scale of the linear systems, involved in subsequent iterations, is reduced significantly. As for obtaining (5.36), it is only needed to accumulate Gauss transformation and permutation factors on (5.35). To obtain $\bar{y}$, it is only needed to accumulate rows, associated with tableau (5.38), hence only touching upon the corresponding components of factors. Alternatively, these factors can be stored, and required vectors can be calculated from them and the original data. Note that G_2 includes $(n-m) \times (n-m)$ unit matrix, corresponding to zeros in the bottom row. Therefore, there is no need for computing the associated $n-m$ columns. Taking the unit matrix as an initial basis, one can get a two-phase primal or dual simplex algorithm started directly.

Example 5.7.1 Solve the following LP problem via the dual elimination:

$$\begin{aligned} \min f &= 3x_1 - x_2 + 2x_3 - x_4 + x_5 - x_7 - 2x_8, \\ \text{s.t. } x_1 + x_3 - x_6 - 2x_8 &= -2, \\ -x_2 + 4x_4 + 2x_6 - 3x_8 &= 9, \\ +x_3 - x_5 + x_6 - 2x_7 &= -5, \\ 2x_1 - 5x_4 - x_8 &= -18, \\ +x_2 - 3x_5 + x_7 &= 3, \\ -2x_1 - x_3 + 4x_5 + 2x_6 - 7x_8 &= -13, \\ x_j &\geq 0, \quad j = 1, \dots, 8. \end{aligned}$$

Answer Construct an initial tableau of form (5.35).

y1	y2	y3	y4	y5	y6	z1	z2	z3	z4	z5	z6	z7	z8	rhs
1			2		-2	1								3
	-2			1			1							-1
1		1			-1			1						2
	4		-5						1					-1
		-1		-3	4					1				1
-1	2	1			2						1			-1
		-2		1								1		-2
2	-3		-1		-7								1	-2
2	-9	5	18	-3	13									

Turn the first 6 columns to upper triangular by Gauss elimination. The resulting tableau, consisting of the first 6 rows, and the tableau in form (5.38) are as follows, respectively:

y1	y2	y3	y4	y5	y6	z1	z2	z3	z4	z5	z6	z7	z8	rhs
1			2		-2	1								3
	-2			1			1							-1
		1	-2		1	-1		1						-1
			-5	2		0	2		1					-3
				-19/5	5	-1	-4/5	1	-2/5	1				6/5
					46/19	25/19	39/19	-6/19	10/19	13/19	1			27/19

z1	z2	z3	z4	z5	z6	z7	z8	rhs
-5/2	-5/2	2	-1	-1/2	-1/2	1		-3/2
761/92	867/92*	-36/23	143/46	289/92	433/92		1	947/92
-513/92	-723/92	-58/23	63/46	-333/92	-561/92			-1739/92

There are only 3 rows in the preceding tableau, compared with the original problem that involves 7 rows. Therefore, it should be easier to handle.

5.8 Bilevel LP: Intercepting Optimal Set

In the application of LP, there would be a need for the optimal (solution) set or the largest optimal face rather than just a single optimal solution, e.g., when one faces the following *bilevel LP model*:

$$\begin{aligned} \min f_1 &= c_1^T x, \\ \min f_2 &= c_2^T x, \\ \text{s.t. } Ax &= b, \quad x \geq 0, \end{aligned} \quad (5.42)$$

which seeks the optimal set of the first level problem with the objective function f_1 subject to the constraints and then solves the second level problem with the objective function f_2 over the optimal set.

By Proposition 2.5.1, it is known that the optimal set exists if so does an optimal solution. Further, Lemma 2.5.3 gives the conceptual characterization that the optimal set is the intersection of the feasible region and the optimal objective isosurface. In this section, we provide an algebraic characterization of it, so that it is easy to obtain the optimal set, provided that a *dual* optimal solution is available.

Let $B = \{j_1, \dots, j_m\}$ and $N = A \setminus B$ be the optimal basis and nonbasis. Assume that the optimal basic solution is

$$\bar{x}_B = B^{-1}b \geq 0, \quad \bar{x}_N = 0,$$

with the optimal value $\bar{f} = c_B^T \bar{x}_B$, and the associated dual optimal solution is

$$\bar{z}_B = 0, \quad \bar{z}_N = c_N - N^T \bar{y} \geq 0, \quad \bar{y} = B^{-T} c_B.$$

Partition the nonbasis N into (N', N'') , such that

$$N' = \{j \in N \mid \bar{z}_j = 0\}, \quad N'' = \{j \in N \mid \bar{z}_j > 0\}. \quad (5.43)$$

Using the preceding notation, we claim the following.

Theorem 5.8.1 *The optimal set can be expressed as*

$$F = \{x \in \mathcal{R}^n \mid Bx_B + N'x_{N'} = b, x_B, x_{N'} \geq 0, x_{N''} = 0\}. \quad (5.44)$$

Proof Let P be the feasible region.

The LP problem is equivalent to

$$\begin{aligned} \min f &= \bar{f} + \bar{z}_N^T x_N \\ \text{s.t. } x_B + B^{-1}N x_N &= \bar{x}_B, \quad x_B, x_N \geq 0. \end{aligned} \quad (5.45)$$

Its objective function can be written as

$$f = \bar{f} + \bar{z}_{N_2}^T x_{N_2}, \quad (5.46)$$

and the constraint system can be written as

$$x_B = \bar{x}_B - B^{-1}N_1x_{N_1} - B^{-1}N_2x_{N_2}.$$

According to Lemma 2.5.3, F is the set of optimal solutions if and only if

$$F = P \cap \{x \in \mathcal{R}^n \mid c^T x = \bar{f}\}.$$

Besides, by (5.46), it holds that

$$c^T x = \bar{f} + \bar{z}_{N_2}^T x_{N_2}, \quad x \in P.$$

From these and the second expression of (5.43), it follows that

$$\begin{aligned} F &= \{x \in \mathcal{R}^n \mid Ax = b, x \geq 0; c^T x = \bar{f}\} \\ &= \{x \in \mathcal{R}^n \mid x_B = B^{-1}b - B^{-1}N_1x_{N_1} - B^{-1}N_2x_{N_2}, x \geq 0; \bar{z}_{N_2}^T x_{N_2} = 0\} \\ &= \{x \in \mathcal{R}^n \mid x_B = B^{-1}b - B^{-1}N_1x_{N_1} - B^{-1}N_2x_{N_2}, x \geq 0; x_{N_2} = 0\} \\ &= \{x \in \mathcal{R}^n \mid x_B = B^{-1}b - B^{-1}N_1x_{N_1}, x_B, x_{N_1} \geq 0; x_{N_2} = 0\}, \end{aligned}$$

which is equal to the right-hand side of (5.44). $\square$

As the largest optimal face, the set F 's dimension is dependent on the set $|N'|$ (or $|N''|$) (Proposition 2.2.1). If any index $q \in N'$ is selected to enter the basis, a row index p can be determined to leave, just as in the simplex context. For the new basis B , the system $x_B = B^{-1}b - B^{-1}N'x_{N'}$ renders a new optimal basic solution, which is truly new if its predecessor is nondegenerate. This process can be used to obtain multiple optimal basic solutions, the number of which is no more than $C_{m+|N'|}^m$.

Corollary 5.8.2 *A feasible solution is optimal for the LP problem if and only if the N'' -indexed components of it are all zero.*

Corollary 5.8.3 *If the set N' is empty, there is a unique optimal solution/basis to the LP problem.*

The preceding implies that a primal optimal solution is unique if the associated dual optimal solution is nondegenerate.

Example 5.8.1 Solve the following bilevel LP problem:

$$\begin{aligned}
 \min f_1 &= -3x_5 + 5x_6 \\
 \min f_2 &= -x_1 - 4x_2 + 2x_3 - 6x_4 + x_5 + 2x_6 \\
 \text{s.t. } &-3x_1 - x_2 + x_3 + 6x_4 - 4x_6 + x_7 = 1 \\
 &5x_1 - 2x_3 + x_4 - x_5 + x_8 = 4 \\
 &2x_2 + 3x_5 + x_6 + x_9 = 3 \\
 &x_j \geq 0, \quad j = 1, \dots, 9.
 \end{aligned}$$

Answer The initial feasible simplex tableau is

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	RHS
-3	-1	1	6		-4	1			1
5		-2	1	-1			1		4
	2			3*	1			1	3
-1	-4	2	-6	1	2				
				-3	5				

where the bottom and second bottom rows correspond to the first and second objective functions, respectively.

Call Algorithm 3.1.1 to solve the first level problem.

Iteration 1:

1. $\min\{0, 0, 0, 0, -3, 5, 0, 0, 0\} = -3$, $q = 5$.
3. $I = \{3\} \neq \emptyset$.
4. $\min\{3/3\}$, $p = 3$.
5. Multiply row 3 by $1/3$, and then add $1, -1, 3$ times of row 3 to rows 2,4,5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	RHS
-3	-1	1	6*		-4	1			1
5	2/3	-2	1		1/3		1	1/3	5
	2/3			1	1/3			1/3	1
-1	-6	2	-6		1			-1	-1
	2				6			1	3

Iteration 2:

1. $\min\{0, 2, 0, 0, 6, 1\} \geq 0$. Therefore, the first level problem is minimized with optimal value $f_1 = -3$ ($N' = \{1, 3, 4\}$, $N'' = \{2, 6, 9\}$). The second objective value equals to $f_2 = 1$. Dropping columns corresponding to N'' and the bottom

row gives the feasible simplex tableau to the second level problem, i.e.,

x_1	x_3	x_4	x_5	x_7	x_8	RHS
-3	1	6*		1		1
5	-2	1			1	5
			1			1
-1	2	-6				-1

Call the simplex algorithm 3.1.1.

Iteration 3:

1. $\min\{-1, 2, -6\} = -6$, $q = 4$.
3. $I = \{1, 2\} \neq \emptyset$.
4. $\min\{1/6, 5/1\} = 1/6$, $p = 1$.
5. Multiply row 1 by $1/6$, and then add $-1, 6$ times of row 1 to rows 2, 4, respectively:

x_1	x_3	x_4	x_5	x_7	x_8	RHS
-1/2	1/6	1		1/6		1/6
11/2*	-13/6			-1/6	1	29/6
			1			1
-4	3			1		

Iteration 4:

1. $\min\{-4, 3, 1\} = -4$, $q = 1$.
3. $I = \{2\} \neq \emptyset$.
4. $\min\{(29/6)/(11/2)\}$, $p = 2$.
5. Multiply row 2 by $2/11$, and then add $1/2; 4$ times of row 2 to rows 1, 4, respectively:

x_1	x_3	x_4	x_5	x_7	x_8	RHS
	-1/33	1		5/33	1/11	20/33
1	-13/33			-1/33	2/11	29/33
			1			1
	47/33			29/33	8/11	116/33

Iteration 5:

1. $\min\{47/33, 29/33, 8/11\} \geq 0$.
2. The second level problem is solved with the optimal basic solution $\bar{x} = (29/33; 0; 0; 20/33; 1; 0; 0; 0; 0)^T$, corresponding to the first and second objective values $f_1 = -3$ and $f_2 = -116/33$, respectively.

5.9 Notes on Duality

The concept and theorems of duality were first proposed by famous mathematician von Neumann. In October 1947, he made foundational discussions on the topic in a talk with George B. Dantzig and in a working paper, finished a few weeks later. In 1948, Dantzig provided a rigorous proof on the duality theorems in a report. Subsequently, (Gale et al. 1951) formulated the duality theorems and proved them using Farkas Lemma, independently. Goldman and Tucker (1956a,b) discussed theoretical properties of the dual problem systematically.

As was mentioned, the simplex tableau is just a concise expression of a LP problem itself, and all such tableaus created by the primal or dual simplex algorithm are equivalent in the sense of their representation of the LP problem. Then the following question arises:

Are the dual problems corresponding to the simplex tableaus equivalent too?

Consider the dual problem, corresponding to tableau (3.6), i.e.,

$$\begin{aligned} & \max \bar{f} + \bar{b}^T y' \\ & \text{s.t.} \quad \begin{pmatrix} I \\ \bar{N}^T \end{pmatrix} y' + \begin{pmatrix} z'_B \\ z'_N \end{pmatrix} = \begin{pmatrix} 0 \\ \bar{z}_N \end{pmatrix}, \quad z'_B, z'_N \geq 0. \end{aligned}$$

As their $\bar{b}$, $\bar{N}$, $\bar{z}_N$ are not the same, the dual problems corresponding to different simplex tableaus are also different. However, such differences are not essential. In fact, making variable transformations $y' = B^T y - c_B$, $z' = z$ and noting

$$\bar{b} = B^{-1}b, \quad \bar{N} = B^{-1}N, \quad \bar{z}_N = c_N - N^T B^{-T} c_B, \quad \bar{f} = c_B^T B^{-1}b,$$

the dual problem can be converted into

$$\begin{aligned} & \max b^T y, \\ & \text{s.t.} \quad \begin{pmatrix} B^T \\ N^T \end{pmatrix} y + \begin{pmatrix} z_B \\ z_N \end{pmatrix} = \begin{pmatrix} c_B \\ c_N \end{pmatrix}, \quad z_B, z_N \geq 0, \end{aligned}$$

which is the original dual problem. Therefore, all the generated simplex tableaus can be regarded as equivalent.

In summary, elementary transformations generate equivalent simplex tableaus. On the primal side, the right-hand sides give primal basic solutions, and the bottom

rows give primal reduced objective functions. On the dual side, the right-hand sides render dual reduced objective functions, and the bottom rows dual basic solutions.

Based on duality, the following is also valid.

Proposition 5.9.1 *If it has a dual solution $(\bar{y}, \bar{z})$, the standard problem (5.1) is equivalent to*

$$\begin{aligned} \max \quad & f = \bar{y}^T b + \bar{z}^T x, \\ \text{s.t.} \quad & Ax = b, \quad x \geq 0. \end{aligned} \quad (5.47)$$

Proof $(\bar{y}, \bar{z})$ satisfies $A^T \bar{y} + \bar{z} = c$, or equivalently,

$$c^T = \bar{y}^T A + \bar{z}^T.$$

Substituting the preceding to the objective of the standard problem and noting the constraint system give the objective of (5.47), i.e.,

$$f = c^T x = \bar{y}^T A x + \bar{z}^T x = \bar{y}^T b + \bar{z}^T x,$$

and vice versa. □

The preceding says that the cost vector c in the standard problem can be replaced by any dual solution $\bar{z}$, with only a constant difference in objective value.

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Chapter 6

Primal-Dual Simplex Method



Methods perform very differently when solving the same problem. It is a common case that a problem that is solved slowly by the simplex method would be solved fast by the dual simplex method, and vice versa. Consequently, LP packages often include multiple options, as it seems to be impossible to predetermine which method would be better to solve a given problem. Therefore, scholars have attempted to fuse the two methods into one for years.

In this chapter, we present selected algorithms of both the primal and dual features, whose common advantage is Phase-I free, including the mixed two-phase simplex algorithm, primal-dual simplex algorithm, self-dual parametric algorithm, criss-cross simplex algorithm using the most-obtuse-angle rule, and perturbation primal-dual simplex algorithm. In the final section, we make some remarks on the criss-cross simplex algorithm.

Like the simplex algorithm, the finiteness of these algorithms is not guaranteed; even the objective value may not change monotonically in the solution process. But it can still be expected that cycling rarely happens in practice.

6.1 Mixed Two-Phase Simplex Algorithm

It has not yet attracted attention that the dual simplex algorithm itself can be used as a Phase-I to get the simplex algorithm started. In fact, if the former terminates, it must stop at a feasible basic solution unless detecting the infeasibility of the problem. Moreover, from careful examination, we can see that the former not only realizes the primal feasibility but also tends to achieve the dual feasibility. The same is true with the simplex algorithm as a Phase-I for the dual simplex algorithm.

Therefore, in principle, there is no need for any other Phase-I procedures, not to mention some of them need to introduce the auxiliary function and artificial variables. Surprisingly enough, this fact has been ignored by the community for

so many years although it is still unknown whether this is a perfect approach computationally.

We demonstrate this approach via examples.

(A) **Dual-primal two-phase scheme:** The dual simplex algorithm is used as Phase-I for the simplex algorithm.

Example 6.1.1 Solve the following problem by the dual-primal two-phase scheme:

$$\begin{aligned} \min \quad & x_8 = -3x_1 - 2x_2 + 4x_3 + 2x_4, \\ \text{s.t.} \quad & x_1 + 2x_2 - x_3 + 2x_4 + x_5 = 4, \\ & 2x_1 + 3x_3 - 4x_4 + x_6 = 1, \\ & -x_1 + 2x_2 - 3x_4 + x_7 = -3, \\ & x_j \geq 0, \quad j = 1, \dots, 7. \end{aligned}$$

Answer Initial tableau is

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	rhs
1	2	-1	2	1				4
2		3	-4		1			1
-1	2		-3			1		-3
-3	-2	4	2				-1	

Phase-I: Call the dual simplex Algorithm 5.4.1.

Iteration 1:

1. $\min\{4, 1, -3\} = -3 < 0$, $p = 3$.
3. $J = \{1, 4\}$.
4. $\min\{-(-3)/(-1), -2/(-3)\} = -3$, $q = 1$.
5. Multiply row 3 by -1 , and then add $-1, -2, 3$ times of row 3 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	rhs
	4	-1	-1	1		1		1
	4	3	-10		1	2		-5
1	-2		3			-1		3
	-8	4	11			-3	-1	9

Iteration 2:

1. $\min\{1, -5, 3\} = -5 < 0$, $p = 2$.
3. $J = \{4\}$.
4. $\min\{-(11)/(-10)\}$, $q = 4$.

5. Multiply row 2 by -10 , and then add $1, -3, -11$ times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	rhs
	18/5	-13/10		1	-1/10	4/5		3/2
	-2/5	-3/10	1		-1/10	-1/5		1/2
1	-4/5	9/10			3/10	-2/5		3/2
	-18/5	73/10			11/10	-4/5	-1	7/2

The feasibility is achieved.

Phase-II: Call the simplex Algorithm 3.1.1.

Iteration 3:

1. $\min\{-18/5, 73/10, 11/10, -4/5\} = -18/5 < 0, q = 2$.
3. $I = \{1, 2\} \neq \emptyset$.
4. $\min\{(3/2)/(18/5)\} = 5/12, p = 1$.
5. Multiply row 1 by $18/5$, then add $2/5, 4/5, 18/5$ times of row 1 to rows 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	rhs
	1	-13/36		5/18	-1/36	2/9		5/12
		-4/9	1	1/9	-1/9	-1/9		2/3
1		11/18		2/9	5/18	-2/9		11/6
		6		1	1		-1	5

The reduced costs $(6, 1, 1)^T \geq 0$, the optimality is achieved. The optimal basic solution and optimal value:

$$x^* = (11/6, 5/12, 0, 2/3, 0, 0, 0)^T, \quad f^* = -5.$$

(B) **Primal-dual two-phase scheme:** Phase-I: The primal reduced simplex algorithm.

Example 6.1.2 Solve the same Example 6.1.1 by the primal-dual two-phase scheme.

Answer Phase-I: Call the simplex Algorithm 3.1.1, starting from the first initial tableau.

Iteration 1:

1. $\min\{-3, -2, 4, 2\} = -3 < 0, q = 1$.
3. $I = \{1, 2\} \neq \emptyset$.
4. $\min\{4/1, 1/2\} = 1/2, p = 2$.

5. Multiply row 2 by 2, then add $-1, 1, 3$ times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	rhs
	2	$-5/2$	4	1	$-1/2$			$7/2$
1		$3/2$	-2		$1/2$			$1/2$
	2	$3/2$	-5		$1/2$	1		$-5/2$
	2	$17/2$	-4		$3/2$		-1	$3/2$

Iteration 2:

1. $\{2, , 17/2, -4, 3/2\} = -4 < 0, q = 4$.
3. $I = \{1\} \neq \emptyset$.
4. $\min\{(7/2)/4\} = 7/8, p = 1$.
5. Multiply row 1 by 4, then add 2, 5, 4 times of row 1 to rows 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	rhs
	$1/2$	$-5/8$	1	$1/4$	$-1/8$			$7/8$
1	1	$1/4$		$1/2$	$1/4$			$9/4$
	$9/2$	$-13/8$		$5/4$	$-1/8$	1		$15/8$
		6		1	1		-1	5

The reduced costs $(6, 1, 1)^T \geq 0$, the dual feasibility is achieved; is so the primal feasibility. Therefore, the optimality is achieved without any Phase-II. The optimal basic solution obtained is different from the preceding one:

$$x^* = (9/4, 0, 0, 7/8, 0, 0, 15/8)^T, \quad f^* = -5.$$

6.2 Primal-Dual Simplex Algorithm

In this section, the primal-dual algorithm (Dantzig et al. 1956) is presented, which is an extension of the homonymous algorithm (Ford and Fulkerson 1956) for solving transportation problems.

This algorithm can be regarded as a variant of the dual simplex algorithm. It also gets started from a dual feasible solution and proceeds to achieve primal feasibility while maintaining dual feasibility and complementarity. Nevertheless, it gets rid of artificial variables from the auxiliary problem to fulfill $Ax = b$ while fulfilling the sign restriction $x \geq 0$, compared with the dual simplex algorithm, which seeks toward $x \geq 0$ while keeping $Ax = b$ satisfied.

We are concerned with the standard LP problem (1.8). Let $(\bar{y}, \bar{z})$ be current dual feasible solution.

To obtain a primal solution matching $(\bar{y}, \bar{z})$, consider the auxiliary problem (3.17), written as

$$\begin{aligned} \min \quad & \zeta = e^T x_a \\ \text{s.t.} \quad & Ax + x_a = b, \quad x, x_a \geq 0, \end{aligned} \quad (6.1)$$

where $x_a = (x_{n+1}, \dots, x_{n+m})^T$ is an artificial variable vector. It would be well to assume $b \geq 0$. Introducing index set

$$Q = \{j \in A \mid \bar{z}_j = 0\}, \quad (6.2)$$

we define the so-called *restricted problem*:

$$\begin{aligned} \min \quad & \zeta = e^T x_a \\ \text{s.t.} \quad & Ax + x_a = b, \quad x_a \geq 0, \\ & x_j \geq 0, \quad \forall j \in Q, \\ & x_j = 0, \quad \forall j \notin Q. \end{aligned} \quad (6.3)$$

Since $b \geq 0$, it is clear that the feasible region of the preceding problem is nonempty, and hence there is an optimal solution to it. The restricted problem may be viewed as one formed by all artificial columns and columns indexed by j belonging to Q .

Assume that $(\bar{x}, \bar{x}_a)$ is an optimal solution to (6.3) with value $\bar{\zeta}$, and that $\bar{w}$ is the associated optimal simplex multiplier.

Theorem 6.2.1 *If the optimal value $\bar{\zeta}$ vanishes, $\bar{x}$ and $(\bar{y}, \bar{z})$ are a pair of primal and dual optimal solutions.*

Proof $\bar{\zeta} = e^T \bar{x}_a = 0$ and $\bar{x}_a \geq 0$ together imply that $\bar{x}_a = 0$. Thus, $\bar{x}$ is a feasible solution to the original problem (5.1). By definition of Q , moreover, it holds that $\bar{x}^T \bar{z} = 0$, exhibiting complementarity. Therefore, $\bar{x}$ and $(\bar{y}, \bar{z})$ are a pair of primal and dual optimal solutions. $\square$

When $\bar{\zeta} > 0$, otherwise, $\bar{x}$ could be regarded as the closest to feasibility among all those complementary with $(\bar{y}, \bar{z})$. Nevertheless, $\bar{x}$ is not feasible to the original problem because it does not satisfy $Ax = b$ but $x \geq 0$ only. In other words, it should be possible to improve $(\bar{y}, \bar{z})$ by increasing the associated dual objective value. To do so, consider the dual problem of (6.3), i.e.,

$$\begin{aligned} \min \quad & b^T w \\ \text{s.t.} \quad & a_j^T w + s_j = 0, \quad s_j \geq 0, \quad j \in Q, \\ & w \leq e. \end{aligned} \quad (6.4)$$

As the simplex multiplier $\bar{w}$ is just an optimal solution to the preceding problem, it follows from the duality that

$$b^T \bar{w} = \bar{\zeta} > 0,$$

which implies that $\bar{w}$ is an ascent direction with the dual objective $b^T y$. This leads to the following line search for updating $(\bar{y}, \bar{z})$:

$$\hat{y} = \bar{y} + \beta \bar{w}, \quad \hat{z} = c - A^T \hat{y}. \quad (6.5)$$

For being an improved dual feasible solution, it must satisfy the dual constraints for some $\beta > 0$, i.e.,

$$\hat{z} = c - A^T(\bar{y} + \beta \bar{w}) = \bar{z} + \beta \bar{s} \geq 0, \quad \bar{s} = -\beta A^T \bar{w}. \quad (6.6)$$

Since $\bar{z} \geq 0$, and $\bar{w}$ satisfies constraints of (6.4), it is known that

$$\bar{s}_j = -a_j^T \bar{w} \geq 0, \quad \forall j \in Q.$$

Therefore, if the index set

$$J = \{j \in A \setminus Q \mid \bar{s}_j = -a_j^T \bar{w} < 0\} \quad (6.7)$$

is empty, then (6.6) holds for all $\beta \geq 0$, giving a class of dual feasible solutions. Since $\bar{\zeta} > 0$, the associated dual objective value, i.e.,

$$b^T \hat{y} = b^T \bar{y} + \beta b^T \bar{w} = b^T \bar{y} + \beta \bar{\zeta}$$

tends to $+\infty$ as β increases, which implies dual unboundedness or primal infeasibility.

If, otherwise, there is some $j \in \hat{Q}$ such that $\bar{s}_j = -a_j^T \bar{w} < 0$, then (6.6) holds for the largest possible stepsize β such that

$$\beta = -\frac{\bar{z}_q}{\bar{s}_q} = \min \left\{ -\frac{\bar{z}_j}{\bar{s}_j} \mid \bar{s}_j < 0, j \in A \setminus Q \right\} > 0. \quad (6.8)$$

Hence, the resulting dual solution is feasible, corresponding to a strictly larger dual objective value, which will be used for the next iteration.

Let B be the optimal basis of the restricted problem. If a column of B is not artificial, it must be indexed by some $j \in Q$ such that $\bar{z}_j = 0$. Since the associated reduced cost is zero, i.e., $\bar{s}_j = 0 - a_j^T \bar{w} = 0$, it holds that

$$\hat{z}_j = \bar{z}_j + \beta \bar{s}_j = 0,$$

implying that j also belongs to the next Q . Therefore, the optimal basis of the restricted problem can be used as a starting basis for the next iteration. In addition, it is seen from (6.8) that there exists at least one index (e.g., q) in $\hat{Q}$, which belongs to the next Q , and the associated reduced cost is negative, i.e., $\bar{s}_q < 0$. In other words, there exist new candidates to enter the basis in the next iteration. In each iteration,

therefore, the restricted problem can be solved by the primal simplex method. As an artificial variable leaves the basis, it is dropped from the original auxiliary problem immediately.

Optimality of the restricted problem is achieved if $Q \cap N = \emptyset$. In case when the initial set Q is empty, for instance, all the artificial columns just form an optimal basis and the optimal multiplier is $\bar{w} = e$; therefore, no simplex steps are needed.

The steps are summarized into the following.

Algorithm 6.2.1 (Primal-Dual Simplex Algorithm) *Initial: a dual feasible solution $(\bar{y}, \bar{z})$, and associated Q defined by (6.2). $B = \{n+1, \dots, n+m\}$, $N = \{1, \dots, n\}$. This algorithm solves the standard LP problem (1.8).*

1. Carry out the simplex algorithm to solve the restricted auxiliary problem (6.1).
2. Stop if the optimal value of the restricted problem vanishes (optimality achieved).
3. Stop if J defined by 6.7 is empty. (infeasible problem)
4. Compute β by (6.8).
5. Update $(\bar{y}, \bar{z})$ by (6.5).
6. Update Q by (6.2)
7. Go to step 1.

Although the simplex method is used to solve the restricted problem, any method for solving it will apply. The primal-dual simplex method seems to be amenable to certain network flow problems because an initial dual feasible solution can be obtained easily and the associated restricted problem can be efficiently solved by a so-called *labeling* method (see Papadimitriou and Steiglitz 1982).

It is noted that the objective value, corresponding to the dual feasible solution, increases monotonically iteration by iteration. Therefore, the primal-dual method will terminate if each restricted problem encountered is solved within a finite number of iterations.

Example 6.2.1 Solve the following problem by Algorithm 6.2.1:

$$\begin{aligned}
 &\min 2x_1 + 5x_2 + x_3 + 4x_4 + 8x_5 \\
 \text{s.t. } &-x_1 + 4x_2 - 2x_3 + 2x_4 - 6x_5 = -1 \\
 &x_1 + 2x_2 + 2x_3 \quad - 4x_5 = 8 \\
 &-x_1 + x_2 \quad + 2x_4 + 2x_5 = 2 \\
 &x_j \geq 0, \quad j = 1, \dots, 5.
 \end{aligned}$$

Answer Construct the auxiliary problem below:

$$\begin{aligned}
 &\min \zeta = x_6 + x_7 + x_8 \\
 \text{s.t. } &x_1 - 4x_2 + 2x_3 - 2x_4 + 6x_5 + x_6 = 1 \\
 &x_1 + 2x_2 + 2x_3 \quad - 4x_5 + x_7 = 8 \\
 &-x_1 + x_2 \quad + 2x_4 + 2x_5 + x_8 = 2 \\
 &x_j \geq 0, \quad j = 1, \dots, 8.
 \end{aligned}$$

Initial: $B = \{6, 7, 8\}$, $N = \{1, \dots, 5\}$. Since the costs of the original problem are positive, a feasible dual solution ($\bar{y} = (0, 0, 0)^T$, $\bar{z} = (2, 5, 1, 4, 8)^T$) is available, with $Q = \emptyset$.

Iteration 1:

1. Since $Q = \emptyset$, no simplex step is needed.
2. The optimal value of the restricted problem is positive, and the optimal simplex multiplier is $\bar{w} = (1, 1, 1)^T$.
3. $\bar{s}_J = (-1, -4, -4)^T$, $J = \{1, 3, 5\} \neq \emptyset$.
4. $\bar{z}_J = (2, 1, 8)^T$, $\theta = \min\{2/1, 1/4, 8/4\} = 1/4$, $q = 3$.
5. $\bar{y} = (0, 0, 0)^T + 1/4(1, 1, 1)^T = (1/4, 1/4, 1/4)^T$,

$$\bar{z}_N = \begin{pmatrix} 2 \\ 5 \\ 1 \\ 4 \\ 8 \end{pmatrix} - \begin{pmatrix} 1 & -4 & 2 & -2 & 6 \\ 1 & 2 & 2 & & -4 \\ -1 & 1 & 2 & 2 & \end{pmatrix}^T \begin{pmatrix} 1/4 \\ 1/4 \\ 1/4 \end{pmatrix} = \begin{pmatrix} 7/4 \\ 21/4 \\ 0 \\ 4 \\ 7 \end{pmatrix}.$$

6. $Q = \{3\}$.

Iteration 2:

1. Carry out restricted simplex steps of Algorithm 3.3.1:

subiteration 1:

- (2) Column selection is restricted to $Q \cap N = \{3\}$. x_3 enters the basis.
- (4) $\bar{a}_3 = a_3 = (2, 2, 0)^T \not\leq 0$.
- (6) $\bar{x}_B = (1, 8, 2)^T$, $\alpha = \min\{1/2, 8/2\} = 1/2$, $p = 1$, x_6 leaves the basis, and is dropped.
- (7) $\bar{x}_B = (1, 8, 2)^T - 1/2(2, 2, 0)^T = (0, 7, 2)^T$. $\bar{x}_3 = 1/2$.
- (8) $B^{-1} = \begin{pmatrix} 1/2 & & \\ -1 & 1 & \\ & & 1 \end{pmatrix}$.
- (9) $B = \{3, 7, 8\}$, $N = \{1, 2, 4, 5\}$.

Subiteration 2:

$$(1) \bar{w} = \begin{pmatrix} 1/2 & & \\ -1 & 1 & \\ & & 1 \end{pmatrix}^T \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix},$$

$$\bar{s}_N = - \begin{pmatrix} 1 & -4 & -2 & 6 \\ 1 & 2 & & -4 \\ -1 & 1 & 2 & 2 \end{pmatrix}^T \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -7 \\ -4 \\ 8 \end{pmatrix}.$$

- (2) $Q \cap N = \{3\} \cap \{1, 2, 4, 5\} = \emptyset$.

2. The optimal value of the restricted problem is positive.
3. $\bar{s}_J = \{-7, -4\}^T$, $J = \{2, 4\} \neq \emptyset$.

$$4. \bar{z}_J = (21/4, 4)^T, \beta = \min\{(21/4)/7, 4/4\} = 21/28, q = 2.$$

$$5. \bar{y} = (1/4, 1/4, 1/4)^T + 21/28(-1, 1, 1)^T = (-1/2, 1, 1)^T,$$

$$\bar{z}_N = \begin{pmatrix} 2 \\ 5 \\ 4 \\ 8 \end{pmatrix} - \begin{pmatrix} 1 & -4 & -2 & 6 \\ 1 & 2 & & -4 \\ -1 & 1 & 2 & 2 \end{pmatrix}^T \begin{pmatrix} -1/2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5/2 \\ 0 \\ 1 \\ 13 \end{pmatrix}.$$

$$6. Q = \{3, 2\}.$$

Iteration 3:

1. Carry out simplex steps of Algorithm 3.3.1 restricted:

subiteration 1:

(2) Column selection is restricted to $Q \cap N = \{2\}$. x_2 enters the basis.

$$(4) \bar{a}_2 = \begin{pmatrix} 1/2 \\ -1 & 1 \\ 1 \end{pmatrix} \begin{pmatrix} -4 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \\ 1 \end{pmatrix} \not\leq 0.$$

(6) $\bar{x}_B = (1/2, 7, 2)^T$, $\alpha = \min\{7/6, 2/1\} = 7/6$, $p = 2$, x_7 leaves the basis, and is dropped.

$$(7) \bar{x}_B = (1/2, 7, 2)^T - 7/6(-2, 6, 1)^T = (17/6, 0, 5/6)^T, \bar{x}_2 = 7/6.$$

$$(8) B^{-1} = \begin{pmatrix} 1 & 1/3 \\ & 1/6 \\ & -1/6 & 1 \end{pmatrix} \begin{pmatrix} 1/2 \\ -1 & 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/6 & 1/3 \\ -1/6 & 1/6 \\ 1/6 & -1/6 & 1 \end{pmatrix}.$$

$$(9) B = \{3, 2, 8\}, N = \{1, 4, 5\}.$$

Subiteration 2:

$$(1) \bar{w} = \begin{pmatrix} 1/6 & 1/3 \\ -1/6 & 1/6 \\ 1/6 & -1/6 & 1 \end{pmatrix}^T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/6 \\ -1/6 \\ 1 \end{pmatrix}$$

$$\bar{s}_N = - \begin{pmatrix} 1 & -2 & 6 \\ 1 & & -4 \\ -1 & 2 & 2 \end{pmatrix}^T \begin{pmatrix} 1/6 \\ -1/6 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -5/3 \\ -11/3 \end{pmatrix}.$$

$$(2) Q \cap N = \{3, 2\} \cap \{1, 4, 5\} = \emptyset.$$

2. The optimal value of the restricted problem is positive.

$$3. s_J = (-5/3, -11/3)^T, J = \{4, 5\} \neq \emptyset.$$

$$4. \bar{z}_J = (1, 13)^T, \beta = \min\{1/(5/3), 13/(11/3)\} = 3/5, q = 4.$$

$$5. \bar{y} = (-1/2, 1, 1)^T + 3/5(1/6, -1/6, 1)^T = (-2/5, 9/10, 8/5)^T,$$

$$\bar{z}_N = \begin{pmatrix} 2 \\ 4 \\ 8 \end{pmatrix} - \begin{pmatrix} 1 & -2 & 6 \\ 1 & & -4 \\ -1 & 2 & 2 \end{pmatrix}^T \begin{pmatrix} -2/5 \\ 9/10 \\ 8/5 \end{pmatrix} = \begin{pmatrix} 31/10 \\ 0 \\ 54/5 \end{pmatrix}.$$

$$6. Q = \{3, 2, 4\}.$$

Iteration 4:

1. Carry out simplex steps of Algorithm 3.3.1 restricted:

subiteration 1:

(2) Column selection is restricted to $Q \cap N = \{3, 2, 4\} \cap \{1, 4, 5\}$. x_4 enters the basis.

$$(4) \bar{a}_4 = \begin{pmatrix} 1/6 & 1/3 \\ -1/6 & 1/6 \\ 1/6 & -1/6 & 1 \end{pmatrix} \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} -1/3 \\ 1/3 \\ 5/3 \end{pmatrix} \not\leq 0.$$

(6) $\bar{x}_B = (17/6, 7/6, 5/6)^T$, $\alpha = \min\{(7/6)/(1/3), (5/6)/(5/3)\} = 1/2$, $p = 3$, x_8 leaves the basis, and dropped.

(7) $\bar{x}_B = (17/6, 7/6, 5/6)^T - 1/2(-1/3, 1/3, 5/3)^T = (3, 1, 0)^T$, $\bar{x}_4 = 1/2$.

$$(8) B^{-1} = \begin{pmatrix} 1 & 1/5 \\ & -1/5 \\ & 1 & 3/5 \end{pmatrix} \begin{pmatrix} 1/6 & 1/3 \\ -1/6 & 1/6 \\ 1/6 & -1/6 & 1 \end{pmatrix} = \begin{pmatrix} 1/5 & 3/10 & 1/5 \\ -1/5 & 1/5 & -1/5 \\ 1/10 & -1/10 & 3/5 \end{pmatrix}.$$

(9) $B = \{3, 2, 4\}$, $N = \{1, 5\}$.

2. The optimal value of the restricted problem is zero, optimality achieved.

The optimal solution and objective value are

$$x^* = (0, 1, 3, 1/2, 0)^T, \quad f^* = 10.$$

6.3 Self-Dual Parametric Simplex Algorithm

This section presents a simplex variant, which solves the standard LP problem via the parameterization of the objective function as well as the right-hand side. This method is closely related to Orchard-Hays' work (1954b, 1956) and has been used by Smale (1983b) for investigating the worst-case complexity of the simplex method.

Consider the following parametric problem:

$$\begin{aligned} \min f &= (c + \theta c')^T x \\ \text{s.t. } Ax &= b + \theta b', \quad x \geq 0. \end{aligned} \tag{6.9}$$

The standard problem is just the preceding problem with $\theta = 0$.

The presented method belongs to a more general *homotopy* approach. Generating a continuous problem deformation, it converts the given problem to a trivially solved one and then proceeds backward from the latter to the original by solving all problems between them. It can solve the LP problem in a single phase by starting from any basis.

Assume the availability of a simplex tableau to the LP problem, which is neither primal nor dual feasible. It is a simple matter to determine the $\theta = \theta_2 > 0$ such that the objective row and the right-hand side both become nonnegative after adding it properly. This amounts to adding some terms $\theta c'$ and $\theta b'$ respectively to the objective function and the right-hand side of the original problem, corresponding to $\theta = \theta_1 = 0$. Then, θ is decreased from θ_2 down to 0 while maintaining optimality. If primal feasibility is violated first in this process, a row index p and a new θ_2 are determined; then a column index q is determined by the dual simplex ratio test. If otherwise, dual feasibility is violated first, a column index q and a new θ_2 are determined; a row index p is determined by the primal simplex ratio test. Subsequent operations are just a normal basis change.

Assume that the current simplex tableau is optimal to $\theta = \theta_2$, i.e.,

x_B	x_N	RHS
I	$\bar{N}$	$\bar{b} + \bar{b}'\theta$,
	$\bar{z}_N^T + (\bar{z}')_N^T \theta$	$-\bar{f}$

(6.10)

The procedure is summarized into the following, where the parametric problem with $\theta = 0$ corresponds to the standard LP problem.

Algorithm 6.3.1 (Self-Dual Parametric Algorithm: Tableau Form) *Given $\theta_2 > 0$. Initial: a simplex tableau of the form (6.10), which is optimal with $\theta = \theta_2$. This algorithm solves the standard LP problem.*

1. If $\bar{z}'_N \leq 0$, set $\beta = 0$; else, determine q and β such that

$$\alpha = -\bar{z}_q / \bar{c}'_q = \max\{-\bar{z}_j / \bar{z}'_j \mid \bar{z}'_j > 0, j \in N\}.$$

1. If $\bar{b}' \leq 0$, set $\alpha = 0$; else, determine p and α such that

$$\beta = -\bar{b}_p / \bar{b}'_p = \max\{-\bar{b}_i / \bar{b}'_i \mid \bar{b}'_i > 0, i = 1, \dots, m\}.$$

3. If $\alpha \geq \beta$, do the following

- (1) Stop with $\theta_2 = 0$ if $\alpha \leq 0$; (optimality achieved)
- (2) Stop if $\bar{a}_q \leq 0$; (unbounded)
- (3) Determine row index p such that

$$(\bar{b}_p + \bar{b}'_p \theta) / \bar{a}_{pq} = \min\{(\bar{b}_i + \bar{b}'_i \theta) / \bar{a}_{iq} \mid \bar{a}_{iq} > 0, i = 1, \dots, m\},$$

where θ is close to θ_2 ;

else

- (4) Stop with $\theta_2 = 0$ if $\beta \leq 0$; (optimality achieved)
- (5) Stop if $J = \{j \in N \mid \bar{a}_{pj} < 0\}$; (infeasible)

(6) Determine column index q such that

$$-(\bar{z}_q + \bar{z}'_q \theta) / \bar{a}_{pq} = \min_{j \in J} -(\bar{z}_j + \bar{z}'_j \theta) / \bar{a}_{pj},$$

where θ is close to θ_2 .

4. If $\alpha \geq \beta$, set $\theta_2 = \alpha$ else set $\theta_2 = \beta$.
5. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
6. Go to step 1.

An advantage of the preceding Algorithm is that it solves LP problems in a single phase by starting from any basis. It might be said *criss-cross* because of shuttling between primal and dual sides, depending on which of α and β is larger (see step 3). Therefore, it seems critical to scale the costs and the right-hand side for an equilibrium of their sizes beforehand. On the other hand, the algorithm requires more computational effort per iteration, compared with the simplex algorithm. As a homotopy algorithm, it seems to be more suitable for solving hard problems. Moreover, it stands well as a tool for handling the parametric program (6.9) itself.

The revised version of it is omitted.

Example 6.3.1 Solve the following problem by Algorithm 6.3.1:

$$\begin{aligned} \min \quad & -2x_1 - 3x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 + x_4 = 2 \\ & -2x_1 - x_2 + x_3 = -1 \\ & -3x_1 + 4x_2 + x_5 = -3 \\ & x_j \geq 0, \quad j = 1, \dots, 4. \end{aligned}$$

Answer Put the problem into the following tableau with the costs and the right-hand side both parameterized

x_1	x_2	x_3	x_4	x_5	RHS
1	2		1		2
-2	-1	1			$-1 + \theta$
-3	4*			1	$-3 + \theta$
$-2 + \theta$	$-3 + \theta$				

Given $\theta_2 = 4 > 0$.

Iteration 1:

1. $\alpha = \max\{-(-2)/1, -(-3)/1\} = 3, \quad q = 2$.
2. $\beta = \max\{-(-1)/1, -(-3)/1\} = 3, \quad p = 3$.
3. $\alpha \geq \beta$.
- (3) $\min\{(-3 + \theta)/4\} = (-3 + \theta)/4, \quad p = 3$, where θ is close to 4.

4. Set $\theta_2 = 3$.
5. Taking $q = 2$, $p = 3$, the associated basis change leads to

x_1	x_2	x_3	x_4	x_5	RHS
$5/2$			1	$-1/2$	$7/2 - 1/2\theta$
$-11/4$		1		$1/4$	$-7/4 + 5/4\theta$
$-3/4^*$	1			$1/4$	$-3/4 + 1/4\theta$
$-17/4 + 7/4\theta$			$3/4 - 1/4\theta$		$-9/4 + 3/2\theta - 1/4\theta^2$

Iteration 2:

1. $\alpha = \max\{-(-17/4)/(7/4)\} = 17/7$, $q = 1$.
2. $\beta = \max\{-(-7/4)/(5/4), -(-3/4)/(1/4)\} = 3$, $p = 3$.
3. $\alpha \not\geq \beta$.
- (6) $\min\{-(-17/4 + 7/4\theta)/(-3/4)\} = -17/3 + 7/3\theta$, $q = 1$, where θ is close to 3.
4. Set $\theta_2 = 3$ (a degenerate step).
5. Taking $p = 3$, $q = 1$, then the corresponding basis change leads to

x_1	x_2	x_3	x_4	x_5	RHS
	$10/3^*$		1	$1/3$	$1 + 1/3\theta$
	$-11/3$	1		$-2/3$	$1 + 1/3\theta$
1	$-4/3$			$-1/3$	$1 - 1/3\theta$
	$-17/3 + 7/3\theta$		$-2/3 + 1/3\theta$		$2 - 5/3\theta + 1/3\theta^2$

Iteration 3:

1. $\alpha = \max\{-(-17/3)/(7/3), -(-2/3)/(1/3)\} = 17/7$, $q = 2$.
2. $\beta = \max\{-1/(1/3), -1/(1/3)\} = -3$, $p = 1$.
3. $\alpha > \beta$.
- (3) $\min\{(1 + 1/3\theta)/(10/3)\}$, $p = 1$, where θ is close to 3.
4. Set $\theta_2 = 17/7$.
5. Taking $q = 2$, $p = 1$, the associated basis change leads to

x_1	x_2	x_3	x_4	x_5	RHS
	1		$3/10$	$1/10^*$	$3/10 + 1/10\theta$
		1	$11/10$	$-3/10$	$21/10 + 7/10\theta$
1			$2/5$	$-1/5$	$7/5 - 1/5\theta$
		$17/10 - 7/10\theta$	$-1/10 + 1/10\theta$		$37/10 - 9/5\theta + 1/10\theta^2$

Iteration 4:

1. $\alpha = \max\{-(-1/10)/(1/10)\} = 1$, $q = 5$.

2. $\beta = \max\{-(3/10)/(1/10), -(21/10)/(7/10)\} = -3, p = 1.$
3. $\alpha > \beta.$
- (3) $\min\{(3/10 + 1/10\theta)/(10/3)\}, p = 1,$ where θ is close to $17/7.$
4. Set $\theta_2 = 1.$
5. Taking $q = 5, p = 1$ as the pivot, the associated basis change leads to

x_1	x_2	x_3	x_4	x_5	RHS
	10		3	1	$3 + \theta$
	3	1	2		$3 + \theta$
1	2		1		2
	$1 - \theta$		$2 - \theta$		$4 - 2\theta$

Iteration 5:

1. $\alpha = 0.$
2. $\beta = \max\{-3/1, -3/1\} = -3, p = 1.$
3. $\alpha > \beta.$
- (1) $\theta_2 = 0.$ The optimal basic solution and associated objective value:

$$x^* = (2, 0, 3, 0, 3)^T, \quad f^* = -4.$$

6.4 Criss-Cross Algorithm Using Most-Obtuse-Angle Rule

Zionts (1969) proposed the so-called *criss-cross algorithm*, which carries out the primal and dual simplex iterations alternately. A defect of such doing seems that while performing the primal iteration, the reduced costs would already be close to nonnegative, but the negative components of the right-hand side are still large in magnitude; in other words, the dual iteration should be more favorable to carry out, and vice versa.

The criss-cross algorithm (Yan and Pan 2009) utilizes the most-obtuse-angle column and row rules (Part II) in the primal and dual iterations, respectively. Therefore, no minimum-ratio test is involved. It does not mechanically perform primal and dual iterations alternately, but instead, performs the primal or dual iteration according to the current degree of the primal and dual infeasibility. To do so, a predetermined threshold value is used to control switching between the two sides, so that the primal (dual) iteration is taken if the primal (dual) infeasibility is stronger than the dual (primal) infeasibility in a sense.

A slight variant of the algorithm is formulated as follows. Involving no threshold, the variant solves the LP problem in a single phase.

Algorithm 6.4.1 (Most-Obtuse-Angle Criss-Cross Algorithm: Tableau Form)

Initial: the simplex tableau of form (1.8). This algorithm solves the standard LP problem.

1. Determine column index $q \in \arg \min_{j \in N} \bar{z}_j$.
2. Determine row index $p \in \arg \min\{\bar{b}_i \mid i = 1, \dots, m\}$.
3. Compute $\mu = \min\{\bar{z}_q, \bar{b}_p\}$.
4. Stop if $\mu \geq 0$ (optimality achieved).
5. If $\mu = \bar{z}_q$, then:
 - (1) Redetermine row index $p \in \arg \max\{\bar{a}_{iq} \mid i = 1, \dots, m\}$;
 - (2) Stop if $\bar{a}_{pq} \leq 0$ (infeasible or unbounded problem);
else
 - (3) Redetermine column index $q \in \arg \min_{j \in N} \bar{a}_{pj}$;
 - (4) Stop if $\bar{a}_{pq} \geq 0$ (infeasible problem).
6. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
7. Go to step 1.

Note As step 3 involves a comparison between the right-hand side and reduced costs, it is important to scale the original data beforehand. If the data are not well-scaled, it would be better to set a suitable threshold value $\tau > 0$, and use

$$\mu = \min\{\bar{z}_q, \tau \bar{b}_p\}$$

in step 3 instead.

For simplicity of description, we used the classical prima pivot rule in step 1 and the dual pivot rule in step 2 in the preceding Algorithm. In practice, other rules, introduced in Part II of this book, should be more favorable.

The following is the revised version of Algorithm 6.4.1.

Algorithm 6.4.2 (The Most-Obtuse-Angle Criss-Cross Algorithm) Given $\tau > 0$. initial : (B, N) ; B^{-1} ; $\bar{x}_B = B^{-1}b$, $\bar{z}_N = c_N - N^T B^{-T} c_B$ and $\bar{f} = c_B^T \bar{x}_B$. This algorithm solves the standard LP problem.

1. Determine $q \in \arg \min_{j \in N} \bar{z}_j$.
2. Determine $p \in \arg \min\{\bar{x}_{j_i} \mid i = 1, \dots, m\}$.
3. Compute $\mu = \min\{\bar{x}_{j_p}, \bar{z}_q\}$.
4. Stop if $\mu \geq 0$ (optimality achieved).
5. If $\mu = \bar{z}_q$, then:
 - (1) Compute $\bar{a}_q = B^{-1} a_q$.
 - (2) Stop if $\bar{a}_q \leq 0$, (infeasible or unbounded problem).
 - (3) Redetermine row index

$$p \in \arg \min\{\bar{x}_{j_i} \mid \bar{a}_{iq} \geq \tau \theta, i = 1, \dots, m\}, \text{ where } \theta = \max\{\bar{a}_{iq} \mid i = 1, \dots, m\} > 0.$$

- else
- (4) Compute $\sigma_N = N^T B^{-T} e_p$.
 - (5) Stop if $\sigma_N \geq 0$ (infeasible problem).
 - (6) Redetermine column index
 $q \in \arg \min\{\bar{z}_j \mid \bar{a}_{pj} \leq \tau\theta, j \in N\}$, where $\theta = \min_{j \in N} \bar{a}_{pj} < 0$.
 7. Update $\bar{x}_B = \bar{x}_B + \alpha \bar{a}_q$, and set $\bar{x}_q = \alpha$, $\bar{f} = \bar{f} - \alpha \bar{z}_q$, where $\alpha = -\bar{x}_{j_p} / \bar{a}_{pq}$.
 8. Update $\bar{z}_N = \bar{z}_N + \beta \sigma_N$, $\bar{z}_{j_p} = \beta$, $\bar{f} = \bar{f} - \beta \bar{x}_{j_p}$, where $\beta = -\bar{z}_q / \bar{a}_{pq}$.
 9. Update B^{-1} .
 10. Update (B, N) by exchanging j_p and q .
 11. Go to step 1.

The computational experiments, reported in Yan and Pan (2009), should well reflect the performance of Algorithm 6.4.2 although still preliminary.

Only the 25 smallest standard NETLIB problems without bounds and ranges were tested (Appendix C: Table C.4: Problems AFIRO-SCAGR25). Programmed in Fortran without exploiting sparsity, the criss-cross code was compared with MINOS 5.51 in terms of the number of iterations required:

The code outperformed MINOS 5.51 with a total iteration ratio of 1.24.

Since there was no minimum-ratio test involved in each iteration, the CPU time margin between it and the simplex algorithm should be even larger.

Finally, we bring up the following example to demonstrate the solution process.

Example 6.4.1 Solve the following problem by Algorithm 6.4.1:

$$\begin{aligned}
 \min \quad & f = -2x_1 - x_2, \\
 \text{s.t.} \quad & x_1 - x_2 + x_3 = 2, \\
 & x_1 + 2x_2 + x_4 = 8, \\
 & -x_1 - x_2 + x_5 = -3, \\
 & x_j \geq 0, \quad j = 1, \dots, 5.
 \end{aligned}$$

Answer Initial simplex tableau:

x_1	x_2	x_3	x_4	x_5	RHS
1	-1	1			2
1	2		1		8
-1*	-1			1	-3
-2	-1				

Iteration 1:

1. $\min\{-2, -1\} = -2 < 0$, $q = 1$.
2. $\min\{2, 8, -3\} = -3 < 0$, $p = 3$.
3. $\mu = \min\{-2, -3\} = -3$.

5. Take $p = 3$, and carry out the dual iteration.

$$5(3) \min\{-1, -1\} = -1, q = 1.$$

6. Multiply row 3 by -1 , and then add $-1, -1, 2$ times of row 3 to rows 1,2,4, respectively:

x_1	x_2	x_3	x_4	x_5	RHS
	-2	1		1*	-1
	1		1	1	5
1	1			-1	3
	1			-2	6

Iteration 2:

1. $\min\{1, -2\} = -2 < 0, q = 5.$

2. $\min\{-1, 5, 3\} = -1 < 0, p = 1.$

3. $\mu = \min\{-2, -1\} = -2.$

5. Take $q = 5$, and carry out the primal iteration.

$$5(1) \max\{1, 1, -1\} = 1, p = 1.$$

6. Add $-1, 1, 2$ times of row 1 to rows 2,3,4, respectively:

x_1	x_2	x_3	x_4	x_5	RHS
	-2	1		1	-1
	3*	-1	1		6
1	-1	1			2
	-3	2			4

Iteration 3:

1. $\min\{-3, 2\} = -3 < 0, q = 2.$

2. $\min\{-1, 6, 2\} = -1 < 0, p = 1.$

3. $\mu = \min\{-3, -1\} = -3.$

5. Take $q = 2$, and carry out the primal iteration.

$$5(1) \max\{-2, 3, -1\} = 3, p = 2.$$

6. Multiply row 2 by $1/3$, and then add $2; 1; 3$ times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	RHS
		1/3	2/3	1	3
	1	-1/3	1/3		2
1		2/3	1/3		4
		1	1		10

The preceding simplex tableau is optimal. The basic optimal solution and optimal value:

$$x^* = (4, 2, 0, 0, 3)^T, \quad f^* = -10.$$

6.5 Perturbation Primal-Dual Simplex Algorithm

Although degeneracy hardly causes cycling in practice, it often leads to stalling, and degrades the efficiency of the simplex method. The perturbation method aims to eliminate this defect (Pan 1999, 2000).

This method starts from any basis, which is usually neither primal nor dual feasible. It performs the normal primal (dual) algorithm but with the negative components of each current primal (dual) solution perturbed to some small positive numbers. Actually, it solves another problem with its right-hand side b (reduced cost c) modified slightly, and finally recovers and solves the original problem with the dual (primal) simplex algorithm. If the changes are small enough, there will be no need for performing the latter simplex algorithm.

There are the primal-dual and dual-primal schemes. In this section, we demonstrate the former only. The overall steps are put into the following algorithm.

Algorithm 6.5.1 (Perturbation Simplex Algorithm 1: Tableau Form) Initial: a simplex tableau of form (1.8), perturbation parameters $\delta_j \geq 0$, $j = 1, \dots, n$. This algorithm solves the standard LP problem.

1. Call the simplex Algorithm 3.1.1: Before each minimum-ratio test, perturb the right-hand side:

$$\bar{b}_i = \delta_{j_i}, \quad i \in \{i \mid \bar{b}_i < \delta_{j_i}, \quad i = 1, \dots, m\}.$$
2. Stop if returning from step 3 (the problem is lower unbounded).
3. If returning from step 2, then:
 - (1) Reset $\bar{b} = B^{-1}b$.
 - (2) Stop if $\bar{b} \geq 0$ (optimality achieved).
4. Call the dual simplex Algorithm 5.4.1: Before each minimum-ratio test, perturb the reduced cost:

$$\bar{z}_j = \delta_j, \quad j \in \{j \in N \mid \bar{z}_j < \delta_j\}.$$
5. Stop if returning from step 3 (the problem is infeasible).
6. If returning from step 2, then
 - (1) Reset $\bar{z}_N = c_N - N^T B^{-1} c_B$.
 - (2) Stop if $\bar{z}_N \geq 0$ (optimality achieved).
7. Go to step 1.

The resetting of the right-hand side in step 1 is to ensure primal feasibility as well as nondegeneracy in primal simplex iterations. At the beginning of step 4, dual feasibility is already achieved, and the resetting of the reduced cost is to ensure the dual nondegeneracy in dual simplex iterations, as might be suitable for solving large-scale and highly degenerate problems. In our preliminary tests, there were few switches between primal and dual iterations required.

The finiteness of the algorithm is not guaranteed since the objective value does not strictly monotonically change in the solution process. Based on our experience, however, we expect that it is more difficult to construct cycling instances to the perturbation algorithm, if possible. There should be a less possibility for stalling, and the degeneracy should never be a problem.

Preliminary computational experiments were done with the preceding Algorithm vs the classical two-phase simplex algorithm. There were more than 100 small test problems involved, each with no more than tens of inequality constraints and decision variables. In the tests, our private classical code requires the iterations about 2–3 times of those with the code based on Algorithms 6.5.1 (Pan, 1999, 2000). It appears that the perturbation algorithm is significantly efficient, compared with the classical simplex algorithm, at least for small problems.

We describe the following simple variant of the preceding algorithm, where only the corresponding perturbation parameters are used in the primal (dual) minimum-ratio test, whereas the right-hand side (reduced cost) itself is untouched.

Algorithm 6.5.2 (Perturbation Simplex Algorithm 2: Tableau Form) Initial: a simplex tableau of form (1.8), perturbation parameters $\delta_j \geq 0$, $j = 1, \dots, n$. This algorithm solves the standard LP problem.

1. Call the simplex Algorithm 3.1.1 but use δ_{j_i} instead of $\bar{b}_i$ in the minimum-ratio test (3.11) if $\bar{b}_i < \delta_{j_i}$, $i = 1, \dots, m$.
2. Stop if returning from step 3. (The problem is lower unbounded.)
3. Stop if $\bar{b} \geq 0$, if returning from step 2. (optimality achieved)
4. Call the dual simplex Algorithm 5.4.1 but use δ_j instead of $\bar{z}_j$ in the minimum-ratio test (5.25) if $\bar{z}_j < \delta_j$ $j \in N$.
5. Stop if returning from step 3 (infeasible problem).
6. Assume returning from step 2. Stop if $\bar{z}_N \geq 0$ (optimality achieved).
7. Go to step 1.

Example 6.5.1 Solve following problem by Algorithm 6.5.2:

$$\begin{aligned}
 \min \quad & f = -2x_1 - 5x_2 + 4x_3, \\
 \text{s.t.} \quad & -2x_1 + 3x_2 - 5x_3 + x_4 = -17, \\
 & -4x_1 + 8x_2 - 2x_3 + x_5 = 1, \\
 & 7x_1 - 4x_2 - x_3 + x_6 = -6, \\
 & x_j \geq 0, \quad j = 1, \dots, 6.
 \end{aligned}$$

Answer Set $\delta_j = 1$, $j = 1, \dots, 6$. Initial simplex tableau:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-2	3	-5	1			-17
-4	8*	-2		1		1
7	-4	-1			1	-6
-2	-5	4				

1. Call simplex Algorithm 3.1.1.

Iteration 1:

1. $\min\{-2, -5, 4\} = -5 < 0$, $q = 2$.
4. In the minimum-ratio test, $\bar{b}_1$ is replaced by 1 ($\bar{b}_1 = -17 < 1$):
 $\min\{1/3, 1/8\} = 1/8$, $p = 2$.
5. Multiply row 2 by $1/8$, and then add $-3, 4, 5$ times of row 2 to rows 1,3,4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-1/2		-17/4	1	-3/8		-139/8
-1/2	1	-1/4		1/8		1/8
5*		-2		1/2	1	-11/2
-9/2		11/4		5/8		5/8

Iteration 2:

1. $\min\{-9/2, 11/4, 5/8\} = -9/2 < 0$, $q = 1$.
4. $\bar{b}_3 = -11/2 < 1$ is replaced by 1: $\min\{1/5\} = 1/5$, $p = 3$.
5. Multiply row 3 by $1/5$, and then add $1/2; 1/2; 9/2$ times of row 3 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
		-89/20*	1	-13/40	1/10	-717/40
	1	-9/20		7/40	1/10	-17/40
1		-2/5		1/10	1/5	-11/10
		19/20		43/40	9/10	-173/40

1. $\min\{19/20, 43/40, 9/10\} \geq 0$,
3. Returned from step 2.
4. Call dual simplex Algorithm 5.4.1.

Iteration 3:

1. $\min\{-717/40, -17/40, -11/10\} = -717/40 < 0$, $p = 1$.

4. $\min\{-(19/20)/(-89/20), -(43/40)/(-13/40)\}$
 $= -(19/20)/(-89/20) = 19/89, q = 3.$
5. Multiply row 1 by $-20/89$, and then add $9/20; 2/5; -19/20$ times of row 1 to rows 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
		1	$-20/89$	$13/178$	$-2/89$	$717/178$
	1		$-9/89$	$37/178$	$8/89$	$247/178$
1			$-8/89$	$23/178$	$17/89$	$91/178$
			$19/89$	$179/178$	$82/89$	$-1451/178$

Iteration 4:

1. $\min\{717/178, 247/178, 91/178\} \geq 0.$
6. Returned from step 2. $\bar{z}_N = (19/89, 179/178, 82/89)^T \geq 0.$

The basic optimal solution and optimal value:

$$x^* = (91/178, 247/178, 717/178, 0, 0, 0)^T, \quad f^* = 1451/178.$$

It is noted that the preceding problem was solved by calling the primal and dual simplex algorithm once each.

Although the perturbation parameters δ_j were set to 1, it seems to be favorable to take different values, no more than 10^{-3} .

We formulate the revised version of Algorithm 6.5.2 as follows.

Algorithm 6.5.3 (Perturbation Simplex Algorithm 2) Given perturbation parameter $\delta_j \geq 0, j = 1, \dots, n$. Initial: $(B, N), B^{-1}, \bar{x}_B = B^{-1}b, \bar{z}_N = c_N - N^T B^{-T} c_B$. This algorithm solves the standard LP problem.

1. Call simplex Algorithm 3.3.1: For the minimum-ratio test, use δ_{j_i} in place of $\bar{x}_{j_i}$ if
 $\bar{x}_{j_i} < \delta_{j_i}, i = 1, \dots, m.$
2. Stop if returning from step 5 (infeasible or unbounded problem).
3. Stop if $\bar{x}_B \geq 0$, when returning from step 3. (optimality achieved).
4. Call the dual simplex Algorithm 3.3.2: For the minimum-ratio test, use δ_j in place of $\bar{z}_j$ if
 $\bar{z}_j < \delta_j, j \in N.$
5. Stop if returning from step 4 (infeasible problem).
6. Stop if $\bar{z}_N \geq 0$, when returning from step 2. (optimality achieved).
7. Go to step 1.

6.6 Notes on Criss-Cross Simplex Algorithm

Talacko and Rockafellar (1960), Balinski and Gomory (1963), and Llewellyn (1964) first suggested switching between the primal and dual iterations under a certain mechanism, but not supported by numerical results.

Later, Zionts (1969) proposed a so-called *criss-cross algorithm*, carrying out the primal and dual simplex iterations alternately. In the primal iteration, the pivot is selected by the primal rule, and the associated basis change is made; in the dual iteration, the pivot is selected by the dual rule, then the associated basis change is made. In the case when the minimum ratio tends to infinity in the row (column) selection, it turns to the dual (primal) iteration, and so on. If no iteration can be carried out, it is asserted that there exists no optimal solution; otherwise, if primal (dual) feasibility is achieved first, then the primal (dual) simplex algorithm is executed subsequently to achieve optimality. Therefore, the algorithm is actually a primal or dual Phase-I method, depending on which type of feasibility is attained first. Like the standard simplex algorithm, the finiteness of Zionts algorithm is not guaranteed.

A decade later or more, Terlaky (1985), and Wang (1987) proposed a purely combinatorial criss-cross variant independently. Their algorithm performs the primal or dual iteration according to the following rule: It determines the minimum among indices, associated with all negative components of $\bar{x}_B$ and $\bar{z}_N$. If the obtained index is associated with $\bar{z}_N$ ($\bar{x}_B$), then the primal (dual) iteration is carried out. To show the finiteness, the algorithm uses a criteria simpler than that of Zionts algorithm to detect the nonexistence of an optimal solution.

However, the performance of all these algorithms cannot compete with the simplex algorithm.

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Chapter 7

Sensitivity Analysis and Parametric LP



After solving an LP problem by the simplex method, there is often the need for solving a new problem, yielding from changing some part of the old problem. This chapter discusses how to deal with such problems effectively, as changing cost, the right-hand side, and coefficient matrix, and adding/dropping variable/constraint, and replacing a row/column. This theme, termed *sensitivity analysis*, has important applications in integer and/or mixed-integer linear programming. In addition, we introduce the closely related parametric LP, such as parameterizing the costs c and/or the right-hand side b .

The basic strategy used is the so-called *warm start*. In fact, even for a new problem differing significantly from its predecessor, it is often a shortcut to start from the final basis obtained. Indeed, it is conceivable that the optimal basis even remains unchanged if the original problem changes within a sufficiently small range.

To begin with, assume that the simplex tableau (3.6) is optimal, associated with the pair of primal and dual optimal solutions below:

$$\begin{aligned}\bar{x}_B = \bar{b} = B^{-1}b \geq 0, \quad \bar{x}_N = 0; \\ \bar{z}_B = 0, \quad \bar{z}_N = c_N - N^T \bar{y} \geq 0, \quad \bar{y} = B^{-T} c_B.\end{aligned}\tag{7.1}$$

7.1 Change in Cost

Assume that the cost c is changed to c' .

For the resulting problem, the primal and dual basic solutions corresponding to the basis B are respectively

$$x'_B = B^{-1}b = \bar{x}_B \geq 0, \quad z'_N = c'_N - N^T y', \quad y' = B^{-T} c'_B.$$

As the primal basic solution does not change at all, primal feasibility remains. If $z'_N \geq 0$, therefore, the pair (7.1) is just optimal to the new problem.

Now assume that $z'_N < 0$. In case when changes in costs are slight, corresponding changes in the dual basic solution should be tiny too; therefore, it could be expected that the number of required iterations for solving the new problem is low if starting from B . In particular, changes in nonbasic costs only cause changes in the associated reduced costs. Specifically, assume that

$$c'_j = c_j + \Delta c_j, \quad j \in T \subset N.$$

In this case, it is clear that $y' = \bar{y}$, and hence only the following reduced costs need to be computed:

$$z'_j = c'_j - a_j^T \bar{y} = \bar{z}_j + \Delta c_j, \quad j \in T.$$

If they are all nonnegative, then optimality is already achieved; otherwise, the simplex algorithm is carried out from this point.

In some applications, one needs to determine a change range to maintain the optimality of B with changes in costs. More precisely, assume that

$$c' = c + \tau \Delta c, \quad (7.2)$$

where $\Delta c \neq 0$ is a given vector and τ is a scalar to be determined.

It is clear that B is an optimal basis to the resulting problem if the new reduced costs are nonnegative, i.e.,

$$z'_N = c_N + \tau \Delta c_N - N^T B^{-T} (c_B + \tau \Delta c_B) = \bar{z}_N + \tau \Delta z_N \geq 0, \quad (7.3)$$

where

$$\Delta z_N = \Delta c_N - N^T B^{-T} \Delta c_B. \quad (7.4)$$

Introducing

$$\begin{aligned} \beta &= \min\{-\bar{z}_j / \Delta z_j \mid \Delta z_j < 0, j \in N\}, \\ \alpha &= \min\{\bar{z}_j / \Delta z_j \mid \Delta z_j > 0, j \in N\}, \end{aligned}$$

then the nonnegative values of τ , satisfying inequality (7.3), are determined by

$$\tau \leq \beta,$$

while the nonpositive values of τ satisfy

$$|\tau| \leq \alpha.$$

Consequently, it follows that the optimal basis/solution remains optimal whenever τ satisfies

$$-\alpha \leq \tau \leq \beta. \quad (7.5)$$

Example 7.1.1 Find the change range of the cost, corresponding to x_1 , such that the optimal solution remains optimal:

$$\begin{aligned} \min \quad & f = -4x_1 - 3x_2 - 5x_3, \\ \text{s.t.} \quad & 2x_1 + x_2 + 3x_3 + x_5 = 15, \\ & x_1 + x_2 + x_3 + x_4 = 12, \\ & -2x_1 + x_2 - 3x_3 + x_7 = 3, \\ & 2x_1 + x_2 + x_6 = 9, \\ & x_j \geq 0, \quad j = 1, \dots, 7. \end{aligned}$$

Answer Example 3.1.1 already gives an optimal simplex tableau to this problem, i.e.,

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		1		1/3	-1/3		2
			1	-7/12	-1/6	-1/4	1
	1			1/2		1/2	9
1				-1/4	1/2	-1/4	
				13/6	1/3	1/2	37

The optimal basis and nonbasis are $B = \{3, 4, 2, 1\}$, $N = \{5, 6, 7\}$, corresponding to optimal solution and objective value $x^* = (0, 9, 2, 1, 0, 0, 0)^T$, $f^* = -37$, $c^* = (-4, -3, -5, 0, 0, 0, 0)^T$, $z_N^* = (13/6, 1/3, 1/2)^T$.

Setting $\Delta c = (1, 0, 0, 0, 0, 0, 0)^T$, then $\Delta c_B = (0, 0, 0, 1)^T$, $\Delta c_N = (0, 0, 0)^T$. By (7.4), it holds that

$$\Delta z_N = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}^T \begin{pmatrix} 3 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 \\ -3 & 0 & 1 & -2 \\ 0 & 0 & 1 & 2 \end{pmatrix}^{-T} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/4 \\ -1/2 \\ 1/4 \end{pmatrix}.$$

$$\alpha = \min\{(13/6)/(1/4), (1/2)/(1/4)\} = 2, \quad \beta = \min\{-(1/3)/(-1/2)\} = 2/3.$$

By (7.5), the optimal solution remains optimal for

$$-2 \leq \tau \leq 2/3.$$

Further, from (7.2) and $\Delta c = (1, 0, 0, 0, 0, 0, 0)^T$, it is known that the optimal solution remains optimal while the cost c_1 of x_1 satisfies

$$-4 - 2 = -6 \leq c'_1 \leq -4 + 2/3 = -10/3.$$

For instance, when the cost of x_1 is set to the lower bound -6 , it holds by (7.3) that

$$z'_N = (13/6, 1/3, 1/2)^T + (-2)(1/4, -1/2, 1/4)^T = (5/3, 4/3, 0)^T \geq 0,$$

and the optimal simplex tableau is then

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		1		1/3	-1/3		2
			1	-7/12	-1/6	-1/4	1
	1			1/2		1/2	9
1				-1/4	1/2	-1/4	
				5/3	4/3		37

7.2 Change in Right-Hand Side

Let the right-hand vector b become

$$b' = b + \eta \Delta b, \quad (7.6)$$

where η is a scalar, and Δb is a given vector.

Corresponding to basis B , the primal and dual basic solutions of the resulting problem are

$$x'_B = B^{-1}(b + \eta \Delta b) = \bar{x}_B + \eta B^{-1} \Delta b, \quad z'_N = c_N - N^T \bar{y} = \bar{z}_N \geq 0, \quad \bar{y} = B^{-T} c_B. \quad (7.7)$$

It is clear that the dual solution remains unchanged, so does the dual feasibility. If $x'_B \geq 0$, then the primal solution is also feasible, and hence the two are just a pair of primal and dual optimal solutions for the resulting problem.

Now assume that $x'_B \not\geq 0$. When $|\eta|$ is small, the change in x'_B is small too, it can be expected to solve the resulting problem quickly by the dual simplex method, starting from B . In fact, B is an optimal basis for the resulting problem if η satisfies the following condition:

$$\bar{x}_B + \eta v \geq 0, \quad v = B^{-1} \Delta b, \quad (7.8)$$

The nonpositive values of η , satisfying the preceding inequality, are determined by

$$|\eta| \leq \alpha = \min\{\bar{x}_i/v_i \mid v_i > 0, i = 1, \dots, m\},$$

while nonnegative values of η are determined by

$$\eta \leq \beta = \min\{-\bar{x}_i/v_i \mid v_i < 0, i = 1, \dots, m\}.$$

Therefore, (7.7) gives primal and dual basic optimal solutions of the new problem whenever η satisfies

$$-\alpha \leq \eta \leq \beta.$$

Example 7.2.1 Find the change range of the first component of the right-hand side such that the optimal basis remains optimal:

$$\begin{aligned} \min f &= x_1 + 2x_2 + x_3, \\ \text{s.t. } -2x_1 - x_2 - x_3 + x_4 &= -1, \\ x_1 - 4x_2 - x_3 + x_5 &= -2, \\ x_1 + 3x_2 + x_6 &= 4, \\ x_j &\geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Answer Example 5.4.1 gives the following optimal simplex tableau to the preceding problem:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		1/3	-4/9	1/9		2/9
	1	1/3	-1/9	-2/9		5/9
		-4/3	7/9	5/9	1	19/9
			2/3	1/3		-4/3

which corresponds to the optimal solution and objective value $x^* = (2/9, 5/9, 0, 0, 0, 19/9)^T$, $f^* = 4/3$.

The optimal basis and nonbasis are $B = \{1, 2, 6\}$, $N = \{3, 4, 5\}$.

Set $\Delta b = (1, 0, 0)^T$. By the second expression of (7.8), it holds that

$$v = \begin{pmatrix} -2 & -1 & 0 \\ 1 & -4 & 0 \\ 1 & 3 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -4/9 \\ -1/9 \\ 7/9 \end{pmatrix},$$

$$\alpha = \min\{(19/9)/(7/9)\} = 19/7, \quad \beta = \min\{-(2/9)/(-4/9), -(5/9)/(-1/9)\} = 1/2.$$

Therefore, the optimal solution remains optimal whenever

$$-19/7 \leq \eta \leq 1/2.$$

Further, from (7.6) and $\Delta b = (1, 0, 0)^T$, it is known that the optimal basis remains optimal whenever the first component b'_1 of the right-hand side satisfies

$$-1 - (19/7) = -26/7 \leq b'_1 \leq -1 + 1/2 = -1/2.$$

If setting $b'_1 = -1/2$, then it holds by (7.7) that

$$x'_B = (2/9, 5/9, 19/9)^T + (1/2)(-4/9, -1/9, 7/9)^T = (0, 1/2, 5/2)^T \geq 0.$$

The optimal simplex tableau to the new problem is accordingly

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		1/3	-4/9	1/9		0
	1	1/3	-1/9	-2/9		1/2
		-4/3	7/9	5/9	1	5/2
			2/3	1/3		-1

7.3 Change in Coefficient Matrix

There are several cases to be discussed separately.

7.3.1 Dropping Variable

Assume that the basic variable x_p , in row p , is to be dropped. It is clear that if the component $\bar{x}_p$ of the optimal solution is equal to zero, deleting it just gives an optimal solution to the new problem.

Now assume that $\bar{x}_p > 0$. Thus, it holds that

$$\bar{x}_p = \bar{b}_p = e_p^T B^{-1} b > 0.$$

Consider the nonbasic entries in row p , i.e.,

$$\bar{a}_{pj} = e_p^T B^{-1} a_j, \quad j \in N.$$

Proposition 7.3.1 *The new problem is infeasible if*

$$J' = \{j \in N \mid \bar{a}_{pj} > 0\} = \emptyset.$$

Proof The feasibility of the new problem implies that the original problem has a feasible solution with the p th component vanishing. Assume that the feasible solution is $\hat{x} \geq 0$, $\hat{x}_p = 0$. Then it satisfies the equality, corresponding to row p of the optimal tableau of the original problem, i.e.,

$$\hat{x}_p + \sum_{j \in N} \bar{a}_{pj} \hat{x}_j = \bar{b}_p,$$

the left side of which is less than or equal to 0, but the right side of which is strictly greater than 0, as is a contradiction. Therefore, the new problem is infeasible. $\square$

If J' is nonempty, determine column index q such that

$$q \in \arg \min_{j \in J'} \bar{z}_j / \bar{a}_{pj},$$

where $\bar{z}_j$, $j \in J'$ are the reduced costs of the optimal simplex tableau of the original problem. So $\bar{a}_{pq}$ is the pivot. Carry out elementary transformations to drop p from and enter q to the basis. Then deleting the column, corresponding to the nonbasic variable x_p , results in a dual feasible simplex tableau to the new problem.

Example 7.3.1 Consider the problem

$$\begin{aligned} \min f &= x_1 + x_2 - 3x_3, \\ \text{s.t. } -2x_1 - x_2 + 4x_3 + x_5 &= -4, \\ x_1 - 2x_2 + x_3 + x_6 &= 5, \\ -x_1 + 2x_3 + x_4 &= -3, \\ x_j &\geq 0, \quad j = 1, \dots, 6. \end{aligned} \tag{7.9}$$

to which there is the optimal simplex tableau below:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	-4/3		-1/3		2/3	13/3
	-2/3	1	1/3		1/3	2/3
	-1		-2	1		2
	1/3		4/3		1/3	-7/3

Solve the problem, resulting from (7.9) by dropping variable x_1 .

Answer From the optimal tableau, it is known that $\bar{x}_1 = 13/3 > 0$, $J' = \{6\} \neq \emptyset$, $\min\{(1/3)/(2/3)\} = 1/2$, $q = 6$.

The following dual feasible simplex tableau is obtained by multiplying row 1 by $3/2$, and then adding $-1/3$, $-1/3$ times of row 1 to rows 2,4, respectively:

x_2	x_3	x_4	x_5	x_6	RHS
-2		-1/2		1	13/2
0	1	1/2			-3/2
-1		-2	1		2
1		3/2			-9/2

Call dual simplex Algorithm 5.4.1.

Iteration 1:

1. $\min\{13/2, -3/2, 2\} = -3/2 < 0$, $p = 2$.
3. $J = \emptyset$, detecting dual unboundedness. Therefore, the new problem is infeasible.

7.3.2 Adding Variable

Assume that the variable x_{n+1} is to be added. Let c_{n+1} and a_{n+1} be the associated cost and column, respectively. The new problem is then

$$\begin{aligned} \min \quad & c^T x + c_{n+1} x_{n+1}, \\ \text{s.t.} \quad & Ax + a_{n+1} x_{n+1} = b, \quad x, x_{n+1} \geq 0. \end{aligned}$$

It is clear that B is a feasible basis to the new problem, corresponding to the feasible basic solution

$$\begin{pmatrix} \bar{x} \\ \bar{x}_{n+1} \end{pmatrix} = \begin{pmatrix} \bar{x} \\ 0 \end{pmatrix}.$$

Since the variable x_{n+1} is nonbasic, the preceding is actually a optimal basic solution if associated reduced cost

$$\bar{z}_{n+1} = c_{n+1} - a_{n+1}^T \bar{y}$$

is nonnegative. If this not the case, solve the new problem by the simplex method, starting from B .

Example 7.3.2 It is known that to problem

$$\begin{aligned} \min f &= -x_1 - 2x_2 - 4x_4 + 3x_5, \\ \text{s.t. } 2x_1 + x_3 &= 3, \\ x_1 + x_4 &= 2, \\ -x_2 + x_5 &= 0, \\ x_j &\geq 0, \quad j = 1, \dots, 5, \end{aligned}$$

there is the optimal simplex tableau below:

x_1	x_2	x_3	x_4	x_5	RHS
2		1			3
1			1		2
	-1			1	
3	1				8

Solve the problem, resulting from the problem by adding variable x_6 , with $a_6 = (1, -2, 3)^T$, $c_6 = -2$.

Answer The optimal basis for the original problem is $B = \{3, 4, 5\}$. $\bar{a}_6 = B^{-1}a_6 = (1, -2, 3)^T$, $\bar{y} = B^{-T}c_B = (0, -4, 3)^T$, $\bar{z}_6 = -2 - (1, -2, 3)(0, -4, 3)^T = -19$.

We solve the new problem by the tableau simplex algorithm. The initial simplex tableau is obtained by modifying the optimal tableau to the original problem, i.e.,

x_1	x_2	x_3	x_4	x_5	x_6	RHS
2		1			1	3
1			1		-2	2
	-1			1	3*	
3	1				-19	8

Call Algorithm 3.1.1.

Iteration 1:

1. $\min\{3, 1, -19\} = -19 < 0$, $q = 6$.
2. $I = \{1, 3\} \neq \emptyset$.
3. $\min\{3/1, 0/3\} = 0$, $p = 3$.
4. Multiply row 3 by $1/3$, then add $-1/3$; $2/3$; $19/3$ times of row 3 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
2	$1/3^*$	1		$-1/3$		3
1	$-2/3$		1	$2/3$		2
	$-1/3$			$1/3$	1	
3	$-16/3$			$19/3$		8

Iteration 2:

1. $\min\{3, -16/3, 19/3\} = -16/3 < 0$, $q = 2$,
3. $I = \{1\} \neq \emptyset$.
4. $\min\{3/(1/3)\}$, $p = 1$.
5. Multiply row 1 by 3, then add $2/3$; $1/3$; $16/3$ times of row 1 to rows 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
6	1	3		-1		9
5		2	1			8
2		1			1	3
35		16		1		56

Iteration 3:

1. $\min\{35, 16, 1\} \geq 0$. The optimal basic solution and objective value:

$$x^* = (0, 9, 0, 8, 0, 3)^T, \quad f^* = -56.$$

7.3.3 Dropping Constraint

After an LP problem is solved, it is straightforward to find whether or not a constraint is active at the optimal solution (equality constraints are active). According to Theorem 5.3.6, the optimal basis/solution remains unchanged after dropping inactive constraints, and there is nothing needed to be done in this case. We might as well assume that some active constraints are dropped.

To solve the new problem, it is advantageous to start with a basis, formed by using the old optimal basis as far as possible. We bring up the following to demonstrate this principle.

Example 7.3.3 Drop the third equality constraint from the problem

$$\begin{aligned}
 \min \quad & f = -2x_1 - x_2, \\
 \text{s.t.} \quad & x_1 - x_2 + x_3 = 2, \\
 & x_1 + 2x_2 + x_4 = 8, \\
 & -x_1 - x_2 + x_5 = -3, \\
 & -3x_1 - 4x_2 + x_4 + 3x_5 = -14, \\
 & x_j \geq 0, \quad j = 1, \dots, 5
 \end{aligned}$$

and solve the resulting problem.

Answer The tableau below is from the relaxation problem.

x_1	x_2	x_3	x_4	x_5	RHS
1	-1	1			2
1	2*		1		8
-3	-4		1	3	-14
-2	-1				

It is known from Example 7.3.5 that the final basis for the original problem is $B = \{5, 2, 1, 3\}$. Taking $\{5, 2, 3\}$ as an initial basis for the relaxation problem, we convert the tableau to a simplex one by multiplying row 2 by $1/2$, and then adding 1, 4, 1 times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	RHS
3/2		1	1/2		6
1/2	1		1/2		4
-1			3	3	2
-3/2			1/2		4

Multiplying row 3 by $1/3$ leads to

x_1	x_2	x_3	x_4	x_5	RHS
3/2*		1	1/2		6
1/2	1		1/2		4
-1/3			1	1	2/3
-3/2			1/2		4

which is a feasible simplex tableau to the relaxation problem. Call Algorithm 3.1.1. Iteration 1:

1. $\min\{-3/2, 1/2\} = -3/2$, $q = 1$.
3. $I = \{1, 2\}$.
4. $\min\{6/(3/2), 4/(1/2)\} = 4$, $p = 1$.
5. Multiply row 1 by $2/3$, then add $-1/2$, $1/3$, $3/2$ times of row 1 to rows 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	RHS
1		$2/3$	$1/3$		4
	1	$-1/3$	$1/3$		2
		$2/9$	$10/9$	1	2
		1	1		10

Optimality achieved. The optimal basic solution and associate objective value are

$$x^* = (4, 2, 0, 0, 2)^T, \quad f^* = -10.$$

7.3.4 Adding Constraint

According to Theorem 5.3.6, adding inactive constraints at an optimal solution does not matter with its optimality. As in dropping constraints, nothing needs to do in this case either. Now assume that added is an active constraint.

There are two cases:

Case 1. Adding an inequality constraint $v^T x \leq \rho$ such that $v^T \bar{x} = v_B^T \bar{x}_B > \rho$.

Introduce slack variable x_{n+1} to turn the added inequality to equality. That is,

$$v_B^T x_B + v_N^T x_N + x_{n+1} = \rho.$$

Inserting the associated data to the optimal simplex tableau to the old problem as row $m + 1$ gives the following tableau for the new problem:

x_B	x_N	x_{n+1}	f	RHS
I	$\bar{N}$			$\bar{b}$
v_B^T	v_N^T	1		ρ
	$\bar{z}_N^T$		-1	$-\bar{f}$

For $i = 1, \dots, m$, add relevant times of row i to row $m + 1$ to eliminate v_B^T . The resulting simplex tableau is of form

x_B	x_N	x_{n+1}	f	RHS
I	$\bar{N}$			$\bar{b}$
	$\bar{v}_N^T$	1		$\bar{\rho}$
	$\bar{z}_N^T$		-1	$-\bar{f}$

(7.10)

It is noted that only row $m + 1$ changed. Thus, the preceding is a dual feasible tableau to the new problem, and can be taken to get Algorithm 5.4.1 started. Note that only component $\bar{\rho} = \bar{x}_{n+1}$ of the right-hand side is negative, since the infeasible solution to the new problem, i.e.,

$$(\bar{x}, \bar{x}_{n+1} = \rho - v_B^T \bar{x}_B < 0),$$

satisfies the equality associated with row $m + 1$.

Example 7.3.4 Solve the problem, generated by adding constraint $-2x_1 + x_3 - 3x_4 + 2x_6 \leq -25$ to Example 7.3.1.

Answer Example 7.3.1 gives the optimal basic solution $\bar{x} = (13/3, 0, 2/3, 0, 2, 0)^T$, which does not satisfy the added constraint, because

$$-2 \times (13/3) + 2/3 = -24/3 > -25.$$

Introduce slack variable x_7 to the added constraint, and insert a row in accordance with the resulting equality to the optimal simplex tableau to the old problem, i.e.,

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1	-4/3		-1/3		2/3		13/3
	-2/3	1	1/3		1/3		2/3
	-1		-2	1			2
-2		1	-3		2	1	-25
	1/3		4/3		1/3		-7/3

Multiply row 1 by 2, then add it to row 4:

5. Multiply row 2 by -1 , then add it to row 4:

Now the preceding is a dual feasible simplex tableau to the new problem. Call dual simplex Algorithm 5.4.1.

Iteration 1:

1. $\min\{13/3, 2/3, 2, -17\} = -17 < 0$, $p = 4$.
3. $J = \{2, 4\} \neq \emptyset$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1	$-4/3$		$-1/3$		$2/3$		$13/3$
	$-2/3$	1	$1/3$		$1/3$		$2/3$
	-1		-2	1			2
	$-8/3$	1	$-11/3$		$10/3$	1	$-49/3$
	$1/3$		$4/3$		$1/3$		$-7/3$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1	$-4/3$		$-1/3$		$2/3$		$13/3$
	$-2/3$	1	$1/3$		$1/3$		$2/3$
	-1		-2	1			2
	-2^*		-4		3	1	-17
	$1/3$		$4/3$		$1/3$		$-7/3$

4. $\min\{-(1/3)/-2, -(4/3)/-4\} = 1/6, q = 2$.
5. Multiply row 4 by $-1/2$, then add $4/3, 2/3, 1, -1/3$ times of row 4 to rows 1, 2, 3, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1			$7/3$		$-4/3$	$-2/3$	$47/3$
		1	$5/3$		$-2/3$	$-1/3$	$19/3$
				1	$-3/2$	$-1/2$	$21/2$
	1		2		$-3/2$	$-1/2$	$17/2$
			$2/3$		$5/6$	$1/6$	$-31/6$

Iteration 2:

1. $\min\{47/3, 19/3, 21/2, 17/2\} \geq 0$. The optimal basic solution and associated objective value:

$$\bar{x} = (47/3, 17/2, 19/3, 0, 21/2, 0, 0)^T, \quad \bar{f} = 31/6.$$

Case 2. Adding equality constraint $v^T x = \rho$ such that $v^T \bar{x} \neq \rho$.

It might be well to assume

$$v^T \bar{x} < \rho. \quad (7.11)$$

Consider the auxiliary problem, obtained from adding constraint $v^T x + x_{n+1} = \rho$ to the old problem. Just as in case (i), a dual feasible simplex tableau,

say (7.10), to the auxiliary problem can be obtained. Since $\bar{x}, \bar{x}_{n+1} = \rho - v_B^T \bar{x}_B > 0$ satisfies the equality, associated with added row $m + 1$, it holds that

$$\bar{\rho} = \bar{x}_{n+1} > 0. \quad (7.12)$$

Therefore, the tableau (7.10) is optimal to the auxiliary problem.

Proposition 7.3.2 Assume that (7.11) holds and that $\bar{v}_N^T$ are nonbasics in row $m + 1$ of tableau (7.10). If

$$J' = \{j \in N \mid \bar{v}_j > 0\}$$

is empty, then the new problem is infeasible.

Proof That there is a feasible solution to the new problem implies that there is a feasible solution to the auxiliary problem, whose $(n + 1)$ -indexed component vanishes. Assume that $(\hat{x} \geq 0, \hat{x}_{n+1} = 0)$ is such a feasible solution, satisfying the equality, associated with row $m + 1$ of (7.10), i.e.,

$$\hat{x}_{n+1} + \sum_{j \in N} \bar{v}_j \hat{x}_j = \bar{\rho}.$$

Since $J' = \emptyset$ and $\hat{x}_N, \hat{x}_{n+1} \geq 0$, the left-hand side of the preceding equality is less than or equal to zero. But it is known by (7.12) that the right-hand side of the preceding is positive, as a contradiction. Therefore, the new problem is infeasible. $\square$

Assume now that J' is nonempty. Select column index q such that

$$q \in \arg \min_{j \in J'} \bar{z}_j / \bar{a}_{m+1 j}.$$

Taking $\bar{a}_{m+1 q}$ as the pivot, we manipulate tableau (7.10) by elementary transformations to drop $n + 1$ from and enter q to the basis. It is clear that deleting the column, associated with x_{n+1} , in the resulting tableau leads to a feasible simplex tableau to the new problem. Consequently, the dual simplex method can get itself start from it.

Example 7.3.5 It is known that associated to problem

$$\begin{aligned} \min f &= -2x_1 - x_2, \\ \text{s.t. } x_1 - x_2 + x_3 &= 2, \\ x_1 + 2x_2 + x_4 &= 8, \\ -x_1 - x_2 + x_5 &= -3, \\ x_j &\geq 0, \quad j = 1, \dots, 5 \end{aligned}$$

there is the following optimal tableau:

x_1	x_2	x_3	x_4	x_5	RHS
		1/3	2/3	1	3
	1	-1/3	1/3		2
1		2/3	1/3		4
		1	1		10

Solve the problem, resulting from the preceding problem by adding constraint $-3x_1 - 4x_2 + x_4 + 3x_5 = -14$.

Answer The optimal solution $\bar{x} = (4, 2, 0, 0, 3)^T$ to the preceding problem does not satisfy the added constraint, because

$$-3 \times 4 - 4 \times 2 + 3 \times 3 = -11 > -14.$$

Inserting row 4, associated with $3x_1 + 4x_2 - x_4 - 3x_5 + x_6 = 14$, to the optimal tableau gives the following tableau to the auxiliary problem:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
		1/3	2/3	1		3
	1	-1/3	1/3			2
1		2/3	1/3			4
3	4		-1	-3	1	14
		1	1			10

Executing relevant elementary transformations gives the simplex tableau below, corresponding to basis $B = \{5, 2, 1, 6\}$:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
		1/3	2/3	1		3
	1	-1/3	1/3			2
1		2/3	1/3			4
		1/3*	-4/3		1	3
		1	1			10

$J' = \{3\} \neq \emptyset$, $\min\{1/(1/3)\} = 3$, $q = 3$. Multiply row 4 by 3, then add $-1/3$; $1/3$; $-2/3$; -1 times of row 4 to rows 1, 2, 3, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
			2	1	-1	
	1		-1		1	5
1			3		-2	-2
		1	-4		3	9
			5		-3	1

Deleting the column associated with x_6 from the preceding leads to a dual feasible simplex tableau to the new problem, i.e.,

x_1	x_2	x_3	x_4	x_5	RHS
			2	1	
	1		-1		5
1			3		-2
		1	-4		9
			5		1

Call Algorithm 5.4.1.

Iteration 1:

1. $\min\{0, 5, -2, 9\} = -2 < 0$, $p = 3$.
3. $J = \emptyset$, detecting infeasibility of the new problem.

7.3.5 Replacing Row/Column

In case when a row of the constraint matrix of a problem is replaced by another, the resulting problem can be handled via the following two phases:

The first phase is to solve the problem, obtained by dropping the equality constraint corresponding to the row. The second is to solve the new problem, obtained by adding the corresponding constraint to the solved problem. It is clear that the techniques presented in the previous two sections apply.

Similarly, a problem, resulting from replacing a column of the constraint matrix, can be handled by dropping and adding the columns in two phases. In the rest of this section, a direct approach to such a problem will be presented.

Assume that column a_k is replaced by a'_k .

If $k \in N$, then the basis B , and hence the related basic solution are not affected, but the reduced costs become

$$z'_k = c_k - (a'_k)^T \bar{y}, \quad \bar{y} = B^{-T} c_B.$$

If $z'_k \geq 0$, it is clear that the optimal solution and basis remain optimal to the new problem.

If, otherwise, $k \in B$, we solve the following auxiliary problem:

$$\begin{aligned} \min \quad & x_{n+1}, \\ \text{s.t.} \quad & a_1x_1 + \cdots + a_{k-1}x_{k-1} + a'_kx_k + a_{k+1}x_{k+1} + \cdots + a_nx_n + a_kx_{n+1} = b, \\ & x, x_{n+1} \geq 0. \end{aligned} \tag{7.13}$$

The new problem has a feasible solution if and only if the optimal objective value to the auxiliary problem is equal to 0. Moreover, the optimal basis B to the old problem is feasible to the auxiliary problem and hence can be taken as an initial basis to get the simplex method started.

Example 7.3.6 From Example 4.5.1, the optimal basis $B = \{1, 2, 6\}$ is known to problem

$$\begin{aligned} \min \quad & f = x_1 + 2x_2 + x_3, \\ \text{s.t.} \quad & -2x_1 - x_2 - x_3 + x_4 = -1, \\ & x_1 - 4x_2 - x_3 + x_5 = -2, \\ & x_1 + 3x_2 + x_6 = 4, \\ & x_j \geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Solve the problem, resulting from replacing the first column of the coefficient matrix by $(1, 1, -2)^T$.

Answer The auxiliary problem of form (7.13) has the following tableau (keep the original objective row, while positioning the auxiliary objective row at the bottom).

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1	-1	-1	1			-2*	-1
1	-4	-1		1		1	-2
-2	3				1	1	4
1	2	1					
						1	

Firstly, find the simplex tableau, corresponding to basis $B = \{7, 2, 6\}$.

1. Multiply row 1 by $-1/2$, then add $-1, -1, -1$ times of row 1 to rows 2, 3, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-1/2$	$1/2$	$1/2$	$-1/2$			1	$1/2$
$3/2$	$-9/2^*$	$-3/2$	$1/2$	1			$-5/2$
$-3/2$	$5/2$	$-1/2$	$1/2$		1		$7/2$
1	2	1					
$1/2$	$-1/2$	$-1/2$	$1/2$				$-1/2$

2. Multiply row 2 by $-2/9$, then add $-1/2$, $-5/2$, -2 , $1/2$ times of row 2 to rows 1, 3, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-1/3$		$1/3^*$	$-4/9$	$1/9$		1	$2/9$
$-1/3$	1	$1/3$	$-1/9$	$-2/9$			$5/9$
$-2/3$		$-4/3$	$7/9$	$5/9$	1		$19/9$
$5/3$		$1/3$	$2/9$	$4/9$			$-10/9$
$1/3$		$-1/3$	$4/9$	$-1/9$			$-2/9$

Call Algorithm 3.1.1.

Iteration 1:

- $\min\{1/3, -1/3, 4/9, -1/9\} = -1/3 < 0$, $q = 3$,
- $I = \{1, 2\} \neq \emptyset$.
- $\min\{(2/9)/(1/3), (5/9)/(1/3)\} = 2/3$, $p = 1$.
- Multiply row 1 by 3, then add $-1/3$, $4/3$, $-1/3$, $1/3$ times of row 1 to rows 2, 3, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
-1		1	$-4/3$	$1/3$		3	$2/3$
	1		$1/3$	$-1/3$		-1	$1/3$
-2			-1	1	1	4	3
2			$2/3$	$1/3$		-1	$-4/3$
						1	

It is seen that the optimal value of the auxiliary problem vanishes. Thus, deleting x_7 column and the bottom row gives a feasible simplex tableau to the new problem, i.e.,

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-1		1	-4/3	1/3		2/3
	1		1/3	-1/3		1/3
-2			-1	1	1	3
2			2/3	1/3		-4/3

Since the objective row is nonnegative, the preceding is optimal to the new problem, with optimal basic solution and objective value as follows:

$$x^* = (0, 1/3, 2/3, 0, 0, 3)^T, \quad f^* = 4/3.$$

7.4 Parameterizing Objective Function

The LP problem involving some parameter is termed *parametric LP problem*. In this section, we consider such type of problem in which a parameter, say θ , is in the objective function in linear form.

Given a vector c' , consider the following parametric variant of the standard LP problem:

$$\begin{aligned} \min f &= (c + \theta c')^T x, \\ \text{s.t. } Ax &= b, \quad x \geq 0. \end{aligned} \tag{7.14}$$

Without loss of generality, we are assumed to solve the preceding problem for $\theta \in [0, +\infty)$; the other case $\theta \in [-\infty, 0)$ can be handled by setting $c' := -c'$ instead. It is noted that the parameter has nothing to do with feasibility of the problem.

We begin with the following example:

Example 7.4.1 Solve the following problem for parameter $\theta \in [0, +\infty)$:

$$\begin{aligned} \min & -(3 + 2\theta)x_1 - (5 - \theta)x_2 \\ \text{s.t. } & 3x_1 + 2x_2 \quad \quad \quad + x_5 = 18 \\ & \quad \quad 2x_2 \quad \quad + x_4 = 12 \\ & x_1 \quad \quad + x_3 = 4 \\ & x_j \geq 0, \quad j = 1, \dots, 5. \end{aligned}$$

Answer At first, the problem with $t = 0$ is solved with Algorithm 3.1.1. The following is the optimal simplex tableau, with parameter terms added as the bottom row afterward:

x_1	x_2	x_3	x_4	x_5	RHS
1			$-1/3$	$1/3$	2
	1		$1/2$		6
		1	$1/3$	$-1/3$	2
					36
-2θ	θ				

Eliminate parameter entries, corresponding to basis, by elementary transformations to convert the preceding into simplex tableau below:

x_1	x_2	x_3	x_4	x_5	RHS
1			$-1/3$	$1/3$	2
	1		$1/2$		6
		1	$1/3^*$	$-1/3$	2
					36
			$-7/6\theta$	$2/3\theta$	-2θ

To be optimal to the parametric problem, the nonbasic reduced costs have to be nonnegative, i.e.,

$$\bar{z}_4 = 3/2 - 7/6\theta \geq 0, \quad \bar{z}_5 = 1 + 2/3\theta \geq 0.$$

As coefficient $2/3$ of θ in the second expression is nonnegative, $\bar{z}_5$ maintains nonnegative as θ increases from 0. But coefficient $-7/6$ of θ in the first is negative, the increase of θ is restricted for $\bar{z}_4$ to be nonnegative. The maximum θ value allowed is $(3/2)/(7/6) = 9/7$. Therefore, the simplex tableau remains optimal while

$$\theta \in [0, 9/7],$$

corresponding to the same basis $B = \{1, 2, 3\}$ and optimal basic solution $\bar{x} = (2, 6, 2, 0, 0)^T$.

That $\bar{z}_4 < 0$ and $\theta > 9/7$ indicates that x_4 should enter basis; so take $q = 4$. To maintain primal feasibility, perform the minimum-ratio test:

$$\min\{2/(1/3), 6/(1/2)\} = 6, \quad p = 3.$$

Thus x_3 is determined to leave basis. Then, carrying out the corresponding elementary transformations results in

x_1	x_2	x_3	x_4	x_5	RHS
1		1		0	4
	1	-3/2		1/2*	3
		3	1	-1	6
				-9/2	5/2
				7/2 θ	-1/2 θ
					27
					5 θ

Now consider the inequalities $\bar{z}_3 = -9/2 + 7/2\theta \geq 0$, $\bar{z}_5 = 5/2 - 1/2\theta \geq 0$. The maximum value of θ is only determined by the second inequality, so that is 5. Therefore, the tableau maintains optimal for

$$\theta \in [9/7, 5],$$

corresponding to the same basis $B = \{1, 2, 4\}$ and optimal basic solution $\bar{x} = (4, 3, 0, 6, 0)^T$.

That $\bar{z}_5 < 0$ and $\theta > 5$ suggests entering x_5 to basis ($q = 5$). From minimum-ratio test $\min\{3/(1/2)\}$, $p = 2$, x_2 is determined to leave the basis. Then associated elementary transformations lead to

x_1	x_2	x_3	x_4	x_5	RHS
1	0	1			4
	2	-3		1	6
	2		1		12
		-5	3		12
		θ	2 θ		8 θ

It is clear that $\bar{z}_2 = -5 + \theta \geq 0$, $\bar{z}_3 = 3 + 2\theta \geq 0$ holds for all $\theta > 5$, corresponding to optimal basis $B = \{1, 4, 5\}$ and optimal basic solution $\bar{x} = (4, 0, 0, 12, 6)^T$. In summary, the results are put in the following table:

Range of θ	Optimal basic solution	Optimal value
$[0, 9/7]$	$(2, 6, 2, 0, 0)^T$	$-36 + 2\theta$
$(9/7, 5]$	$(4, 3, 0, 6, 0)^T$	$-27 - 5\theta$
$(5, +\infty)$	$(4, 0, 0, 12, 6)^T$	$-12 - 8\theta$

From the preceding table, it is seen that there are 3 ranges of θ , each of which corresponds to a fixed optimal basic solution and a varying optimal value in linear function of θ . It can be shown that the optimal value of the problem is a continuous piecewise linear convex function of θ , in general.

From the preceding illustration, it is not difficult to form an algorithm for solving the problem with parameterized costs. Starting from an optimal basic solution with $\theta = 0$, it uses a column rule to clarify the problem's optimality over the whole interval $(0, +\infty)$, subinterval by subinterval, from the left to the right.

Algorithm 7.4.1 *Initial: (B, N) , B^{-1} are optimal for $\theta = \theta_1 = 0$. $\bar{b} = B^{-1}b \geq 0$. This algorithm solves the parametric problem (7.14).*

1. Compute $\bar{z}_N = c_N - N^T \bar{y} \geq 0$, where $\bar{y} = B^{-T} c_B$ and $\bar{z}'_N = c'_N - N^T \bar{y}'$, where $\bar{y}' = B^{-T} c'_B$.
2. Stop if $\bar{z}'_N \geq 0$. (optimal basis with $\theta \in (\theta_1, +\infty)$)
3. Determine q and θ_2 such that

$$\theta_2 = -\bar{z}_q / \bar{c}'_q = \min\{-\bar{z}_j / \bar{z}'_j \mid \bar{z}'_j < 0, j \in N\}.$$

(optimal basis with $\theta \in (\theta_1, \theta_2]$)

4. Compute $\bar{a}_q = B^{-1} a_q$.
5. Stop if $\bar{a}_q \leq 0$ (unbounded problem with $\theta \in [\theta_1, +\infty)$).
6. Determine stepsize α and pivot row index p such that
 $\alpha = \bar{b}_p / \bar{a}_{pq} = \min\{\bar{b}_i / \bar{a}_{iq} \mid \bar{a}_{iq} > 0; i = 1, \dots, m\}.$
7. Update B^{-1} by (3.21).
8. Update (B, N) by exchanging j_p and q and set $\theta_1 = \theta_2$.
9. Compute $\bar{b} = B^{-1} b$.
10. Go to step 1.

Proofs for finiteness of the preceding algorithm under nondegeneracy and meanings of its exits are omitted.

For a better understanding, take a look at the only parameterized part, that is, the objective function

$$f = (c + \theta c')^T x,$$

which is a positive combination of $c^T x$ and $c'^T x$. As weight θ increases, the parametric problem shifts to the problem with objective $c'^T x$ and the same constraints. For large enough θ , therefore, its optimal basis coincides with the optimal basis of the normal problem, if any. If $c' = c$, in particular, the optimal basis with $\theta = 0$ is also optimal with $\theta \in [0, +\infty)$, although the corresponding optimal values are not constant.

It is seen that 3 linear systems have to be solved in each iteration of the preceding Algorithm, compared with 2 systems involved in simplex contexts. As they share the same coefficient matrix, however, systems $B^T y = c_B$ and $B^T y' = c'_B$ can be solved cheaply.

7.5 Parameterizing Right-Hand Side

Consider the following problem with parameterized right-hand side:

$$\begin{aligned} \min \quad & f = c^T x \\ \text{s.t.} \quad & Ax = b + \theta b', \quad x \geq 0. \end{aligned} \quad (7.15)$$

where $b' \in \mathcal{R}^m$, and parameter $\theta \in [0, +\infty)$. The other case of $\theta \in [-\infty, 0)$ can be handled by setting $b' = -b'$.

Again, we begin with an example.

Example 7.5.1 Solve the following problem for parameter $\theta \in [0, +\infty)$:

$$\begin{aligned} \min \quad & -2x_1 - 3x_2 \\ \text{s.t.} \quad & x_1 - x_2 + x_4 = 6 - 2\theta \\ & -x_1 + 4x_2 + x_3 = 3 \\ & -3x_1 + 2x_2 + x_5 = -2 + \theta \\ & x_j \geq 0, \quad j = 1, \dots, 4. \end{aligned}$$

Answer Put the problem into the following tableau, where a combination of the last two columns gives the right-hand side of the parametric problem, i.e.,

x_1	x_2	x_3	x_4	x_5	$\bar{b}$	θ
1	-1		1		6	-2
-1	4	1			3	
-3	2			1	-2	1
-2	-3					

The problem for $\theta = 0$ is solved by Algorithm 3.1.1, resulting in optimal simplex tableau below:

x_1	x_2	x_3	x_4	x_5	$\bar{b}$	θ
1		1/3	4/3		9	-8/3
	1	1/3	1/3		3	-2/3
		1/3	10/3	1	19	-17/3
		5/3	11/3		27	-22/3

The increase of θ should maintain nonnegativity of the right-hand side, such that

$$\bar{b}_1 = 9 - 8/3\theta \geq 0, \quad \bar{b}_2 = 3 - 2/3\theta \geq 0, \quad \bar{b}_3 = 19 - 17/3\theta \geq 0.$$

The maximum possible θ is determined by

$$\min\{-9/(-8/3), -3/(-2/3), -19/(-17/3)\} = -19/(-17/3) = 57/17, \quad p = 3.$$

Therefore, the current basis and solution are optimal for $\theta \in [0, 57/17]$.

$\bar{b}_3 < 0$ and $\theta > 57/17$ together suggest that x_5 should leave basis, so take $p = 3$. But the nonbasic entries in the third row are all nonnegative, which detects dual unboundedness or primal infeasibility of the parametric problem with $\theta \in (57/17, +\infty)$.

The results are put to the table below:

Range of θ	Optimal basic solution	Optimal value
$[0, 57/17]$	$(9 - 8/3\theta, 3 - 2/3\theta, 0, 0, 19 - 17/3\theta)^T$	$-27 + 22/3\theta$
$(57/17, +\infty)$	infeasible	—

We formulate the following algorithm for solving the parametric problem. It can be viewed as a variant of the dual simplex algorithm; but starting from an optimal basis with $\theta = 0$, it uses a different row rule.

Algorithm 7.5.1 *Initial: (B, N) , B^{-1} , $\bar{z}_N \geq 0$, $\bar{b} = B^{-1}b \geq 0$ and $\bar{b}' = B^{-1}b'$. This algorithm solves the parametric problem (7.15).*

1. Stop if $\bar{b}' \geq 0$. (optimal basis with $\theta \in (\theta_1, +\infty)$)
2. Determine p and θ_2 such that

$$\theta_2 = -\bar{b}_p/\bar{b}'_p = \min\{-\bar{b}_i/\bar{b}'_i \mid \bar{b}'_i < 0, i = 1, \dots, m\}.$$

(optimal basis with $\theta \in (\theta_1, \theta_2]$)

3. Compute $\sigma_N = N^T B^{-T} e_p$.
4. Stop if $J = \{j \in N \mid \sigma_j < 0\} = \emptyset$ (infeasible problem with $\theta \in [\theta_1, +\infty)$).
5. Determine β and column index q such that

$$\beta = -\bar{z}_q/\sigma_q = \min_{j \in J} -\bar{z}_j/\sigma_j.$$
6. Update $\bar{z}_N = \bar{z}_N + \beta\sigma_N$, $\bar{z}_{j_p} = \beta$.
7. Compute $\bar{a}_q = B^{-1}a_q$.
8. Update B^{-1} by (3.21).
9. Update (B, N) by exchanging j_p and q .
10. Compute $\bar{b} = B^{-1}b$ and $\bar{b}' = B^{-1}b'$, and set $\theta_1 = \theta_2$.
11. Go to step 1.

Discussions can be made similarly to those with Algorithm 7.4.1. For θ large enough, for instance, the parametric problem amounts to minimization of the same objective function subject to $Ax = b'$ (in terms of the same optimal basis, if any).

Although the previous two Algorithms handle parametric programs by starting from the left-hand end, 0, of the interval of θ , it is of course possible, without any essential difficulty, to start from the other end of the interval.

Chapter 8

Generalized Simplex Method



There are various problems from practice, and all can be put into the following general form:

$$\begin{aligned} \min \quad & f = c^T x, \\ \text{s.t.} \quad & a \leq Ax \leq b, \\ & l \leq x \leq u : \end{aligned} \quad (8.1)$$

where $A \in \mathcal{R}^{m \times n}$, $c, l, u \in \mathcal{R}^n$, $a, b \in \mathcal{R}^m$, $m < n$, $\text{rank } A = m$, and a, b, l, u are given vectors. Such types of problems not only have upper and lower bounds on variables, but also ranges of Ax .

Ranges involved in the problems can be eliminated by introducing new variables. Setting $w = Ax$, the preceding problem can be converted to

$$\begin{aligned} \min \quad & f = c^T x, \\ \text{s.t.} \quad & Ax - w = 0, \\ & l \leq x \leq u, \\ & a \leq w \leq b. \end{aligned} \quad (8.2)$$

Components of x are said to be *structural variables*, whereas those of w be *logical variables*.

Therefore, practical LP problems boil down to the following so-called *bounded-variable problem*:

$$\begin{aligned} \min \quad & f = c^T x, \\ \text{s.t.} \quad & Ax = b, \quad l \leq x \leq u, \end{aligned} \quad (8.3)$$

where $A \in \mathcal{R}^{m \times n}$, $c, l, u \in \mathcal{R}^n$, $b \in \mathcal{R}^m$, $\text{rank } A = m$, $m < n$. Unless indicated otherwise, it is assumed that l, u are finite, and $l_j < u_j$. Infinite upper or lower bounds can be represented by sufficiently large/small reals.

This problem can be converted to the standard form by variable transformations. However, such doing increases the problem's scales. In this chapter, we generalize the simplex and dual simplex algorithms to solve the bounded-variable problem directly. In addition, we introduce the generalized dual simplex algorithm with bound-flipping incorporated.

In the sequel, the following sign function will be useful:

$$\text{sign}(t) = \begin{cases} 1, & \text{if } t > 0, \\ -1, & \text{if } t < 0, \\ 0, & \text{if } t = 0. \end{cases} \quad (8.4)$$

Again, assume that the current basis B is as follows:

$$B = \{j_1, \dots, j_m\}, \quad N = A \setminus B. \quad (8.5)$$

8.1 Generalized Simplex Algorithm

Almost all terms for standard LP problems apply to bounded-variable problems. Similarly, a solution to $Ax = b$ is said to be basic if its nonbasic components attain one of the associated upper/lower bounds. Therefore, a basic solution is not necessarily unique, just as the basic solution in the standard LP context.

In the sequel, we state results without proof.

Lemma 8.1.1 *If there exists a feasible solution to the bounded-variable problem, so does a feasible basic solution; if there exists an optimal solution to it, so does an optimal basic solution.*

Therefore, it is possible to find an optimal basic solution in the feasible region.

Let $\bar{x}$ be a feasible basic solution, associated with B , i.e.,

$$\bar{x}_j = l_j \text{ or } u_j, \quad j \in N, \quad (8.6)$$

$$l_B \leq \bar{x}_B = B^{-1}b - B^{-1}N\bar{x}_N \leq u_B. \quad (8.7)$$

The reduced costs and objective value are then

$$\bar{z}_N = c_N - N^T B^{-T} c_B, \quad \bar{f} = c_B^T B^{-1} b + \bar{z}_N^T \bar{x}_N. \quad (8.8)$$

Define the index set

$$\Gamma = \{j \in N \mid \bar{x}_j = l_j\}, \quad \Pi = \{j \in N \mid \bar{x}_j = u_j\}. \quad (8.9)$$

Then it holds that

$$\Gamma \cup \Pi = N, \quad \Gamma \cap \Pi = \emptyset.$$

Without confusion, Γ and Π also denote submatrices consisting of corresponding columns.

Lemma 8.1.2 *A feasible solution $\bar{x}$ is optimal if the following set is empty:*

$$J = \{j \in \Gamma \mid \bar{z}_j < 0\} \cup \{j \in \Pi \mid \bar{z}_j > 0\}. \quad (8.10)$$

Proof Let x' be any feasible solution. Then it holds that

$$l_j \leq x'_j \leq u_j, \quad j \in N.$$

It is known that

$$\bar{z}_j \geq 0, \quad j \in \Gamma; \quad \bar{z}_j \leq 0, \quad j \in \Pi.$$

Hence for any $j \in N$, there are the following two cases:

(i) $j \in \Gamma$. It follows from $x'_j \geq l_j = \bar{x}_j$ that

$$\bar{z}_j x'_j \geq \bar{z}_j \bar{x}_j; \quad (8.11)$$

(ii) $j \in \Pi$. From $x'_j \leq u_j = \bar{x}_j$, (8.11) follows. Therefore

$$\sum_{j \in N} \bar{z}_j x'_j \geq \sum_{j \in N} \bar{z}_j \bar{x}_j,$$

which implies that

$$c_B^T B^{-1} b + \bar{z}_N^T x'_N \geq c_B^T B^{-1} b + \bar{z}_N^T \bar{x}_N.$$

The preceding means that the objective value at x' is no less than that at $\bar{x}$, therefore $\bar{x}$ is optimal.

□

Now assume that J is nonempty, and hence a column index q can be determined by

$$q \in \arg \max_{j \in J} |\bar{z}_j|.$$

Assuming $q = j_t$, define the vector

$$\Delta x \triangleq \begin{pmatrix} \Delta x_B \\ \Delta x_N \end{pmatrix} = \text{sign}(\bar{z}_q) \begin{pmatrix} -B^{-1}a_q \\ e_{t-m} \end{pmatrix}, \quad (8.12)$$

where e_{q-m} is the $(n-m)$ -dimensional unit vector with the $(q-m)$ th component 1.

Proposition 8.1.1 *The vector Δx satisfies*

$$A\Delta x = 0, \quad c^T \Delta x > 0.$$

Proof It is known by (8.12) that

$$A\Delta x = B\Delta x_B + N\Delta x_N = \text{sign}(\bar{z}_q)(-a_q + a_q) = 0.$$

From the first formula of (8.8) together with (8.1) and (8.12), it follows that

$$-c^T \Delta x = \text{sign}(\bar{z}_q)(a_q^T B^{-1}c_B - c_q) = -\text{sign}(\bar{z}_q)\bar{z}_q = \mu(q)\bar{z}_q = -|\bar{z}_q| < 0. \quad (8.13)$$

□

The preceding Lemma says that $-\Delta x$ is a descent direction concerning the objective $c^T x$.

Let $\alpha \geq 0$ be a stepsize from $\bar{x}$. The new iterate is then

$$\hat{x} = \bar{x} - \alpha \Delta x. \quad (8.14)$$

Since $\bar{x}$ is feasible, it holds for any $\alpha \geq 0$ that

$$A\hat{x} = A\bar{x} - \alpha[B, N]\Delta x = A\bar{x} = b.$$

The value of stepsize α should enable $\hat{x}$ to satisfy $l \leq \hat{x} \leq u$. Consequently, the maximum possible stepsize is

$$\alpha = \min\{u_q - l_q, \min\{\alpha_i \mid i = 1, \dots, m\}\}, \quad (8.15)$$

where

$$\alpha_i = \begin{cases} (\bar{x}_{j_i} - u_{j_i})/\Delta x_{j_i}, & \text{if } \Delta x_{j_i} < 0, \\ (\bar{x}_{j_i} - l_{j_i})/\Delta x_{j_i}, & \text{if } \Delta x_{j_i} > 0, \\ \infty, & \text{if } \Delta x_{j_i} = 0, \end{cases} \quad i = 1, \dots, m. \quad (8.16)$$

There are the following two cases:

- (i) $\alpha = u_q - l_q$. In this case, $\hat{x}_q = u_q$ if $\bar{x}_q = l_q$, and $\hat{x}_q = l_q$ if $\bar{x}_q = u_q$. Since new solution $\hat{x}$ is feasible basic, corresponding to the same basis, there is no need for any basis change.
- (ii) $\alpha < u_q - l_q$. Determine row index $p \in \{1, \dots, m\}$ such that

$$\alpha = \alpha_p. \quad (8.17)$$

Then $\hat{x}_{j_p}$ attains its lower bound l_{j_p} or upper bound u_{j_p} . In this case, the new basis follows from B and N by exchanging j_p and q . In addition, it is verified that $\hat{x}$ is a basic solution, corresponding to the new basis.

By (8.13) and (8.14), it is known that the new objective value is

$$\hat{f} = c^T \hat{x} = c^T \bar{x} - \alpha c^T \Delta x = c^T \bar{x} - \alpha |\bar{z}_q| \leq c^T \bar{x},$$

which strictly decreases if $\alpha > 0$. The preceding expression leads to the recurrence formula of the objective value, i.e.,

$$\hat{f} = \bar{f} - \alpha |\bar{z}_q|,$$

However, we will not use the preceding formula in each iteration, but compute the final objective value from original data and the end basis.

Definition 8.1.1 A feasible solution is degenerate if a basic component of it is on one of its bounds.

Concerning stepsize α , the following two points should be noted:

- (i) In case when a basic solution is degenerate, α would vanish, and the basic solution remains unchanged even if the basis changes.
- (ii) In practice, the problem should be deemed unbounded if the value of α exceeds some predetermined large number.

From the discussions made above, we make the following conclusions.

Lemma 8.1.3 *Let $\bar{x}$ be a feasible basic solution. Then the new solution, determined by (8.12), (8.14), (8.15) and (8.16), is a basis feasible solution. The corresponding objective value does not increase; or strictly decreases if nondegeneracy is assumed.*

The overall steps are put into the following.

Algorithm 8.1.1 (Generalized Simplex Algorithm) Initial: (B, N) , B^{-1} and associated feasible basic solution $\bar{x}$. This algorithm solves bounded-variable problem (8.3).

1. Compute $\bar{z}_N = c_N - N^T \bar{y}$, where $\bar{y} = B^{-T} c_B$.
2. Compute $\bar{f} = c^T \bar{x}$, and stop if set J defined by (8.10) is empty.
3. Select column index q such that $q \in \max_{j \in J} |\bar{z}_j|$.

4. Compute $\Delta x_B = -\text{sign}(\bar{z}_q)B^{-1}a_q$.
5. Determine stepsize α by (8.15) and (8.16).
6. Update $\bar{x}$ by (8.14) and (8.12).
7. Go to step 1 if $\alpha = u_q - l_q$; else determine row index $p \in \{1, \dots, m\}$ such that $\alpha = \alpha_p$.
8. Update B^{-1} by (3.21).
9. Update (B, N) by exchanging j_p and q .
10. Go to step 1.

Theorem 8.1.4 Algorithm 8.1.1 generates a sequence of feasible basic solutions. Assuming nondegeneracy throughout the solution process, it terminates at step 2, giving an optimal basic solution.

Proof Its validity comes from Lemmas 8.1.2 and 8.1.3. □

Example 8.1.1 Solve the following problem by Algorithm 8.1.1:

$$\begin{aligned}
 \min f &= -x_1 + 3x_2, \\
 \text{s.t. } 2 &\leq 2x_1 - 3x_2 \leq 10, \\
 1 &\leq x_1 - x_2 \leq 5, \\
 &\quad -x_1 + 2x_2 \leq 0, \\
 0 &\leq x_1 \leq 6, \quad -2 \leq x_2.
 \end{aligned}$$

Answer Introduce x_3, x_4, x_5 to convert the preceding to

$$\begin{aligned}
 \min f &= x_1 - 3x_2, \\
 \text{s.t. } -2x_1 + 3x_2 + x_3 &= 0, \\
 -x_1 + x_2 + x_4 &= 0, \\
 x_1 - 2x_2 + x_5 &= 0, \\
 0 \leq x_1 \leq 6, \quad -2 \leq x_2 \leq \infty, \quad 2 \leq x_3 \leq 10, \quad 1 \leq x_4 \leq 5, \quad -\infty \leq x_5 \leq 0.
 \end{aligned}$$

In the following, the unbounded variables will be handled as $-\infty$ or ∞ , upon which only the determination of stepsize touches.

Initial: $B = \{3, 4, 5\}$, $N = \{1, 2\}$, $B^{-1} = I$, $\bar{x}_N = (0_{(-)}, -2_{(-)})^T$, $\bar{x}_B = (6, 2, -4)^T$, $\bar{f} = 6$. The initial solution is feasible basic, where the subscript “(–)” labels a component on its lower bound and the superscript “(+)” labels one on its upper bound (the same below).

Iteration 1:

1. $y = B^{-T}c_B = (0, 0, 0)^T$, $\bar{z}_N = (1, -3)^T$.
2. $J = \{2\}$.
3. $\max_J |\bar{z}_j| = 3$, $q = 2$, x_2 enters the basis.
4. $\bar{a}_2 = B^{-1}a_2 = (3, 1, -2)^T$.
5. $\alpha_1 = (6 - 2)/3 = 4/3$, $\alpha_2 = (2 - 1)/1 = 1$,
 $\alpha_3 = (-4 - 0)/-2 = 2$; $\alpha = \min\{\infty, 4/3, 1, 2\} = 1$.

6. $\bar{x}_B = (6, 2, -4)^T - 1 \times (3, 1, -2)^T = (3, 1, -2)^T$,
 $\bar{x}_N = (0_{(-)}, -2)^T - 1 \times (0, -1) = (0_{(-)}, -1)^T$.
7. $p = 2$, x_4 leaves the basis.
8. $B^{-1} = \begin{pmatrix} 1 & -3 \\ & 1 \\ & 2 & 1 \end{pmatrix}$.
9. $B = \{3, 2, 5\}$, $N = \{1, 4\}$; $\bar{x}_B = (3, -1, -2)^T$, $\bar{x}_N = (0_{(-)}, 1_{(-)}^T)$.

Iteration 2:

1. $y = B^{-T}c_B = (0, -3, 0)^T$, $\bar{z}_N = c_N - N^T y = (1, 0)^T - (3, -3)^T = (-2, 3)^T$.
2. $J = \{1\}$.
3. $\max_J |\bar{z}_j| = 2$, $q = 1$, x_1 enters the basis.
4. $\bar{a}_1 = B^{-1}a_1 = (1, -1, -1)^T$.
5. $\alpha_1 = (3 - 2)/1 = 1$, $\alpha_2 = (-1 - \infty)/-1 = \infty$,
 $\alpha_3 = (-2 - 0)/-1 = 2$, $\alpha = \min\{6 - 0, 1, \infty, 2\} = 1$.
6. $\bar{x}_B = (3, -1, -2)^T - 1 \times (1, -1, -1)^T = (2, 0, -1)^T$,
 $\bar{x}_N = (0, 1)^T - 1 \times (-1, 0) = (1, 1)^T$.
7. $p = 1$, x_3 leaves the basis.
8. $B^{-1} = \begin{pmatrix} 1 & & \\ 1 & 1 & \\ 1 & & 1 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ & 1 \\ & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -3 \\ 1 & -2 \\ 1 & -1 & 1 \end{pmatrix}$.
9. $B = \{1, 2, 5\}$, $N = \{3, 4\}$; $\bar{x}_B = (1, 0, -1)^T$, $\bar{x}_N = (2_{(-)}, 1_{(-)})^T$.

Iteration 3:

1. $y = B^{-T}c_B = (-2, 3, 0)^T$, $\bar{z}_N = c_N - N^T y = (0, 0)^T - (-2, 3)^T = (2, -3)^T$.
2. $J = \{4\}$.
3. $\max_J |\bar{z}_j| = 3$, $q = 4$, x_4 enters the basis.
4. $\bar{a}_4 = B^{-1}a_4 = (-3, -2, -1)^T$.
5. $\alpha_1 = (1 - 6)/-3 = 5/3$, $\alpha_2 = (0 - \infty)/-2 = \infty$,
 $\alpha_3 = (-1 - 0)/-1 = 1$; $\alpha = \min\{5 - 1, 5/3, \infty, 1\} = 1$.
6. $\bar{x}_B = (1, 0, -1)^T - 1 \times (-3, -2, -1)^T = (4, 2, 0)^T$,
 $\bar{x}_N = (2, 1)^T - 1 \times (0, -1) = (2, 2)^T$.
7. $p = 3$, x_5 leaves the basis.
8. $B^{-1} = \begin{pmatrix} 1 & -3 \\ & 1 & -2 \\ & & -1 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 1 & -2 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} -2 & -3 \\ -1 & -2 \\ -1 & 1 & -1 \end{pmatrix}$.
9. $B = \{1, 2, 4\}$, $N = \{3, 5\}$; $\bar{x}_B = (4, 2, 2)^T$, $\bar{x}_N = (2_{(-)}, 0^{(+)})^T$.

Iteration 4:

1. $y = B^{-T}c_B = (1, 0, 3)^T$, $\bar{z}_N = c_N - N^T y = (0, 0)^T - (1, 3)^T = (-1, -3)^T$.
2. $J = \{3\}$.
3. $\max_J |\bar{z}_j| = 1$, $q = 3$, x_3 enters the basis.
4. $\bar{a}_3 = B^{-1}a_3 = (-2, -1, -1)^T$.

5. $\alpha_1 = (4 - 6)/-2 = 1, \alpha_2 = (2 - \infty)/-1 = \infty,$
 $\alpha_3 = (2 - 5)/-1 = 3; \alpha = \min\{10 - 2, 1, \infty, 3\} = 1.$
6. $\bar{x}_B = (4, 2, 2)^T - 1 \times (-2, -1, -1)^T = (6, 3, 3)^T,$
 $\bar{x}_N = (2, 0)^T - 1 \times (-1, 0) = (3, 0)^T.$
7. $p = 1, x_1$ leaves the basis.
8. $B^{-1} = \begin{pmatrix} -1/2 & \\ -1/2 & 1 \\ -1/2 & 1 \end{pmatrix} \begin{pmatrix} -2 & -3 \\ -1 & -2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3/2 \\ & -1/2 \\ 1 & 1/2 \end{pmatrix}.$
9. $\bar{x}_B = (3, 3, 3)^T, B = \{3, 2, 4\}; \bar{x}_N = (6^{(+)}, 0^{(+)})^T, N = \{1, 5\}.$

Iteration 5:

1. $y = B^{-T} c_B = (0, 0, 3/2)^T,$
 $\bar{z}_N = c_N - N^T y = (1, 0)^T - (3/2, 3/2)^T = (-1/2, -3/2)^T.$
2. $J = \emptyset.$ The optimal basic solution and associated objective value:

$$\bar{x} = (6, 3, 3, 3, 0)^T, \quad \bar{f} = 6 - 3 \times 3 = -3.$$

As for the tableau version of Algorithm 8.1.1, the associated simplex tableau is the same as the conventional, except there is no need for RHS column to display corresponding basic solution. We add three additional rows $(u, \bar{x}, l)$, respectively, listing upper bounds, variable values, and lower bounds. The simplex tableau is of the form below:

	x_B^T	x_N^T
	I	$\bar{N}$
		$\bar{z}_N$
u	u_B^T	u_N^T
$\bar{x}$	$\bar{x}_B^T$	$\bar{x}_N^T$
l	l_B^T	l_N^T

Based on Table 3.2, Algorithm 8.1.1 can be revised to a tableau form. As $\bar{a}_q = B^{-1} a_q$, (8.16) should be replaced by

$$\alpha_i = \begin{cases} (u_{j_i} - \bar{x}_{j_i})/\text{sign}(\bar{z}_q) \bar{a}_{i q}, & \text{if } \text{sign}(\bar{z}_q) \bar{a}_{i q} > 0, \\ (l_{j_i} - \bar{x}_{j_i})/\text{sign}(\bar{z}_q) \bar{a}_{i q}, & \text{if } \text{sign}(\bar{z}_q) \bar{a}_{i q} < 0, \\ \infty, & \text{if } \bar{a}_{i q} = 0, \end{cases} \quad i = 1, \dots, m. \quad (8.18)$$

Algorithm 8.1.2 (Generalized Simplex Algorithm: Tableau Form) Initial: feasible tableau in form (8.1), associated with $\bar{x}$. This algorithm solves the bounded-variable problem (8.3).

1. Compute $\bar{f} = c^T \bar{x}$ and stop (optimality achieved) if J defined by (8.10) is empty.
2. Select column index q such that $q \in \max_{j \in J} |\bar{z}_j|$.

3. Determine stepsize α by (8.15), where α_i defined by (8.18).
4. Set $\bar{x}_q = -\text{sign}(\bar{z}_q)\alpha$ and update $\bar{x}_B = \bar{x}_B + \alpha \text{sign}(\bar{z}_q)\bar{a}_q$ if $\alpha \neq 0$.
5. If $\alpha = u_q - l_q$, go to step 1; else, determine row index $p \in \{1, \dots, m\}$ such that $\alpha = \alpha_p$.
6. Convert $\bar{a}_{pq}$ to 1 and eliminate the other nonzeros in the column by elementary transformations.
7. Go to step 1.

Note The last 3 rows in the tableau should be updated in each iteration.

8.1.1 Generalized Phase-I

Assume that B is the basis at current iteration, associated with basic solution $\bar{x}$.

Introduce the following index sets:

$$I_1 = \{i = 1, \dots, m \mid \bar{x}_{j_i} < l_{j_i}\},$$

$$I_2 = \{i = 1, \dots, m \mid \bar{x}_{j_i} > u_{j_i}\},$$

$$I = \{1, \dots, m\} \setminus (I_1 \cup I_2).$$

If $I_1 \cup I_2$ is empty, then $\bar{x}$ is feasible. Assume that it is not the case.

Construct the following auxiliary problem:

$$\begin{aligned} \min \quad & w = -\sum_{i \in I_1} x_{j_i} + \sum_{i \in I_2} x_{j_i}, \\ \text{s.t.} \quad & Bx_B = b - Nx_N, \\ & l_I \leq x_I \leq u_I, \quad l_N \leq x_N \leq u_N, \end{aligned}$$

where the objective function is termed *infeasible-sum*. Then solution $\bar{x}$ is feasible to the preceding problem. From $\bar{x}$, one iteration of Algorithm 8.1.2 can be performed with its pivot rule slightly modified. The auxiliary problem is formed iteration by iteration with the change of $\bar{x}$, until achieving feasibility or detecting infeasibility.

8.2 Generalized Dual Simplex Algorithm: Tableau Form

The dual version of the generalized simplex method can be derived using the same simplex tableau. Assume that (8.1) is the simplex tableau, associated with B and N given by (8.5).

Let index sets Γ and Π be defined by (8.9). It is known that solution ($\bar{z}_B = 0, \bar{z}_N$) is dual feasible (as well as complementary with the primal solution) if the following condition holds:

$$\bar{z}_\Gamma \geq 0, \quad \bar{z}_\Pi \leq 0. \quad (8.19)$$

Since components of l and u are finite, it is always possible to set $\bar{x}_N$ to fulfill the preceding condition. The associated $\bar{x}_B$ and objective value are then

$$\bar{x}_B = \bar{b} - \bar{N}\bar{x}_N, \quad \bar{f} = c^T \bar{x}.$$

If $l_B \leq x_B \leq u_B$ also holds, then $(\bar{x}, \bar{z})$ are a pair of primal and dual optimal basic solutions.

Introduce *bound-violation* quantity below:

$$\rho_i = \begin{cases} l_{j_i} - \bar{x}_{j_i}, & \text{If } \bar{x}_{j_i} < l_{j_i}, \\ u_{j_i} - \bar{x}_{j_i}, & \text{If } \bar{x}_{j_i} > u_{j_i}, \\ 0, & \text{If } l_{j_i} \leq \bar{x}_{j_i} \leq u_{j_i}, \end{cases} \quad i = 1, \dots, m, \quad (8.20)$$

and determine row index p by the following rule:

$$p \in \arg \max\{|\rho_i| \mid i = 1, \dots, m\}. \quad (8.21)$$

It is clear that optimality is achieved if $\rho_p = 0$. Now assume that $\rho_p \neq 0$: then $\rho_p > 0$ indicates that $\bar{x}_p$ violates the lower bound while $\rho_p < 0$ indicates violation of the upper bound. Introduce index set

$$J = \{j \in \Gamma \mid \text{sign}(\rho_p)\bar{a}_{pj} < 0\} \cup \{j \in \Pi \mid \text{sign}(\rho_p)\bar{a}_{pj} > 0\}. \quad (8.22)$$

It is not difficult to show that the original problem is infeasible if $J = \emptyset$; else, a column index q and β can be determined such that

$$\beta = -\bar{z}_q / (\text{sign}(\rho_p)\bar{a}_{pq}) = \min_{j \in J} -\bar{z}_j / (\text{sign}(\rho_p)\bar{a}_{pj}) \geq 0, \quad (8.23)$$

which is the maximum possible stepsize in y -space to maintain dual feasibility (also see Sect. 8.3).

Taking $\bar{a}_{pq}$ as pivot, convert the simplex tableau by relevant elementary transformations. Then the resulting simplex tableau corresponds to the new basis as follows:

$$B = \{j_1, \dots, j_{p-1}, q, j_{p+1}, \dots, j_m\}, \quad N = N \setminus q \cup \{j_p\}.$$

Let us use (8.1) again to denote the new tableau, and $\hat{x}$ to denote the associated basic solution. Since the new tableau is equivalent to the old, $\hat{x}$ and $\bar{x}$ satisfy

$$\bar{x}_B = -\bar{N}\bar{x}_N, \quad \hat{x}_B = -\bar{N}\hat{x}_N. \quad (8.24)$$

Now set the leaving component $\hat{x}_{j_p}$ to the violated bound, i.e.,

$$\hat{x}_{j_p} = \bar{x}_{j_p} + \rho_p, \quad (8.25)$$

and maintain other nonbasic components unchanged, i.e.,

$$\hat{x}_j = \bar{x}_j, \quad j \in N, \quad j \neq j_p.$$

Then subtraction of the two equalities in (8.24) gives the following updating formula:

$$\hat{x}_B = \bar{x}_B - \rho_p \bar{a}_{j_p}. \quad (8.26)$$

It is not difficult to show that the new simplex tableau with $\hat{x}$ is still dual feasible.

Noting that $\bar{z}_{j_p} = \text{sign}(\rho_p)\beta$ holds for the new tableau, the following recurrence formula of the objective value can be derived from $\hat{x}$ and $\bar{x}$, satisfying (8.25), and the equality associated with the bottom row of the tableau:

$$\hat{f} = \bar{f} + \bar{z}_{j_p}(\hat{x}_{j_p} - \bar{x}_{j_p}) = \bar{f} + \rho_p \bar{z}_{j_p} = \bar{f} + \beta \geq \bar{f},$$

which means that the objective value increases.

If all components of $\bar{z}_N$ are nonzero, the simplex tableau is said to be *dual nondegenerate*. In this case, β given by (8.23) is strictly positive, and hence the objective value increases strictly.

Overall steps are put into the following, in which the objective value is calculated at the end.

Algorithm 8.2.1 (Generalized Dual Simplex Algorithm : Tableau Form) Initial: dual feasible tableau in form (8.1), associated with $\bar{x}$. This algorithm solves the bounded-variable problem (8.3).

1. Select row index p by (8.21) together with (8.20).
2. If $\rho_p = 0$, compute $\bar{f} = c^T \bar{x}$, and stop (optimality achieved).
3. Stop if J defined by (8.22) is empty (infeasible problem).
4. Determine the column index q by (8.23).
5. Convert $\bar{a}_{pq}$ to 1, and eliminate other nonzeros in the column by the elementary transformations.
6. Update $\bar{x}$ by (8.25) and (8.26).
7. Go to step 1.

Note The last 3 rows in the simplex tableau should be updated in each iteration.

The proof of meanings of the algorithm's exits is delayed to the derivation of its revised version.

Example 8.2.1 Solve the following problem by Algorithm 8.2.1:

$$\begin{aligned} \min f &= 2x_1 - x_2 + 3x_3 - 6x_4, \\ \text{s.t. } -2x_1 + 3x_2 - 4x_3 + 2x_4 + x_5 &= 14, \\ -3x_1 + 4x_2 - 5x_3 + 6x_4 + x_6 &= 16, \\ x_1 - 2x_2 + 2x_3 - 7x_4 + x_7 &= -15, \\ -15 \leq x_1 \leq 30, \quad -12 \leq x_2 \leq 20, \quad -17 \leq x_3 \leq 10, \\ -8 \leq x_4 \leq 15, \quad -10 \leq x_5 \leq 26, \quad -13 \leq x_6 \leq 34, \\ 0 \leq x_7 \leq 19. \end{aligned}$$

Answer Initial tableau:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7
	-2	3	-4	2	1		
	-3	4*	-5	6		1	
	1	-2	2	-7			1
	2	-1	3	-6			
u	30	20	10	15	26	34	19
$\bar{x}$	-15	20	-17	15	-174	-284	179
l	-15	-12	-17	-8	-10	-13	0

Take

$$\bar{x}_N = (-15_{(-)}, 20^{(+)}, -17_{(-)}, 15^{(+)})^T (N = \{1, 2, 3, 4\}),$$

$$\bar{x}_B = \bar{b} - \bar{N}\bar{x}_N = (-174, -284, 179)^T (B = \{5, 6, 7\}), \bar{f} = -191.$$

Iteration 1:

- $\rho_1 = -10 - (-174) = 164, \rho_2 = -13 - (-284) = 271,$
 $\rho_3 = 19 - 179 = -160. \max\{|164|, |271|, |-160|\} = 271 \neq 0, p = 2, j_2 = 6.$
- $J = \{1, 2, 3, 4\} \neq \emptyset.$
- $\min\{-2/(-3), -(-1)/4, -3/(-5), -(-6)/6\} = 1/4, q = 2.$
- Multiply row 2 by $1/4$, and then add $-3, 2, 1$ times of row 2 to rows 1,3,4, respectively.
- $\bar{x}_6 = -284 + 271 = -13.$

$$\bar{x}_B = (-174, 20, 179)^T - 271(-3/4, 1/4, 1/2, -1/4)^T.$$

$$= (117/4, -191/4, 87/2)^T, B = \{5, 2, 7\}.$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7
	1/4		-1/4	-5/2	1	-3/4*	
	-3/4	1	-5/4*	3/2		1/4	
	-1/2		-1/2	-4		1/2	1
	5/4		7/4	-9/2		1/4	
u	30	20	10	15	26	34	19
$\bar{x}$	-15	-191/4	-17	15	117/4	-13	87/2
l	-15	-12	-17	-8	-10	-13	0

Iteration 2:

1. $\rho_1 = 26 - 117/4 = -13/4$, $\rho_2 = -12 - (-191/4) = 143/4$,
 $\rho_3 = 19 - 87/2 = -49/2$. $\max\{|-13/4|, |143/4|, |-49/2|\} = 143/4 \neq 0$,
 $p = 2$, $j_2 = 2$.
3. $J = \{1, 3, 4\} \neq \emptyset$.
4. $\min\{-(5/4)/(-3/4), -(7/4)/(-5/4), -(-9/2)/(3/2)\} = 7/5$, $q = 3$.
5. Multiply row 2 by $-4/5$, and then add $1/4$, $1/2$, $-7/4$ times of row 2 to rows 1,3,4, respectively.
6. $\bar{x}_2 = -191/4 + 143/4 = -12$,
 $\bar{x}_B = (117/4, -17, 87/2)^T - (143/4)(-1/5, -4/5, -2/5)^T$.
 $= (182/5, 58/5, 289/5)^T$, $B = \{5, 3, 7\}$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7
	2/5	-1/5		-14/5	1	-4/5	
	3/5	-4/5	1	-6/5		-1/5	
	-1/5	-2/5		-23/5*		2/5	1
	1/5	7/5		-12/5		3/5	
u	30	20	10	15	26	34	19
$\bar{x}$	-15	-12	58/5	15	182/5	-13	289/5
l	-15	-12	-17	-8	-10	-13	0

Iteration 3:

1. $\rho_1 = 26 - 182/5 = -52/5$, $\rho_2 = 10 - 58/5 = -8/5$,
 $\rho_3 = 19 - 289/5 = -194/5$. $\max\{|-52/5|, |-8/5|, |-194/5|\}$
 $= 194/5 \neq 0$, $p = 3$, $j_3 = 7$.
3. $J = \{4, 6\} \neq \emptyset$.
4. $\min\{-(-12/5)/(-23/5), -(3/5)/(2/5)\} = 12/23$, $q = 4$.
5. Multiply row 3 by $-5/23$, and then add $14/5$, $6/5$, $12/5$ times of row 3 to rows 1,2,4, respectively.
6. $\bar{x}_7 = 289/5 - 194/5 = 19$.
 $\bar{x}_B = (182/5, 58/5, 15)^T - (-194/5)(-14/23, -6/23, -5/23)^T$.
 $= (294/23, 34/23, 151/23)^T$, $B = \{5, 3, 4\}$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7
	12/23	1/23			1	-24/23	-14/23
	15/23	-16/23	1			-7/23	-6/23
	1/23	2/23		1		-2/23	-5/23
	7/23	37/23				9/23	-12/23
u	30	20	10	15	26	34	19
$\bar{x}$	-15	-12	34/23	151/23	294/23	-13	19
l	-15	-12	-17	-8	-10	-13	0

Iteration 4:

1. $\rho_1 = \rho_2 = \rho_3 = 0$. The optimal basic solution and optimal value are

$$x^* = (-15, -12, 34/23, 151/23, 294/23, -13, 19)^T,$$

$$f^* = (2, -1, 3, -6)(-15, -12, 34/23, 151/23)^T = -1218/23.$$

8.2.1 Generalized Dual Phase-I

A dual Phase-I algorithm is needed to get the generalized dual simplex algorithm started. In this aspect, the computational experiments carried out by Koberstein and Suhl (2007) are noteworthy.

To clarify the computational efficiency, they realized several commonly used dual Phase-I algorithms, including the generalization of the dual Phase-I using the most-obtuse-angle row rule (named PAN), within MOPS,¹ and tested with 46 typical large-scale sparse problems, the maximum among which involves more than 500,000 constraints and 1,000,000 variables. The computational results showed that for most tested problems, PAN required the minimum number of iterations; only for a few most difficult problems, the required iterations exceeded an acceptable amount. In the latter case, the code turned to the dual infeasibility-sum Phase-I, named SDI. Such a combination, PAN+SDI, turned out to be the best among all the dual Phase-I algorithms tested and was finally taken as the default option for MOPS' dual simplex algorithm.

Based on the preceding, the author suggests replacing (8.15) with

$$\alpha = \min\{u_q - l_q, \min\{\alpha_i \mid |\bar{a}_{iq}| \geq \tau\theta, i = 1, \dots, m\}\}, \quad (8.27)$$

$$\theta = \max\{|\bar{a}_{iq}| \mid i = 1, \dots, m\},$$

where $0 < \tau \leq 1$, α_i , $i = 1, \dots, m$, are defined by (8.18). The main idea is to limit stepsizes to a certain extent.

This consideration leads to the following algorithm.

Algorithm 8.2.2 (Tableau Generalized Dual Phase-I: The Most-Obtuse-Angle Rule) Given $0 < \tau \leq 1$. Initial: simplex tableau in form (8.1) associated with $\bar{x}$. This algorithm solves the bounded-variable problem (8.3).

1. If J defined by (8.10) is empty, compute $\bar{f} = c^T \bar{x}$, and stop (optimality achieved).
2. Select column index q such that $q \in \max_{j \in J} |\bar{z}_j|$.
3. Determine stepsize α by (8.27).
4. Set $\bar{x}_q = -\text{sign}(\bar{z}_q) \alpha$, and update $\bar{x}_B = \bar{x}_B + \alpha \text{sign}(\bar{z}_q) \bar{a}_q$ if $\alpha \neq 0$.

¹ MOPS is a package developed by Suhl et al. of College of Production Information Economy and Operations Research of Berlin Free University, see Suhl (1994, 1999).

5. If $\alpha = u_q - l_q$, go to step 1; else, determine row index $p \in \{1, \dots, m\}$ such that $\alpha = \alpha_p$.
6. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
7. Go to step 1.

8.3 Generalized Dual Simplex Algorithm

According to Table 3.2, which gives the correspondence between items of the simplex and the revised tableau, it is not difficult to formulate the revised version of Algorithm 8.2.1. However, we will not do so but derive by dealing with the dual bounded-variable problem directly.

We will use the same symbols as those in the previous section, e.g., primal basic solution $\bar{x}$, satisfying

$$\begin{aligned}\bar{x}_s &= l_s \text{ or } u_s, \quad s \in N, \\ \bar{x}_B &= B^{-1}b - B^{-1}N\bar{x}_N,\end{aligned}\tag{8.28}$$

sets Γ and Π defined by (8.9), dual solution $(\bar{z}_B = 0, \bar{z}_N)$ satisfying (8.19), and ρ_i defined by (8.20), etc.

Assume that row index p was determined by (8.21). According to Theorem 5.3.6, we turn to the relaxed problem by ignoring inactive constraints at $\bar{x}$, i.e.,

$$\begin{aligned}\min f &= c^T x, \\ \text{s.t. } Ax &= b, \\ l_\Gamma &\leq x_\Gamma, \\ x_\Pi &\leq u_\Pi, \\ l_{j_p} &\leq \bar{x}_{j_p}, \text{ if } \rho_p > 0, \\ \bar{x}_{j_p} &\leq u_{j_p}, \text{ if } \rho_p < 0.\end{aligned}\tag{8.29}$$

Using notation

$$h_p = \begin{cases} l_{j_p}, & \text{If } \rho_p > 0, \\ u_{j_p}; & \text{If } \rho_p < 0, \end{cases}\tag{8.30}$$

the relaxed problem can be written

$$\begin{aligned}\max b^T y &- u_\Pi^T v_\Pi + l_\Gamma^T w_\Gamma + h_p z_{j_p}, \\ \text{s.t. } B^T y &+ z_{j_p} e_p = c_B, \\ \Gamma^T y &+ w_\Gamma = c_\Gamma, \\ \Pi^T y &- v_\Pi = c_\Pi, \\ \rho_p z_{j_p}, v_\Pi, w_\Gamma &\geq 0.\end{aligned}$$

Based on the equality constraints, eliminating v_Π , w_Γ , and combining (8.9) and (8.10) reduce the objective function to

$$\begin{aligned} (b - \Pi u_\Pi - \Gamma l_\Gamma)^T y + h_p z_p &= c_\Pi^T \bar{x}_\Pi + c_\Gamma^T \bar{x}_\Gamma + (b - \Pi \bar{x}_\Pi - \Gamma \bar{x}_\Gamma)^T y + h_p z_{j_p} \\ &= c_N^T \bar{x}_N + (b - N \bar{x}_N)^T y + h_p z_{j_p}. \end{aligned} \quad (8.31)$$

Then setting $z_\Gamma = w_\Gamma$, $z_\Pi = -v_\Pi$, we transform the relaxed problem to the following, equivalently,

$$\begin{aligned} \max g(y, z) &= c_N^T \bar{x}_N + (b - N \bar{x}_N)^T y + h_p z_{j_p}, \\ \text{s.t. } B^T y + z_{j_p} e_p &= c_B, \\ N^T y + z_N &= c_N, \\ \rho_p z_{j_p} &\geq 0, \quad z_\Pi \leq 0, \quad z_\Gamma \geq 0. \end{aligned} \quad (8.32)$$

Now, define

$$\bar{y} = B^{-T} c_B, \quad (8.33)$$

$$\bar{z}_N = c_N - N^T \bar{y}, \quad \bar{z}_B = 0 \quad (8.34)$$

and assume that the following condition holds:

$$\bar{z}_\Pi \leq 0, \quad \bar{z}_\Gamma \geq 0. \quad (8.35)$$

Therefore, it is verified that the primal objective value at $\bar{x}$ and the dual objective value at $(\bar{y}, \bar{z})$ are equal, i.e., $\bar{f} = \bar{g}$.

Using the preceding notation, moreover, the following is valid.

Lemma 8.3.1 *$(\bar{y}, \bar{z})$ is feasible basic solution to the relaxed problem, which exhibits complementarity with $\bar{x}$.*

Proof It is clear that $(\bar{y}, \bar{z})$ is the basic solution to (8.32), satisfying the sign constraints at the bottom. So, it is only needed to show

$$B^T \bar{y} + \bar{z}_{j_p} e_p = c_B, \quad N^T \bar{y} + \bar{z}_N = c_N, \quad (8.36)$$

$$(\bar{x}_\Gamma - l_\Gamma)^T \bar{z}_\Gamma = 0, \quad (u_\Pi - \bar{x}_\Pi)^T \bar{z}_\Pi = 0, \quad (8.37)$$

$$(\bar{x}_{j_p} - l_{j_p}) \bar{z}_{j_p} = 0, \quad \text{if } \rho_p > 0, \quad (8.38)$$

$$(\bar{x}_{j_p} - u_{j_p}) \bar{z}_{j_p} = 0, \quad \text{if } \rho_p < 0. \quad (8.39)$$

From (8.33) and the second expression of (8.34), the first expression of (8.35) follows. By the first expression of (8.34), it holds that

$$\Pi^T \bar{y} + \bar{z}_\Pi = \Pi^T \bar{y} + c_\Pi - \Pi^T \bar{y} = c_\Pi. \quad (8.40)$$

Similarly that

$$\Gamma^T \bar{y} + \bar{z}_\Gamma = c_\Gamma. \quad (8.41)$$

Therefore, (8.36) is valid.

By (8.9), on the other hand, it is clear that (8.37) holds; and it is known from the second expression of (8.34) that (8.38) or (8.39) holds. $\square$

By setting

$$\bar{v}_B = 0, \quad \bar{w}_B = 0, \quad (8.42)$$

$$\bar{v}_\Pi = -\bar{z}_\Pi, \quad \bar{w}_\Pi = 0, \quad (8.43)$$

$$\bar{v}_\Gamma = 0, \quad \bar{w}_\Gamma = \bar{z}_\Gamma, \quad (8.44)$$

it is not difficult to verify that $(\bar{y}, \bar{v}, \bar{w})$ is a feasible basic solution to the dual problem of (8.3) and exhibits complementarity with $\bar{x}$.

Lemma 8.3.2 *If $l_B \leq \bar{x}_B \leq u_B$ holds, then $\bar{x}$ is an optimal basic solution.*

Proof Since $l_B \leq \bar{x}_B \leq u_B$ holds, $\bar{x}$ is clearly a feasible basic solution to the original problem (8.3) as well as the relaxed problem (8.29). By Lemma 8.3.1, it is known that $(\bar{y}, \bar{z})$ is dual feasible, exhibiting complementarity with $\bar{x}$. Therefore, the two are primal and dual optimal solutions with respect to the relaxed problem. According to Theorem 5.3.6, $\bar{x}$ is optimal basic solution to (8.3). $\square$

Now we find a new dual solution to improve the objective value. Define search direction

$$h = -\text{sign}(\rho_p) B^{-T} e_p, \quad \sigma_{j_p} = \text{sign}(\rho_p), \quad (8.45)$$

$$\sigma_N = -N^T h. \quad (8.46)$$

Lemma 8.3.3 *The preceding search direction satisfies the following conditions:*

$$B^T h + \sigma_{j_p} e_p = 0, \quad N^T h + \sigma_N = 0, \quad (8.47)$$

$$(b - N\bar{x}_N)^T h + h_p \sigma_{j_p} > 0. \quad (8.48)$$

Proof The first expression is easily verified. We only show (8.48).

From (8.45) and the second expression of (8.28), it follows that

$$\begin{aligned} h^T (b - N\bar{x}_N) + h_p \sigma_{j_p} &= -\text{sign}(\rho_p) e_p^T (B^{-1} b - B^{-1} N\bar{x}_N) + \text{sign}(\rho_p) h_p \\ &= -\text{sign}(\rho_p) (e_p^T \bar{x}_B - h_p). \end{aligned}$$

Then from (8.4) and (8.30), it follows that the right-hand side of the preceding equals

$$l_{j_p} - \bar{x}_{j_p} > 0$$

in case of $\rho_p > 0$, and equals

$$\bar{x}_{j_p} - u_{j_p} > 0$$

in case of $\rho_p < 0$. □

Consider the following line search scheme:

$$\hat{y} = \bar{y} + \beta h, \quad \hat{z}_{j_p} = \bar{z}_{j_p} + \beta \sigma_{j_p} = \text{sign}(\rho_p) \beta, \quad (8.49)$$

$$\hat{z}_N = \bar{z}_N + \beta \sigma_N. \quad (8.50)$$

Introduce the index set

$$J = \{j \in \Gamma \mid \sigma_j < 0\} \cup \{j \in \Pi \mid \sigma_j > 0\}. \quad (8.51)$$

Assume that $J \neq \emptyset$. Then from (8.50) and sign condition $\hat{z}_\Pi \leq 0$, $\hat{z}_\Gamma \geq 0$, it is known that the maximum possible stepsize β and pivot column index q satisfy the minimum-ratio test

$$\beta = -\bar{z}_q / \sigma_q = \min_{j \in J} -\bar{z}_j / \sigma_j \geq 0. \quad (8.52)$$

If all components of $\bar{z}_N$ are nonzero, then the solution is dual nondegenerate, hence the determined stepsize is positive.

Lemma 8.3.4 *If $J \neq \emptyset$, the new solution, determined by (8.49) and (8.50) together with (8.52), is a feasible basic solution to the dual relaxed problem. The respective objective value increases, and strictly increases if dual nondegeneracy is assumed.*

Proof From (8.49), the first expression of (8.47) and (8.33), it is known for any $\beta \geq 0$ that

$$B^T \hat{y} + \hat{z}_{j_p} e_p = B^T \bar{y} + \beta (B^T h + \sigma_{j_p} e_p) = B^T \bar{y} = c_B. \quad (8.53)$$

From the first expression of (8.49), (8.50), the second expression of (8.47) and (8.34), it follows that

$$N^T \hat{y} + \hat{z}_N = N^T \bar{y} + \beta N^T h + \bar{z}_N + \beta \sigma_N = (N^T \bar{y} + \bar{z}_N) + \beta (N^T h + \sigma_N) = c_N. \quad (8.54)$$

In addition, by (8.49), (8.50) and (8.47), it is known that $\hat{z}_{j_p}, \hat{z}$ satisfies the sign condition at the bottom of problem (8.32), hence the new solution is feasible basic

solution, associated with objective value increasing to

$$\begin{aligned}
 \hat{g} &= (b - N\bar{x}_N)^T \hat{y} + h_p \hat{z}_{j_p} \\
 &= (b - N\bar{x}_N)^T \bar{y} + \beta((b - N\bar{x}_N)^T h + h_p \sigma_{j_p}) \\
 &\geq \bar{g},
 \end{aligned} \tag{8.55}$$

where the inequality comes from (8.48) and $\beta \geq 0$. Under dual nondegeneracy, $\beta > 0$ holds, and hence the strict inequality holds, which means a strict increase in the objective value. $\square$

When J is empty, (8.47) is not well-defined but the following is valid.

Lemma 8.3.5 *If $J = \emptyset$, the original problem (8.3) is infeasible.*

Proof $J = \emptyset$ implies that

$$\sigma_\Pi \leq 0, \quad \sigma_\Gamma \geq 0. \tag{8.56}$$

Combining the preceding two expressions together with $\bar{z}_\Pi \leq 0$ and $\bar{z}_\Gamma \geq 0$ leads to

$$\hat{z}_\Pi = \bar{z}_\Pi + \beta \sigma_\Pi \leq 0, \quad \hat{z}_\Gamma = \bar{z}_\Gamma + \beta \sigma_\Gamma \geq 0 \quad \forall \beta > 0.$$

Similarly to the proof of Theorem 8.3.4, it can be shown that $(\hat{y}, \hat{z})$ satisfies the other constraints, with the objective value denoted again by (8.55). Thus, noting (8.48), it is known that

$$\hat{g} \rightarrow \infty \quad \text{as} \quad \beta \rightarrow \infty.$$

Thus, the dual relaxed problem is unbounded, or the relaxed problem is infeasible. It is then clear that the original problem is infeasible. $\square$

Now we need to determine a primal solution that is complementary with the dual solution. In order to change the value of basic variable x_{j_p} from $\bar{x}_p$ to the violated bound, it is necessary to change the value of nonbasic variable x_q from $\bar{x}_q$ by the following range:

$$\Delta x_q = \begin{cases} -\rho_p/|\sigma_q|, & \text{If } \bar{x}_q = l_q, \\ \rho_p/|\sigma_q|, & \text{If } \bar{x}_q = u_q. \end{cases} \tag{8.57}$$

Therefore, the new values are

$$\hat{x}_B = \bar{x}_B - \Delta x_q \bar{a}_q, \quad \hat{x}_q = \bar{x}_q + \Delta x_q, \quad \hat{x}_j = \bar{x}_j, \quad j \in N, \quad j \neq q, \tag{8.58}$$

where $\bar{a}_q = B^{-1}a_q$, associated with the new objective value

$$\hat{f} = \bar{f} + |\Delta x_q \bar{z}_q| \geq \bar{f}. \quad (8.59)$$

Note that all components of $\hat{x}_N$ are the same as those of $\bar{x}_N$, except for $\hat{x}_q$. From the first expression of (8.58) and the second expression of (8.49), it is known that if $\rho_p > 0$, then $\hat{x}_{j_p} = l_{j_p}$ and $\hat{z} \geq 0$ hold, while if $\rho_p < 0$, then $\hat{x}_{j_p} = u_{j_p}$ and $\hat{z}_{j_p} \leq 0$ hold. Therefore, after updating the basis by exchanging p and q , the $\hat{x}$ and $(\hat{y}, \hat{z})$ exhibit complementarity, and the latter remains dual feasible. Then, we are ready to carry out the next iteration.

Overall steps are summarized into the following revision of Algorithm 8.2.1.

Algorithm 8.3.1 (Generalized Dual Simplex Algorithm) Initial: (B, N) , B^{-1} ; $\bar{y}$, $\bar{z}$, $\bar{x}$ satisfying (8.28), (8.33), and (8.34). This algorithm solves the bounded-variable problem (8.3).

1. Select a pivot row index $p \in \arg \max\{|\rho_i| \mid i = 1, \dots, m\}$, where ρ_i is defined by (8.20).
2. If $\rho_p = 0$, compute $\bar{f} = c^T \bar{x}$, and stop.
3. Compute $\sigma_N = -N^T h$, where $h = -\text{sign}(\rho_p) B^{-T} e_p$.
4. Stop if J defined by (8.51) is empty.
5. Determine β and pivot column index q by (8.52).
6. Compute Δx_q by (8.57).
7. Compute $\bar{a}_q = B^{-1}a_q$.
8. Update $\bar{x}$ by (8.58).
9. Update $\bar{y}$, $\bar{z}_N$, $\bar{z}_{j_p}$ by (8.49) and (8.50).
10. Update B^{-1} by (3.21).
11. Update (B, N) by exchanging j_p and q .
12. Go to step 1.

Theorem 8.3.6 Algorithm 8.3.1 generates a sequence of primal and of dual basic solutions. Assuming nondegeneracy, it terminates either at

- (i) Step 2, giving a pair of primal and dual basic optimal solutions; or at
- (ii) Step 4, detecting the infeasibility of the problem.

Proof The validity comes from Lemmas 8.3.2, 8.3.4, and 8.3.5, and related discussions, made preceding Algorithm 8.3.1. $\square$

Example 8.3.1 Solve the following problem by Algorithm 8.3.1:

$$\begin{aligned}
 \min \quad & f = x_1 + 2x_2 - 2x_3, \\
 \text{s.t.} \quad & -2x_1 + x_2 + x_3 + x_4 = 0, \\
 & -x_1 - x_2 + x_3 + x_5 = 0, \\
 & x_1 - x_2 - 2x_3 + x_6 = 0, \\
 & 1 \leq x_1 \leq 5, \quad -2 \leq x_2 \leq \infty, \quad -3 \leq x_3 \leq 0, \quad 2 \leq x_4 \leq 5, \\
 & 0 \leq x_5 \leq 6, \quad -3 \leq x_6 \leq 0.
 \end{aligned}$$

Answer Initial: $B = \{4, 5, 6\}$, $N = \{1, 2, 3\}$, $B^{-1} = I$, $\bar{x}_N = (1_{(-)}, -2_{(-)}, 0^{(+)})^T$, $\bar{x}_B = (4, -1, -3)^T$, $\bar{y} = (0, 0, 0)$, $\bar{z}_N = (1, 2, -2)^T$, $\bar{f} = -3$.

Iteration 1:

1. $\max\{0, |0 - (-1)|, 0\} = 1$, $p = 2$, x_5 leaves the basis.
3. $h = -\text{sign}(\rho_2)B^{-T}e_2 = (0, -1, 0)^T$, $\sigma_N = -N^Th = (-1, -1, 1)^T$.
4. $J = \{1, 2, 3\} \neq \emptyset$.
5. $\beta = \min\{-1/(-1), -2/(-1), -(-2)/1\} = 1$, $q = 1$.
6. $\Delta x_1 = \rho_2/|\sigma_1| = 1$.
7. $\bar{a}_1 = B^{-1}a_1 = (-2, -1, 1)^T$.
8. $\bar{x}_B = \bar{x}_B - \Delta x_1 \bar{a}_1 = (4, -1, -3)^T - (-2, -1, 1)^T = (6, 0, -4)^T$.
 $\bar{x}_N = \bar{x}_N + \Delta x_1 e_1 = (1, -2, 0)^T + (1, 0, 0)^T = (2, -2, 0)^T$.
9. $\bar{y} = \bar{y} + \beta h = (0, -1, 0)^T + 1 \times (0, -1, 0)^T = (0, -2, 0)^T$,
 $\bar{z}_N = \bar{z}_N + \beta \sigma_N = (1, 2, -2)^T + 1(-1, -1, 1)^T = (0, 1, -1)^T$,
 $\bar{z}_5 = \text{sign}(\rho_2)\beta = 1$.
10. Update $B^{-1} = \begin{pmatrix} 1 & -2 \\ -1 & \\ 1 & 1 \end{pmatrix}$.
11. $\bar{x}_B = (6, 2, -4)^T$, $B = \{4, 1, 6\}$; $\bar{x}_N = (0_{(-)}, -2_{(-)}, 0^{(+)})^T$, $\bar{z}_N = (1, 1, -1)^T$,
 $N = \{5, 2, 3\}$.

Iteration 2:

1. $\max\{|5 - 6|, 0, (-3) - (-4)\} = 1$, $p = 1$, x_4 leaves the basis.
3. $h = -\text{sign}(\rho_1)B^{-T}e_1 = (1, -2, 0)^T$, $\sigma_N = -N^Th = (2, -3, 1)^T$.
4. $J = \{2, 3\} \neq \emptyset$.
5. $\beta = \min\{-1/(-3), -(-1)/1\} = 1/3$, $q = 2$.
6. $\Delta x_2 = -\rho_1/|\sigma_2| = 1/3$.
7. $\bar{a}_2 = B^{-1}a_2 = (3, 1, -2)^T$.
8. $\bar{x}_B = (6, 2, -4)^T - (1/3)(3, 1, -2)^T = (5, 5/3, -10/3)^T$,
 $\bar{x}_N = \bar{x}_N + \Delta x_2 e_2 = (0, -2, 0)^T + (0, 1/3, 0)^T = (0, -5/3, 0)^T$.
9. $\bar{y} = \bar{y} + \beta h = (0, -2, 0)^T$,
 $\bar{z}_N = \bar{z}_N + \beta \sigma_N = (1, 1, -1)^T + (1/3)(2, -3, 1)^T = (5/3, 0, -2/3)^T$,
 $\bar{z}_4 = \text{sign}(\rho_1)\beta = -1/3$.
10. Update $B^{-1} = \begin{pmatrix} 1/3 & & \\ -1/3 & 1 & \\ 2/3 & & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -1 & \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1/3 & -2/3 \\ -1/3 & -1/3 \\ 2/3 & -1/3 & 1 \end{pmatrix}$.
11. $B = \{2, 1, 6\}$, $N = \{5, 4, 3\}$, $\bar{x}_B = (-5/3, 5/3, -10/3)^T$,
 $\bar{x}_N = (0_{(-)}, 5^{(+)}, 0^{(+)})^T$,
 $\bar{z}_N = (5/3, -1/3, -2/3)^T$.

Iteration 3:

1. $\max\{0, 0, |(-3) - (-10/3)|\} = 1/3$, $p = 3$, x_6 leaves the basis.
3. $h = -\text{sign}(\rho_3)B^{-T}e_3 = (-2/3, 1/3, -1)^T$, $\sigma_N = -N^T h = (-1/3, 2/3, -5/3)^T$.
4. $J = \{1, 2\} \neq \emptyset$.
5. $2 = q \in \min\{-(5/3)/(-1/3), -(-1/3)/(2/3)\}$, $\beta = 1/2$, x_4 enters the basis.
6. $\Delta x_2 = \rho_3/|\sigma_2| = -(1/3)/(2/3) = -1/2$.
7. $\bar{a}_2 = B^{-1}a_2 = (1/3, -1/3, 2/3)^T$.
8. $\bar{x}_B = (-5/3, 5/3, -10/3)^T - (-1/2)(1/3, -1/3, 2/3)^T = (-3/2, 3/2, -3)^T$,
 $\bar{x}_N = \bar{x}_N + \Delta x_2 e_2 = (0, 5, 0)^T + (0, -1/2, 0)^T = (0, 9/2, 0)^T$.
9. $\bar{y} = (0, -2, 0)^T + (1/2)(-2/3, 1/3, -1)^T = (-1/3, -11/6, -1/2)^T$,
 $\bar{z}_N = \bar{z}_N + \beta \sigma_N = (5/3, -1/3, -2/3)^T + (1/2)(-1/3, 2/3, -5/3)^T$
 $= (3/2, 0, -3/2)^T$,
 $\bar{z}_6 = \text{sign}(\rho_3)\beta = 1/2$.
10. Update $B^{-1} = \begin{pmatrix} 1 & -1/2 \\ 1 & 1/2 \\ -3/2 \end{pmatrix} \begin{pmatrix} 1/3 & -2/3 \\ -1/3 & -1/3 \\ 2/3 & -1/3 & 1 \end{pmatrix} = \begin{pmatrix} -1/2 & -1/2 \\ -1/2 & 1/2 \\ -1 & 1/2 & -3/2 \end{pmatrix}$.
11. It is satisfied that $l_B \leq \bar{x}_B \leq u_B$. The optimal solution and value are

$$x^* = (3/2, -3/2, 0, 9/2, 0, -3)^T, \quad f^* = 3/2 + 2(-3/2) = -3/2.$$

8.4 Generalized Dual Simplex Algorithm with Bound-Flipping

A so-called *bound-flipping* tactic can improve effectiveness of the generalized dual simplex algorithm significantly. In fact, it might be the main cause for the dual simplex algorithm to outperform its primal counterpart at present (Kirillova et al. 1979; Fourer 1994; Kostina 2002; Maros 2003a,b; Koberstein 2008).

Let $(\bar{y}, \bar{z})$ be the current dual feasible basic solution and let $\bar{x}$ be the associated primal solution. Assume that a row index p has been determined by (8.21) and that a column index q determined by the minimum-ratio test (8.52). Let the nonbasic variable x_q change from the current value $\bar{x}_q$ (going up or down) toward the other bound, while keeping the other nonbasic variables unchanged. For the basic variable x_{j_p} to attain the violated bound, the value of x_q could fall either within the range between the lower and upper bounds or beyond the other bound. In the latter case, it is favorable to execute the following so-called *bound-flipping* tactic:

Fix the value of x_q on the other bound and update values of basic variables accordingly; then find new column index q that attains the second minimum ratio, and do the same thing again, until the value of x_{j_p} attains the violated bound while the current value of x_q falls within the range between its lower and upper bounds.

Then, a normal step is taken by dropping x_{j_p} from and entering x_q to the basis, and the primal and dual solutions are updated. It is seen that the dual feasibility is maintained.

The bound-flipping tactic is embedded in the following subalgorithm.

Subalgorithm 8.4.1 (Bound-Flipping Subalgorithm) This algorithm provides the pivot column index q , dual stepsize β , and carries out related computations.

1. Set $j = 0$, $v = 0$, and compute $r_j = -\bar{z}_j/\sigma_j$, $\forall j \in J$.
2. Set $j = j + 1$.
3. Set $v = v + \delta a_{jq}$.
4. Set $\bar{x}_q = \begin{cases} u_q, & \text{If } \bar{x}_q = l_q, \\ l_q, & \text{If } \bar{x}_q = u_q. \end{cases}$
5. Update: $\rho_p = \rho_p - |\delta \sigma_q|$.
6. Determine q and r_q such that $r_q = \min_{j \in J} r_j$.
7. Compute Δx_q by (8.57).
8. Update: $J = J \setminus \{q\}$.
9. Go to step 13 if $J = \emptyset$.
10. Compute $\delta = \begin{cases} u_q - l_q, & \text{If } \bar{x}_q = l_q, \\ l_q - u_q, & \text{If } \bar{x}_q = u_q. \end{cases}$
11. Go to step 2 if $|\Delta x_q| \geq |\delta|$.
12. Set $\beta = r_q$.
13. Compute $u = B^{-1}v$, and update: $\bar{x}_B = \bar{x}_B - u$.
14. Return.

The following master algorithm is a slight modification of Algorithm 8.3.1.

Algorithm 8.4.1 (Generalized Dual Simplex Algorithm: Bound-Flipping) Initial: (B, N) , B^{-1} , $\bar{y}$, $\bar{z}$, $\bar{x}$ satisfying (8.28), (8.33), and (8.34). This algorithm solves the bounded-variable problem (8.3).

1. Select row index p by (8.21) together with (8.20).
2. Stop if $\rho_p = 0$ (optimality achieved).
3. Compute σ_N by (8.46) together with (8.45).
4. Stop if J defined by (8.51) is empty (Dual unbounded or primal infeasible).
5. Determine column index q by (8.52).
6. Compute Δx_q by (8.57).
7. Set $J = J \setminus \{q\}$.
8. If $J = \emptyset$, go to step 12.
9. Compute $\delta = \begin{cases} u_q - l_q, & \text{If } \bar{x}_q = l_q, \\ l_q - u_q, & \text{If } \bar{x}_q = u_q. \end{cases}$
10. If $|\Delta x_q| \geq |\delta|$, call Algorithm 8.4.1.
11. Compute $\bar{a}_q = B^{-1}a_q$.
12. Update $\bar{x}$ by (8.58).
13. Update $\bar{y}$, $\bar{z}_N$, $\bar{z}_{j_p}$ by (8.49) together with (8.50).
14. Update B^{-1} by (3.21).
15. Update (B, N) by exchanging j_p and q .

16. Go to step 1.

The bound-flipping adds extra work, involving a linear system (in step 13 of Algorithm 8.4.1). This is tolerable, compared with the return. Since the associated dual stepsize is usually much larger than that without bound-flipping, the increment in objective value is also much larger, and hence the number of iterations required usually reduces significantly. The bound-flipping tactic is essential in the current dual simplex codes.

Example 8.4.1 Solve the following problem by Algorithm 8.4.1:

$$\begin{aligned}
 \min \quad & f = -x_1 + 2x_3 + 3x_4, \\
 \text{s.t.} \quad & -2x_1 + x_2 + x_3 + x_4 = -2, \\
 & x_1 - x_3 + x_4 + x_5 = 1, \\
 & x_1 - 2x_3 - 3x_4 + x_6 = 0, \\
 & 0 \leq x_1 \leq 2, \quad -6 \leq x_2 \leq 10, \quad 0 \leq x_3 \leq 7, \\
 & 1 \leq x_4 \leq 5, \quad 2 \leq x_5 \leq 6, \quad -1 \leq x_6 \leq 6.
 \end{aligned}$$

Answer Initial: $B = \{2, 5, 6\}$, $N = \{1, 3, 4\}$, $B^{-1} = I$, $\bar{x}_N = (2^{(+)}, 0_{(-)}, 1_{(-)})^T$, $\bar{x}_B = (1, -2, 1)^T$, $\bar{y} = (0, 0, 0)$, $\bar{z}_N = (-1, 2, 3)^T$, $\bar{f} = 1$.

Iteration 1:

1. $\max\{0, 2 - (-2), 0\} = 4$, $p = 2$, x_5 leaves the basis.
3. $h = -\text{sign}(\rho_2)B^{-T}e_2 = (0, -1, 0)^T$, $\sigma_N = -N^Th = (1, -1, 1)^T$.
4. $J = \{1, 2\}$.
5. $\beta = \min\{-(-1)/1, -2/-1\} = 1$, $q = 1$.
6. $\Delta x_1 = -\rho_2/|\sigma_1| = -4/1 = -4$.
7. $J = J \setminus \{1\} = \{2\} \neq \emptyset$.
9. $\delta = l_1 - u_1 = 0 - 2 = -2$.
10. $|\Delta x_1| > |\delta|$, so call Algorithm 8.4.1.
 - 1) $j = 0$, $v = 0$, $r_2 = -2/-1 = 2$;
 - 2) $j = j + 1 = 1$;
 - 3) $v = v + \delta a_1 = (-2)(-2, 1, 1)^T = (4, -2, -2)^T$;
 - 4) $\bar{x}_1 = l_1 = 0$;
 - 5) $\rho_2 = \rho_2 - |\delta\sigma_1| = 4 - 2 \times 1 = 2$;
 - 6) $q = 2$;
 - 7) $\Delta x_2 = -\rho_2/|\sigma_2| = -2/(-1) = 2$;
 - 8) $J = J \setminus \{2\} = \emptyset$;
- 12) $\beta = r_2 = 2$;
- 13) $u = B^{-1}v = (4, -2, -2)^T$;
- $\bar{x}_B = \bar{x}_B - u = (1, -2, 1)^T - (4, -2, -2)^T = (-3, 0, 3)^T$;
- 14) Return.
11. $\bar{a}_2 = B^{-1}a_2 = (1, -1, -2)^T$.
12. $\bar{x}_B = (-3, 0, 3)^T - 2(1, -1, -2)^T = (-5, 2, 7)^T$,
 $\bar{x}_N = (0, 0, 1)^T + (0, 2, 0)^T = (0, 2, 1)^T$.

13. $\bar{y} = (0, 0, 0)^T + 2(0, -1, 0)^T = (0, -2, 0)^T$,
 $\bar{z}_N = \bar{z}_N + \beta\sigma_N = (-1, 2, 3)^T + 2(1, -1, 1)^T = (1, 0, 5)^T$, $\bar{z}_{j_2} = \text{sign}(\rho_2)\beta = 2$.
14. $B^{-1} = \begin{pmatrix} 1 & 1 \\ -1 & -2 \\ 1 & 1 \end{pmatrix}$.
16. $\bar{x}_B = (-5, 2, 7)^T$, $\bar{x}_N = (0_{(-)}, 2_{(-)}, 1_{(-)})^T$, $\bar{z}_N = (1, 2, 5)^T$,
 $B = \{2, 3, 6\}$, $N = \{1, 5, 4\}$.

Iteration 2:

1. $\max\{0, 0, 7 - 6\} = 1$, $p = 3$, x_6 leaves the basis.
3. $h = -\text{sign}(\rho_3)B^{-T}e_3 = (0, -2, 1)^T$, $\sigma_N = -N^Th = (1, 2, 5)^T$.
4. $J = \emptyset$, hence the problem is dual unbounded, or primal infeasible.

If bound-flipping had not been used, there would be much more iterations required.

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Chapter 9

Decomposition Method



Solving large-scale LP problems is a challenging task, putting forward high requirements on the algorithms' efficiency, storage, and numerical stability. The decomposition method divides a large-scale LP problem into relatively small ones to cope with normal LP solvers.

This can be done because the objective function and nonnegativity constraints are variable separable, i.e., can be divided into several independent groups. If, in addition, the system of constraint equations is also variable-separable, then the large-scale problem can be handled by solving smaller problems. This chapter presents D-W decomposition and its illustration, its economic interpretation, Benders decomposition and its illustration, and dual Benders decomposition. A new decomposition principle will be introduced in the final chapter of Part II.

Let us bring up the following example:

$$\begin{aligned}
 \min \quad & f = (c^1)^T x^1 + \cdots + (c^k)^T x^k, \\
 \text{s.t.} \quad & D_1 x^1 = h^1, \\
 & D_2 x^2 = h^2, \\
 & \vdots \\
 & D_k x^k = h^k, \\
 & x^1, \dots, x^k \geq 0,
 \end{aligned} \tag{9.1}$$

where the coefficient matrix is of a block diagonal structure: orders of submatrices D_j and dimensions of subvectors $c^j, x^j, h^j, j = 1, \dots, k$, are consistent, that is, each pair of c_j and x_j share the same dimension, equal to the number of columns of D_j , and the dimension of b_j is equal to the number of rows of D_j .

The feasible region of the problem is

$$P' = \{((x^1)^T, \dots, (x^k)^T)^T \mid D_j x^j = h^j, x^j \geq 0, j = 1, \dots, k\},$$

or, otherwise, denoted in the form of the “Cartesian product”:

$$P' = P^1 \times \dots \times P^k; \quad P^j = \{x^j \mid D_j x^j = h^j, x^j \geq 0\}, \quad j = 1, \dots, k.$$

Since variables and constraints of the problem can be separated into independent k sets, the problem comes down to solving k smaller problems, i.e.,

$$\begin{aligned} & \min (c^j)^T x^j, \\ & \text{s.t. } D_j x^j = h^j, \quad x^j \geq 0, \quad j = 1, \dots, k. \end{aligned}$$

If there is a feasible basic solution $\bar{x}^j$, associated objective value $\bar{f}^j$, to each of the small problems, there is a feasible basic solution, associated with corresponding objective value, to the original problem, i.e.,

$$\bar{x} = ((\bar{x}^1)^T, \dots, (\bar{x}^k)^T)^T, \quad \bar{f} = \bar{f}^1 + \dots + \bar{f}^k. \quad (9.2)$$

The same is valid with optimal basic solutions and objective values. On the other hand, if some of the small problems are unbounded, the original is unbounded too; if required, it is not difficult to determine the associated descent extreme direction (Proposition 3.3.2).

Although such types of problems would not be encountered in practice very often, there are problems with some partially separable structures, which can be solved by methods, such as D-W decomposition and Benders decomposition, as introduced in this chapter.

9.1 D-W Decomposition

The Dantzig–Wolfe (D-W) decomposition (1960, 1961) partitions constraints into two sets and handles a *master* problem and a subprogram alternately. The master problem is expressed in terms of vertices and extreme directions of the feasible region, associated with one of the sets. Throughout the solution process, the subprogram takes the set as its constraints with its objective function varying iteration by iteration. It is solved to determine a column of the master problem to enter the basis or claim optimality of the original problem achieved. As a result, the LP problem is solved by dealing with two smaller programs. Further division is possible if the set of constraints of the subprogram has a separable structure.

By partitioning the constraints into two sets, the standard LP problem can be written as

$$\begin{aligned} & \min f = c^T x, \\ & \text{s.t. } Ax = b, \\ & \quad Hx = h, \quad x \geq 0, \end{aligned} \quad (9.3)$$

where $A \in \mathcal{R}^{m \times n}$, $H \in \mathcal{R}^{m^1 \times n}$; $b \in \mathcal{R}^m$; $h \in \mathcal{R}^{m^1}$.

Introduce polyhedral convex set

$$P = \{x \in \mathcal{R}^n \mid Hx = h, x \geq 0\}.$$

Let P be nonempty and possess the following sets of vertices and extreme directions, respectively:

$$U = \{u^1, \dots, u^s\}, \quad V = \{v^1, \dots, v^t\}.$$

According to the Representation Theorem 2.4.1, it holds that

$$P = \{x \mid x = \sum_{i=1}^s \alpha_i u^i + \sum_{j=1}^t \beta_j v^j; \sum_{i=1}^s \alpha_i = 1, \alpha_i \geq 0, i = 1, \dots, s, \beta_j \geq 0, j = 1, \dots, t\}.$$

Substituting the expression of x into $\min\{c^T x \mid Ax = b\}$, we convert (9.3) into the standard problem with respect to variables α_i, β_j , i.e.,

$$\begin{aligned} \min_{\alpha_i, \beta_j} f &= \sum_{i=1}^s (c^T u^i) \alpha_i + \sum_{j=1}^t (c^T v^j) \beta_j, \\ \text{s.t.} \quad &\sum_{i=1}^s (A u^i) \alpha_i + \sum_{j=1}^t (A v^j) \beta_j = b, \\ &\sum_{i=1}^s \alpha_i = 1, \\ &\alpha_i, \beta_j \geq 0, \quad i = 1, \dots, s, j = 1, \dots, t, \end{aligned} \tag{9.4}$$

which is called *master problem*.

The relation between feasible solutions to the master problem and to the original problem is determined by:

$$x = \sum_{i=1}^s \alpha_i u^i + \sum_{j=1}^t \beta_j v^j, \tag{9.5}$$

corresponding to equal feasible values. Respectively, the feasible/optimal solutions of the master problem correspond to those of the original problem. In general, however, the latter is not a basic solution even if the former is.

Let us focus on the master problem alone at the moment. The difficulty seems that the number $s + t$ of columns of the master problem is usually much larger than the number $m + 1$ of its rows, even if the number of rows would be much less than the number $m + m^1$ of the rows of the original problem; moreover, vertices and extreme directions of P are used in the expression of the coefficient matrix and the objective

of the master problem. Fortunately, D-W decomposition can generate an entering column (with a leaving index determined), which allows simplex iteration to make basis change. For this reason, the method is often termed *column generation*.

Without loss of generality, assume that the current basis matrix is

$$B = \begin{pmatrix} Au^1 & \cdots & Au^{s'} & Av^1 & \cdots & Av^{t'} \\ 1 & \cdots & 1 & 0 & \cdots & 0 \end{pmatrix}, \quad (9.6)$$

the number of columns of which is $s' + t' = m + 1$. The basic components of the associated feasible solution are defined by system

$$B(\alpha_1, \dots, \alpha_{s'}, \beta_1, \dots, \beta_{t'})^T = \begin{pmatrix} b \\ 1 \end{pmatrix}.$$

Denote the feasible solution by:

$$\bar{\alpha}_i \geq 0, \quad i = 1, \dots, s'; \quad \bar{\beta}_j, \quad j = 1, \dots, t'.$$

Then the associated feasible solution and objective value of the original problem are

$$\bar{x} = \sum_{i=1}^{s'} \bar{\alpha}_i u^i + \sum_{j=1}^{t'} \bar{\beta}_j v^j, \quad \bar{f} = \sum_{i=1}^{s'} (c^T u^i) \bar{\alpha}_i + \sum_{j=1}^{t'} (c^T v^j) \bar{\beta}_j. \quad (9.7)$$

The simplex multipliers $(\bar{y}, \bar{\gamma})$, corresponding to the first m rows and the bottom row, are defined by system

$$B^T \begin{pmatrix} y \\ \gamma \end{pmatrix} = \hat{c}_B, \quad (9.8)$$

where $\hat{c}_B$ consists of costs of the master problem, corresponding to basic columns. Check numbers (including the vanished reduced costs) are then

$$\begin{aligned} c^T u^i - (Au^i)^T \bar{y} - \bar{\gamma}, & \quad i = 1, \dots, s, \\ c^T v^j - (Av^j)^T \bar{y}, & \quad j = 1, \dots, t. \end{aligned}$$

If the reduced costs are all nonnegative, then optimality of the original problem is achieved, and (9.7) gives its optimal solution and objective value; otherwise, some nonbasic column corresponding to a negative reduced cost is selected to enter the basis.

To avoid computing all the reduced costs, D-W decomposition solves a subprogram with P as its feasible region instead, i.e.,

$$\begin{aligned} \min \quad & \zeta = (c - (A)^T \bar{y})^T x - \bar{\gamma}, \\ \text{s.t.} \quad & Hx = h, \quad x \geq 0. \end{aligned} \quad (9.9)$$

Let us examine the relationship between the subprogram and the master as well as the original problems. If the feasible region P of the subprogram is empty, the master and the original programs are both infeasible. Otherwise, we have the following result.

Lemma 9.1.1 *Let f^* be the optimal value of the original problem. If $\bar{f}$ is the current objective value of the master problem, and ζ^* is the optimal value of the subprogram, then it holds that*

$$\bar{f} + \zeta^* \leq f^* \leq \bar{f}. \quad (9.10)$$

Proof It is clear that $f^* \leq \bar{f}$, and hence it is only needed to show

$$\bar{f} + \zeta^* \leq f^*. \quad (9.11)$$

Let x^* be optimal solution of the original problem. Since it is also a feasible solution to the subprogram, we have

$$(c - (A)^T \bar{y})^T x^* - \bar{\gamma} \geq \zeta^*.$$

Note that $B^{-1}(h^T, 1)^T$ is a basic solution. From the preceding expression, $Ax^* = b$ and $(\bar{y}, \bar{\gamma})$ satisfying system (9.8), it follows that

$$f^* = c^T x^* \geq \bar{y}^T Ax^* + \bar{\gamma} + \zeta^* = (\bar{y}^T h + \bar{\gamma}) + \zeta^* = \hat{c}_B^T B^{-1} \begin{pmatrix} b \\ 1 \end{pmatrix} + \zeta^* = \bar{f} + \zeta^*,$$

which gives (9.11). $\square$

In each iteration, therefore, it is possible to estimate the optimal value of the original problem by giving lower and upper bounds by (9.10). It holds by (9.11) that

$$\bar{f} - f^* \leq -\zeta^*.$$

If $\bar{f}$ is taken as an approximately optimal value of the original problem, therefore, the associated absolute error bound is $-\zeta^*$. If $\bar{f} \neq 0$, it follows further that

$$(\bar{f} - f^*)/|\bar{f}| \leq -\zeta^*/|\bar{f}|,$$

the right-hand side of which gives a relative error bound.

Theorem 9.1.2 *If the optimal value of the subprogram vanishes, optimality of the original problem is achieved.*

Proof By Lemma 9.1.1 and $\bar{f} - f^* \geq 0$, it is known that $\zeta^* \leq 0$, that is, the optimal value of the subprogram is less than or equal to 0. If it is equal to zero, then $\bar{f} \leq f^* \leq \bar{f}$, which implies that optimality of the master and of the original problem is achieved. $\square$

If the feasible region P of the subprogram is nonempty, consequently, solving it by the simplex method leads to one of the following two cases:

- (i) A vertex optimal solution, say u^* , of the subprogram, is reached.

If the optimal value vanishes, optimality of the original problem is achieved, and we are done; otherwise, a column

$$w' = \begin{pmatrix} Au^* \\ 1 \end{pmatrix} \quad (9.12)$$

is generated to enter the basis.

- (ii) It is detected that the subprogram is unbounded. Proposition 3.3.2 In this case, a descent extreme direction, say $v^* \in V$, of P can be determined such that (Proposition 3.3.2)

$$(c - (A)^T \bar{y})^T v^* < 0,$$

implying that the associated reduced costs of the master problem are negative. Then, a column

$$w' = \begin{pmatrix} Av^* \\ 0 \end{pmatrix} \quad (9.13)$$

is generated to enter the basis. To avoid repetition, other simplex steps, such as the determination of a leaving column, and etc. will not be described here.

The overall steps are summarized into the following algorithm.

Algorithm 9.1.1 (Dantzig–Wolfe Decomposition) Initial: basis B of the master problem, basic components $\bar{\alpha}_i$, $\bar{\beta}_j$ of the feasible basic solution, and associated objective value $\bar{f}$. This algorithm solves LP problem (9.3).

1. Solve system (9.8) for $(\bar{y}, \bar{\gamma})$.
2. Call the simplex algorithm to solve subprogram (9.9).
3. If an optimal solution u^* of the subprogram is obtained, then:
 - (1) If the associated optimal value $\zeta^* < 0$, generate an entering basic column w' by (9.12), and go to step 5.
 - (2) Else compute the optimal solution and objective value of the original problem by (9.7), and stop.
4. If it is detected that the subprogram is unbounded, generate a new basic column w' by (9.13).
5. Solve system $Bw = w'$ for $\bar{w}$.
6. Stop if $\bar{w} \leq 0$ (the original problem is unbounded).
7. Determine a stepsize α and leaving index by the minimum-ratio test.
8. Update B , $\bar{\alpha}_i$, $\bar{\beta}_j$, and $\bar{f}$.
9. Go to step 1.

Note When the optimal value ζ^* of the subprogram is close to 0, terminating the preceding algorithm gives an approximate optimal solution to the original problem (Lemma 9.1.1).

The algorithm terminates if nondegeneracy is assumed.

It is noticeable that the new reduced cost, corresponding to a leaving column, is certainly positive (see Proposition 3.1.1 and its proof). In each iteration of Algorithm 9.1.1, therefore, actually solved is a restricted master problem with $m + 2$ columns, consisting of basic columns and the entering column. The decomposition algorithm can be modified to allow a restricted master problem including more columns. Thus, we redefine the following.

Definition 9.1.1 The problem consisting of k columns of the master problem is a restricted subprogram, where $m + 2 \leq k \leq s + t$, provided these columns have full row rank $m + 1$.

Then we have the following algorithm.

Algorithm 9.1.2 (Modified Dantzig–Wolfe Decomposition) *The same as Algorithm 9.1.1, except the restricted master problem is solved before it is updated (by generating a new entering and determining a leaving index).*

There are multiple ways to construct such an initial restricted master problem. For example, taking any $m + 1$ independent columns to form an initial basis, we add new entering columns successively until the number of columns attaining k or, otherwise, add the nonbasic columns, corresponding to the smallest $k - m - 1$ reduced costs, after the number of columns is still less than k but close to the computer's storage limit.

9.1.1 Starting-Up of D-W Decomposition

At first, we attempt to find a vertex, say u^1 , of $P = \{x \in \mathcal{R}^n \mid Hx = h, x \geq 0\}$. If it does not exist, the original LP problem is clearly infeasible. Assume that u^1 is available. We introduce m artificial variables σ_l , $l = 1, \dots, m$ to construct the following auxiliary Phase-I master problem:

$$\begin{aligned}
 \min_{\sigma_l, \alpha_i, \beta_j} \quad & w = \sum_{l=1}^m \sigma_l, \\
 \text{s.t.} \quad & \sum_{l=1}^m \pm e_l \sigma_l + \sum_{i=1}^s (Au^i) \alpha_i + \sum_{j=1}^t (Av^j) \beta_j = b, \\
 & \sum_{i=1}^s \alpha_i = 1, \\
 & \sigma_l, \alpha_i, \beta_j \geq 0, \quad l = 1, \dots, m, \quad i = 1, \dots, s, \quad j = 1, \dots, t,
 \end{aligned} \tag{9.14}$$

where for $l = 1, \dots, m$, if $b_l - (Au^1)_l \geq 0$, $+e_l$ is taken; else, $-e_l$ taken. Thus, α_l together with σ_l , $l = 1, \dots, m$, constitute a set of basic variables. Basic components of the feasible basic solution are

$$\bar{\alpha}_l = 1, \quad \bar{\sigma}_l = |b_l - (Au^1)_l| \geq 0, \quad l = 1, \dots, m.$$

Then Algorithm 9.1.1 can get itself started from the basis consisting of columns associated with these basic components.

It is clear that there exists an optimal solution, corresponding to a nonnegative optimal value, to the auxiliary problem: if the optimal value is greater than zero, the original problem is infeasible; if, otherwise, the optimal value vanishes, a feasible solution of the master problem comes from deleting all artificial components from the auxiliary optimal solution, giving a feasible basic solution to the master problem, which can be taken as an initial one for Phase-II, if artificial components are all nonbasic. In the case when some artificial variables are basic, additional iterations are needed to force them to leave the basis, just as *following-up steps* described in Sect. 3.2.

9.2 Illustration of D-W Decomposition

In principle, D-W decomposition can handle general standard LP problems, and the bisection of constraints can be arbitrary. In terms of effectiveness, nevertheless, its performance varies with problems of different structures. Many problems coming from practice are often of some (or partial) structures that are amenable to D-W decomposition.

Models involving block structure may belong to this category, as illustrated by the example below.

Example 9.2.1 Solve the following problem by the D-W decomposition:

$$\begin{aligned} \min \quad & f = -x_2 + x_3 - 2x_4 - 3x_5 + x_6 - x_7 + x_8, \\ \text{s.t.} \quad & x_1 - 3x_2 - x_3 + x_4 + 2x_5 + x_6 - x_7 - 3x_8 = 2, \\ & x_1 + 4x_3 + x_4 = 1, \\ & x_2 + x_3 - x_4 = 4, \\ & x_5 + 3x_7 - x_8 = 1, \\ & x_6 + x_7 - 3x_8 = 2, \\ & x_j \geq 0, \quad j = 1, \dots, 8. \end{aligned}$$

Answer The first equality constraint is a coupling one, corresponding to a master problem of form (9.19), while the others correspond to a subproblem of the form (9.21).

The second and third equality constraints are

$$\begin{aligned}x_1 + 4x_3 + x_4 &= 1, \\x_2 + x_3 - x_4 &= 4, \quad x_j \geq 0, \quad j = 1, \dots, 4,\end{aligned}$$

which has a basic solution $x_1 = 1, x_2 = 4, x_3 = x_4 = 0$, and the fourth and fifth are

$$\begin{aligned}x_5 + 3x_7 - x_8 &= 1, \\x_6 + x_7 - 3x_8 &= 2, \quad x_j \geq 0, \quad j = 5, \dots, 8,\end{aligned}$$

which has a basic solution $x_5 = 1, x_6 = 2, x_7 = x_8 = 0$. Therefore, there is a feasible basic solution $u^1 = (1, 4, 0, 0, 1, 2, 0, 0)^T$ to the subprogram.

Accordingly,

$$b_1^1 - (Au^1)_1 = 2 - (1, -3, -1, 1, 2, 1, -1, -3)(1, 4, 0, 0, 1, 2, 0, 0)^T = 2 + 7 = 9 \geq 0.$$

Thereby, we construct an auxiliary master problem of form (9.14), i.e.,

$$\begin{aligned}\min \quad & \sigma_1, \\ \text{s.t.} \quad & \sigma_1 - 7\alpha_1 + \sum_{i=2}^s (Au^i)\alpha_i + \sum_{j=1}^t (Av^j)\beta_j = 2, \\ & \sum_{i=1}^s \alpha_i = 1, \\ & \sigma_1, \alpha_i, \beta_j \geq 0, \quad i = 1, \dots, s, \quad j = 1, \dots, t,\end{aligned} \tag{9.15}$$

where $A = (1, -3, -1, 1, 2, 1, -1, -3)$.

The basis matrix is $B = \begin{pmatrix} 1 & -7 \\ 0 & 1 \end{pmatrix}$. Solving system $B(\sigma_1, \alpha_1)^T = (2, 1)^T$ leads

to the feasible basic solution, whose basic components are $\begin{pmatrix} \bar{\sigma}_1 \\ \bar{\alpha}_1 \end{pmatrix} = \begin{pmatrix} 9 \\ 1 \end{pmatrix}$, with auxiliary objective value $\bar{w} = 1$.

Phase-I: Call D-W Decomposition 9.1.1 to solve the auxiliary master problem (9.15).

Iterations 1:

1. Solving $B^T(y, \gamma)^T = (1, 0)^T$ gives simplex multipliers $(\bar{y}, \bar{\gamma}) = (1, 7)$.
2. The objection function of the subprogram is

$$\begin{aligned}\zeta &= (c - (A)^T \bar{y})^T x - \bar{\gamma} \\ &= ((0, -1, 1, -2, -3, 1, -1, 1) - (1, -3, -1, 1, 2, 1, -1, -3))x - 7 \\ &= (-1, 2, 2, -3, -5, 0, 0, 4)x - 7.\end{aligned}$$

To be solved separately, the subprogram is decomposed into the following two smaller programs:

(i)

$$\begin{aligned} \min & -x_1 + 2x_2 + 2x_3 - 3x_4, \\ \text{s.t. } & x_1 + 4x_3 + x_4 = 1, \\ & x_2 + x_3 - x_4 = 4, \quad x_j \geq 0, \quad j = 1, \dots, 4. \end{aligned}$$

Taking basis and nonbasis $B = \{1, 2\}$, $N = \{3, 4\}$ ($B^{-1} = I$) determines the feasible basic solution $\bar{x}_B = (1, 4)^T \geq 0$, $\bar{x}_N = (0, 0)^T$, and the reduced costs

$$\bar{z}_N = c_N - N^T B^{-1} c_B = \begin{pmatrix} 2 \\ -3 \end{pmatrix} - \begin{pmatrix} 4 & 1 \\ 1 & -1 \end{pmatrix}^T \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}.$$

Since $\min\{4, 0\} = 0 \geq 0$, the following optimal vertex solution with objective value is obtained:

$$\bar{x} = (1, 4, 0, 0)^T, \quad \zeta_1 = 7.$$

(ii)

$$\begin{aligned} \min & -5x_5 + 4x_8, \\ \text{s.t. } & x_5 + 3x_7 - x_8 = 1, \\ & x_6 + x_7 - 3x_8 = 2, \quad x_j \geq 0, \quad j = 5, \dots, 8. \end{aligned}$$

Taking basis and nonbasis $B = \{5, 6\}$, $N = \{7, 8\}$ ($B^{-1} = I$) determines feasible basic solution $\bar{x}_B = (1, 2)^T \geq 0$, $\bar{x}_N = (0, 0)^T$, and the reduced costs

$$\bar{z}_N = c_N - N^T B^{-1} c_B = \begin{pmatrix} 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 & -1 \\ 1 & -3 \end{pmatrix}^T \begin{pmatrix} -5 \\ 0 \end{pmatrix} = \begin{pmatrix} 15 \\ -1 \end{pmatrix}.$$

Thus $\min\{15, -1\} = -1 < 0$, $q = 8$. Since $B^{-1}a_8 = (-1, -3)^T \leq 0$, the subprogram is unbounded.

4. The feasible region of the subprogram has descent extreme direction $v^1 = (0, 0, 0, 0, 1, 3, 0, 1)^T$.

Generate entering column $w' = (2, 0)^T$ by (9.13).

5. $\bar{w} = (2, 0)^T \not\leq 0$.
7. $\alpha = \min\{9/2\}$, $p = 1$.
8. $(\bar{\sigma}_1, \bar{\alpha}_1)^T = (9, 1)^T - (9/2)(2, 0)^T = (0, 1)^T$.

$$B = \begin{pmatrix} 2 & -7 \\ 0 & 1 \end{pmatrix}. (\bar{\beta}_1, \bar{\alpha}_1)^T = (9/2, 1)^T,$$

$$\begin{aligned}\bar{f} &= (0, -1, 1, -2, -3, 1, -1, 1)(0, 0, 0, 0, 1, 3, 0, 1)^T(9/2) \\ &+ (0, -1, 1, -2, -3, 1, -1, 1)(1, 4, 0, 0, 1, 2, 0, 0)^T = -1/2.\end{aligned}$$

The only artificial variable σ_1 has left the basis, hence Phase-I is finished.

Phase-II: Call D-W Decomposition 9.1.1 to solve the master problem.

Iterations 2:

1. $\hat{c}_B = (c^T v^1, c^T u^1)^T = (1, -5)^T$.
Solve $B^T(y, \gamma)^T = (1, -5)^T$ for simplex multipliers $(\bar{y}, \bar{\gamma}) = (1/2, -3/2)$.
2. The objective function of the subprogram is

$$\begin{aligned}\zeta &= (c - (A)^T \bar{y})^T x - \bar{\gamma} \\ &= ((0, -1, 1, -2, -3, 1, -1, 1) - (1, -3, -1, 1, 2, 1, -1, -3)(1/2))x + 3/2 \\ &= (-1/2, 1/2, 3/2, -5/2, -4, 1/2, -1/2, 5/2)x + 3/2.\end{aligned}$$

(i)

$$\begin{aligned}\min & -1/2x_1 + 1/2x_2 + 3/2x_3 - 5/2x_4, \\ \text{s.t. } & x_1 + 4x_3 + x_4 = 1, \\ & x_2 + x_3 - x_4 = 4, \quad x_j \geq 0, \quad j = 1, \dots, 4,\end{aligned}$$

which has the optimal vertex solution $(\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4)^T = (0, 5, 0, 1)^T$ with optimal value $\zeta_1 = 0$.

(ii)

$$\begin{aligned}\min & -4x_5 + 1/2x_6 - 1/2x_7 + 5/2x_8, \\ \text{s.t. } & x_5 + 3x_7 - x_8 = 1, \\ & x_6 + x_7 - 3x_8 = 2, \quad x_j \geq 0, \quad j = 5, \dots, 8\end{aligned}$$

has optimal vertex solution $(\bar{x}_5, \bar{x}_6, \bar{x}_7, \bar{x}_8)^T = (1, 2, 0, 0)^T$ with optimal value $\zeta_1 = -3$.

Thereby, the subprogram has the following optimal solution and optimal value:

$$u^2 = (0, 5, 0, 1, 1, 2, 0, 0)^T, \quad \zeta^* = 0 - 3 + 3/2 = -3/2 < 0.$$

3(1). $\zeta^* < 0$. Generate entering column $w' = (-10, 1)^T$ by (9.12).

$$5. \bar{w} = (-3/2, 1)^T \not\leq 0.$$

$$7. \alpha = \min\{1/1\}, \quad p = 2.$$

$$8. (\bar{\beta}_1, \bar{\alpha}_1)^T = (9/2, 1)^T - (-3/2, 1)^T = (6, 0)^T. B = \begin{pmatrix} 2 & -10 \\ 0 & 1 \end{pmatrix}.$$

$$(\bar{\beta}_1, \bar{\alpha}_2)^T = (6, 1)^T. \bar{f} = \bar{f} + \alpha\zeta^* = -1/2 - 3/2 = -2.$$

Iterations 3:

$$1. \hat{c}_B = (c^T v^1, c^T u^2)^T = (1, -8)^T.$$

Solve $B^T(y, \gamma)^T = (1, -8)^T$ for simplex multipliers $(\bar{y}, \bar{\gamma}) = (1/2, -3)$.

2. The objective of the subprogram is

$$\begin{aligned} \zeta &= (c - (A)^T \bar{y})^T x - \bar{\gamma} \\ &= ((0, -1, 1, -2, -3, 1, -1, 1) - (1, -3, -1, 1, 2, 1, -1, -3)(1/2))x + 3 \\ &= (-1/2, 1/2, 3/2, -5/2, -4, 1/2, -1/2, 5/2)x + 3. \end{aligned}$$

(i) The problem is the same as that in the previous iteration, with optimal vertex solution and objective value

$$(\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4)^T = (0, 5, 0, 1)^T, \quad \zeta_1 = 0.$$

(ii) The problem is the same as that in the previous iteration, with optimal vertex solution

$$(\bar{x}_5, \bar{x}_6, \bar{x}_7, \bar{x}_8)^T = (1, 2, 0, 0)^T \text{ and optimal value } \zeta_1 = -3.$$

The optimal vertex solution and optimal value to the subprogram are

$$u^3(0, 5, 0, 1, 1, 2, 0, 0)^T, \quad \zeta^* = 0 - 3 + 3 = 0.$$

3(2) Since $\zeta^* = 0$, an optimal solution and optimal value to the original problem are obtained, i.e.,

$$\begin{aligned} x^* &= (0, 0, 0, 0, 1, 3, 0, 1)^T(6) + (0, 5, 0, 1, 1, 2, 0, 0)^T \\ &= (0, 5, 0, 1, 7, 20, 0, 0)^T, \quad f^* = \bar{f} = -2. \end{aligned}$$

Another suitable type of problems have staircase structure close to block, like the following instance, which can be solved by the D-W decomposition in a nested manner:

$$\begin{aligned} \min f &= (c^1)^T x^1 + (c^2)^T x^2 + (c^3)^T x^3 + (c^4)^T x^4, \\ \text{s.t. } D_{11}x^1 &= h^1, \\ A_{21}x^1 + D_{22}x^2 &= b^2, \\ A_{32}x^2 + D_{33}x^3 &= b^3, \\ A_{43}x^3 + D_{44}x^4 &= b^4, \\ x^1, x^2, x^3, x^4 &\geq 0, \end{aligned} \tag{9.16}$$

where x^j , $j = 1, 2, 3, 4$ are vectors of appropriate dimensions consistent with the preceding matrices.

Partition the equality constraints of the above problem into two parts, denoted by $\{t = 1\}$ and $\{t = 2, 3, 4\}$, respectively. Let $\{u^i\}$ and $\{v^j\}$ be, respectively, the vertex set and extreme direction set of the set of solutions to the latter part $\{t = 2, 3, 4\}$, i.e.,

$$\begin{aligned} A_{21}x^1 + D_{22}x^2 &= b^2, \\ A_{32}x^2 + D_{33}x^3 &= b^3, \\ A_{43}x^3 + D_{44}x^4 &= b^4, \\ x^1, x^2, x^3, x^4 &\geq 0. \end{aligned}$$

According to the Representation Theorem 2.4.1, the solution to the preceding can be expressed as

$$x = \sum_{i=1}^s \alpha_i u^i + \sum_{j=1}^t \beta_j v^j, \quad \sum_{i=1}^s \alpha_i = 1, \quad \alpha_i, \beta_j \geq 0, \quad i = 1, \dots, s, \quad j = 1, \dots, t. \quad (9.17)$$

Introduce partition

$$x = \begin{pmatrix} x^1 \\ x^2 \\ x^3 \\ x^4 \end{pmatrix}; \quad u^i = \begin{pmatrix} u^{i1} \\ u^{i2} \\ u^{i3} \\ u^{i4} \end{pmatrix}, \quad i = 1, \dots, s; \quad v^j = \begin{pmatrix} v^{j1} \\ v^{j2} \\ v^{j3} \\ v^{j4} \end{pmatrix}, \quad j = 1, \dots, t.$$

Substituting (9.17) to the first part $\{t = 1\}$ leads to D-W master problem

$$\begin{aligned} \min_{\alpha_i, \beta_j} \quad & f = \sum_{i=1}^s ((c^1)^T u^{i1} + (c^2)^T u^{i2} + (c^3)^T u^{i3} + (c^4)^T u^{i4}) \alpha_i \\ & + \sum_{j=1}^t ((c^1)^T v^{j1} + (c^2)^T v^{j2} + (c^3)^T v^{j3} + (c^4)^T v^{j4}) \beta_j, \\ \text{s.t.} \quad & \sum_{i=1}^s D_{11} u^{i1} \alpha_i + \sum_{j=1}^t D_{11} v^{j1} \beta_j = b^1, \\ & \sum_{i=1}^s \alpha_i = 1, \\ & \alpha_i, \beta_j \geq 0, \quad i = 1, \dots, s, \quad j = 1, \dots, t. \end{aligned}$$

Assume that the current restricted master problem has been solved, and $(\bar{y}_1, \bar{\gamma}_1)$ is the optimal simplex multiplier. The constructed subprogram

$$\begin{aligned} \min & (c^1 - (D_{11})^T \bar{y}_1)^T x^1 + (c^2)^T x^2 + (c^3)^T x^3 - \bar{\gamma}_1, \\ \text{s.t. } & A_{21}x^1 + D_{22}x^2 = b^2, \\ & A_{32}x^2 + D_{33}x^3 = b^3, \\ & A_{43}x^3 + D_{44}x^4 = b^4, \\ & x^1, x^2, x^3, x^4 \geq 0. \end{aligned}$$

just likes the original problem, but the number of stairs reduces by 1. Then apply D-W decomposition again by partitioning it into two parts, $\{t = 2\}$ and $\{t = 3, 4\}$, and do as previously until the stairs are reduced to 1, and hence the subprogram is easier to solve.

9.3 Economic Interpretation of D-W Decomposition

The D-W decomposition can be interpreted differently, depending on various application backgrounds. Using this method to realize the optimal allocation of limited resources is a typical example.

Assume that a company has k plants. It is required to determine an output vector x^j for each plant j , whose production activities depend on restrictions upon available limited resources, such as manpower, facilities, materials, storage, etc., but not on inner restrictions of the other plants. In mathematical terms, x^j should satisfy constraints

$$D_j x^j = h^j, \quad x^j \geq 0, \quad j = 1, \dots, k,$$

where $D_j \in \mathcal{R}^{m_j \times n_j}$ ($m_j < n_j$), termed *consumption* matrix, reflects the consumption for a unit product, and h^j is vector of available resource. This part of constraints is of block structure with variables and constraints separable. In addition, there are common resources to be shared by the k plants, which can be expressed by a set of so-called *coupling constraints*:

$$A_1 x^1 + \dots + A_k x^k = b,$$

where $A_j \in \mathcal{R}^{m \times n_j}$, $j = 1, \dots, k$.

To minimize the total cost of production, we solve the following LP model:

$$\begin{aligned} \min & f = (c^1)^T x^1 + \dots + (c^k)^T x^k, \\ \text{s.t. } & A_1 x^1 + \dots + A_k x^k = b, \\ & D_j x^j = h^j, \quad x^j \geq 0, \quad j = 1, \dots, k. \end{aligned} \tag{9.18}$$

It is advantageous to use D-W decomposition if the preceding model is of large scale, especially when there are a large number of plants associated with large consumption matrices: the company can focus on a smaller problem to realize an optimal allocation of resources by ignoring the details of the inner restrictions on the plants.

For simplicity, one might assume that the feasible regions

$$P^j \triangleq \{x^j \mid D_j x^j = h^j, x^j \geq 0\}, \quad j = 1, \dots, k,$$

corresponding to the plants, are all bounded (actually, available resources are always limited). Introduce notation

$$A = (A_1, \dots, A_k), \quad H = \begin{pmatrix} D_1 & & \\ & \ddots & \\ & & D_K \end{pmatrix}, \quad h = \begin{pmatrix} h^1 \\ \vdots \\ h^k \end{pmatrix}.$$

Thus, it is clear that

$$P \triangleq \{x \in \mathcal{R}^n \mid Hx = h, x \geq 0\} = P^1 \times \dots \times P^k$$

is bounded. Denote the vertex set of P by $U = \{u^1, \dots, u^s\}$. Based on Representation Theorem 2.4.1, we obtain the following master problem:

$$\begin{aligned} \min_{\alpha_i} \quad & f = \sum_{i=1}^s (c^T u^i) \alpha_i, \\ \text{s.t.} \quad & \sum_{i=1}^s (A u^i) \alpha_i = b, \\ & \sum_{i=1}^s \alpha_i = 1, \\ & \alpha_i \geq 0, \quad i = 1, \dots, s. \end{aligned} \tag{9.19}$$

The number of rows of the preceding is less than the number of rows of the original problem, but the number of its columns would be quite large, compared with the original problem, letting alone unrealistic computation of the whole vertex set U .

Nevertheless, the company can deal with a restricted master problem instead, involving only some k ($m + 2 \leq k \ll s$) columns (with rank $m + 1$) of the master problem, ignoring most of the inner restrictions of the plants. Now denote by $\widehat{U}$ the vertex set, resulting from U by deleting vertices corresponding to the k columns.

Assume that optimality of the restricted master problem was achieved and that $(\bar{y}, \bar{\gamma})$ is the optimal simplex multipliers, associated with the first m rows and the bottom row of the constraint system. In economic terms, $(\bar{y}, \bar{\gamma})$ is the shadow price

of resources (Sect. 5.6), from which the reduced costs corresponding to set $\widehat{U}$ follow, i.e.,

$$c^T u^i - (A u^i)^T \bar{y} - \bar{\gamma} = (c - (A)^T \bar{y})^T u^i - \bar{\gamma} : u^i \in \widehat{U}. \quad (9.20)$$

A negative reduced cost implies that increasing the associated nonbasic variable from 0 results in decrement of the total cost of production. Typically, the most negative reduced cost is selected to enter the basis. The difficulty is that computing the reduced costs directly by (9.20) requires all vertices of P available, while the company does not know the inner details of the plants.

The trick to get rid of this difficulty is to solve the following subprogram by the simplex algorithm, i.e.,

$$\begin{aligned} \min \quad & \zeta = (c - (A)^T \bar{y})^T x - \bar{\gamma}, \\ \text{s.t.} \quad & x \in P. \end{aligned} \quad (9.21)$$

where the objective is related to the shadow price $(\bar{y}, \bar{\gamma})$.

Since the P is bounded, solving (9.21) renders an optimal vertex solution, say u^* , associated with the optimal value, say ζ^* , equal to the minimum reduced cost of the restricted master problem. If $\zeta^* \geq 0$, optimality of the master problem as well as the original problem is achieved, so is an optimal allocation scheme of resources. If, otherwise, $\zeta^* < 0$, then u^* corresponds to the most negative reduced cost, hence the restricted master problem can be updated by entering to basis the master problem's column corresponding to u^* , and dropping from basis the column determined by the minimum-ratio test. Then an iteration is complete.

It is noted that subprogram (9.21) can be further decomposed into k smaller programs as

$$\begin{aligned} \min \quad & (c^j - A_j^T \bar{y})^T x^j - \bar{\gamma}, \\ \text{s.t.} \quad & x^j \in P^j, \end{aligned} \quad j = 1, \dots, k.$$

Once optimal basic solutions to these programs are obtained, an optimal vertex solution of form (9.2) to the subprogram is readily available.

9.4 Benders Decomposition

The D-W decomposition presented previously converts an LP problem into two smaller ones by partitioning its rows into two parts. In contrast, Benders (1962) converts a problem into two smaller ones by partitioning its columns (variables) into two parts, one of which is an LP problem, and the other can belong to another category. It can be used to solve large-scale mixed LP programs, especially mixed-integer LP programs. Subsequently, Benders decomposition is widely applied to

dealing with stochastic programming programs and is further generalized to handle nonlinear programming programs (Geoffrion 1972).

Consider

$$\begin{aligned} & \max_{\pi, y} f(\pi) + b^T y, \\ & \text{s.t.} \quad F(\pi) + A^T y \leq c, \\ & \quad \pi \in \Pi, \end{aligned} \tag{9.22}$$

where $A \in \mathcal{R}^{m \times n}$, $c \in \mathcal{R}^n$, $b \in \mathcal{R}^m$. Scalar function $f(\pi)$ and vector-valued function $F(\pi) \in \mathcal{R}^n$ and their domain $\Pi \subset \mathcal{R}^m$ will be clarified later.

For any fixed π , (9.22) is a LP problem with respect to variable y . Therefore, Benders decomposition firstly deems π as a parameter to realize a partition of variables.

Lemma 9.4.1 *Problem (9.22) is equivalent to the following problem:*

$$\begin{aligned} & \max_{\pi} f(\pi) + \max_y \{b^T y \mid A^T y \leq c - F(\pi)\}, \\ & \text{s.t.} \quad \pi \in \Pi \cap S, \end{aligned} \tag{9.23}$$

where

$$S = \{\pi \mid A^T y + F(\pi) \leq c\}.$$

Proof It is clear that (9.22) is infeasible and unbounded and has an optimal solution if and only if (9.23) is infeasible and unbounded and has an optimal solution accordingly.

For any fixed π , the objective value of problem (9.23) is equal to the sum of $f(\pi)$ and the optimal value of the following subprogram:

$$\begin{aligned} D(\pi) : & \max_y b^T y, \\ & \text{s.t.} \quad A^T y \leq c - F(\pi). \end{aligned}$$

Let $(\bar{\pi}, \bar{y})$, $\hat{\pi}$, and $\hat{y}$ be an optimal solution to (9.22), (9.23), and $D(\hat{\pi})$, respectively. Then it is clear that $(\hat{y}, \hat{\pi})$ is a feasible solution to (9.22), and hence to

$$f(\hat{\pi}) + b^T \hat{y} \leq f(\bar{\pi}) + b^T \bar{y}. \tag{9.24}$$

On the other hand, it is known that

$$f(\hat{\pi}) + b^T \hat{y} \geq f(\bar{\pi}) + \max_y \{b^T y \mid A^T y \leq c - F(\bar{\pi})\},$$

and $\bar{y}$ is clearly an optimal solution to $D(\bar{\pi})$. Therefore, it holds that

$$f(\bar{\pi}) + \max_y \{b^T y \mid A^T y \leq c - F(\bar{\pi})\} = f(\bar{\pi}) + b^T \bar{y}.$$

Combining the previous two formulas leads to

$$f(\hat{\pi}) + b^T \hat{y} \geq f(\bar{\pi}) + b^T \bar{y},$$

which along with (9.24) gives

$$f(\hat{\pi}) + b^T \hat{y} = f(\bar{\pi}) + b^T \bar{y}.$$

Therefore programs (9.22) and (9.23) share the same optimal solutions and optimal value. $\square$

Therefore, problem (9.22) is decomposed into the two smaller programs $D(\pi)$ and (9.23), respectively, involving p and q variables. Nevertheless, they are defined implicitly. To go further, consider the dual problem of $D(\pi)$, i.e.,

$$\begin{aligned} P(\pi) : \min_x \quad & \zeta = (c - F(\pi))^T x, \\ \text{s.t.} \quad & Ax = b, \quad x \geq 0. \end{aligned}$$

Proposition 9.4.1 *If the feasible region*

$$X = \{x \in \mathcal{R}^n \mid Ax = b, x \geq 0\}$$

of $P(\pi)$ is empty, then (9.22) is infeasible or unbounded.

Proof $X = \emptyset$ means that $P(\pi)$ is infeasible for any $\pi \in \mathcal{R}^p$. According to the duality theory, problem $D(\pi)$ is infeasible or unbounded, hence (9.22) is infeasible or unbounded. $\square$

In case when $X = \emptyset$, one can solve

$$\begin{aligned} \min_x \quad & (c - F(\pi))^T x, \\ \text{s.t.} \quad & Ax = 0, \quad x \geq 0, \end{aligned}$$

by the simplex algorithm to determine whether (9.22) is infeasible or unbounded, if needed. If the preceding is unbounded, it can be asserted that (9.22) is infeasible, and if optimality is achieved, (9.22) is unbounded (see discussions made in Sect. 5.2).

From now on, it is assumed that $X \neq \emptyset$, whose vertex and extreme direction sets are, respectively, denoted by

$$U = \{u^1, \dots, u^s\}, \quad V = \{v^1, \dots, v^t\}.$$

Using preceding notation, we have the following result.

Lemma 9.4.2 *Given π , there is an optimal solution to problem $P(\pi)$ if and only if*

$$(c - F(\pi))^T v^j \geq 0, \quad v^j \in V. \quad (9.25)$$

Proof It is seen that (9.25) implies nonexistence of a descent unbounded direction in set X . Therefore, there exists an optimal solution to $P(\pi)$. According to Theorem 2.5.5, there is an optimal vertex solution to it.

Conversely, that there is an optimal solution to $P(\pi)$ implies that (9.23) holds, because if there is some $v^k \in V$ such that

$$(c - F(\pi))^T v^k < 0,$$

then v^k is a descent extreme direction, hence $P(\pi)$ is unbounded below, which contradicts that there exists an optimal solution. $\square$

Define the *dual Benders master problem* as follows (it will be clear why it is prefixed with “dual”):

$$\begin{aligned} & \max_{\pi, g} \quad f(\pi) + g, \\ & \text{s.t.} \quad g \leq (c - F(\pi))^T u^i, \quad u^i \in U, \\ & \quad \quad (c - F(\pi))^T v^j \geq 0, \quad v^j \in V, \\ & \quad \quad \pi \in \Pi \cap S. \end{aligned} \tag{9.26}$$

Theorem 9.4.1 $(\bar{\pi}, \bar{y})$ is an optimal solution to (9.22) if and only if $\bar{y}$ is an optimal solution to $D(\bar{\pi})$ and $(\bar{\pi}, \bar{g})$ is an optimal solution to the dual Benders master problem.

Proof It is clear that there is an optimal solution to (9.22) if and only if there is an optimal solution to the dual Benders master problem.

According to Lemma 9.4.1, problem (9.22) is equivalent to (9.23). On the other hand, that there is optimal solution to problem (9.22) means that so is to $D(\pi)$. By the strong dual theorem, there is optimal solution to $P(\pi)$ with the same optimal value, i.e.,

$$\max_y \{b^T x \mid A^T y \leq c - F(\pi)\} = \min_x \{(c - F(\pi))^T x \mid Ax = b, x \geq 0\}.$$

Substitute the preceding to (9.23) leads to

$$\begin{aligned} & \max_{\pi} \quad f(\pi) + \min_x \{(c - F(\pi))^T x \mid Ax = b, x \geq 0\}, \\ & \text{s.t.} \quad \pi \in \Pi \cap S. \end{aligned} \tag{9.27}$$

By Theorem 2.5.5, if there is optimal solution to $P(\pi)$, then there is a vertex optimal solution to it. Therefore it holds that

$$\min_x \{(c - F(\pi))^T x \mid Ax = b, x \geq 0\} = \min_{u^i \in U} (c - F(\pi))^T u^i. \tag{9.28}$$

Substituting the preceding to (9.27) and applying Lemma 9.4.2 lead to

$$\begin{aligned} & \max_{\pi} f(\pi) + \min_{u^i \in U} (c - F(\pi))^T u^i, \\ \text{s.t.} \quad & (c - F(\pi))^T v^j \geq 0, \quad v^j \in V, \\ & \pi \in \Pi \cap S, \end{aligned}$$

which gives (9.26). These equivalent relations ensure validity of Theorem 9.4.1. $\square$

According to the preceding theorem, what we should do next is to solve (9.26). However, it is not realistic to handle (9.26) directly, because it requires the availability of all vertices and extreme directions of X , let alone constraint $\pi \in S$ is given in implicit impression. To overcome this difficulty, a possible approach is to handle a relaxed problem, yielding from ignoring some of the constraints. Its optimal solution is also an optimal solution to the master problem if it is a feasible solution to the latter; otherwise, the relaxed problem is updated by adding some constraints.

Assume that at some iteration, subsets of the vertex and extreme direction sets of X are known as

$$U' \subset U, \quad V' \subset V.$$

Define the *dual Benders restricted master problem*¹ by:

$$\begin{aligned} & \max_{\pi, g} f(\pi) + g, \\ \text{s.t.} \quad & g \leq (c - F(\pi))^T u^i, \quad u^i \in U', \\ & (c - F(\pi))^T v^j \geq 0, \quad v^j \in V', \\ & \pi \in \Pi. \end{aligned} \tag{9.29}$$

Theorem 9.4.4 *Let $(\bar{\pi}, \bar{g})$ be an optimal solution to the dual Benders restricted master problem.*

If u^ and ζ^* are an optimal vertex solution and optimal value to the subprogram $P(\bar{\pi})$, then:*

- (i) *If $\bar{g} > \zeta^*$, then u^* is a new vertex generated.*
- (ii) *If $\bar{g} = \zeta^*$, then $(\bar{\pi}, \bar{g})$ is an optimal solution to the dual Benders master problem.*

If $P(\bar{\pi})$ is unbounded, then a new extreme direction is given.

Proof Since u^* and ζ^* are optimal vertex solution and optimal value to the subprogram $P(\bar{\pi})$, it holds that

$$\zeta^* = (c - F(\bar{\pi}))^T u^* = \min\{(c - F(\bar{\pi}))^T u^i \mid u^i \in U\}. \tag{9.30}$$

¹ The term by Dantzig–Wolfe is employed here.

Besides, $\bar{\pi}$ satisfies

$$(c - F(\bar{\pi}))^T v^j \geq 0, \quad v^j \in V, \quad (9.31)$$

because otherwise there is an ascent extreme direction in X , hence $P(\bar{\pi})$ is unbounded, which contradicts the existence of optimal solution. Moreover, it also holds that

$$\bar{\pi} \in \Pi \cap S. \quad (9.32)$$

In fact, it is clear that $\bar{\pi} \in \Pi$; and if $\bar{\pi} \notin S$, then it is derived that $D(\bar{\pi})$ is infeasible, hence $P(\bar{\pi})$ is either infeasible or unbounded, which contradicts the existence of an optimal solution to $P(\bar{\pi})$.

Since $(\bar{\pi}, \bar{g})$ is an optimal solution to (9.29), it is obtained from (9.30) that

$$\begin{aligned} f(\bar{\pi}) + \bar{g} &= f(\bar{\pi}) + \min\{(c - F(\bar{\pi}))^T u^i \mid u^i \in U'\} \\ &\geq f(\bar{\pi}) + \min\{(c - F(\bar{\pi}))^T u^i \mid u^i \in U\} \\ &= f(\bar{\pi}) + \zeta^*, \end{aligned}$$

hence $\bar{g} \geq \zeta^*$.

Consequently, only the following two cases arise:

(i) $\bar{g} > \zeta^*$.

In this case, from

$$\bar{g} > \zeta^* = (c - F(\bar{\pi}))^T u^*$$

and

$$\bar{g} = \min\{(c - F(\bar{\pi}))^T u^i \mid u^i \in U'\}$$

it follows that

$$(c - F(\bar{\pi}))^T u^* < \min\{(c - F(\bar{\pi}))^T u^i \mid u^i \in U'\},$$

implying that $u^* \notin U'$ is a new vertex.

(ii) $\bar{g} = \zeta^*$.

In view of (9.30), it can be obtained in this case that

$$\bar{g} = \zeta^* \leq (c - F(\bar{\pi}))^T u^i, \quad u^i \in U,$$

which together with (9.31) and (9.32) implies that $(\bar{\pi}, \bar{g})$ is a feasible solution to the dual Benders master problem. It is actually optimal because it is optimal to the relaxed problem of the dual Benders master problem.

If subprogram $P(\bar{\pi})$ is unbounded, then an extreme direction v^* is determined such that (Proposition 3.3.2)

$$(c - F(\bar{\pi}))^T v^* < 0.$$

Thus, from $(\bar{\pi}, \bar{g})$ satisfying constraints of the master problem, i.e.,

$$(c - F(\bar{\pi}))^T v^j \geq 0, \quad v^j \in V',$$

it is known that $v^* \notin V'$ is a new extreme direction. $\square$

Proposition 9.4.2 *If the Benders restricted master problem is infeasible, then problem (9.22) is infeasible.*

Proof If $\Pi = \emptyset$, (9.22) is clearly infeasible. If $\Pi \neq \emptyset$, then infeasibility of the Benders restricted master problem implies that for any $\pi \in \Pi$, there is some $v^k \in V' \subset V$ such that $(c - F(\pi))^T v^k < 0$, hence for any $\pi \in \Pi$, problem $P(\pi)$ is unbounded and $D(\pi)$ is infeasible, and the original problem is therefore infeasible. $\square$

The dual Benders master problem (9.26) can be converted to

$$\begin{aligned} & \max_{\pi, g} f(\pi) + g, \\ & \text{s.t.} \quad (u^i)^T F(\pi) + g \leq (u^i)^T c, \quad u^i \in U, \\ & \quad (v^j)^T F(\pi) \leq (v^j)^T c, \quad v^j \in V, \\ & \quad \pi \in \Pi \cap S. \end{aligned}$$

Accordingly, the dual Benders restricted master problem (9.29) becomes

$$\begin{aligned} & \max_{\pi, g} f(\pi) + g, \\ & \text{s.t.} \quad (u^i)^T F(\pi) + g \leq (u^i)^T c, \quad u^i \in U', \\ & \quad (v^j)^T F(\pi) \leq (v^j)^T c, \quad v^j \in V', \\ & \quad f(\pi) + g \leq M, \quad (*) \quad \pi \in \Pi, \end{aligned} \tag{9.33}$$

where the constraint, marked by $*$, is added to rule out possibility of unboundedness, and M is a parameter large enough, so that such addition does not affect vertices of the original feasible region. We call the added constraint *standby*, because it is active only when the problem is unbounded. The parameter M can be dropped technically in real computations (see the example below).

Based on Theorem 9.4.4 and Proposition 9.4.2, we proceed similarly as in Sect. 9.2 as follows:

In each iteration, a Benders restricted master problem of form (9.33) is handled. If it is infeasible, the original problem is infeasible; if an optimal solution $(\bar{\pi}, \bar{g})$ is obtained, then an associated subprogram $P(\bar{\pi})$ is solved. When the subprogram is unbounded, a new extreme direction is generated (Proposition 3.3.2); when an optimal vertex u^* and optimal value ζ^* are reached, it can be determined whether

$(\bar{\pi}, \bar{g})$ is an optimal solution to the Benders master problem; if it is not, a new vertex u^* is generated; if it is, then $(\bar{\pi}, \bar{y})$ is an optimal solution to the original problem (9.22), where $\bar{y}$ is optimal to $D(\bar{\pi})$. In fact, there is no need for solving $D(\bar{\pi})$ actually, because $\bar{y}$, as a dual solution associated with u^* , can be obtained at the same time.

In the case when optimality of (9.22) is not achieved, the restricted master problem is updated by adding a constraint, corresponding to the new vertex or extreme direction, and the next iteration is carried out. In such an iteration, the restricted master problem has one more constraint (this is why the method is often called *row generation*). Note that subprogram $P(\bar{\pi})$ is a standard LP problem, only its objective function changes in the solution process (with the constraints remaining unchanged).

The overall steps are summarized in the following algorithm.

Algorithm 9.4.1 (Benders Decomposition) Initial: dual Benders restricted master problem in form (9.33). This algorithm solves problem (9.22).

1. Solve dual Benders restricted master problem.
2. Stop if it is infeasible.
3. If its optimal solution $(\bar{\pi}, \bar{g})$ is obtained, solve subprogram $P(\bar{\pi})$ by the simplex algorithm.
4. If an optimal vertex solution u^* and optimal value ζ^* to $P(\bar{\pi})$ are obtained, then:
 - (1) Update (9.33) by adding the constraint, corresponding to the new vertex u^* , and go to step 1 if $\bar{g} > \zeta^*$.
 - (2) Stop if $\bar{g} = \zeta^*$.
5. If subprogram $P(\bar{\pi})$ is unbounded, update (9.33) by adding the constraint, associated with the generated new extreme direction.
6. Go to step 1.

Theorem 9.4.5 Assume that the dual Benders restricted master problem either has an optimal solution or is infeasible. If it and subprogram $P(\bar{\pi})$ are solved in finitely many iterations, Algorithm 9.4.1 terminates, at either:

- (i) Step 2, detecting that the original problem is infeasible or
- (ii) Step 4(2), achieving an optimal solution $(\bar{\pi}, \bar{y})$ to the original problem, where $\bar{y}$ and u^* are complementary

Proof The meaning of exit step 2 is directly from Proposition 9.4.2. According to Theorem 9.4.4, termination at step 4(2) gives optimal solution $(\bar{\pi}, \bar{g})$ to the dual Benders master problem; further, it is known by Theorem 9.4.1 that $(\bar{\pi}, \bar{y})$ is an optimal solution to the original problem, where $\bar{y}$ is the dual solution, associated with u^* . Therefore, it is only needed to show termination of Algorithm 9.4.1.

By Theorem 9.4.4, the restricted master problem has one more added constraint, associated with a vertex or extreme direction in each iteration. Since vertices and extreme directions in X are finite, the restricted master problem must become full

master problem in finitely many iterations, if the solution process does not terminate earlier. $\square$

Although the number of constraints of the dual Benders restricted master problem increases steadily, fortunately, it hardly becomes full dual Benders master problem. Usually, the process terminates when it is very far from the latter.

9.5 Illustration of Benders Decomposition

Efficiency in solution of the dual Benders restricted master problem is crucial to Algorithm 9.4.1, since it depends on characteristic of $f(\pi)$ as well as the algorithm used to solve it. At present, there are quite good algorithms for the following cases, concerning (9.22):

- (i) $f(\pi) = h^T \pi$, $F(\pi) = H^T \pi$, $h \in \mathcal{R}^m$, $H \in \mathcal{R}^{m \times n}$, and $\Pi \subset \mathcal{R}^m$ is a (bounded) polyhedron.

In this case, the original problem (9.22) is a linear one. If π is a free variable ($\Pi = \mathcal{R}^m$), in fact, (9.22) is clearly a standard dual problem. As the dual restricted master problem and subprogram are linear, the simplex algorithm is a suitable tool for this purpose.

- (ii) The same as (i), except for adding some integer requirement to components of π .

In this case, the original problem (9.22) is a mixed-integer LP problem, and hence the dual restricted master problem is a pure integer LP problem, except involving a free variable g . Since one inequality constraint is added to the restricted master problem in each iteration, the cutting plane method is amenable to be used (Sect. 11.2).

- (iii) $f(\pi)$, $F(\pi)$ are continuously differentiable functions on a bounded and closed convex set $\Pi \subset \mathcal{R}^m$.

In this case, the dual restricted master problem is a smooth convex problem, for which efficient algorithms are available (see, e.g., Sun and Yuan 2006).

Cases (ii) and (iii) are beyond the scope of this book. The following discussions are focused on (i), related to LP.

As for how to construct an initial dual restricted master problem, a direct way is to start from the feasible region

$$X = \{x \in \mathcal{R}^n \mid Ax = b, x \geq 0\}$$

of $P(\bar{\pi})$. If $X = \emptyset$, the original problem is infeasible or unbounded (Proposition 9.4.1); otherwise, a feasible vertex, say $u^1 \in X$, can be determined. Therefore, allowing

$$U' = \{u^1\}, \quad V' = \emptyset,$$

leads to an initial dual restricted master problem in form (9.33).

Benders decomposition is suitable for some programs, involving block diagonals, such as

$$\begin{aligned} \min \quad & h^T \pi + (b^1)^T y^1 + \cdots + (b^k)^T y^k, \\ \text{s.t.} \quad & A_1^T \pi + D_1^T y^1 \leq c_1, \\ & A_2^T \pi + D_2^T y^2 \leq c_2, \\ & \vdots \\ & A_k^T \pi + D_k^T y^k \leq c^k, \end{aligned}$$

where orders of submatrices D_i and dimensions of subvectors b^i, y^i, c^i , $i = 1, \dots, k$, are consistent, i.e., b_i and y_i are of dimension equal to the number of rows of D_i , and c_i of dimension equal to the number of columns of D_i ; π is of dimension equal to the number of rows of A_i .

If one partitions the coefficient matrix as

$$H^T = \begin{pmatrix} A_1^T \\ A_2^T \\ \vdots \\ A_k^T \end{pmatrix}, \quad A^T = \begin{pmatrix} D_1^T & & & \\ & D_2^T & & \\ & & \ddots & \\ & & & D_k^T \end{pmatrix},$$

then subprogram $P(\bar{\pi})$ can be decomposed into k smaller programs, corresponding to D_k .

Let us bring up the following example.

Example 9.5.1 Solve the following problem by Benders Decomposition 9.4.1:

$$\begin{aligned} \max \quad & 2\pi + y_1 + 4y_2 + y_3 + 2y_4, \\ \text{s.t.} \quad & \pi + y_1 \leq 0, \\ & -3\pi + y_2 \leq -1, \\ & -\pi + 4y_1 + y_2 \leq 1, \\ & \pi + y_1 - y_2 \leq -2, \\ & 2\pi + y_3 \leq -3, \\ & \pi + y_4 \leq 1, \\ & -\pi + 3y_3 + y_4 \leq -1, \\ & -3\pi - y_3 - 3y_4 \leq 1. \end{aligned}$$

Answer Denote the problem by:

$$\begin{aligned} \max_{\pi, y} \quad & h^T \pi + b^T y, \\ \text{s.t.} \quad & H^T \pi + A^T y \leq c, \end{aligned}$$

where

$$h^T = (2), b = (1, 4, 1, 2)^T, c = (0, -1, 1, -2, -3, 1, -1, 1)^T.$$

$$H = (1, -3, -1, 1, 2, 1, -1, -3), \quad A = \begin{pmatrix} 1 & 0 & 4 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & -3 \end{pmatrix}.$$

There is a vertex in feasible region $X = \{x \mid Hx = b, x \geq 0\}$ of Subprogram $P(\pi)$, i.e.,

$$u^1 = (1, 4, 0, 0, 1, 2, 0, 0)^T.$$

$$(u^1)^T H^T \pi = (1, 4, 0, 0, 1, 2, 0, 0)(1, -3, -1, 1, 2, 1, -1, -3)^T \pi = -7\pi.$$

$$(u^1)^T c = (1, 4, 0, 0, 1, 2, 0, 0)(0, -1, 1, -2, -3, 1, -1, 1)^T = -5.$$

The associated initial restricted master problem is

$$\begin{aligned} & \max_{\pi, g} \quad 2\pi + g, \\ & \text{s.t.} \quad -7\pi + g \leq -5, \\ & \quad \quad 2\pi + g \leq M, \end{aligned} \quad (*)$$

where (*) marks the *standby* constraint.

Iteration 1:

1. Solve the Benders restricted master problem by the simplex algorithm.

Taking basis $B = \begin{pmatrix} -7 & 2 \\ 1 & 1 \end{pmatrix}$, $B^{-1} = \begin{pmatrix} -1/9 & 2/9 \\ 1/9 & 7/9 \end{pmatrix}$ gives the dual feasible solution

$$\begin{pmatrix} \bar{\pi} \\ \bar{g} \end{pmatrix} = B^{-T} \begin{pmatrix} -5 \\ M \end{pmatrix} = \begin{pmatrix} 5/9 + 1/9M \\ -10/9 + 7/9M \end{pmatrix}. \quad (9.34)$$

The associated primal solution is feasible, i.e., $B^{-1}(2, 1)^T = (0, 1)^T \geq 0$. Therefore optimality of the restricted master problem is achieved.

3. The objective of Subprogram $P(\bar{\pi})$ is

$$\begin{aligned} (c - H^T \bar{\pi})^T &= (0, -1, 1, -2, -3, 1, -1, 1) - (5/9 + 1/9M)(1, -3, -1, 1, 2, 1, -1, -3) \\ &= (-5/9 - 1/9M, 2/3 + 1/3M, 14/9 + 1/9M, -23/9 - 1/9M, \\ &\quad -37/9 - 2/9M, 4/9 - 1/9M, -4/9 + 1/9M, 8/3 + 1/3M). \end{aligned}$$

As parameter M is large enough, the terms without M in the preceding can be omitted. For simplicity, the common factor M can be omitted as well; in other words, (9.34) can be replaced by

$$\begin{pmatrix} \bar{\pi} \\ \bar{g} \end{pmatrix} = B^{-T} \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Subprogram $P(\bar{\pi})$ is decomposed into the following two problems, as solved separately:

(i)

$$\begin{aligned} \min \quad & -1/9x_1 + 1/3x_2 + 1/9x_3 - 1/9x_4, \\ \text{s.t.} \quad & x_1 + 4x_3 + x_4 = 1, \\ & x_2 + x_3 - x_4 = 4, \quad x_j \geq 0, \quad j = 1, \dots, 4. \end{aligned}$$

Using basis and nonbasis $B_1 = \{1, 2\}$, $N_1 = \{3, 4\}$. $B_1^{-1} = I$ gives feasible basic solution $\bar{x}_{B_1} = (1, 4)^T \geq 0$, $\bar{x}_{N_1} = (0, 0)^T$.

$$\bar{y}_1 = B_1^{-T} c_{B_1} = \begin{pmatrix} -1/9 \\ 1/3 \end{pmatrix},$$

$$\bar{z}_{N_1} = c_{N_1} - N_1^T \bar{y}_1 = \begin{pmatrix} 1/9 \\ -1/9 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 1 & -1 \end{pmatrix}^T \begin{pmatrix} -1/9 \\ 1/3 \end{pmatrix} = \begin{pmatrix} 2/9 \\ 1/3 \end{pmatrix},$$

$\min\{2/9, 1/3\} \geq 0$. The reduced costs are all nonnegative, optimality is hence achieved.

The optimal vertex solution is $(\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4)^T = (1, 4, 0, 0)^T$, with optimal value $\zeta_1 = 11/9$.

(ii)

$$\begin{aligned} \min \quad & -2/9x_5 - 1/9x_6 + 1/9x_7 + 1/3x_8, \\ \text{s.t.} \quad & x_5 + 3x_7 - x_8 = 1, \\ & x_6 + x_7 - 3x_8 = 2, \quad x_j \geq 0, \quad j = 5, \dots, 8. \end{aligned}$$

Take the basis and nonbasis $B_1 = \{5, 6\}$, $N_2 = \{7, 8\}$. $B_1^{-1} = I$.

The feasible basic solution is

$$\bar{x}_{B_1} = (1, 2)^T \geq 0, \quad \bar{x}_{N_2} = (0, 0)^T,$$

$$\bar{y}_2 = B_1^{-T} c_{B_1} = \begin{pmatrix} -2/9 \\ -1/9 \end{pmatrix},$$

$$\bar{z}_{N_2} = c_{N_2} - N_2^T \bar{y}_2 = \begin{pmatrix} 1/9 \\ 1/3 \end{pmatrix} - \begin{pmatrix} 3 & -1 \\ 1 & -3 \end{pmatrix}^T \begin{pmatrix} -2/9 \\ -1/9 \end{pmatrix} = \begin{pmatrix} 8/9 \\ -2/9 \end{pmatrix}.$$

The minimum reduced cost is $\min\{8/9, -2/9\} = -2/9$, $q = 8$.

Since $\bar{a}_8 = B_1^{-1}(-1, -3)^T = (-1, -3)^T \leq 0$, $P(\bar{\pi})$ is unbounded. The descent extreme direction is $v^* = (0, 0, 0, 0, 1, 3, 0, 1)^T$.

$$5. (v^*)^T H^T \pi = (0, 0, 0, 0, 1, 3, 0, 1)(1, -3, -1, 1, 2, 1, -1, -3)^T \pi = 2\pi.$$

$$(v^*)^T c = (0, 0, 0, 0, 1, 3, 0, 1)(0, -1, 1, -2, -3, 1, -1, 1)^T = 1.$$

The restricted master problem is then updated to

$$\begin{aligned} & \max_{\pi, g} \quad 2\pi + g, \\ & \text{s.t.} \quad -7\pi + g \leq -5, \\ & \quad \quad 2\pi \leq 1, \\ & \quad \quad 2\pi + g \leq M. \end{aligned} \quad (*)$$

Iteration 2:

1. Solve the restricted master problem. The basis $B = \begin{pmatrix} -7 & 2 \\ 1 & 0 \end{pmatrix}$,

$$B^{-1} = \begin{pmatrix} 0 & 1 \\ 1/2 & 7/2 \end{pmatrix}, \text{ corresponds to dual feasible solution}$$

$$\begin{pmatrix} \bar{\pi} \\ \bar{g} \end{pmatrix} = B^{-T} \begin{pmatrix} -5 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -3/2 \end{pmatrix}.$$

The associated primal solution is feasible, i.e., $B^{-1}(2, 1)^T = (1, 9/2)^T \geq 0$, hence optimality of the dual restricted master problem is achieved.

3. The objective of the subprogram $P(\bar{\pi})$ is

$$\begin{aligned} (c - H^T \bar{\pi})^T &= (0, -1, 1, -2, -3, 1, -1, 1) - (1/2)(1, -3, -1, 1, 2, 1, -1, -3) \\ &= (-1/2, 1/2, 3/2, -5/2, -4, 1/2, -1/2, 5/2). \end{aligned}$$

And $P(\bar{\pi})$ can be decomposed into two programs, which are solved separately:

(i)

$$\begin{aligned} & \min \quad -1/2x_1 + 1/2x_2 + 3/2x_3 - 5/2x_4, \\ & \text{s.t.} \quad x_1 + 4x_3 + x_4 = 1, \\ & \quad \quad x_2 + x_3 - x_4 = 4, \quad x_j \geq 0, \quad j = 1, \dots, 4. \end{aligned}$$

(1) Take the basis and nonbasis $B_1 = \{1, 2\}$, $N_1 = \{3, 4\}$. $B_1^{-1} = I$.

The feasible basic solution is $\bar{x}_{B_1} = (1, 4)^T \geq 0$, $\bar{x}_{N_1} = (0, 0)^T$.

$$\bar{y}_1 = B_1^{-T} c_{B_1} = \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix},$$

$$\bar{z}_{N_1} = c_{N_1} - N_1^T \bar{y}_1 = \begin{pmatrix} 3/2 \\ -5/2 \end{pmatrix} - \begin{pmatrix} 4 & 1 \\ 1 & -1 \end{pmatrix}^T \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix} = \begin{pmatrix} 3 \\ -3/2 \end{pmatrix}.$$

The minimum reduced cost is $\min\{3, -3/2\} = -3/2$, $q = 4$.

$\bar{a}_4 = B_1^{-1}(1, -1)^T = (1, -1)^T$, $p = 1$. The minimum-ratio is $\min\{1/1\} = 1$, $p = 1$.

(2) Update the basis:

$$B_1 = \{4, 2\}, N_1 = \{3, 1\}. B_1^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The feasible basic solution is

$$\bar{x}_{B_1} = B_1^{-1}(1, 4)^T = (1, 5)^T \geq 0, \quad \bar{x}_{N_1} = (0, 0)^T,$$

$$\bar{y}_1 = B_1^{-T}c_{B_1} = B_1^{-T} \begin{pmatrix} -5/2 \\ 1/2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1/2 \end{pmatrix},$$

$$\bar{z}_{N_1} = c_{N_1} - N_1^T \bar{y}_1 = \begin{pmatrix} 3/2 \\ -1/2 \end{pmatrix} - \begin{pmatrix} 4 & 1 \\ 1 & 0 \end{pmatrix}^T \begin{pmatrix} -2 \\ 1/2 \end{pmatrix} = \begin{pmatrix} 9 \\ 3/2 \end{pmatrix}.$$

The minimum reduced cost is $\min\{9, 3/2\} \geq 0$, and the optimal solution and associated objective value are

$$(\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4)^T = (0, 5, 0, 1)^T, \quad \zeta_1 = (1/2)5 - 5/2 = 0.$$

(ii)

$$\begin{aligned} \min \quad & -4x_5 + 1/2x_6 - 1/2x_7 + 5/2x_8, \\ \text{s.t.} \quad & x_5 + 3x_7 - x_8 = 1, \\ & x_6 + x_7 - 3x_8 = 2, \quad x_j \geq 0, \quad j = 5, \dots, 8. \end{aligned}$$

Take the basis and nonbasis $B_1 = \{5, 6\}$, $N_2 = \{7, 8\}$. $B_1^{-1} = I$.

The feasible basic solution is $\bar{x}_{B_1} = (1, 2)^T \geq 0$, $\bar{x}_{N_2} = (0, 0)^T$.

$$\bar{y}_2 = B_1^{-T}c_{B_1} = \begin{pmatrix} -4 \\ 1/2 \end{pmatrix},$$

$$\bar{z}_{N_2} = c_{N_2} - N_2^T \bar{y}_2 = \begin{pmatrix} -1/2 \\ 5/2 \end{pmatrix} - \begin{pmatrix} 3 & -1 \\ 1 & -3 \end{pmatrix}^T \begin{pmatrix} -4 \\ 1/2 \end{pmatrix} = \begin{pmatrix} 11 \\ 0 \end{pmatrix}.$$

The minimum reduced cost is $\min\{11, 0\} \geq 0$, and the optimal solution and objective value are

$$(\bar{x}_5, \bar{x}_6, \bar{x}_7, \bar{x}_8)^T = (1, 2, 0, 0)^T, \quad \zeta_2 = -4 + (1/2)2 = -3.$$

Summarizing (i) and (ii) gives an optimal vertex solution and optimal objective value to the subprogram below:

$$u^* = (0, 5, 0, 1, 1, 2, 0, 0)^T, \quad \zeta^* = \zeta_1 + \zeta_2 = 0 + (-3) = -3. \quad (9.35)$$

4. (1) $\bar{g} = -3/2 > \zeta^* = -3$.

$$(u^*)^T H^T \pi = (0, 5, 0, 1, 1, 2, 0, 0)(1, -3, -1, 1, 2, 1, -1, -3)^T \pi = -10\pi.$$

$$(u^*)^T c = (0, 5, 0, 1, 1, 2, 0, 0)(0, -1, 1, -2, -3, 1, -1, 1)^T = -8.$$

Update the restricted master problem:

$$\begin{aligned}
 & \max_{\pi, g} \quad 2\pi + g, \\
 & \text{s.t.} \quad -7\pi + g \leq -5, \\
 & \quad \quad 2\pi \leq 1, \\
 & \quad \quad -10\pi + g \leq -8, \\
 & \quad \quad 2\pi + g \leq M. \quad (*)
 \end{aligned}$$

Iteration 3:

1. Solve the restricted master problem.

Column $(-10, 1)^T$ enters basis, $q = 3$.

$B^{-1}(-10, 1)^T = (1, -3/2)^T$. The minimum-ratio test: $\min\{1/1\} = 1$, $p = 1$.

The basis is updated as

$$B = \begin{pmatrix} -10 & 2 \\ 1 & 0 \end{pmatrix}, \quad B^{-1} = \begin{pmatrix} 0 & 1 \\ 1/2 & 5 \end{pmatrix},$$

corresponding to the dual feasible solution

$$\begin{pmatrix} \bar{\pi} \\ \bar{g} \end{pmatrix} = B^{-T} \begin{pmatrix} -8 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -3 \end{pmatrix}.$$

The associated primal solution is also feasible, i.e., $B^{-1}(2, 1)^T = (1, 6)^T \geq 0$.

Thus optimality of the restricted master problem is achieved.

3. The objective function of the subprogram $P(\bar{\pi})$ is

$$\begin{aligned}
 (c - H^T \bar{\pi})^T &= (0, -1, 1, -2, -3, 1, -1, 1) - (1/2)(1, -3, -1, 1, 2, 1, -1, -3) \\
 &= (-1/2, 1/2, 3/2, -5/2, -4, 1/2, -1/2, 5/2),
 \end{aligned}$$

which is the same as that in Iteration 2, hence the optimal solution and associated objective value are given by (9.35).

4. (2) $\bar{g} = \zeta^* = -3$, optimality of the Benders master problem is achieved, and the optimal solution and objective value to the original problem are as follows:

$$\bar{\pi} = 1/2, \quad \bar{y}^T = (\bar{y}_1^T, \bar{y}_2^T) = (-2, 1/2, -4, 1/2), \quad \bar{f} = -2.$$

9.6 Dual Benders Decomposition

The decomposition presented in the preceding section handles the dual problem of a standard problem. It is also possible to handle a standard problem directly in a similar manner, resulting in a *dual Benders decomposition*. This section offers related results without detailed derivation.

Partition the variables of a problem into two parts, one of which are in linear form, i.e.,

$$\begin{aligned} \min_{w,x} \quad & f = f(w) + c^T x, \\ \text{s.t.} \quad & F(w) + Ax = b, \\ & w \in W, x \geq 0, \end{aligned} \tag{9.36}$$

where $A \in \mathcal{R}^{m \times n}$, $c \in \mathcal{R}^n$, $b \in \mathcal{R}^m$. Real-valued function $f(\pi)$ and vector-valued function $F(\pi) \in \mathcal{R}^n$ and their domain $W \subset \mathcal{R}^p$ satisfy certain conditions. If $f(w)$ and $F(w)$ are linear, and $W = \{w \in \mathcal{R}^p \mid w \geq 0\}$ holds, the preceding is clearly a standard LP problem.

Viewing w as a parameter, introduce subprogram

$$\begin{aligned} P(w) : \min_x \quad & c^T x, \\ \text{s.t.} \quad & Ax = b - F(w), \quad x \geq 0. \end{aligned}$$

The dual problem of the preceding is then

$$\begin{aligned} D(w) : \max_y \quad & \zeta = (b - F(w))^T y, \\ \text{s.t.} \quad & A^T y \leq c. \end{aligned}$$

Assume nonempty of the feasible region of $D(w)$, i.e.,

$$Y = \{y \mid A^T y \leq c\} \neq \emptyset,$$

whose vertex and extreme direction sets are denoted by

$$U = \{u^1, \dots, u^s\}, \quad V = \{v^1, \dots, v^t\}.$$

Define the *dual Benders master problem* as follows:

$$\begin{aligned} \min_{w,g} \quad & f(w) + g, \\ \text{s.t.} \quad & (u^i)^T F(w) + g \geq (u^i)^T b, \quad u^i \in U, \\ & (v^j)^T F(w) \geq (v^j)^T b, \quad v^j \in V, \\ & w \in W \cap T, \end{aligned} \tag{9.37}$$

where

$$T = \{w \mid F(w) + Ax = b, x \geq 0\}.$$

As for relationship between the dual Benders master problem and the original problem (9.36), we have the following result.

Theorem 9.6.1 $(\bar{w}, \bar{x})$ is an optimal solution to (9.36) if and only if $\bar{x}$ is an optimal solution to $P(w)$ and $(\bar{w}, \bar{g})$ is an optimal solution to the dual Benders master problem.

Thereby, one can solve problem (9.36) by handling the dual Benders master problem (9.37). Now turn to a relaxation problem yielded from dropping some constraints. Assume that subsets of the vertex and extreme direction sets of Y are known at some iteration, i.e.,

$$U' \subset U, \quad V' \subset V.$$

Define *dual Benders restricted master problem* as follows:

$$\begin{aligned} & \min_{w, g} f(w) + g, \\ & \text{s.t.} \quad (u^i)^T F(w) + g \geq (u^i)^T b, \quad u^i \in U', \\ & \quad (v^j)^T F(w) \geq (v^j)^T b, \quad v^j \in V', \\ & \quad w \in W. \end{aligned} \tag{9.38}$$

Note A standby constraint $f(w) + g \geq -M$ may be added to the preceding problem technically, where M is a number large enough.

Theorem 9.6.2 Let $(\bar{w}, \bar{g})$ be an optimal solution to problem (9.38).

If u^* and ζ^* are an optimal vertex solution and optimal value to subprogram $D(\bar{w})$, then:

- (i) If $\bar{g} < \zeta^*$, u^* is a new vertex generated.
- (ii) If $\bar{g} = \zeta^*$, $(\bar{w}, \bar{g})$ is an optimal solution to the dual Benders master problem.

If $D(\bar{w})$ is unbounded, then a new extreme direction is generated.

Proposition 9.6.1 If dual Benders restricted master problem is infeasible, so is problem (9.36).

Algorithm 9.6.1 (Dual Benders Decomposition) Initial: dual Benders restricted master problem in form (9.38). This algorithm solves the restricted master problem (9.36).

1. Call the simplex algorithm to solve (9.38).
2. Stop if it is infeasible (the original problem is infeasible).
3. If its optimal solution $(\bar{w}, \bar{g})$ is obtained, solve subprogram $D(\bar{w})$.
4. If optimal vertex solution u^* and optimal value ζ^* to $D(\bar{w})$ are obtained, then:
 - (1) Update (9.38) by adding the constraint, associated with u^* , and go to step 1 if $\bar{g} < \zeta^*$.
 - (2) Stop if $\bar{g} = \zeta^*$.

The $(\bar{w}, \bar{x})$ is an optimal solution to (9.36), where $\bar{x}$ is complementary with u^* .

5. If subprogram $D(\bar{w})$ is unbounded, update (9.38) by adding the constraint associated with the new extreme direction.
6. Go to step 1.

We are interested in the linear case only. So, consider a special case of dual Benders decomposition: $f(w)$, $F(w)$ is a linear function.

Assuming

$$f(w) = h^T w, \quad F(w) = Hw, \quad W = \{w \in \mathcal{R}^n \mid w \geq 0\},$$

where $H \in \mathcal{R}^{m \times n^1}$, $h \in \mathcal{R}^{n^1}$, then (9.36) becomes standard LP problem

$$\begin{aligned} \min_{w,x} \quad & f = h^T w + c^T x, \\ \text{s.t.} \quad & Hw + Ax = b, \\ & w, x \geq 0. \end{aligned} \tag{9.39}$$

Viewing w as a parameter, we handle the subprogram below:

$$\begin{aligned} \min_{w,x} \quad & f = c^T x, \\ \text{s.t.} \quad & Ax = b - Hw, \\ & x \geq 0. \end{aligned} \tag{9.40}$$

For simplicity, assume that the set

$$Y = \{y \in \mathcal{R}^m \mid A^T y \leq c\} \neq \emptyset \tag{9.41}$$

is bounded, whose vertex set is denoted by:

$$U = \{u^1, \dots, u^s\}. \tag{9.42}$$

Thus, the dual Benders restricted master problem (9.38) becomes

$$\begin{aligned} \min_{w,g} \quad & h^T w + g, \\ \text{s.t.} \quad & (u^i)^T Hw + g \geq (u^i)^T b, \quad u^i \in U'. \\ & w \geq 0. \end{aligned} \tag{9.43}$$

Assuming that there is an optimal solution $(\bar{w}, \bar{g})$ to the preceding problem, solve the following subprogram:

$$\begin{aligned} D(w) : \max_y \quad & \zeta = (b - Hw)^T y, \\ \text{s.t.} \quad & y \in Y. \end{aligned}$$

If optimal value ζ^* to $D(w)$ is obtained, test for optimality of the original problem by the following criterion:

$$\bar{g} = \zeta^*.$$

Another way to derive the dual Benders decomposition is to use D-W decomposition and duality by dealing with the dual problem of (9.39):

$$\begin{aligned} \max \quad & b^T y, \\ \text{s.t.} \quad & H^T y \leq h, \\ & A^T y \leq c. \end{aligned} \quad (9.44)$$

Let Y and U be defined by (9.41) and (9.42), respectively. Express y in a convex combination of vertices in Y . Then, the D-W master problem follows from (9.44), i.e.,

$$\begin{aligned} \max \quad & \sum_{i=1}^s (b^T u^i) \alpha_i, \\ \text{s.t.} \quad & \sum_{i=1}^s (H^T u^i) \alpha_i \leq h, \\ & \sum_{i=1}^s \alpha_i = 1, \\ & \alpha_i \geq 0, \quad i = 1, \dots, s. \end{aligned} \quad (9.45)$$

Thus, the dual Benders decomposition results from solving its dual problem, which is just the dual Benders restricted master problem (9.43).

On the other hand, solving the dual problem of the D-W master problem of the original problem leads to the Benders decomposition.

Finally, let us turn to the program with the staircase structure. We will not show that some of such programs can be solved by the Benders decomposition in a nested manner but show the so-called *forward nested* manner of the dual Benders decomposition by taking (9.16) again.

Partition the variables into $\{x^1\}$ and $\{x^2, x^3, x^4\}$. According to the dual Benders decomposition, we need to solve the subprogram with x^1 as its parameter, i.e.,

$$\begin{aligned} \min \quad & (c^2)^T x^2 + (c^3)^T x^3 + (c^4)^T x^4, \\ \text{s.t.} \quad & D_{22}x^2 = b^2 - A_{21}x^1, \\ & A_{32}x^2 + D_{33}x^3 = b^3, \\ & A_{43}x^3 + D_{44}x^4 = b^4, \\ & x^2, x^3, x^4 \geq 0. \end{aligned} \quad (9.46)$$

Thus, we turn to the dual problem of the preceding. In fact, it is easier to handle since only its objective function is related to parameter x^1 . In each iteration, the dual Benders restricted master problem is solved to provide a parameter value $\bar{x}^1$ to the subprogram. Then (9.46) has a staircase structure similar to that of the original problem (9.16), with stairs reduced by 1. So, the dual Benders decomposition applies. Such a process is repeated until only one star is left, and the problem is easier to solve.

References

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Chapter 10

Interior-Point Method



As it is known, the simplex method moves on the underlying polyhedron, from vertex to adjacent vertex along edges, until attaining an optimal vertex unless the lower unboundedness is detected. Nevertheless, it could go through an exponential number of vertices of the polyhedron and even stay at a vertex forever because of cycling (Sect. 3.4).

On the other hand, it is possible to achieve optimality directly, if one goes across the interior of the polyhedron. Indeed, there exists a ray, emanating from any point, to an optimal vertex, if any; in other words, a single iteration is enough to solve the LP problem if a “right” descent direction is known. Although unrealistic, this idea would be the motivation of the so-called *interior-point method*.

To seek an optimal point, this method moves from interior point to interior point in the feasible region. Therefore, it can get rid of degeneracy that is closely related to vertices. Algorithms of this type mainly fall into three categories: potential function algorithms, affine algorithms, and path-following algorithms based on some log barrier function. This chapter is devoted to typical/efficient interior-point algorithms, such as the Karmarkar algorithm, affine interior-point algorithm, dual affine interior-point algorithm, path-following interior-point algorithm, including primal–dual algorithm, infeasible primal–dual algorithm, predictor–corrector primal–dual algorithm, and homogeneous and self-dual algorithm.

Thereafter the following basic assumptions are made with the standard LP problem:

A1: $\text{rank } A = m$.

A2: $P^+ = \{x \in \mathcal{R}^n \mid Ax = b, x > 0\} \neq \emptyset$.

A3: $D^+ = \{(y, z) \in \mathcal{R}^m \times \mathcal{R}^n \mid A^T y + z = c, z > 0\} \neq \emptyset$.

A4: $c \notin \text{range } A^T$.

A4 is equivalent to the absence of a vector $\bar{y}$ such that $c = A^T \bar{y}$, in which case the objective function is constant over the feasible region and vice versa if the feasible region is nonempty. Therefore, the optimality can only be attained on the boundary, if any.

10.1 Karmarkar Algorithm

The Karmarkar algorithm (1984) might be the most well-known interior-point algorithm, as it is of the polynomial-time complexity of the order lower than Khachiyan's ellipsoid algorithm. the Karmarkar algorithm can be classified into the category of potential function methods. The basic idea behind it inspired the birth of some efficient algorithms of such type, changing the state of the art of LP to a larger extent.

Consider the so-called *Karmarkar standard problem*:

$$\begin{aligned} \min \quad & f = c^T x \\ \text{s.t.} \quad & Ax = 0, \\ & e^T x = 1, \quad x \geq 0, \end{aligned} \tag{10.1}$$

where $A \in \mathcal{R}^{m \times n}$, $c \in \mathcal{R}^n$, $\text{rank } A = m$, $m < n$, $n \geq 2$.

As a convex hull of n points $e_1, \dots, e_n$ in the n -dimensional space, the polyhedral

$$\Gamma = \left\{ x \in \mathcal{R}^n \mid \sum_{j=1}^n x_j = 1, x_j \geq 0 \right\}$$

is a $(n - 1)$ -dimensional regular simplex. It is noted that the n vertices of Γ lie in symmetric positions, and its center is e/n . The radius of its inscribed sphere and circumscribed sphere are, respectively,

$$r = 1/\sqrt{n(n-1)} < 1, \quad R = \sqrt{n-1}/\sqrt{n} < 1.$$

The feasible region of problem (10.1) is the intersection of simplex Γ and null space, $\{x \in \mathcal{R}^n \mid Ax = 0\}$, of A . It is clear that there is an optimal solution if the feasible region is nonempty.

Concerning problem (10.1), Karmarkar makes the following two assumptions:

1. $Ae = 0$. So the center e/n of the simplex is an interior point of the feasible region.
2. The optimal solution, say x^* , satisfies $c^T x^* = 0$, i.e., the optimal value is 0.

The preceding assumptions imply that $c^T x > 0$ holds for all interior point x , in particular, the center e/n .

It is not difficult to convert a standard LP problem to the Karmarkar standard problem, we will not go into details though. Interested readers are referred to the literature, e.g., Tomlin (1987), Gay (1987), and de Ghellinck and Vial (1986).

10.1.1 Projective Transformation

Let $\bar{x}$ be an interior point, satisfying $A\bar{x} = 0$, $e^T\bar{x} = 1$ and $\bar{x} > 0$. Denote by $\bar{X}$ the diagonal matrix, whose diagonals are components of $\bar{x}$, i.e.,

$$\bar{X} = \text{diag}(\bar{x}_1, \dots, \bar{x}_n).$$

Consider transformation

$$x' = \frac{\bar{X}^{-1}x}{e^T\bar{X}^{-1}x} \triangleq T(x), \quad (10.2)$$

whose inverse transformation is

$$x = T^{-1}(x') = \frac{\bar{X}x'}{e^T\bar{X}x'}. \quad (10.3)$$

$T(x)$, termed *projective transformation*, is a 1-1 mapping from Γ to Γ itself. In fact, for any $x \in \Gamma$, it holds that $x' = T(x) \in \Gamma$; conversely, for any $x' \in \Gamma$, it holds that $x = T^{-1}(x') \in \Gamma$. Under T , in particular, each vertex e_j ($j = 1, \dots, n$) of Γ corresponds to itself, and so does each edge; most importantly, $\bar{x}$ corresponds to the center e/n of Γ , whereas any interior point corresponds to interior point.

Using T , (10.1) is transformed to problem

$$\begin{aligned} \min f &= \frac{c^T\bar{X}x'}{e^T\bar{X}x'}, \\ \text{s.t. } A\bar{X}x' &= 0, \\ e^Tx' &= 1, \quad x' \geq 0. \end{aligned} \quad (10.4)$$

which is no longer a LP problem though, because its objective function is not linear. However, it is known from (10.2) and $e^Tx = 1$ that when x is close to $\bar{x}$, the denominator in the objective function can approximately be regarded as a positive constant, i.e.,

$$e^T\bar{X}x' = \frac{e^Tx}{e^T\bar{X}^{-1}x} = 1 / \sum_{j=1}^n x_j / \bar{x}_j \approx 1/n.$$

Karmarkar therefore employs $c^T \bar{X}x'$ to replace the objective function approximately. Precisely speaking, he used the following subproblem in each iteration:

$$\begin{aligned} \min f &= c^T \bar{X}x', \\ \text{s.t. } A\bar{X}x' &= 0, \\ e^T x' &= 1, \quad x' \geq 0. \end{aligned} \tag{10.5}$$

The preceding and (10.4) have the same feasible region and initial interior point e/n .

Proposition 10.1.1 *Let x^* be an optimal solution to problem (10.1). Then for any interior point $\bar{x}$, there is an optimal solution $(x^*)' = T(x^*)$, associated with objective value zero, to subproblem (10.5).*

Proof Note that for any interior point $\bar{x} > 0$ to (10.1) and any feasible point $x' \geq 0$ to (10.4), it holds that

$$e^T \bar{X}x' = \sum_{j=1}^n \bar{x}_j x'_j \geq e^T x' \min\{\bar{x}_j \mid j = 1, \dots, n\} > 0.$$

It is known from $(x^*)' = T(x^*)$ and Assumption 2 that $(x^*)'$ is an optimal solution to (10.4), satisfying $c^T \bar{X}(x^*)' = 0$. In other words, there is a feasible solution $(x^*)'$ to (10.5) with objective value zero. Assume that $\tilde{x}'$ is its feasible solution with a negative objective value, i.e., $c^T \bar{X}\tilde{x}' < 0$. Then it follows that

$$\frac{c^T \bar{X}\tilde{x}'}{e^T \bar{X}\tilde{x}'} < 0,$$

which contradicts that the optimal value of (10.4) is zero. Therefore, $(x^*)'$ is an optimal solution to (10.5) with objective value zero. $\square$

10.1.2 Karmarkar Algorithm

Assume that a descent direction has been determined. If some components of $\bar{x}$ are close to zero, then the stepsize along the direction could be very small, and the associated improvement in objective function would be negligible. Problem (10.5) helps avoid this situation, to some extent, since $\bar{x}$'s image, $\bar{x}' = T(\bar{x}) = e/n$, in x' space is now at the center of the simplex, the distance from which to each coordinate plane is the same. Of course, we do not really want to solve (10.5) itself, but only use it as a subproblem to determine a “good” search direction and associated stepsize.

To this end, denote the coefficient matrix of subproblem (10.5) by:

$$F = \begin{pmatrix} A\bar{X} \\ \vdots \\ e^T \end{pmatrix}. \quad (10.6)$$

Then the orthogonal projection matrix from x' space to the null of F is

$$P = I - F^T(F F^T)^{-1}F. \quad (10.7)$$

Thus, the orthogonal projection of the objective gradient $\bar{X}c$ is

$$\Delta x = P\bar{X}c = (I - F^T(F F^T)^{-1}F)\bar{X}c. \quad (10.8)$$

Proposition 10.1.2 *Vector Δx is nonzero, satisfying $F\Delta x = 0$ and $(\bar{X}c)^T\Delta x > 0$.*

Proof Assume $\Delta x = 0$. It is known from (10.8) that there is $h \in \mathcal{R}^m, h_{m+1}$ such that

$$\bar{X}c = F^T(h^T, h_{m+1})^T.$$

Therefore, for any feasible solution $x' \geq 0$ to subproblem (10.5), it holds that

$$c^T\bar{X}x' = (h^T, h_{m+1})F x' = h^T A\bar{X}x' + h_{m+1}e^T x' = h_{m+1}.$$

In other words, the feasible value is constant. Note that this is the case for $\bar{X}$, constructed from any interior point $\bar{x}$. In particular, for $\bar{x} = e/n$, subproblem's objective value at the center e/n of the simplex is $c^T e/n^2$; on the other hand, it is known from Proposition 10.1.1 that the optimal value is zero. Hence it follows that $c^T e/n^2 = 0$, which implies that e/n is an optimal solution, contradicting that any interior point is not optimal. Therefore, $\Delta x \neq 0$. Finally, from (10.8) and $P^2 = P$ together with $P^T = P$, both $F\Delta x = 0$ and $(\bar{X}c)^T\Delta x > 0$ follow. $\square$

The preceding Proposition says that $-\Delta x \neq 0$ is a descent feasible direction. Further, it is not difficult to show that $-\Delta x$, which is within the null of F , forms the maximum possible angle with the objective gradient, that is, $-\Delta x$ is the steepest descent feasible vector in the null. As a result, if the stepsize from the center along the direction is less than the radius of the inscribed sphere of the simplex, then the new interior point in x' space must lie in the interior of the feasible region, hopefully leading to a satisfactory decrease in the objective value. More precisely, the new interior point in x' space is determined by:

$$\hat{x}' = \frac{e}{n} - \alpha \rho \frac{\Delta x}{\|\Delta x\|}, \quad (10.9)$$

where $\alpha \in (0, 1)$, $\rho \in (0, r]$, and r is the radius of the inscribed sphere. Thus mapping it to the original x -space via the inverse transformation gives

$$\hat{x} = \bar{X}\hat{x}'/e^T\bar{X}\hat{x}'. \quad (10.10)$$

Then, an iteration is complete.

The overall steps are summarized into the following algorithm.

Algorithm 10.1.1 (Karmarkar Algorithm) Given tolerance $\epsilon > 0$. Initial: $k = 1$, $\bar{x} = e/n$.

This algorithm solves the problem (10.1).

1. Compute $\Delta x = P\bar{X}c = (I - F^T(FF^T)^{-1}F)\bar{X}c$, where F is defined by (10.6).
2. Compute $\hat{x}' = e/n - \alpha\rho \Delta x / \|\Delta x\|$.
3. Compute $\bar{x} = \bar{X}\hat{x}'/e^T\bar{X}\hat{x}'$.
4. Stop if $c^T\bar{x} < \epsilon$.
5. Set $k = k + 1$.
6. Go to step 1.

The Karmarkar algorithm's inspiration is perhaps more valuable than itself, which stimulated coming out of some very successful algorithms of practical use. The idea that determining a search direction in the image space and then transforming it back to the original space has influenced the development of interior-point methods greatly. At present, search direction of a typical interior-point method is usually a combination of a descent direction, stemming from the negative gradient, and a centering direction, generated via some transformation. Such a combination, which allows the algorithm to go farther in each iteration, is the real cause of the success of the interior-point methods. As a “by-product,” in addition, approaching an optimal solution by a sequence of interior points overcomes the troublesome degeneracy encountered by the simplex method.

10.1.3 Convergence

To analyze Algorithm 10.1.1, define the following *potential function* over Γ of problem (10.1):

$$f(x) = f(x, c) \triangleq n \ln(c^T x) - \sum_{j=1}^n \ln x_j = \sum_{j=1}^n \ln \frac{c^T x}{x_j}. \quad (10.11)$$

The corresponding potential function over Γ of problem (10.5) is then

$$f'(x') \triangleq f(x', \bar{X}c) = n \ln(c^T \bar{X}x') - \sum_{j=1}^n \ln x'_j = \sum_{j=1}^n \ln \frac{c^T \bar{X}x'}{x'_j}. \quad (10.12)$$

Proposition 10.1.3 *The potential function values at any point $x \in \Gamma$ and at its image x' differ by a same constant that is dependent on $\bar{x}$.*

Proof From (10.3), (10.11), and (10.12), it follows that

$$f(x) = f(T^{-1}(x')) = \sum_{j=1}^n \ln \frac{c^T \bar{X} x'}{x'_j} - \ln(\Pi_{j=1}^n \bar{x}_j) = f'(x') - \ln(\Pi_{j=1}^n \bar{x}_j).$$

□

It is therefore seen that if a $\hat{x}'$ is determined such that $f'(\hat{x}')$ is less than $f'(e/n)$ by a constant, then the corresponding $f(\hat{x})$ is less than $f(\bar{x})$ by the same amount. We will estimate

$$f'(\hat{x}') = n \ln(c^T \bar{X} \hat{x}') - \sum_{j=1}^n \ln \hat{x}'_j. \quad (10.13)$$

Lemma 10.1.1 *If Δx and $\hat{x}'$ are, respectively, determined by (10.8) and (10.9), then it holds that*

$$\ln c^T \bar{X} \hat{x}' \leq \ln \frac{c^T \bar{X} e}{n} - \alpha \rho. \quad (10.14)$$

Proof From $P^2 = P$, $P^T = P$ and (10.8), it follows that

$$c^T \bar{X} \Delta x = c^T \bar{X} P \bar{X} c = c^T \bar{X} P^2 \bar{X} c = (P \bar{X} c)^T (P \bar{X} c) = \|\Delta x\|^2 > 0, \quad (10.15)$$

which together with (10.9) gives

$$c^T \bar{X} \hat{x}' = \frac{c^T \bar{X} e}{n} - \alpha \rho \frac{c^T \bar{X} \Delta x}{\|\Delta x\|} = \frac{c^T \bar{X} e}{n} - \alpha \rho \|\Delta x\|. \quad (10.16)$$

Now consider

$$\begin{aligned} \min f &= c^T \bar{X} x', \\ \text{s.t. } A \bar{X} x' &= 0, \\ e^T x' &= 1, \\ \|x' - e/n\| &\leq R, \end{aligned} \quad (10.17)$$

where R is the radius of the circumscribed sphere, i.e.,

$$R = \sqrt{n-1}/\sqrt{n} < 1.$$

Noting that $-\Delta x$ is the orthogonal projection of the negative objective gradient onto the null of F , and

$$\{x \in \mathcal{R}^n \mid \|x' - e/n\| \leq R\}$$

is the spherical domain with radius R and the simplex's center as its center, it is not difficult to show that the optimal solution is

$$x'(R) = \frac{e}{n} - R \frac{\Delta x}{\|\Delta x\|}.$$

On the other hand, since $\|x' - e/n\| \leq R$ is the circumscribed sphere of Γ , the feasible region of (10.17) includes the feasible region of subproblem (10.5), therefore the optimal value of the former is no more than the optimal value zero of the latter. So, considering (10.15) leads to

$$c^T \bar{X} x'(R) = \frac{c^T \bar{X} e}{n} - R \|\Delta x\| \leq 0,$$

or

$$-\|\Delta x\| \leq -\frac{c^T \bar{X} e}{nR},$$

which, combined with (10.16), $R < 1$, and $c^T \bar{x} > 0$, gives

$$c^T \bar{X} \hat{x}' \leq \frac{c^T \bar{X} e}{n} - \alpha \rho \frac{c^T \bar{X} e}{nR} = (1 - \alpha \rho / R) \frac{c^T \bar{X} e}{n} \leq (1 - \alpha \rho) \frac{c^T \bar{X} e}{n}.$$

Consequently, taking logarithm of the two sides of the preceding together with

$$\ln(1 - \alpha \rho) \leq -\alpha \rho$$

leads to (10.14). □

Lemma 10.1.2 *If $x' \in \Gamma$ satisfies $\|x' - e/n\| \leq \alpha \rho$, then*

$$-\sum_{j=1}^n \ln x'_j \leq -\sum_{j=1}^n \ln(1/n) + \frac{(n\alpha\rho)^2}{2(1 - \alpha n\rho)^2}. \quad (10.18)$$

Proof Since $\|x' - e/n\| \leq \alpha \rho$, it holds that

$$1/n - \alpha \rho \leq x'_j \leq 1/n + \alpha \rho, \quad j = 1, \dots, n,$$

or

$$1 - \alpha n \rho \leq nx'_j \leq 1 + \alpha n \rho, \quad j = 1, \dots, n. \quad (10.19)$$

Based on Taylor formula, $\ln(nx'_j)$ can be expressed as

$$\ln(nx'_j) = \ln 1 + (nx'_j - 1) - (nx'_j - 1)^2 / (2\theta_j^2),$$

where θ_j is between 1 and nx'_j , hence it is seen from (10.19) that $\theta_j \geq 1 - \alpha n \rho$. Therefore, the following inequality holds:

$$\ln(nx'_j) \geq (nx'_j - 1) - \frac{(nx'_j - 1)^2}{2(1 - \alpha n \rho)^2}. \quad (10.20)$$

In addition, it is known from the assumption that

$$\sum_{j=1}^n (nx'_j - 1) = ne^T x' - n = 0$$

and

$$\sum_{j=1}^n (nx'_j - 1)^2 = \|nx' - e\|^2 = n^2 \|x' - e/n\|^2 \leq (\alpha n \rho)^2.$$

Then, it follows from (10.20) that

$$\sum_{j=1}^n \ln(nx'_j) \geq \frac{-(\alpha n \rho)^2}{2(1 - \alpha n \rho)^2},$$

which implies (10.18). □

Theorem 10.1.3 *If $\rho = 1/n < r$, $\alpha \in (0, 1/2)$, then the complexity of the number of iterations required by the Karmarkar algorithm is $O(nL)$, where L is the binary digits of all input data.*

Proof Combining (10.13), (10.14), and (10.18) gives

$$f'(\hat{x}') \leq n \ln \frac{c^T \bar{X} e}{n} - \alpha n \rho - \sum_{j=1}^n \ln(1/n) + \frac{(\alpha n \rho)^2}{2(1 - \alpha n \rho)^2} = f'(e/n) - \delta(n, \rho, \alpha), \quad (10.21)$$

where a lower bound of the decrement of the potential is

$$\delta(n, \rho, \alpha) = \alpha n \rho - \frac{(\alpha n \rho)^2}{2(1 - \alpha n \rho)^2}.$$

Substituting $\rho = 1/n$ into the preceding gives a function, only related to α , i.e.,

$$\delta(\alpha) = \alpha - \frac{\alpha^2}{2(1-\alpha)^2} = \frac{\alpha(\alpha-2)(\alpha-1/2)}{(1-\alpha)^2}, \quad (10.22)$$

for which it is clear that

$$\delta(\alpha) \begin{cases} > 0, & 0 < \alpha < 1/2, \\ = 0, & \alpha = 1/2, \\ < 0, & 1/2 < \alpha < 1. \end{cases}$$

Therefore, from the assumption of this Proposition, it is known that $\delta(\alpha) > 0$.

Let $\hat{x}$ be the original image of $\hat{x}'$, determined by (10.10). According to Proposition 10.1.3, there is a constant μ related to $\bar{x}$ such that

$$f(\bar{x}) = f'(e/n) + \mu, \quad f(\hat{x}) = f'(\hat{x}') + \mu.$$

The preceding together with (10.21) gives

$$f(\hat{x}) - f(\bar{x}) = f'(\hat{x}') - f'(e/n) \leq -\delta(\alpha),$$

and hence, for each iteration, it holds that

$$f(\hat{x}) \leq f(\bar{x}) - \delta(\alpha).$$

It is noted that the initial interior point is e/n . Therefore, if $\hat{x}$ is the interior point, generated at iteration $k = 1, 2, \dots$ (step 3 of Algorithm 10.1.1), then

$$f(\hat{x}) \leq f(e/n) - k\delta(\alpha).$$

By (10.11), therefore, it is valid that

$$n \ln(c^T \hat{x}) - \sum_{j=1}^n \ln \hat{x}_j \leq n \ln(c^T e/n) - \sum_{j=1}^n \ln(1/n) - k\delta(\alpha),$$

from which, $\hat{x} \in \Gamma$ and function $\sum_{j=1}^n \ln x_j$ attaining the highest value at e/n over Γ , it follows that

$$n \ln(c^T \hat{x}) \leq n \ln(c^T e/n) - k\delta(\alpha).$$

The preceding is equivalent to

$$c^T \hat{x} \leq e^{-k\delta(\alpha)/n} (c^T e/n). \quad (10.23)$$

Let ϵ be a given tolerance on the optimal value. Then it is clear that there is a $\tau > 0$ such that

$$2^{-\tau L} < \frac{\epsilon}{c^T e/n}. \quad (10.24)$$

Once the number k of iterations attains the smallest integer such that

$$k > (\tau/\delta(\alpha))nL,$$

then considering (10.24) gives

$$e^{-k\delta(\alpha)/n} \leq e^{-\tau L} < 2^{-\tau L} < \frac{\epsilon}{c^T e/n},$$

which and (10.23) together lead to

$$c^T \hat{x} < \epsilon.$$

Under the assumption, therefore, the complexity of the number of iterations, required by the Karmarkar algorithm, is $O(nL)$. $\square$

Since the computational complexity of a single iteration is $O(n^3)$, the overall computational complexity of the Karmarkar algorithm is $O(n^4L)$. Further, Karmarkar improves the complexity of a single iteration to $O(n^{2.5})$, hence the overall complexity reduces to $O(n^{3.5}L)$.

In principle, it is desirable to have a larger lower bound on the decrement of the potential function. To this end, setting the derivative of (10.22) to 0 leads to its maximum $\alpha^* \approx 0.3177$ with value $\delta(\alpha)^* \approx 0.2093$. Therefore, taking $\alpha = 0.3177$ in the Karmarkar algorithm appears to be a good choice. However, it is not the case in practice, experience shows that an α close to 1 often accelerates convergence, even if the polynomial complexity cannot be guaranteed when $\alpha \geq 1/2$, in theory.

10.2 Affine Interior-Point Algorithm

The Karmarkar algorithm is not very convenient for practical use since it is only amenable to the Karmarkar standard problem. To handle the standard LP problem directly, the so-called *affine* algorithm was soon proposed as a variant of the Karmarkar algorithm (Barnes 1986; Cavalier and Soyster 1985; Karmarkar and Ramakrishnan 1985; Vanderbei et al. 1986). However, it is immediately found that the same algorithm was proposed by Dikin as early as in 1967 (without convergence proof), but not noted by the academic community, unfortunately.

Thereafter, we are concerned with the pair of the standard LP problems (5.1) and (5.2).

10.2.1 Formulation of the Algorithm

Let $\bar{x}$ be the current interior point. Again denote by $\bar{X}$ the diagonal matrix, whose diagonals are components of $\bar{x}$. It is clear that affine transformation

$$x' = \bar{X}^{-1}x$$

maps the positive octant to itself, and $\bar{x}$ to its *center* e . This transformation turns problem (5.1) into

$$\begin{aligned} \min \quad & c^T \bar{X} x', \\ \text{s.t.} \quad & A \bar{X} x' = b, \quad x' \geq 0. \end{aligned} \quad (10.25)$$

The orthogonal projection from x' space onto the null of $A \bar{X}$ is

$$P = I - \bar{X} A^T (A \bar{X}^2 A^T)^{-1} A \bar{X}, \quad (10.26)$$

and the orthogonal projection of objective gradient $\bar{X}c$ is then

$$\Delta x' = P \bar{X} c = (I - \bar{X} A^T (A \bar{X}^2 A^T)^{-1} A \bar{X}) \bar{X} c. \quad (10.27)$$

Lemma 10.2.1 *Under the basic assumption A4, it holds that $\Delta x' \neq 0$.*

Proof Equation (10.27) can be written as

$$\Delta x' = \bar{X}(c - A^T \bar{y}),$$

where

$$\bar{y} = (A \bar{X}^2 A^T)^{-1} A \bar{X}^2 c.$$

Thus $\Delta x' = 0$ implies $c = A^T \bar{y}$, which contradicts assumption A4. Therefore $\Delta x' \neq 0$. $\square$

To determine a new interior point in x' space, take $-\Delta x'$ as search direction:

$$\hat{x}' = e - \mu \Delta x' / \|\Delta x'\|.$$

If $\mu \in (0, 1)$ is taken, then $\hat{x}'$ lies within interior of the sphere with radius 1 at e . The original image of $\hat{x}'$ in x space is then

$$\hat{x} = \bar{x} - \mu \bar{X} \Delta x' / \|\Delta x'\|,$$

which is the *short stepsize* iteration scheme of the affine algorithm.

It is verified that the point

$$\hat{x} = \bar{x} - \alpha \bar{X} \Delta x' \quad (10.28)$$

satisfies $Ax = b$ for any α . In particular, for the short stepsize $\alpha = \mu / \|\Delta x'\| > 0$, it holds that $\hat{x} > 0$, hence the new iterate is again an interior point. Combining $P = P^2$, $P^T = P$, and (10.27) leads to

$$c^T \hat{x} = c^T \bar{x} - \alpha c^T \bar{X} P \bar{X} c = c^T \bar{x} - \alpha \|\Delta x'\|^2 < c^T \bar{x}, \quad (10.29)$$

which implies that the objective value strictly decreases.

It is desirable to go further without violating constraints. It turns out that letting the new interior point closer to the boundary leads to a bigger decrement of the objective value and significantly enhances the efficiency of the algorithm. To this end, the following result should be noted first.

Lemma 10.2.2 *If $\Delta x' \leq 0$, the standard LP problem (5.1) is unbounded below.*

Proof Under the assumption, $\hat{x}$ defined by (10.28) is clearly an interior point for any $\alpha > 0$. It is seen from (10.29) that

$$c^T \hat{x} = c^T \bar{x} - \alpha \|\Delta x'\|^2 \rightarrow -\infty \quad (\alpha \rightarrow \infty).$$

Therefore, the original problem is unbounded below. $\square$

Assume now that $\Delta x' \not\leq 0$. In this case, the maximum stepsize attaining the boundary is $1 / \max(\Delta x')$, leading to the following “long stepsize” scheme:

$$\hat{x} = \bar{x} - \lambda \bar{X} \Delta x' / \max(\Delta x'), \quad (10.30)$$

where $\lambda \in (0, 1)$ is a *stepsize*.

Accompanying the original interior point $\bar{x}$, a dual estimate $(\bar{y}, \bar{z})$ is expected. If ignoring the nonnegative requirement for $\bar{z}$ but fulfilling the complementarity condition as much as possible, $(\bar{y}, \bar{z})$ should be the solution to the following least-squares problem:

$$\begin{aligned} \min & (1/2) \|\bar{X} \bar{z}\|^2, \\ \text{s.t. } & z = c - A^T \bar{y}. \end{aligned}$$

It is not difficult to solve the preceding, leading to

$$\bar{y} = (A \bar{X}^2 A^T)^{-1} A \bar{X}^2 c, \quad \bar{z} = \bar{X}^{-1} P \bar{X} c. \quad (10.31)$$

Thus (10.27) can be written alternatively as

$$\Delta x' = \bar{X}(c - A^T \bar{y}) = \bar{X} \bar{z}, \quad (10.32)$$

which reveals the relation between the search direction and the dual estimate. It is noted that $\bar{X}\bar{z}$ is just the associated dual gap (see Sect. 5.3).

The overall steps with long stepsize are put into the following algorithm (without generating dual estimates). It is known from (10.32) that the dual gap tends to zero as the procedure converges. Therefore, optimality is achieved approximately whenever $\|\Delta x'\|$ becomes small enough.

Algorithm 10.2.1 (Affine Interior-Point Algorithm) Given $\lambda \in (0, 1)$, $\epsilon > 0$. Initial: interior point $\bar{x} > 0$.

This algorithm solves the standard LP problem (5.1).

1. Compute $\Delta x' = P\bar{X}c = (I - \bar{X}A^T(A\bar{X}^2A^T)^{-1}A\bar{X})\bar{X}c$.
2. Stop if $\|\Delta x'\| < \epsilon$ (optimality achieved approximately).
3. Stop if $\Delta x' \leq 0$ (unbounded below).
4. Update by $\bar{x} = \bar{x} - \lambda\bar{X}\Delta x' / \max(\Delta x')$.
5. Go to step 1.

It is seen from (10.27) that the algorithm's major work in each iteration is the computation of projection $\Delta x'$, concerning the solution of $m \times m$ system below:

$$(A\bar{X}^2A^T)y = A\bar{X}^2c.$$

In contrast to the Karmarkar algorithm, the affine algorithm is simpler and yet performs better. In fact, the latter is the earliest interior-point algorithm found to possibly outperform the simplex method.

10.2.2 Convergence and Starting-Up

Firstly, we state two convergence results without proof. The first, given by Dikin (1974), is under the nondegeneracy assumption (see also Gonzaga 1990; Vanderbei and Lagarias 1990).

Theorem 10.2.3 *Assume the existence of an optimal solution to the standard problem (5.1). If all feasible solutions are nondegenerate, then the sequence of interior points, generated by (10.30) ($\lambda \in (0, 1)$), and sequence of dual estimates, generated by (10.31), respectively, converge to relative interior points of the primal and dual optimal faces.*

If the range of λ is restricted properly, the long stepsize affine algorithm converges without the nondegeneracy assumption, as the following theorem says (Tsuchiya and Muramatsu 1995).

Theorem 10.2.4 *Assume the existence of an optimal solution to the standard problem (5.1). Then the sequence of interior points, generated by (10.30) ($\lambda \in (0, 2/3]$), converges to a relative interior point of the primal optimal face, and the*

sequence of dual estimates, generated by (10.31), converges to the relative analytic center of the dual optimal face, with asymptotic descent rate $1 - \lambda$ for the objective value.

When $\lambda > 2/3$, however, the convergence of the affine algorithm is not guaranteed. Counterexamples have been found, showing that $2/3$ is the maximum value of λ for ensuring convergence (Hall and Vanderbei 1993). As for practice, nevertheless, the case is just the contrary: it is faster and more reliable to take a λ value close to 1 (e.g., $\lambda \in [0.9, 0.99]$). On the other hand, it is shown that even if $\lambda \leq 2/3$, the algorithm solves the Klee–Minty problem by traversing neighborhoods of all the 2^n vertices and hence is not of polynomial complexity (Megiddo and Shub 1989).

Now turn to Phase-1 to provide an initial interior point to the affine algorithm.

Given any n -dimensional vector $h > 0$ such that $b - Ah \neq 0$ ($h = e$ seems to be a good choice). Introducing an artificial variable x_{n+1} and a normalized vector

$$a_{n+1} = (b - Ah)/\|b - Ah\|, \quad (10.33)$$

construct the following auxiliary problem:

$$\begin{aligned} \min \quad & x_{n+1}, \\ \text{s.t.} \quad & Ax + a_{n+1}x_{n+1} = b, \quad x, x_{n+1} \geq 0. \end{aligned} \quad (10.34)$$

There is an optimal solution to the preceding problem. Moreover, $\bar{x} = h$, $\bar{x}_{n+1} = \|b - Ah\|$ is an interior-point solution available. If the optimal value is strictly greater than 0, then the original problem is infeasible.

Assume that solving the auxiliary problem by the affine algorithm with $\lambda \in (0, 2/3]$ produces a sequence of interior points converging to x^∞ with objective value 0. Thus, x^∞ is a relative interior point of the feasible region of the original problem.

Define

$$B = \{j \in A \mid x_j^\infty > 0\}, \quad N = \{j \in A \mid x_j^\infty = 0\}.$$

Then, the original problem reduces to

$$\begin{aligned} \min \quad & c_B^T x_B, \\ \text{s.t.} \quad & Bx_B = b, \quad x \geq 0, \end{aligned}$$

which has an interior point $x_B^\infty > 0$ available. Then, the affine algorithm can get itself started.

10.3 Dual Affine Interior-Point Algorithm

The basic idea behind the affine method is applied to the dual problem to derive its dual counterpart (Andersen et al. 1996).

Consider the dual problem (5.2). Assume that $(\bar{y}, \bar{z})$ is the current interior point, satisfying dual constraints. Denote by $\bar{Z}$ the diagonal matrix with components of $\bar{z}$ as its diagonals. The affine transformation $z = \bar{Z}z'$ turns the dual problem to

$$\begin{aligned} \max \quad & b^T y, \\ \text{s.t.} \quad & A^T y + \bar{Z}z' = c, \quad z' \geq 0 \end{aligned}$$

or equivalently,

$$\begin{aligned} \max \quad & b^T y, \\ \text{s.t.} \quad & \bar{Z}^{-1} A^T y + z' = \bar{Z}^{-1} c, \quad z' \geq 0. \end{aligned} \tag{10.35}$$

Note that the affine transformation $z = \bar{Z}z'$ maps the positive octant of z -space to itself, $\bar{z}$ to its *center* e . Since A has full row rank, $\bar{Z}^{-1} A^T$ has full column rank.

Assume that the QR factorization of $\bar{Z}^{-1} A^T$ is

$$\bar{Z}^{-1} A^T = [Q_1, Q_2] \begin{pmatrix} R \\ 0 \end{pmatrix} = Q_1 R, \tag{10.36}$$

where $[Q_1, Q_2]$ is orthogonal, $Q_1 \in \mathcal{R}^{n \times m}$ and $Q_2 \in \mathcal{R}^{n \times (n-m)}$, and $R \in \mathcal{R}^{m \times m}$ are upper triangular.

Substituting (10.36) into the equality constraint of (10.35) gives

$$Q_1 R y + z' = \bar{Z}^{-1} c.$$

Premultiplying the preceding by $R^{-1} Q_1^T$, noting $Q_1^T Q_1 = I$, and rearranging lead to

$$y = R^{-1} Q_1^T \bar{Z}^{-1} c - R^{-1} Q_1^T z'.$$

Combining the preceding and

$$Q_1 Q_1^T + Q_2 Q_2^T = I, \quad Q_2^T Q_2 = I,$$

we convert problem (10.35) into the following standard form with respect to z' :

$$\begin{aligned} \min \quad & b^T R^{-1} Q_1^T z', \\ \text{s.t.} \quad & Q_2^T z' = Q_2^T \bar{Z}^{-1} c, \quad z' \geq 0, \end{aligned} \tag{10.37}$$

where the constant $-b^T R^{-1} Q_1^T \bar{Z}^{-1} c$ was omitted from the objective function.

Proposition 10.3.1 *The orthogonal projection of the objective gradient of (10.37) to the null space of Q_2^T is equal to*

$$\Delta z' = \bar{Z}^{-1} A^T (A \bar{Z}^{-2} A^T)^{-1} b. \quad (10.38)$$

Proof By $Q_2^T Q_1 = 0$, the orthogonal projection of the objective gradient to the null becomes

$$\Delta z' = (I - Q_2(Q_2^T Q_2)^{-1} Q_2^T) Q_1 R^{-T} b = Q_1 R^{-T} b.$$

By $Q_1^T Q_1 = I$, the preceding is equal to

$$\Delta z' = Q_1 (R R^{-1}) R^{-T} b = Q_1 R (R^T R)^{-1} b = Q_1 R (R^T Q_1^T Q_1 R)^{-1} b,$$

which together with (10.36) gives (10.38). $\square$

Thereby, $-\Delta z'$ is a favorable search direction in z' -space. For a new iterate to satisfy the equality constraint of (10.35), $\Delta z'$ and the associated search direction Δy in y -space should fulfill the condition

$$\bar{Z}^{-1} A^T \Delta y + \Delta z' = 0,$$

premultiplying which by $A \bar{Z}^{-1}$ and using (10.38) gives

$$\Delta y = -(A \bar{Z}^{-2} A^T)^{-1} A \bar{Z}^{-1} \Delta z' = -(A \bar{Z}^{-2} A^T)^{-1} b. \quad (10.39)$$

Via $z = \bar{Z} z'$, vector

$$\Delta z' = -\bar{Z}^{-1} A^T \Delta y$$

can be transformed back to z -space, i.e.,

$$\Delta z = -A^T \Delta y. \quad (10.40)$$

The long stepsize line search scheme is then

$$\hat{y} = \bar{y} - \lambda \beta \Delta y, \quad \hat{z} = \bar{z} - \lambda \beta \Delta z, \quad (10.41)$$

where $\lambda \in (0, 1)$, and

$$\beta = \min\{\bar{z}_j / \Delta z_j \mid \Delta z_j > 0, j = 1, \dots, n\}. \quad (10.42)$$

In the case when $\Delta z \leq 0$, the preceding scheme is not well-defined, and the dual problem is unbounded, or the primal problem is infeasible.

As for an estimate, say $\bar{x}$, of the associated primal solution, it can be determined as follows. If ignoring the nonnegative requirements but attempting to satisfy complementarity condition, the $\bar{x}$ should solve the following least-squares problem:

$$\begin{aligned} \min \quad & \|\bar{Z}x\|, \\ \text{s.t.} \quad & A\bar{Z}^{-1}(\bar{Z}x) = b. \end{aligned}$$

The solution to which can be obtained by taking a look at the minimum 2-norm solution to $A\bar{Z}^{-1}w = b$ and (10.39), i.e.,

$$\bar{x} = \bar{Z}^{-2}A^T(A\bar{Z}^{-2}A^T)^{-1}b = \bar{Z}^{-2}\Delta z.$$

Consequently, when $\bar{x} \geq 0$ and dual gap $c^T\bar{x} - b^T\bar{y}$ becomes sufficiently small, the procedure, generating the sequence of dual interior points, can be terminated.

The overall steps are put into the following algorithm.

Algorithm 10.3.1 (Dual Affine Algorithm) Given $\lambda \in (0, 1)$, $\epsilon > 0$. Initial: interior point $(\bar{y}, \bar{z})$.

This algorithm solves the standard LP problem (5.1).

1. Compute $(\Delta y, \Delta z)$ by (10.39) and (10.40).
2. Stop if $\Delta z \leq 0$ (dual unbounded or primal infeasible).
3. Compute $\bar{x} = \bar{Z}^{-2}\Delta z$.
4. Stop if $\bar{x} \geq 0$ and $c^T\bar{x} - b^T\bar{y} < \epsilon$ (optimality achieved approximately).
5. Update $(\bar{y}, \bar{z})$ by (10.41) and (10.42).
6. Go to step 1.

The preceding algorithm starts from a dual interior point. ($\bar{y} = 0, \bar{z} = c$) is available one in case of $c > 0$. Otherwise, an approach similar to that described in the last half of Sect. 10.2.2 can be employed to generate an initial dual interior point by solving the following auxiliary problem:

$$\begin{aligned} \max \quad & -y_{m+1}, \\ \text{s.t.} \quad & [A^T : c - e] \begin{pmatrix} y \\ y_{m+1} \end{pmatrix} + z = c, \quad z, y_{m+1} \geq 0. \end{aligned}$$

The preceding can be handled by the dual affine algorithm since it is upper bounded and has an interior point available, i.e.,

$$\bar{y} = 0, \quad \bar{y}_{m+1} = 1; \quad \bar{z} = e.$$

We point out that primal and dual affine algorithms can be derived from each other and share similar convergence properties (Tsuchiya 1992), but the latter performs better in practice.

10.4 Path-Following Interior-Point Algorithm

These types of methods perform very well in practice. Such a method can be regarded as one using a *homotopy* strategy, in which one creates a trivial problem, compared with a hard problem to be solved, so that there is a continuum of problems between them and handles backward from the trivial problem to the hard problem by solving the problems in between (exactly or approximately).

Taking the standard LP problem (5.1) as the *hard* problem, we construct the *trivial* problem, whose optimal solution is an interior point.

As the border of the positive octant, coordinate planes $x_j = 0$, $j = 1, \dots, n$, defining the scope of the feasible region, might well be deemed as a “wall.” We have seen that the Karmarkar algorithm and/or the affine algorithm employ transformations to create a “centering force” to push the iterate leaving the wall. The combination of the *centering force* and the *pushing force* along the direction of the negative objective gradient gives a favorable feasible descent search direction. Another approach for generating “centering force” is via logarithmic barrier function, which was initially employed by Frisch (1955) to solve nonlinear problems. After the Karmarkar algorithm emerged, the logarithmic barrier function draws high attention from the community and is applied in the path-following method to construct the *trivial* problem.

As nonnegative constraints $x \geq 0$ are inactive for interior points, we remove them and add logarithmic barrier terms to the objective function to construct the problem below:

$$\begin{aligned} (\mathbf{P}_\mu) \min f(x) &= c^T x - \mu \sum_{j=1}^n \ln x_j, \\ \text{s.t. } Ax &= b, \end{aligned} \quad (10.43)$$

where $\mu > 0$ is called *barrier parameter*. Implying $x \geq 0$, (10.43) characterizes a class of problems with parameter $\mu > 0$. Because of the barrier terms, $f \rightarrow +\infty$ as $x_j \rightarrow 0^+$, so that minimization of f “bounces” points from the “wall” to prevent them from leaving the interior of the feasible region.

Related to the *hard* problem by μ , the *trivial* problem (10.43) is nonlinear but not difficult to handle. The gradient and Hessian matrix of the objective $f(x)$ are, respectively,

$$\nabla f(x) = c - \mu X^{-1} e, \quad \nabla^2 f(x) = \mu X^{-2},$$

where $X = \text{diag}(x_1, \dots, x_n)$. It is clear that $f(x)$ is a strict convex function over region $x > 0$. The Lagrange function of problem $(\mathbf{P}_\mu)$ is then

$$L(x, y) = c^T x - \mu \sum_{j=1}^n \ln x_j - y^T (Ax - b).$$

Therefore, the point $x > 0$ is an optimal solution to (P_μ) if and only if there is a $y \in \mathcal{R}^m$ such that

$$Ax - b = 0, \quad (10.44)$$

$$c - \mu X^{-1}e - A^T y = 0. \quad (10.45)$$

Using the notation

$$z = c - A^T y, \quad Z = \text{diag}(z_1, \dots, z_n),$$

the preceding condition becomes the following system in variables x, y, z :

$$Ax = b, \quad (10.46)$$

$$A^T y + z = c, \quad (10.47)$$

$$Xz = \mu e. \quad (10.48)$$

Megiddo (1989) shows that:

- (i) For each $\mu > 0$, problem (P_μ) either has a unique optimal solution or is lower unbounded.
- (ii) If there exists a solution $x > 0$ to the system for some $\mu > 0$, then so is a solution $x(\mu)$ to the system for all $\mu > 0$, and $x(\mu)$ is a continuous curve. There is a limit $\lim x(\mu)$ as $\mu \rightarrow 0^+$, which is an optimal solution to the problem (P).

If the interior of the feasible region of (P) is nonempty and lower bounded, (10.46) ~ (10.48) determine a unique path $x(\mu)$ to an optimal solution. Such a $x(\mu)$ is termed *central path* or *trajectory*. The common idea of various path-following algorithms is to find an optimal solution by following the path approximately.

On the other hand, the central path can also be derived in another way. We can handle the standard LP problem directly via the optimality condition (5.15), viewed as a nonlinear system of x, y, z subject to $x, z \geq 0$. To generate interior points across the interior of the feasible region and approach an optimal solution on the boundary, it is only possible to fulfill the complementarity condition approximately by replacing $Xz = 0$ with $Xz = v$, where $v > 0$ is a parameter vector. The associated solution $x(v)$ to the system is expected to tend to an optimal solution as v tends to 0. In view of the symmetry of $x_j z_j = v_j$, $\forall j = 1, \dots, n$, it is natural to allow components of v to be equal by setting $v = \mu e$, which leads to (10.46) ~ (10.48) consequently. So, the “hard” problem is the optimality condition, while the *trivial* problem is system (10.46–10.48). Of course, it is not realistic to solve all systems for continuous μ , but for some sequence $\{\mu_k\}$ instead, as will be seen in the next section.

10.4.1 Primal–Dual Interior-Point Algorithm

A realization of the path-following strategy is to design a monotone descent sequence $\{\mu_k\}$ with limit 0, and for every fixed parameter $\mu = \mu_k$, $k = 1, \dots$, solve system (10.46)~(10.48) for its solution $x(\mu_k)$. In general, it is difficult to directly obtain the exact solution to the nonlinear system (10.48). Instead, we employ the Newton method to find an approximate one. It is unnecessary to pay a high price for the accuracy since what we are interested in is not the solutions themselves but their limit.

Thus, only a single Newton step is taken to obtain an approximate solution in each iteration, and the monotone descent μ_k are generated one by one.

Assume that the interior point $(\bar{x}, \bar{y}, \bar{z})$ satisfies (10.46) and (10.47) and approximately satisfies (10.48). We need to determine $(-\Delta x, -\Delta y, -\Delta z)$ such that $(\bar{x} - \Delta x, \bar{y} - \Delta y, \bar{z} - \Delta z)$ is a new approximate solution, which satisfies (10.46)~(10.48), i.e.,

$$\begin{aligned} A(\bar{x} - \Delta x) &= b, \\ A^T(\bar{y} - \Delta y) + (\bar{z} - \Delta z) &= c, \\ (\bar{X} - \Delta X)(\bar{z} - \Delta z) &= \mu e. \end{aligned}$$

The preceding expressions are equivalent to

$$A\Delta x = A\bar{x} - b, \quad (10.49)$$

$$A^T\Delta y + \Delta z = A^T\bar{y} + \bar{z} - c, \quad (10.50)$$

$$\bar{Z}\Delta x + \bar{X}\Delta z = \bar{X}\bar{z} + \Delta X\Delta z - \mu e, \quad (10.51)$$

where $\Delta X = \text{diag}(\Delta x_1, \dots, \Delta x_n)$. It is known from the assumptions that the right-hand sides of (10.49) and (10.50) are equal to zero. Then, omitting the second-order term $\Delta X\Delta z$ of the right-hand side of (10.51) gives the so-called *Newton equation*:

$$A\Delta x = 0, \quad (10.52)$$

$$A^T\Delta y + \Delta z = 0, \quad (10.53)$$

$$\bar{Z}\Delta x + \bar{X}\Delta z = \bar{X}\bar{z} - \mu e. \quad (10.54)$$

Introduce notation

$$D = \bar{Z}^{-1}\bar{X}. \quad (10.55)$$

Premultiplying (10.54) by $A\bar{Z}^{-1}$ and noting (10.52) lead to

$$AD\Delta z = AD\bar{z} - \mu A\bar{Z}^{-1}e,$$

and premultiplying (10.53) by $A\bar{Z}^{-1}\bar{X}$ gives

$$ADA^T\Delta y + AD\Delta z = 0.$$

From the preceding two equations, it follows that

$$\Delta y = -(ADA^T)^{-1}A(D\bar{z} - \mu\bar{Z}^{-1}e). \quad (10.56)$$

By (10.53) and (10.54), on the other hand, it holds that

$$\begin{aligned} \Delta z &= -A^T\Delta y, \\ \Delta x &= D(\bar{z} - \Delta z) - \mu\bar{Z}^{-1}e. \end{aligned}$$

The preceding three expressions together determine the so-called *Newton direction*.

Considering that the Newton point, determined along the direction by taking stepsize 1 probably violates the nonnegative conditions, the new iterate is defined as

$$\hat{x} = \bar{x} - \alpha\Delta x, \quad \hat{y} = \bar{y} - \alpha\Delta y, \quad \hat{z} = \bar{z} - \alpha\Delta z, \quad (10.57)$$

with the stepsize α determined by:

$$\alpha = \lambda \min\{\alpha_p, \alpha_d\}, \quad (10.58)$$

where $\lambda \in (0, 1)$,¹ and

$$\begin{aligned} \alpha_p &= \min\{\bar{x}_j/\Delta x_j \mid \Delta x_j > 0, j = 1, \dots, n\}, \\ \alpha_d &= \min\{\bar{z}_j/\Delta z_j \mid \Delta z_j > 0, j = 1, \dots, n\}. \end{aligned} \quad (10.59)$$

Note that the new interior point satisfies conditions (10.46) and (10.47), but only approximately satisfies (10.48).

Since the iterate is primal and dual feasible, the iteration process can be terminated when the dual gap becomes sufficiently small, e.g., by using the following criterion:

$$\bar{x}^T\bar{z}/(1 + |b^T\bar{y}|) < \epsilon,$$

where $0 < \epsilon \ll 1$ is the precision tolerance. The end iterate is then regarded as an approximate optimal solution.

As a key factor, the barrier parameter μ affects the behavior of the algorithm greatly, as it determines the magnitude of the *centripetal force*, compared with that of the “pushing down force.” It turns out that iterates too close to the central path

¹ $\lambda = 0.99995$ may be a suitable choice—a long stepsize performs better than a short stepsize.

are unfavorable and degrade computational efficiency. For each given μ , it is only needed to go by one step along the Newton direction. As the larger the μ , the greater the rebound to the point, so it is necessary to reduce μ gradually as iterates approach an optimal solution. It is difficult to find an “ideal” way to do so, however. In this aspect, extensive research has been done, and schemes have been suggested. Some schemes make the algorithm have polynomial time complexity (e.g., Kojima et al. 1989; Jansen et al. 1996 offered a unified form).

The first practicable scheme, proposed by McShane et al. (1989), monotonically reduces μ with the varying dual gap in the solution process.

From (10.57), it follows that

$$c^T \hat{x} - b^T \hat{y} = c^T \bar{x} - b^T \bar{y} - \delta.$$

So, the decrement of the dual gap is

$$\delta = \alpha(c^T \Delta x - b^T \Delta y). \quad (10.60)$$

It is known by $A(\bar{x} - \Delta x) = b$ that

$$b^T(\bar{y} - \Delta y) = (\bar{x} - \Delta x)^T A^T(\bar{y} - \Delta y),$$

and by $A^T(\bar{y} - \Delta y) + (\bar{z} - \Delta z) = c$ that

$$c^T(\bar{x} - \Delta x) = (\bar{y} - \Delta y)^T A(\bar{x} - \Delta x) + (\bar{z} - \Delta z)^T(\bar{x} - \Delta x).$$

From the preceding two equalities, it follows that

$$c^T(\bar{x} - \Delta x) - b^T(\bar{y} - \Delta y) = (\bar{z} - \Delta z)^T(\bar{x} - \Delta x) = \bar{z}^T \bar{x} - \Delta z^T \bar{x} - \bar{z}^T \Delta x + \Delta z^T \Delta x.$$

In addition, by (10.54) it holds that

$$\bar{z}^T \Delta x + \bar{x}^T \Delta z = \bar{x}^T \bar{z} - n\mu.$$

Combining the preceding two equalities gives

$$c^T(\bar{x} - \Delta x) - b^T(\bar{y} - \Delta y) = n\mu + \Delta z^T \Delta x,$$

which, by omitting the second-order term, can be written as

$$-(c^T \Delta x - b^T \Delta y) \approx -c^T \bar{x} + b^T \bar{y} + n\mu.$$

Substituting the preceding to (10.60) leads to

$$\delta \approx \alpha(c^T \bar{x} - b^T \bar{y} - n\mu).$$

In order for dual gap's decrement to be positive, therefore, μ should satisfy

$$\mu < (c^T \bar{x} - b^T \bar{y})/n,$$

at least. On the other hand, too big μ is not suitable, since it may cause numerical instability. In this aspect, one might take

$$\mu = (c^T \bar{x} - b^T \bar{y})/n^2.$$

Overall steps are summarized into the following algorithm.

Algorithm 10.4.1 (Primal–Dual Interior-Point Algorithm) Given $0 < \epsilon \ll 1$, $\lambda \in (0, 1)$. Initial: interior point $(\bar{x}, \bar{y}, \bar{z})$. This algorithm solves the standard LP problem.

1. Stop if $\bar{x}^T \bar{z} / (1 + |b^T \bar{y}|) < \epsilon$ (optimality achieved approximately).
2. Compute $\mu = (c^T \bar{x} - b^T \bar{y})/n^2$.
3. Solve system $(ADA^T)\Delta y = -A(D\bar{z} - \mu \bar{Z}^{-1}e)$.
4. Compute $\Delta z = -A^T \Delta y$, $\Delta x = D(\bar{z} - \Delta z) - \mu \bar{Z}^{-1}e$.
5. Update $\bar{y}$, $\bar{z}$, $\bar{x}$ by (10.57), where α is determined by (10.58) and (10.59).
6. Go to step 1.

10.4.2 Infeasible Primal–Dual Algorithm

Since it was not easy to obtain an initial interior point to get it started, the use of the primal–dual method was limited, as prompted a simple variant to appear.

Terms *infeasible interior point* and *interior point* differ only from that the former does not necessarily satisfy primal and dual equality constraints. To get started from an infeasible interior point, Lustig (1990) and Tanabe (1990) modify the primal–dual method presented in the previous section. The basic idea is similar to that behind the primal–dual simplex method (Sect. 6.2) but is realized in a different framework.

Assume that the current iterate $(\bar{x}, \bar{y}, \bar{z})$ be an infeasible interior point, satisfying $\bar{x}, \bar{z} > 0$ but the residuals

$$r_p = A\bar{x} - b, \quad r_d = A^T \bar{y} + \bar{z} - c$$

are not equal to 0, in general.

Omitting the second-order term $\Delta X \Delta z$ from (10.49)~(10.51) gives the Newton equation, i.e.,

$$A \Delta x = r_p, \tag{10.61}$$

$$A^T \Delta y + \Delta z = r_d, \tag{10.62}$$

$$\bar{Z}\Delta x + \bar{X}\Delta z = \bar{X}\bar{z} - \mu e. \quad (10.63)$$

Using notation D defined by (10.55), the solution to Newton equation can be written as

$$\Delta y = (ADA^T)^{-1}(r_p - AD(\bar{z} - r_d) + \mu A\bar{Z}^{-1}e), \quad (10.64)$$

$$\Delta z = r_d - A^T\Delta y, \quad (10.65)$$

$$\Delta x = D(\bar{z} - \Delta z) - \mu\bar{Z}^{-1}e. \quad (10.66)$$

In fact, the last two expressions can be derived from (10.62) and (10.63). In addition, premultiplying (10.63) by $A\bar{Z}^{-1}$ and noting (10.61) give

$$A\bar{Z}^{-1}\bar{X}\Delta z = -r_p + A\bar{Z}^{-1}(\bar{X}\bar{z} - \mu e),$$

and premultiplying (10.62) by $A\bar{Z}^{-1}\bar{X}$ leads to

$$A\bar{Z}^{-1}\bar{X}A^T\Delta y + A\bar{Z}^{-1}\bar{X}\Delta z = A\bar{Z}^{-1}\bar{X}r_d.$$

Then, (10.64) follows from the above two equations.

Except for different search directions, determination of the new iterate is the same as that for the primal–dual method. The infeasible primal–dual method also uses updating formula (10.57), where stepsize α is determined by (10.58)~(10.59), with $\lambda \in (0, 1)$ (usually, $\lambda = 0.99995$ is taken).

The barrier parameter is determined by Lustig et al. (1991):

$$\mu = (c^T\bar{x} - b^T\bar{y} + M\gamma)/\phi(n), \quad (10.67)$$

where

$$\gamma = \|A\bar{x} - b\| / \|Ax^0 - b\| + \|A^T\bar{y} + \bar{z} - c\| / \|A^Ty^0 + z^0 - c\|, \quad (10.68)$$

$$M = \xi\phi(n) \max\{\max_{j=1}^n |c_j|, \max_{i=1}^m |b_i|\}, \quad (10.69)$$

$$\phi(n) = \begin{cases} n^2, & \text{If } n \leq 5000, \\ n^{3/2}, & \text{If } n > 5000. \end{cases} \quad (10.70)$$

Initial (x^0, y^0, z^0) and ξ are determined by another algorithm (see discussions following Algorithm 10.4.3).

Overall steps are put into the following algorithm.

Algorithm 10.4.2 (Infeasible Primal–Dual Interior-Point Algorithm) Given $0 < \epsilon, \epsilon_p, \epsilon_d \ll 1$, $\lambda \in (0, 1)$. Initial: $(\bar{x}, \bar{y}, \bar{z})$ satisfying $\bar{x}, \bar{z} > 0$. This algorithm

solves the standard LP problem.

1. Compute $r_p = A\bar{x} - b$, $r_d = A^T\bar{y} + \bar{z} - c$.
2. Stop if $\bar{x}^T\bar{z}/(1 + |b^T\bar{y}|) < \epsilon$, $r_p < \epsilon_p$, $r_d < \epsilon_d$ (optimality achieved approximately).
3. Solve system $(ADA^T)\Delta y = r_p - AD(\bar{z} - r_d) + \mu A\bar{Z}^{-1}e$, where μ is determined by (10.67).
4. Compute $\Delta z = r_d - A^T\Delta y$, $\Delta x = D(\bar{z} - \Delta z) - \mu\bar{Z}^{-1}e$.
5. Update $\bar{y}$, $\bar{z}$, $\bar{x}$ by (10.57), where α is determined by (10.58) and (10.59).
6. Go to step 1.

Convergence of the preceding algorithm is not guaranteed in theory, and there is no device to detect the infeasibility or unboundedness either. Nevertheless, this algorithm performs well and inspires an even better variant, introduced in the next section.

10.4.3 Predictor–Corrector Primal–Dual Algorithm

The Newton equation (10.61)~(10.63), used in the infeasible primal–dual method, is yielded from (10.49)~(10.51) by omitting the second-order term $\Delta X\Delta z$. Mehrotra’s (1992) predictor–corrector primal–dual method introduces an additional correction step, involving the second-order term, to improve the search direction (for simplicity, the prefix “infeasible” will be omitted).

It is noted that the left-hand side of (10.49)~(10.51) is linear in Δx , Δy , Δz , while only the right-hand side involves the second-order term $\Delta X\Delta z$. The system can therefore be regarded as an implicit one. We can determine a predictor solution, say $(\Delta x', \Delta y', \Delta z')$, to approach its solution first, then substitute it to the right-hand side, and solve the resulting system for a corrector solution.

Despite the Newton direction (10.64)~(10.66) is a clear choice for being predictor solution, Mehrotra uses the solution to the linear system below:

$$\begin{aligned} A\Delta x' &= r_p, \\ A^T\Delta y' + \Delta z' &= r_d, \\ \bar{Z}\Delta x' + \bar{X}\Delta z' &= \bar{X}\bar{z}. \end{aligned}$$

The only difference between the preceding and Newton equation is that its right-hand side no longer involves term $-\mu e$. It can be viewed as one derived from the complementarity condition $Xz = 0$ rather than $Xs = \mu e$. So, its solution can be obtained easily from (10.64)~(10.66) by setting $\mu = 0$, i.e.,

$$\Delta y' = (ADA^T)^{-1}(r_p - AD(\bar{z} - r_d)), \quad (10.71)$$

$$\Delta z' = r_d - A^T\Delta y', \quad (10.72)$$

$$\Delta x' = D(\bar{z} - \Delta z'). \quad (10.73)$$

Substituting $\Delta z'$ and $\Delta x'$ to the right-hand side of (10.51), we solve

$$\begin{aligned} A\Delta x &= r_p, \\ A^T\Delta y + \Delta z &= r_d, \\ \bar{Z}\Delta x + \bar{X}\Delta z &= \bar{X}\bar{z} + \Delta X'\Delta z' - \mu e, \end{aligned}$$

to obtain a corrector solution. Only the right-hand sides of the preceding system and (10.61)~(10.63) differ, hence the derivation of the corrector solution is similar to that of (10.64)~(10.66), i.e.,

$$\Delta y = \Delta y' - (ADA^T)^{-1}(A\bar{Z}^{-1}\Delta X'\Delta z' - \mu A\bar{Z}^{-1}e), \quad (10.74)$$

$$\Delta z = r_d - A^T\Delta y, \quad (10.75)$$

$$\Delta x = D(\bar{z} - \Delta z) + \bar{Z}^{-1}\Delta X'\Delta z' - \mu\bar{Z}^{-1}e. \quad (10.76)$$

The predictor solution may also be used to determine barrier parameter μ . Mehrotra (1991) uses the following formula:

$$\mu = (g'/\bar{x}^T\bar{z})^2(g'/n), \quad (10.77)$$

where

$$\begin{aligned} g' &= (\bar{x} - \alpha'_p\Delta x')^T(\bar{z} - \alpha'_d\Delta z'), \\ \alpha'_p &= 0.99995 \min\{\bar{x}_j/\Delta x'_j \mid \Delta x'_j > 0, j = 1, \dots, n\}, \\ \alpha'_d &= 0.99995 \min\{\bar{z}_j/\Delta z'_j \mid \Delta z'_j > 0, j = 1, \dots, n\}. \end{aligned}$$

It is noted that g' is a predicted dual gap, whereas $\bar{x}^T\bar{z}$ is the current dual gap. Their ratio (less than 1) may be regarded as a predicted improvement at the iteration. So, a small μ is taken when the predicted improvement is significant, while a large μ is taken in the other case. This is appropriate because small improvement implies that the tendency toward the *center* should be enhanced.

Mehrotra's work draws the attention of the community immediately. To evaluate the predictor–corrector primal–dual method vs. the pure infeasible primal–dual method, Lustig et al. (1992) conducted further computational experiments, involving 86 Netlib standard test problems. Their numerical results indicated that the former outperformed the latter with 85 out of the 86 problems, in terms of iteration counts; and most importantly, in terms of CPU time, the former defeated the latter with 71 out of the 86, exceeding 82%. Overall, it is well accepted that the predictor–corrector primal–dual method is superior to the infeasible primal–dual method.

In order to avoid solving ill-conditioned problems and instability when iterates are close to optimal solution, they modified the determination of the barrier parameter as follows:

Use (10.77) only when $\bar{x}^T \bar{z} \geq 1$; otherwise, use

$$\mu = \bar{x}^T \bar{z} / \phi(n), \quad (10.78)$$

where $\phi(n)$ is still defined by (10.70).

Overall steps are put into the following algorithm.

Algorithm 10.4.3 (Predictor–Corrector Primal–Dual Algorithm) Given $0 < \epsilon, \epsilon_p, \epsilon_d \ll 1$, $\lambda \in (0, 1)$. Initial: $(\bar{x}, \bar{y}, \bar{z})$ satisfying $\bar{x}, \bar{z} > 0$. This algorithm solves the standard LP problem.

1. Compute $r_p = A\bar{x} - b$, $r_d = A^T \bar{y} + \bar{z} - c$.
2. Stop if $\bar{x}^T \bar{z} / (1 + |b^T \bar{y}|) < \epsilon$, $r_p < \epsilon_p$, $r_d < \epsilon_d$ (optimality achieved approximately).
3. Compute μ by (10.77) if $\bar{x}^T \bar{z} \geq 1$; else, by (10.78).
4. Compute $\Delta z'$, $\Delta x'$ by (10.71)~(10.73).
5. Compute Δy , Δz , Δx by (10.74)~(10.76).
6. Update $\bar{y}$, $\bar{z}$, $\bar{x}$ by (10.57), where α is determined by (10.58) and (10.59).
7. Go to step 1.

Now let us turn to how to obtain an initial point for the infeasible primal–dual method and the predictor–corrector primal–dual method, etc.

It is attractive that the initial infeasible point seems to be quite free to choose. Since it can affect the merit of the algorithm considerably, the initial point should be nearly primal and dual feasible as well as close to the *center*. It turns out that there are still numerical difficulties if it is not properly centered even if being close to an optimal solution. Mehrotra (1992) suggests obtaining the initial point by solving a quadratic programming problem. Subsequently, Andersen, Gondzio, Mészáros, and Xu derive a variant, handling the following convex quadratic programming problem:

$$\begin{aligned} \min \quad & c^T x + (\eta/2)x^T x, \\ \text{s.t.} \quad & Ax = b, \end{aligned}$$

where $\eta > 0$ is a weight factor. The Lagrange function of the preceding problem is

$$L(x, w) = c^T x + (\eta/2)x^T x - w^T (Ax - b).$$

An explicit expression of the solution to $\nabla L(x, w) = 0$ is then

$$\bar{x} = (1/\eta)(A^T(AA^T)^{-1}Ac - c).$$

To be an initial solution, $\bar{x}$'s components less than some positive number δ are modified to δ (e.g., $\delta = 1$). An initial dual solution $(\bar{y}, \bar{z})$ is determined similarly.

Although it works well in practice, the above heuristic approach is scaling dependent and does not certainly generate a well-centered initial point. For Algorithm 10.4.3, moreover, it is assumed that there exists an optimal solution since there is no reliable method to detect infeasibility/unboundedness, as is a serious drawback to practice.

10.4.4 Homogeneous and Self-Dual Algorithm

The so-called *homogeneous and self-dual algorithm* might be the most efficient interior-point method at present. It overcomes the two drawbacks of the predictor–corrector primal–dual algorithm, presented in the previous subsection, by a so-called *homogeneous and self-dual model*, which is based on *skew-symmetric self-dual artificial model* introduced by Ye et al. (1994), though it was subsequently found that Goldman and Tucker (1956) and Tucker (1956) studied such types of models even earlier. We will focus on main elements of it here in this subsection; for more details, the reader is referred to the relevant literature, such as Jansen et al. (1994), Xu and Ye (1995).

We begin with the self-dual problem (5.9), written as

$$\begin{aligned} \min \quad & c^T x - b^T y, \\ \text{s.t.} \quad & Ax - b = 0, \quad x \geq 0, \\ & -A^T y + c - z = 0, \quad z \geq 0, \end{aligned} \tag{10.79}$$

which is equivalent to the pair of primal and dual standard problems (5.1) and (5.2), in the sense that they share the same optimal solutions. We attempt to modify the preceding homogeneous and self-dual problem to a better “trivial” one, whose solution is infeasible to the original problem.

Since the expected iterates will satisfy the nonnegative but not equality constraints, we add

$$b^T y - c^T x - \sigma = 0, \quad \sigma \geq 0,$$

to constraints and put the preceding problem into the following problem in variables (x, y, z, σ, τ) :

$$\begin{aligned} \min \quad & 0 \\ \text{s.t.} \quad & Ax - b\tau = 0, \\ & -A^T y + c\tau - z = 0, \\ & b^T y - c^T x + \sigma = 0, \\ & x, z, \sigma, \tau \geq 0, \end{aligned} \tag{10.80}$$

which is associated with (10.79) by σ, τ . According to Theorem 5.3.3, there exists a strictly complementary optimal solution, say $(x^*, y^*, z^*, \tau^*, \sigma^*)$, to (10.80), which can be shown to satisfy the following systems:

$$\begin{aligned} Xz &= 0 \quad \tau\sigma = 0, \\ x + z &> 0 \quad \tau + \sigma > 0. \end{aligned} \quad (10.81)$$

The following can be shown:

- (i) Case $\tau^* > 0$. Then $(x^*/\tau^*$ and $(y^*/\tau^*, z^*/\tau^*)$ are a pair of strictly complementary optimal solutions to (5.1) and (5.2).
- (ii) Case $\tau^* = 0$. This implies $\sigma^* > 0$, and hence $b^T y^* - c^T x^* > 0$. Thus, at least one of $b^T y^* > 0$ and $-c^T x^* > 0$ holds: if $b^T y^* > 0$, the primal problem (5.1) is infeasible; if $c^T x^* < 0$, the dual problem (5.2) is infeasible, and if $b^T y^* > 0$ and $c^T x^* < 0$, then the pair problems are both infeasible.

The required strictly complementary optimal solution to (10.79) can be achieved in the limit via solving a series of “trivial” problems, modified from (10.80). To this end, introduce the feasibility and the average complementarity residual functions below:

$$\begin{aligned} r_p &= b\tau - Ax, \\ r_d &= A^T y - c\tau + z, \\ r_g &= c^T x - b^T y - \sigma, \\ \mu &= (x^T z + \tau\sigma)/(n+1). \end{aligned} \quad (10.82)$$

Denote by the symbol “ $\bar{\bullet}$ ” the quantity at the current iterate ($\bar{y}, \bar{x} > 0, \bar{\tau} > 0, \bar{z} > 0, \bar{\sigma} > 0$). Problem (10.80) is modified by putting the complementarity residual functions and logarithmic barrier terms in its objective and laying current residuals in the place of the right-hand side of the constraints. As in the primal-dual method, the logarithmic barrier terms are for centering and keeping iterate’s positiveness. The resulting problem is then

$$\begin{aligned} \min \quad & z^T x + \sigma\tau - \lambda\mu^{(0)} \sum_{j=1}^n (\ln x_j + \ln z_j) - \lambda\mu^{(0)} (\ln \tau + \ln \sigma), \\ \text{s.t.} \quad & Ax - b\tau = -\lambda\bar{r}_p, \\ & -A^T y + c\tau - z = -\lambda\bar{r}_d, \\ & b^T y - c^T x + \sigma = -\lambda\bar{r}_g, \end{aligned} \quad (10.83)$$

where $\lambda \in [0, 1]$ is called the *path parameter*. It is noted that the current iterate is feasible to (10.83).

The first optimality conditions for (10.83) are

$$\begin{aligned}
 Ax - b\tau &= -\lambda\bar{r}_p, \\
 -A^T y + c\tau - z &= -\lambda\bar{r}_d, \\
 b^T y - c^T x + \sigma &= -\lambda\bar{r}_g, \\
 Xz &= \lambda\bar{\mu}e, \\
 \tau\sigma &= \lambda\bar{\mu}, \\
 x, z, \sigma, \tau &> 0.
 \end{aligned} \tag{10.84}$$

Similar to the primal–dual method, the search direction is determined by applying Newton’s method to (10.84). Denoting by $\gamma \in [0, 1]$ a predetermined reduction rate of the path parameter, the resulting Newton equation for the search direction $(\Delta x, \Delta y, \Delta z, \Delta\sigma, \Delta\tau)$ can be written as

$$\begin{pmatrix}
 A & -b \\
 -A^T & c & -I \\
 b^T & -c^T & -1 \\
 \bar{Z} & \bar{X} \\
 \bar{\sigma} & \bar{\tau}
 \end{pmatrix}
 \begin{pmatrix}
 \Delta y \\
 \Delta x \\
 \Delta \tau \\
 \Delta z \\
 \Delta \sigma
 \end{pmatrix}
 =
 \begin{pmatrix}
 (1 - \gamma)\bar{r}_p \\
 (1 - \gamma)\bar{r}_d \\
 (1 - \gamma)\bar{r}_g \\
 \gamma\bar{\mu}e - \bar{X}\bar{z} \\
 \gamma\bar{\mu} - \bar{\tau}\bar{\sigma}
 \end{pmatrix}.$$

Setting $\gamma = 1$ gives a pure centering direction, and setting $\gamma = 0$ gives an affine direction. Such a system is solved in each iteration, constituting the major computational task in the method.

Once the Newton direction has been computed, a stepsize is determined by using the same formulas as in the primal–dual method so that the new iterates are strictly positive. It is self-evident that the termination criterion should be set properly.

10.5 Notes on Interior-Point Algorithm

It is known so far that the earliest interior-point algorithm is proposed by famous mathematician von Neumann in a talk with G.B. Dantzig in 1948 (Dantzig and Thapa 2003). It is designed, without convergence proof, to generate a feasible solution to a special LP problem, which turns out to be just the *Karmarkar standard problem*. An elegant proof was given by Dantzig in a letter to J. von Neumann.

Originally, the logarithmic barrier function was used by Frisch (1955) to design interior-point methods for solving nonlinear programming problems. This nonlinear tool was applied to LP only after the publication of the Karmarkar algorithm (1984), inspiring a boom of interior-point methods. It was utilized by the path-following method to create the search direction that is a combination of a descent direction and a centering direction. The other interior-point methods presented in this chapter can also be derived alternatively in such a way (Gill et al. 1985–1986; Shanno and Bagchi 1990).

To explain, we begin with the nonlinear programming problem (10.43). The gradient and Hessian matrix of its objective function at current interior point, say $\bar{x}$, are, respectively,

$$\nabla f(\bar{x}) = c - \mu \bar{X}^{-1}e, \quad \nabla^2 f(\bar{x}) = \mu \bar{X}^{-2}.$$

It is noted that $\nabla^2 f(\bar{x})$ is positive definite. Setting $x = \bar{x} - \Delta x$, the problem becomes one with respect to Δx , and the latter can be approximated by the following strict convex quadratic programming problem:

$$\begin{aligned} \min \quad & f(\bar{x}) - \nabla f(\bar{x})^T \Delta x + (1/2) \Delta x^T \nabla^2 f(\bar{x}) \Delta x \\ \text{s.t.} \quad & A \Delta x = 0. \end{aligned}$$

Thus, what needs to do is to find the stationary point of the following Lagrange function:

$$L(\Delta x, y) = -\nabla f(\bar{x})^T \Delta x + (1/2) \Delta x^T \nabla^2 f(\bar{x}) \Delta x - y^T A \Delta x,$$

where y is the Lagrange multiplier vector, by solving system

$$\mu \bar{X}^{-2} \Delta x - A^T y = c - \mu \bar{X}^{-1}e, \quad (10.85)$$

$$A \Delta x = 0. \quad (10.86)$$

It holds by (10.85) that

$$\Delta x = (1/\mu)(\bar{X}^2 A^T y + \bar{X}^2 c - \mu \bar{X}e), \quad (10.87)$$

substituting which to (10.86) gives

$$y = -(A \bar{X}^2 A^T)^{-1} A \bar{X}(\bar{X}c - \mu e).$$

Then, substituting the preceding to (10.87) leads to

$$\Delta x = (1/\mu) \bar{X} P(\bar{X}c - \mu e), \quad (10.88)$$

where P is the projection matrix, defined by (10.26). $-\Delta x$ is the Newton direction of problem (10.43) at $\bar{x}$. It is seen by comparing with (10.27) that the vector is just a positive combination of $\bar{X} P e$ and the search direction $-\bar{X} \Delta x' = -\bar{X} P \bar{X} c$, used by the primal affine algorithm.

Assume that the feasible region of the original problem is nonempty. Consider problem

$$\begin{aligned} \min \quad & \sum_{j=1}^n \ln x_j, \\ \text{s.t.} \quad & Ax = b, \end{aligned}$$

to which there is a unique optimal solution that can be regarded as the center of the feasible region. It is not difficult to show that $\bar{X}Pe$ is just the Newton direction of the preceding problem at $\bar{x}$. Therefore, the search direction, defined by (10.88), reflects the common effect of a “descent force” and a “centering force.” The size of the barrier parameter determines the proportion of the two tendencies. Algorithms with polynomial time complexity can be obtained by properly selecting the parameters and stepsize (Gonzaga 1987). On the other hand, the search direction tends to the one, used by the primal affine algorithm, as μ tends to zero. The latter is not of polynomial-time complexity.

With the same logarithmic barrier function, it is possible to derive the search direction, used by the Karmarkar algorithm. The resulting algorithm is equivalent to the Karmarkar algorithm if a proper barrier parameter and stepsize are taken.

As for the dual problem, construct the following nonlinear programming problem:

$$\max g(y) = b^T y + \mu \sum_{j=1}^n \ln(c_j - a_j^T y),$$

where a_j denotes j -indexed column of A . Introduce

$$\bar{z} = c - A^T \bar{y}.$$

The gradient and Hessian matrix of the objective function at the current point $\bar{y}$ are, respectively,

$$\nabla g(\bar{y}) = b - \mu A \bar{Z}^{-1} e, \quad \nabla^2 g(\bar{y}) = -\mu A \bar{Z}^{-2} A^T.$$

The Newton direction is then

$$\Delta y = -(A \bar{Z}^{-2} A^T)^{-1} (b/\mu + A \bar{Z}^{-1} e),$$

which is a positive combination of $-A \bar{Z}^{-1} e$ and the search direction, used by the dual affine algorithm, and the former is Newton direction, somehow pointing to the neighborhood of the *center* of the dual feasible region of the following problem:

$$\max \sum_{j=1}^n \ln(c_j - a_j^T y).$$

Another nonlinear tool applicable to interior-point method is ordinary differential equations (ODE). Let us bring up the affine algorithm. Its search direction

$$-X(I - XA^T(AX^2A^T)^{-1}AX)Xc$$

actually determines a vector field for the feasible region. Taking it as right-hand side, we construct the following autonomous (ODE) system:

$$\frac{dx(t)}{dt} = -X(I - XA^T(AX^2A^T)^{-1}AX)Xc, \quad x(0) = \bar{x}, \quad (10.89)$$

where $\bar{x}$ is an initial interior point. Adler et al. (1989) realized the dual affine method and reported remarkable computational results. They defined a trajectory by ordinary differential equations (ODEs) and used the power series numerical integration of the first and second order to obtain an approximate optimal solution.

As for the ODE method, it is worth making further remarks not just because it is successful in realizing the dual affine method but also because it is of general significance. In fact, for any linear or smooth nonlinear programming methods, the ODE system can be introduced similarly. Under mild conditions, there exists a trajectory to such a system, whose limit point is an optimal solution (Pan 1992a; 1992b). More precisely, let $x(t)$, $0 \leq t < \gamma$, be the right-hand half of (10.89)'s trajectory, starting from $\bar{x}$, such that

$$x(t) \rightarrow x^* \quad \text{as} \quad t \rightarrow \gamma,$$

where x^* is the optimal solution. It is not difficult to solve it approximately by some numerical integration method.

According to the author, the earliest ODE method was proposed by Arrow and Hurwicz (1956) for solving equality constraint nonlinear programming problems. There were many related papers published subsequently (e.g., Fiacco and McCormick 1968; Abadie and Corpentier 1969; Evtushenko 1974; Botsaris 1974; Tanabe 1977). A common belief seems that the computational cost related to the ODE is quite high, compared with standard methods, such as SQP, even if Brown and Bartholomew-Biggs (1987) reported very favorable numerical results with the former. The author himself has also experienced the success of the ODE method (Pan 1982, 1992b) and is surprised why it has been ignored by the academic community.

After all, the normal line search is only of the first order in certain sense; taking small stepsize in Algorithm 10.2.1 amounts to executing numerical integration with the first-order power series method, or Euler method. In particular, realizing curve search via the second-order power series integration should apply to nonlinear programming problems. Based on this idea, the algorithm using *higher order* search scheme outperformed both DFP and BFGS methods significantly in computational experiments (Shi and Pan 2011).

Finally, we conclude this chapter with the following remarks.

The interior-point method is very successful. Some scholars even believe that it is better than the simplex method for solving large-scale sparse LP problems. The homogeneous and self-dual method has already been implemented in some commercial codes, such as CPLEX, and is deemed as one of the most powerful methods. Such types of methods can be well parallelized, compared with the simplex method.

Despite all these, the interior-point method has been seriously limited in applications due to its inherent weakness. It only provides an approximate optimal solution and hence needs an additional *purification* procedure in applications requiring an optimal vertex solution. Moreover, it cannot be *warmly* started, compared with the simplex method that achieves optimality quickly if starting from some basis reached in a previous solution process. Consequently, the interior-point method does not apply to integer or mixed-integer LP programming, since solving such problems yields a large number of correlated LP problems, which can only be left for the simplex method. At present, therefore, the dominant position of the simplex method is difficult to shake.

Now a race between the simplex method and the interior-point method seems to continue. No single method applies to all LP problems. As a result, there are usually multiple options of methods included in professional codes. To take the advantage of both types, some scholars first apply the interior-point method, and then the simplex method (Bixby et al. 1992), although it seems too mechanized. In Part II of this book, we will introduce recently developed pivotal interior-point methods in a natural manner.

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Chapter 11

Integer Linear Programming (ILP)



The feasible region of the LP problem is continuous since each variable is restricted to a continuous interval. If variables (or a part of variables) are further restricted to integer values, it becomes an *integer LP (ILP)* (or mixed-integer (MILP)) problem that differs from LP problem essentially. This chapter highlights topics, such as graphic approach and basic idea, cutting-plane method, branch-and-bound method, controlled-cutting method, controlled-branch method, including depth-oriented strategy and breadth-oriented strategy. The final section introduces an actualization of the controlled-branch method with the reduced simplex framework, which is dealt with in detail in Part II.

A vector/point is called *integer* if all of its components are integer. The definition of the *feasible set* of the ILP problem is obvious.

In the next section, a graphic approach to ILP is illustrated with a 2-dimensional instance to introduce basic ideas behind ILP solvers. The following two sections are devoted to cutting-plane and branch-and-bound methods, respectively. Then, we develop new ILP methods, named *controlled-cutting* and *controlled-branch* in the other two sections. We present a realization of the controlled-branch method in the final section.

In this chapter, we denote the maximum integer not greater than $\bullet$ by

$$\lfloor \bullet \rfloor$$

and the minimum integer not less than $\bullet$ by

$$\lceil \bullet \rceil.$$

11.1 Graphic Approach

The convex hull of the feasible set is termed *integer hull* of the ILP problem. It can be shown that vertices of the integer hull are integers. Therefore, the task is boiled down to solving the corresponding relaxation LP problem over the integer hull.

This can be illustrated by solving the following 2-dimensional instance graphically:

$$\begin{aligned}
 \min f &= -2x_1 - 5x_2 \\
 (ILP) \text{ s.t. } & 2x_1 + 3x_2 \leq 12 \\
 & x_1 + x_2 \leq 5 \\
 & x_2 \leq 3 \\
 & \text{integer } x_1, x_2 \geq 0.
 \end{aligned} \tag{11.1}$$

The optimal solution to the LP relaxation of the preceding is $x_1 = 1.5, x_2 = 3$, with optimal value $f = -18$ (see Sect. 2.6).

As sketched in Fig. 11.1, the feasible region of the LP relaxation is the area enclosed by polygon OABCD. The dots constitute the feasible set of (11.1), and the area enclosed by polygon OAEFD is the integer hull. We solve the problem over the integer hull by the following LP graphic approach.

In the figure, the isoline $-2x_1 - 5x_2 = 0$ of the objection function passes through the origin. The objective value reduces from 0 as the line shifts parallel to the upper-right direction. It is seen that the farthest possible position, with respect to the area of OAEFD, is line $-2x_1 - 5x_2 = -17$, passing through integer vertex E . Therefore, the optimal solution to ILP problem (11.1) is $x_1^* = 1, x_2^* = 3, f^* = -17$.

Although the graphic approach is only of conceptual sense, it would give some idea about the creation of some ILP solvers.

Definition 11.1.1 The LP relaxation of an ILP problem is the LP problem resulting from dropping its integer restriction on variables.

ILP problem and its LP relaxation form a 1-to-1 correspondence. For example, the LP relaxation of the ILP problem (11.1) is over the feasible region, the area surrounded by OABCD.

The validity of the following is obvious.

Proposition 11.1.1 *If the optimal solution to the LP relaxation is an integer solution, it is optimal for the ILP problem.*

The preceding forms a common ground of the existing ILP solvers, like the *cutting-plane* and *branch-and-bound* methods. At first, it solves the LP relaxation of the ILP problem. It is done if the obtained is an integer optimal solution. Otherwise, it deals with the LP relaxation according to its idea by solving a series of LP relaxations, which are so rich in practice, as to form a major application area of LP at present.

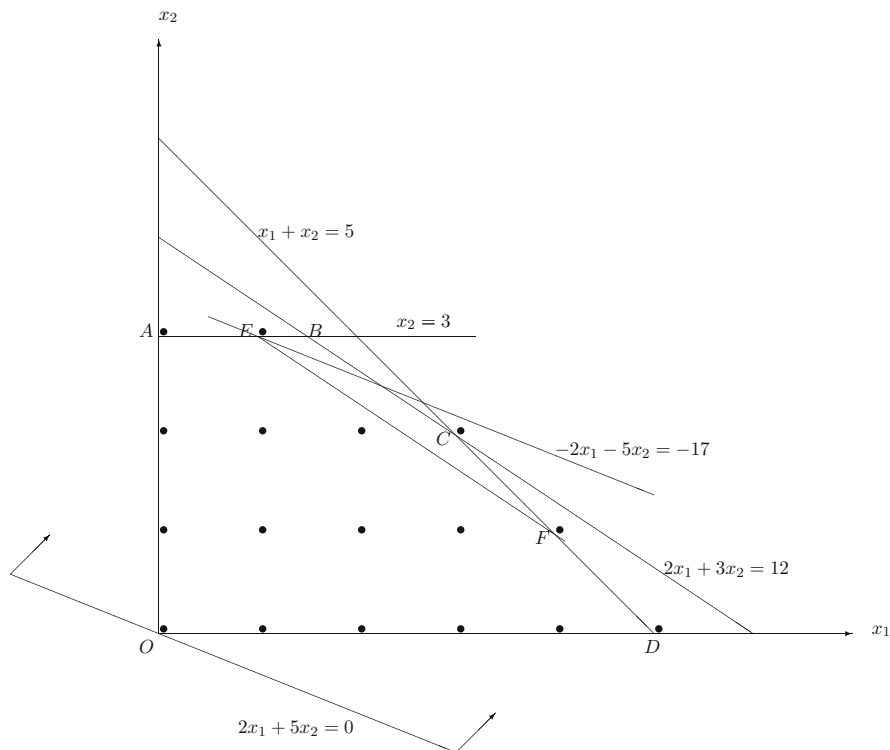


Fig. 11.1 Graphic solution to ILP problem 11.1

It turns out that only a finite number of LP relaxations need to be solved. However, the finiteness is of theoretical value only. In practice, any of the solvers alone is far from satisfactory. Therefore, some combination of them is now widely used in the ILP area. We will not go into details though.

In addition, it is observed that if one cuts off some part of the feasible region OABCD, which does not include the feasible set of the ILP problem, then the rest may be closer to the integer hull OAEFD; in the algebraic words, adding some appropriate inequalities to the constraints would lead to a better LP relaxation. This is another common point of the existing ILP solvers.

The *valid cut* used is defined as follows.

Definition 11.1.2 A valid cut of an ILP problem is an inequality, satisfied by all feasible solutions to it but violated by the optimal solution to its LP relaxation.

On the other hand, one can divide OABCD into several parts and seek integer optimal solutions over each of them. This means solving the corresponding so-called LP *subrelaxations* and comparing their end products. This is the basic idea behind the branch-and-bound method.

The resulting parts are called *branches* of the feasible region of the LP relaxation. The feasible region of the ILP problem is the largest (trivial) branch. Without confusion, the corresponding LP subrelaxations as well as ILP subproblems are also said to be *branches*.

In this aspect, the basic tool is the so-called *valid pair* of integer cuts, which divides OABCD into two branches. The pair cuts off a narrow area with a thickness of one unit without affecting the feasible set of the ILP problem.

11.1.1 Basic Idea Behind New ILP Solvers

Assume that the coefficients of the objective function are all integers and hence so are the objective values associated with the feasible (integer) solutions.

From the Fig. 11.1, it is observed that the wanted optimal integer solution lies on the isoline, $-2x_1 - 5x_2 = -17$, of the objective function of the LP relaxation, which is a line *close* to its lowest isoline passing through vertex B , i.e., $-2x_1 - 5x_2 = -18$.

Definition 11.1.3 Let f^- be an integer lower bound on the optimal value of the ILP problem. The inequality

$$c^T x \geq f^- \quad (11.2)$$

is an objective cut.

Then, the validity of the following claim is obvious.

Lemma 11.1.1 *Let f^- be an integer lower bound on the optimal value of the ILP problem. If an optimal solution to an LP subrelaxation is an integer solution that attains the objective value equaling to f^- , it is optimal for the ILP problem.*

The preceding forms the footstone of the new ILP solvers.

But, how to obtain a tight integer lower bound on the optimal value? If the optimal value, $\tilde{f}$, of the LP relaxation is available, we take the following:

$$f^- = \lceil \tilde{f} \rceil. \quad (11.3)$$

Thus, the following reveals the basic idea behind the new solvers.

Proposition 11.1.2 (Heuristic Characteristic of the Optimal Value) *The optimal value of the ILP problem is towards $\tilde{f}$ and tends to reach $f^- = \lceil \tilde{f} \rceil$.*

It is clear that if the range of the objective value of the relaxation is no more than one unit, the optimal solution of the LP relaxation modified by adding the objective cut (11.2) must attain the objective value being equal to f^- . If further, it is an integer solution, it is optimal for the ILP problem, according to Lemma 11.1.1. Otherwise, we proceed by adding some cuts and solving corresponding LP subrelaxations.

11.2 Cutting-Plane Method

Many authors contributed to the cutting-plane method. The method presented in this section essentially follows Gomory (1958). It solves a series of LP relaxation problems associated with ILP subproblems, generated from adding valid cut successively. It terminates when an optimal solution to some LP relaxation is an integer, which is also optimal to the ILP problem. Thus, the idea of the method is quite simple.

The big problem is how to generate *good* cuts to speed up the solution process. Extensive research has been done in this aspect. In this section, only the so-called *fractional cut* is utilized for tasting.

Let $B = \{j_1, \dots, j_m\}$ and $N = A \setminus B$ be optimal basis and nonbasis of the LP relaxation, respectively. Assume that the associated optimal basic solution is noninteger. Specifically, assume that the equation corresponding to the i th row of the optimal simplex tableau, i.e.,

$$x_{j_i} + \sum_{j \in N} \bar{a}_{i,j} x_j = \bar{b}_i \quad (11.4)$$

is of noninteger right-hand side $\bar{b}_i$.

Introduce the following quantities:

$$0 \leq \alpha_{i,j} = \bar{a}_{i,j} - \lfloor \bar{a}_{i,j} \rfloor, \quad 0 < \beta_i = \bar{b}_i - \lfloor \bar{b}_i \rfloor < 1$$

by which, (11.4) can be written

$$x_{j_i} + \sum_{j \in N} (\lfloor \bar{a}_{i,j} \rfloor + \alpha_{i,j}) x_j = \lfloor \bar{b}_i \rfloor + \beta_i > 0.$$

Converting the preceding by moving the terms with integer coefficients to the left-hand side and those with fractions to the right-hand side gives

$$x_{j_i} + \sum_{j \in N} (\lfloor \bar{a}_{i,j} \rfloor) x_j - \lfloor \bar{b}_i \rfloor = \beta_i - \sum_{j \in N} \alpha_{i,j} x_j.$$

For any integer feasible solution x , the left-hand side of the preceding is an integer, and hence so is the right-hand side. In addition, since

$$\alpha_{i,j} \geq 0, \quad x_j \geq 0, \quad \forall j \in N,$$

the right-hand is no more than $0 < \beta_i < 1$. Therefore, both sides must be an integer less than and equal to 0, i.e.,

$$\text{integer} \quad \beta_i - \sum_{j \in N} \alpha_{i,j} x_j \leq 0. \quad (11.5)$$

It is then asserted that the associated optimal basic solution does not satisfy (11.5). In fact, if it is substituted to (11.5), the left-hand side becomes $\beta_i > 0$ because all its

nonbasic components are zero. Therefore, since it includes the whole feasible set, (11.5) is a valid cut.

The ILP problem is modified by adding such a cut, and the associated LP relaxation is solved then. If the yielded solution is still noninteger, a new valid cut is determined, and added to the constraints, and so on until a resulting solution is an integer, which is an optimal solution to the original ILP problem.

Optimal solution to LP relaxation would have multiple noninteger components, and hence there are multiple choices of valid cut. In this case, it seems to be favorable to form a cut cutting deepest into the feasible region of the LP relaxation.

To this end, let I be the set of row index corresponding to noninteger right-hand side. For any $i \in I$,

$$\beta_i / \sqrt{\sum_{j \in N} \alpha_{i,j}^2}$$

is the Euclidean distance from the associated solution to the LP relaxation to the boundary of the area defined by (11.5) (Sect. 2.1). Unfortunately, the associated computations are cumbersome. Instead, we may use residuals approximately by determining row index i' such that

$$i' \in \arg \max_{i \in I} \beta_i.$$

Example 11.2.1 Solve the following ILP problem by the cutting-plane method:

$$\begin{aligned} \min f &= -48x_1 - 40x_2 \\ \text{s.t. } 8x_1 &+ x_3 - 3x_4 &= 2 \\ -2x_1 + 2x_2 &+ 2x_4 &= 12 \\ \text{integers } x_j &\geq 0, \quad j = 1, \dots, 4. \end{aligned} \tag{11.6}$$

Answer Call simplex Algorithm 3.1.1, yielding

x_1	x_2	x_3	x_4	RHS
1		1/8	-3/8	1/4
	1	1/8	5/8	25/4
		11/8	7/8	131/4

Arbitrarily, take the first row with noninteger right-hand side. The corresponding equation is

$$x_1 + 1/8x_3 - 3/8x_4 = 1/4,$$

or equivalently

$$x_1 + 1/8x_3 + (-1 + 5/8)x_4 = 1/4.$$

The preceding can be written

$$x_1 - x_4 = 1/4 - 1/8x_3 - 5/8x_4.$$

From which the valid cut follows, i.e.,

$$1/4 - 1/8x_3 - 5/8x_4 \leq 0.$$

Insert the row corresponding to $-1/8x_3 - 5/8x_4 + x_5 = -1/4$ to the preceding tableau as the second bottom row, giving

x_1	x_2	x_3	x_4	x_5	RHS
1		1/8	-3/8		1/4
	1	1/8	5/8		25/4
		-1/8	-5/8	1	-1/4
		11/8	7/8		131/4

which is a dual feasible simplex tableau. Call the dual simplex Algorithm 5.4.1, yielding

x_1	x_2	x_3	x_4	x_5	RHS
1		1/5		-3/5	2/5
	1	0		1	6
		1/5	1	-8/5	2/5
		6/5		7/5	162/5

Arbitrarily, take the third row with noninteger right-hand side. The corresponding equation is

$$1/5x_3 + x_4 - 8/5x_5 = 2/5,$$

or equivalently

$$1/5x_3 + x_4 + (-2 + 2/5)x_5 = 2/5.$$

The preceding can be written

$$x_4 - 2x_5 = 2/5 - 1/5x_3 - 2/5x_5.$$

From which the valid cut follows, i.e.,

$$2/5 - 1/5x_3 - 2/5x_5 \leq 0$$

Insert the row corresponding to $-1/5x_3 - 2/5x_5 + x_6 = -2/5$ to the preceding tableau as the second bottom row, giving which is dual feasible. Call Algo-

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		1/5		-3/5		2/5
	1	0		1		6
		1/5	1	-8/5		2/5
		-1/5		-2/5	1	-2/5
		6/5		7/5		162/5

rithm 5.4.1, yielding

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		1/2			-3/2	1
	1	-1/2			5/2	5
		1	1		-4	2
		1/2		1	-5/2	1
		1/2			7/2	31

which give an integer solution, which is optimal to the original ILP problem, i.e.,

$$x_1^* = 1, \quad x_2^* = 5, \quad f^* = 31.$$

11.3 Branch-and-Bound Method

Let P and $\bar{x}$ be the feasible region and the optimal solution to the LP relaxation problem of the ILP problem.

If $\bar{x}$ is noninteger, we need to divide the LP relaxation into a pair of LP subrelaxations. To do so, introduce the following.

Definition 11.3.1 Integer cut is inequality corresponding to hyperplane passing through integer point in an axis, and being perpendicular to the axis.

Let $\bar{x}_j$ be fractional component of $\bar{x}$. Then cuts $x_j \leq \lfloor \bar{x}_j \rfloor$ and $x_j \geq \lceil \bar{x}_j \rceil$ are said *valid pair* of integer cuts, and the variable x_j is *cutting variable*. Adding the two cuts divides P into a pair of LP subrelaxations with

$$P \cap \{x \mid x_j \leq \lfloor \bar{x}_j \rfloor\} \quad \text{and} \quad P \cap \{x \mid x_j \geq \lceil \bar{x}_j \rceil\}$$

as their feasible regions, respectively, including the feasible set of the ILP problem, but cutting out the area

$$\lfloor \bar{x}_j \rfloor < x_j < \lceil \bar{x}_j \rceil.$$

LP subrelaxation problem or the associated ILP subproblem is also called *branch*.

The branch-and-bound method was developed by Land and Doig (1960). It solves a series of branches, yielding from adding integer cuts to seek an optimal solution to the ILP problem, as described as follows.

Set $f^+ = +\infty$ as the initial upper bound on optimal value f^* to the ILP problem.

Each time, one usually faces multiple branches. One of them is solved to find its optimal solution, and the others are left pending, including those deemed as *fathomed*. The former and related integer cut are said *active*.

The *fathomed* means that there is no need for the branch to be divided further, as is in one of the following cases:

- (i) The branch is infeasible.
- (ii) The solution to the branch is an integer.

It is clearly feasible to the ILP problem. Update $f^+ = \bar{f}$ if the corresponding objective value $\bar{f}$ is strictly less than the current upper bound.

- (iii) The solution is noninteger, but it holds that $\lceil \bar{f} \rceil \geq f^+$.

The ILP problem is solved when all branches are *fathomed*.

The branch-and-bound method generates a series of branches that form the so-called *enumeration tree*. There are *breadth-oriented* and *depth-oriented* strategies to develop enumeration tree. Once a branch is found not fathomed, the former solves the nearest pending branch (in the same level), and the latter solves one of the branches at a deeper level while leaving the other branch pending, until reaching a fathomed branch. Then, the pending branches are handled one by one, from the nearest (deepest) to the top level, until all branches are examined.

We take the example (11.1) to show details of the branch-and-bound method with depth-oriented strategy.

The solution to the LP relaxation of (11.1) is $\bar{x}_1 = 3/2$, $\bar{x}_2 = 3$, $\bar{f} = -18$. Since component $\bar{x}_1 = 1\frac{1}{2}$ is noninteger, we add the valid pair of cuts, $x_1 \leq 1$ and $x_1 \geq 2$ to generate the following pair of ILP subprograms:

$\begin{array}{ll} \min & f = -2x_1 - 5x_2 \\ \text{s.t.} & 2x_1 + 3x_2 \leq 12 \\ (ILP1) & x_1 + x_2 \leq 5 \\ & x_2 \leq 3 \\ & x_1 \leq 1 \\ & \text{integer } x_1, x_2 \geq 0, \end{array}$	$\begin{array}{ll} \min & f = -2x_1 - 5x_2 \\ \text{s.t.} & 2x_1 + 3x_2 \leq 12 \\ (ILP2) & x_1 + x_2 \leq 5 \\ & x_2 \leq 3 \\ & x_1 \geq 2 \\ & \text{integer } x_1, x_2 \geq 0, \end{array}$
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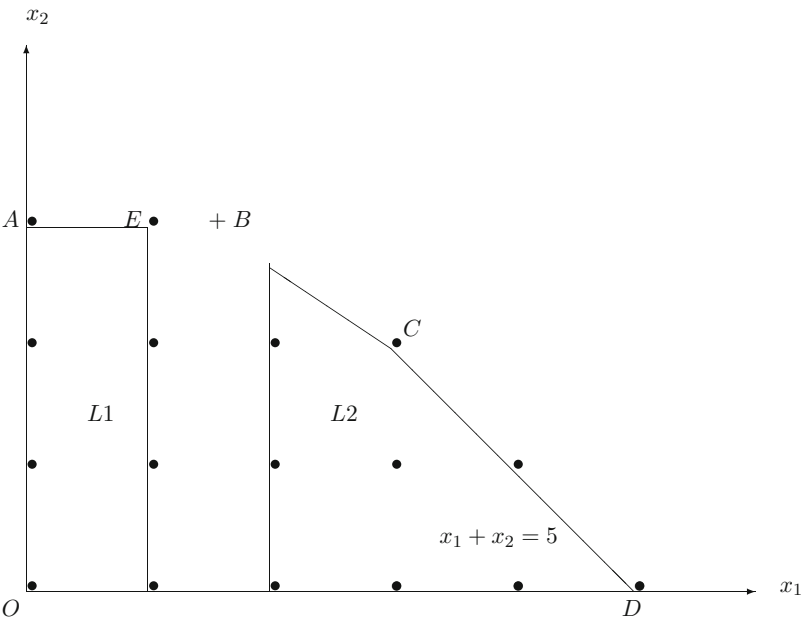


Fig. 11.2 Feasible sets for the two ILP subprograms

Figure 11.2 depicts the corresponding two branches with $L1$ and $L2$ as their feasible regions, respectively. It is noted that the solution B to the LP relaxation is excluded from $L1 \cap L2$.

Therefore, solving the ILP problem (11.1) is boiled down to solving subprograms (ILP1) and (ILP2), separately. We arbitrarily select, and solve the branch (ILP2), and leave (ILP1) pending, and so on. The solution process is illustrated via the simplex framework, as is follows.

Example 11.3.1 Solve ILP problem (11.1) by the branch-and-bound method.

Answer Set initial upper bound on the optimal value by $f^+ = +\infty$. Convert the problem to the standard form by introducing slack variable x_3, x_4, x_5 . The initial simplex tableau of the associated LP relaxation is then

x_1	x_2	x_3	x_4	x_5	RHS
2	3	1			12
1	1		1		5
	1			1	3
-2	-5				

(1) Call simplex Algorithm 3.1.1, yielding

x_1	x_2	x_3	x_4	x_5	RHS
1		1/2		-3/2	3/2
		-1/2	1	1/2	1/2
	1			1	3
		1		2	18

(2) Selecting integer cut $x_1 \geq 2$, add the row, corresponding to $-x_1 + x_6 = -2$, to the preceding as the second bottom row.

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		1/2		-3/2		3/2
		-1/2	1	1/2		1/2
	1			1		3
-1					1	-2
		1		2		18

Convert the preceding to the simplex tableau, and call dual simplex Algorithm 5.4.1, yielding

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1					-1	2
		-1/3	1		1/3	1/3
	1	1/3			2/3	8/3
		-1/3		1	-2/3	1/3
		5/3			4/3	52/3

(3) Selecting integer cut $x_2 \leq 2$, add the row corresponding to $x_2 + x_7 = 2$ to the preceding as the second bottom row.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1					-1		2
		-1/3	1		1/3		1/3
	1	1/3			2/3		8/3
		-1/3		1	-2/3		1/3
	1					1	2
		5/3			4/3		52/3

Convert the preceding to a simplex tableau, and call dual simplex Algorithm 5.4.1, yielding

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1		1/2				-3/2	3
		-1/2	1			1/2	
	1					1	2
				1		-1	1
		1/2			1	-3/2	1
		1				2	16

which gives a feasible (integer) solution to the ILP problem (fathomed branch). Updated the upper bound by $f^+ = -16$.

- (4) Selecting integer cut $x_2 \geq 3$, add the row corresponding to $-x_2 + x_7 = -3$ to the end tableau of step (2) as the second bottom row.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1					-1		2
		-1/3	1		1/3		1/3
	1	1/3			2/3		8/3
		-1/3		1	-2/3		1/3
	-1					1	-3
		5/3			4/3		52/3

Convert the preceding to a simplex tableau, and call dual simplex Algorithm 5.4.1, yielding

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1					-1		2
		-1/3	1		1/3		1/3
	1	1/3			2/3		8/3
		-1/3		1	-2/3		1/3
		1/3			2/3	1	-1/3
		5/3			4/3		52/3

which indicates that there is no feasible solution to the subprogram (fathomed branch).

- (5) Selecting integer cut $x_1 \leq 1$, add the row corresponding to $x_1 + x_6 = 1$ to the end tableau of step (1) as the second bottom row.

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		1/2		-3/2		3/2
		-1/2	1	1/2		1/2
	1			1		3
1					1	1
		1		2		18

Convert the preceding to a simplex tableau, and call dual simplex Algorithm 5.4.1, yielding

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1					1	1
			1	-1	-1	1
	1			1		3
		1		-3	-2	1
				5	2	17

which gives a feasible (integer) solution to the original ILP problem. A comparison with step (3) indicates that it is optimal, i.e.,

$$x^* = (1, 3, 1, 1, 0, 0)^T, \quad f^* = -17.$$

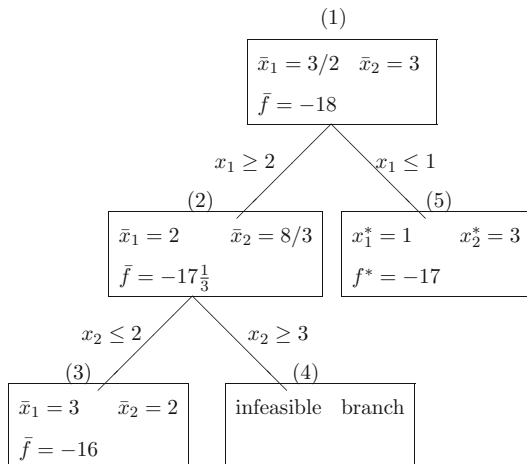
Figure 11.3 gives the corresponding enumerate tree, showing branches (1) through (5), handled in the solution process.

It is clear that enumerate tree is not unique. The branch-and-bound method would still generate an enumeration tree with an unacceptable large number of branches. It could perform unsatisfactorily or even fail with some relatively small ILP problems.

The issue involves a double decision: how to select a cutting variable, and how to select one from the valid pair of integer cuts. Different decisions lead to different computational costs. The classical branch-and-bound method makes a decision arbitrarily, as was shown in the instance. Nevertheless, the effect of a decision cannot be ignored, and it is wise to do according to some appropriate criteria rather than blindly.

Let J be the index set of noninteger components of the current solution $\bar{x}$. To select a cutting variable, the following rule depends on the contribution to the optimal value.

Fig. 11.3 An enumerate tree to ILP problem (11.1)



Rule 11.3.1 Determine cutting variable $x_{j'}$ such that

$$j' \in \arg \max_{j \in J} |c_j \bar{x}_j|. \quad (11.7)$$

Another rule depends on objective coefficients only.

Rule 11.3.2 Determine cutting variable $x_{j'}$ such that

$$j' \in \arg \max_{j \in J} |c_j|. \quad (11.8)$$

An alternative consideration is to cut out deepest into the feasible region of the LP relaxation, as is follows.

For any $j \in J$,

$$\alpha_j = \bar{x}_j - \lfloor \bar{x}_j \rfloor$$

is the distance from $\bar{x}$ to the boundary of $\{x \mid x_j \leq \lfloor \bar{x}_j \rfloor\}$, and

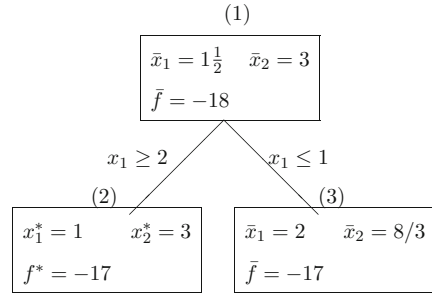
$$\beta_j = \lceil \bar{x}_j \rceil - \bar{x}_j$$

is the distance from $\bar{x}$ to the boundary of $x_j \geq \lceil \bar{x}_j \rceil$ (Definition 2.1.2). Determine j' such that

$$\delta_{j'} = \max\{\max_{j \in J} \alpha_j, \max_{j \in J} \beta_j\},$$

and take $x_{j'}$ as branching variable. Then, if $\delta_{j'} = \alpha_{j'}$, choose cut $x_j \leq \lfloor \bar{x}_j \rfloor$; otherwise, choose cut $x_j \geq \lceil \bar{x}_j \rceil$.

Fig. 11.4 Another enumerate tree to ILP problem (11.1)



Based on the most-obtuse-angle heuristic (Proposition 2.7.1), we recommend the following as the tentative criteria.

Rule 11.3.3 *From the valid pair of variable cuts, choose one as active cut, whose normal direction (pointing to the interior of the half-space) forms an obtuse angle with the negative objective gradient.*

Specifically, assume that index j' is determined. Then, if $c_{j'} < 0$, select $x_{j'} \leq \lfloor \bar{x}_{j'} \rfloor$; and if $c_{j'} > 0$, select $x_{j'} \geq \lceil \bar{x}_{j'} \rceil$. In case of a tie, turn to the deepest cutting or the maximum $|\bar{x}_{j'}|$.

According to the preceding rule, branch (ILP1), rather than (ILP2), should be chosen to be active.

Solution $\bar{x}_1 = 1$, $\bar{x}_2 = 3$, $\bar{f} = -17$ to branch (ILP1) is integer, and hence it is fathomed with setting upper bound $f^+ = -17$. Besides, since the solution to branch (ILP2) is $\bar{x}_1 = 2$, $\bar{x}_2 = 2\frac{2}{3}$, $\bar{f} = -17\frac{1}{3}$, and it holds that $\lceil \bar{f} \rceil \geq f^+$, (ILP2) is also fathomed. Therefore, it can be asserted that $\bar{x}_1 = 1$, $\bar{x}_2 = 3$, $\bar{f} = -17$ is optimal to the ILP problem.

Consequently, Example (11.1) is solved by 3 calls of LP solver, while solved by 5 calls previously, as is a considerable difference.

Figure 11.4 gives the corresponding enumerate tree with less branches.

11.4 Controlled-Cutting Method

From now on, it is assumed that coefficients of the objective function are integers.

This method utilizes the objective cutting in addition to the normal valid cutting, compared with the cutting-plane method that uses the latter only.

Let $\bar{f}$ be the objective value of the obtained optimal solution to the LP relaxation of the ILP problem. If $\bar{f}$ is noninteger, we take the integer lower bound f^- on the optimal value by (11.3) and add the associated objective cut (11.2), i.e.,

$$c^T x \geq f^-$$

to the constraints. Then, solve the resulting LP relaxation. It is obvious that the objective value of its optimal solution must be equal to f^- . If the solution is integer, it is also optimal for the original ILP problem.

If the solution is noninteger, add a normal valid cut, and solve the resulting LP relaxation, and so on, until achieving an optimal integer solution attaining an integer lower bound on the optimal value of the ILP problem.

The objective cutting and valid cutting can be used alternately or flexibly.

Example 11.4.1 Solve the following ILP problem by the controlled-cutting method:

$$\begin{aligned} \min \quad & f = -5x_1 + x_2 \\ \text{s.t.} \quad & -7x_1 + x_2 + x_3 = 4 \\ & 2x_1 + 5x_2 + x_4 = 7 \\ & \text{integer } x_j \geq 0, \quad j = 1, \dots, 4. \end{aligned}$$

Answer (1) Call simplex Algorithm 3.1.1, yielding

x_1	x_2	x_3	x_4	RHS
	37/2	1	7/2	57/2
1	5/2		1/2	7/2
	27/2		5/2	35/2

(2) $\bar{f} = -35/2$, $f^- = -17$. Add the row corresponding to objective cut $5x_1 - x_2 + x_5 = 17$ to the preceding as its second bottom row:

x_1	x_2	x_3	x_4	x_5	RHS
	37/2	1	7/2		57/2
1	5/2		1/2		7/2
5	-1			1	17
	27/2		5/2		35/2

After converting the preceding to a simplex tableau, call dual simplex Algorithm 5.4.1, yielding

x_1	x_2	x_3	x_4	x_5	RHS
		1	2/27	37/27	751/27
1			1/27	5/27	92/27
	1		5/27	-2/27	1/27
				1	17

- (3) Add the row associated with the fractional cut $-2/27x_4 - 10/27x_5 + x_6 = -22/27$, formed from the first row of the preceding, as the second bottom row, leading to

x_1	x_2	x_3	x_4	x_5	x_6	RHS
		1	$2/27$	$37/27$		$751/27$
1			$1/27$	$5/27$		$92/27$
	1		$5/27$	$-2/27$		$1/27$
			$-2/27$	$-10/27$	1	$-22/27$
				1		17

Call dual simplex Algorithm 5.4.1, yielding

x_1	x_2	x_3	x_4	x_5	x_6	RHS
		1	1		$7/2$	25
	1				$1/2$	3
		-1		1	$-5/2$	2
		5	1		-1	1
	1				$5/2$	15

which gives an integer solution, which is also an optimal solution to the ILP problem, i.e.,

$$x^* = (3, 0, 25, 1)^T, \quad f^* = -15.$$

11.5 Controlled-Branch Method

As was mentioned, the branch-and-bound method is far from satisfactory, although current commercial/professional ILP solvers are all rooted in it. We introduce the so-called *controlled-branch* method in this section.

Its distinguishing feature is the use of objective cutting in addition to a valid pair of integer cutting.

As usual, the method solves the LP relaxation of the ILP problem at first. If the optimal solution is integer, it is optimal for the ILP problem. Assume that it is noninteger with objective value $\bar{f}$.

The method sets the integer upper bound on the optimal value of the ILP problem by

$$f^+ = \lfloor \bar{f} \rfloor.$$

At the same time, it takes the integer lower bound on the optimal value by

$$f^- = \lceil \bar{f} \rceil.$$

It adds the objective cut

$$c^T x \geq f^-$$

to the constraints, and solve the resulting LP relaxation. It is clear that the objective value of its optimal solution is equal to f^- (assuming the range of its feasible values is no more than one unit). If the solution itself is integer, it is also optimal for the original ILP problem. Otherwise, we determine a cutting variable and add the associated valid pair of integer cuts to get two branches.

Then, there are the two strategies to proceed, as follows.

11.5.1 Depth-Oriented Strategy

Facing a pair of branches, this strategy solves one of them and leaves the other pending. Then, it handles the active former just as before by further objective cutting; and so on. A series of branches are solved until reaching a so-called *fathomed* branch, as is in one of the following cases:

- (i) Current branch is infeasible.
- (ii) The solution to the current branch is an integer.
The solution is feasible to the ILP problem. If the corresponding objective value $\bar{f}$ is strictly less than f^+ , update $f^+ = \bar{f}$ as well as the associated objective cut.
- (iii) The solution is noninteger and it holds that $\lceil \bar{f} \rceil \geq f^+$.

If the latter condition does not hold, update $f^- = \lceil \bar{f} \rceil$,¹ and continue.

The ILP problem is solved when all branches are fathomed.

Using the simplex framework, the following example demonstrates the realization of the controlled-branch method with the depth-oriented strategy.

Example 11.5.1 Solve the following ILP problem by the controlled-branch method with the depth-oriented strategy:

$$\begin{aligned} \min f &= 6x_1 + x_2 + 3x_3 \\ \text{s.t. } 2x_1 + 5x_3 &\leq 9 \\ 4x_1 + 3x_2 + x_3 &\leq 5 \\ \text{integer } x_j &\geq 0, \quad j = 1, 2, 3. \end{aligned}$$

¹ f^- is only an integer lower bound on the optimal value of the current ILP subproblem, not the original ILP problem.

Answer Introduce slack variable x_4, x_5 . The optimal simplex tableau of its LP relaxation is

(1)

x_1	x_2	x_3	x_4	x_5	RHS
$2/5$		1	$-1/5$		$9/5$
$6/5$	1		$1/15$	$-1/3$	$16/15$
$18/5$			$8/15$	$1/3$	$-97/15$

(2) Add objective cut $-6x_1 - x_2 - 3x_3 + x_6 = -7$ to the preceding as its second bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
$2/5$		1	$-1/5$			$9/5$
$6/5$	1		$1/15$	$-1/3$		$16/15$
-6	-1	-3			1	-7
$18/5$			$8/15$	$1/3$		$-97/15$

the optimal simplex tableau of which is

x_1	x_2	x_3	x_4	x_5	x_6	RHS
		1	$-7/27$	$-1/27$	$1/9$	$47/27$
	1		$-1/9$	$-4/9$	$1/3$	$8/9$
1			$4/27$	$5/54$	$-5/18$	$4/27$
					1	-7

(3) Arbitrarily selecting cutting variable x_3 , add the row corresponding to integer cut $x_3 + x_7 = 1$ to the preceding as the third bottom row:]

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		1	$-7/27$	$-1/27$	$1/9$		$47/27$
	1		$-1/9$	$-4/9$	$1/3$		$8/9$
		1			0	1	1
1			$4/27$	$5/54$	$-5/18$		$4/27$
					1		-7

the optimal simplex tableau to which is The integer solution is obtained, and the

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		1				1	1
	-3		-2	1		-9	4
	-1		-3		1	12	8
1			-1/2			-5/2	2
	1		3			12	-15

branch is fathomed. Set upper bound $f^+ = 15$

- (4) Examine the pending branch. Add the row corresponding to integer cut $x_3 \geq 2$ to the end tableau of (2) as its third bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		1	-7/27	-1/27	1/9		47/27
	1		-1/9	-4/9	1/3		8/9
		-1			0	1	-2
1			4/27	5/54	-5/18		4/27
					1		-7

the optimal tableau to which is which gives integer solution with objective value

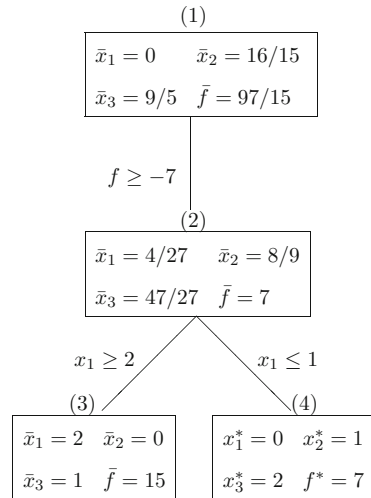
x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		1				-1	2
	1			-3/7	2/7	-3/7	1
			1	1/7	-3/7	-27/7	1
1				1/14	-3/14	4/7	
					1		-7

equaling to f^- , and hence is optimal solution to the ILP problem, i.e.,

$$x^* = (0, 1, 2)^T, \qquad f^* = 7.$$

The corresponding enumeration tree is given in Fig. 11.5.

Fig. 11.5 Enumerate tree to ILP problem (11.5.1)



11.5.2 Breadth-Oriented Strategy

Each time when facing a pair of branches (LP subrelaxations), this strategy solves both of them and takes a branch as *active* that has the minimum objective value, say $\bar{f}$, among the solved branches (excluding fathomed ones). It is obvious that the value

$$f^- = \lceil \bar{f} \rceil$$

is an integer lower bound on the optimal value of the ILP problem. The strategy adds the objective cut

$$c^T x \geq f^-$$

to the active branch and solves it. If the obtained optimal solution is integer, it is also optimal for the ILP problem. Otherwise, a valid pair of integer cuts are added to the active branch to yield a lower level pair of branches; and so on, until the optimal solution of a branch is integer as well as attains the current integer lower bound f^- .

A branch is *fathomed* in one of the following cases:

- (i) Current branch is infeasible.
- (ii) The solution to the current branch is an integer.

If it is a valid solution attaining f^- , the solution is optimal to the ILP problem.

Otherwise, if $\bar{f}$ is strictly less than the current upper bound on f^* , update $f^+ = \bar{f}$ as well as the corresponding objective cut.

- (iii) The solution is noninteger and it holds that $\lceil \bar{f} \rceil$ is greater than and equal to f^+ .
If the preceding condition is not fulfilled, update $f^- = \lceil \bar{f} \rceil$, and continue.

A tight integer lower bound on the optimal value of the ILP problem is certainly desirable. Since the optimal value of Example 11.5.1 is just equal to $f^- = \lceil \bar{f} \rceil$, it was solved in few iterations. Unfortunately, this is not the case in general.

The following example would allow us to learn something about how the strategy performs with an ILP problem that has its optimal value far away from expected.

Example 11.5.2 Solve the following ILP problem by the controlled-branch method with breadth-oriented strategy:

$$\begin{aligned} \min f &= -7x_1 - 5x_2 + 9x_3 \\ \text{s.t. } 3x_1 - x_2 + 4x_3 &\leq 6 \\ x_1 + 7x_2 + 5x_3 &\leq 35 \\ \text{integer } x_j &\geq 0, \quad j = 1, 2, 3. \end{aligned}$$

Answer Introduce slack variables x_4, x_5 to the inequalities. The initial tableau is

x_1	x_2	x_3	x_4	x_5	RHS
3	-1	4	1		6
1	7	5		1	35
-7	-5	9			

The optimal simplex tableau is
(1)

x_1	x_2	x_3	x_4	x_5	RHS
1		3/2	7/22	1/22	7/2
	1	1/2	-1/22	3/22	9/2
		22	2	1	47

- (2) Since $\bar{f} = -47$, we do not add $f \geq f^- = \bar{f}$, but add $x_1 \leq 3$.

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		3/2	7/22	1/22		7/2
	1	1/2	-1/22	3/22		9/2
1					1	3
		22	2	1		47

the optimal tableau to which is

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1					1	3
	1	5/7		1/7	-1/7	32/7
		33/7	1	1/7	-22/7	11/7
		88/7		5/7	44/7	307/7

(3) Adding $x_1 \geq 4$ to (1) gives

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		3/2	7/22	1/22		7/2
	1	1/2	-1/22	3/22		9/2
-1					1	-4
		22	2	1		47

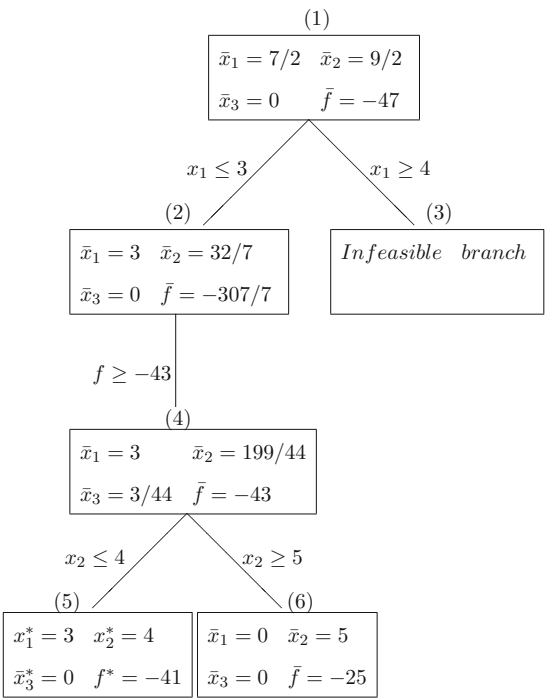
dealing with which leads to the following infeasible tableau (fathomed):

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		3/2	7/22	1/22		7/2
	1	1/2	-1/22	3/22		9/2
		3/2	7/22	1/22	1	-1/2
		22	2	1		47

(4) Adding $f \geq f^- = -43$ to (2) gives

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1					1		3
	1	5/7		1/7	-1/7		32/7
		33/7	1	1/7	-22/7		11/7
7	5	-9				1	43
		88/7		5/7	44/7		307/7

Fig. 11.6 Enumerate tree to ILP problem (11.5.2)



whose optimal simplex tableau is

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1				1			7	0
	1						-1	5
		-5		-1	1		-7	3
		-11	1	-3			-22	11
		-44		-7		1	-44	18
		44		7			44	25

The corresponding enumeration tree is given in Fig. 11.6.

11.6 ILP: with Reduced Simplex Framework²

Firstly, each of the ILP methods presented previously solves the LP relaxation of the ILP problem by the simplex method. If the solution is noninteger, it solves a series of LP relaxations (branches) by the dual simplex method. Since each LP

² The reader is referred to Sect. 18.1, Part II.

relaxation yields from its predecessor by adding a cut, the number of constraints increases in the solution process, as is unfavorable for solving large-scale ILP problems.

This defect can be eliminated if the branch-and-bound method is actualized with the generalized simplex framework, which adds an integer cut by simply changing the associated bound. But the latter is not very relevant to the controlled-branch method. In this aspect, the generalized *reduced* simplex framework turns out to be desirable (see Sect. 18.1). Along this line, we will actualize the controlled-branch method with the breadth-oriented strategy.

Consider the variable-bounded ILP problem in the reduced form below:

$$\begin{aligned} \min \quad & f, \\ \text{s.t.} \quad & A \begin{pmatrix} x \\ f \end{pmatrix} = b, \quad l \leq \text{integer } x \leq u. \end{aligned} \quad (11.9)$$

Let $\bar{x}$ be the current optimal solution to the LP relaxation of the ILP problem, corresponding to the objective value $\bar{f} = f^-$, where f^- is an integer lower bound on the optimal value of the ILP problem.

We put the problem into the following form:

$$\begin{array}{|c|c|c|c|} \hline x_B & x_N & f & \bar{f} = f^- \\ \hline I & \bar{N} & \bar{a}_{n+1} & \bar{x}_B \\ \hline \end{array} \quad (11.10)$$

which is termed reduced simplex tableau.

Assume that solution $\bar{x}$ is feasible. If the *bound violations* defined by

$$\rho_i = \begin{cases} l_{j_i} - \bar{x}_{j_i}, & \text{if } \bar{x}_{j_i} < l_{j_i}, \\ u_{j_i} - \bar{x}_{j_i}, & \text{if } \bar{x}_{j_i} > u_{j_i}, \\ 0, & \text{if } l_{j_i} \leq \bar{x}_{j_i} \leq u_{j_i}, \end{cases} \quad i = 1, \dots, m+1, \quad (11.11)$$

are all zero, then $\bar{x}$ attains $\bar{f} = f^-$ for the current branch. Since a branch is generated from the solved predecessor by adding a cut, only a single bound is violated at first.

To handle a branch, we put forward the following subalgorithm in tableau form.

Subalgorithm 11.6.1 (ILP Subalgorithm: Tableau Form) Initial: Integer lower bound f^- on the optimal value and $\bar{f} = f^-$. A reduced tableau in form (11.10), with basic solution $(\bar{x}, \bar{f})$. This procedure seeks a solution, attaining f^- , to the branch.

1. Select row index

$$p \in \arg \max \{ |\rho_i| \mid i = 1, \dots, m+1 \},$$

where ρ_i is defined by (11.11).

2. Return if $\rho_p = 0$.
3. Go to step 8 if

$$J = \{j \in \Gamma \mid \text{sign}(\rho_p)\bar{a}_{pj} < 0\} \cup \{j \in \Pi \mid \text{sign}(\rho_p)\bar{a}_{pj} > 0\} = \emptyset,$$

where Γ and Π are defined by (18.9)

4. Determine column index $q \in \arg \max_{j \in J} |\bar{a}_{pj}|$.
5. Convert $\bar{a}_{pq}$ to 1 and eliminate the other nonzeros in the column by elementary transformations (without touching the $\bar{x}_B$ column except for its p th component replaced by the value of $\bar{x}_q$).
6. Update $\bar{x}_B = \bar{x}_B - \rho_p \bar{a}_{jp}$, $\bar{x}_{jp} = \bar{x}_{jp} + \rho_p$.
7. Go to step 1.
8. Compute $\alpha = -\rho_p / \bar{a}_{p, n+1}$.
9. Return if $\alpha < 0$.
10. If $\alpha = 0$, set $\alpha = 1$; otherwise, set $\alpha = \lceil \alpha \rceil$.
11. Add $-\alpha$ times of f column to $\bar{x}_B$ column, and set $\bar{f} = \bar{f} + \alpha$.
12. Return.

The preceding Subalgorithm has the following three exits:

- Step 2: The end $\bar{x}$ is a valid solution. If it is an integer, the solution is optimal to the ILP problem. If it is noninteger, execute the Subalgorithm on a deeper branch, yielded from adding an integer cut.
- Step 9: The branch is infeasible, and hence fathomed.
- Step 12: There is no valid solution to the branch. The end $\bar{x}$ is optimal to the branch with objective value $\bar{f} > f^-$.

Assume that an integer valid solution is not attained in the process eventually. If all the branches are infeasible, so is the original ILP problem. In the other case, determine the solution associated with the minimum objective value, say $\bar{f}$, among end objective values of the branches, excluding fathomed ones. If it is an integer, the solution is optimal to the original ILP problem; otherwise, update the objective cut with $f^- = \lceil \bar{f} \rceil$ to the branch, and execute the Subalgorithm, and so on, until an optimal solution to the ILP problem is found, or infeasibility is detected.

The following is an illustrative instance:

Example 11.6.1 Solve the following ILP problem:

$$\begin{aligned} \min \quad & f = -5x_1 + 8x_2 - x_3, \\ \text{s.t.} \quad & 4x_1 - 5x_2 + x_3 + x_4 = 5, \\ & -3x_1 + 7x_2 - 2x_3 + x_5 = 2, \\ & 0 \leq \text{integer } x_j \leq 6, \quad j = 1, \dots, 5. \end{aligned} \tag{11.12}$$

Answer

(1) Initial tableau:

	x_1	x_2	x_3	x_4	x_5	f	$\bar{x}_B$
	4	-5	1	1			5
	-3	7	-2		1		2
	-5*	8	-1			-1	0
u	6	6	6	6	6	$+\infty$	
$\bar{x}$	0	0	0	5	2	0	
l	0	0	0	0	0	$-\infty$	

Iteration 1:

Convert the preceding to reduced simplex tableau by pivoting on the entry in row 3 and column 1, and call the generalized reduced simplex Algorithm 18.1.2 (Part II), yielding an optimal reduced simplex tableau to the LP branch below:

	x_1	x_2	x_3	x_4	x_5	f	$\bar{f} = -25/4$
		7/5	1/5	1		-4/5	0
		11/5	-7/5		1	3/5	23/4
	1	-8/5	1/5			1/5	5/4
u	6	6	6	6	6	$+\infty$	
$\bar{x}$	5/4	0	0	0	23/4	-25/4	
l	0	0	0	0	0	$-\infty$	

(2) Change the lower bound on the objective variable f to the suspected-optimal value -6 in the tableau. Increase the objective value to -6 by giving increment $1/4$, and accordingly update the solution by $\bar{x}_B = (023/45/4)^T - (1/4)(-4/53/51/5)^T = (1/5, 28/5, 6/5)^T$.

	x_1	x_2	x_3	x_4	x_5	f	$\bar{f} = -6$
		7/5	1/5	1		-4/5	1/5
		11/5	-7/5		1	3/5	28/5
	1	-8/5*	1/5			1/5	6/5
u	6	6	6	6	6	$+\infty$	
$\bar{x}$	6/5	0	0	1/5	28/5	-6	
l	0	0	0	0	0	$[-6]$	

Call Subalgorithm 11.6.1. It returns from step 2 with a feasible tableau with the objective value $\bar{f} = -6$.

- (3) As $6/5c_1 > 1/5c_4 = 28/5c_5 = 0$, select x_1 as the cutting variable ($1 < \bar{x}_1 = 6/5 < 2$). Add integer cut $x_1 \geq 2$ by changing the lower bound on x_1 to 2 (while leaving the branch with $x_1 \leq 1$ pending).

Call Subalgorithm 11.6.1.

Iteration 2:

1. $p = 3, \rho_3 = 2 - 6/5 = 4/5$.
3. $J = \{2\}$.
4. $q = 2$.
5. Multiply row 3 by $-5/8$, and add $-7/5, -11/5$ times of row 3 to rows 1 and 2, respectively (without touching $\bar{x}_B$ columns, but its third component is replaced by value, 0, of $\bar{x}_2$).
6. $\bar{x}_B = (1/5, 28/5, 0)^T - (4/5)(7/8, 11/8, -5/8)^T = (-1/2, 9/2, 1/2)^T$.

	x_1	x_2	x_3	x_4	x_5	f	$\bar{f} = -6$
	7/8		3/8	1		-5/8	-1/2
	11/8		-6/8		1	7/8	9/2
	-5/8	1	-1/8			-1/8	1/2
u	6	6	6	6	6	$+\infty$	
$\bar{x}$	2	1/2	0	-1/2	6/2	-6	
l	[2]	0	0	0	0	[-6]	

Iteration 3: 1. $p = 1, \rho_1 = 1/2, j_p = 4$.

3. $J = \emptyset$.

8. $\alpha = -(1/2)/(-5/8) = 4/5$.

10. $\alpha = 1$.

11. $\bar{x}_B = (-1/2, 9/2, 1/2)^T - (-5/8, 7/8, -1/8)^T = (1/8, 29/8, 5/8)^T$;
 $\bar{f} = -6 + 1 = -5$.

Returned from step 12. Leave the branch pending.

- (4) Handle the pending ILP subprogram by adding cut $x_1 \leq 1$ to the end tableau of (2). Change the upper bound on x_1 to 1.

Call Subalgorithm 11.6.1.

Iteration 4:

1. $p = 3, \rho_3 = 1 - 6/5 = -1/5$.
3. $J = \{3\}$.
4. $q = 3$.

	x_1	x_2	x_3	x_4	x_5	f	$\bar{f} = -6$
		7/5	1/5	1		-4/5	1/5
		11/5	-7/5		1	3/5	28/5
	1	-8/5	1/5*			1/5	6/5
u	[1]	6	6	6	6	$+\infty$	
$\bar{x}$	6/5	0	0	1/5	28/5	-6	
l	0	0	0	0	0	[-6]	

5. Multiply row 3 by 5, and add $-1/5, 7/5$ times of row 3 to rows 1 and 2, respectively (without touching the $\bar{x}_B$ columns but is the third component replaced by the value, 0, of $\bar{x}_3$).
6. $\bar{x}_B = (1/528/50)^T + (1/5)(-1, 7, 5)^T = (0, 7, 1)^T$.

	x_1	x_2	x_3	x_4	x_5	f	$\bar{f} = -6$
	-1	3		1		-1	0
	7	-9			1	2	7
	5	-8	1			1	1
u	[1]	6	6	6	6	$+\infty$	
$\bar{x}$	1	0	1	0	7	-6	
l	0	0	0	0	0	[-6]	

Iteration 5:

1. $p = 2, \rho_2 = 6 - 7 = -1, j_p = 5$.
3. $J = \emptyset$.
8. $\alpha = 1/2$.
10. $\alpha = 1$.
11. $\bar{x}_B = (0, 7, 1)^T - (-1, 2, 1)^T = (1, 5, 0)^T, \bar{f} = -6 + 1 = -5$.
- Returned from step 12. The ILP subprogram is stalled.

	x_1	x_2	x_3	x_4	x_5	f	$\bar{f} = -5$
	-1	3		1		-1	1
	7	-9			1	2	5
	5	-8	1			1	0
u	[1]	6	6	6	6	$+\infty$	
$\bar{x}$	1	0	0	1	5	-5	
l	0	0	0	0	0	[-5]	

It is seen that (3) and (4) give the same lower objective bound -5 . We finally grasp the end solution of (4) since it is a feasible integer, and hence optimal to the original ILP problem, i.e.,

$$x^* = (1, 0, 0, 1, 5)^T, f^* = -5.$$

It is imageable that more iterations would be required if one uses some “bad” integer lower bound on the optimal value. See that following instance with binary (0 or 1) variables.

Example 11.6.2 Solve the following 0-1 ILP problem:

$$\begin{aligned} \min \quad & f = 4x_1 + 2x_2 + 3x_3 + 3x_4 + 2x_5, \\ \text{s.t.} \quad & 3x_1 - 4x_2 - 7x_3 - 3x_4 \leq -2, \\ & x_1 + 2x_2 - x_3 - x_4 - x_5 \leq 1, \\ & -3x_2 + 11x_3 - 3x_4 - 6x_5 \leq -1, \\ & x_j \in \{0, 1\}, \quad j = 1, \dots, 5. \end{aligned}$$

Answer Replacing $x_j \in \{0, 1\}$ by $0 \leq \text{integer } x_j \leq 1$, the 0-1 problem is converted to a normal ILP problem, to which the controlled-branch method applies.

Introducing slack variables x_j , $j = 6, 7, 8$, solve its LP relaxation.
Initial tableau: $\bar{x} = (0, 1, 0, 0, 1, 2, 0, 8)^T$, $\bar{x}_9 = 4$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 4$
	3	-4	-7	-3		1				2
	1	2	-1	-1	-1		1			0
		-3	11	-3	-6			1		8
	4	2*	3	3	2				-1	
u	1	1	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	0	1	0	0	1	2	0	8	4	
l	0	0	0	0	0	0	0	0	$-\infty$	

Converting the preceding tableau to reduced one by multiplying row 4 by 1/2, and adding 4, -2, 3 times of row 4 to rows 1,2,3 gives the following tableau (without touching the last column—only its fourth component changed accordingly):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 4$
	11		-1	3	4*	1			-2	2
	-3		-4	-4	-3		1		1	0
	6		31/2	3/2	-3			1	-3/2	8
	2	1	3/2	3/2	1				-1/2	1
u	1	1	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	0	1	0	0	1	2	0	8	4	
l	0	0	0	0	0	0	0	0	$-\infty$	

(1) Call generalized reduced simplex Algorithm 18.1.2.

Iteration 1:

1. $\alpha = 1, p = 1$.
2. $\bar{x}_B = (2, 0, 8, 1)^T + 1 \times (-2, 1, -3/2, -1/2)^T = (0, 1, 13/2, 1/2)^T, \bar{x}_9 = 4 - 1 = 3$.
4. $q = 5$
5. Multiplying row 1 by 1/4, and adding 3, 3, -1 times of row 1 to rows 2,3,4, respectively

(without touching the $\bar{x}_B$ column (the first entry is replaced by the value 1 of $\bar{x}_5$)).

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 3$
	11/4		$-1/4^*$	3/4	1	1/4			$-1/2$	1
	21/4		$-19/4$	$-7/4$		3/4	1		$-1/2$	1
	57/4		59/4	15/4		3/4		1	-3	13/2
	$-3/4$	1	7/4	3/4		$-1/4$			0	1/2
u	1	1	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	0	1/2	0	0	1	0	1	13/2	3	
l	0	0	0	0	0	0	0	0	$-\infty$	

Iteration 2:

1. $\alpha = 2, p = 1$.
2. $\bar{x}_B = (1, 1, 13/2, 1/2)^T + 2 * (-1/2, -1/2, -3, 0) = (0, 0, 1/2, 1/2)^T, \bar{x}_9 = 3 - 2 = 1$.
4. $q = 3$.
5. Multiplying row 1 by -4 , and adding $19/4, -59/4, -7/4$ times of row 1 to rows 2,3,4, respectively

(without touching the $\bar{x}_B$ column but its first component replaced by the same value, 0, of $\bar{x}_3$).

The optimal value of the LP relaxation is strictly greater than 0. Therefore, the current solution attains the integer lower bound, $\bar{f} = 1$, on the optimal value of the ILP problem. So, add objective cut $x_9 \geq f^- = 1$.

(2) Add cut $x_2 \geq 1$ and leave $x_2 \leq 0$ pending.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 1$
	-11		1	-3	-4	-1			2	0
	-47			-16	-19	-4	1		9	0
	353/2			48	59	31/2		1	$-65/2$	1/2
	37/2	1		6	7	3/2			$-7/2$	1/2
u	1	1	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	0	1/2	0	0	0	0	0	1/2	1	
l	0	[1]	0	0	0	0	0	0	[1]	

Iteration 5:

1. $p = 3, \rho_3 = 0 - (-158/37) = 158/37, j_p = 8.$
3. $J = \{4, 5\}.$
4. $q = 4, \max\{|-342/37|, |6|, |-288/37|\} = 342/37.$
5. Multiply row 3 by $-37/342$, and then add $-21/37, 28/37, -12/37$ times of row 3 to rows 1, 2, 4, respectively (without touching the $\bar{x}_B$ column but its second component is replaced by value, 0, of $\bar{x}_4$).
6. $\bar{x}_B = (11/37, 47/37, 0, 1/37)^T - (158/37)(7/114, -14/171, -37/342, 2/57)^T = (2/57, 277/171, 79/171, -7/57)^T, \bar{x}_8 = -158/37 + 158/37 = 0.$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 1$
		1/114	1		-6/19	-2/57		7/114	-1/38	2/57
		568/171			-11/19	-49/171	1	-14/171	2/57	277/171
		353/342		1	16/19	-22/171		-37/342	-11/114	79/171
	1	-16/57			2/19	7/57		2/57	-3/19	-7/57
u	1	[0]	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	-7/57	0	2/57	79/171	0	0	277/171	0	1	
l	0	0	0	0	0	0	0	0	[1]	

Iteration 6:

1. $p = 4, \rho_4 = 0 - (-7/57) = 7/57, j_p = 1.$
3. $J = \emptyset.$
8. $\alpha = -(7/57)/(-3/19) = 7/9 > 0.$
10. $\alpha = \lceil 7/9 \rceil = 1. (11) \bar{x}_B = (2/57, 277/171, 79/171, -7/57)^T - 1 \times (-1/38, 2/57, -11/114, -3/19)^T = (7/114, 271/171, 191/342, 2/57)^T, \bar{x}_9 = 1 + 1 = 2 > f^- = 1.$

The branch is pending.

Comparing the end objective values reached in branches (2) and (3) gives a new integer lower bound, $f^- = 2$, on the optimal value of the ILP problem.

(4) We continue to handle (2) by adding objective cut $x_9 \geq f^- = 2$ first.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 2$
	-11		1	-3	-4	-1			2	-2
	-47			-16	-19	-4	1		9	-9
	353/2			48	59	31/2		1	-65/2	33
	37/2*	1		6	7	3/2			-7/2	4
u	1	1	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	0	4	-2	0	0	0	-9	33	2	
l	0	[1]	0	0	0	0	0	0	[2]	

Call Subalgorithm 11.6.1 ($f^- = 2$, $x_2 \geq 1$).

Iteration 7:

1. $p = 4$, $\rho_3 = 1 - 4 = -3$, $j_p = 2$.
3. $J = \{1, 4, 5, 6\}$.
4. $q = 1$, $\max\{|37/2|, |6|, |7|, |3/2|\} = 37/2$.
5. Multiply row 4 by $2/37$, and then add $11, 47, -353/2$ times of row 4 to rows 1, 2, 3, respectively (without touching the $\bar{x}_B$ column but its second component is replaced by value, 0, of $\bar{x}_1$).
6. $\bar{x}_B = (-2, -9, 33, 0)^T + 3(22/37, 94/37, -353/37, 2/37)^T = (-8/37, -51/37, 162/37, 6/37)^T$, $\bar{x}_2 = 4 + (-3) = 1$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 2$
		22/37	1	21/37	6/37	-4/37			-3/37	-8/37
		94/37		-28/37	-45/37*	-7/37	1		4/37	-51/37
		-353/37		-342/37	-288/37	44/37		1	33/37	162/37
	1	2/37		12/37	14/37	3/37			-7/37	6/37
u	1	1	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	6/37	1	-8/37	0	0	0	-51/37	162/37	2	
l	0	[1]	0	0	0	0	0	0	[2]	

Iteration 8:

1. $p = 2$, $\max\{|0 - (-8/37)|, |0 - (-51/37)|\} = 51/37$, $\rho_2 = 51/37$, $j_p = 7$.
3. $J = \{4, 5, 6\}$.
4. $q = 5$, $\max\{|-28/37|, |-45/37|, |-7/37|\} = 45/37$.
5. Multiply row 2 by $-37/45$, and then add $-6/37, 288/37, -14/37$ times of row 2 to rows 1, 3, 4, respectively (without touching the $\bar{x}_B$ column but its second component is replaced by value, 0, of $\bar{x}_5$).
6. $\bar{x}_B = (-8/37, 0, 162/37, 6/37)^T - (51/37)(2/15, -37/45, -32/5, 14/45)^T = (-2/5, 17/15, 66/5, -4/15)^T$, $\bar{x}_7 = 0$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 2$
		14/15	1	7/15		-2/15*	2/15		-1/15	-2/5
		-94/45		28/45	1	7/45	-37/45		-4/45	17/15
		-129/5		-22/5		12/5	-32/5	1	1/5	66/5
	1	38/45		4/45		1/45	14/45		-7/45	-4/15
u	1	1	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	-4/15	1	-2/5	0	17/15	0	0	66/5	2	
l	0	[1]	0	0	0	0	0	0	[2]	

Iteration 9:

1. $p = 1$, $\max\{|0 - (-2/5)|, |1 - (17/15)|, |0 - (-4/15)|\} = 2/5$, $\rho_1 = 2/5$, $j_p = 3$.
3. $J = \{6\}$.
4. $q = 6$.
5. Multiply row 1 by $-15/2$, and then add $-7/45$, $-12/5$, $-1/45$ times of row 1 to rows 2, 3, 4, respectively (without touching the $\bar{x}_B$ column but its first component is replaced by value, 0, of f).
6. $\bar{x}_B = (0, 17/15, 66/5, -4/15)^T - (2/5)(-15/2, 7/6, 18, 1/6)^T = (3, 2/3, 6, -1/3)^T$, $\bar{x}_3 = 0$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_B$
		-7	-15/2	-7/2		1	-1		1/2	3
		-1	7/6	7/6	1		-2/3		-1/6	2/3
		-9	18	4			-4	1	-1	6
	1	1	1/6	1/6			1/3		-1/6	-1/3
u	1	1	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	-1/3	1	0	0	2/3	3	0	6	2	
l	0	[1]	0	0	0	0	0	0	[2]	

Iteration 10:

1. $p = 4$, $\rho_4 = 1/3$, $j_p = 1$.
 3. $J = \emptyset$.
 8. $\alpha = -(1/3)/(-1/6) = 2 > 0$.
 10. $\alpha = 2$.
 11. $\bar{x}_B = (3, 2/3, 6, -1/3)^T - 2(1/2, -1/6, -1, -1/6)^T = (2, 1, 8, 0)^T$, $\bar{x}_9 = 2 + 2 = 4 > f^- = 2$. This branch is pending.
- (5) We continue to handle (3) by adding objective cut $x_9 \geq f^- = 2$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 2$
		1/114	1		-6/19	-2/57		7/114	-1/38	7/114
		568/171			-11/19	-49/171	1	-14/171	2/57	271/171
		353/342		1	16/19	-22/171		-37/342	-11/114	191/342
	1	-16/57			2/19	7/57		2/57	-3/19	2/57
u	1	[0]	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	2/57	0	7/114	191/342	0	0	271/171	0	2	
l	0	0	0	0	0	0	0	0	[2]	

Call Subalgorithm 11.6.1 ($f^- = 2$, $x_2 \leq 0$).

Iteration 11:

1. $\rho_i = 0, i = 1, 2, 3, 4$. Returning from step (2) gives a valid solution:
 $j = \arg \max\{4(2/57), 3(7/114), 3(191/342)\} = 3$. Add cut $x_4 \geq 1$ and let $x_4 \leq 0$ pend.

Call Subalgorithm 11.6.1 ($f^- = 2, x_2 \leq 0, x_4 \geq 1$).

Iteration 12:

1. $p = 3, \rho_3 = 151/342, j_p = 4$.
3. $J = \{6, 8\}$.
4. $q = 6, \max\{|-22/171|, |-37/342|\} = 22/171$.
5. Multiply row 3 by $-171/22$, and then add $2/57, 49/171, -7/57$ times of row 3 to rows 1, 2, 4, respectively (without touching the $\bar{x}_B$ column but its third component is replaced by value, 0, of $\bar{f}$).
6. $\bar{x}_B = (7/114, 271/171, 0, 2/57)^T - (151/342)(-3/11, -49/22, -171/22, 21/22)^T = (2/11, 113/44, 151/44, -17/44)^T, \bar{x}_4 = 1$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 2$
		$-3/11$	1	$-3/11$	$-6/11$			$1/11$	0	$2/11$
		$45/44$		$-49/22$	$-27/11$		1	$7/44$	$1/4$	$113/44$
		$-353/44$		$-171/22$	$-72/11$	1		$37/44$	$3/4$	$151/44$
	1	$31/44$		$21/22$	$10/11$			$-3/44^*$	$-1/4$	$-17/44$
u	1	[0]	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	$-17/44$	0	$2/11$	1	0	$151/44$	$113/44$	0	2	
l	0	0	0	[1]	0	0	0	0	[2]	

Iteration 13:

1. $p = 4, \rho_4 = 17/44, j_p = 1$.
3. $J = \{8\}$. (4) $q = 8$.
5. Multiply row 4 by $-44/3$, and then add $-1/11, -7/44, -37/44$ times of row 4 to rows 1, 2, 3, respectively (without touching the $\bar{x}_B$ column but its fourth component is replaced by value, 0, of $\bar{x}_8$).
6. $\bar{x}_B = (2/11, 113/44, 151/44, 0)^T - (17/44)(4/3, 7/3, 37/3, -44/3)^T = (-1/3, 5/3, -4/3, 17/3)^T, \bar{x}_1 = 0$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	$x_9(f)$	$\bar{x}_9 = 2$
	4/3	2/3	1	1	2/3				-1/3	-1/3
	7/3	8/3		0	-1/3		1		-1/3	5/3
	37/3	2/3		4	14/3	1			-7/3	-4/3
	-44/3	-31/3		-14	-40/3			1	11/3	17/3
u	1	[0]	1	1	1	$+\infty$	$+\infty$	$+\infty$	$+\infty$	
$\bar{x}$	0	0	-1/3	1	0	-4/3	5/3	17/3	2	
l	0	0	0	[1]	0	0	0	0	[2]	

Iteration 14:

1. $p = 3$, $\rho_3 = 4/3$, $j_p = 6$.
3. $J = \emptyset$.
8. $\alpha = -(4/3)/(-7/3) = 4/7 > 0$.
10. $\alpha = 1$.
11. $\bar{x}_B = (-1/3, 5/3, -4/3, 17/3)^T - 1 \times (-1/3, -1/3, -7/3, 11/3)^T = (0, 2, 1, 2)^T$, $\bar{x}_9 = 2 + 1 = 3 > f^- = 2$.

This branch is pending.

- (6) The end objective value 3 reached by branch (5) is less than the value 4 by branch (4). So $f^- = 3$ is the new integer lower bound on the optimal value. Continue to handle (5) by adding objective cut $f \geq f^- = 3$.

Call Subalgorithm 11.6.1.

1. $\rho_j = 0$, $j = 1, 2, 3, 4$. Since it is integer, the valid solution is optimal to the original 0-1 problem. The optimal solution and optimal value are as follow:

$$x^* = (0, 0, 0, 1, 0)^T, \quad f^* = 3.$$

The preceding instance is just for illustration of the controlled-branch method. In fact, it is simpler to solve the 0-1 problem otherwise by the following *implicit enumeration* approach:

First, $\bar{x} = (0, 0, 0, 0, 0)^T$ is ruled out to be optimal because it is infeasible.

Then, compare objective values associated with solutions having only a single 1-valued component, i.e., the coefficients, 4, 2, 3, 3, 2, of the objective function.

Corresponding to value $\min\{4, 2, 3, 3, 2\} = 2$, both solutions $(0, 1, 0, 0, 0)^T$ and $(0, 0, 0, 0, 1)^T$ are ignored since they are infeasible.

Therefore, turn to $(0, 0, 1, 0, 0)^T$ and $(0, 0, 0, 1, 0)^T$, corresponding to the second minimum 3. It is then clear that the later is the only optimal solution to the 0-1 problem.

A tight integer lower bound f^- on the optimal value of the ILP problem is conducive to the controlled-branch method; it is even desirable if the optimal value of the ILP problem is available. For some ILP problems, like the 0-1 problem, it would be possible to determine a tight integral lower bound on the optimal value beforehand.

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Part II

Advances

Chapter 12

Pivot Rule



Consider the following standard LP problem:

$$\begin{aligned} \min f &= c^T x \\ \text{s.t. } Ax &= b, \quad x \geq 0, \end{aligned} \quad (12.1)$$

where $A \in \mathcal{R}^{m \times n}$, $c \in \mathcal{R}^n$, $b \in \mathcal{R}^m$, $m < n$, and $\text{rank}(A) = m$, unless indicated otherwise.

The pivot rule¹ plays an important role in the simplex method. Starting from a vertex of the feasible region, the method moves from a vertex to vertex until reaching an optimal vertex. The *path* consists of a series of edges. In essence, it is the pivot rule that determines the number of edges, or the number of iterations required. Indeed, the pivot rule used embodies the soul of the simplex method.

Having long been used in practice, Dantzig's original rule selects a pivot column index q such that

$$q \in \arg \min_{j \in N} \bar{z}_j.$$

In principle, however, any index associated with a negative reduced cost is eligible to be chosen. This provides room for improving the simplex method.

As early as the beginning of LP, Dantzig noted that the original rule was far from ideal, and turned to a so-called *most-improvement rule*. According to (3.10) and (3.11), the objective decrement is

$$\bar{f} - \hat{f} = -\bar{z}_q (\bar{b}_p / \bar{a}_{pq}).$$

¹ The pivot rule is referred to as the column pivot rule, as after a pivot column is determined, the choice of a row pivot is very limited.

The most-improvement rule selects q, p such that the preceding is the largest possible. Although this seems to be attractive, the associated computational effort is too high to realize. It turned out that this rule performed poorly (Kuhn and Quandt 1953; Jeroslow 1973) in practice. In fact, the objective decrement is not only dependent on $\bar{z}_q$ but also on the stepsize α . If α is too small, so is the decrement. In fact, the decrement would even vanish in the presence of degeneracy.

It seems to be attractive to pursue an “ideal” pivot rule that determines the shortest path of all paths connecting a given initial vertex to an optimal one. Therefore, the pivot rule has received great attention in the past. Various pivot rules have been proposed from time to time. However, all efforts have failed, and it is not known whether there is a rule that turns the method to polynomial time. Practically, the method can stall at a vertex for too long a time before exiting it in the presence of degeneracy when the number of hyperplanes meeting at a vertex exceeds the dimension of the feasible region, let alone the cycling possibility.

In fact, infinite rather than finite rules are widely applied in practice. It is essentially a practical issue to judge the merits of the rules: their vitality or value is determined by the performance only.

In this chapter, therefore, we introduce some favorable pivot rules, ignoring their theoretical aspect.

Assume that the current simplex tableau is

$$\begin{array}{c|c|c|c}
 x_B & x_N & f & \text{RHS} \\
 \hline
 I & \bar{N} & & \bar{b} \\
 \hline
 & \bar{z}_N^T & -1 & -\bar{f}
 \end{array} \tag{12.2}$$

For simplicity, assume that the associated basis and nonbasis are as follows:

$$B = \{1, \dots, m\}, \quad N = \{m+1, \dots, n\}. \tag{12.3}$$

12.1 Partial Pricing

The conventional method to obtain the reduced costs used in pivoting is to solve the system

$$B^T y = c_B, \tag{12.4}$$

for $\bar{y}$ first and then calculate the reduced costs (pricing), i.e.,

$$\bar{z}_N = c_N - N^T \bar{y}. \tag{12.5}$$

Computing all components of $\bar{z}_N$ by the preceding formula requires $(n - m) \times m$ multiplications, termed *full pricing*. Consequently, a large proportion of running time is spent on pricing by the simplex method, especially when $n - m \gg m$.

Such a situation hastens some variants by a so-called *partial pricing* strategy, that is, only a portion of $\bar{z}_N$ is calculated in each iteration.

Pricing options in MINOS (Murtagh and Saunders 1998) include *full pricing*, *sectional pricing*, and *multiple pricing*, the last two of which belong to the partial pricing category (see Benichou et al. 1977; Chvatal 1983; Maros 2003; Nocedal and Wright 1999).

(A) Sectional pricing.

As a typical partial pricing, it partitions all column indices into p (e.g., $p = 20$) sections of approximately equal number of indices. In order to treat each index equally, the m logical indices are almost equally distributed into the p sections.

In each iteration, pricing starts with the section that follows the section from which the previous pivot column index is selected (excluding basic variables). If a reduced cost is found in the section that is lower than a dynamically changing negative threshold, the related index is selected to enter the basis; otherwise, pricing is continued similarly with the subsequent section, and so on.

If all the p sections are priced and the minimum reduced cost is still greater than the threshold, optimality is achieved numerically if it is greater than the negative optimality tolerance (normally -10^{-6}); otherwise, the index associated with the minimum reduced cost is selected to enter the basis, and the size of the minimum reduced cost is recorded.

Subsequently, the threshold is decreased (e.g., to 0.1 times of itself) whenever the size is not sufficiently greater than (e.g., lower than 1.1 times of) the threshold. An initial value of the threshold is set to a very large value (e.g., 10^{20}) by MINOS. So, full pricing is actually carried out in the first iteration, at least.

(B) Multiple pricing.

Firstly, it determines a small set, consisting of nonbasic indices, associated with negative reduced costs. Then a pivot index is determined by some rule, e.g., by choosing an index associated with the minimum cost among them. If the cost is lower than some dynamically changing negative threshold, enter the according index to the basis. In the next iteration, computed are only the reduced costs associated with the rest indices in the set, from which a pivot index is determined, and so on.

When all reduced costs associated with the set are greater than the negative optimal tolerance, a new set of nonbasic indices is determined, and the preceding procedure is repeated. Optimality is achieved if no such a set is found.

It is initially anticipated that partial pricing requires more iterations than full pricing. What is expected is only that the overall efficiency is improved due to the reduction of computational effort per iteration by partial pricing. Surprisingly enough, it turns out that the number of iterations as well as the overall running time required by partial pricing reduces in general (see, e.g., Harris 1973; Pan 1997,

2008). Extensive numerical experiments confirm the superiority of partial pricing to full pricing. This fact would suggest searching for better pivot rules.

12.2 Steepest-Edge Rule

In the second half of the eighties and the first half of nineties of the twentieth century, when interior-point methods climaxed, some scholars thought of such methods being superior to simplex methods for solving large-scale LP problems. During the same period, on the other hand, the simplex methods got active strength with computational results, reported on the *steepest-edge pivot rule* (Forrest and Goldfarb 1992). Consequently, the two types of methods inextricably involved in fierce competition (Nemhauser 1994; Bixby 2002).

Recall the assumption that the current basis B consists of the first m columns of A . The $n - m$ edge directions, emanating from the current vertex, are as follows:

$$d^j = \begin{pmatrix} -B^{-1}a_j \\ e_{j-m} \end{pmatrix}, \quad j \in N, \quad (12.6)$$

where $e_{j-m} \in \mathcal{R}^{n-m}$ is the unit vector with its $(j - m)$ th component equal to 1. For each $j \in N$, the reduced cost

$$\bar{z}_j = c_j - c_B^T B^{-1} a_j$$

is equal to the inner product of the objective gradient c and edge direction d^j , i.e.,

$$\bar{z}_j = c^T d^j, \quad j \in N.$$

If $\bar{z}_j < 0$, then d^j is a descent edge direction. Thus, Dantzig's original standard rule selects a column index

$$q \in \arg \min_{j \in N} \bar{z}_j,$$

associated with a descent edge direction, along which the corresponding decrement of the objective value attains the maximum amount for a unit increment in the entering variable.

The following pivot rule uses normalized reduced costs:

Rule 12.2.1 (Steepest-Edge Column Rule) *Select the column index*

$$q \in \arg \min \{ \bar{z}_j / \|d^j\| \mid j \in N \}.$$

It is noted that the $\bar{z}_j / \|d^j\|$, $\forall j \in N$, differs from the cosine of the angle between c and d^j by the same constant factor $1/\|c\|$. Therefore, the preceding rule actually

selects a descent edge direction that forms the maximum angle with the objective gradient, which is why the rule is prefixed by the *steepest-edge*. While it seems to be attractive, computing the norms $\|d^j\|$, $\forall j \in N$, is too expensive to realize.

Goldfarb and Reid (1977) offered a scheme, in which the $\|d^j\|^2$, $\forall j \in N$, were calculated by recurrence formulas in each iteration. Further, they gave several variants of the rule and reported remarkable numerical results, obtained in extensive experiments. This work received wide attention from the academic community.

The recurrence formulas can be derived based on the basic change, i.e.,

$$\hat{d}^p = -(1/\sigma_q)d^q, \quad (12.7)$$

$$\hat{d}^j = d^j - (\sigma_j/\sigma_q)d^q, \quad j \in N, \quad j \neq q, \quad (12.8)$$

where σ_j denotes entries in the pivot row, i.e.,

$$\sigma_j = a_j^T B^{-T} e_p, \quad j \in N, \quad (12.9)$$

which along with (12.7) and (12.8) gives the following formulas:

$$\|\hat{d}^p\|^2 = (1/\sigma_q^2)\|d^q\|^2, \quad (12.10)$$

$$\|\hat{d}^j\|^2 = \|d^j\|^2 - 2(\sigma_j/\sigma_q)a_j^T v + (\sigma_j/\sigma_q)^2\|d^q\|^2, \quad j \in N, \quad j \neq q, \quad (12.11)$$

where

$$B\bar{a}_q = a_q, \quad B^T v = \bar{a}_q. \quad (12.12)$$

In fact, only the second system of the preceding is needed to solve because $\bar{a}_q$ has already been calculated. Moreover, $\bar{a}_q$ can also be employed to directly compute

$$\|d^q\|^2 = 1 + \|\bar{a}_q\|^2.$$

It is therefore seen that updating squares of norms of edge directions involves 3 linear systems: besides the two typical systems, another system is

$$B^T h = e_p. \quad (12.13)$$

Since the simplex algorithm also involves $B\bar{a}_q = a_q$ and (12.13), it is only needed to solve an additional system; that is,

$$B^T v = \bar{a}_q.$$

Table 12.1 CPU time and ratio

Pivot rule	Dantzig	Devex	Steepest-edge	Dynamic steepest-edge	Projective steepest-edge
Hours	110.19	7.76	4.90	4.08	3.89
Time ratio	28.33	1.99	1.26	1.05	1.00

In (12.11), in addition, only those inner products $a_j^T v$ that correspond to nonzero σ_j are needed to calculate. Fortunately, most of σ_j often vanish in sparse computations.

The steepest-edge rule requires more computational effort per iteration but fewer total iterations. Numerical results with the steepest rule are listed in Table 12.1 in the next section for a comparison with other rules.

A shortcoming is that the rule has to compute initial squares of the norms of edge directions corresponding to all the nonbasic indices. According to the *restarting strategy*, this must be carried out periodically to avoid overaccumulation of the rounding error. Consequently, the rule involves a considerable amount of computational effort for large-scale problems, especially when it is not possible to take advantages of computer structure (like Forrest and Goldfarb 1992) to solve linear systems efficiently. Moreover, it is clear that the partial pricing strategy is not amenable in such a context.

12.3 Approximate Steepest-Edge Rule

In this section, the notation of the previous section will still be used to present three approximate variants of the steepest-edge rule.

(A) Devex row rule.

Enlightened on the graphic approach to solving LP problems of 2-dimension, Harris (1973) proposes an idea of searching along an approximate steepest-edge. The resulting rule, named *Devex*, is simple, involving the pivot row and column only, and performs well in computational experiments.

Let a set of $n - m$ column indices, termed “reference framework,” be given. Assume that the subvector $\hat{d}^j$, $j \in N$, consists of components, located on the reference framework; the weight t_j is endowed to the reduced cost $\bar{z}_j$, as an approximation of $\|\hat{d}^j\|$.

Rule 12.3.1 (Devex Row Rule) *Select the pivot column index*

$$p \in \arg \min \{\bar{z}_j / t_j \mid j \in N\},$$

where the weights t_j are determined as follows, so that the preceding rule may be regarded as selecting the steepest-edge under the reference framework.

Initially, the set of current nonbasic indices is taken as a reference framework. Set $t_j = 1$ for all indices j in it. As a result, the Devex rule coincides with the Dantzig's original rule in the first iteration.

The weights t_j are updated in subsequent iterations. It is clear that (12.7) and (12.8) are still valid if vectors involved in them are replaced by vectors, consisting of components located only on the reference framework. Thus, we have the following updating formulas:

$$\bar{t}_p = \max\{1, \|\hat{d}^q\|/|\sigma_q|\}, \quad (12.14)$$

$$\bar{t}_j = \max\{t_j, |\sigma_j/\sigma_q|\|\hat{d}^q\|\}, \quad j \in N, \quad j \neq q, \quad (12.15)$$

where the last formula comes from using the larger size of the vectors

$$\hat{d}^j \quad \text{and} \quad (\sigma_j/\sigma_q)\hat{d}^q$$

to replace the size of their sum

$$\hat{d}^j - (\sigma_j/\sigma_q)\hat{d}^q.$$

Since $\bar{a}_q$ is independent of the weight updating in each iteration, $\|\hat{d}^q\|$ is easy to compute as d^q is available. When q does not belong to the reference framework, on the other hand, $\|\hat{d}^q\|$ could become very small even though it holds that $\|d^q\| \geq 1$. In this aspect, (12.14) ensures that all weights t_j are no less than 1. It is therefore clear that their weights never decrease even if indices stay nonbasic for a long time.

When the rounding error accumulates too much, it is necessary to determine a new reference framework and set all weights to 1 again. As the weight $\bar{t}_q$ is calculated directly, it is convenient to monitor errors in the process, which should be restarted when the calculated value differs from the updated value by a relatively large margin, e.g., when the former exceeds a multiple of the latter.²

(B) Projective steepest-edge rule.

Another scheme is to directly use $\|\hat{d}^j\|$ themselves rather than their approximation $\bar{t}_j$. Since (12.7) and (12.8) are still valid if vectors d^j in them are replaced by subvectors $\hat{d}^j$, recurrence formulas for squares of $\hat{d}^j$'s norms can be derived from (12.10) and (12.11) by replacing d^j with $\hat{d}^j$. It is still possible to directly compute

$$\|\hat{d}^q\|^2 = \delta + \|\bar{a}_q\|^2,$$

where δ is equal to 1 or 0, depending on whether q belongs to the reference framework or not. All $\bar{a}_j$ involved in the formulas should be regarded as subvectors,

² Harris uses twice as much.

corresponding to the reference framework. Pivot rules using these formulas are termed *projective steepest-edge rule*.

(C) Dynamic steepest-edge rule.

A further variant of the projective steepest-edge rule is to expend the reference framework when resetting it, by adding the index of the current leaving variable to the reference framework if the index is not already in it. Therefore, only minor changes are needed in the recurrence formulas. The resulting pivot rule is named *dynamic steepest-edge rule*.

We quote computational results from Forrest and Goldfarb (1992) as follows:

There were 20 test problems involved in their tests, including all 14 Netlib standard test problems that involve more than 10,000 nonzeros and six larger and more difficult problems collected by the authors. CPU times required by five codes based on Dantzig, Devex, the steepest-edge, dynamic steepest-edge, and projective steepest-edge to solve all the 20 problems are listed in the first row of Table 12.1. Time ratios of the first four rules to the projective steepest-edge rule are listed in the bottom row.

It is seen from the preceding that Dantzig's conventional rule is the slowest while the projective steepest-edge rule is the fastest for solving these test problems: their total time ratio is as high as 28.33. However, the authors indicate that the tests are favorable to the last three rules because they take advantages of the structures of computer IBM RISC system/6000 in pricing and solving systems. These numerical results firmly establish big superiority of the steepest-edge rule and approximate steepest-edge rules, such as Devex, over the conventional rule.

12.4 Largest-Distance Rule

Like the steepest-edge rule or Devex rule, the so-called *largest-distance* rule is also based on normalized reduced costs although it is simpler and easier to realize.

We explore the (primal) pivot rule from the dual side. Consider the dual problem

$$\begin{aligned} \max \quad & g = b^T y \\ \text{s.t.} \quad & A^T y \leq c \end{aligned}$$

to gain some insight. In fact, a negative reduced cost

$$\bar{z}_j = c_j - a_j^T B^{-T} c_B$$

implies that the current dual solution $\bar{y} = B^{-T} c_B$ violates the dual constraint $a_j^T y \leq c_j$. Entering the associated variable x_j to the basis means forcing the violated dual inequality constraint to be satisfied as an equality (binding), i.e., $a_j^T y = c_j$.

Therefore, it is reasonable to enter the variable associated with the mostly violated dual constraint to enter the basis.

It is not surprising that the conventional rule is unsatisfactory, as it takes the nonbasic variable associated with the most-negative reduced cost to enter the basis, but the variable does not necessarily correspond to the mostly violated dual constraint. In fact, the signed distance of $\bar{y}$ to the boundary $a_j^T y = c_j$ is (Sect. 2.1 of Part I)

$$d = \bar{z}_j / \|a_j\|.$$

Even if $\bar{z}_j$ is the most negative, the absolute value of d may still be very small when $\|a_j\|$ is large, so that the current dual solution is not far from the boundary.

The following rule selects the variable, associated with the mostly violated dual constraint, to enter the basis (Pan 2008a).

Rule 12.4.1 (Largest-Distance Column Rule) *Select the column index*

$$q \in \arg \min \{ \bar{z}_j / \|a_j\| \mid j \in N \}.$$

The preceding actually represents a class of pivot rules if $\|a_j\|$ is regarded as any norm of a_j although the Euclidean norm seems to be preferable.

We quote the associated computational results from Pan (2008a):

The actualization of the largest-distance rule is easy. The norms $\|a_j\|$, $j = 1, \dots, n$, of the columns of the coefficient matrix remain unchanged in solution process. If these norms are calculated at the beginning, the number of divisions required by the pivot rule will not exceed $n - m$ in each iteration. It is even better to normalize the columns of A first, so that the largest-distance rule is just the same as Dantzig's rule because any reduced cost $\bar{z}_j$ itself is equal to the signed distance of $\bar{y}$ to the boundary.

Computational experiments were carried out on an IBM PC with Windows XP 2002 system, processor 1.86GHz, 1.00GB internal storage, about 16 digits precision, and visual Fortran 5.0 compiler. There were three codes involved:

1. Devex: Devex rule
2. SCL1: Euclidean norm largest-distance rule
3. SCL2: ∞ -norm Largest-distance rule

MINOS 5.51 was used as a platform for the preceding three codes: only the pivot rule is replaced by the tested rules, respectively. Altogether, three sets of 80 large-scale sparse problems were tested (see Appendix C: Tables C.1, C.2, C.3):

1. 47 (according to $m + n$) largest Netlib problems (SCRS8-STOCFOR3)
2. All 16 Kennington problems (KEN07-KEN18)
3. All 17 BPMPD problems larger than 500KB (in compressed format) (RAT7A-DBIC1)

Table 12.2 gives iteration and CPU time ratios of these codes.

Table 12.2 Iteration and time ratios

Problem	Devex/SCL2		Devex/SCL1		SCL2/SCL1	
	Iters	Time	Iters	Time	Iters	Time
Netlib(47)	0.26	0.33	0.29	0.34	1.09	1.03
Kennington(16)	1.75	2.18	4.95	4.33	2.84	1.98
BPMPD(17)	1.29	2.33	1.88	3.64	1.46	1.56
Average(80)	1.00	2.06	1.43	3.24	1.43	1.58

It is seen from the bottom row that the total iteration and CPU time ratios of Devex to SCL1 are 1.43 and 3.24, respectively. Therefore, SCL1 outperformed Devex significantly. It is more important that the gap between their CPU times required are larger. SCL2 also defeated Devex with CPU time ratio 2.06. Moreover, it can be expected that the largest-distance rule using partial pricing will perform even better.

As for the two largest-distance rules, SCL1 outperformed SCL2 with iteration ratio 1.43 and time ratio 1.58. Therefore, the rule using Euclidean norm seems to be preferable.

Besides the remarkable performance of the largest-distance rule, the author himself was surprised that it is relatively easy to realize: SCL1 and SCL2 were yielded from MINOS 5.51 by only inserting a single sentence for normalizing columns of the coefficient matrix in its scaling subroutine.

12.5 Nested Rule

The nested pivot rule may be classified into the partial pricing category. In particular, it would be somehow close to multiple pricing. The key difference from the latter is that the former focuses on, and eliminate the most stubborn nonbasic indices that do not fulfil the the optimality condition (within the dual tolerance).

Let $\epsilon > 0$ be the dual feasibility tolerance. At the beginning of each iteration, a set N_1 of nonbasic indices is given. Pricing is carried out on N_1 to determine a column index, associated with a negative reduced cost by, e.g., the conventional column rule. If the associated reduced cost is less than $-\epsilon$, then the index is selected to enter the basis, and N_1 for the next iteration is generated from the current N_1 by including all indices, associated with reduced costs less than $-\epsilon$. In the other case, do the same thing with $N_2 = N \setminus N_1$; if there is still no reduced cost less than $-\epsilon$, then optimality is achieved.

As an instance, the nested counterpart of the classical rule is described below (Pan 2008b).

Rule 12.5.1 (Nested-Dantzig Rule) *Given a dual tolerance ϵ . Set $N_1 = N$ and $N_2 = \emptyset$.*

1. *Go to step 4 if $N_1 = \{j \in N_1 \mid \bar{z}_j < -\epsilon\} \neq \emptyset$.*
2. *Go to step 4 if $N_1 = \{j \in N_2 \mid \bar{z}_j < -\epsilon\} \neq \emptyset$.*

3. *Stop (optimality achieved).*
4. *Select a pivot column index $q \in \arg \min\{\bar{z}_j \mid j \in N_1\}$.*
5. *Update: $N_1 = N_1 \setminus q$, $N_2 = N \setminus N_1$.*

Such a rule is prefixed by *nested* since the current set N_1 , which undergoes pricing with priority, is a proper subset of its predecessor N_1 . The idea behind it can be explained further as follows.

In the initial iteration or iterations in which N_2 is touched, a full pricing is carried out. After that, a series of nested pricings follow, which might be called a “circle.” As each N_1 is a proper subset of its predecessor in a circle, the computational effort for pricing decreases monotonically iteration by iteration. In the k th iteration of a circle, moreover, the reduced costs associated with N_1 are less than $-\epsilon$ all the time. Therefore, it is reasonable to enter such a “stubborn” nonbasic index to the basis.

In fact, the index set N_1 generated from the first iteration corresponds to nonnegative constraints, gradients of which form acute angles with the negative reduced gradient. Based on the most-obtuse-angle heuristic, one should try to make these constraints inactive; in other words, one should give indices associated with these constraints priority to enter basis.

Any conventional pivot rule may be modified to a nested one. The *nested steepest-edge rule* can be obtained from Rule 12.5.1 by using

$$q \in \arg \min \{\bar{z}_j / \|d^j\| \mid j \in N_1\}$$

in step 4 instead, and the nested Devex rule obtained by using

$$q = \arg \min \{\bar{z}_j / t_j \mid j \in N_1\}.$$

We quote the computational results from Pan (2008c) below:

The nested pivot rules are easy for actualization. Associated computational experiments involve the following five codes:

1. Dantzig: MINOS 5.51 (full pricing option)
2. Devex: Devex rule
3. P-Dantzig: MINOS 5.51 (default sectional pricing option)
4. N-Devex: Nested Devex rule
5. N-Dantzig: Rule 12.5.1

These codes are all based on MINOS 5.51 as a platform, differing from the pivot rule used only. The software and hardware environments are the same as those described in the previous section, as well as the same 80 test problems (Appendix C: Tables C.1 (AFIRO-STANDMPS), C.2, C.3). Table 12.3 lists total iteration and CPU time ratios of the first four codes to the fifth (N-Dantzig).

It is seen from the preceding table that the N-Dantzig and N-Devex, based on the nested pivot rules, are significantly superior to the classical ones, and N-Dantzig is the best. In fact, the N-Dantzig rule outperformed Devex rule by total iteration ratio

Table 12.3 Iteration and time ratios to N-Dantzig

Problem	Dantzig		Devex		P-Dantzig		N-Devex	
	Iters	Time	Iters	Time	Iters	Time	Iters	Time
Netlib(47)	5.16	5.95	1.20	1.21	4.65	4.00	1.04	0.95
Kennington(16)	5.63	5.65	5.56	5.55	3.51	2.64	1.00	0.91
BPMPD(17)	8.29	12.86	3.83	6.54	5.04	5.20	1.18	1.22
Average(80)	6.78	9.75	3.48	5.73	4.57	4.20	1.10	1.09

3.48 and time ratio 5.73 and outperformed Dantzig’s rule by total iteration ratio 6.78 and time ratio 9.75!

Further computational experiments were conducted with nested pricing extensively. Contained in Appendix D, associated numerical results can be outlined below (see Table D.16):

For 77 test problems, the nested-Dantzig rule defeated the steepest-edge rule by total time ratio as high as 25.22, even though the former required more iterations (with total iteration ratio 0.34). Even the largest-distance rule was not comparable with the nested-Dantzig rule. However, the nested steepest-edge rule was inferior to the steepest-edge rule, since the former restarted from scratch too many times. It should be pointed out that these tests were somehow unfair to steepest-edge rules, as they neither use recurrence formulas (thus the additional two systems were solved in each iteration) nor take the advantage of the computer’s structure to solve systems more efficiently.

In summary, nested pivot rules are superior to their conventional counterparts by a large margin (except for the nested steepest-edge rule), and the nested-Dantzig rule is superior to the nested Devex rule.

12.6 Nested Largest-Distance Rule

Largest-distance rules can be improved further by combining the nesting strategy (Pan 2008c). The resulting rule is as follows.

Rule 12.6.1 (Nested Largest-Distance Column Rule) *The same as Rule 12.5.1, except for its step 4 using the following instead:*

$$q \in \arg \min \{\bar{z}_j / \|a_j\| \mid j \in N\}.$$

Computational results (Pan 2010):
The following three codes were involved:

1. Devex: Devex rule
2. LDN1: Nested largest-distance rule with Euclidean norm
3. LDN2: Nested largest-distance rule with ∞ norm

Table 12.4 Iteration ratio and time ratio

Problem	Devex/LDN2		Devex/LDN1		LDN2/LDN1	
	Iters	Time	Iters	Time	Iters	Time
Netlib(47)	1.20	1.17	1.15	1.16	0.96	0.99
Kennington(16)	5.63	5.42	7.83	5.77	1.39	1.06
BPMPD(17)	4.32	7.43	5.43	10.17	1.26	1.37
Average(80)	3.69	6.08	4.34	7.27	1.18	1.20

The software and hardware environments are the same as those given in the previous two sections, with the same 80 test problems (Appendix C: Tables C.1, C.2, C.3). Table 12.4 lists total iteration and CPU time ratios.

It is seen that the nested largest-distance rules outperformed Devex rules unambiguously in terms of either iterations or CPU time. As for the two nested ones, the nested largest-distance rules with Euclidean norm is better, as defeated Devex rule by iteration ratio 4.34 and time ratio 7.27. The margins are even larger than that between the nested-Dantzig rule and Devex rule.

In view of the numerical results reported in Sects. 12.4–12.6, we conclude that the nested largest-distance rules with Euclidean norm is the best among existing pivot rules, at least in terms of the 80 tested problems.

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Chapter 13

Dual Pivot Rule



A pivot rule used in the dual simplex method is termed *dual pivot rule*.¹ Like in the primal simplex context, a dual pivot rule plays an important role in the dual algorithm. This chapter will present very promising dual pivot rules.

Assume that $\bar{x} = B^{-1}b < 0$. The dual Dantzig's rule selects a row index $p \in B$ such that $\bar{x}_p$ is the minimum among components of the basic solution. Thus, the dual objective value will increase the most possible for a unit stepsize. This rule is far from ideal, like its primal counterpart. The so-called *most-improvement* rule determines the pivot such that the corresponding dual objective increment attains the maximum possible in the current iteration. Unfortunately, this rule is still unsatisfactory, just as its primal counterpart.

It might be well to assume that the current basis matrix consists of the first m columns of A , thus

$$B = \{1, \dots, m\}, \quad N = \{m+1, \dots, n\}.$$

13.1 Dual Steepest-Edge Rule

Consider the dual problem

$$\begin{array}{ll} \max & g = b^T y, \\ \text{s.t.} & A^T y \leq c. \end{array}$$

¹ The rule is termed “row” pivot rule since after a pivot row is determined, there is a very limited choice of a pivot column.

Let $\bar{y}$ be the current dual basic feasible solution, satisfying

$$B^T \bar{y} = c_B, \quad (13.1)$$

$$N^T \bar{y} \leq c_N. \quad (13.2)$$

Define

$$\begin{aligned} y(\beta) &= \bar{y} - \beta h^i, \\ h^i &= B^{-T} e_i, \quad i = 1, \dots, m. \end{aligned}$$

From the preceding two expressions and (13.1), for any $i \in \{1, \dots, m\}$ and $\beta \geq 0$ it holds that

$$\begin{aligned} a_i^T y(\beta) &= a_i^T \bar{y} - \beta = c_i - \beta \leq c_i, \\ a_k^T y(\beta) &= a_k^T \bar{y} = c_k, \quad k = 1, \dots, m, \quad k \neq i. \end{aligned}$$

It is known that $-h^i$, $i = 1, \dots, m$, is an edge direction, emanating from the vertex $\bar{y}$ of the feasible region $\{y | A^T y \leq c\}$. The determination of row index i implies that the basic variable x_i leaves the basis, and hence the constraint $a_i^T y \leq c_i$ can be satisfied as a strict inequality. Since

$$-b^T h^i = -e_i^T B^{-1} b = -\bar{x}_i,$$

when $\bar{x}_i < 0$, the edge direction $-h^i$ forms an acute angle with the dual objective gradient b , as is an uphill direction. Therefore, the objective value will never decrease if a row index $p \in \{1, \dots, m\}$ is selected such that $\bar{x}_p < 0$; it strictly increases if dual nondegeneracy is assumed. In particular, the following rule selects the edge that forms the smallest angle with b (Forrest and Goldfarb 1992).

Rule 13.1.1 (Dual Steepest-Edge Row Rule) *Select the pivot row index*

$$p \in \arg \min \{\bar{x}_i / \|h^i\| \mid i = 1, \dots, m\}.$$

Like in the primal context, practicability of the preceding rule lies in computing

$$\|h^i\|^2, \quad i = 1, \dots, m,$$

in a recurrent manner.

Assume that x_p is selected to leave and x_q to enter the basis. Then the inverse of the basis matrix is updated by (Sect. 3.3 of Part I):

$$\widehat{B}^{-1} = B^{-1} - \frac{(\bar{a}_q - e_p) e_p^T B^{-1}}{\bar{a}_{pq}},$$

where $\bar{a}_q = B^{-1}a_q$. Premultiplying the preceding expression by e_i^T and transposing it lead to the recurrence for edge directions, i.e.,

$$\tilde{h}^p = (1/\bar{a}_{pq})h^p, \quad (13.3)$$

$$\tilde{h}^i = h^i - (\bar{a}_{iq}/\bar{a}_{pq})h^p, \quad i = 1, \dots, m; \quad i \neq p, \quad (13.4)$$

from which the recurrence formulas for squares of norms of edge directions follow:

$$\|\tilde{h}^p\|^2 = (1/\bar{a}_{pq})^2 \|h^p\|^2, \quad (13.5)$$

$$\begin{aligned} \|\tilde{h}^i\|^2 &= \|h^i\|^2 - 2(\bar{a}_{iq}/\bar{a}_{pq})u_i + (\bar{a}_{iq}/\bar{a}_{pq})^2 \|h^p\|^2, \\ i &= 1, \dots, m, \quad i \neq p, \end{aligned} \quad (13.6)$$

where

$$B^T h^p = e_p, \quad Bu = h^p. \quad (13.7)$$

Note that $\|h^p\|^2 = (h^p)^T h^p$ can be directly calculated, because h^p and $\bar{a}_q$ are obtained otherwise independently.

Computations, involved in Rule 13.1.1, are usually cheaper than that in the (primal) steepest-edge rule. They both solve an additional system (13.6), hence $Bu = h^p$. It is noted that formula (12.11) needs to solve $B^T v = \bar{a}_q$. In contrast to (12.11), which is expensive when the pivot row $e_p^T B^{-1}N$ is dense, there is no any inner product involved in the middle term of (13.6). In practice, it is often the case when $n - m \gg m$, in which initially computing squares of norms of edge directions in the primal rule is cumbersome, compared with those in the dual rule. As for taking advantage of sparsity, handling $B^{-1}a_j$ is also inferior to $B^{-T}e_i$.

Now let us turn to the dual problem of form

$$\begin{aligned} \max \quad & g = b^T y, \\ \text{s.t.} \quad & A^T y + z = c, \quad z \geq 0. \end{aligned} \quad (13.8)$$

A new pivot rule can be derived if the steepest-edge direction is considered in (y, z) -space.

It is clear that the dual basic feasible solution

$$(\bar{y}, \bar{z}_N, \bar{z}_B) = (B^{-T}c_B, c_N - N^T B^{-T}c_B, 0)$$

is actually the unique solution to the $(m + n) \times (m + n)$ system

$$\begin{pmatrix} B^T & 0 & I \\ N^T & I & 0 \\ 0 & 0 & I \end{pmatrix} \begin{pmatrix} y \\ z_N \\ z_B \end{pmatrix} = \begin{pmatrix} c_B \\ c_N \\ 0 \end{pmatrix},$$

which is, geometrically, a vertex of the polyhedron

$$\{(y, z) \in \mathcal{R}^m \times \mathcal{R}^n \mid A^T y + z = c, z \geq 0\}.$$

The inverse of the coefficient matrix of the system is

$$\begin{pmatrix} B^{-T} & 0 & -B^{-T} \\ -N^T B^{-T} & I & N^T B^{-T} \\ 0 & 0 & I \end{pmatrix}.$$

It is easy to verify that the last m columns of the preceding are just edge directions, emanating from the vertex, i.e.,

$$h^i = \begin{pmatrix} -B^{-T} \\ N^T B^{-T} \\ I \end{pmatrix} e_i, \quad i = 1, \dots, m. \quad (13.9)$$

Recurrence formulas of these edge directions are of the form (13.3) and (13.4), and squares of norms of them are of form (13.5) and (13.6), though (13.7) should be replaced by:

$$B^T h = e_p, \quad Bu = h + \sum_{j=m+1}^n \sigma_j a_j, \quad (13.10)$$

where σ_j is the j th component of the pivot row vector, i.e.,

$$\sigma = A^T h. \quad (13.11)$$

As σ is computed while pricing, the following quantity:

$$\|h^p\|^2 = h^T h + \sigma^T \sigma \quad (13.12)$$

can be directly computed.

In this book, the pivot rule, based on (13.5)–(13.7), is referred to as *dual steepest-edge pivot rule I*, and that based on (13.5), (13.6), and (13.10)–(13.12) referred to as *dual steepest-edge pivot rule II*. Associated numerical results will be given in Table 13.1 in the next section, in a comparison with according approximate rules there.

Table 13.1 Total time and ratio

Dual pivot rule	Dantzig	Devex	Steepest -edge II	Dynamic steepest-edge	Projective steepest-edge	Steepest -edge I
Hours	177.78	67.43	12.72	10.41	7.36	6.36
Time ratio	27.95	10.60	2.00	1.64	1.16	1.00

13.2 Approximate Dual Steepest-Edge Rule

Harris (1973) derived a dual variant of the Devex rule, as may be regarded as an approximation of the dual steepest-edge rule II.

She constructed a “reference framework,” a set of m indices of components of z . Assume that subvector $\hat{h}^i$ consists of components, located on the reference framework, of edge direction h^i . Weights $s_i, i = 1, \dots, m$, are endowed to $\bar{x}_i$ to approximate $\|\hat{h}^i\|$.

Rule 13.2.1 (Dual Devex Row Rule) *Select the pivot row index*

$$p \in \arg \min\{\bar{x}_i/s_i | i = 1, \dots, m\},$$

where the weights $s_i, i = 1, \dots, m$, are determined as follows, so that the rule can be regarded as an approximation of the steepest-edge rule under the reference framework.

At the beginning, index set B is taken as the reference framework, and s_i are set to 1 for all $i = 1, \dots, m$ (see (13.9)). Hence, the dual Devex rule is just the same as the dual Dantzig rule in this case. Subsequently, s_i are updated iteration by iteration. Assume that σ is defined by (13.11) and $\hat{\sigma}$ is the subvector consisting of the components located on the reference framework. As (13.3) and (13.4) are still valid when vectors in them are replaced by subvectors, consisting of components located on the reference framework, we have the following updates:

$$\begin{aligned}\bar{s}_p &= \max\{1, \|\hat{\sigma}\|/|\bar{a}_{pq}|\}, \\ \bar{s}_i &= \max\{s_i, |\bar{a}_{iq}/\bar{a}_{pq}|\|\hat{\sigma}\|\}, \quad i = 1, \dots, m, i \neq p,\end{aligned}$$

the last of which comes from using the larger norm of vectors $\hat{h}^i$ and $-(\bar{a}_{iq}/\bar{a}_{pq})h^p$ instead of the norm of their sum (see (13.9)). In each iteration, $\bar{a}_q = B^{-1}a_q$ is computed separately. As σ is available, $\|\hat{\sigma}\|$ can be computed directly.

Similar to the (primal) Devex rule, when errors caused by repeatedly using the updating formulas accumulate too high, it is necessary to determine a new reference framework and set all weights to 1 again. As weight $\bar{s}_q$ is calculated directly, it is convenient to monitor errors. It should be restarted when the calculated value differs from the updated value by a relatively large gap, e.g., when the former exceeds a multiple of the latter (Harris used double).

When the error, caused by repeatedly using of the updating formula, accumulates too much, it is necessary to determine a new reference framework and set all weights to 1 again, just like the approximate steepest-edge rules. It is also possible to monitor errors: the process should be restarted when the calculated value of $\|\hat{\sigma}\|$ differs from the updated value $\bar{s}_p$ by a relatively large margin.

The so-called *dual projective steepest-edge rule* results from using $\|\hat{h}^i\|$ rather than $\bar{s}_i$. This rule can also be obtained by modifying the dual steepest-edge rule under the reference framework. In fact, replacing h^i in (13.5) and (13.6) by $\hat{h}^i$ leads

to recurrence formulas of $\|\hat{h}^i\|^2$, whereas u in (13.6) can be obtained by solving the system

$$Bu = \sum_{j=m+1}^n \hat{\sigma}_j a_j.$$

A further variant of the dual steepest-edge rule results from expending the reference framework whenever it is reset, that is, by adding the pivot column index to the current reference framework if it is not in it already. Accordingly, the recurrence formulas are modified slightly. Such a variant is called *dual dynamic steepest-edge rule*.

We cite numerical results reported by Forrest and Goldfarb (1992) below. The hardware and software environments as well as 20 test problems were the same as those described for the primal case, presented in Sect. 12.3. Codes based on six dual rules: Dantzig, Devex, steepest-edge II, dynamic steepest-edge, project steepest-edge, and steepest-edge I were tested. Table 13.1 gives the total CPU times, required for solving all the problems, in the first row, and time ratios of the first five rules to the dual projective steepest-edge rule in the second row.

It is seen from the preceding table that the dual Dantzig rule is the slowest, whereas the dual steepest-edge rule I is the fastest, with time ratio as high as 27.95, though the tests are favorable to the last four rules because they take advantages of the structures of computer IBM RISC system/6000 in pricing and solving systems.

In terms of reported numerical results, the dual rules appear to be inferior to their primal counterparts slightly. However, Forrest and Goldfarb indicate that these data are unfair to the dual rules, because their primal codes were optimized, while the dual ones were not; the numbers of hours required by the last four codes in the preceding table should be reduced by 10 % at least, though decrements of the dual Dantzig and Devex are not as so much. Therefore, the dual rules are actually efficient, compared to their primal counterparts.

The steepest-edge rule and Devex approximates as well as their dual variants are superior over the standard rules with large margins and are widely used in many commercial codes, such as CPLEX.

13.3 Dual Largest-Distance Rule

This section derive a dual variant of the largest-distance rule (Sect. 12.4). The basic idea is to determine a leaving variable by finding an inequality constraint, mostly violated by the current solution in the reduced space.

Let $B = \{1, \dots, m\}$, $N = A \setminus B$ be the current basis and nonbasis of the standard problem (12.1). Assume that the associated simplex tableau is dual but not primal feasible, i.e., $\bar{b} = \bar{x}_B \not\geq 0$.

For some $i \in B$, the inequality $\bar{x}_i < 0$ implies that the solution, $\bar{x}_N = 0$, in the reduced space violates the constraint

$$\bar{b}_i - (w^i)^T x_N \geq 0,$$

where $\bar{b} = B^{-1}b$ and

$$w^i = N^T B^{-T} e_i.$$

The signed distance from the point $\bar{x}_N = 0$ to the boundary $\bar{b}_i - (w^i)^T x_N = 0$ is $\bar{x}_i / \|w^i\|$, where any norm is allowed in principle although the Euclidean norm should be preferable.

The dual Dantzig conventional rule is

$$p \in \arg \min \{\bar{x}_i \mid i = 1, \dots, m\},$$

which does not necessarily correspond to the mostly violated inequality constraint, since the according distance would be very small when $\|w^p\|$ is large, as the point $\bar{x}_N = 0$ may not be far away from the boundary actually. Therefore, it should not be expected that this rule performs satisfactorily.

If one determines a leaving variable that corresponds to the mostly violated inequality constraint, the following rule follows.

Rule 13.3.1 (Dual Largest-Distance Row Rule) *Select the pivot row index*

$$p \in \arg \min \{\bar{x}_i / \|w^i\| \mid i = 1, \dots, m\}.$$

Involving $\|w^i\|$, the preceding rule is cumbersome, compared with the (primal) largest-distance rule. Like in the primal context, however, $\|w^i\|^2$, $i = 1, \dots, m$, can be computed recursively, so that the rule would be still practicable.

Consider the set of n -dimensional vectors

$$\sigma^i = A^T B^{-T} e_i, \quad i = 1, \dots, m.$$

Note that w^i is an $(n - m)$ -subvector of σ^i for any $i = 1, \dots, m$. Assume that x_p leaves and x_q enters the basis, then it is not difficult to derive the following recurrence relation:

$$\tilde{\sigma}^p = (1/\bar{a}_{pq})\sigma^p,$$

$$\tilde{\sigma}^i = \sigma^i - (\bar{a}_{iq}/\bar{a}_{pq})\sigma^p, \quad i = 1, \dots, m; \quad i \neq p,$$

where the pivot column $\bar{a}_q = B^{-1}a_q$ is available. Further, recurrence formulas of squares of their norms can be obtained, i.e.,

$$\begin{aligned}\|\tilde{\sigma}^p\|^2 &= (1/\bar{a}_{pq})^2 \|\sigma^p\|^2, \\ \|\tilde{\sigma}^i\|^2 &= \|\sigma^i\|^2 - 2(\bar{a}_{iq}/\bar{a}_{pq})\sigma^{iT}\sigma^p \\ &\quad + (\bar{a}_{iq}/\bar{a}_{pq})^2 \|\sigma^p\|^2, \quad i = 1, \dots, m, \quad i \neq p.\end{aligned}$$

Combining the preceding, $\sigma^{iT}\sigma^p = e_i^T B^{-1} N \sigma^p$, and $\|\sigma^i\|^2 = \|w^i\|^2 + 1$ leads to the following required formulas:

$$\|\tilde{w}^p\|^2 = (1/\bar{a}_{pq})^2 (\|w^p\|^2 + 1) - 1, \quad (13.13)$$

$$\begin{aligned}\|\tilde{w}^i\|^2 &= \|w^i\|^2 - 2(\bar{a}_{iq}/\bar{a}_{pq})u_i \\ &\quad + (\bar{a}_{iq}/\bar{a}_{pq})^2 (\|w^p\|^2 + 1), \quad i = 1, \dots, m, \quad i \neq p,\end{aligned} \quad (13.14)$$

where

$$Bu = \sum_{j=m+1}^n \sigma_j a_j,$$

and σ_j is the j th component of σ^p (i.e., the $(j - m)$ th component of w^p). Note that components of w^p are available entries of the pivot row, and hence $\|w^p\|^2 = (w^p)^T w^p$ can be calculated directly.

13.4 Dual Nested Rule

This section derives a dual variant of the nested rule (Sect. 12.5). Regarding the negative components of the primal basic solution as elimination target, it focuses on the most stubborn ones among them.

Let $\epsilon > 0$ be the primal feasibility tolerance. At the beginning of an iteration, a set I_1 of row indices is given the priority for associated basic indices to leave the basis. The row index in I_1 corresponding to the smallest component of $\bar{x}_B$ is determined. If the component is less than $-\epsilon$, then the corresponding basic index is selected to leave the basis, and the I_1 for the next iteration is born from the current I_1 by including all its row indices, associated with $\bar{x}_B$'s components less than $-\epsilon$. In the other case, do the same as before with $I_2 = B \setminus I_1$. If no basic component is less than $-\epsilon$, then optimality is attained.

The following is a nested variant of Dantzig conventional dual rule.

Rule 13.4.1 (Dual Nested Row Rule: Dantzig) *Given primal tolerance $\epsilon > 0$, set $I_1 = B$ and $I_2 = \emptyset$.*

1. *Go to step 4 if $I_1 = \{i \in I_1 \mid \bar{x}_i < -\epsilon\} \neq \emptyset$.*
2. *Go to step 4 if $I_1 = \{i \in I_2 \mid \bar{x}_i < -\epsilon\} \neq \emptyset$.*
3. *Stop (optimality achieved).*
4. *Select a pivot row index $p \in \arg \min \{\bar{x}_i \mid i \in I_1\}$.*
5. *Update: $I_1 = I_1 \setminus p$, $I_2 = B \setminus I_1$.*

In the first iteration or iterations in which I_2 is touched, the preceding rule actually proceeds the same as the standard dual row rule. After such an iteration, a series of iterations with nested pivoting follow, as might be called a “circle,” where each I_1 is a proper subset of its predecessor. In the k th iteration of a circle, basic components associated with I_1 are less than $-\epsilon$ all the time. It is therefore reasonable to select such a “stubborn” index leaving the basis.

It is possible to turn any full dual rule to a nested version. The dual nested steepest-edge rule can be obtained from Rule 13.4.1 by using

$$p \in \arg \min \{\bar{x}_i / \|h^i\| \mid i \in I_1\}$$

in step 4 instead, the dual nested Devex rule obtained by using

$$p \in \arg \min \{\bar{x}_i / s_i \mid i \in I_1\},$$

and the dual nested largest-distance rule by using

$$p \in \arg \min \{\bar{x}_i / \|w^i\| \mid i \in I_1\}.$$

In view of the performance of their primal counterparts, it might be expected that the dual nested rules and the dual nested largest-distance rules perform satisfactorily. The dual nested rules are easy to implement, fortunately. However, there is no computational experience at present.

References

- Forrest JJH, Goldfarb D (1992) Steepest edge simplex algorithm for linear programming. Math Program 57:341–374
- Harris PMJ (1973) Pivot selection methods of the Devex LP code. Math Program 5:1–28

Chapter 14

Simplex Phase-I Method



The simplex method requires a feasible basis to get itself started. The so-called *Phase-I* procedure is for this purpose.

The Phase-I method in Chap. 3 of Part I is only for textbooks and has not been used in practice at all since it rigidly determines the initial basis and expands the scale of the problem. This chapter will present selected practicable Phase-I methods, which are artificial variable free or involve a single variable only.

14.1 Infeasibility-Sum Algorithm

The auxiliary objective function of this method only involves variables, associated with negative components of the current basic solution. The method is widely used in practice since it is artificial variable free and is workable.

Assume that an initial simplex tableau of form (3.6) is available, associated with basic solution

$$\bar{x}_B = \bar{b}, \quad \bar{x}_N = 0. \quad (14.1)$$

Introduce

$$I = \{i = 1, \dots, m \mid \bar{b}_i < 0\}, \quad \bar{I} = \{1, \dots, m\} \setminus I. \quad (14.2)$$

Assume that $I \neq \emptyset$. Construct the following auxiliary problem:

$$\begin{aligned} \min & -\sum_{i \in I} x_{j_i}, \\ \text{s.t. } & x_B = \bar{b} - \bar{N}x_N, \\ & x_{j_i} \geq 0, \quad i \in \bar{I}; \quad x_N \geq 0, \end{aligned} \quad (14.3)$$

where the objective function is the negative sum of variables associated with negative components of the basic solution. Note that the constraint system of the auxiliary problem is the same as that of the original problem, except for the nonnegativity restriction only imposed on variables, associated with nonnegative components of the basic solution.

It is clear that the solution (14.1) is feasible for the auxiliary problem. The associated feasible simplex tableau can be obtained by eliminating nonzero entries in the objective row by relevant elementary transformations. Consequently, the reduced objective function is

$$-\sum_{i \in I} x_{ji} = w_0 + \bar{z}_N^T x_N, \quad (14.4)$$

where

$$w_0 = -\sum_{i \in I} \bar{b}_i > 0, \quad \bar{z}_j = \sum_{i \in I} \bar{a}_{i,j}, \quad j \in N,$$

where the “>” inequality is from (14.2).

Theorem 14.1.1 (Infeasibility Test) *If the reduced costs of an auxiliary simplex tableau are all nonnegative, the original problem is infeasible.*

Proof Assume that $\bar{z}_N \geq 0$, and $\tilde{x} \geq 0$ is a feasible solution to the original problem. Substituting $\tilde{x}$ into (14.4) gives the system, whose left-hand side is clearly less than or equal to zero, and, by $w_0 > 0$ and $\bar{z}_N \geq 0$, the right-hand side is strictly greater than zero, leading to a contradiction. Therefore, the statement is valid. $\square$

If there is a negative reduced cost, then any existing column rule applies. Assume that a column index q was determined such that

$$\bar{z}_q = \sum_{i \in I} \bar{a}_{iq} < 0. \quad (14.5)$$

Since there is no nonnegativity restriction on the current infeasible basic variables, the auxiliary problem could be unbounded even if the original problem itself bounded. In fact, this happens when $\bar{I} = \emptyset$ or

$$\{i \in \bar{I} | \bar{a}_{iq} > 0\} = \emptyset. \quad (14.6)$$

Thus, the conventional row rule is not applicable. As a remedy, the rule is modified as follows.

Rule 14.1.1 (Auxiliary Row Rule) *Determine the row index p such that*

$$\alpha = \bar{b}_p / \bar{a}_{pq} = \begin{cases} \min\{\bar{b}_i / \bar{a}_{iq} | \bar{a}_{iq} > 0, i \in \bar{I}\}, & \text{if } \{i \in \bar{I} | \bar{a}_{iq} > 0\} \neq \emptyset, \\ \max\{\bar{b}_i / \bar{a}_{iq} | \bar{a}_{iq} < 0, i \in I\}, & \text{in the other case.} \end{cases} \quad (14.7)$$

Lemma 14.1.2 *Assume that the selected column index q satisfies (14.5). Then rule 14.1.1 determines a row index p such that the simplex tableau, resulting from the according basis change, is feasible to the auxiliary problem.*

Proof First, it can be asserted that there is a row index $t \in I$ such that $\bar{a}_{tq} < 0$ since, otherwise, from

$$\bar{a}_{iq} \geq 0, \quad \forall i \in I,$$

it follows that

$$\sum_{i \in I} \bar{a}_{iq} \geq 0,$$

which contradicts the assumption (14.5). Therefore, the second expression of (14.7) is always well-defined, hence so is the row rule.

Take $\bar{a}_{p,q}$ as the pivot. Executing according elementary transformations leads to a new simplex tableau, whose right-hand side can be expressed in terms of its predecessor's entries (Part I), i.e.,

$$\begin{aligned} \hat{b}_i &= \bar{b}_i - \alpha \bar{a}_{iq}, \quad i \in \bar{I}, i \neq p, \\ \hat{b}_p &= \alpha, \end{aligned}$$

where α is determined by (14.7). It is easy to verify that these components are all nonnegative, hence the resulting simplex tableau is again feasible to the auxiliary problem. $\square$

Theorem 14.1.3 *Assume the feasibility of the original problem and nondegeneracy throughout solution process. The number of basic infeasible variables strictly decreases in finitely many iterations.*

Proof It is seen from the proof of Lemma 14.1.2 that the second expression of (14.7) is still well-defined if “max” there is replaced by “min,” i.e.,

$$\alpha_1 = \bar{b}_s / \bar{a}_{sq} = \min\{\bar{b}_i / \bar{a}_{iq} \mid \bar{a}_{iq} < 0, i \in I\} > 0. \quad (14.8)$$

If, in some iteration, the stepsize α is determined by the first expression of (14.7), and it holds that

$$\alpha \geq \alpha_1, \quad (14.9)$$

then the s row-indexed component of the new solution satisfies

$$\hat{x}_{js} = \bar{b}_s - \alpha \bar{a}_{sq} \geq 0,$$

which implies that a basic infeasible variable becomes feasible, at least.

If, in some iteration, the second expression of (14.7) is used, then, for the new solution, not only the basic components, associated with all row indices $i \in \bar{I}$, are feasible, but so are basic components, associated with row indices

$$s \in \{i \in I \mid \bar{a}_{iq} < 0\} \neq \emptyset.$$

This is of course favorable.

Assume that the number of basic infeasible variables does not strictly decrease forever. In each iteration, there is always a negative cost since the original problem is feasible (Theorem 14.1.1), and the stepsize α is determined by the first expression of (14.7), which does not satisfy (14.9). But the number of bases is finite, hence there are some bases that appear infinitely, as cycling occurs. This is impossible because the objective value of the auxiliary problem strictly decreases under the nondegeneracy assumption. Therefore the number of basic infeasible variables strictly decreases in finitely many iterations. $\square$

When, after some iteration, the number of basic infeasible variables strictly decreases, but does not vanish, a new auxiliary problem is constructed, whose objective function (infeasibility-sum) involves less variables than its predecessor, and iterations are carried out, until infeasibility of the original problem is detected or a feasible simplex tableau is reached. Involving a series of infeasibility-sums as objective functions in general, this method is sometimes referred to as *method with piecewise sum of infeasibilities*.

The associated steps are put into the following algorithm.

Algorithm 14.1.1 (Tableau Phase-I: Infeasibility-Sum) Initial: simplex tableau of form (3.6). This algorithm finds a feasible simplex tableau by handling auxiliary programs of form (14.3).

1. Stop if $\bar{b}_i \geq 0, \forall i = 1, \dots, m$ (a feasible simplex tableau reached).
2. For all $j \in N$, compute $\bar{z}_j = \sum_{i \in I} \bar{a}_{i,j}$, where $I = \{i = 1, \dots, m \mid \bar{b}_i < 0\}$.
3. Stop if $\bar{z}_N \geq 0$ (the original problem is infeasible).
4. Form a simplex tableau to the auxiliary problem (14.3).
5. Call one iteration of the simplex algorithm with Rule 14.1.1 being used for row pivoting.
6. Go to step 1.

Example 14.1.1 Solve the following problem by Algorithm 14.1.1 in Phase-I:

$$\begin{aligned} \min f &= -x_1 + x_2 - 2x_3, \\ \text{s.t. } x_1 - 3x_2 - 2x_3 + x_4 &= -4, \\ x_1 - x_2 + 4x_3 - x_5 &= 2, \\ -3x_1 + x_2 + x_3 + x_6 &= 8, \\ x_j &\geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Answer Phase-I. Multiply the second equation by -1 to turn the coefficient of x_5 to 1. Setting $I = \{1, 2\}$, construct auxiliary problem

$$\begin{aligned} \min f &= -x_4 - x_5, \\ \text{s.t. } x_1 - 3x_2 - 2x_3 + x_4 &= -4, \\ -x_1 + x_2 - 4x_3 + x_5 &= -2, \\ -3x_1 + x_2 + x_3 + x_6 &= 8, \\ x_j &\geq 0, \quad j = 1, 2, 3, 6. \end{aligned}$$

Its initial tableau is

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	-3	-2	1			-4
-1	1	-4		1		-2
-3	1	1			1	8
			-1	-1		

Add the first and the second rows to the objective row to eliminate the nonzero basic entries, yielding the simplex tableau below:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	-3	-2	1			-4
-1	1	-4		1		-2
-3	1	1*			1	8
	-2	-6				-6

Iteration 1:

$q = 3$; $\min\{8/1\} = 8$, $p = 3$. So 1 in the x_3 column and the 3rd row is the pivot. Add 2, 4, 5 times of row 3 to rows 1, 2, 4, respectively.

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-5	-1		1		2	12
-13	5			1	4	30
-3	1	1			1	8
-18	4				6	42

The right-hand side of the preceding tableau is nonnegative, hence Phase-I is finished. To go to Phase-II, use coefficients of the original objective function to overwrite the bottom (objective) row.

Add 2 times of row 3 to the bottom row to eliminate nonzero basic entries, obtaining feasible simplex tableau:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-5	-1		1		2	12
-13	5			1	4	30
-3	1	1			1	8
-1	1	-2				

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-5	-1		1		2	12
-13	5			1	4	30
-3	1	1			1	8
-7	3				2	16

Carry out Phase-II from the preceding tableau. By the conventional rule, x_1 column is selected as the pivot column. All components of this column are nonpositive, as detects unboundedness of the original problem.

In practice, used is usually the following revised version of Algorithm 14.1.1.

Algorithm 14.1.2 (Phase-I: Infeasibility-Sum) Initial: (B, N) , B^{-1} , $\bar{x}_B = B^{-1}b$. This algorithm finds a feasible basic solution to the standard LP problem (12.1) by handling auxiliary programs of form (14.3).

1. Stop if $\bar{x}_B \geq 0$ (producing a feasible basic solution).
2. Construct $c_B : c_{j_i} = \begin{cases} -1, & \text{If } \bar{x}_{j_i} < 0, \\ 0, & \text{If } \bar{x}_{j_i} \geq 0, \end{cases} i = 1, \dots, m$.
3. Call one iteration of the simplex algorithm, in which Rule 14.1.1 is used to determine a pivot row index.
4. Stop if it terminates as step 3 (or step 2) (the original problem is infeasible).
5. Go to step 1.

14.2 Single-Artificial-Variable Algorithm

Although the infeasible-sum method is artificial variable free, the row pivot rule used in it is cumbersome, compared with the normal row rule. This section describes a method, which uses the normal row rule and only involves a single artificial variable.

Let (B, N) be basis and nonbasis of the standard problem (12.1). There are two schemes for the method.

Scheme 1: The auxiliary problem is based on the original data of the problem.

Given an m -dimensional vector $h \geq 0$ such that $b - Bh \neq 0$. Introducing artificial variable f and normalized vector

$$a_{n+1} = (b - Bh) / \|b - Bh\|, \quad (14.10)$$

construct the following auxiliary problem:

$$\begin{aligned} \min \quad & f, \\ \text{s.t.} \quad & Ax + fa_{n+1} = b, \quad x, f \geq 0. \end{aligned} \quad (14.11)$$

It is verified that

$$\bar{x}_B = h, \quad \bar{x}_N = 0, \quad \bar{f} = \|b - Bh\|$$

is a feasible solution to the problem, though not necessarily a basic solution. For instance, setting $h = e > 0$ seems to be helpful to avoid zero stepsize, though according a_{n+1} would lose sparsity.

The constraint system of the auxiliary problem is equivalent to

$$x_B = \bar{b} - \bar{a}_{n+1}f - \bar{N}x_N, \quad (14.12)$$

where

$$\bar{b} = B^{-1}b, \quad \bar{a}_{n+1} = B^{-1}a_{n+1}, \quad \bar{N} = B^{-1}N. \quad (14.13)$$

If $\bar{b} \geq 0$, then f is clearly allowed to be zero. Thus, erasing the artificial variable leads to a feasible basic solution to the original problem; otherwise, the following applies.

Proposition 14.2.1 *If $\bar{b} \not\geq 0$, the row index set $\{i = 1, \dots, m \mid \bar{a}_{i,n+1} < 0, \bar{b}_i < 0\}$ is nonempty.*

Proof We show that if $\bar{b}_i < 0$ for some $i \in \{1, \dots, m\}$, then $\bar{a}_{i,n+1} < 0$. Assume it is not the case, i.e.,

$$\bar{a}_{i,n+1} \geq 0.$$

Then from the preceding, (14.10) and (14.13), it is known that

$$0 \leq \bar{a}_{i,n+1} = e_i^T B^{-1}a_{n+1} = e_i^T B^{-1}(b - Bh)/\|b - Bh\| = \bar{b}_i - h_i/\|b - Bh\|,$$

hence $\bar{b}_i \geq h_i \geq 0$, leading to a contradiction. $\square$

Now assume $\bar{b} \not\geq 0$. If $\bar{x}_N = 0$ is fixed and the value of $\bar{x}_{n+1}$ decreases from $\|b - Bh\|$, then $\bar{x}_B$ changes from h accordingly. If $\bar{b}_i < 0$ for some $i \in \{1, \dots, m\}$, the according inequality constraint may block up the value of $\bar{x}_{n+1}$ from decreasing to zero. In fact, the smallest value of f satisfying, inequalities

$$x_B = \bar{b} - \bar{a}_{n+1}f \geq 0, \quad f \geq 0,$$

is

$$\alpha = \bar{b}_r / \bar{a}_{r,n+1} = \min\{\bar{b}_i / \bar{a}_{i,n+1} \mid \bar{a}_{i,n+1} < 0, \bar{b}_i < 0, i = 1, \dots, m\} > 0. \quad (14.14)$$

As the value of $\bar{x}_{n+1}$ decreases, the according value of $\bar{x}_{j_r}$ hits 0 first. Let x_{j_r} leave and f enter the basis, then the resulting basis corresponds to a feasible basic solution to the auxiliary problem, with its objective value strictly decreasing. As it is lower bounded, solving the auxiliary problem will render its optimal solution.

It is not difficult to show that a feasible basic solution or basis to the original problem can be obtained if the optimal value of the auxiliary problem is equal to zero (f leaves the basis). The original problem is infeasible if the optimal value of the auxiliary problem is positive.

Scheme 2: The auxiliary problem is based on a canonical form of the constraint system.

Assume that the canonical form, associated with basis B , is available and that $\bar{b} = B^{-1}b \not\geq 0$. Giving an/the m -dimensional vector $h \geq 0$, and defining

$$\bar{a}_{n+1} = (\bar{b} - h) / \|\bar{b} - h\|, \quad (14.15)$$

construct the following auxiliary problem:

$$\begin{aligned} \min \quad & f \\ \text{s.t.} \quad & x_B = \bar{b} - f\bar{a}_{n+1} - \bar{N}x_N, \\ & x, f \geq 0. \end{aligned} \quad (14.16)$$

It is clear that there is a feasible solution to the problem; that is,

$$\bar{x}_B = h, \quad \bar{x}_N = 0, \quad \bar{f} = \|\bar{b} - h\|.$$

Thus, determining pivot column index $q = n + 1$ and row index p satisfying (14.14), one can obtain a feasible basis by performing according basis change. It is noted that Proposition 14.2.1 is still valid in this case.

As for how to determine h , it is only required that $h \geq 0$, in principle. With respect to practice, however, it is not proper for the associated $\|\bar{a}_{n+1}\|$ being too large. A simple choice is $h = e > 0$. Another choice is setting

$$\delta_i \geq 0, \quad i = 1, \dots, m,$$

and taking

$$h_i = \begin{cases} \delta_i, & \text{if } \bar{b}_i < \delta_i, \\ \bar{b}_i, & \text{otherwise,} \end{cases} \quad i = 1, \dots, m,$$

where $\delta_i \geq 0$, $i = 1, \dots, m$, are a set of constants, which are equal or close to zero. Consequently, (14.15) becomes

$$\bar{a}_{i,n+1} = \begin{cases} \bar{b}_i - \delta_i, & \text{if } \bar{b}_i < \delta_i, \\ 0, & \text{in the other case,} \end{cases} \quad i = 1, \dots, m. \quad (14.17)$$

We prefer the latter because it is better for maintaining sparsity of $\bar{a}_{n+1}$.

Specifically, assume availability of a simplex tableau, say (3.6), of the original problem. Inserting the artificial f column, we obtain the auxiliary simplex tableau below:

$$\begin{array}{c|ccc|c} x_B & x_N & f & & \text{RHS} \\ \hline I & \bar{N} & \bar{a}_{n+1} & & \bar{b} \end{array} \quad (14.18)$$

where the original objective row is excluded. Determine row index r by (14.14). Then drop x_{j_r} from and enter f to the basis by elementary transformations, so that f becomes the r th basic variable. Taking the r th row as the objective row, the auxiliary problem can be solved by the simplex method. If the optimal value vanishes and f has left the basis, a feasible simplex tableau to the original problem is obtained. If the optimal value is nonzero, the original problem is infeasible.

The steps can be summarized into the following algorithm.

Algorithm 14.2.1 (Tableau Phase-I: Single-Artificial-Variable) Initial: a simplex tableau of form (14.18). This algorithm finds a feasible simplex tableau to the standard LP problem (12.1).

1. Select a row index $r \in \arg \min\{\bar{b}_i \mid i = 1, \dots, m\}$.
2. Stop if $\bar{b}_r \geq 0$ (infeasible problem).
3. Determine row index $r \in \arg \min\{\bar{b}_i / \bar{a}_{i,n+1} \mid \bar{a}_{i,n+1} < 0, \bar{b}_i < 0, i = 1, \dots, m\}$.
4. Carry out elementary transformation to convert $\bar{a}_{r,n+1}$ to 1 and eliminate all other nonzeros of the column.
5. Determine column index $q \in \arg \max_{j \in N} \bar{a}_{rj}$.
6. Stop if $\bar{a}_{rj} \leq 0$.
7. Determine row index $p \in I = \arg \min\{\bar{b}_i / \bar{a}_{iq} \mid \bar{a}_{iq} > 0; i = 1, \dots, m\}$.
8. Set $p = r$ if $r \in I$.
9. Carry out elementary transformations to convert $\bar{a}_{pq}$ to 1 and eliminate all other nonzeros of the column.
10. Stop if $p = r$ (feasibility achieved).
11. Go to step 5.

It is not difficult to transform Algorithm 14.2.1 to a revised version, as is omitted.

Example 14.2.1 Find a feasible simplex tableau to the following problem by Algorithm 14.2.1:

$$\begin{aligned} \min f &= x_1 + 3x_2 - 2x_3 + 6x_4, \\ \text{s.t. } &-x_1 + x_2 - x_3 + x_5 = -1, \\ &-3x_1 + x_2 + 2x_3 + x_6 = 2, \\ &x_1 - 3x_2 - 3x_3 + x_4 = -4, \\ &x_j \geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Answer The constraint system is of a canonical form. Set $\delta_1 = \delta_2 = 0$. Introducing artificial variable x_7 , construct an auxiliary problem of form (14.16):

$$\begin{aligned} \min x_7, \\ \text{s.t. } &-x_1 + x_2 - x_3 + x_5 - x_7 = -1, \\ &-3x_1 + x_2 + 2x_3 + x_6 = 2, \\ &x_1 - 3x_2 - 3x_3 + x_4 - 4x_7 = -4, \\ &x_j \geq 0, \quad j = 1, \dots, 7. \end{aligned}$$

In order to turn to Phase-II conveniently after the Phase-I is finished, we put the objective row of the original problem as the bottom row. The initial tableau is

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
-1	1	-1		1		-1*	-1
-3	1	2			1		2
1	-3	-3	1			-4	-4
1	3	-2	6				

Iteration 1:

1. $\min\{-1, 2, -4\} = -4 < 0$.
3. $\max\{-1/-1, -4/-4\} = 1, r = 1$.
4. Multiply row 1 by -1 , then add 4 times of row 1 to row 3:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1	-1	1		-1		1	1
-3	1	2			1		2
5*	-7	1	1	-4			
1	3	-2	6				

5. $\max\{1, -1, 1, -1\} = 1 > 0, q = 1$.
7. $\min\{1/1, 0/5\} = 0, p = 3$.
8. $p \neq 1$.

9. Multiply row 3 by $1/5$, then add $-1, 3, -1$ times of row 3 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	$2/5$	$4/5$	$-1/5$	$-1/5$		1	1
	$-16/5$	$13/5$	$3/5$	$-12/5$	1		2
1	$-7/5$	$1/5^*$	$1/5$	$-4/5$			
	$22/5$	$-11/5$	$29/5$	$4/5$			

Iteration 2:

5. $\max\{2/5, 4/5, -1/5, 1/5\} = 4/5 > 0$, $q = 3$.
 7. $\min\{1/(4/5), 2/(13/5), 0/(1/5)\} = 0$, $p = 3$.
 8. $p \neq 1$.
 9. Multiply row 3 by 5, then add $-4/5, -13/5, 11/5$ times of row 3 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
-4	6		-1	3		1	1
-13	15^*		-2	8	1		2
5	-7	1	1	-4			
11	-11		8	-8			

Iteration 3:

5. $\max\{-4, 6, -1, 3\} = 6 > 0$, $q = 2$.
 7. $\min\{1/6, 2/15\} = 2/15$, $p = 2$.
 8. $p \neq 1$.
 9. Multiply row 2 by $1/15$, then add $-6, 7, 11$ times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$6/5^*$			$-1/5$	$-1/5$	$-2/5$	1	$1/5$
$-13/15$	1		$-2/15$	$8/15$	$1/15$		$2/15$
$-16/15$		1	$1/15$	$-4/15$	$7/15$		$14/15$
$22/15$			$95/15$	$-32/15$	$11/15$		$22/15$

Iteration 4:

5. $\max\{5/6, -1/5, -1/5, -2/5\} = 5/6 > 0$, $q = 1$.
 7. $\min\{(1/5)/(11/5)\} = 1/11$, $p = 1$.
 8. $p = 1$.

9. Multiply row 1 by $5/6$, then add $13/15$; $16/15$; $-6/5$; $-22/15$ times of row 1 to rows 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1			$-1/6$	$-1/6$	$-1/3$	$5/6$	$1/6$
	1		$-5/18$	$7/18$	$-2/9$	$13/18$	$5/18$
		1	$-1/9$	$-4/9$	$1/9$	$8/9$	$10/9$
			$61/9$	$-17/9$	$11/9$	$-11/9$	$11/9$

As the artificial variable x_7 has already left the basis, the resulting is a feasible simplex tableau to the original problem.

It is noted that δ_1 and δ_2 above were set to 0, and hence no progress was made in the first 3 iterations due to degeneracy. Although parameters $\delta_i, i = 1, \dots, m$, are required to be nonnegative theoretically, therefore, they should be set to different positive values practically. Moreover, it seems appropriate to set $\delta_i \ll 1$ so that $\|\bar{a}_{n+1}\|$ is not too large.

14.3 Perturbation of Reduced Cost

As a result of not involving reduced costs, Algorithm 14.4.2 could generate a feasible solution that is far from dual feasible, and hence the number of iterations required subsequently by Phase-II would be quite large. In this aspect, the Phase-I method, presented in this section, seems to be advantageous (Pan 2000).

It solves an auxiliary perturbed problem. Let (3.6) be an infeasible simplex tableau, and let $\delta_j \geq 0, j \in N$, be predetermined perturbation parameters. The auxiliary problem results from perturbing reduced costs to nonnegative values. More precisely, introduce the index set

$$J = \{j \in N | \bar{z}_j < \delta_j\}, \quad \bar{J} = N \setminus J. \quad (14.19)$$

Replacing reduced costs $\bar{z}_j$ in the simplex tableau by

$$\bar{z}'_j = \begin{cases} \delta_j, & \text{if } j \in J, \\ \bar{z}_j, & \text{if } j \in N/J \end{cases} \quad (14.20)$$

leads to a dual feasible tableau, associated with a perturbed problem. As a result, the dual simplex algorithm can get itself started, until optimality of the perturbed problem is achieved, or dual unboundedness of it is detected.

Theorem 14.3.1 *If the auxiliary perturbed problem is dual unbounded, the original problem is infeasible.*

Proof The perturbed problem has the same constraints as the original problem. Unboundedness of the former implies that there is no feasible solution to the former, and hence so to the latter. $\square$

Assume that optimality of the perturbed problem is achieved. We might as well assume that (3.6) is again the final tableau, associated with basis B . Its right-hand side is now non-negative. Consequently, computing $\hat{z}_N = c_N - N^T B^{-1} c_B$ (restoring reduced costs of the original problem) and covering $\bar{z}_N$ with $\hat{z}_N$ lead to a feasible simplex tableau to the original problem.

The overall steps are put into the following algorithm.

Algorithm 14.3.1 (Tableau Phase-I: Reduced Cost Perturbation) Assume we are given the perturbation parameters $\delta_j \geq 0$, $j \in N$. Initial: simplex tableau of form (3.6). This algorithm finds a feasible simplex tableau to the standard LP problem.

1. Perturb: $\bar{z}_j = \delta_j, \forall j \in \{j \in N \mid \bar{z}_j < \delta_j\}$.
2. Call the dual simplex algorithm.
3. Stop if it terminates at step 3 (infeasible problem).
4. If it terminates at step 2, compute $\bar{z}_N = c_N - N^T B^{-1} c_B$.
5. Stop (feasibility achieved).

Note There exists a perturbation-free variant of the preceding algorithm: in each iteration, one uses $\bar{z}_N$ determined by (14.20) in place of $\bar{z}_N$ for the minimum-ratio test, therefore saving computations for restoring reduced costs.

Algorithm 14.3.1 performed very well in preliminary computational experiments (Pan, 2000). A reason seems to be that not all perturbed reduced costs affect the final outcome. Sometimes, reduced costs restored in step 4 of Algorithm 14.3.1 are themselves nonnegative, and hence there is no need for carrying out Phase-II. In fact, it is verified that the manipulations in step 2 amount to solving the following problem:

$$\begin{aligned} \min & b^T y, \\ \text{s.t. } & A^T y \leq \hat{c}, \end{aligned}$$

where

$$\hat{c}_j = \begin{cases} a_j^T B^{-T} c_B + \delta_j, & \text{if } j \in J, \\ \bar{z}_j, & \text{if } j \in N/J. \end{cases}$$

That is to say, perturbing $\bar{z}_j < \delta_j$ to $\bar{z}_j = \delta_j$ amounts to slackening dual constraint $a_j^T y \leq c_j$ to

$$a_j^T y \leq a_j^T B^{-T} c_B + \delta_j = c_j + (\delta_j - c_j + a_j^T B^{-T} c_B) = c_j + (\delta_j - \bar{z}_j).$$

If the slackened constraints are all inactive for dual optimal solutions, or in other words, nonnegative constraints of primal variables, associated with perturbed reduced costs, are active for primal optimal solutions, the primal and dual optimal solutions will not be affected by such perturbations at all (for detailed analysis, see Pan, 2000).

Example 14.3.1 Solve the following problem using Algorithm 14.3.1 in Phase-I:

$$\begin{aligned} \min f &= x_1 + x_2 - 3x_3, \\ \text{s.t. } -2x_1 - x_2 + 4x_3 + x_5 &= -4, \\ x_1 - 2x_2 + x_3 + x_6 &= 5, \\ -x_1 + 2x_3 + x_4 &= -3, \\ x_j &\geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Answer

Phase-I.

Initial: There is an available simplex tableau ($B = \{5, 6, 4\}$) to this problem. Perturb negative reduced cost -3 to 0 , so that it becomes dual feasible. As an illustration for implementation, we add the perturbed reduced costs at the bottom ($\delta_3 = 0$).

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-2^*	-1	4		1		-4
1	-2	1			1	5
-1		2	1			-3
1	1	-3				
1	1					

Taking the bottom row as the objective row, execute the dual simplex algorithm. Note that the second (objective) bottom row does not take a part in pivoting.

Iteration 1:

- $\min\{-4, 5, -3\} = -4 < 0$, $p = 1$.
- $J = \{1, 2\} \neq \emptyset$.
- $\min\{-1/(-2), -1/(-1)\} = 1/2$, $q = 1$.
- Multiply row 1 by $-1/2$, then add $-1; 1; -1; -1$ times of row 1 to rows 2, 3, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	1/2	-2		-1/2		2
	-5/2	3		1/2	1	3
	1/2		1	-1/2*		-1
	1/2	-1		1/2		-2
	1/2	2		1/2		-2

Iteration 2:

1. $\min\{2, 3, -1\} = -1 < 0$, $p = 3$.
3. $J = \{5\} \neq \emptyset$.
4. $\min\{-(1/2)/(-1/2)\} = 1$, $q = 5$.
5. Multiply row 3 by -2 , then add $1/2$, $-1/2$, $-1/2$, $-1/2$ times of row 3 to rows 1, 2, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1		-2	-1			3
	-2	3*	1		1	2
	-1		-2	1		2
	1	-1	1			-3
	1	2	1			-3

Optimality of the perturbed tableau is achieved. Deleting the perturbed (bottom) row gives a feasible simplex tableau of the original problem.

Phase-II.

Iteration 3: call the simplex algorithm.

Select x_3 column as the pivot column and the 2nd row as the pivot row. Multiply row 2 by $1/3$, then add 2 ; 1 ; -2 times of row 2 to rows 1, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	-4/3		-1/3		2/3	13/3
	-2/3	1	1/3		1/3	2/3
	-1		-2	1		2
	1/3		4/3		1/3	-7/3

The reduced costs are all nonnegative, optimality is hence achieved. The optimal basic solution and optimal value:

$$x^* = (13/3, 0, 2/3, 0, 2, 0)^T, \quad f^* = 7/3.$$

Although the perturbation parameter was set to 0 in the preceding example, the positive parameter value should be practically favorable for the sake of anti-

degeneracy. It seems to be suitable to use

$$\delta_j \geq \epsilon > 0,$$

where ϵ is the feasibility tolerance ($\epsilon = 10^{-6}$, or so). Computational experiences indicate that the perturbation method is not sensitive to the magnitude of the parameters although exceeding 10^{-1} seems to be inadvisable.

14.4 Using Most-Obtuse-Angle Column Rule

It is somehow surprising that the dual pivot rule can be used to achieve primal feasibility and perform well in preliminary computational experiments (Pan 1994). That is to say, whether the reduced costs are all nonnegative or not, the classical dual pivot rule can be applied. The Phase-I method presented in this section is even simpler, as it involves neither the minimum-ratio test nor reduced costs.

Assume that a simplex tableau, say (3.6), is infeasible, and a pivot row index p has been selected by the conventional rule

$$p \in \arg \min \{\bar{b}_i \mid i = 1, \dots, m\}. \quad (14.21)$$

Therefore, it holds that $\bar{b}_p < 0$.

The key of the presented Phase-I method is to use the following column rule.

Rule 14.4.1 (Most-Obtuse-Angle Column Rule) *Determine the pivot column index*

$$q \in \arg \min_{j \in N} \bar{a}_{p j}. \quad (14.22)$$

In case of $\bar{a}_{p q} < 0$, after the basis change, the basic infeasible variable x_{j_p} becomes nonbasic (hence feasible), and the p th component of the right-hand side of the tableau changes from negative to positive, that is, $\hat{b}_p = \bar{b}_p / \bar{a}_{p q} > 0$.

The associated steps can be summarized into the following algorithm.

Algorithm 14.4.1 (Tableau Phase-I: The Most-Obtuse-Angle Column Rule)

Initial: simplex tableau of form (3.6). This algorithm finds a feasible simplex tableau to the standard LP problem (12.1).

1. Select row index $p \in \arg \min \{\bar{b}_i \mid i = 1, \dots, m\}$.
2. Stop if $\bar{b}_p \geq 0$.
3. Determine column index $q \in \arg \min_{j \in N} \bar{a}_{p j}$.
4. Stop if $\bar{a}_{p, q} \geq 0$.
5. Carry out elementary transformations to convert $\bar{a}_{p, q}$ to 1 and eliminate all other nonzeros of the column.
6. Go to step 1.

It is noted that only the right-hand side and the pivot row of the simplex tableau are used in each iteration of the preceding Algorithm.

Theorem 14.4.1 *Assume termination of Algorithm 14.4.1. It terminates either at*

- (i) *Step 2, obtaining a feasible basic solution or at*
- (ii) *Step 4, detecting the infeasibility of the problem*

Proof It is clear that if it terminates at step 2, then $\bar{b} \geq 0$, and hence a feasible simplex tableau is achieved. In case of the termination at step 4, it holds that $\bar{b}_p < 0$ and that $\bar{a}_{pj} \geq 0, \forall j \in N$. According to Theorem 3.3.1 of Part I, there is no feasible solution to the problem. $\square$

The algorithm is not a monotone one, that is, the objective value would not monotonically change in the solution process. Its finiteness is hence not guaranteed even if nondegeneracy is assumed. Although Guerrero-Garcia and Santos-Palomo (2005) offer a cycling instance for this algorithm, it may be still expected that cycling hardly happens in practice.

The geometric meaning of the most-obtuse-angle column rule can be revealed from the dual space. Let $(\bar{y}, \bar{z})$ be the current dual basic solution. The $(m+n)$ -dimensional vector

$$d \triangleq \begin{pmatrix} h \\ \sigma_N \\ \sigma_B \end{pmatrix} = \begin{pmatrix} -B^{-T} \\ N^T B^{-T} \\ I \end{pmatrix} e_p \quad (14.23)$$

is an ascent direction. In fact, it is verified that d is in the null space of coefficient matrix of the dual constraint equalities, satisfying

$$A^T d + \sigma = 0,$$

and forms an acute angle with the dual objective gradient, i.e.,

$$(b^T, 0, 0)d = b^T h = -\bar{x}_{j_p} > 0.$$

The negative σ_q implies that d forms the maximum possible obtuse angle with the gradient e_{m+q} of the dual nonnegative constraint $z_q \geq 0$. Therefore, if the ascent direction is close to the dual objective gradient, then e_{m+q} tends to form the most obtuse angle with the dual objective gradient. According to the heuristic characteristic of optimal basis (Sect. 2.7 of Part I), it is favorable to let the dual constraint $z_q \geq 0$ be satisfied as equality, accordingly entering x_q to the basis.

We stress that the use of the conventional dual rule in step 1 is just for simplicity, which does not mean it is the best choice. In fact, it should be much better to use the rules presented in Chap. 13 instead. From a geometric point of view, the most-obtuse-angle column rule should be best matched by the dual steepest-edge rule II.

Nevertheless, Rule 14.4.1 does not employ information associated with the objective function at all. Considering the extent to which the current vertex violates dual constraints, we suggest the following variant.

Rule 14.4.2 (Variant of Most-Obtuse-Angle Column Rule) Assume we are given a constant τ such that $0 < \tau \leq 1$. Select pivot column index

$$q \in \arg \min \{\bar{z}_j \mid \bar{a}_{pj} \leq \tau \theta, j \in N\}, \quad \theta = \min_{j \in N} \bar{a}_{pj} < 0.$$

In order to widen the range of choice, τ should be close to 1, so that in the case when $\bar{z}_N$ has negative components, the dual constraint inequality, that is violated the most, will be satisfied as equality. It might be suitable to set $\sigma = 0.95$, or so.

The Phase-I algorithm seems very promising, not to mention its dual algorithm, which performs very well in solving large and sparse problems in computational experiments (Sect. 15.4). It is the author's favorite Phase-I algorithm not only because of its simplicity and good numerical stability but also because of its geometric meaning based on the most-obtuse-angle heuristic of the optimal solution.

Example 14.4.1 Find a feasible simplex tableau to the following problem by Algorithm 14.4.1:

$$\begin{aligned} \min f &= 2x_1 - x_2, \\ \text{s.t. } 2x_1 - x_2 + x_3 &= -2, \\ x_1 + 2x_2 + x_4 &= 3, \\ -8x_1 + x_2 - x_4 + x_5 &= -4, \\ x_j &\geq 0, \quad j = 1, \dots, 5. \end{aligned}$$

Answer Initial: adding the second equality constraint to the third to eliminate x_4 . This results in the following simplex tableau:

x_1	x_2	x_3	x_4	x_5	RHS
2	-1*	1			-2
1	2		1		3
-7	3			1	-1
2	-1				

Iteration 1:

1. $\min\{-2, 3, -1\} = -2 < 0, p = 1$.
3. $\min\{2, -1\} = -1 < 0, q = 2$.
5. Multiply row 1 by -1 , then add $-2, -3, 1$ times of row 1 to rows 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	RHS
-2	1	-1			2
5		2	1		-1
-1*		3		1	-7
		-1			2

Iteration 2:

1. $\min\{2, -1, -7\} = -7 < 0$, $p = 3$.
3. $\min\{-1, 3\} = -1 < 0$, $q = 1$.
5. Add 2, 5 times of row 3 to rows 1, 2, respectively:

x_1	x_2	x_3	x_4	x_5	RHS
	1	-7			16
		17	1		34
1		-3		-1	7
		-1			2

The right-hand side of the above tableau is nonnegative, hence the obtained is a feasible simplex tableau, from which Phase-II can get itself started.

In practice, usually used is the following revised version of the preceding algorithm.

Algorithm 14.4.2 (Phase-I: Most-Obtuse-Angle Column Rule) Initial: (B, N) , B^{-1} , $\bar{x}_B = B^{-1}b$. This algorithm finds a feasible basic solution to the standard LP problem (12.1).

1. Select pivot row index $p \in \arg \min\{\bar{x}_{j_i} \mid i = 1, \dots, m\}$.
2. Stop if $\bar{x}_{j_p} \geq 0$ (feasibility achieved).
3. Compute $h = B^{-T}e_p$, $\sigma_N = N^T h$.
4. Determine pivot column index $q \in \arg \min_{j \in N} \sigma_j$.
5. Stop if $\sigma_q \geq 0$ (infeasible problem).
6. Compute $\bar{a}_q = B^{-1}a_q$ and $\alpha = \bar{x}_{j_p}/\sigma_q$.
7. Set $\bar{x}_q = \alpha$, and update: $\bar{x}_B = \bar{x}_B - \alpha \bar{a}_q$ if $\alpha \neq 0$.
8. Update B^{-1} .
9. Update (B, N) by exchanging j_p and q .
10. Go to step 1.

References

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Chapter 15

Dual Simplex Phase-I Method



The mission of a *dual Phase-I* procedure is to provide an initial dual feasible basis to get the dual simplex method started.

In the dual simplex context, one can establish a dual Phase-I by introducing $n - m$ dual artificial variables, with or without penalty terms in the objective function, just as in the primal context. We will not touch these methods because they are not suitable for applications (interested readers are referred to, e.g., Padberg 1995).

This chapter is devoted to practicable dual Phase-I methods, which can be regarded as dual counterparts of the simplex Phase-I methods.

15.1 Dual Infeasibility-Sum Algorithm

This section will present a dual version of the infeasibility-sum method. Appearing in the early days, the basic idea of the method draws new attention and is put into effect in practice in recent years because of its good performance (see, e.g., Koberstain and Suhl, 2007).

Assume that dual basic solution

$$\bar{y} = B^{-T}c_B; \quad \bar{z}_N = c_N - N^T\bar{y}, \quad \bar{z}_B = 0 \quad (15.1)$$

is infeasible, i.e., $\bar{z}_N \not\geq 0$. Introduce the index set

$$J = \{j \in N \mid \bar{z}_j < 0\}. \quad (15.2)$$

Taking the second term of the sum of the dual infeasible variables, i.e.,

$$\sum_{j \in J} z_j = \sum_{j \in J} c_j - \left(\sum_{j \in J} a_j \right)^T y$$

as objective function, we construct the following auxiliary problem

$$\begin{aligned} \max & -(\sum_{j \in J} a_j)^T y, \\ \text{s.t. } & B^T y + z_B = c_B, \\ & N^T y + z_N = c_N, \\ & z_B \geq 0; \quad z_j \geq 0, \quad j \in N \setminus J, \end{aligned} \tag{15.3}$$

whose nonnegativity constraints only impose on the variables corresponding to the nonnegative components of the dual basic solution.

Clearly, there is a basic solution to the primal problem of the auxiliary problem; i.e.,

$$\bar{x}_B = -B^{-1} \sum_{j \in J} a_j, \quad \bar{x}_N = 0, \tag{15.4}$$

where $\bar{x}_B$ is termed *auxiliary right-hand side*.

Theorem 15.1.1 (Dual Infeasibility Test) *If $\bar{x}_B \geq 0$, the original problem is infeasible or unbounded.*

Proof Assume that y' is a feasible solution to the original dual problem such that

$$z' = c_N - N^T y' \geq 0.$$

Thus, it holds that

$$\sum_{j \in J} z'_j = \sum_{j \in J} c_j - \left(\sum_{j \in J} a_j \right)^T y' \geq 0. \tag{15.5}$$

On the other hand, it is known from the assumption that (15.4) gives a feasible basic solution $\bar{x}$ to the primal problem of (15.3), and (15.1) gives a complementary feasible solution $(\bar{y}, \bar{z})$ of (15.3), hence the two are a pair of optimal solutions. It is clear that (y', z') is a feasible solution to (15.3), associated with objective value no more than the optimal value, hence

$$\sum_{j \in J} z'_j \leq \sum_{j \in J} \bar{z}_j < 0,$$

where the last inequality comes from (15.2). This contradicts (15.5). Therefore, there is no feasible solution to the original dual problem when $\bar{x}_B \geq 0$. According to the duality principle, the original (primal) problem is infeasible or unbounded. $\square$

In case when $\bar{x}_B \not\geq 0$, carry out one dual simplex iteration with a modified column pivot rule. To this end, assume that row index p has been determined such that

$$\bar{x}_{j_p} < 0. \quad (15.6)$$

Consider the following scheme for updating dual solution:

$$\hat{y} = \bar{y} - \beta B^{-T} e_p, \quad \hat{z} = \bar{z} + \beta v, \quad v = A^T B^{-T} e_p, \quad (15.7)$$

where stepsize β is defined by the rule below:

Rule 15.1.1 (Auxiliary Column Rule) *Select pivot column index q such that*

$$\beta = -\bar{z}_q/v_q = \begin{cases} \min\{-\bar{z}_j/v_j \mid v_j < 0, j \in \bar{J}\} & \text{if } \{j \in N \setminus J \mid v_j < 0\} \neq \emptyset, \\ \max\{-\bar{z}_j/v_j \mid v_j > 0, j \in J\} & \text{otherwise.} \end{cases} \quad (15.8)$$

The following results are counterparts of Lemma 14.1.2 and Theorem 14.1.3 in Sect. 14.1, proofs of which are therefore omitted.

Lemma 15.1.2 *Let p be a row index, satisfying (15.6). Then (15.8) is well-defined, and the expression (15.7) gives a new feasible basic to the auxiliary problem.*

Theorem 15.1.3 *Assume that the original dual problem is feasible. If the dual feasible variables are all nondegenerate, the number of dual infeasible variables strictly decreases in finitely many iterations.*

If the number of infeasible variables reduces but does not vanish after an iteration, the process is carried out with an updated auxiliary dual problem, until dual feasibility is attained or dual infeasibility or unboundedness of the original problem is detected.

The steps are put in the following algorithm.

Algorithm 15.1.1 (Dual Phase-I: Dual Infeasibility-Sum) Initial: (B, N) , B^{-1} , $\bar{y}$, $\bar{z}$. This algorithm finds a dual feasible solution by handling auxiliary programs of form (15.3).

1. If $\bar{z} \geq 0$, compute $\bar{x}_B = B^{-1}b$, and stop (dual feasibility achieved).
2. Compute $\bar{x}_B = -B^{-1} \sum_{j \in J} a_j$, where $J = \{j \in N \mid \bar{z}_j < 0\}$.
3. Stop if $\bar{x}_B \geq 0$ (the original dual problem is infeasible or unbounded).
4. Call one iteration of the dual simplex algorithm with the Rule 15.1.1 used for column pivoting.
5. Go to step 1.

Example 15.1.1 Find a dual feasible basic solution of the following problem by Algorithm 15.1.1:

$$\begin{aligned} \min f &= -x_1 + x_2 - 2x_3, \\ \text{s.t. } x_1 - 3x_2 - 2x_3 + x_4 &= -4, \\ -x_1 + x_2 - 4x_3 + x_5 &= -2, \\ -3x_1 + x_2 + x_3 + x_6 &= 8, \\ x_j &\geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Answer Taking $J = \{1, 3\}$, construct an auxiliary right-hand side $\bar{x}_B = -(-1, -5, -2)^T$. Since components of it are all nonnegative, it is asserted that the problem is infeasible or unbounded. In fact, the problem is just the same as that in Exam. 14.1.1, where the infeasibility of the problem is declared in Phase-II.

Example 15.1.2 Find a dual feasible basic solution to the following problem by Algorithm 15.1.1:

$$\begin{aligned} \min f &= -5x_1 - 7x_2 + x_4, \\ \text{s.t. } x_1 + 2x_2 + x_3 &= 3, \\ 2x_1 + x_2 - x_4 + x_5 &= -2, \\ -x_1 + x_2 + x_4 + x_6 &= -1, \\ x_j &\geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Answer Taking $J = \{1, 2\}$ and $N \setminus J = \{3, 4, 5, 6\}$, construct the auxiliary problem. $B = \{3, 5, 6\}$, $N = \{1, 2, 4\}$, $B^{-1} = I$, $\bar{z}_N = (-5, -7, 1)^T$. The auxiliary right-hand side is $\bar{x}_B = (-3, -3, 0)^T$.

Iteration 1: Carry out a single iteration of the revised dual algorithm; in which (15.8) is used for column pivoting instead.

1. $\min\{-3, -3, 0\} = -3$, $p = 1$, so x_3 leaves the basis.
3. $\sigma_N = N^T B^{-T} e_p = (1, 2, 0)^T$.
5. $\beta = \max\{-\bar{z}_j/\sigma_j \mid \sigma_j < 0, j = 1, 2, 3\} = \max\{5/1, 7/2\} = 5$, $q = 1$, so x_1 enters the basis.
6. $\bar{z}_N = \bar{z}_N + \beta\sigma_N = (-5, -7, 1)^T + 5 \times (1, 2, 0)^T = (0, 3, 1)^T$, $\bar{z}_{j_p} = 5$.
7. $\bar{a}_q = B^{-1}a_q = (1, 2, -1)^T$.
8. $\alpha = \bar{x}_{j_p}/\sigma_q = -3/1 = -3$,
 $\bar{x}_B = \bar{x}_B - \alpha\bar{a}_q = (-3, -3, 0)^T - (-3) \times (1, 2, -1)^T = (0, 3, -3)^T$, $\bar{x}_1 = \alpha = -3$.
9. $B^{-1} = \begin{pmatrix} 1 & & \\ -2 & 1 & \\ & & 1 \end{pmatrix}$.
10. $B = \{1, 5, 6\}$, $N = \{3, 2, 4\}$, $\bar{z}_N = (5, 3, 1)^T$, $J = \emptyset$, $\bar{x}_B = B^{-1}b = (3, -8, 2)^T$. Dual feasibility attained.

Algorithm 15.1.1 is easily converted to its tableau version. Note only that p th row of the simplex tableau is $v^T = e_p^T B^{-1} A$, x_j column is equal to $B^{-1} a_j$, and in step 2 the following formula is used, instead:

$$\bar{x}_B = - \sum_{j \in J} \bar{a}_j, \quad J = \{j \in N | \bar{z}_j < 0\}.$$

We illustrate with the preceding example.

Example 15.1.3 Find a dual feasible simplex tableau for Example 15.1.2 by the tableau version of Algorithm 15.1.1.

Answer Initial: Clearly, there is a simplex tableau to the problem; i.e.,

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1*	2	1				3
2	1		-1	1		-2
-1	1		1		1	-1
-5	-7		1			

Iteration 1:

$$J = \{1, 2\}, \quad \bar{J} = \{4\}, \quad \bar{x}_B = -((1, 2, -1)^T + (2, 1, 1)^T) = (-3, -3, 0)^T$$

$$p = 1; \beta = \max\{-(-5)/1, -(-7)/2\} = 5, \quad q = 1.$$

Respectively add -2, 1, 5 times of row 1 to rows 2,3,4, resulting in the following tableau:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1*	2	1				3
	-3	-2	-1	1		-8
	3	1	1		1	2
	3	5	1			15

which is dual feasible.

15.2 Dual Single-Artificial-Variable Algorithm

In this section, the idea of the single-artificial-variable method, described in Sect. 14.2, is applied to the dual problem to derive a dual Phase-I method. To this end, we introduce an artificial variable to the dual problem. The derived method has been used as dual Phase-I for the so-called *dual projective pivot algorithm* (Pan 2005).

As in the primal context, there are two schemes:

Scheme 1: The auxiliary dual problem is based on the original data of the problem.

Given any vector $v \geq 0$. Introducing the artificial variable y_{m+1} , we construct the following auxiliary dual problem from constraints of the dual problem:

$$\begin{aligned} \max \quad & y_{m+1}, \\ \text{s.t.} \quad & A^T y + (v - c)y_{m+1} + z = c, \quad z \geq 0 \end{aligned} \quad (15.9)$$

which has the feasible solution

$$\begin{pmatrix} \hat{y} \\ \hat{y}_{m+1} \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \begin{pmatrix} \hat{z} \\ \hat{z}_{n+1} \end{pmatrix} = \begin{pmatrix} v \\ 1 \end{pmatrix}$$

Theorem 15.2.1 *There is an optimal solution, associated with a nonpositive objective value, to the auxiliary dual problem (15.9). If the optimal value is equal to zero, then the (y, z) part of its optimal basic solution is feasible basic to the original dual problem; otherwise, the latter is infeasible.*

Proof To the auxiliary dual problem (15.9), there is a feasible solution with objective value bounded above by 0, hence there is an optimal solution with nonpositive optimal value. Let $(y', y'_{m+1}, z', z'_{n+1})$ be its optimal solution, where $\bar{z}'_B, \bar{z}'_N \geq 0$. Assume that the optimal value is equal to zero, i.e., $y'_{m+1} = 0$. Since the optimal solution satisfies equality constraints of (15.9), it holds that

$$\begin{pmatrix} B^T \\ N^T \\ 0 \end{pmatrix} y' + \begin{pmatrix} z'_B \\ z'_N \\ z'_{n+1} \end{pmatrix} = \begin{pmatrix} c_B \\ c_N \\ 0 \end{pmatrix}$$

Thus, (y', z') is a feasible basic solution to the original dual problem.

Assume now that the optimal value is less than 0, i.e., $y'_{m+1} < 0$. If the original dual problem has a feasible solution (y'', z'') , then (y'', z'') together with $y''_{m+1} = z''_{n+1} = 0$ is a feasible solution to (15.9), therefore leading to $y'_{m+1} \geq y''_{m+1} = 0$, which contradict $y'_{m+1} < 0$. Therefore, the original dual problem is infeasible if the optimal value of (15.9) is less than 0. $\square$

According to the preceding Theorem, if the dual simplex method is used to solve the auxiliary dual problem, a dual feasible basic solution to the original problem can be obtained, if any.

Scheme 2: The auxiliary dual problem is based on a canonical form of the constraint system.

Let (B, N) be basis and nonbasis of the standard LP problem, associated with a simplex tableau, say (3.6). Given any $\hat{z}_N \geq 0$. Introducing artificial variable x_{n+1} ,

construct the following auxiliary problem:

$$\begin{aligned} \min f &= \bar{f} + \bar{z}_N^T x_N, \\ \text{s.t. } \begin{pmatrix} I & \bar{N} & 0 \\ 0 & (\hat{z}_N - \bar{z}_N)^T & 1 \end{pmatrix} \begin{pmatrix} x_B \\ x_N \\ x_{n+1} \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad x \geq 0, \quad x_{n+1} \geq 1, \end{aligned} \quad (15.10)$$

where

$$\bar{N} = B^{-1}N, \quad \bar{z}_N = c_N - N^T B^{-T} c_B, \quad \bar{f} = b^T B^{-T} c_B.$$

Clearly, $(\bar{x} = 0, \bar{x}_{n+1} = 1)$ is a solution to the preceding problem. We are interested in its dual problem; i.e.,

$$\begin{aligned} \max w &= \bar{f} + y_{m+1}, \\ \text{s.t. } \begin{pmatrix} z_B \\ z_N \\ z_{n+1} \end{pmatrix} &= \begin{pmatrix} 0 \\ \bar{z}_N \\ 0 \end{pmatrix} - \begin{pmatrix} I & 0 \\ \bar{N}^T & \hat{z}_N - \bar{z}_N \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y \\ y_{m+1} \end{pmatrix}, \quad z, z_{n+1} \geq 0. \end{aligned} \quad (15.11)$$

The above is just the auxiliary dual problem we want, as it is much simpler than (15.9), and has a feasible solution

$$\begin{pmatrix} \hat{y} \\ \hat{y}_{m+1} \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \begin{pmatrix} \hat{z}_B \\ \hat{z}_N \\ \hat{z}_{n+1} \end{pmatrix} = \begin{pmatrix} 0 \\ \hat{z}_N \\ 1 \end{pmatrix} \geq 0. \quad (15.12)$$

Like Theorem 15.2.1, it can be shown that if (15.11) is solved with a zero optimal value, a dual feasible basic solution to the original problem is obtained; otherwise, there is no dual feasible solution, or the original problem is infeasible or unbounded. From the last equality constraint, $z_{n+1} = -y_{m+1}$, of (15.11), it is known that $\max y_{m+1}$ and $\min z_{n+1}$ are equal, if any.

Like the dual simplex method, it is also possible to solve (15.11) via the simplex tableau. In fact, problem (15.10) has the following simplex tableau, which is not dual feasible:

x_B	x_N	x_{n+1}	f	RHS
I	$\bar{N}$			
	$(\hat{z}_N - \bar{z}_N)^T$	1		1
	$\bar{z}_N^T$		-1	$-\bar{f}$

Adding the $(m + 1)$ th row as the bottom row gives

x_B	x_N	x_{n+1}	f	RHS
I	$\bar{N}$			
	$(\hat{z}_N - \bar{z}_N)^T$	1		1
	$\hat{z}_N^T$	$\hat{z}_{n+1}$	-1	$-\bar{f} + 1$

(15.13)

where $\hat{z}_{n+1} = 1$. The preceding tableau, corresponding to the dual feasible solution defined by (15.12), can be taken as an auxiliary initial tableau though not a simplex one. To turn it to a simplex one, one more basic variable is required. To this end, determine a dual stepsize

$$\beta = \hat{z}_q / (\hat{z}_q - \bar{z}_q) = \min\{\hat{z}_j / (\hat{z}_j - \bar{z}_j) \mid \hat{z}_j - \bar{z}_j > 0, j \in N\}. \quad (15.14)$$

If $\beta \geq 1$, set $\beta = 1$, $q = n + 1$. Converting the entry in the $(m + 1)$ th row and q column to 1, and eliminating the other nonzeros in the column by elementary transformations, so that q -indexed column becomes basic. If it happens that $q = n + 1$, then deleting the $(m + 1)$ th row and x_{n+1} column, and putting $\bar{b}$ in the RHS column gives a dual feasible simplex tableau to the original problem. In a general case of $q \neq n + 1$, obtained is only a dual feasible tableau to the auxiliary problem, which can be solved by the dual simple method.

Theoretically, any nonnegative vector $\hat{z}_N$ can be used to construct an auxiliary problem. A simple way is to set $\hat{z}_N = e$. But another way would be more suitable for sparse computations: for given $\delta_j \geq 0$, $j \in N$. Setting

$$\hat{z}_j = \begin{cases} \delta_j, & \text{if } \bar{z}_j < \delta_j, \\ 0, & \text{otherwise,} \end{cases} \quad j \in N.$$

The associated steps can be summarized into the following algorithm.

Algorithm 15.2.1 (Tableau Dual Phase-I: Single-Artificial-Variable) Initial: auxiliary tableau of form (15.13). This algorithm finds a dual feasible tableau to the standard LP problem.

1. Determine β and q such that

$$\beta = \hat{z}_q / (\hat{z}_q - \bar{z}_q) = \min\{\hat{z}_j / (\hat{z}_j - \bar{z}_j) \mid \hat{z}_j - \bar{z}_j > 0, j \in N\}.$$

2. If $\beta \geq 1$, set $\beta = 1$, $q = n + 1$.
3. Convert the entry in the $(m + 1)$ th row and q column to 1, and eliminate all other nonzeros in the column by elementary transformations.
4. If $q = n + 1$, delete the $(m + 1)$ th row and x_{n+1} column, and stop (dual feasibility achieved).
5. Call the dual simplex algorithm (the $(n + 1)$ -indexed column is taken as nonbasic at the beginning).

Note In the initial tableau, the first $m + 1$ components of the x_{n+1} column and the RHS column are equal. When carrying out the preceding algorithm, the $(n + 1)$ -indexed column may be also deemed as the auxiliary right-hand side, and the right-hand side $\bar{b}$ of the original problem is filled in RHS column (but does not take a part in pivoting). In this way, once x_{n+1} enters the basis, eliminating the $(m + 1)$ th row and $(n + 1)$ -indexed column gives a dual feasible tableau to the original problem. The f column can be omitted, as it does not change at all in the solution process.

It is not difficult to transfer Algorithm 15.2.1 to its revised version, which is omitted here.

Example 15.2.1 Solve the following problem using Algorithm 15.2.1 as dual Phase-I:

$$\begin{aligned} \min \quad & f = 5x_1 - 2x_2 - x_3 + x_4, \\ \text{s.t.} \quad & 2x_1 - x_2 + x_3 + x_4 = 5, \\ & -5x_1 + 3x_2 + 2x_3 + x_5 = -2, \\ & -x_1 + 2x_2 - x_3 + x_6 = -1, \\ & x_j \geq 0, \quad j = 1, \dots, 6 \end{aligned}$$

Answer From the first constraint equality, it follows that

$$x_4 = 5 - 2x_1 + x_2 - x_3.$$

Substitute it to the objective function to eliminate x_4 :

$$f = 5 + 3x_1 - x_2 - 2x_3.$$

Setting $\hat{z}_1 = 0$, $\hat{z}_2 = \hat{z}_3 = 1$, construct the following auxiliary initial simplex tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
2	-1	1	1				5
-5	3	2		1			-2
-1	2	-1			1		-1
-3	$1 + 1$	$1 + 2^*$				1	1
	1	1				1	$-5 + 1$

Phase-I: call Algorithm 15.2.1. $m + 1 = 4$.

Iteration 1:

1. $\beta = \min\{1/(1+1), 1/(1+2), \dots\} = 1/3 < 1, q = 3$.
3. Multiply row 4 by $1/3$, then add $-1, -2, 1, -1$ times of row 4 to rows 1, 2, 3, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
3	$-5/3$		1			$-1/3$	$14/3$
-3^*	$5/3$			1		$-2/3$	$-8/3$
-2	$8/3$				1	$1/3$	$-2/3$
-1	$2/3$	1				$1/3$	$1/3$
1	$1/3$					$2/3$	$-13/3$

5. Call the dual simplex algorithm (tableau form). Note that x_7 column also represents the auxiliary right-hand side.

Iteration 2:

1. $\min\{-1/3, -2/3, 1/3, 1/3\} = -2/3 < 0, p = 2$.
4. $\min\{-1/(-3), (-2/3)/(-2/3)\} = 1/3, q = 1$.
5. Multiply row 2 by $-1/3$, then add $-3, 2, 1, -1$ times of row 2 to rows 1, 3, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
			1	1		-1^*	2
1	$-5/9$			$-1/3$		$2/9$	$8/9$
	$14/9$			$-2/3$	1	$7/9$	$10/9$
	$1/9$	1		$-1/3$		$5/9$	$11/9$
	$8/9$			$1/3$		$4/9$	$-47/9$

Iteration 3:

1. $\min\{-1, 2/9, 7/9, 5/9\} = -1 < 0, p = 1$.
4. $\min\{-(4/9)/(-1)\} = 4/9, q = 7$.
5. Multiply row 1 by -1 , then add $-2/9, -7/9, -5/9, -4/9$ times of row 1 to rows 2, 3, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
			-1	-1		1	-2
1	-5/9		2/9	-1/9			4/3
	14/9		7/9	1/9	1		8/3
	1/9	1	5/9	2/9			7/3
	8/9		4/9	7/9			-13/3

Optimality of the auxiliary problem is achieved with the artificial variable x_7 entering the basis. Deleting the first row and the seventh column leads to a dual feasible simplex tableau to the original problem; i.e.,

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	-5/9		2/9	-1/9		4/3
	14/9		7/9	1/9	1	8/3
	1/9	1	5/9	2/9		7/3
	8/9		4/9	7/9		-13/3

$\min\{4/3, 8/3, 7/3\} \geq 0$. Optimality is achieved, hence there is no need for dual Phase-II. The optimal basic solution and objective value are

$$x^* = (4/3, 0, 7/3, 0, 0, 8/3)^T, \quad f^* = 13/3.$$

15.3 Perturbation of the Right-Hand Side

The basic idea of this method is simple. It perturbs the right-hand side of the simplex tableau to nonnegative and converts the resulting tableau to an optimal one via the simplex method; then, it recomputes the right-hand side from the original data to obtain a dual feasible simplex tableau.

Let (3.6) be an initial simplex tableau and let the perturbation parameter be given by

$$\delta_i \geq 0, \quad i = 1, \dots, m.$$

To weaken effect of degeneracy, we recommend that δ_i take positive values, different from each other. The method replaces $\bar{b}$ by

$$\bar{b}'_i = \begin{cases} \delta_i, & \text{if } \bar{b}_i \leq \delta_i, \\ \bar{b}_i, & \text{otherwise,} \end{cases} \quad i = 1, \dots, m, \quad (15.15)$$

leading to a feasible simplex tableau.

We claim the following without proof.

Theorem 15.3.1 *If the perturbed problem is unbounded, the original problem is infeasible or unbounded.*

We might as well denote again by (3.6) the final tableau of the perturbed problem, associated with basis B . Assume that the final tableau is optimal, i.e., $\bar{z}_j \geq 0$, $j \in N$. To restore a simplex tableau to the original problem, compute $\bar{b} = B^{-1}b$ and $\bar{f} = -c_B^T \bar{b}$ to overwrite the old $\bar{b}$ and $\bar{f}$, resulting in

$$\begin{array}{c|c} \bar{A} & \bar{b} \\ \hline \bar{z}^T & -\bar{f} \end{array}$$

which is clearly a dual feasible tableau to the original problem.

The overall steps are put into the following algorithm.

Algorithm 15.3.1 (Tableau Dual Phase I: Right-Hand Side Perturbation)

Given perturbation parameter $\delta_i > 0$, $i = 1, \dots, m$, a simplex tableau of form (3.6). The algorithm finds a dual feasible simplex tableau to the standard LP problem.

1. Perturbation: $\bar{b}_i = \delta_i$, $\forall i \in \{i = 1, \dots, m \mid \bar{b}_i \leq \delta_i\}$.
2. Call the simplex algorithm.
3. Stop if termination occurs at step 3 (Infeasible or unbounded).
4. Cover $\bar{b}$, $\bar{f}$ by $\bar{b} = B^{-1}b$, $\bar{f} = -c_B^T \bar{b}$ if termination occurs at step 2.
5. Stop (dual feasibility achieved).

Note A variant of this algorithm results from not perturbing the right-hand side, but directly using $\bar{b}'$, defined by (15.15), in place of $\bar{b}$ for the minimum-ratio test, as therefore saves the restoration.

Intuitively, the perturbation of the right-hand side amounts to relaxing nonnegativity restrictions on basic variables. If these constraints are inactive at the optimal solution, then the perturbation is independent of the determination of the optimal solution. In some cases, an LP problem can be solved by Algorithm 15.3.1 alone. In fact, we can claim the following (Pan, 1999).

Theorem 15.3.2 *Assume the dual nondegeneracy of optimal solutions. If the perturbed components of the right-hand side correspond to optimal basic variables, Algorithm 15.3.1 generates an optimal simplex tableau.*

Example 15.3.1 Find a dual feasible tableau by Algorithm 15.3.1:

$$\begin{aligned} \min \quad & f = -5x_1 - 7x_2 + x_4, \\ \text{s.t.} \quad & x_1 + 2x_2 + x_3 = 3, \\ & 2x_1 + x_2 - x_4 + x_5 = -2, \\ & -x_1 + x_2 + x_4 + x_6 = -1, \\ & x_j \geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Answer Initial: A simplex tableau can be obtained from the problem. Take $\delta_2 = 1/6$, $\delta_3 = 1/12$, and put the perturbed right-hand side at the right end of the tableau.

x_1	x_2	x_3	x_4	x_5	x_6	RHS	RHS1
1	2	1				3	3
2	1		-1	1		-2	1/6
-1	1*		1		1	-1	1/12
-5	-7		1				

Call the simplex algorithm.

Iteration 1:

1. $\min\{-5, -7, 1\} = -7 < 0$, $q = 2$.
3. $I = \{1, 2, 3\} \neq \emptyset$.
4. $\min\{3/2, (1/6)/1, (1/12)/1\} = 1/12$, $p = 3$.
5. Add $-2, -1, 7$ times of row 3 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS	RHS1
3		1	-2		-2	5	17/6
3*			-2	1	-1	-1	1/12
-1	1		1		1	-1	1/12
-12			8		7	-7	-

Iteration 2:

1. $\min\{-12, 8, 7\} = -12 < 0$, $q = 1$.
3. $I = \{1, 2\} \neq \emptyset$.
4. $\min\{(17/6)/3, (1/12)/3\} = 1/36$, $p = 2$.
5. Multiply row 2 by $1/3$, then add $-3, 1, 12$ times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	RHS	RHS1
		1		-1	-1	6	11/4
1			-2/3	1/3	-1/3	-1/3	1/36
	1		1/3	1/3	2/3	-4/3	1/9
				4	3	-11	-

all reduced costs of which are nonnegative. Finally, deleting the end column gives a dual feasible simplex tableau.

15.4 Using Most-Obtuse-Angle Row Rule

In this section, we present an artificial-variable free dual Phase-I method, involving neither minimum-ratio test, nor reduced costs. It uses the so-called *most-obtuse-angle* row rule, which is the dual counterpart of the most-obtuse-angle column rule (Sect. 14.4).

Assume that the current dual solution is infeasible, and a pivot column index $q \in N$ is selected (e.g., by the conventional column rule) such that

$$\bar{z}_q = c_q - c_B^T B^{-1} a_q < 0. \quad (15.16)$$

Lemma 15.4.1 *If (15.16) holds and $B^{-1} a_q \leq 0$, then the original problem is infeasible or unbounded.*

Proof Assume that $\bar{x}$ is a feasible solution. If we define/Defining

$$d_B = -B^{-1} a_q; \quad d_q = 1; \quad d_j = 0, \quad j \in N, j \neq q \quad (15.17)$$

the vector

$$\hat{x} = \bar{x} + \alpha d$$

is a feasible solution for any $\alpha \geq 0$; in fact, it holds that

$$A\hat{x} = A\bar{x} + \alpha(Bd_B + Nd_N) = b,$$

and, since $\bar{x} \geq 0$ and $d \geq 0$, it holds that $\hat{x} \geq 0$. It is known from (15.16) that the vector d satisfies

$$c^T d = (c_B^T d_B + c_N^T d_N) = c_q - c_B^T \bar{a}_q = \bar{z}_q < 0, \quad (15.18)$$

and hence the corresponding feasible value tends to ∞ as α tends to ∞ , i.e.,

$$c^T \hat{x} = c^T \bar{x} + \alpha c^T d \rightarrow -\infty, \quad \alpha \rightarrow \infty,$$

indicating the lower unboundedness of the original problem. Therefore, the original problem is either infeasible, or feasible but unbounded below. $\square$

The key of this dual Phase-I method is to use the following row rule (Pan 1994, 1997).

Rule 15.4.1 (Most-Obtuse-Angle Row Rule) *Select row index*

$$p \in \arg \max \{\bar{a}_{iq} \mid i = 1, \dots, m\}. \quad (15.19)$$

Assume that $\bar{a}_{pq} > 0$. Taking $\bar{a}_{pq}$ as the pivot, carry out according elementary transformations. As a result, the negative reduced cost, indexed by q , becomes zero, whereas the zero reduced cost, indexed by j_p , becomes positive, i.e., $-\bar{z}_q/\bar{a}_{p,q} > 0$.

Overall steps can be summarized into the following algorithm.

Algorithm 15.4.1 (Dual Phase-I: The Most-Obtuse-Angle Row Rule) Initial: (B, N) , B^{-1} . This algorithm finds a dual feasible basic solution to the standard LP problem.

1. Compute $\bar{z}_N = c_N - N^T \bar{y}$, where $\bar{y} = B^{-T} c_B$.
2. Select pivot column index q such that $q \in \arg \min_{j \in N} \bar{z}_j$.
3. If $\bar{z}_q \geq 0$, compute $\bar{x}_B = B^{-1} b$, $\bar{x}_N = 0$, and stop.
4. Compute $\bar{a}_q = B^{-1} a_q$.
5. Stop if $\bar{a}_q \leq 0$.
6. Determine pivot row index $p \in \arg \max\{\bar{a}_{iq} \mid i = 1, \dots, m\}$.
7. Update B^{-1} .
8. Update (B, N) by exchanging j_p and q .
9. Go to step 1.

Theorem 15.4.2 *If Algorithm 15.4.1 terminates at*

- (i) *step 3, then a dual feasible basic solution is obtained; or*
- (ii) *step 5, detecting infeasibility or lower unboundedness of the problem.*

Proof Termination at step 3 clear implies the current solution is dual feasible basic. When terminating from step 5, then it holds that $\bar{z}_N < 0$ and $\bar{a}_q \leq 0$; by Lemma 15.4.1, infeasibility or lower unboundedness of the problem is asserted. $\square$

Let us investigate the most-obtuse-angle row rule geometrically. For search vector d defined by (15.17), it holds that $-\bar{a}_{pq} < 0$, and d forms the most-obtuse-angle with the gradient, e_p , of the constraint $x_{j_p} \geq 0$. In addition, (15.18) indicates that d is a descent direction with respect to the objective function $c^T x$. If d is close to the negative objective gradient $-c$, therefore, then e_p tends to form the most-obtuse-angle with $-c$. By the heuristic characteristic of the optimal basis (Part I), it is reasonable to let the constraint $x_{j_p} \geq 0$ be satisfied as equality, accordingly dropping x_{j_p} from the basis.

This dual Phase-I method is advantageous for its remarkable simplicity and stability. Initially, it performed very well for solving a set of small test problems (Pan 1994). Taking MINOS 5.3 (Murtagh and Saunders 1998) as a platform, subsequently, a code based on it outperformed MINOS 5.3 (with default options) on 48 Netlib test problems with a CPU time ratio 1.37 (Pan 1997).

Koberstein and Suhl (2007) extended this method to solve more general LP problems. They reported extensive computational results, which confirmed significant advantages of this method compared with some commonly used dual Phase-I methods. It was found, however, that the method performed a little unsatisfactorily on a few of the most difficult problems.

Considering the extent to which the current vertex violates nonnegative constraints, we propose the following variant to improve.

Rule 15.4.2 (Variant of Most-Obtuse-Angle Row Rule) Given constant $0 < \tau \leq$

1. Select the pivot row index

$$p \in \arg \min\{\bar{x}_{j_i} \mid \bar{a}_{i_q} \geq \tau\theta, i = 1, \dots, m\}, \quad \theta = \max\{\bar{a}_{i_q} \mid i = 1, \dots, m\} > 0.$$

In case of $\tau = 1$, the preceding is just Rule 15.4.1 itself. Practically, it should be close to 1 to expand the range of choices, so that when there exist negative components of $\bar{x}_B$, the nonnegative constraint that is violated the most is satisfied as equality. It may require around $\sigma = 0.95$.

Though Algorithm 15.4.2 uses Dantzig's rule for column pivoting, rules presented in Chap. 12 all apply. It has not been known which is the best. Intuitively, the most-obtuse-angle row rule seems to be best matched by the steepest-edge rule. But no numerical results are available at this stage.

Note that the objective value does not necessarily monotonically change in the solution process. In addition, the finiteness of the algorithm is not guaranteed even under the degeneracy assumption. Guerrero-Garcia and Santos-Palomo (2005) offered a cycling instance. Even so, cycling has not been reported so far in practice.

Example 15.4.1 Find a dual feasible simplex basis by Algorithm 15.4.1:

$$\begin{aligned} \min f &= -5x_1 - 7x_2 + x_4, \\ \text{s.t. } x_1 + 2x_2 + x_3 &= 3, \\ 2x_1 + x_2 - x_4 + x_5 &= -2, \\ -x_1 + x_2 + x_4 + x_6 &= -1, \\ x_j &\geq 0, \quad j = 1, \dots, 6. \end{aligned}$$

Answer Initial: $B = \{3, 5, 6\}$, $N = \{1, 2, 4\}$, $B^{-1} = I$.

Iteration 1:

1. $\bar{y} = B^{-T}c_B = (0, 0, 0)^T$, $\bar{z}_N = c_N - N^T y = (-5, -7, 1)^T$.
2. $\min\{\bar{z}_j \mid j = 1, 2, 4\} = \min\{-5, -7, 1\} = -7$, $q = 2$, hence x_2 enters the basis.
4. $\bar{a}_2 = B^{-1}a_2 = (2, 1, 1)^T$.
6. $\max\{\bar{a}_{i2} \mid i = 1, 2, 3\} = \max\{2, 1, 1\} = 2$, $p = 1$, so x_3 leaves the basis.
7. Update $B^{-1} = \begin{pmatrix} 1/2 \\ -1/2 & 1 \\ -1/2 & 1 \end{pmatrix}$.
8. $B = \{2, 5, 6\}$, $N = \{1, 3, 4\}$.

Iteration 2:

1. $\bar{y} = (-7/2, 0, 0)^T$, $\bar{z}_N = (-5, 0, 1)^T - (-7/2, -7/2, 0)^T = (-3/2, 7/2, 1)^T$.
2. $\min\{-3/2, 7/2, 1\} = -3/2$, $q = 1$, so x_1 enters the basis.
4. $\bar{a}_1 = (1/2, 3/2, -3/2)^T$.
6. $\max\{1/2, 3/2, -3/2\} = 3/2$, $p = 2$, so x_5 leaves the basis.

$$7. \text{ Update } B^{-1} = \begin{pmatrix} 1 & -1/3 \\ 0 & 2/3 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1/2 \\ -1/2 & 1 \\ -1/2 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \\ -1 & 1 & 1 \end{pmatrix}.$$

$$8. B = \{2, 1, 6\}, N = \{5, 3, 4\}.$$

Iteration 3:

1. $\bar{y} = (-3, -1, 0)^T$, $\bar{z}_N = (0, 0, 1)^T - (-1, -3, 1)^T = (1, 3, 0)^T$.
2. $3 = q \in \arg \min\{1, 3, 0\}$.
3. $\bar{z}_{j_3} = 0$, Dual feasibility is achieved. According primal basic solution is
 $\bar{x}_B = B^{-1}b = (4/3, 1/3, -2)^T$, $\bar{x}_N = (0, 0, 0)^T$.

The tableau version of Algorithm 15.4.1 is of a simple form.

Algorithm 15.4.2 (Tableau Dual Phase-I: The Most-Obtuse-Angle Row Rule)

Initial: simplex tableau of form (3.6). This algorithm finds a dual feasible simplex tableau.

1. Select pivot column index $q \in \arg \min_{j \in N} \bar{z}_j$.
2. Stop if $\bar{z}_q \geq 0$ (A dual feasible tableau obtained).
3. Stop if $\bar{a}_{iq} \leq 0$ (infeasibility or lower unbounded).
4. Determine pivot row index $p \in \arg \max\{\bar{a}_{iq} \mid i = 1, \dots, m\}$.
5. Carry out elementary transformations to convert $\bar{a}_{pq}$ to 1, and eliminate all other nonzeros of the column.
6. Go to step 1.

References

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Chapter 16

Reduced Simplex Method



Consider the standard LP problem (12.1) with the additional assumption that the cost c is not a linear combination of rows of A . This is appropriate, since, otherwise, there exists a vector y such that $c = A^T y$, leading to

$$c^T x = y^T A x = y^T b,$$

which implies that the objective function is constant over the feasible region if any. Redefining the notation

$$A := \begin{pmatrix} A \\ c^T \end{pmatrix}, \quad b := \begin{pmatrix} b \\ 0 \end{pmatrix},$$

we convert the standard problem into the so-called *reduced* problem below:

$$\begin{aligned} & \min f \\ & \text{s.t. } (A \mid -e_{m+1}) \begin{pmatrix} x \\ f \end{pmatrix} = b, \quad x \geq 0, \end{aligned} \quad (16.1)$$

where $(A \mid -e_{m+1}) \in \mathcal{R}^{(m+1) \times (n+1)}$, $b \in \mathcal{R}^{m+1}$, $m < n$, and $\text{rank } A = m + 1$. The objective variable f is viewed as the $(n + 1)$ th variable, corresponding to column a_{n+1} , i.e.,

$$f \triangleq x_{n+1}, \quad a_{n+1} \triangleq -e_{m+1}.$$

The reduced problem is a special standard problem with a single free variable f as its objective. As a result, computations related to the simplex algorithm can be simplified as follows.

Let (B, N) be a partition, where B is basis with the $(m + 1)$ th basic column $-e_{m+1}$.

The reduced cost is then

$$\bar{z}_N = N^T \bar{y}, \quad B^T \bar{y} = e_{m+1}.$$

If $B = LU$ is available, the second system of the preceding can be converted to the two triangular systems below:

$$U^T v = e_{m+1}, \quad L^T \bar{y} = v.$$

The solution to the first system is readily available, i.e., $v = (1/\mu)e_{m+1}$, where μ is the $(m+1)$ th diagonal of U . Therefore, only a single triangular system

$$L^T \bar{y} = (1/\mu)e_{m+1}$$

needs to be solved to obtain the simplex multiplier $\bar{y}$. In other words, only the three triangular systems are solved in each iteration, compared with the four systems in the simplex algorithm. However, the preceding tactic is not new since some LP codes, such as MINOS, already involve it.

In this chapter, we introduce the so-called reduced simplex method. It is the first simplex method that searches along the *objective-edge* direction. Specifically, we put forward the reduced simplex algorithm and introduce its dual version with a dual Phase-I procedure in the next section. In the subsequent sections, we introduce several fusion versions of the primal and dual reduced algorithms, including the crisscross, perturbation, and bisection reduced simplex algorithms. In the final section, some notes are made on the birth of the reduced simplex algorithm.

16.1 Reduced Simplex Algorithm

The main feature of the introduced method is to use objective variable f as a special nonbasic variable, compared with the simplex method that always takes it as a basic variable.

Let (B, N) be a pair of basis and nonbasis:

$$B = \{j_1, \dots, j_{m+1}\}, \quad N = A \setminus B, \quad n+1 \notin B. \quad (16.2)$$

Assume that the equality constraint of (16.1) is converted to the following canonical form

$$x_B = \bar{b} - \bar{N}x_N - f\bar{a}_{m+1}, \quad (16.3)$$

where

$$\bar{b} = B^{-1}b, \quad \bar{N} = B^{-1}N \quad \bar{a}_{n+1} = -B^{-1}e_{m+1} \neq 0. \quad (16.4)$$

Proposition 16.1.1 Assume that $(\bar{x}, \bar{f})$ with $\bar{x}_N = 0$ is a solution to the system (16.3). If there is a row index r such that

$$\bar{x}_{j_r} = 0, \quad \bar{a}_{r, n+1} \neq 0, \quad (16.5)$$

then $\bar{x}$ is a basic solution to the original problem. If, in addition,

$$\bar{x}_B \geq 0, \quad (16.6)$$

it is a feasible basic solution.

Proof If (16.5) holds, $\bar{x}$ is just the basic solution corresponding to the simplex tableau, resulting from entering f to and dropping x_{j_r} from B . If, in addition, (16.6) holds, it is clearly a feasible basic solution. $\square$

Consider the objective edge $(\Delta x, \Delta f)$ defined by

$$\Delta x_B = \bar{a}_{n+1}, \quad \Delta f = -1, \quad \Delta x_N = 0. \quad (16.7)$$

It is clear that the objective value reduces as it moves from $\bar{x}$ down along the edge. In general, however, it is not allowed to reduce infinitely to maintain the $\hat{x}_B$ nonnegative.

The largest possible α and associated blocking row can be determined by the following rule.

Rule 16.1.1 (Pivot Row Rule) Assume that $\bar{a}_{n+1} \not\geq 0$. Select the pivot row index p such that

$$\alpha = -\bar{x}_{j_p}/\bar{a}_{p, n+1} = \min\{-\bar{x}_{j_i}/\bar{a}_{i, n+1} \mid \bar{a}_{i, n+1} < 0, \quad i = 1, \dots, m+1\} \geq 0. \quad (16.8)$$

It is clear that the preceding rule is well-defined under its assumption.

The corresponding new solution attained is as follows:

$$\hat{x}_B = \bar{x}_B + \alpha \bar{a}_{n+1}, \quad \hat{f} = \bar{f} - \alpha, \quad \hat{x}_N = 0. \quad (16.9)$$

We can claim the following.

Lemma 16.1.1 Assume that $\bar{x}$ is the current feasible basic solution, and stepsize α is determined by (16.8). Then new solution $\hat{x}$ given by (16.9) is a feasible basic solution.

Proof By (16.8), it holds that

$$\hat{x}_{j_p} = 0, \quad \bar{a}_{p, n+1} < 0, \quad (16.10)$$

and hence $\hat{x}$ is a basic solution. It is only needed to show that $\hat{x}_B$ is nonnegative. Note that $\bar{x}_B \geq 0$ and $\alpha \geq 0$.

If $\bar{a}_{i, n+1} \geq 0$ for some row index i , it is clear that

$$\hat{x}_{j_i} = \bar{x}_{j_i} + \alpha \bar{a}_{i, n+1} \geq 0.$$

Now assume that row index i satisfies $\bar{a}_{i, n+1} < 0$. By (16.8), it is known that

$$-\alpha \geq \bar{x}_{j_i} / \bar{a}_{i, n+1}.$$

Multiplying the preceding by $\bar{a}_{i, n+1} < 0$ gives

$$-\alpha \bar{a}_{i, n+1} \leq \bar{x}_{j_i},$$

or equivalently,

$$\bar{x}_{j_i} + \alpha \bar{a}_{i, n+1} \geq 0,$$

which together with the first expression of (16.9) leads to

$$\hat{x}_{j_i} \geq 0.$$

Consequently, it holds that $\hat{x}_B \geq 0$, and $\hat{x}$ is therefore a feasible basic solution. $\square$

Note that the stepsize α determined by (16.8) is equal to the decrement of the objective value. However, if the $\bar{x}_{j_p}$ is zero, so is the decrement.

If some component of $\bar{x}_B$ is zero, the feasible basic solution is said to be *degenerate*.

It is obvious that the stepsize α vanishes only if degeneracy occurs. On the other hand, the α can be allowed to be infinite under a certain condition, as the following says.

Lemma 16.1.2 *If $\bar{a}_{n+1} \geq 0$, problem (16.1) is infeasible or lower unbounded.*

Proof Assume that the feasible region of the problem is nonempty. Let $(\bar{x}, \bar{f})$ be a feasible basic solution, satisfying

$$\bar{x}_B = \bar{b} - \bar{f} \bar{a}_{n+1} \geq 0, \quad \bar{x}_N = 0. \quad (16.11)$$

We show that for any $\alpha > 0$, the corresponding new solution given by (16.9) is a feasible solution.

Substituting

$$\hat{f} = \bar{f} - \alpha,$$

into the first expression of (16.9) gives

$$\hat{x}_B = \bar{b} - \hat{f}\bar{a}_{n+1} \quad (16.12)$$

$$= (\bar{b} - \bar{f}\bar{a}_{n+1}) + \alpha\bar{a}_{n+1} \quad (16.13)$$

$$= \bar{x}_B + \alpha\bar{a}_{n+1} \geq 0, \quad (16.14)$$

where the inequality comes from (16.11) and $\alpha > 0$ and the assumption $\bar{a}_{n+1} \geq 0$. Therefore, the $(\hat{x}, \hat{f})$ is a feasible solution such that

$$\hat{f} = \bar{f} - \alpha \rightarrow -\infty \quad \text{as} \quad \alpha \rightarrow \infty.$$

This means that the problem is lower unbounded, and the proof is complete. $\square$

Once row index p is determined, the following column rule applies.

Rule 16.1.2 (Column Rule) *Select the pivot column index by*

$$q \in \arg \min_{j \in N} \bar{a}_{p,j}. \quad (16.15)$$

Theorem 16.1.3 *Under the same assumption of Lemma 16.1.1, if q is determined by (16.15) and $\bar{a}_{p,q} \geq 0$, then $\hat{x}$ is an optimal basic solution.*

Proof According to Lemma 16.1.1, $\hat{x}$ is a feasible basic solution. Thus, it is only needed to show that $\hat{f} \leq f$ over the feasible region.

From $\bar{a}_{p,q} \geq 0$ and (16.15), it is clear that the p th row of $\bar{N}$ is nonnegative, i.e.,

$$e_p^T \bar{N} \geq 0. \quad (16.16)$$

The equation corresponding to the p th row can be written

$$x_{j_p} = \bar{b}_p - e_p^T \bar{N} x_N - \bar{f} \bar{a}_{p,n+1}. \quad (16.17)$$

From the first expressions of (16.12) and of (16.10), it follows that

$$\bar{b}_p = \hat{f} \bar{a}_{p,n+1}.$$

Substituting the preceding into (16.17) gives

$$x_{j_p} = \hat{f} \bar{a}_{p,n+1} - e_p^T \bar{N} x_N - \bar{f} \bar{a}_{p,n+1}.$$

Multiplying the preceding by $1/\bar{a}_{p,n+1}$ and moving terms lead to the following expression of the objective function:

$$f = \hat{f} - (1/\bar{a}_{p,n+1})e_p^T \bar{N} x_N - (1/\bar{a}_{p,n+1})x_{j_p}. \quad (16.18)$$

From (16.16) and the second expression of (16.10), it follows that the entries of the right-hand side of (16.18) are all nonnegative. Since the x is nonnegative over the feasible region, therefore, it can be asserted that $\hat{x}$ is an optimal solution with objective value $\hat{f}$. $\square$

Starting from a feasible solution,¹ in each iteration, the algorithm determines the row index p and stepsize α , updates the solution, and determines column index q . If $\bar{a}_{p,q} \geq 0$, optimality is achieved; otherwise, it makes the basis change and carries out the next iteration.

These steps can be actualized with the so-called *reduced simplex tableau*, corresponding to the canonical system

$$x_B + \bar{N}x_N + f\bar{a}_{n+1} = \bar{b},$$

that is,

x_B	x_N	f	$f = \bar{f}$
I	$\bar{N}$	$\bar{a}_{n+1}$	$\bar{x}_B$

(16.19)

which is related to the objective value $\bar{f}$, corresponding to $\bar{x}_B = \bar{b} - \bar{f}\bar{a}_{n+1}$.

The overall steps can be summarized into the following algorithm.

Algorithm 16.1.1 (Reduced Simplex Algorithm: Tableau Form) *Initial: reduced tableau in form (16.19), with feasible solution $(\bar{x}, \bar{f})$. This algorithm solves (16.1).*

1. Stop if $\bar{a}_{n+1} \geq 0$.
2. Determine row index p and α such that

$$\alpha = -\bar{x}_{j_p}/\bar{a}_{p,n+1} = \min\{-\bar{x}_{j_i}/\bar{a}_{i,n+1} \mid \bar{a}_{i,n+1} < 0, i = 1, \dots, m+1\}.$$

3. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha\bar{a}_{n+1}$ and $\bar{f} = \bar{f} - \alpha$.
4. Determine column index $q \in \arg \min_{j \in N} \bar{a}_{p,j}$.
5. Stop if $\bar{a}_{p,q} \geq 0$.
6. Convert $\bar{a}_{p,q}$ to 1, and eliminate the other nonzeros in the pivot column by elementary transformations.
7. Go to step 1.

¹ It is clearly not necessary to get started from a basic solution.

Note The elementary transformation in step 6 does not affect the $\bar{x}_B$ column.

Theorem 16.1.4 Under the assumption of nondegeneracy, Algorithm 16.1.1 terminates either at

- (i) step 1, detecting lower unboundedness of the problem, or at
- (ii) step 5, giving an optimal basic solution $\bar{x}$.

Proof According to Lemma 16.1.1, each iteration yields a feasible basic solution.

Under nondegeneracy, stepsize α is positive in each iteration. Therefore, the objective value decreases strictly monotonically, and the corresponding feasible basic solution never appears repeatedly. Since there are a finite number of such solutions, the algorithm must terminate.

The meanings of the exits are obvious by Lemma 16.1.2 and Theorem 16.1.3. $\square$

The preceding algorithm needs a Phase-I procedure to get it started. For this purpose, the generalized reduced simplex Phase-I Algorithm 18.1.3 should be a good choice. However, for standard LP problems, it seems preferable to use Algorithm 14.4.1 instead.

Example 16.1.1 Solve the following problem:

$$\begin{aligned}
 \min f &= 6x_1 + 4x_2 - 6x_3 - 4x_4, \\
 \text{s.t. } 3x_1 + x_2 + 3x_3 + 5x_4 + x_5 &= 3, \\
 -4x_1 - x_2 + 5x_3 - 2x_4 + x_6 &= 2, \\
 -6x_1 - 7x_2 - x_3 + 4x_4 + x_7 &= -5, \\
 x_1 - 3x_2 - 2x_3 - x_4 + x_8 &= -1, \\
 x_j &\geq 0, \quad j = 1, \dots, 8.
 \end{aligned}$$

Answer Phase-I: Call Algorithm 14.4.1.

Initial tableau is

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	RHS
3	1	3	5	1					3
-4	-1	5	-2		1				2
-6	-7	-1	4			1			-5
1	-3	-2	-1				1		-1
6	4	-6*	-4					-1	

Iteration 1:

1. $\min \{3, 2, -5, -1\} = -5, p = 3.$
3. $\min \{-6, -7, -1, 4\} = -7, q = 2.$

5. Multiply row 3 by $-1/7$, and add $-1, 1, 3, -4$ times of row 3 to rows 1, 2, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	RHS
15/7		20/7	39/7	1		1/7			16/7
-22/7		36/7	-18/7		1	-1/7			19/7
6/7	1	1/7	-4/7			-1/7			5/7
25/7		-11/7	-19/7			-3/7	1		8/7
18/7		-46/7	-12/7			4/7		-1	-20/7

The preceding tableau is feasible. Convert it to the reduced simplex tableau:
Iteration 2:

1. $p = 5$, $\min\{18/7, -46/7, -12/7, 4/7\} = -46/7 < 0$, $q = 3$.
2. Multiply row 5 by $-46/7$, and add $-20/7, -36/7, -1/7, 11/7$ times of row 5 to rows 1, 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 0$
75/23			111/23	1		9/23		-10/23	24/23
-26/23			-90/23		1	7/23		-18/23	11/23
21/23	1		-14/23			-3/23		-1/46	15/23
68/23			-53/23			-13/23	1	11/46	42/23
-9/23		1	6/23			-2/23		7/46	10/23

Phase-II. Call Algorithm 16.1.1.

Iteration 3:

1. $\bar{a}_9 \not\geq 0$.
2. $\alpha = \min\{-(24/23)/(-10/23), -(11/23)/(-18/23), -(15/23)/(-1/46)\}$
 $= (11/23)/(-18/23) = 11/18$, $p = 2$.
3. $\bar{x}_B = (7/9, 0, 23/36, 71/36, 19/36)^T$, $\bar{f} = 0 - 11/18$.
4. $\min\{-26/23, -90/23, 7/23\} = -90/23 < 0$, $q = 4$.
6. Multiply row 2 by $-23/90$, and add $-111/23; 14/23; 53/23; -6/23$ times of row 2 to rows 1, 3, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -11/18$
28/15				1	37/30	23/30		-7/5	7/9
13/45			1		-23/90	-7/90		1/5	
49/45	1				-7/45	-8/45		1/10	23/36
163/45					-53/90	-67/90	1	7/10	71/36
-7/15		1			1/15	-1/15		1/10	19/36

Iteration 4:

1. $\bar{a}_9 \not\geq 0$.
2. $\alpha = \min\{-(7/9)/(-7/5)\} = 5/9$, $p = 1$.
3. $\bar{x}_B = (0, 1/9, 25/36, 85/36, 7/12)^T$, $\bar{f} = -11/18 - 5/9 = -7/6$:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -7/6$
28/15				1	37/30	23/30		-7/5	
13/45			1		-23/90	-7/90		1/5	1/9
49/45	1				-7/45	-8/45		1/10	25/36
163/45					-53/90	-67/90	1	7/10	85/36
-7/15		1			1/15	-1/15		1/10	7/12

4. $\min\{28/15, 37/30, 23/30\} \geq 0$.
5. The optimal basic solution and optimal objective value:

$$x^* = (0, 25/36, 7/12, 1/9, 0, 0, 0, 85/36)^T, \quad f^* = -7/6.$$

Let B be the current basis. Corresponding to the reduced simplex tableau (16.19), the revised tableau is

$$\begin{array}{c|c} \begin{array}{ccc} x_B & x_N & f \\ I & B^{-1}\bar{N} & -B^{-1}e_{m+1} \end{array} & \begin{array}{c} f = \bar{f} \\ \bar{x}_B \end{array} \end{array} \quad (16.20)$$

Based on the equivalence between the two tableaus, it is not difficult to transform Algorithm 16.1.1 into its revised version.

Algorithm 16.1.2 (Reduced Simplex Algorithm) *Initial* : (B, N) , B^{-1} , $\bar{x}_B = \bar{b} - \bar{f}\bar{a}_{n+1} \geq 0$, where $\bar{b} = B^{-1}b$, $\bar{a}_{n+1} = -B^{-1}e_{m+1}$. This algorithm solves reduced problem (16.1).

1. Stop if $\bar{a}_{n+1} \geq 0$ (unbounded problem).
2. Determine α and row index p such that

$$\alpha = -\bar{x}_{j_p}/\bar{a}_{p,n+1} = \min\{-\bar{x}_{j_i}/\bar{a}_{i,n+1} \mid \bar{a}_{i,n+1} < 0, i = 1, \dots, m+1\}.$$

3. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha\bar{a}_{n+1}$, $\bar{f} = \bar{f} - \alpha$.
4. Compute $\sigma_N = N^T B^{-T} e_p$.
5. Determine column index $q \in \arg \min_{j \in N} \sigma_j$.
6. Stop if $\sigma_q \geq 0$ (optimality achieved).
7. Compute $\bar{a}_q = B^{-1}a_q$.
8. Update $\bar{a}_{n+1} = \bar{a}_{n+1} + v(\bar{a}_q - e_p)$, where $v = -\bar{a}_{p,n+1}/\sigma_q$.
9. Update B^{-1} .

10. Update (B, N) by exchanging j_p and q .

11. Go to step 1.

16.2 Dual Reduced Simplex Algorithm

The dual version of the reduced simplex algorithm can be obtained similarly to the derivation of the classical dual simplex algorithm; that is, using the same reduced simplex tableau (16.19), we seek primal feasibility while maintaining dual feasibility, until attaining optimality.

The reduced simplex tableau is said to be *dual feasible* if the following condition is satisfied for some row index p :

$$e_p^T \bar{N} \geq 0, \quad \bar{a}_{p, n+1} < 0. \quad (16.21)$$

The associated basic solution $(\bar{x}, \bar{f})$ is such that

$$\bar{f} = \bar{b}_p / \bar{a}_{p, n+1}, \quad \bar{x}_B = \bar{b} - \bar{f} \bar{a}_{n+1}, \quad \bar{x}_N = 0, \quad (16.22)$$

with

$$\bar{x}_{j_p} = 0. \quad (16.23)$$

The p th row of the tableau is termed the *objective row*. It corresponds to the equation

$$x_{j_p} = \bar{b}_p - e_p^T \bar{N} x_N - f \bar{a}_{p, n+1},$$

or equivalently,

$$f = \bar{b}_p / \bar{a}_{p, n+1} - (1 / \bar{a}_{p, n+1}) e_p^T \bar{N} x_N - x_{j_p} / \bar{a}_{p, n+1}. \quad (16.24)$$

Consider the following rule.

Rule 16.2.1 (Dual Row Rule) *Select the row index r such that*

$$\bar{x}_{j_r} = \min\{\bar{x}_{j_i} \mid i = 1, \dots, m+1; \quad i \neq p\}.$$

According to Theorem 16.1.3, $(\bar{x}, \bar{f})$ is an optimal basic solution if $\bar{x}_{j_r} \geq 0$, and hence $\bar{x}_B \geq 0$.

Now assume the opposite, i.e.,

$$\bar{x}_{j_r} < 0. \quad (16.25)$$

Then we have the following.

Lemma 16.2.1 *Assume the satisfaction of (16.21), (16.22), and (16.25). The LP problem is infeasible if it holds that*

$$J = \{j \in N \mid \bar{a}_{r,j} < 0\} = \emptyset, \quad \bar{a}_{r,n+1} \geq 0. \quad (16.26)$$

Proof There are only the following two cases:

(i) $\bar{a}_{r,n+1} = 0$.

In this case, it holds that

$$x_{j_r} = \bar{b}_r - e_r^T \bar{N} x_N, \quad (16.27)$$

substituting $(\bar{x}, \bar{f})$ into which gives

$$\bar{x}_{j_r} = \bar{b}_r < 0, \quad (16.28)$$

where the inequality is from (16.25). Suppose that there is a feasible solution (x', f') . Then, it satisfies (16.27), i.e.,

$$x'_{j_r} = \bar{b}_r - e_r^T \bar{N} x'_N \leq \bar{b}_r,$$

where the inequality is from $x'_N \geq 0$ and the first expression of (16.26). From the preceding and (16.28), it follows that $x'_{j_r} < 0$, contradicting that (x', f') is feasible solution. Therefore, there exists no feasible solution.

(ii) $\bar{a}_{r,n+1} > 0$.

By the second expression of (16.22), it holds that

$$\bar{x}_{j_r} = \bar{b}_r - \bar{f} \bar{a}_{r,n+1},$$

which along with (16.25) and $\bar{a}_{r,n+1} > 0$ gives

$$\bar{f} = \bar{b}_r / \bar{a}_{r,n+1} - \bar{x}_{j_r} / \bar{a}_{r,n+1} > \bar{b}_r / \bar{a}_{r,n+1}. \quad (16.29)$$

In addition, the equation corresponding to the r th row is

$$x_{j_r} = \bar{b}_r - e_r^T \bar{N} x_N - f \bar{a}_{r,n+1},$$

or equivalently,

$$f = \bar{b}_r / \bar{a}_{r,n+1} - (1 / \bar{a}_{r,n+1}) e_r^T \bar{N} x_N - x_{j_r} / \bar{a}_{r,n+1}.$$

Suppose that there is a feasible solution (x', f') . Then it satisfies the preceding, i.e.,

$$f' = \bar{b}_r / \bar{a}_{r,n+1} - (1 / \bar{a}_{r,n+1}) e_r^T \bar{N} x'_N - x'_{j_r} / \bar{a}_{r,n+1}.$$

From the preceding, $\bar{a}_{r,n+1} > 0$ and the first expression of (16.26), it follows that

$$f' \leq \bar{b}_r / \bar{a}_{r,n+1},$$

which along with (16.29) leads to

$$\bar{f} > \bar{b}_r / \bar{a}_{r,n+1} \geq f'. \quad (16.30)$$

On the other hand, substituting (x', f') into (16.24) leads to

$$f' = \bar{b}_p / \bar{a}_{p,n+1} - (1 / \bar{a}_{p,n+1}) e_p^T \bar{N} x'_N - x'_{j_p} / \bar{a}_{p,n+1}.$$

The preceding with (16.21) and the first expression of (16.22) gives

$$f' \geq \bar{b}_p / \bar{a}_{p,n+1} = \bar{f},$$

which contradicts (16.10). Therefore, there exists no feasible solution. $\square$

The following rule is well-defined.

Rule 16.2.2 Assume that the set J defined by (16.26) is nonempty. Select column index q such that

$$\beta = -\bar{a}_{p,q} / \bar{a}_{r,q} = \min_{j \in J} -\bar{a}_{p,j} / \bar{a}_{r,j} \geq 0. \quad (16.31)$$

To make the corresponding basis change, multiply the r th row by $1 / \bar{a}_{r,q}$, and add $-\bar{a}_{i,q}$ times of the r th row to the i th row, for $i = 1, \dots, m+1, i \neq r$. Then, update the column index sets:

$$B = B \setminus \{j_r\} \cup \{q\}, \quad N = N \setminus \{q\} \cup \{j_r\}. \quad (16.32)$$

Note that the p th row of the resulting tableau yields from adding β times of the r row to the old p th row. Therefore, it can be denoted by

x_{j_p}	x_N	f	Solution
1	$e_p^T \widehat{N}$	$\hat{a}_{p,n+1}$	$\hat{x}_{j_p}$

(16.33)

where

$$\hat{a}_{p,n+1} = \bar{a}_{p,n+1} + \beta \bar{a}_{r,n+1}, \quad \hat{x}_{j_p} = \beta \bar{x}_{j_r}, \quad (16.34)$$

with the nonbasic coefficients omitted.

If $\beta \neq 0$ and hence $\hat{x}_{j_p} \neq 0$, the resulting is clearly not a dual reduced simplex tableau yet. In this case, we add

$$\alpha = -\hat{x}_{j_p}/\hat{a}_{p,n+1} \quad (16.35)$$

times of the f column to the $\hat{x}_B$ column, which corresponds to updating the objective value by

$$\hat{f} = \bar{f} - \alpha.$$

If the $\bar{a}_{p,q}$ is zero, so is the β defined by (16.31); therefore, from (16.34) and (16.35), it follows that the α vanishes. Consequently, the new objective value $\hat{f}$ is not really new, but the same as the old.

If $e_p^T \bar{N} > 0$, the dual reduced simplex tableau is said to be *dual nondegenerate*.

Under the assumption of nondegeneracy, the objective value increases strictly.

Then, we can claim that the end tableau is as follows.

Lemma 16.2.2 *Under the same assumption of Rule 16.2.2, there is no feasible solution if $\hat{a}_{p,n+1} \geq 0$.*

Proof In this case, it is verified that $e_p^T \hat{N} \geq 0$. Therefore, the claim can be shown as the same as that for Lemma 16.26. $\square$

Based on the preceding discussions, the validity of the following result is obvious.

Lemma 16.2.3 *Under the same assumption of Rule 16.2.2, if $\hat{a}_{p,n+1} < 0$, the new reduced simplex tableau is a dual feasible tableau with the p th objective row and the objective value $\hat{f}$.*

Overall steps can be summarized into the following.

Algorithm 16.2.1 (Dual Reduced Simplex Algorithm: Tableau Form) *Initial : reduced simplex tableau (16.19), where $e_p^T \bar{N} \geq 0$, $\bar{a}_{p,n+1} < 0$, $\bar{f} = \bar{b}_p/\bar{a}_{p,n+1}$, $\bar{x}_B = \bar{b} - \bar{f}\bar{a}_{n+1}$. This algorithm solves the reduced problem (16.1).*

1. Determine row index $r \in \arg \min\{\bar{x}_{j_i} \mid i = 1, \dots, m+1, i \neq p\}$.
2. Stop if $\bar{x}_{j_r} \geq 0$ (optimality achieved).
3. If $J = \{j \in N \mid \bar{a}_{rj} < 0\} = \emptyset$, set $p = r$, and go to step 6.
4. Determine β and column index q such that $\beta = -\bar{a}_{pq}/\bar{a}_{rq} = \min_{j \in J} -\bar{a}_{pj}/\bar{a}_{rj}$.
5. Convert $\bar{a}_{r,q}$ to 1, and eliminate the other nonzeros of the column by elementary transformations.
6. Stop if $\bar{a}_{p,n+1} \geq 0$ (infeasible problem).
7. Compute $\alpha = -\bar{x}_{j_p}/\bar{a}_{p,n+1}$.
8. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha\bar{a}_{n+1}$ and $\bar{f} = \bar{f} - \alpha$.
9. Go to step 1.

Theorem 16.2.4 Under dual nondegeneracy, Algorithm 16.2.1 terminates. It terminates at

- (i) step 2, giving an optimal basic solution $\bar{x}$; or at
- (ii) step 6, detecting the infeasibility of the problem.

Proof Since there exist only a finite number of bases, the algorithm does not terminate implies that some bases are repeated infinitely many times, which contradicts that the objective value increases strictly monotonically in the solution process, under the dual nondegeneracy. Therefore, the algorithm terminates.

According to the Lemmas 16.2.1, 16.2.2, 16.2.3, and Theorem 16.1.3, the meanings of the exits of the algorithm are obvious. $\square$

Example 16.2.1 Solve the following problem by Algorithm 16.2.1 :

$$\begin{aligned}
 \min \quad & f = 4x_1 + 2x_2 + 3x_3 + 8x_4 + 5x_5, \\
 \text{s.t.} \quad & -x_1 + 5x_2 - 4x_4 - 2x_5 + x_6 = -3, \\
 & -3x_1 - 2x_2 - 6x_3 + x_4 - x_5 + x_7 = -1, \\
 & -x_2 + 4x_3 - 6x_4 + 4x_5 + x_8 = 5, \\
 & 5x_1 + 3x_2 - 3x_3 + 3x_4 + 5x_5 + x_9 = 2, \\
 & x_j \geq 0, \quad j = 1, \dots, 9.
 \end{aligned}$$

Answer Initial tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	f	$\bar{f} = 0$
-1	5		-4	-2	1					-3
-3	-2	-6	1	-1		1				-1
	-1	4	-6	4			1			5
5	3	-3	3	5				1		2
4	2*	3	8	5					-1	

Iteration 1:

To convert the preceding tableau to the reduced simplex tableau, take the smallest 2 among the first 5 entries in the bottom row (in x_2 column) as the pivot.

Multiply row 5 by 1/2, and add -5, 2, 1, -3 times of row 5 to rows 1, 2, 3, 4, respectively.

Take $p = 5$, $e_5 \bar{N} \geq 0$, $\bar{a}_{5,10} = -1/2 < 0$. $\bar{x}_B = (-3, -1, 5, 2, 0)^T$:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	f	$\bar{f} = 0$
-11		-15/2	-24*	-29/2	1				5/2	-3
1		-3	9	4		1			-1	-1
2		11/2	-2	13/2			1		-1/2	5
-1		-15/2	-9	-5/2				1	3/2	2
2	1	3/2	4	5/2					-1/2	

Call Algorithm 16.2.1.

Iteration 2:

1. $\min\{-3, -1, 5, 2, 0\} = -3 < 0$, $r = 1$.
3. $J = \{1, 3, 4, 5\} \neq \emptyset$.
4. $\beta = \min\{-2/(-11), -(3/2)/(-15/2), -4/-24, -(5/2)/(-29/2)\}$
 $= -4/-24 = 1/6$, $q = 4$.
5. Multiply row 1 by $-1/24$, and add $-9, 2, 9, -4$ times of row 1 to rows 2, 3, 4, 5, respectively:
6. $\bar{a}_{5,10} = -1/12 < 0$.
7. $\alpha = -(-1/2)/(-1/12) = -6$.
8. $\bar{x}_B = (3/4, -7/4, 19/2, -1/4, 0)^T$, $\bar{f} = 0 - (-6) = 6$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	f	$\bar{f} = 6$
11/24		5/16	1	29/48	-1/24				-5/48	3/4
-25/8		-93/16*		-23/16	3/8	1			-1/16	-7/4
35/12		49/8		185/24	-1/12		1		-17/24	19/2
25/8		-75/16		47/16	-3/8			1	9/16	-1/4
1/6	1	1/4		1/12	1/6				-1/12	

Iteration 3:

1. $\min\{3/4, -7/4, 19/2, -1/4\} = -7/4 < 0$, $r = 2$.
3. $J = \{1, 3, 5\} \neq \emptyset$.
4. $\beta = \min\{-(1/6)/(-25/8), -(1/4)/(-93/16), -(1/12)/(-23/16)\}$
 $= -(1/4)/(-93/16) = 4/93$, $q = 3$.
5. Multiply row 2 by $-16/93$, and add $-5/16; -49/8; 75/16; -1/4$ times of row 2 to rows 1, 3, 4, 5, respectively:
6. $\bar{a}_{5,10} = -7/93 < 0$.
7. $\alpha = -(-7/93)/(-8/93) = -7/8$.
8. $\bar{x}_B = (3/4, 7/24, 25/3, 5/8, 0)^T$ column, and set $\bar{f} = 6 - (-7/8) = 55/8$:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	f	$\bar{f} = 55/8$
9/31			1	49/93	-2/93	5/93			-10/93	3/4
50/93		1		23/93	-2/31	-16/93			1/93	7/24
-35/93				192/31	29/93	98/93	1		-24/31	25/3
175/31				127/31	-21/31	-25/31		1	19/31	5/8
1/31	1			2/93	17/93	4/93			-8/93	

Iteration 4:

1. $\min\{3/4, 7/24, 25/3, 5/8\} \geq 0$.
2. The optimal basic solution and optimal value:

$$x^* = (0, 0, 7/24, 3/4, 0, 0, 0, 25/3, 5/8)^T, \quad f^* = 3 \times (7/24) + 8 \times (3/4) = 55/8.$$

Based on the equivalence between the reduced tableau (16.19) and the revised tableau (16.20), the following revision of Algorithm 16.2.1 can be obtained.

Algorithm 16.2.2 (Dual Reduced Simplex Algorithm) *Initial:* $(B, N), B^{-1}$. $\sigma = e_p^T B^{-1} N \geq 0$, $\bar{a}_{p,n+1} < 0$, $\bar{f} = \bar{b}_p / \bar{a}_{p,n+1}$, $\bar{x}_B = \bar{b} - \bar{f} \bar{a}_{n+1} \geq 0$, where $\bar{b} = B^{-1} b$, $\bar{a}_{n+1} = -B^{-1} e_{m+1}$. This algorithm solves the reduced problem (16.1).

1. Determine row index $r \in \arg \min\{\bar{x}_{j_i} \mid i = 1, \dots, m+1, i \neq p\}$.
2. Stop if $\bar{x}_{j_r} \geq 0$ (optimality achieved).
3. Compute $\omega_N = N^T B^{-T} e_r$.
4. If $J = \{j \in N \mid \omega_j < 0\} = \emptyset$, set $p = r$, $\sigma_N = \omega_N$, and go to step 11.
5. Determine β and column index q such that $\beta = -\sigma_q / \omega_q = \min_{j \in J} -\sigma_j / \omega_j$.
6. Compute $\bar{a}_q = B^{-1} a_q$.
7. Update $\bar{a}_{n+1} = \bar{a}_{n+1} + \nu(\bar{a}_q - e_r)$, where $\nu = -\bar{a}_{r,n+1} / \omega_q$.
8. Update $\bar{x}_B = \bar{x}_B + \tau(\bar{a}_q - e_r)$, where $\tau = -\bar{x}_{j_r} / \omega_q$.
9. Update B^{-1} .
10. Update (B, N) by exchanging j_p and q .
11. Stop if $\bar{a}_{p,n+1} \geq 0$ (infeasible problem).
12. Compute $\alpha = -\bar{x}_{j_p} / \bar{a}_{p,n+1}$.
13. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha \bar{a}_{n+1}$, $\bar{f} = \bar{f} - \alpha$.
14. Go to step 1.

16.2.1 Dual Reduced Simplex Phase-I: Most-Obtuse-Angle

Based on the most-obtuse-angle heuristic, the overall steps can be put into the following algorithm.

Algorithm 16.2.3 (Dual Reduced Phase-I: Tableau Form) *Initial :* reduced simplex tableau in form (16.19), where $\bar{x}_B = \bar{b} - \bar{f} \bar{a}_{n+1}$. This algorithm finds a dual feasible reduced tableau.

1. Determine row index $p \in \arg \min\{\bar{a}_{i,n+1} \mid i = 1, \dots, m+1\}$.
2. Stop if $\bar{a}_{p,n+1} \geq 0$ (infeasible or unbounded problem).
3. Compute $\alpha = -\bar{x}_{j_p} / \bar{a}_{p,n+1}$.
4. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha \bar{a}_{n+1}$ and $\bar{f} = \bar{f} - \alpha$.
5. Determine column index $q \in \arg \min_{j \in N} \bar{a}_{p,j}$.
6. Stop if $\bar{a}_{p,q} \geq 0$ (dual feasibility achieved).

7. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
8. Go to step 1.

Note Step 7 is independent of the $\bar{x}_B$ column.

Example 16.2.2 Solve the following problem using Algorithm 16.2.3 as dual Phase-I :

$$\begin{aligned}
 \min f &= -4x_1 - 7x_2 + 2x_3 + 3x_4, \\
 \text{s.t. } -2x_1 + 1x_2 - 3x_3 - x_4 + x_5 &= -11, \\
 3x_1 + 3x_2 + x_3 + 5x_4 + x_6 &= 35, \\
 5x_1 + 4x_2 - x_3 - 2x_4 + x_7 &= 2, \\
 -x_1 + 6x_2 - 7x_3 + 4x_4 + x_8 &= 7, \\
 x_j &\geq 0, \quad j = 1, \dots, 8.
 \end{aligned}$$

Answer Initial tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 0$
-2	1	-3	-1	1					-11
3	3	1	5		1				35
5	4	-1	-2			1			2
-1	6	-7	4				1		7
-4	-7*	2	3					-1	

where $\bar{x}_B = (-11, 35, 2, 7)^T$, $\bar{f} = 0$.

Dual Phase-I: call Algorithm 16.2.3.

Iteration 1:

1. $\min\{0, 0, 0, 0, -1\} = -1 < 0$, $p = 5$.
3. $\alpha = 0 / -1 = 0$.
5. $\min\{-4, -7, 2, 3\} = -7 < 0$, $q = 2$.
7. Multiply row 5 by $-1/7$, and add $-1, -3, -4, -6$ times of row 5 to rows 1, 2, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 0$
-18/7		-19/7	-4/7	1				-1/7	-11
9/7		13/7	44/7		1			-3/7	35
19/7		1/7	-2/7			1		-4/7	2
-31/7		-37/7*	46/7				1	-6/7	7
4/7	1	-2/7	-3/7					1/7	

Iteration 2:

1. $\min\{-1/7, -3/7, -4/7, -6/7, 1/7\} = -6/7 < 0$, $p = 4$.
3. $\alpha = -7/(-6/7) = 49/6$.
4. $\bar{x}_B = (-73/6, 63/2, -8/3, 0, 7/6)^T$, $\bar{f} = -49/6$.
5. $\min\{-31/7, -37/7, 46/7, 0, 7/6\} = -37/7 < 0$, $q = 3$.
7. Multiply row 4 by $-7/37$, and add $19/7, -13/7, -1/7, 2/7$ times of row 4 to rows 1, 2, 3, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -49/6$
$-11/37$			$-146/37$	1			$-19/37$	$11/37$	$-73/6$
$-10/37^*$			$318/37$		1		$13/37$	$-27/37$	$63/2$
$96/37$			$-4/37$			1	$1/37$	$-22/37$	$-8/3$
$31/37$		1	$-46/37$				$-7/37$	$6/37$	
$30/37$	1		$-29/37$				$-2/37$	$7/37$	$7/6$

Iteration 3:

1. $\min\{11/37, -27/37, -22/37, 6/37, 7/37\} = -27/37 < 0$, $p = 2$.
3. $\alpha = -(63/2)/(-27/37) = 259/6$.
4. $\bar{x}_B = (2/3, 0, , -85/3, 7, 28/3)^T$, $\bar{f} = -49/6 - 259/6 = -154/3$.
5. $\min\{-10/37, 318/37, 13/37\} = -10/37 < 0$, $q = 3$.
7. Multiply row 2 by $-37/10$, and add $11/37, -96/37, -31/37, -30/37$ times of row 2 to rows 1, 3, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -154/3$
			$-67/5$	1	$-11/10$		$-9/10$	$11/10$	$2/3$
1			$-159/5$		$-37/10$		$-13/10$	$27/10$	
			$412/5$		$48/5$	1	$17/5$	$-38/5$	$-85/3$
		1	$127/5$		$31/10$		$9/10$	$-21/10$	7
	1		25		3		1	-2	$28/3$

Iteration 4:

1. $\min\{11/10, 27/10, -38/5, -21/10, -2\} = -38/5 < 0$, $p = 3$.
3. $\alpha = -(-85/3)/(-38/5) = -425/114$.
4. $\bar{x}_B = (-261/76, -765/76, 0, 1127/76, 319/19)^T$, $\bar{x}_9 = -154/3 + 425/114 = -1809/38$.
5. $\min\{412/5, 48/5, 17/5\} \geq 0$.
6. Dual feasibility is achieved:

Phase-II: call Algorithm [16.2.1](#)

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -1809/38$
			$-67/5$	1	$-11/10$		$-9/10$	$11/10$	$-261/76$
1			$-159/5^*$		$-37/10$		$-13/10$	$27/10$	$-765/76$
			$412/5$		$48/5$	1	$17/5$	$-38/5$	
		1	$127/5$		$31/10$		$9/10$	$-21/10$	$1127/76$
	1		25		3		1	-2	$319/19$

Iteration 5:

1. $\min\{-261/76, -765/76, 1127/76, 319/19\} = -765/76 < 0, r = 2.$
3. $J = \{4, 6, 8\} \neq \emptyset.$
4. $\beta = \min\{-(412/5)/(-159/5), -(48/5)/(-37/10), -(17/5)/(-13/10)\}$
 $= (412/5)/(159/5) = 412/159, q = 4.$
5. Multiply row 2 by $-5/159$, and add $67/5, -412/5, -127/5, -25$ times of row 2 to rows 1,3,4,5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -1809/38$
$-67/159$				1	$73/159$		$-56/159$	$-2/53$	$813/1007$
$-5/159$			1		$37/318$		$13/318$	$-9/106$	$358/1131$
$412/159$					$2/159$	1	$5/159$	$-32/53$	$-4747/182$
$127/159$		1			$23/159$		$-22/159$	$3/53$	$4311/635$
$125/159$	1				$29/318$		$-7/318$	$13/106$	$6950/783$

6. $\bar{a}_{3,9} = -32/53 < 0.$
7. $\alpha = -(-4747/182)/(-32/53) = -8683/201.$
8. $\bar{x}_B = (39/16, 255/64, 0, 139/32, 229/64)^T, \bar{f} = -1809/38 - (-8683/201) = -16898/3835:$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -16898/3835$
$-67/159$				1	$73/159$		$-56/159$	$-2/53$	$39/16$
$-5/159$			1		$37/318$		$13/318$	$-9/106$	$255/64$
$412/159$					$2/159$	1	$5/159$	$-32/53$	
$127/159$		1			$23/159$		$-22/159$	$3/53$	$139/32$
$125/159$	1				$29/318$		$-7/318$	$13/106$	$229/64$

The optimal basic solution and optimal value:

$$x^* = (0, 229/64, 139/32, 255/64, 39/16, 0, 0, 0)^T,$$

$$f^* = -16898/3835.$$

The following is a revision of Algorithm 16.2.3.

Algorithm 16.2.4 (Dual Reduced Phase-I: Most-Obtuse-Angle) *Initial* : $(B, N), B^{-1}$. $\bar{x}_B = \bar{b} - \bar{f}\bar{a}_{n+1}$, where $\bar{b} = B^{-1}b$, $\bar{a}_{n+1} = -B^{-1}e_{m+1}$. This algorithm finds a dual feasible basis to (16.1).

1. Determine row index $p \in \arg \min\{\bar{a}_{i, n+1} \mid i = 1, \dots, m+1\}$.
2. Stop if $\bar{a}_{p, n+1} \geq 0$ (infeasible or unbounded problem).
3. Compute $\alpha = -\bar{x}_{j_p} / \bar{a}_{p, n+1}$.
4. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha\bar{a}_{n+1}$, $\bar{f} = \bar{f} - \alpha$.
5. Compute $\sigma_N = N^T B^{-T} e_p$.
6. Determine column index $q \in \arg \min_{j \in N} \sigma_j$.
7. Stop if $\sigma_q \geq 0$ (dual feasibility achieved).
8. Compute $\bar{a}_q = B^{-1}a_q$.
9. Update $\bar{a}_{n+1} = \bar{a}_{n+1} + v(\bar{a}_q - e_p)$, where $v = -\bar{a}_{p, n+1} / \sigma_q$.
10. Update B^{-1} .
11. Update (B, N) by exchanging j_p and q .
12. Go to step 1.

16.3 Perturbation Reduced Simplex Algorithm

As it is known, the simplex method is not finite. Degeneracy often leads to stalling and therefore takes too much time in the solution process although cycling hardly happens in practice. The perturbation simplex method (Pan, 1999, 2000) aims to eliminate this defect by taking strictly positive stepsizes.

The degeneracy in the reduced simplex context causes similar problems. As a remedy, in this section, we realize the same idea to introduce the so-called *perturbation reduced simplex algorithm*. Starting from any basis, it performs the normal reduced (or dual reduced) simplex algorithm but with the negative components of each solution perturbed to some small positive numbers. Actually, it solves another problem with the right-hand side b (reduced cost c) modified and finally recovers and solves the original problem with the dual (primal) reduced simplex algorithm. If the changes are small enough, there will be no need for performing the latter reduced simplex algorithm.

The overall steps are put into the following algorithm.

Algorithm 16.3.1 (Perturbation Reduced Simplex Algorithm: Tableau Form) *Given* perturbation parameters $\delta_j = 1$, $j = 1, \dots, n$. *Initial*: reduced tableau of form (16.19), where $\bar{x}_B = \bar{b} - \bar{f}\bar{a}_{n+1}$. This algorithm solves the reduced problem (16.1).

1. Call the reduced simplex Algorithm 16.1.1 but use δ_{j_i} instead of $\bar{x}_{j_i}$ in the minimum-ratio test (16.8) if $\bar{x}_{j_i} < \delta_{j_i}$, $i = 1, \dots, m$.
2. Stop if returning from step 1 (infeasible problem).

3. Returning from step 5: stop if $\bar{x}_B \geq 0$ (optimality achieved).
4. Call the dual reduced simplex Algorithm 16.2.1 but use δ_j instead of $\bar{a}_{p,j}$ in the minimum-ratio test (16.35) if $\bar{a}_{p,j} < \delta_j$.
5. Stop if returning from step 6 (infeasible problem).
6. Returning from step 2: stop if $\bar{a}_{p,j} \geq 0$, $j \in N$ (optimality achieved).
7. Go to step 1.

Example 16.3.1 Solve the following problem by Algorithm 16.3.1:

$$\begin{array}{ll}
 \min & f = -2x_1 + 7x_2 - 5x_3 - 3x_4, \\
 \text{s.t.} & -x_1 + 4x_2 - 5x_3 - 2x_4 + x_5 = 12, \\
 & x_1 - 3x_2 + 3x_3 - 5x_4 + x_6 = 2, \\
 & 7x_1 - 6x_2 - x_3 - 4x_4 + x_7 = -15, \\
 & -3x_1 + x_2 + 2x_3 - x_4 + x_8 = -2, \\
 & x_j \geq 0, \quad j = 1, \dots, 8.
 \end{array}$$

Answer Set perturbation parameters $\delta_j = 1$, $j = 1, \dots, 8$.

1. Call Algorithm 16.1.1. Start from the initial tableau: $\bar{f} = 0$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 0$
-1	4	-5	-2	1					12
1	-3	3	5		1				2
7	-6	-1	-4			1			-15
-3	1	2	-1				1		-2
-2	7	-5*	-3					-1	

Iteration 1:

1. $\bar{a}_{95} = -e_5 \neq 0$.
2. $\alpha = 0/(-1) = 0$, $p = 5$.
4. $\min\{-2, 7, -5, -3\} = -5 < 0$, $q = 3$.
6. Multiply row 5 by $-1/5$, then add 5, -3 , 1, -2 times of row 5 to rows 1, 2, 3, 4, respectively.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 0$
1	-3		1	1				1	12
-1/5	6/5		16/5		1			-3/5	2
37/5	-37/5		-17/5			1		1/5	-15
-19/5*	19/5		-11/5				1	-2/5	-2
2/5	-7/5	1	3/5					1/5	

Iteration 2:

2. Determine row index by replacing the right-hand side term -2 by $1: \min\{-2/(-3/5), -1/(-2/5)\} = 5/2$, $p = 4$, $\alpha = -(-2)/(-2/5) = -5$.
3. $\bar{x}_B = (7, 5, -16, 0, -1)^T$, $\bar{f} = 0 - (-5) = 5$.
4. $\min\{-19/5, 19/5, -11/5\} = -19/5 < 0$, $q = 1$.
6. Multiply row 4 by $-5/19$, then add $-1, 1/5, -37/5, -2/5$ times of row 4 to rows 1, 2, 3, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 5$
	-2		8/19	1			5/19	17/19	7
	1		63/19		1		-1/19	-11/19	5
			-146/19*			1	37/19	-11/19	-16
1	-1		11/19				-5/19	2/19	
	-1	1	7/19				2/19	3/19	-1

Iteration 3:

2. Determine row index by replacing the right-hand side term -16 by 1:

$$\alpha = \min\{-5/(-11/19), -1/(-11/19)\} = 19/11, \quad p = 3; \quad \alpha = -(-16)/(-11/19) = -304/11.$$

3. $\bar{x}_B = (-195/11, 21, 0, -32/11, -59/11)^T$, $\bar{f} = 5 - (-304/11) = 359/11$.
4. $\min\{0, -146/19, 37/19\} = -146/19 < 0$, $q = 4$.
6. Multiply row 3 by $-19/146$, then add $-8/19, -63/19, -11/19, -7/19$ times of row 3 to rows 1, 2, 4, 5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 359/11$
	-2			1		4/73	27/73	63/73	-195/11
	1				1	63/146	115/146	-121/146	21
			1			-19/146	-37/146	11/146	
1	-1					11/146	-17/146	9/146	-32/11
	-1	1				7/146	29/146	19/146	-59/11

Iteration 4:

2. $\alpha = \min\{-21/(-121/146)\} = 3066/121$, $p = 2$.
3. $\bar{x}_B = (501/121, 0, 21/11, -163/121, -250/121)^T$, $\bar{f} = 359/11 - (3066/121) = 883/121$.
4. $\min\{1, 63/146, 115/146\} \geq 0$.
3. Returning from step 5, but $\bar{x}_B \not\geq 0$.
4. Call Algorithm 16.2.1.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 883/121$
	-2			1		4/73	27/73	63/73	501/121
	1				1	63/146	115/146	-121/146	
			1			-19/146	-37/146	11/146	21/11
1	-1					11/146	-17/146	9/146	-163/121
	-1*	1				7/146	29/146	19/146	-250/121

Iteration 5:

- 1. $\min\{501/121, 0, 21/11, -163/121, -250/121\} = -250/121 < 0, r = 5.$
- 3. $J = \{2\} \neq \emptyset.$
- 4. $\min\{-1/(-1)\} = 1, q = 2.$
- 5. Multiply row 5 by -1 , then add $2, -1, 1$ times of row 5 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	RHS
		-2		1		-3/73	-2/73	44/73	91/11
		1			1	35/73	72/73	-51/73	-250/121
			1			-19/146	-37/146	11/146	21/11
1		-1				2/73	-23/73	-5/73	87/121
	1	-1				-7/146	-29/146	-19/146	250/121

- 6. $\bar{a}_{2,9} = -51/73 < 0.$
- 7. $\alpha = -(-250/121)/(-51/73) = -1943/657.$
- 8. $\bar{x}_B = (331/51, 0, 86/51, 47/51, 121/51)^T, \bar{f} = 883/121 - (-1943/657) = 523/51.$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 523/51$
		-2		1		-3/73	-2/73	44/73	331/51
		1			1	35/73	72/73	-51/73	
			1			-19/146	-37/146	11/146	86/51
1		-1				2/73	-23/73	-5/73	47/51
	1	-1				-7/146	-29/146	-19/146	125/51

Iteration 6:

1. $\min\{331/51, 0, 86/51, 47/51, 125/51\} \geq 0$.
6. Returning from step 2, and $\bar{a}_{2j} \geq 0$, $j \in N$. The optimal basic solution and optimal value:

$$x^* = (47/51, 125/51, 0, 86/51, 331/51, 0, 0, 0)^T, \quad f^* = 523/51.$$

Although the preceding problem was solved by calling the primal and dual reduced simplex algorithms once each, there would be more calls required for solving large-scale and highly degenerate problems.

16.4 Bisection Reduced Simplex Algorithm

In some applications, one can estimate the optimal value to some extent. In this case, it would be possible to speed up the solution process. If the optimal value is known to be γ , e.g., after adding the extra constraint $c^T x = \gamma$, any feasible solution to the modified problem is clearly an optimal solution to the original problem. Therefore, only a Phase-I procedure is needed to solve the problem. Although this is not the case in general, it would be possible to determine a so-called *existence interval*, including the optimal value within. It is clear that the optimal value divides the interval into two subintervals: the left-hand one consists of dual feasible values while the right-hand one consists of primal feasible values.

This is the basic idea behind the *bisection simplex algorithm* Pan (1991, 1996). Let $[\lambda, \mu]$ be an existence interval. The algorithm seeks a feasible solution attaining the objective value

$$\tilde{f} = (\lambda + \mu)/2,$$

using the subalgorithm similar to the *infeasibility-sum Phase-I* (Sect. 14.1).

The outcome will be one of the following cases:

- (i) Such a feasible solution is found, hence the associated objective value $\tilde{f}$ is a feasible value. We shift it down to the objective value corresponding to a feasible *basic* solution. If the dual feasibility is also attained, so is the optimality; otherwise, its objective value is a new upper bound on the optimal value and hence is set to μ .
- (ii) There is no such feasible solution, hence the reached solution is dual feasible. We shift it up to the objective value corresponding to a dual feasible *basic* solution. If the primal feasibility is reached, so is the optimality; otherwise, its objective value is a new lower bound on the optimal value and hence is set to λ .

Consequently, a new existence interval $[\lambda, \mu]$ is obtained, contracted by half of the previous length, at least. This process is repeated until the length of the interval becomes so short that an optimal solution can be found.

Just recently, the author was surprised to find that the ‘second type of simplex tableau’ used in the bisection algorithm is the same as the reduced simplex tableau. In this section, we introduce the algorithm simplified by using the most-obtuse-angle column rule (Sect. 14.4). Moreover, we use the following approach to produce the initial existence interval $[\lambda, \mu]$:

Firstly, the desirable Phase-I (Algorithm 14.4.1) is called to seek the primal feasibility. If the solution reached is feasible, its objective value is set to μ . Then, starting from the end basis, the dual reduced Phase-1 Algorithm 16.2.3 is called to seek the dual feasibility. If the solution reached is dual feasible, its objective value is set to λ . Therefore, the interval (λ, μ) can be determined unless the problem is found to be infeasible or lower unbounded; and the length of the interval might not be too large.

The overall steps are summarized into the following algorithm (without proofs).

Algorithm 16.4.1 (Bisection Reduced Simplex Algorithm: Tableau Form) *Initial: Reduced simplex tableau of form (16.19) with $\bar{f}$ and $\bar{x}_B = \bar{b} - \bar{f}\bar{a}_{n+1}$. This algorithm solves the reduced problem (16.1).*

Step 1. Call Algorithm 14.4.1 to achieve feasibility:

(1) set $\mu = \bar{f}$ if returned from step 4;

(2) stop if returned from step 2 (the feasibility is attained).

Step 2. Call Algorithm 16.2.3 to seek the primal feasibility:

(1) stop if returned from step 2 (lower unbounded problem);

(2) set $\lambda = \bar{f}$ if returned from step 6.

Step 3. Compute $\hat{f} = (\lambda + \mu)/2$, $\alpha = \bar{f} - \hat{f}$.

Step 4. Update $\bar{x}_B = \bar{x}_B + \alpha\bar{a}_{n+1}$ and $\bar{f} = \hat{f}$.

Step 5. Determine row index $p \in \arg \min\{\bar{x}_{j_i} \mid i = 1, \dots, m+1\}$.

Step 6. If $\bar{x}_{j_p} \geq 0$ then:

(1) determine α and row index p such that

$$\alpha = -\bar{x}_{j_p}/\bar{a}_{p,n+1} = \min\{-\bar{x}_{j_i}/\bar{a}_{i,n+1} \mid \bar{a}_{i,n+1} < 0, \quad i = 1, \dots, m+1\};$$

(2) update $\bar{x}_B = \bar{x}_B + \alpha\bar{a}_{n+1}$ and $\bar{f} = \bar{f} - \alpha$;

(3) determine column index $q \in \arg \min_{j \in N} \bar{a}_{p,j}$;

(4) stop if $\bar{a}_{p,q} \geq 0$ (optimality achieved);

(5) set $\mu = \bar{f}$, and go to step 3;

else

(6) determine column index $q \in \arg \min_{j \in N} \bar{a}_{p,j}$;

(7) if $\bar{a}_{p,q} < 0$, convert $\bar{a}_{p,q}$ to 1, eliminate the other nonzeros in the column by elementary transformations, and go to Step 5.

(8) compute $\alpha = -\bar{x}_{j_p}/\bar{a}_{p,n+1}$;

(9) update $\bar{x}_B = \bar{x}_B + \alpha\bar{a}_{n+1}$ and $\bar{f} = \bar{f} - \alpha$;

(10) determine $p \in \arg \min\{\bar{x}_{j_i}, i = 1, \dots, m+1\}$.

(11) stop if $\bar{x}_{j_p} \geq 0$ (optimality achieved);

(12) set $\lambda = \bar{f}$, and go to step 3.

end if

Example 16.4.1 Solve the following problem by Algorithm 16.4.1 :

$$\begin{aligned}
 \min f &= 3x_1 - 4x_2 + 8x_3 + x_4, \\
 \text{s.t.} \quad &x_1 + 2x_2 + x_3 + 5x_4 + x_5 = 3, \\
 &3x_1 + 4x_2 - 7x_3 - x_4 + x_6 = 20, \\
 &-2x_1 - x_2 + x_3 + 2x_4 + x_7 = -5, \\
 &-x_1 + x_2 - 2x_3 - x_4 + x_8 = -2, \\
 &-6x_1 + 8x_2 - 16x_3 - 2x_4 + x_9 = 50, \\
 &x_j \geq 0, \quad j = 1, \dots, 9.
 \end{aligned}$$

Answer Initial tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	f	$\bar{f} = 0$
1	2	1	5	1						3
-3	4	-7	-1		1					20
-2*	-1	1	2			1				-5
-1	1	-2	-1				1			-2
-6	8	-16	-2					1		50
3	-4	8	1						-1	

Step 1 Call Algorithm 14.4.1.

Iteration 1.

1. $\min \{3, 20, -5, -2, 50\} = -5$, $p = 3$.

3. $\min \{-2, -1, 1, 2\} = -2$, $q = 1$

5. Multiply row 3 by -2 , and add $-1, 3, 1, 6, -3$ times of row 3 to rows 1, 2, 4, 5, 6, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	f	$\bar{f} = 0$
	3/2	3/2	6	1		1/2				1/2
	11/2	-17/2	-4		1	-3/2				55/2
1	1/2	-1/2	-1			-1/2				5/2
	3/2	-5/2	-2			-1/2	1			1/2
	11	-19	-8			-3		1		65
	-11/2	19/2	4			3/2			-1	-15/2

The primal feasibility is attained.

Step 1(2) return with $\mu = \bar{f} = 15/2$.

Iteration 2.

Transform the preceding tableau into the reduced simplex tableau by taking $p = 6$, $q = 2$.

Change the end entry of the bottom line to the value, 0, of $\bar{x}_2$. Multiply the bottom row by $-11/2$, and add $-3/2$, $-11/2$, $-1/2$, $-3/2$, -11 times of row 6 to rows 1–5, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	f	$\bar{f} = 15/2$
		45/11	78/11	1		10/11			$-3/11$	$1/2$
		1	0		1	0			-1	$55/20$
1		4/11	$-7/11$			$-4/11$			$-1/11$	$5/2$
		1/11	$-10/11$			$-1/11$	1		$-3/11$	$1/2$
		0	0			0		1	-2	65
	1	$-19/11$	$-8/11$			$-3/11$			0	

Step 2 Call Algorithm 16.2.3 to seek the dual feasibility.

1. $\min\{-3/11, -1, -1/11, -3/11, -2, 2/11\} = -2 < 0$, $p = 5$.
3. $\alpha = -(65)/(-2) = 65/2$.
4. $\bar{x}_B = \bar{x}_B + (65/2)\bar{a}_{10} = (-92/11, -5, -5/11, -92/11, 0, 65/11)^T$,
 $\bar{f} = 15/2 - 65/2 = -25$;
5. $\min\{0, 0, 0\} \geq 0$;

Step 2(2) Return with $\lambda = -25$.

Step 3 $\hat{f} = (-25 + 15/2)/2 = -35/2$, $\alpha = -25 - (-35/2) = -15/2$.

Step 4 $\bar{x}_B = \bar{x}_B + (-15/2)\bar{a}_{10} = (-139/22, 5/2, 5/22, -139/22, 15, 50/11)^T$,
 $\bar{f} = -35/2$.

Step 5 $\min\{-139/22, 5/2, 5/22, -139/22, 15, 50/11\} = -139/22 < 0$, $p = 1$.

Step 6(6) $\min\{45/11, 78/11, 10/11\} = 10/11 \geq 0$, $q = 7$.

(8) $\alpha = -(-139/22)/(-3/11) = -139/6$;

(9) $\bar{x}_B = \bar{x}_B + (-139/6)\bar{a}_{10} = (0, 77/3, 7/3, 0, 184/3, 1/3)^T$, $\bar{f} = -35/2 + 139/6 = 17/3$;

(10) $\min\{0, 77/3, 7/3, 0, 184/3, 1/3\} \geq 0$;

(11) the optimal basic solution and optimal value are:

$$x^* = (7/3, 1/3, 0, 0, 0, 77/3, 0, 0, 184/3)^T, \quad f^* = 17/3.$$

The revised version of Algorithms 16.4.1 is omitted.

16.5 Notes on Reduced Simplex Algorithm

We make the following remarks on the reduced simplex algorithm:

First of all, it is the first simplex algorithm that searches along the objective-edge direction. In contrast to the simplex algorithm, it determines the pivot row first and the pivot column after. Moreover, it determines its pivot without selection in essence, compared with the simplex algorithm that selects the pivot from multiple candidates, which is unprecedented.

Secondly, while its computational effort per iteration is the same as the simplex algorithm, the reduced simplex algorithm has much better numerical stability than the former. In fact, it tends to determine a large pivot in magnitude, compared with the former that would select a pivot arbitrarily close to zero.

In addition, it is worth mentioning that the generalization of the reduced simplex algorithm turns out to be a desirable framework for the controlled-branch ILP solver (Part I), which would have potential significance for its applications.

The birth of the reduce simplex algorithm has been very exciting to the author. In the first edition of this book, the original derivation of its prototype was introduced in the chapter same-named and was then transformed into the “improved” version in the subsequent chapter. When preparing the second edition of this book, the review of the bisection simplex algorithm inspired us to find the concise derivation, so we can introduce it in a single chapter.

As early as 30 years ago, in the paper entitled “A Simplex-like algorithm with bisection for linear programming” (p. 724), the author indicated:

(Actually, the proposed is) an approach to improving feasible solutions with the dual type of canonical form; in a manner similar to that in the classical (simplex) algorithm.

We are now aware of the so-called dual type of canonical form being just the reduced simplex tableau. At that time, we seemed to be only one step away from the current version of the reduced simplex algorithm, but this step has been taken for 30 years!

In the last paragraph of the paper (p. 742), the author wrote:

Finally, as the classical (simplex method), the new method is also simplex-like, a purely linear one. Because the problems encountered are linear, I feel it is more natural to pursue an ‘ideal’ linear method, and it might be wise not to ignore this line while pursuing another (nonlinear) method for linear programming.

Obviously, the preceding was aimed at the upsurge of the interior-point algorithm research at that time. A brief dip into the past 30 years serves to confirm this view.

References

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- Pan P-Q (1999) A new perturbation simplex algorithm for linear programming. *J Comput Math* 17:233–242

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Chapter 17

D-Reduced Simplex Method



Consider the following special form of the standard LP problem:

$$\begin{array}{ll} \min & c^T x, \\ \text{s.t.} & Ax = e_r, \quad x \geq 0, \end{array} \quad (17.1)$$

which is termed *D-reduced problem*. Its right-hand side is a unit vector, and hence the dual problem of it is an analogue of the reduced simplex problem. Therefore, it should be advantageous to deal with this problem from the dual side.

If a standard LP problem has a nonzero right-hand term $b_r \neq 0$, the problem can be transformed into a D-reduced problem by multiplying the r th equality by $1/b_r$, and adding an appropriate multiple of it to the other equalities.

There are usually multiple choices of row index r . A too small b_r is clearly not appropriate. Considering numerical stability, it seems to be favorable to take the largest in magnitude. On the other hand, a row involving the minimum number of nonzeros should be chosen to reduce fill-ins. Numerical stability and sparsity should be balanced in large-scale sparse computations.

In this chapter, we introduce the D-reduced simplex tableau in the next section and describe the dual D-reduced simplex algorithm with the Phase-I procedure in the subsequent section. In the last two sections, we present the D-reduced simplex algorithm with the Phase-I procedure and the bisection dual D-reduced simplex algorithm, respectively.

As it plays a specially role in the new method, the r -indexed row is separated off to handle. To this end, introduce the following column and row index sets:

$$B = \{j_1, \dots, j_{m-1}\}, \quad N = A \setminus B, \quad R = \{1, \dots, m\} \setminus \{r\} = \{i_1, \dots, i_{m-1}\}. \quad (17.2)$$

17.1 D-Reduced Simplex Tableau

Let $I \in \mathcal{R}^{(m-1) \times (m-1)}$ be the unit matrix. Assume that the initial tableau of problem (17.1) is converted to the following so-called *D-reduced (simplex) tableau* by the Gaussian-Jordan elimination with row and column exchanges:

x_B	x_N	f	RHS
I	$\bar{N}_R$		
	$\bar{\omega}_N^T$		1
	$\bar{z}_N^T$	-1	$-\bar{f}$

(17.3)

where the m th (r -indexed) row is termed *datum row*. Of course, a D-reduced tableau represents the original problem itself, just as a conventional simplex tableau although now the RHS column alone does not give a basic solution.

It is noted that the datum row represents the equation

$$\bar{\omega}_N^T x_N = 1, \quad (17.4)$$

whereas the bottom (objective) row represents equation

$$\bar{z}_N^T x_N - f = -\bar{f}. \quad (17.5)$$

Lemma 17.1.1 *If the nonbasic entries in the datum row are no more than zero, there is no feasible solution.*

Proof Since $\bar{\omega}_N \leq 0$, if $\hat{x} \geq 0$ is a feasible solution, then it follows that the left side of (17.4) is less than or equal to 0, as contradicts the right-hand side 1. Therefore, there is no feasible solution. $\square$

Denote by $\bar{a}_q(R)$ the subvector of $\bar{a}_q$, indexed by set R .

Definition 17.1.1 If it holds that

$$\bar{\omega}_q \neq 0, \quad \bar{z}_q = 0, \quad (17.6)$$

the q th column is a datum column of the D-reduced tableau (17.3).

Proposition 17.1.1 *The D-reduced tableau with the q th datum column gives the basic solution*

$$\bar{x}_B = -\bar{x}_q \bar{a}_q(R), \quad \bar{x}_q = 1/\bar{\omega}_q, \quad \bar{x}_j = 0, \quad q \neq j \in N, \quad (17.7)$$

with objective value $\bar{f}$.

Proof The D-reduced tableau (17.3) corresponds to problem

$$\begin{aligned} \max \quad & f = \bar{f} + \bar{z}_N^T x_N, \\ \text{s.t.} \quad & x_B + \bar{N}_R x_N = 0, \\ & \bar{\omega}_N^T x_N = 1, \quad x_B, x_N \geq 0. \end{aligned} \quad (17.8)$$

Substituting $\bar{x}_j = 0$, $q \neq j \in N$, $\bar{x}_q = 1/\bar{\omega}_q$ into the equality constraint gives (17.7). Carrying out elementary transformations, by taking $\bar{\omega}_q$ as pivot, will convert (17.3) to a conventional simplex tableau, associated with a basic solution, which is just that given by (17.7). If $\bar{z}_q = 0$, the bottom row remains unchanged, and hence the according objective value of the solution is equal to $\bar{f}$. $\square$

It is seen that a D-reduced tableau could give multiple basic solutions because its datum column can be specified differently.

The D-reduced tableau satisfying

$$\bar{z}_q = 0, \quad \bar{\omega}_q > 0; \quad \bar{a}_{iq} \leq 0, \quad i \in R, \quad (17.9)$$

is called *feasible*, whereas satisfying

$$\bar{z}_N \geq 0, \quad (17.10)$$

called *dual feasible*.

Lemma 17.1.2 *A feasible D-reduced tableau corresponds to a feasible basic solution. A dual feasible D-reduced tableau satisfying (17.6) corresponds to a dual feasible basic solution.*

Proof Consider the classical simplex tableau, resulting from the basis change by taking $\bar{\omega}_q \neq 0$ as pivot. If (17.9) holds, the basic solution $\bar{x} \geq 0$, defined by (17.7), is clearly a feasible solution. If (17.6) and 12.12b both hold, the tableau corresponds to a dual basic feasible solution $\bar{z}_B = 0$, $\bar{z}_N \geq 0$. $\square$

Lemma 17.1.3 *A D-reduced tableau satisfying*

$$\bar{\omega}_q > 0, \quad \bar{a}_{iq} \leq 0, \quad i \in R, \quad (17.11)$$

$$\bar{z}_q = 0, \quad \bar{z}_N \geq 0, \quad (17.12)$$

corresponds to a pair of primal and dual optimal basic solutions.

Proof Under the assumption, conditions (17.9) and (17.10) are both satisfied. By Lemma 17.1.2, the D-reduced tableau corresponds to a primal and a dual feasible basic solution which are complementary. $\square$

Lemma 17.1.4 Let (17.3) be a feasible D-reduced tableau. If, for some $q' \in N$, it holds that

$$\bar{z}_{q'} < 0, \quad \bar{a}_{i, q'} \leq 0, \quad i \in R, \quad \bar{\omega}_{q'} \leq 0,$$

then the problem is lower unbounded.

Proof The feasibility of the D-reduced tableau implies satisfaction of (17.9) for some $q \in N$. Carrying out elementary transformations, with $\bar{\omega}_q > 0$ as the pivot, will lead to a feasible simplex tableau with the sign of the q' column remaining unchanged; there is no change to the objective row, and hence to $\bar{z}_{q'} < 0$. Therefore the problem is lower unbounded (Lemma 3.1.2). $\square$

Lemma 17.1.5 If a D-reduced tableau satisfies

$$\bar{z}_q < 0 : \quad \bar{a}_{iq} \leq 0, \quad i \in R, \quad (17.13)$$

then the problem has no feasible solution or has no dual feasible solution with objective value $\bar{f}$.

Proof Consider the dual problem, represented by D-reduced tableau (17.3), i.e.,

$$\begin{aligned} & \max \bar{f} + g, \\ & \text{s.t.} \quad \begin{pmatrix} I \\ \bar{N}_R^T \quad \bar{\omega}_N \end{pmatrix} \begin{pmatrix} y \\ g \end{pmatrix} + \begin{pmatrix} z_B \\ z_N \end{pmatrix} = \begin{pmatrix} 0 \\ \bar{z}_N \end{pmatrix}, \quad z_B, z_N \geq 0, \quad y \in R^{m-1}, g \in R, \end{aligned} \quad (17.14)$$

where the objective function involves a single variable g . Setting $g = 0$ turns this problem to

$$\begin{aligned} & \max \bar{f}, \\ & \text{s.t.} \quad \begin{pmatrix} I \\ \bar{N}_R^T \end{pmatrix} y + \begin{pmatrix} z_B \\ z_N \end{pmatrix} = \begin{pmatrix} 0 \\ \bar{z}_N \end{pmatrix}, \quad z_B, z_N \geq 0, \end{aligned} \quad (17.15)$$

where the objective value equals $\bar{f}$. On the other hand, deleting the m th row of the tableau gives a classical simplex tableau of (17.15). Since (17.13) holds, by Lemma 15.4.1 the problem either has a feasible solution or is lower unbounded. If there is no feasible solution, the D-reduced tableau has no feasible solution: if lower unbounded, then (17.15) has no feasible solution, hence there is no dual feasible solution, associated with objective value $\bar{f}$. $\square$

It will be seen a little later that the datum row corresponds to the dual reduced gradient, and the basis change, carried out on the D-reduced tableau, only touch the R -indexed rows, except the objective value $\bar{f}$. Desired changes in objective value will be realized by adding appropriate times of the datum row to the bottom row.

It is clear that an initial D-reduced tableau may be obtained from a (classical) simplex tableau as follows. If $\bar{b}_r$ of the right-hand side is nonzero, multiplying the r -indexed row by $1/\bar{b}_r$, and then adding appropriate times of the row to the other rows respectively gives a D-reduced tableau with the r -indexed row being datum row. The j_r -indexed column can be taken as the datum column, as it becomes nonbasic, i.e.,

$$\hat{a}_{r, j_r} = 1/\bar{b}_r; \quad \hat{a}_{i, j_r} = -\bar{b}_i/\bar{b}_r, \quad i \in R,$$

corresponding to zero reduced cost. Further, if the conventional tableau is feasible, i.e., $\bar{b} \geq 0$, then the resulting is a feasible D-reduced tableau.

17.2 Dual D-Reduced Simplex Algorithm

Starting from a dual feasible D-reduced tableau, the method presented in this section generates a sequence of such tableaus to achieve primal feasibility until achieving a pair of primal and dual optimal basic solutions.

Let (17.3) be such a tableau and let (17.10) hold. It is noted by Preposition 17.1.1 that the objective value of the basic solution $\bar{x}$ is $\bar{f}$, which is equal to the opposite number of the south-eastern entry of the tableau.

If entries in the datum row are all less than or equal to 0, the problem has no feasible solution (Lemma 17.1.1). This is usually not the case, and hence the following rule is well-defined.

Rule 17.2.1 (Dual Column Rule) *Determine the column index q such that*

$$\beta = \bar{z}_q/\bar{\omega}_q = \min\{\bar{z}_j/\bar{\omega}_j \mid \bar{\omega}_j > 0, j \in N\} \geq 0, \quad (17.16)$$

where β is termed *dual stepsize*.

If $\beta \neq 0$, add $-\beta$ times of the datum row to the bottom row to convert $\bar{z}_q$ to 0, then apply the following row rule.

Rule 17.2.2 (Dual Row Rule) *Select the row index*

$$p \in \arg \max_{i \in R} \bar{a}_{iq}.$$

Since $\bar{\omega}_q > 0$, an optimal D-reduced tableau is already obtained if $\bar{a}_{pq} \leq 0$ (by taking column q as datum column); otherwise, convert it to 1, and eliminate the other nonzeros of column q by elementary transformations. Thus, an iteration is complete; consequently, column q enters, and column j_p leaves the basis. It is noted that the latter becomes the new datum column.

The resulting tableau remains dual feasible. In fact, Rule 17.2.2 ensures a largest possible stepsize β subject to dual constraints. Now the south-eastern entry of the

tableau becomes $-\bar{f} + \beta$, so the increment of the objective value is just equal to $\beta \geq 0$. If β vanishes, the objective value will remain unchanged.

If it holds that $\bar{z}_N > 0$, the D-reduced tableau is said to be *dual nondegenerate*. In this case, β given by (17.16) is positive, and hence the objective value strictly increases.

We repeat the preceding steps until optimality is achieved or infeasibility detected. It is not difficult to show that, the leaving column j_p will not enter the basis immediately, and the entering column q will not leave immediately either. The basis change will not touch the objective value; only adding a multiple of the datum row to the bottom row changes it. Since the f column remains unchanged in the process, it can be omitted from the tableau.

The overall steps can be put into the following model.

Algorithm 17.2.1 (Dual D-Reduced Simplex Algorithm: Tableau Form) Initial: dual feasible D-reduced tableau of form (17.3). This algorithm solves the D-reduced problem (17.1).

1. Stop if $J = \{j \in N \mid \bar{\omega}_j > 0\} = \emptyset$.
2. Determine β and column index q such that $\beta = \bar{z}_q / \bar{\omega}_q = \min_{j \in J} \bar{z}_j / \bar{\omega}_j$.
3. If $\beta > 0$, add $-\beta$ times of datum row to the bottom row.
4. Determine row index $p \in \arg \max_{i \in R} \bar{a}_{iq}$.
5. Stop with $\bar{x}_q = 1 / \bar{\omega}_q$, $\bar{x}_B = -\bar{x}_q \bar{a}_q(R)$ if $\bar{a}_{pq} \leq 0$.
6. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
7. Go to step 1.

Theorem 17.2.1 Under the dual nondegeneracy assumption, Algorithm 17.2.1 terminates either at

- (i) Step 1, detecting infeasibility of the problem, or at
- (ii) Step 5, giving a basic optimal solution.

Proof The proof of the termination under the dual nondegeneracy is the same as in the classical simplex context. Meanings of the exits come from Lemmas 17.1.1 and 17.1.3, as well as discussion preceding the Algorithm $\square$

This Algorithm can be viewed as solving the dual problem through the tableau. To explain, return to the dual problem (17.14).

It is clear that there is the following solution to the dual problem:

$$\begin{aligned} y &= 0, & g &= 0, \\ z_B &= 0, & z_N &= \bar{z}_N, \end{aligned}$$

which is feasible when $\bar{z}_N \geq 0$. The reduced gradient of the objective function is given by

$$\Delta y = 0, \quad \Delta g = 1,$$

along which the objective value in the reduced space increases the fastest. The associated vector in z space is then

$$\Delta z_B = 0, \quad \Delta z_N = -\bar{\omega}_N.$$

This leads to the following update:

$$\hat{z}_N = \bar{z}_N - \beta \bar{\omega}_N, \quad \hat{g} = \beta. \quad (17.17)$$

It is noted that the dual stepsize $\beta \geq 0$ equals the according increment of the objective value. If $\bar{\omega}_N \not\leq 0$, the greatest possible value, maintaining dual feasibility, is

$$\beta = \bar{z}_q / \bar{\omega}_q = \min\{\bar{z}_j / \bar{\omega}_j \mid \bar{\omega}_j > 0, j \in N\}. \quad (17.18)$$

It is then seen that the dual D-reduced simplex algorithm actually deals with the dual problem according to the idea of the reduced simplex algorithm.

Example 17.2.1 Solve the following problem by Algorithm 17.2.1:

$$\begin{aligned} \min \quad & 3x_1 + 5x_2 + 2x_3 + 4x_4, \\ \text{s.t.} \quad & 2x_1 + 5x_2 - 3x_3 - 4x_4 + x_5 = -2, \\ & -4x_1 - 3x_2 + 6x_3 - 2x_4 + x_6 = 4, \\ & -x_1 - 4x_2 - x_3 + 2x_4 + x_7 = -5, \\ & x_j \geq 0, \quad j = 1, \dots, 7. \end{aligned}$$

Answer Initial tableau

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
2	5	-3	-4	1			-2
-4	-3	6	-2		1		4
-1	-4	-1	2			1	-5*
3	5	2	4				

Iteration 1:

$\max\{|-2|, |4|, |-5|\} = 5$, $r = 3$, take the third row as datum row. Multiply row 3 by $-1/5$, and add 2, -4 times of row 3 to rows 1, 2, respectively, resulting in the following D-reduced tableau:

which is dual feasible, as the bottom line is nonnegative.

Call Algorithm 17.2.1.

Iteration 2:

1. $J = \{1, 2, 3\} \neq \emptyset$.
2. $\beta = \min\{3/(1/5), 5/(4/5), 2/(1/5)\} = 5/(4/5)$, $q = 2$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
12/5	33/5*	-13/5	-24/5	1		-2/5	
-24/5	-31/5	26/5	-2/5		1	4/5	
1/5	4/5	1/5	-2/5			-1/5	1
3	5	2	4				

3. Add $-5/(4/5)$ times of row 3 to the bottom row.
4. $\max\{33/5, -31/5\} = 33/5 > 0$, $p = 1$.
6. Multiply row 1 by $5/33$, and add $31/5$, $-4/5$ times of row 1 to rows 2 and 3, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
4/11	1	-13/33	-8/11	5/33		-2/33	
-28/11		91/33*	-54/11	31/33	1	14/33	
-1/11		17/33	2/11	-4/33		-5/33	1
7/4		3/4	13/2			5/4	-25/4

Iteration 3:

1. $J = \{3, 4\} \neq \emptyset$.
2. $\beta = \min\{(3/4)/(17/33), (13/2)/(2/11)\} = (3/4)/(17/33)$, $q = 3$.
3. Add $-(3/4)/(17/33)$ times of row 3 to the bottom row.
4. $\max\{-13/33, 91/33\} = 91/33 > 0$, $p = 2$.
6. Multiply row 2 by $33/91$, and add $13/33$, $-17/33$ times of row 2 to rows 1 and 3, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	1		-10/7	2/7	1/7		
-12/13		1	-162/91	31/91	33/91	2/13	
5/13			100/91	-27/91	-17/91	-3/13	1
32/17			106/17	3/17		25/17	-131/17

Iteration 4:

1. $J = \{1, 4\} \neq \emptyset$.
2. $\beta = \min\{(32/17)/(5/13), (106/17)/(100/91)\} = (32/17)/(5/13)$, $q = 1$.
3. Add $(32/17)/(5/13)$ times of row 3 to the bottom row:
4. $\max\{-12/13, 0\} \leq 0$.
5. $\bar{x}_1 = 13/5$. The optimal basic solution and optimal value:

$$x^* = (13/5, 0, 12/5, 0, 0, 0, 0)^T, \quad f^* = 63/5.$$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	1		$-10/7$	$2/7$	$1/7$		
$-12/13$		1	$-162/91$	$31/91$	$33/91$	$2/13$	
$5/13$			$100/91$	$-27/91$	$-17/91$	$-3/13$	1
			$6/7$	$57/35$	$32/35$	$13/5$	$-63/5$

Further, we give the revised version of Algorithm 17.2.1. Denote by B_R the submatrix consisting of entries associated with B , R (similarly below). Assume that row $r = m$ is the datum row. It is not difficult to derive the following revised tableau which is equivalent to the D-reduced tableau (17.3):

$$\begin{array}{c|c|c|c}
 x_B^T & x_N^T & f & \text{RHS} \\
 \hline
 I & B_R^{-1} N_R & & \\
 & e_r^T N - e_r^T B B_R^{-1} N_R & & 1 \\
 \hline
 & c_N^T - c_B^T B_R^{-1} N_R & -1 &
 \end{array} \quad (17.19)$$

Assume the column q enters the basis. Introduce $\bar{a}_q(R) = B_R^{-1} a_q(R)$, where $a_q(R)$ is a_q 's subvector associated with row index set R . The formula for updating the inverse of the basis matrix is

$$\widehat{B}_R^{-1} = E_s B_R^{-1}, \quad (17.20)$$

where

$$E_p = \begin{pmatrix} 1 & -\bar{a}_{i_1, q} / \bar{a}_{i_s, q} & & & \\ & \ddots & \vdots & & \\ & & -\bar{a}_{i_{s-1}, q} / \bar{a}_{i_s, q} & & \\ & & 1 / \bar{a}_{i_s, q} & & \\ & & -\bar{a}_{i_{s+1}, q} / \bar{a}_{i_s, q} & & \\ & & \vdots & \ddots & \\ & & -\bar{a}_{i_{m-1}, q} / \bar{a}_{i_s, q} & & 1 \end{pmatrix} \quad (17.21)$$

Based on equivalence between (17.3) and (17.19), a revised version of Algorithm 17.2.1 is stated as follows.

Algorithm 17.2.2 (Dual D-Reduced Simplex Algorithm) Initial: B, R, N defined by (17.2). B_R^{-1} and $\bar{z}_N \geq 0$. This algorithm solves the D-reduced problem (17.1).

1. Compute $\bar{\omega}_N = N^T e_m - N_R^T B_R^{-T} B^T e_m$.
2. Stop if $J = \{j \in N \mid \bar{\omega}_j > 0\} = \emptyset$ (dual unbounded problem).

3. Determine β and column index q such that $\beta = \bar{z}_q / \bar{\omega}_q = \min_{j \in J} \bar{z}_j / \bar{\omega}_j$.
4. If $\beta > 0$, update by $\bar{z}_N = \bar{z}_N - \beta \bar{\omega}_N$.
5. Compute $\bar{a}_q(R) = B_R^{-1} a_q(R)$.
6. Select $p \in \arg \max_{i \in R} \bar{a}_{i q}$ ($p = i_s$).
7. If $\bar{a}_{p, q} \leq 0$, compute

$$\bar{x}_q = 1 / \bar{\omega}_q, \quad \bar{x}_B = -\bar{x}_q \bar{a}_q(R), \quad \bar{f} = c_q \bar{x}_q + c_B^T \bar{x}_B.$$

8. Stop (optimality achieved).
9. Update B_R^{-1} by (17.20).
10. Update (B, N) by exchanging j_s and q .
11. Go to step 1.

17.2.1 Dual D-Reduced Phase-I

The dual D-reduced algorithm has to start from a dual feasible tableau/solution. If $\bar{z}_N = c_N - N_R^T B_R^{-T} c_B \geq 0$, such a starting point is readily available; otherwise, there is a need for dual Phase-I procedure to provide it. An initial dual feasible tableau for Algorithm 17.2.1 can be obtained by the following modification of D-reduced simplex Algorithm 17.3.1.

Algorithm 17.2.1 (Dual D-Reduced Phase-I: Tableau Form) *Initial : D-reduced tableau of form (17.3). This algorithm finds a dual feasible D-reduced tableau.*

1. Determine column index $q \in \arg \min_{j \in N} \bar{z}_j$.
2. Stop if $\bar{z}_q \geq 0$.
3. If $\bar{\omega}_q \neq 0$, add $\beta = -\bar{z}_q / \bar{\omega}_q$ times of the datum row to the bottom row.
4. Determine row index $p \in \arg \max_{i \in R} \bar{a}_{i q}$.
5. Go to 7 if $\bar{a}_{p q} > 0$.
6. Stop if $\bar{\omega}_q = 0$.
7. Convert $\bar{a}_{p q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
8. Go to step 1.

Theorem 17.2.2 *Assume termination of Algorithm 17.2.1. It terminates either at*

- (i) *step 2, giving a dual feasible D-reduced tableau, or at*
- (ii) *step 6, detecting infeasibility or lower unboundedness.*

Proof Validity of the meaning of exit 2 is clear. Assume that it returns from exit 6. Then $\bar{z}_q < 0$, $\bar{a}_q \leq 0$ and $\bar{\omega}_q = 0$ hold. If the problem is feasible, by Lemma 17.1.1 there is $q' \in N$ such that $\bar{\omega}_{q'} > 0$. Taking $\bar{\omega}_{q'}$ as pivot, a (conventional) simplex tableau is obtained by carrying out elementary transformations, without changing the q column. As components of the column are positive, the problem is infeasible or lower unbounded, by Lemma 15.4.1. $\square$

Example 17.2.2 Solve following problem using Algorithm 17.2.1 as Phase-I :

$$\min -3x_1 - 5x_2 + 4x_3 + 8x_4,$$

$$\text{s.t.} \quad 6x_1 - 2x_2 + 3x_3 + 2x_4 + x_5 = 2,$$

$$\quad \quad -2x_1 + 4x_2 - x_3 - 8x_4 + x_6 = -4,$$

$$\quad \quad 4x_1 + 5x_2 + 2x_3 + x_4 + x_7 = 6,$$

$$x_j \geq 0, \quad j = 1, \dots, 7.$$

Answer Initial tableau

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
6	-2	3	2	1			2
-2	4	-1	-8		1		-4*
4	5	2	1			1	6
-3	-5	4	8				

Iteration 1:
Arbitrarily take row 2 to be as the datum row. Multiply the row by $-1/4$, and add $-2, -6$ times of it to rows 1,3, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
5		5/2	-2	1	1/2		
1/2	-1	1/4	2		-1/4		1
1	11*	1/2	-11		3/2	1	
-3	-5	4	8				

Dual Phase-I: call Algorithm 17.2.1. $r = 2$.

Iteration 2:

1. $\min\{-3, -5, 4, 8\} = -5 < 0, \quad q = 2.$

3. Add $\beta = -(-5)/(-1) = -5$ times of row 2 to the bottom row.

4. $\max\{0, 11\} = 11 > 0, \quad p = 3.$

7. Add $1/11$ times of row 3 to the bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
5*		5/2	-2	1	1/2		
13/22		13/44	1		-5/44	1/11	1
1/11	1	1/22	-1		3/22	1/11	
-11/2		11/4	-2		5/4		-5

Iteration 3:

1. $\min\{-11/2, 11/4, -2, 5/4, 0\} = -11/2 < 0$, $q = 1$.
3. Add $\beta = -(-11/2)/(13/22)$ times of row 2 to the bottom row.
4. $\max\{5, 1/11\} = 5 > 0$, $p = 1$.
7. Multiply row 1 by $1/5$, and add $-13/22$, $-1/11$ times of row 1 to rows 2 and 3, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1		1/2	-2/5	1/5	1/10		
			68/55	-13/110	-19/110	1/11	1
	1		-53/55	-1/55	7/55	1/11	
		11/2	95/13		5/26	11/13	56/13

Iteration 4:

1. $\min\{11/2, 95/13, 0, 5/26, 11/13\} \geq 0$, $q = 1$.
2. Dual feasibility is achieved.

Dual Phase-II: call Algorithm 17.2.1: $r = 2$, $q = 2$.

Iteration 5:

1. $J = \{4, 7\}$.
2. $\beta = \max\{-(95/13)/(68/55), -(11/13)/(1/11)\} = -(95/13)/(68/55)$, $q = 4$.
3. Add β times of row 2 to the bottom row.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1		1/2	-2/5	1/5	1/10		
			68/55	-13/110	-19/110	1/11	1
	1		-53/55	-1/55	7/55	1/11	
		11/2		95/136	165/136	21/68	-109/68

4. $\max\{-2/5, -53/55\} \leq 0$.
5. Take $\bar{x}_4 = 55/68$. The optimal basic solution and optimal value:

$$x^* = (11/34, 53/68, 0, 55/68, 0, 0, 0, 1)^T, \quad f^* = 109/68.$$

Based on the equivalence between the reduced tableau (17.3) and the revised tableau (17.19), the tableau Algorithm 17.2.1 can be turned to the following revision.

Algorithm 17.2.2 (Dual D-Reduced Phase-I) *Initial: B, R, N defined by (17.2). B_R^{-1} ; $\bar{z}_N = c_N - N_R^T B_R^{-T} c_B$. This algorithm finds a dual feasible D-reduced*

basis.

1. Determine column index $q \in \arg \min_{j \in N} \bar{z}_j$.
2. Stop if $\bar{z}_q \geq 0$ (dual feasibility achieved).
3. Compute $\bar{\omega}_N = N^T e_m - N_R^T (B_R^{-T} B e_m)$.
4. If $\bar{\omega}_q \neq 0$, update $\bar{z}_N = \bar{z}_N + \beta \bar{\omega}_N$, where $\beta = -\bar{z}_q / \bar{\omega}_q$.
5. compute $\bar{a}_q(R) = B_R^{-1} a_q(R)$.
6. Determine $p \in \arg \max_{i \in R} \bar{a}_{iq} (p = i_s)$.
7. Go to 9 if $\bar{a}_{p,q} > 0$.
8. Stop if $\bar{\omega}_q = 0$ (infeasible or lower unbounded).
9. Update B_R^{-1} by (17.20).
10. Update (B, N) by exchanging j_s and q .
11. Go to step 1.

There have been no numerical results with the algorithms, described in this section. It seems to be better to obtain a (conventional) dual feasible simplex tableau (basis) first, and then convert it to a D-reduced tableau (basis).

17.3 D-Reduced Simplex Algorithm

Starting from a feasible D-reduced tableau, the so-called D-reduced simplex algorithm, described in this section, proceeds toward dual feasibility, iteration by iteration, while maintaining primal feasibility.

Let (17.3) be a feasible D-reduced tableau with column q as datum column, satisfying conditions (17.6) and (17.9). Column pivoting is carried out first.

Rule 17.3.1 (Column Rule) *Select the column index*

$$q' \in \arg \min \{\bar{z}_j \mid j \in N, j \neq q\}. \quad (17.22)$$

If $\bar{z}_{q'} \geq 0$, it is clear that optimality is already achieved. Now assume that

$$\bar{z}_{q'} < 0. \quad (17.23)$$

According to whether

$$\bar{a}_{iq'} \leq 0, \quad i \in R \quad (17.24)$$

holds or not, one of the following two cases is handled:

(i) Condition (17.24) holds.

If $\bar{\omega}_{q'} \leq 0$, the problem is lower unbounded (Lemma 17.1.4). If $\bar{\omega}_{q'} > 0$, then set $q = q'$ and compute

$$\beta = -\bar{z}_{q'} / \bar{\omega}_{q'}, \quad (17.25)$$

and add β times of the datum row to the bottom row, so that $\bar{z}_q$ is zeroed. Taking column q as the new datum column, it is clear that the D-reduced tableau is still feasible, thus an iteration is complete. As $\beta > 0$, the objective value $\bar{f}$, associated with the tableau, strictly decreases.

(ii) Condition (17.24) does not hold.

Then the following rule is well-defined:

Rule 17.3.2 (Row Rule) Determine the row index p such that

$$\alpha = -\bar{a}_{pq}/\bar{a}_{pq'} = \min\{-\bar{a}_{iq}/\bar{a}_{iq'} \mid \bar{a}_{iq'} > 0, i \in R\}. \quad (17.26)$$

It is known from (17.9) and (17.26) that

$$\bar{a}_{pq} \leq 0, \quad \bar{a}_{pq'} > 0. \quad (17.27)$$

Therefore, $\alpha \geq 0$. When the strict inequality in (17.9) holds, the D-reduced tableau is said to be *nondegenerate*. In this case, $\alpha > 0$.

Lemma 17.3.1 Assume that (17.23), (17.6), and (17.9) hold, and q' and p are determined by (17.22) and (17.26), respectively. Then the problem is lower unbounded if

$$\bar{\omega}_{q'} < 0, \quad \alpha \geq -\bar{\omega}_q/\bar{\omega}_{q'}. \quad (17.28)$$

Proof Multiply the datum row of the D-reduced tableau by $1/\bar{\omega}_q$, then for $i \in R$, add $-\bar{a}_{iq}$ times of the datum row to the i th row. The resulting is a (conventional) feasible simplex tableau. The q' column's component at the datum row of this tableau is

$$\tilde{\omega}_{q'} = \bar{\omega}_{q'}/\bar{\omega}_q < 0, \quad (17.29)$$

where the inequality comes from the first expression of (17.6) and of (17.28). Besides, the other components of q' column are

$$\tilde{a}_{iq'} = \bar{a}_{iq'} - \bar{a}_{iq}(\bar{\omega}_{q'}/\bar{\omega}_q), \quad i \in R. \quad (17.30)$$

If $\bar{a}_{iq'} \leq 0$, $i \in R$, then from (17.9), (17.29), and (17.30), it is known that $\tilde{a}_{iq'} \leq 0$; and if $\bar{a}_{iq'} > 0$, $i \in R$, from the second expression of (17.28) and (17.26), it follows that

$$-\bar{a}_{iq}/\bar{a}_{iq'} \geq -\bar{\omega}_q/\bar{\omega}_{q'} > 0, \quad i \in R, \quad (17.31)$$

where the strict inequality is from the first expression of (17.6) and of (17.28). From $\bar{a}_{iq'} > 0$ and (17.28), it follows that $\bar{a}_{iq} < 0$ and

$$-\bar{a}_{iq'}/\bar{a}_{iq} \leq -\bar{\omega}_{q'}/\bar{\omega}_q.$$

Multiplying the preceding by $-\bar{a}_{iq} > 0$ and combining the resulting expression with (17.30) lead to $\bar{a}_{iq'} \leq 0$. Thus, it was shown that the q' column of the feasible simplex tableau is less than or equal to zero. By Lemma 3.1.2, the problem is lower unbounded. $\square$

Now assume that (17.28) does not hold. Execute elementary transformations on the feasible D-reduced tableau to convert $\bar{a}_{pq'}$ to 1, and eliminate the other nonzeros in the column (including those in the bottom row). Thus, x_{j_p} leaves and $x_{q'}$ enters the basis. As $p \in R$, the objective value $\bar{f}$ remains unchanged. If the q -indexed entry of the resulting bottom row is nonzero, add appropriate times of the datum row to the bottom row to eliminate it. For the resulting tableau, the following result stands good.

Proposition 17.3.1 *Assume that (17.23), (17.6), and (17.9) hold, and q' and p are determined by (17.22) and (17.26), respectively. If $\bar{\omega}_{q'} \geq 0$, or*

$$\bar{\omega}_{q'} < 0, \quad \alpha < -\bar{\omega}_q/\bar{\omega}_{q'}, \quad (17.32)$$

then the new tableau is a feasible D-reduced tableau, with column q as the datum column. The according objective value never increases and strictly decreases under nondegeneracy assumption.

Proof Mark entries of the new tableau by $\hat{\bullet}$. The following inequality will be shown first:

$$\hat{a}_{iq} \leq 0, \quad i \in R. \quad (17.33)$$

The p ($\neq r$)th component of the q column of the new tableau is

$$\hat{a}_{pq} = \bar{a}_{pq}/\bar{a}_{pq'} \leq 0, \quad (17.34)$$

where the inequality follows from (17.9) and the first expression of (17.27). The other components of the column are

$$\hat{a}_{iq} = \bar{a}_{iq} - \bar{a}_{iq'}(\bar{a}_{pq}/\bar{a}_{pq'}), \quad i \in R, \quad i \neq p. \quad (17.35)$$

For $i \in R, i \neq p$, if $\bar{a}_{iq'} \leq 0$, then by (17.9), (17.34), and (17.35), it is known that $\hat{a}_{iq} \leq 0$; and if $\bar{a}_{iq'} > 0$, by (17.26) and (17.35), it is known that $\hat{a}_{iq} \leq 0$. Therefore, (17.33) hold.

On the other hand, the q column's component at the datum row of the new tableau is

$$\hat{\omega}_q = \bar{\omega}_q - \bar{\omega}_{q'}(\bar{a}_{pq}/\bar{a}_{pq'}). \quad (17.36)$$

If $\bar{\omega}_{q'} \geq 0$, from the first expression of (17.6), (17.34) and (17.36), it follows that

$$\hat{\omega}_q > 0. \quad (17.37)$$

If, otherwise, (17.32) holds, from the second expression of (17.6) and (17.26), it follows that

$$-\bar{a}_{pq}/\bar{a}_{pq'} < -\bar{\omega}_q/\bar{\omega}_{q'}.$$

Multiplying the preceding by $\bar{\omega}_{q'} < 0$ gives

$$-\bar{\omega}_{q'}(\bar{a}_{pq}/\bar{a}_{pq'}) > -\bar{\omega}_q,$$

combining which and (17.36) leads to (17.37).

Note that $\bar{z}_q = 0$. After a basis change by taking $\bar{a}_{pq'}$ as pivot, the bottom row's entry at the q column equals

$$\nu = \bar{z}_{q'}\alpha \leq 0,$$

where the inequality follows from (17.26) and (17.34). If $\nu = 0$, then it is known by (17.33) and (17.37) that resulting is a feasible D-reduced tableau with q column as the datum one, and the according objective value does not increase. If nondegeneracy is assumed, then $\alpha > 0$ and $\nu < 0$. Add $-\nu/\hat{\omega}_q$ times of the datum row of the current tableau to the bottom row to eliminate its entry at the q column, the right-hand side entry at the bottom row becomes

$$-(\bar{f} + \nu/\hat{\omega}_q).$$

Thus, from $\nu < 0$ and $\hat{\omega}_q > 0$, it is known that the associate objective value strictly decreases. $\square$

The preceding steps are repeated until an optimal tableau is obtained or the lower unboundedness of the problems detected. The overall steps are put into the following algorithm.

Algorithm 17.3.1 (D-Reduced Simplex Algorithm: Tableau Form) Initial: feasible D-reduced tableau of form (17.3) with q -indexed column as the datum one. This algorithm solves D-reduced problem (17.1).

1. Select column index $q' \in \arg \min\{\bar{z}_j \mid j \in N, j \neq q\}$.
2. Stop with $\bar{x}_q = 1/\bar{\omega}_q$, $\bar{x}_B = -\bar{x}_q\bar{a}_q(R)$ if $\bar{z}_{q'} \geq 0$.
3. Go to step 6 if $I = \{i \in R \mid \bar{a}_{iq'} > 0\} \neq \emptyset$.
4. Stop if $\bar{\omega}_{q'} \leq 0$.
5. Set $q = q'$, and go to step 9.
6. Determine α and row index p such that $\alpha = -\bar{a}_{pq}/\bar{a}_{pq'} = \min\{-\bar{a}_{iq}/\bar{a}_{iq'} \mid \bar{a}_{iq'} > 0, i \in R\}$.
7. Stop if $\bar{\omega}_{q'} < 0$ and $\alpha \geq -\bar{\omega}_q/\bar{\omega}_{q'}$.
8. Convert $\bar{a}_{pq'}$ to 1 and eliminate the other nonzeros in the column by elementary transformations.
9. Add $\beta = -\bar{z}_q/\bar{\omega}_q$ times of the datum row to the bottom row.
10. Go to step 1.

Theorem 17.3.2 *Under nondegeneracy assumption, Algorithm 17.3.1 terminates either at*

- (i) *Step 2, giving a basic optimal solution, or at*
- (ii) *Step 4 or 7, detecting lower unboundedness of the problem.*

Proof The proof of the termination of the Algorithm under the assumption of nondegeneracy is the same as that in the conventional context. Meanings of its exits follow from Lemmas 17.1.3, 17.1.4, and 17.3.1, as well as the discussions preceding the Algorithm. $\square$

Example 17.3.1 Solve the following problem by Algorithm 17.3.1:

$$\begin{aligned} \min \quad & 5x_1 - 4x_2 + 7x_3 - x_4, \\ \text{s.t.} \quad & -3x_1 - 2x_2 + 5x_3 - 4x_4 + x_5 = 4, \\ & -6x_1 - 4x_2 + 3x_3 + 2x_4 + x_6 = 2, \\ & x_1 + 5x_2 - 4x_3 + x_4 + x_7 = 3, \\ & x_j \geq 0, \quad j = 1, \dots, 7. \end{aligned}$$

Answer Initial tableau

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
-3	-2	5	-4	1			4*
-6	-4	3	2		1		2
1	5	-4	1			1	3
5	-4	7	-1				

Iteration 1:

$\max\{|4|, |2|, |3|\} = 4$, $r = 1$. Take the first row as pivot row. Multiply the row by $1/4$, and add -2 , -3 times of it to rows 2, 3, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
-3/4	-1/2	5/4	-1	1/4			1
-9/2	-3	1/2	4	-1/2	1		
13/4	13/2*	-31/4	4	-3/4		1	
5	-4	7	-1				

x_5 becomes a nonbasis variable. For the tableau, $\bar{a}_{1,5} = 1/4 > 0$, $\bar{a}_{2,5} = -1/2 \leq 0$, $\bar{a}_{3,5} = -3/4 \leq 0$, hence it is a feasible D-reduced tableau.

Call Algorithm 17.3.1. $r = 1$, $q = 5$.

Iteration 2:

1. $\min\{5, -4, 7, -1\} = -4 < 0$, $q' = 2$.
3. $I = \{3\} \neq \emptyset$.
6. $\alpha = \min\{-(-3/4)/(13/2)\} = 3/26$, $p = 3$.
7. $\bar{a}_{1,2} = -1/2 < 0$, $3/26 < -(1/4)/(-1/2) = 1/2$.
8. Multiply row 3 by $2/13$, and add $1/2, 3, 4$ times of row 3 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-1/2$		$17/26$	$-9/13$	$5/26$		$1/13$	1
-3		$-40/13$	$76/13$	$-11/13$	1	$6/13$	
$1/2$	1	$-31/26$	$8/13$	$-3/26$		$2/13$	
7		$29/13$	$19/13$	$-6/13$		$8/13$	

9. Add $\beta = -(-6/13)/(5/26)$ times of row 1 to the bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-1/2$		$17/26$	$-9/13$	$5/26$		$1/13$	1
-3		$-40/13$	$76/13^*$	$-11/13$	1	$6/13$	
$1/2$	1	$-31/26$	$8/13$	$-3/26$		$2/13$	
$29/5$		$19/5$	$-1/5$			$4/5$	$12/5$

Iteration 3:

1. $\min\{29/5, 19/5, -1/5, 4/5\} = -1/5 < 0$, $q' = 4$.
3. $I = \{2, 3\} \neq \emptyset$.
6. $\alpha = \min\{-(-11/13)/(76/13), -(-3/26)/(8/13)\}$
 $= -(-11/13)/(76/13) = 11/76$, $p = 2$.
7. $\bar{a}_{1,4} = -9/13 < 0$, $11/76 < -(5/26)/(-9/13) = 5/18$.
8. Multiply row 2 by $13/76$, and add $9/13, -8/13, 1/5$ times of row 2 to rows 1, 3, and 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-65/76$		$11/38$		$7/76$	$9/76$	$5/38$	1
$-39/76$		$-10/19$	1	$-11/76$	$13/76$	$3/38$	
$31/38$	1	$-33/38$		$-1/38$	$-2/19$	$2/19$	
$433/76$		$351/95$		$-11/380$	$13/380$	$31/38$	$12/5$

9. Add $\beta = -(-11/380)/(7/76)$ times of row 1 to the bottom row.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
-65/76		11/38		7/76	9/76	5/38	1
-39/76		-10/19	1	-11/76	13/76	3/38	
31/38	1	-33/38		-1/38	-2/19	2/19	
38/7		53/14			1/14	6/7	19/7

Iteration 4:

1. $\min\{38/7, 53/14, 1/14, 6/7\} \geq 0$.
2. Set $\bar{x}_5 = 76/7$. The optimal basic solution and optimal value:

$$x^* = (0, 2/7, 0, 11/7, 76/7, 0, 0, 1)^T, \quad f^* = -19/7.$$

A revised version of Algorithm 17.3.1 can be obtained by deriving the search direction using the inverse of the basis matrix, as done for the derivation of the (conventional) revised simplex algorithm. But, for simplicity, here we directly apply the equivalence between the D-reduced tableau (17.3) and revised tableau (17.19).

Denote a vector $\bullet$'s subvector, corresponding to the row index set R , by $\bullet(R)$. Then the revised version of Algorithm 17.3.1 can be described as follows.

Algorithm 17.3.2 (D-Reduced Simplex Algorithm) Initial: B, R, N defined by (17.2). B_R^{-1} ; $\bar{z}_N = c_N - N_R^T B_R^{-T} c_B$, $\bar{\omega}_N = N^T e_m - N_R^T (B_R^{-T} B e_m)$. $\bar{a}_q(R) = B_R^{-1} a_q(R) \leq 0$, $\bar{\omega}_q > 0$, $\bar{z}_q = 0$. This algorithm solves the D-reduced problem (17.1).

1. Determine column index $q' \in \arg \min\{\bar{z}_j \mid j \in N, j \neq q\}$.
2. If $\bar{z}_{q'} \geq 0$, compute

$$\bar{x}_q = 1/\bar{\omega}_q, \quad \bar{x}_B = -\bar{x}_q \bar{a}_q(R), \quad \bar{f} = c_q \bar{x}_q + c_B^T \bar{x}_B.$$

3. Stop (optimality achieved).
4. Compute $\bar{a}_{q'}(R) = B_R^{-1} a_{q'}(R)$.
5. Go to step 7 if $I = \{i \in R \mid \bar{a}_{i q'} > 0\} \neq \emptyset$.
6. Stop if $\bar{\omega}_{q'} \leq 0$ (infeasible problem).
7. Set $q = q'$, and go to step 12.
8. Determine α and p such that $\alpha = -\bar{a}_{p q}/\bar{a}_{p q'} = \min_{i \in I} -\bar{a}_{i q}/\bar{a}_{i q'} \ (p = i_s)$.
9. Stop if $\bar{\omega}_{q'} < 0$ and $\alpha \geq -\bar{\omega}_q/\bar{\omega}_{q'}$ (unbounded problem).
10. Compute $\sigma_N = N_R^T B_R^{-T} e_s$.
11. Update $\bar{\omega}_N = \bar{\omega}_N + \nu \sigma_N$, $\bar{\omega}_{j_p} = \nu$, where $\nu = -\bar{\omega}_{q'}/\sigma_{q'}$.
12. Update $\bar{z}_N = \bar{z}_N + \beta_1 \sigma_N$, $\bar{z}_{j_p} = \beta_1$, where $\beta_1 = -\bar{z}_{q'}/\sigma_{q'}$.
13. Update $\bar{z}_N = \bar{z}_N + \beta_2 \bar{\omega}_N$, where $\beta_2 = -\bar{z}_q/\bar{\omega}_q$.
14. Update B_R^{-1} by (17.20).

15. Update (B, N) by exchanging j_s and q .
16. Go to step 1.

17.3.1 D-Reduced Phase-I: Most-Obtuse-Angle Rule

As usual, the D-reduced simplex method, described in the previous section, is only a Phase-II procedure. Any conventional Phase-I method can be slightly modified to generate a starting point (Chap. 14). Dropping the initial condition $\bar{z}_N \geq 0$, in addition, the dual Algorithm 17.2.1 can be directly used to achieve primal feasibility.

According to the analysis made at the second half of Sect. 17.2, the reduced gradient of the dual objective function corresponds to the negative datum row in z -space. Based on the most-obtuse-angle heuristic, the dual nonnegativity constraint (e.g., $z_q \geq 0$), whose gradient and the negative datum row form the most obtuse angle, should be binding, that is, the associated index be chosen to enter the basis. Based on this idea, the following algorithm can be obtained by modifying the dual Algorithm 17.2.1.

Algorithm 17.3.3 (Tableau D-Reduced Phase-I: Most-Obtuse-Angle) Initial: D-reduced tableau of form (17.3). This algorithm finds a feasible D-reduced tableau.

1. Determine column index $q \in \arg \max_{j \in N} \bar{\omega}_j$.
2. Stop if $\bar{\omega}_q \leq 0$.
3. Add $\beta = -\bar{z}_q / \bar{\omega}_q$ times of the datum row to the bottom row.
4. Determine row index $p \in \arg \max \{\bar{a}_{iq} \mid i \in R\}$.
5. Stop if $\bar{a}_{p,q} \leq 0$.
6. Convert $\bar{a}_{p,q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
7. Go to step 1.

Theorem 17.3.3 Assume termination of Algorithm 17.3.3. It terminates at

- (i) Step 2, detecting infeasibility of the problem, or at
- (ii) Step 5, giving a feasible D-reduced tableau.

Proof It follows from Lemmas 17.1.1 and 17.2.1. □

Example 17.3.2 Solve the following problem using Algorithm 17.3.3 as Phase-I:

$$\begin{aligned}
 \min \quad & f = -x_1 + 5x_2 - 3x_3 - x_4, \\
 \text{s.t.} \quad & x_1 + 1x_2 + 2x_3 + 5x_4 + x_5 = 3, \\
 & -7x_1 + 3x_2 + 4x_3 - x_4 + x_6 = -5, \\
 & -2x_1 - x_2 + x_3 - x_4 + x_7 = -2, \\
 & -5x_1 - 6x_2 + 8x_3 - 2x_4 + x_8 = 6, \\
 & x_j \geq 0, \quad j = 1, \dots, 8.
 \end{aligned}$$

Answer Initial tableau

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1	1	2	5	1				3*
-7	3	4	-1		1			-5
-2	-1	1	-1			1		-2
-5	-6	8	-2				1	6
-1	5	-3	-1					

Iteration 1:

Arbitrarily take the first row to be as the datum row. Multiply the row by $1/3$, and add $5/3$; 2 ; -6 times of it to rows 2, 3, 4, respectively, yielding the following D-reduced tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
$1/3$	$1/3$	$2/3$	$5/3$	$1/3$				1
$-16/3$	$14/3$	$22/3$	$22/3^*$	$5/3$	1			
$-4/3$	$-1/3$	$7/3$	$7/3$	$2/3$		1		
-7	-8	4	-12	-2			1	
-1	5	-3	-1	$1/5$				

Phase-I: Call Algorithm 17.3.3. $r = 1$.

Iteration 2:

- $\max\{1/3, 1/3, 2/3, 5/3, 1/3\} = 5/3 > 0$, $q = 4$.
- Add $\beta = -(-1)/(5/3) = 3/5$ times of row 1 to the bottom row.
- $\max\{22/3, 7/3, -12\} = 22/3 > 0$, $p = 2$.
- Multiply row 2 by $3/22$, and add $-5/3$, $-7/3$, 12 times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
$17/11$	$-8/11$	-1		$-1/22$	$-5/22$			1
$-8/11$	$7/11$	1	1	$5/22$	$3/22$			
$4/11^*$	$-20/11$			$3/22$	$-7/22$	1		
$-173/11$	$-4/11$	16		$8/11$	$18/11$		1	
$-4/5$	$26/5$	$-13/5$		$1/5$				$3/5$

Iteration 3:

- $\max\{17/11, -8/11, -1, -1/22, -5/22\} = 17/11 > 0$, $q = 1$.
- Add $\beta = -(-4/5)/(17/11)$ times of row 1 to the bottom row.

4. $\max\{-8/11, 4/11, -173/11\} = 4/11 > 0$, $p = 3$.
 6. Multiply row 3 by $11/4$, and add $-17/11$, $8/11$, $173/11$ times of row 3 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	7	-1		-5/8	9/8	-17/4		1
	-3	1	1	1/2	-1/2	2		
1	-5			3/8	-7/8	11/4		
	-79	16		53/8	-97/8	173/4	1	
	82/17	-53/17		3/17	-2/17			19/17

Iteration 4:

1. $\max\{7, -1, -1, -5/8, 9/8, -17/4\} = 7 > 0$, $q = 2$.
 3. Add $\beta = -(82/17)/7$ times of row 1 to the bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	7	-1		-5/8	9/8	-17/4		1
	-3	1*	1	1/2	-1/2	2		
1	-5			3/8	-7/8	11/4		
	-79	16		53/8	-97/8	173/4	1	
		-17/7		17/28	-25/28	41/14		3/7

4. $\max\{-3, -5, -79\} \leq 0$.
 5. Feasibility is achieved.

Phase-II: Call Algorithm 17.3.1. $r = 1$, $q = 2$.

Iteration 5:

1. $\min\{-17/7, 17/28, -25/28, 41/14\} = -17/7 < 0$, $q' = 3$.
 3. $I = \{2, 4\} \neq \emptyset$.
 6. $\alpha = \min\{-(-3)/1, -(-79/16)\} = 3$, $p = 2$.
 7. $\bar{a}_{1,3} = -1 < 0$, $\alpha = 3 < -7/(-1) = 7$.
 8. Add 1, -6, $17/7$ times of row 2 to rows 1, 4, 5, respectively.
 9. Add $-(-51/7)/4$ times of row 1 to the bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	4		1	-1/8	5/8	-9/4		1
	-3	1	1	1/2	-1/2	2		
1	-5			3/8	-7/8	11/4		
	-31		-16	-11/8	-33/8	45/4	1	
			17/4	51/32	-31/32	59/16		9/4

Iteration 6:

1. $\min\{17/4, 51/32, -31/32, 59/16\} = -31/32 < 0$, $q' = 6$.
3. $I = \emptyset$.
4. $5/8 > 0$.
5. $q = 6$.
9. Add $-(-31/32)/(5/8)$ times of row 1 to the bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	4		1	$-1/8$	$5/8$	$-9/4$		1
	-3	1	1	$1/2$	$-1/2$	2		
1	-5			$3/8$	$-7/8$	$11/4$		
	-31		-16	$-11/8$	$-33/8$	$45/4$	1	
	$31/5$		$29/5$	$7/5$		$1/5$		$19/5$

Iteration 7:

1. $\min\{31/5, 29/5, 7/5, 1/5\} \geq 0$.
2. The optimal basic solution and optimal value:

$$x^* = (7/5, 0, 4/5, 0, 0, 8/5, 0, 33/5)^T, \quad f^* = -19/5.$$

Based on equivalence between the reduced tableau (17.3) and the revised tableau (17.19), a revised version of Algorithm 17.3.3 can be stated, matching Algorithm 17.3.2.

Algorithm 17.3.4 (D-Reduced Phase-I) Initial: B, R, N defined by (17.2). B_R^{-1} ; $\bar{z}_N = c_N - N_R^T B_R^{-T} c_B \geq 0$. This algorithm finds a feasible D-reduced basis.

1. Compute $\tilde{\omega}_N = N^T e_m - N_R^T (B_R^{-T} B e_m)$.
2. Determine column index $q \in \arg \max_{j \in N} \tilde{\omega}_j$.
3. Stop if $\tilde{\omega}_q \leq 0$ (infeasible problem).
4. Update $\bar{z}_N = \bar{z}_N + \beta \tilde{\omega}_N$, where $\beta = -\bar{z}_q / \tilde{\omega}_q$.
5. Compute $\tilde{a}_q(R) = B_R^{-1} a_q(R)$.
6. Determine $p \in \arg \max_{i \in R} \tilde{a}_{i,q} \ (p = i_s)$.
7. Stop if $\tilde{a}_{p,q} \leq 0$ (feasible D-reduced basis B_R obtained).
8. Update B_R^{-1} by (17.20).
9. Update (B, N) by exchanging j_s and q .
10. Go to step 1.

17.4 Bisection D-Reduced Simplex Algorithm

In Sect. 16.4, the bisection reduced simplex algorithm is described, where the so-called existence interval $[\lambda, \mu]$, containing the optimal value, is contracted by at least a half of the length in each iteration, until an optimal solution is achieved. The same idea was also realized from the dual side (Shen, 2006). In this section, we show that a more relevant dual bisection algorithm can be derived in the D-reduced simplex framework.

Let $[\lambda, \mu]$ be an existence interval. If a dual feasible solution is found with objective value $\bar{f} = (\lambda + \mu)/2$, the value is switched up to one associated with a dual basic feasible solution; if the latter does not satisfy the optimality condition, the objective value is a new lower bound on the optimal value, and hence it is set to λ . On the other hand, if there exists no dual feasible solution with objective value $\bar{f} = (\lambda + \mu)/2$, the value must be primal feasible. Then, it is switched down to one associated with a dual basic solution; if the latter does not satisfy the optimality condition, the associated value is a new upper bound on the optimal value, and it is then set to μ . Consequently, the resulting $[\lambda, \mu]$ is again an existence interval, contracted by a half of its length, at least.

To create an initial existence interval, it uses the D-reduced Phase-I Algorithm 17.3.3 to produce an upper bound μ of the optimal value, and the dual D-reduced Phase-I Algorithm 17.2.1 to provide a lower bound λ . Then, it bisections the interval, until reaching optimality.

The overall steps are put into the following algorithm.

Algorithm 17.4.1 (Bisection D-Reduced Simplex Algorithm: Tableau Form)

Initial: D-reduced tableau of form (17.3). This algorithm solves the D-reduced problem (17.1).

Step 1. Call Algorithm 17.3.3 to seek the primal feasibility:

- (1) stop if returning from step 2 (infeasible problem);
- (2) Returning from step 5: stop if $\bar{z}_N \geq 0$ (the optimality achieved); else set $\mu = \bar{f}$ and return.

Step 2. Call Algorithm 17.2.1 to seek the dual feasibility:

- (1) stop if returning from step 6 (infeasible or lower unbounded);
- (2) if returning from step 2, determine β and column index q such that

$$\beta = -\bar{z}_q/\bar{\omega}_q = \max\{-\bar{z}_j/\bar{\omega}_j \mid \bar{\omega}_j > 0, j \in N\}.$$

Step 3. if $\beta \neq 0$, add β times of the datum row to the bottom row.

Step 4. stop if $\bar{\omega}_q > 0$; $\bar{a}_{iq} \leq 0, i \in R$ (optimality achieved).

Step 5. set $\lambda = \bar{f}$.

Step 6. set $\beta = (\lambda - \mu)/2$.

Step 7. Add β times of the datum row to the bottom row.

Step 8. Determine column index $q \in \arg \min_{j \in N} \bar{z}_j$.

Step 9. If $\bar{z}_q \geq 0$, then:

- (1) determine β and column index q such that $\beta = -\bar{z}_j/\bar{\omega}_j = \max\{-\bar{z}_j/\bar{\omega}_j \mid \bar{\omega}_j > 0, j \in N\}$;
- (2) if $\beta \neq 0$, add β times of row r to the bottom row;
- (3) stop if $\bar{a}_{i_q} \leq 0, \forall i \in R$ (optimality achieved);
- (4) go to step 7 with $\lambda = \bar{f}$ and $\beta = (\lambda - \mu)/2$;
 else
- (5) determine row index $p \in \arg \max_{i \in R} \bar{a}_{i_q}$;
- (6) if $\bar{a}_{p_q} > 0$, convert $\bar{a}_{p_q}$ to 1, eliminate the other nonzeros in the column by elementary transformations, and go to Step 8;
- (7) add $\beta = -\bar{z}_q/\bar{\omega}_q$ times of row r to the bottom row;
- (8) stop if $\bar{z}_N \geq 0$ (optimality achieved);
- (9) go to Step 7 with setting $\mu = \bar{f}$ and $\beta = (\mu - \lambda)/2$.
 end if

The proof regarding the meanings of the exits of the preceding Algorithm is omitted (interested readers is referred to Yuan Shen and P.-Q. Pan 2006)

We demonstrate the algorithm by the following example.

Example 17.4.1 Solve following problem by Algorithm 17.4.1:

$$\begin{aligned}
 \min f &= 10/7 - 5x_1 + 7x_2 - 6x_3 + 5x_4, \\
 \text{s.t.} \quad &2x_1 + 5x_2 - 3x_3 + 5x_4 + x_5 = 1, \\
 &-4x_1 - 3x_2 + 8x_3 - 2x_4 + x_6 = -1, \\
 &x_1 - x_2 + x_3 + 6x_4 + x_7 = 6, \\
 &-7x_1 + 3x_2 + 4x_3 - 2x_4 + x_8 = -2, \\
 &x_j \geq 0, \quad j = 1, \dots, 8.
 \end{aligned}$$

Answer Initial tableau

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
2	5	-3	5	1				1
-4	-3	8	-2		1			-1
1	-1	1	6			1		6
-7	3	4	-2				1	-2*
-5	7	-6	5					-10/7

Iteration 1:

Arbitrarily take the 4th row as the datum row. Multiply the row by $-1/2$, and add $-1, 1, -6$ times of it to rows 1, 2, and 3, respectively, resulting in D-reduced tableau:

Step 1 Call Algorithm 17.3.3 to pursue primal feasibility. $r = 4$.

Iteration 2:

1. $\max\{7/2, -3/2, -2, 1, -1/2\} = 7/2 > 0$, $q = 1$.
3. Add $\beta = -(-5)/(7/2) = 10/7$ times of row 4 to the bottom row.
4. $\max\{-3/2, -1/2, -20\} = -1/2 \leq 0$.
5. Primal feasibility achieved:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
$-3/2$	$13/2$	-1	4	1			$1/2$	
$-1/2$	$-9/2$	6	-1		1		$-1/2$	
-20	8	13				1	3	
$7/2$	$-3/2$	-2	1				$-1/2$	1
-5	7	-6	5					$-10/7$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
$-3/2$	$13/2$	-1	4	1			$1/2$	
$-1/2$	$-9/2$	6	-1		1		$-1/2$	
-20	8	13^*				1	3	
$7/2$	$-3/2$	-2	1				$-1/2$	1
	$34/7$	$-62/7$	$45/7$				$-5/7$	

Step 1(2) $\mu = 0$.

Step 2. Call Algorithm 17.2.1 to seek the dual feasibility: $r = 4$.

Iteration 3:

1. $\min\{0, 34/7, -62/7, 45/7, -5/7\} = -62/7 < 0$, $q = 3$.
3. Add $\beta = -(-62/7)/(-2)$ times of row 4 to the bottom row.
4. $\max\{-1, 6, 13\} = 13 > 0$, $p = 3$.
7. Multiply row 3 by $1/13$, and add $1, -6, 2$ times of row 3 to rows 1, 2, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
$-79/26$	$185/26$		4	1		$1/13$	$19/26$	
$227/26^*$	$-213/26$		-1		1	$-6/13$	$-49/26$	
$-20/13$	$8/13$	1				$1/13$	$3/13$	
$11/26$	$-7/26$		1			$2/13$	$-1/26$	1
$-31/2$	$23/2$		2				$3/2$	$-31/7$

Iteration 4:

1. $\min\{-31/2, 23/2, 2, 0, 3/2\} = -31/2 < 0$, $q = 1$.
3. Add $\beta = -(-31/2)/(11/26)$ times of row 4 to the bottom row.
4. $\max\{-79/26, 227/26, -20/13\} = 227/26 > 0$, $p = 2$.
7. Multiply row 2 by $26/227$, and add $79/26, 20/13, -11/26$ times of row 2 to rows 1, 3, 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	968/227		829/227	1	79/227	-19/227	17/227	
1	-213/227		-26/227		26/227	-12/227	-49/227	
	-188/227	1	-40/227		40/227	-1/227	-23/227	
	29/227		238/227		-11/227	40/227	12/227	1
	18/11		425/11			62/11	1/11	2480/77

Iteration 5:

1. $\min\{18/11, 425/11, 0, 62/11, 1/11\} \geq 0$.

2. return.

Step 2(2) $\beta = \max\{-(18/11)/(29/227), -(425/11)/(238/227), -(62/11)/(40/227), -(1/11)/(12/227)\} = -(1/11)/(12/227)$, $q = 8$.

Step 3 Add $\beta = -(1/11)/(12/227)$ times of row 4 to the bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	968/227		829/227	1	79/227	-19/227	17/227	
1	-213/227		-26/227		26/227	-12/227	-49/227	
	-188/227	1	-40/227		40/227	-1/227	-23/227	
	29/227		238/227		-11/227	40/227	12/227	1
	17/12		221/6		1/12	16/3		2561/84

Step 4 $\bar{a}_{4,8} = 12/227 > 0$, $\bar{a}_{1,8} = 17/227 > 0$. Step 5 $\lambda = -2561/84$.

Step 6 Set $\beta = (-2561/84 - 0)/2 = -2561/168$.

Step 7 Add β times of row 4 to the bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	968/227		829/227	1	79/227	-19/227	17/227*	
1	-213/227		-26/227		26/227	-12/227	-49/227	
	-188/227	1	-40/227		40/227	-1/227	-23/227	
	29/227		238/227		-11/227	40/227	12/227	1
	-491/925		1814/87		3164/3849	1583/598	-1432/1777	2561/168

Iteration 6:

Step 8 $\min\{-491/925, 1814/87, 3164/3849, 1583/598, -1432/1777\}$
 $= -1432/1777 < 0$, $q = 8$.

Step 9 $\bar{z}_q \not\geq 0$.

(5) $\max\{17/227, -49/227, -23/227\} = 17/227 > 0$, $p = 1$.

(6) Multiply row 1 by 227/17, and add 49/227, 23/227, -12/227, 1432/1777 times of row 1 to rows 2,3,4, respectively:

Iteration 7:

Step 8 $\min\{5488/121, 5293/88, 2561/238, 717/157, 1247/714\} \geq 0$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	968/17		829/17	227/17	79/17	-19/17	1	
1	193/17		177/17	49/17	19/17	-5/17		
	84/17	1	81/17	23/17	11/17	-2/17		
	-49/17		-26/17	-12/17	-5/17	4/17		1
	5488/121		5293/88	2561/238	717/157	1247/714		2561/168

- Step 9 $\bar{z}_N \geq 0$.
- (1) $\beta = \max\{-(1247/714)/(4/17)\}$, $q = 8$.

(2) Add $-(1247/714)/(4/17)$ times of row 4 to the bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	968/17		829/17	227/17	79/17	-19/17	1	
1	193/17		177/17	49/17	19/17	-5/17		
	84/17	1	81/17	23/17	11/17	-2/17		
	-49/17		-26/17	-12/17	-5/17	4/17		1
	267/4		143/2	16	27/4			219/28

- (3) $-19/17, -5/17, -2/17 \leq 0$. Setting $\bar{x}_8 = 17/4$, we obtain the optimal basic solution and optimal value:

$$x^* = (5/4, 0, 1/2, 0, 0, 0, 17/4, 1)^T, \quad f^* = -219/28.$$

Reference

Shen Y, Pan P-Q (2006) Dual besiction simplex algorithm (in Chinese). In: Preceedings of the national conference of operatios research society of China, Shenzhen. Globa-Link Informatics, Hong Kong, pp 168–174

Chapter 18

Generalized Reduced Simplex Method



Although we always consider the standard LP problem, the LP problems from practice are various. The latter can be transformed into a more general form, that is, the so-called *bounded-variable* LP problem. This type of problem can be obtained from the standard problem by replacing the sign restriction with the general bound restriction.

Similarly modifying the reduced problem (16.1), we obtain the *reduced bounded-variable problem* below:

$$\begin{aligned} \min \quad & f, \\ \text{s.t.} \quad & A \begin{pmatrix} x \\ f \end{pmatrix} = b, \quad l \leq x \leq u, \end{aligned} \quad (18.1)$$

where $A \in \mathcal{R}^{(m+1) \times (n+1)}$, $b \in \mathcal{R}^{m+1}$, $\text{rank } A = m + 1$, $m < n$; both l and u are given vectors (may include infinite components). Note that the objective variable f is viewed as the same as the $(n + 1)$ th variable x_{n+1} , corresponding to the end column $a_{n+1} \triangleq -e_{m+1}$.

To cope with the problem (18.1), in this chapter, we generalize the reduced and D-reduced simplex algorithms, deriving the generalized reduced simplex algorithm together with the single-artificial-variable Phase-I, generalized dual reduced simplex algorithm with the dual Phase-I, and generalized dual D-reduced simplex algorithm.

Assume that the current basis and nonbasis are as follows:

$$B = \{j_1, \dots, j_{m+1}\}, \quad N = A \setminus B, \quad (n + 1) \notin B. \quad (18.2)$$

18.1 Generalized Reduced Simplex Algorithm

To begin with, we bring up some useful facts under the generalized reduced simplex framework.

It is noticeable that the constraints of (18.1) are equivalent to the following canonical form:

$$l_B \leq x_B = \bar{b} - \bar{N}x_N - f\bar{a}_{n+1} \leq u_B, \quad l_N \leq x_N \leq u_N,$$

where

$$\bar{b} = B^{-1}b, \quad \bar{N} = B^{-1}N, \quad \bar{a}_{n+1} = B^{-1}a_{n+1}.$$

If $(\bar{x}, \bar{f})$ satisfies

$$\bar{x}_j = l_j \text{ or } u_j, \quad j \in N, \quad (18.3)$$

$$l_B \leq \bar{x}_B = \bar{b} - \bar{N}\bar{x}_N - \bar{a}_{n+1}\bar{f} \leq u_B, \quad (18.4)$$

it is called *feasible basic solution*; if, further, $l_B < \bar{x}_B < u_B$, it is said *nondegenerate*. If the bounded-variable problem has a feasible solution, it has a basic feasible solution, and if it has an optimal solution, then it has an optimal basic solution.

Consider the following line search scheme:

$$\hat{x}_B = \bar{x}_B + \alpha\bar{a}_{n+1}, \quad \hat{f} = \bar{f} - \alpha, \quad \hat{x}_N = \bar{x}_N, \quad (18.5)$$

where the stepsize α is determined by the rule below.

Rule 18.1.1 (Row Rule) Determine the stepsize α and select the row index p such that

$$\alpha = \alpha_p = \min\{\alpha_i \mid i = 1, \dots, m+1\}, \quad (18.6)$$

where

$$\alpha_i = \begin{cases} (u_{j_i} - \bar{x}_{j_i})/\bar{a}_{i,n+1}, & \text{if } \bar{a}_{i,n+1} > 0, \\ (l_{j_i} - \bar{x}_{j_i})/\bar{a}_{i,n+1}, & \text{if } \bar{a}_{i,n+1} < 0, \\ \infty, & \text{if } \bar{a}_{i,n+1} = 0, \end{cases} \quad i = 1, \dots, m+1. \quad (18.7)$$

Lemma 18.1.1 If $(\bar{x}, \bar{f})$ is a basic feasible solution, then the solution $(\hat{x}, \hat{f})$, defined by (18.5), is a basic feasible solution, associated with the objective value not increasing, or strictly decreasing if $\bar{x}$ is nondegenerate.

Proof Note that $\hat{x}_N = \bar{x}_N$. For any real number α , the new solution given by (18.5) satisfies the equality constraints of (18.1); from $(\bar{x}, \bar{f})$ satisfying the equality constraint, in fact, it follows that

$$\begin{aligned} A(\hat{x}^T, \hat{f})^T &= B\hat{x}_B + N\hat{x}_N + \hat{f}a_{n+1} \\ &= B(\bar{x}_B + \alpha\bar{a}_{n+1}) + N\bar{x}_N + (\bar{f} - \alpha)a_{n+1} \\ &= B\bar{x}_B + N\bar{x}_N + \bar{f}a_{n+1} \\ &= b. \end{aligned}$$

On the other hand, the α determined by (18.6) is clearly nonnegative, so that the new solution $\bar{x}$ satisfies bound constraints $l \leq \hat{x} \leq u$. Therefore $(\hat{x}, \hat{f})$ is a basic feasible solution. When $\bar{x}$ is nondegenerate, in addition, it is clear that $\alpha > 0$, and hence

$$\hat{f} = \bar{f} - \alpha < \bar{f},$$

which implies that the objective value decreases. $\square$

It is noted that the decrement of the objective value is equal to the stepsize. The α , given by (18.7), is the largest possible stepsize to maintain the feasibility of $\hat{x}$. In practice, the problem should be deemed as unbounded if α is too large. Note that the j_p -indexed component of the new solution is equal to the associated upper or lower bound, i.e.,

$$\hat{x}_{j_p} = \begin{cases} u_{j_p}, & \text{if } \bar{a}_{p, n+1} > 0, \\ l_{j_p}, & \text{if } \bar{a}_{p, n+1} < 0. \end{cases} \quad (18.8)$$

Introducing the following index sets:

$$\begin{aligned} \Gamma &= \{j \in N \mid \bar{x}_j = l_j\}, \\ \Pi &= \{j \in N \mid \bar{x}_j = u_j\}, \\ \sigma_N &= N^T B^{-T} e_p, \end{aligned} \quad (18.9)$$

we define the set

$$J = \{j \in \Gamma \mid \bar{a}_{p, n+1} \sigma_j > 0\} \cup \{j \in \Pi \mid \bar{a}_{p, n+1} \sigma_j < 0\}. \quad (18.10)$$

Then the following is valid for the new solution.

Lemma 18.1.2 *If the set J is empty, the new solution $(\hat{x}, \hat{f})$, defined by (18.5), is a basic optimal solution.*

Proof By (18.6) and (18.7), it is known that $\bar{a}_{p,n+1} \neq 0$. From the p th equation of the canonical form $x_B = \bar{b} - \bar{N}x_N - \bar{a}_{n+1}f$, it follows that

$$f = (\bar{b}_p - \sigma_N^T x_N - x_{j_p}) / \bar{a}_{p,n+1}. \quad (18.11)$$

Assuming that (x', x'_{n+1}) is any feasible solution, it holds that

$$l_j \leq x'_j \leq u_j, \quad j = 1, \dots, n. \quad (18.12)$$

If $\bar{a}_{p,n+1} > 0$, then $J = \emptyset$ implies that

$$\sigma_\Gamma \leq 0, \quad \sigma_\Pi \geq 0. \quad (18.13)$$

Thus, from the first expression of (18.13) and $x'_\Gamma \geq l_\Gamma = \bar{x}_\Gamma$, it follows that

$$\sigma_\Gamma^T x'_\Gamma \leq \sigma_\Gamma^T \bar{x}_\Gamma,$$

and from the second expression of (18.13) and $x'_\Pi \leq u_\Pi = \bar{x}_\Pi$, it follows that

$$\sigma_\Pi^T x'_\Pi \leq \sigma_\Pi^T \bar{x}_\Pi.$$

Combining the preceding two expressions and $\bar{a}_{p,n+1} > 0$ gives

$$-(\sigma_N^T x'_N) / \bar{a}_{p,n+1} \geq -(\sigma_N^T \bar{x}_N) / \bar{a}_{p,n+1} = -(\sigma_N^T \hat{x}_N) / \bar{a}_{p,n+1}. \quad (18.14)$$

On the other hand, from (18.8), (18.12), and $\bar{a}_{p,n+1} > 0$, it follows that

$$-x'_{j_p} / \bar{a}_{p,n+1} \geq -u_{j_p} / \bar{a}_{p,n+1} = -\hat{x}_{j_p} / \bar{a}_{p,n+1}. \quad (18.15)$$

Therefore, by $(\hat{x}, \hat{f})$ and (x', x'_{n+1}) satisfying (18.11), (18.14), and (18.15), it holds that

$$x'_{n+1} = (\bar{b}_p - \sigma_N^T x'_N - x'_{j_p}) / \bar{a}_{p,n+1} \geq (\bar{b}_p - \sigma_N^T \hat{x}_N - \hat{x}_{j_p}) / \bar{a}_{p,n+1} = \hat{f}.$$

Therefore, by considering Lemma 18.1.1, it is known that $\hat{x}$ is an optimal basic solution.

It can be shown analogously that the statement is valid if $\bar{a}_{p,n+1} < 0$. $\square$

In the case of $J \neq \emptyset$, the following rule is well-defined.

Rule 18.1.2 (Column Rule) *Select the column index*

$$q \in \arg \max_{j \in J} |\sigma_j|. \quad (18.16)$$

Then after the associated basis change, when j_p left from and q entered to the basis, an iteration is complete. It is clear that the new basis corresponds to the new solution $(\hat{x}, \hat{x})$, given by (18.5) together with (18.6) and (18.7).

The overall steps can be put into the following algorithm.

Algorithm 18.1.1 (Generalized Reduced Simplex Algorithm) *Initial:* (B, N) , B^{-1} ; $\bar{a}_{n+1} = -B^{-1}e_{m+1}$; *basic feasible solution* $\bar{x}$. This algorithm solves the reduced bounded-variable problem (18.1).

1. Determine the stepsize α and row index p by (18.6) and (18.7).
2. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha \bar{a}_{n+1}$.
3. Compute $\sigma_N = N^T B^{-T} e_p$.
4. Stop with $\bar{f} = c^T \bar{x}$ if J defined by (18.10) is empty.
5. Determine column index q by (18.16).
6. Compute $\bar{a}_q = B^{-1}a_q$ and $v = -\bar{a}_{p, n+1}/\sigma_q$.
7. If $v \neq 0$, update $\bar{a}_{n+1} = \bar{a}_{n+1} + v(\bar{a}_q - e_p)$.
8. Update B^{-1} .
9. Update (B, N) by exchanging j_p and q .
10. Go to step 1.

Theorem 18.1.3 Algorithm 18.1.1 produces a sequence of basic feasible solutions. Assuming degeneracy, it terminates at step 4, offering a basic optimal solution.

Proof The validity is from Lemmas 18.1.1 and 18.1.2 and the discussions preceding Algorithm 18.1.1. $\square$

Example 18.1.1 Solve the following problem by Algorithm 18.1.1:

$$\begin{aligned}
 \min \quad & f = -x_1 - 3x_2 - 2x_3 + 4x_4 - 5x_5, \\
 \text{s.t.} \quad & -2x_1 + x_2 - 3x_3 - x_4 + x_5 + x_6 = -52, \\
 & 3x_1 - 3x_2 + x_3 + 5x_4 - 2x_5 + x_7 = 26, \\
 & x_1 - 6x_2 - 2x_3 - 4x_4 + 6x_5 + x_8 = 7, \\
 & 2 \leq x_1 \leq 10, \quad -5 \leq x_2 \leq 12, \quad 1 \leq x_3 \leq 15, \\
 & -4 \leq x_4 \leq 27, \quad -3 \leq x_5 \leq 14, \quad -9 \leq x_6 \leq 18, \\
 & -7 \leq x_7 \leq 13, \quad -10 \leq x_8 \leq 21.
 \end{aligned}$$

Given a feasible solution, $\bar{x} = (2, -5, 15, -4, -3, 1, 4, 7)^T$, $\bar{f} = -18$.

Answer The initial tableau is

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -18$
-2	1	-3	-1	1	1				-52
3	-3	1	5	-2		1			26
1	-6	-2	-4	6			1		7
-1*	-3	-2	4	-5				-1	

Iteration 1: Convert the preceding to a reduced tableau by taking $p = 4$ and $q = 1$. Multiply row 4 by -1 , and then add 2, -3 , and -1 times of row 5 to rows 1, 2, and 3, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -18$
	7	1	-9	11	1			2	-52
	-12	-5	17	-17		1		-3	26
	-9	-4		1			1	-1	7
1	3	2	-4	5				1	

Taking the left side of the preceding tableau as original data, call Algorithm 18.1.1.

Initial: $B = \{6, 7, 8, 1\}$, $N = \{2, 3, 4, 5\}$, $B^{-1} = I$, $\bar{x}_N = (-5_{(-)}, 15^{(+)}, -4_{(-)}, -3_{(-)})^T$, $\bar{x}_B = (1, 4, 7, 2)^T$, $\bar{f} = -18$. $\bar{a}_9 = (2, -3, -1, 1)^T$. The $\bar{x}$ is a basic feasible solution.

Iteration 2:

$$1. \alpha = \min\{(18 - 1)/2, (-7 - 4)/(-3), (-10 - 7)/(-1), (10 - 2)/(1)\} \\ = \min\{17/2, 11/3, 17, 8\} = 11/3, \quad p = 2.$$

$$2. \bar{x}_B = (1, 4, 7, 2)^T + (11/3)(2, -3, -1, 1)^T = (25/3, -7, 10/3, 17/3)^T.$$

$$3. \sigma_N = N^T B^{-T} e_2 = (-12, -5, 17, -17)^T.$$

$$4. J = \{2, 5\} \neq \emptyset.$$

$$5. \max\{|-12|, |-17|\} = 17, \quad q = 5.$$

$$6. \bar{a}_5 = B^{-1} a_5 = (11, -17, 1, 5)^T.$$

$$7. v = -(-3)/(-17) = -3/17,$$

$$\bar{a}_{n+1} = (2, -3, -1, 1)^T + (-3/17)(11, -17 - 1, 1, 5)^T \\ = (1/17, 3/17, -20/17, 2/17)^T.$$

$$8. B^{-1} = \begin{pmatrix} 1 & 11/17 & & \\ & -1/17 & & \\ & 1/17 & 1 & \\ & 5/17 & & 1 \end{pmatrix}.$$

$$9. \bar{x}_B = (25/3, -3, 10/3, 17/3)^T, \quad B = \{6, 5, 8, 1\},$$

$$\bar{x}_N = (-5_{(-)}, 15^{(+)}, -4_{(-)}, -7_{(-)})^T, \quad N = \{2, 3, 4, 7\}.$$

Iteration 3:

$$1. \alpha = \min\{(18 - (25/3))/(1/17), (14 - (-3))/(3/17), (-10 - (10/3))/(-20/17), \\ (10 - (17/3))/(2/17)\} = \min\{493/3, 289/3, 34/3, 221/6\} = 34/3, \quad p = 3.$$

$$2. \bar{x}_B = (25/3, -3, 10/3, 17/3)^T + (34/3)(1/17, 3/17, -20/17, 2/17)^T \\ = (9, -1, -10, 7)^T.$$

$$3. \sigma_N = N^T B^{-T} e_3 = (-165/17, -73/17, 1, 1/17)^T.$$

$$4. J = \{2\} \neq \emptyset.$$

$$5. \max\{|-165/17|\} = 165/17, \quad q = 2.$$

$$6. \bar{a}_2 = B^{-1} a_2 = (-13/17, 12/17, -165/17, -9/17)^T.$$

$$7. v = -(-20/17)/(-165/17) = -4/33,$$

$$\begin{aligned}\bar{a}_{n+1} &= (1/17, 3/17, -20/17, 2/17)^T + (-4/33)(-13/17, 12/17, -165/17 - \\ & 1, -9/17)^T \\ &= (5/33, 1/11, 4/33, 2/11)^T.\end{aligned}$$

$$8. B^{-1} = \begin{pmatrix} 1 & -13/165 \\ 1 & 4/55 \\ & -17/165 \\ & -3/55 & 1 \end{pmatrix} \begin{pmatrix} 1 & 11/17 \\ -1/17 & 1 \\ 1/17 & 1 \\ 5/17 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 106/165 & -13/165 \\ -3/55 & 4/55 \\ -1/165 & -17/165 \\ 16/55 & -3/55 & 1 \end{pmatrix}.$$

$$\begin{aligned}9. \bar{x}_B &= (9, -1, -5, 7)^T, B = \{6, 5, 2, 1\}, \\ \bar{x}_N &= (-10_{(-)}, 15^{(+)}, -4_{(-)}, -7_{(-)})^T, N = \{8, 3, 4, 7\}.\end{aligned}$$

Iteration 4:

$$\begin{aligned}1. \alpha &= \min\{(18 - 9)/(5/33), (14 - (-1))/(1/11), (12 - (-5))/(4/33), \\ & (10 - 7)/(2/11)\} \\ &= \min\{297/5, 165, 561/4, 33/2\} = 33/2, p = 4. \\ 2. \bar{x}_B &= (9, -1, -5, 7)^T + (33/2)(5/33, 1/11, 4/33, 2/11)^T = (23/2, 1/2, \\ & -3, 10)^T. \\ 3. \sigma_N &= N^T B^{-T} e_4 = (-3/55, 42/55, 52/55, 16/55)^T. \\ 4. J &= \{4, 7\} \neq \emptyset. \\ 5. \max\{|52/55|, |16/55|\} &= \{52/55, 16/55\} = 52/55, q = 4. \\ 6. \bar{a}_4 &= B^{-1} a_4 = (317/165, -51/55, -17/165, 52/55)^T. \\ 7. v &= -(2/11)/(52/55) = -5/26, \\ \bar{a}_{n+1} &= (5/33, 1/11, 4/33, 2/11)^T + (-5/26)(317/165, -51/55, -17/165, \\ & 52/55 - 1)^T \\ &= (-17/78, 7/26, 11/78, 5/26)^T. \\ 8. B^{-1} &= \begin{pmatrix} 1 & -317/156 \\ 1 & 51/52 \\ 1 & 17/156 \\ & 55/52 \end{pmatrix} \begin{pmatrix} 1 & 106/165 & -13/165 \\ -3/55 & 4/55 \\ -1/165 & -17/165 \\ 16/55 & -3/55 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2/39 & 5/156 & -317/156 \\ 3/13 & 1/52 & 51/52 \\ 1/39 & -17/156 & 17/156 \\ 4/13 & -3/52 & 55/52 \end{pmatrix}. \\ 9. \bar{x}_B &= (23/2, 1/2, -3, -4)^T, B = \{6, 5, 2, 4\}, \\ \bar{x}_N &= (-10_{(-)}, 15^{(+)}, 10^{(+)}, -7_{(-)})^T, N = \{8, 3, 1, 7\}.\end{aligned}$$

Iteration 5:

$$\begin{aligned}1. \alpha &= \min\{(-9 - 23/2)/(-17/78), (14 - (1/2))/(7/26), (12 - (-3))/(11/78), \\ & (27 - (-4))/(5/26)\} \\ &= \min\{1599/17, 351/7, 1170/11, 806/5\} = 351/7, p = 2. \\ 2. \bar{x}_B &= (23/2, 1/2, -3, -4)^T + (351/7)(-17/78, 7/26, 11/78, 5/26)^T \\ &= (4/7, 14, 57/14, 79/14)^T. \\ 3. \sigma_N &= N^T B^{-T} e_2 = (1/52, 19/26, 51/52, 3/13)^T. \\ 4. J &= \{8, 7\} \neq \emptyset. \\ 5. \max\{|1/52|, |3/13|\} &= 3/13 > 0, q = 7. \\ 6. \bar{a}_7 &= B^{-1} a_7 = (2/39, 3/13, 1/39, 4/13)^T. \\ 7. v &= -(7/26)/(3/13) = -7/6, \\ \bar{a}_{n+1} &= (-17/78, 7/26, 11/78, 5/26)^T + (-7/6)(2/39, 3/13 - \\ & 1, 1/39, 4/13)^T\end{aligned}$$

$$\begin{aligned}
 &= (-5/18, 7/6, 1/9, -1/6)^T. \\
 8. \quad B^{-1} &= \begin{pmatrix} 1 & -2/9 & & \\ & 13/3 & & \\ & -1/9 & 1 & \\ & -4/3 & & 1 \end{pmatrix} \begin{pmatrix} 1 & 2/39 & 5/156 & -317/156 \\ & 3/13 & 1/52 & 51/52 \\ & 1/39 & -17/156 & 17/156 \\ & 4/13 & -3/52 & 55/52 \end{pmatrix} = \begin{pmatrix} 1 & 1/36 & -9/4 & \\ & 1 & 1/12 & 17/4 \\ & & -1/9 & \\ & & & -1/12 & -1/4 \end{pmatrix}. \\
 9. \quad \bar{x}_B &= (4/7, -7, 57/14, 79/14)^T, \quad B = \{6, 7, 2, 4\}, \\
 \bar{x}_N &= (-10^{(-)}, 15^{(+)}, 10^{(+)}, 14^{(+)})^T, \quad N = \{8, 3, 1, 5\}.
 \end{aligned}$$

Iteration 6:

$$\begin{aligned}
 1. \quad \alpha &= \min\{(-9 - 4/7)/(-5/18), (13 - (-7))/(7/6), (12 - (57/14))/(1/9), \\
 &\quad (-3 - (79/14))/(-1/6)\} \\
 &= \min\{1206/35, 120/7, 999/14, 363/7\} = 120/7, \quad p = 2. \\
 2. \quad \bar{x}_B &= (4/7, -7, 57/14, 79/14)^T + (120/7)(-5/18, 7/6, 1/9, -1/6)^T \\
 &= (-88/21, 13, 251/42, 39/14)^T. \\
 3. \quad \sigma_N &= N^T B^{-T} e_2 = (1/12, 19/6, 17/4, 13/3)^T. \\
 4. \quad J &= \{8\} \neq \emptyset. \\
 5. \quad \max\{|1/12|\} &= 1/12 > 0, \quad q = 8. \\
 6. \quad \bar{a}_8 &= B^{-1} a_8 = (1/36, 1/12, -1/9, -1/12)^T. \\
 7. \quad v &= -(7/6)/(1/12) = -14, \\
 \bar{a}_{n+1} &= (-5/18, 7/6, 1/9, -1/6)^T + (-14)(1/36, 1/12 - 1, -1/9, -1/12)^T \\
 &= (-2/3, 14, 5/3, 1)^T. \\
 8. \quad B^{-1} &= \begin{pmatrix} 1 & -4/13 & & \\ & 12 & & \\ & 4/3 & 1 & \\ & 1 & & 1 \end{pmatrix} \begin{pmatrix} 1 & 1/36 & -9/4 & \\ & 1 & 1/12 & 17/4 \\ & & -1/9 & \\ & & & -1/12 & -1/4 \end{pmatrix} = \begin{pmatrix} 1 & -4/13 & 1/468 & -185/52 \\ & 12 & 1 & 51 \\ & 4/3 & & 17/3 \\ & 1 & & 4 \end{pmatrix}. \\
 9. \quad \bar{x}_B &= (-88/21, -10, 251/42, 39/14)^T, \quad B = \{6, 8, 2, 4\}, \\
 \bar{x}_N &= (13^{(+)}, 15^{(+)}, 10^{(+)}, 14^{(+)})^T, \quad N = \{7, 3, 1, 5\}.
 \end{aligned}$$

Iteration 7:

$$\begin{aligned}
 1. \quad \alpha &= \min\{(-9 - (-88/21))/(-2/3), (21 - (-10))/14, (12 - (251/42))/(5/3), \\
 &\quad (27 - (39/14))/1\} \\
 &= \min\{101/14, 31/14, 253/70, 339/14\} = 31/14, \quad p = 2. \\
 2. \quad \bar{x}_B &= (-88/21, -10, 251/42, 39/14)^T + (31/14)(-2/3, 14, 5/3, 1)^T \\
 &= (-17/3, 21, 29/3, 5)^T. \\
 3. \quad \sigma_N &= N^T B^{-T} e_2 = (1, 166, 51, 52)^T. \\
 4. \quad J &= \emptyset. \text{ The optimal basic solution and optimal value are}
 \end{aligned}$$

$$\begin{aligned}
 x^* &= (10, 29/3, 15, 5, 14, -17/3, 13, 21)^T, \\
 f^* &= (-1, -3, -2, 4, -5)(10, 29/3, 15, 5, 14)^T = -119.
 \end{aligned}$$

Finally, we formulate the tableau version of Algorithm 18.1.1 based on the relationship between the reduced and the revised reduce simplex tableaus.

Algorithm 18.1.2 (Generalized Reduced Simplex Algorithm: Tableau Form)

Initial: reduced tableau of form (18.19), associated with $(\bar{x}, \bar{f})$ with (18.3) and (18.4) satisfied. This algorithm solves the reduced bounded-variable problem (18.1).

1. Determine stepsize α and row index p by (18.6) and (18.7).
2. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha \bar{a}_{n+1}$ and $\bar{f} = \bar{f} - \alpha$.
3. If $J = \{j \in \Gamma \mid \bar{a}_{p,n+1} \bar{a}_{pj} > 0\} \cup \{j \in \Pi \mid \bar{a}_{p,n+1} \bar{a}_{pj} < 0\} = \emptyset$, compute $\bar{f} = c^T \bar{x}$, and stop (optimality achieved).
4. Determine column index $q \in \arg \max_{j \in J} |\bar{a}_{pj}|$.
5. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations (see the Note below).
6. Go to step 1.

It is noted that the preceding step 5 does not touch the $\bar{x}_B$ column but replaces its p th component by $\bar{x}_q$.

18.1.1 Generalized Reduced Phase-I: Single Artificial Variable

In this section, we derive a generalized reduced Phase-I procedure similar to the single-artificial-variable Algorithm 14.2.1 for the standard problem. Such an algorithm is especially suitable in the reduced simplex framework.

Assume that $\bar{x}$ satisfies

$$\bar{x}_j = l_j \text{ or } u_j, \quad j \in N, \quad (18.17)$$

but not

$$l \leq \bar{b} - \bar{N} \bar{x}_N \leq u.$$

For given $\bar{x}_B$ such that $l_B \leq \bar{x}_B \leq u_B$, set the artificial f' ($\equiv x_{n+1}$) column as follows:

$$\bar{a}_{n+1} = \bar{b} - \bar{N} \bar{x}_N - \bar{x}_B, \quad \bar{f}' = 1. \quad (18.18)$$

Construct the following auxiliary reduced tableau:

x_B	x_N	f'	$f' = \bar{f}'$
I	$\bar{N}$	$\bar{a}_{n+1}$	$\bar{x}_B$

(18.19)

This tableau is clearly feasible, from which the generalized reduced simplex algorithm can get itself started. If the optimal objective value vanishes, then a feasible reduced tableau of the original problem is obtained, and hence it is ready to

go to Phase-II. To pursue zero optimal value, we modify Algorithm 18.1.2, giving the following algorithm.

Algorithm 18.1.3 (Tableau Generalized Reduced Phase-I: Single Artificial Variable) *Initial: reduced tableau in form (18.19). $\bar{x}, \bar{a}_{n+1}, \bar{f}' = 1$ satisfying (18.17) and (18.18). This algorithm finds a feasible reduced tableau for the bounded-variable problem (18.1).*

1. Determine stepsize α and row index p by (18.6) and (18.7).
2. If $\alpha > \bar{f}'$, set $\alpha = \bar{f}'$.
3. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha \bar{a}_{n+1}$ and $\bar{f}' = \bar{f}' - \alpha$.
4. Stop if $\bar{f}' = 0$ (feasibility achieved).
5. Stop if

$$J = \{j \in \Gamma \mid \bar{a}_{p, n+1} \bar{a}_{pj} > 0\} \cup \{j \in \Pi \mid \bar{a}_{p, n+1} \bar{a}_{pj} < 0\} = \emptyset$$

(infeasible problem).

6. Determine column index $q \in \arg \max_{j \in J} |\bar{a}_{pj}|$.
7. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations
(without touching $\bar{x}_B$ column, only its p th component is replaced by $\bar{x}_q$).
8. Go to step 1.

Example 18.1.2 Solve the following problem by Algorithm 18.1.2, supported by Algorithm 18.1.3 in Phase-I:

$$\begin{aligned} \min f &= -7x_1 + 6x_2 + 3x_3 - 4x_4, \\ \text{s.t. } -2x_1 - 3x_2 + 7x_3 - 5x_4 + x_5 &= 2, \\ 3x_1 - 5x_2 - 2x_3 - x_4 + x_6 &= -15, \\ -2x_1 - 4x_2 + 3x_3 - 6x_4 + x_7 &= 0, \\ -5x_1 + 6x_2 - 3x_3 + 3x_4 + x_8 &= 7, \\ -7 \leq x_1 \leq 6, \quad -3 \leq x_2 \leq 9, \quad -5 \leq x_3 \leq 12, \\ -4 \leq x_4 \leq 11, \quad -7 \leq x_5 \leq 5, \quad -8 \leq x_6 \leq 14, \\ -7 \leq x_7 \leq 7, \quad -6 \leq x_8 \leq 10. \end{aligned}$$

Answer $B = \{5, 6, 7, 8\}$, $N = \{1, 2, 3, 4\}$. Set $\bar{x}_B = (0, 0, 0, 0)^T$ and $\bar{x}_N = (-7_{(-)}, 9^{(+)}, 12^{(+)}, -4_{(-)})^T$. Form the auxiliary objective column

$$\bar{a}_9 = (2, -15, 0, 7)^T - (91, -86, 38, 41)^T = (-89, 71, -38, -34)^T,$$

and set the initial auxiliary objective value $\bar{f}' = 1$.

The initial auxiliary reduced tableau is as follows:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f'	$\bar{f}' = 1$
	-2	-3	7*	-5	1				-89	
	3	-5	-2	-1		1			71	
	-2	-4	3	-6			1		-38	
	-5	6	-3	3				1	-34	
	-7	6	3	-4						
u	6	9	12	11	5	14	7	10	1	
$\bar{x}$	-7	9	12	-4					1	
l	-7	-3	-5	-4	-7	-8	-7	-6	0	

Phase-I: call Algorithm 18.1.3.

Iteration 1:

- $\alpha = \min\{(-7 - 0)/(-89), (14 - 0)/(71), (-7 - 0)/(-38), (-6 - 0)/(-34)\};$
 $= \min\{7/89, 14/71, 7/38, 3/17\} = 7/89, p = 1.$
- $7/89 < \bar{f}' = 1.$
- $\bar{x}_B = (12, 497/89, -266/89, -238/89, 0)^T, \bar{f}' = 1 - 7/89 = 82/89.$
- $J = \{1, 3, 4\} \neq \emptyset.$
- $\max\{|-2|, |7|, |-5|\} = 7, q = 3.$
- Multiply row 1 by $1/7$, and then add 2, -3, 3, and -3 times of row 1 to rows 2-5, respectively (without touching $\bar{x}_B$ column, only its first component is replaced by $\bar{x}_3 = 12$):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f'	$\bar{f}' = 82/89$
	-2/7	-3/7	1	-5/7	1/7				-89/7	12
	17/7	-41/7		-17/7	2/7	1			319/7	497/89
	-8/7	-19/7		-27/7	-3/7		1		1/7	-266/89
	-41/7*	33/7		6/7	3/7			1	-505/7	-238/89
	-43/7	51/7		-13/7	-3/7					
u	6	9	12	11	5	14	7	10	1	
$\bar{x}$	-7	9	12	-4	-7	497/89	-266/89	-238/89	82/89	
l	-7	-3	-5	-4	-7	-8	-7	-6	0	

Iteration 2:

- $\alpha = \min\{(-5 - 12)/(-89/7), (14 - 497/89)/(319/7), (7 - (-266/89))/(1/7),$
 $(-6 - (-238/89))/(-505/7)\}$
 $= \min\{119/89, 306/1657, 6223/89, 107/2321\} = 107/2321, p = 4.$
- $107/2321 < \bar{f}' = 82/89.$
- $\bar{x}_B = (5764/505, 5764/505, -1506/505, -7, 0)^T, \bar{f}' = 82/89 - 107/2321 =$
 $442/505.$

5. $J = \{1, 2\} \neq \emptyset$.
6. $\max\{|-41/7|, |33/7|\} = 41/7$, $q = 1$.
7. Multiply row 1 by $-7/42$, and then add $2/7$, $-17/7$, $8/7$, and $43/7$ times of row 1 to rows 2–5, respectively (without touching $\bar{x}_B$ column, only its 4th component is replaced by $\bar{x}_1 = -7$):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f'	$\bar{f}' = 442/505$
		$-27/41$	1	$-31/41$	$5/41$			$-2/41$	$-377/41$	$5764/505$
		$-160/41^*$		$-85/41$	$19/41$	1		$17/41$	$642/41$	$5764/505$
		$-149/41$		$-165/41$	$-21/41$		1	$-8/41$	$583/41$	$-1506/505$
	1	$-33/41$		$-6/41$	$-3/41$			$-7/41$	$505/41$	-7
		$96/41$		$-113/41$	$-36/41$			$-43/41$		
u	6	9	12	11	5	14	7	10	1	
$\bar{x}$	-7	9	$\frac{5764}{505}$	-4	-7	$\frac{3881}{505}$	$-\frac{1506}{505}$	-6	$\frac{442}{505}$	
l	-7	-3	-5	-4	-7	-8	-7	-6	0	

Iteration 3:

1. $\alpha = \min\{(-5 - 5764/505)/(-377/41), (14 - 3881/505)/(642/41), (7 - (-1506/505))/(583/41), (6 - (-7))/(505/41)\}$
 $= \min\{1171/656, 221/548, 245/349, 533/505\} = 221/548$, $p = 2$.
2. $221/548 < \bar{f}' = 442/505$.
3. $\bar{x}_B = (1649/214, 9, 3834/1393, -3169/1559, 0)^T$, $\bar{f}' = 442/505 - 221/548 = 101/214 \neq 0$.
5. $J = \{2, 5, 8\} \neq \emptyset$.
6. $\max\{|-160/41|, |19/41|, |17/41|\} = 160/41$, $q = 2$.
7. Multiply row 2 by $-41/160$, and then add $27/41$, $149/41$, $33/41$, and $-96/41$ times of row 2 to rows 1, 3, 4, and 5, respectively (without touching $\bar{x}_B$ column, only its second component is replaced by $\bar{x}_2 = 9$):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f'	$\bar{f}' = 101/214$
			1	$-13/32$	$7/160$	$-27/160$		$-19/160$	$-947/80$	$1649/214$
		1		$17/32$	$-19/160$	$-41/160$		$-17/160$	$-321/80$	9
				$-67/32$	$-151/160$	$-149/160$	1	$-93/160$	$-29/80$	$3834/1393$
	1			$9/32$	$-27/160$	$-33/160$		$-41/160$	$727/80$	$-3169/1559$
				-4	$-3/5$	$3/5$		$-4/5$		
u	6	9	12	11	5	14	7	10	1	
$\bar{x}$	$-\frac{3169}{1559}$	9	$\frac{1649}{214}$	-4	-7	14	$\frac{3834}{1393}$	-6	$\frac{101}{214}$	
l	-7	-3	-5	-4	-7	-8	-7	-6	0	

Iteration 4:

1. $\alpha = \min\{(-5 - 1649/214)/(-947/80), (-3 - 9)/(-321/80),$
 $(-7 - 3834/1393)/(-29/80), (6 - (-3169/1559))/(727/80)\}$
 $= \min\{3498/3259, 320/107, 3605/134, 5803/6565\} = 5803/6565, p =$
- 2.
2. $\alpha > \bar{f}' = 101/214$; set $\alpha = 101/214$.
3. $\bar{x}_B = [339/160, 1137/160, 4845/1877, 4605/2041]^T, \bar{f}' = \bar{f}' - \alpha = 0$.
4. The feasibility is achieved.

Phase-II:

Replacing the artificial f' column by the f column of the original problem, and setting $\bar{f} = c^T \bar{x} = (-7, 6, 3, -4, 0, 0, 0, 0)(4605/2041, 1137/160, 339/160, -4, -7, 14, 4845/1877, -6)^T = 246/5$, we obtain the following feasible tableau:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -2211/433$
			1	$-13/32$	$7/160$	$-27/160$		$-19/160$		$339/160$
		1		$17/32$	$-19/160$	$-41/160$		$-17/160$		$1137/160$
				$-67/32$	$-151/160$	$-149/160$	1	$-93/160$		$4845/1877$
	1			$9/32$	$-27/160$	$-33/160$		$-41/160$		$4605/2041$
				-4^*	$-3/5$	$3/5$		$-4/5$	-1	-
u	6	9	12	11	5	14	7	10		
$\bar{x}$	$\frac{4605}{2041}$	$\frac{1137}{160}$	$\frac{339}{160}$	-4	-7	14	$\frac{4845}{1877}$	-6		
l	-7	-3	-5	-4	-7	-8	-7	-6		

Take $p = 5$. $J = \{4, 5, 6, 8\} \neq \emptyset$. $\max\{|-4|, |-3/5|, |3/5|, |-4/5|\} = 4$, $q = 4$.

Multiply row 5 by $-1/4$, and then add $13/32$, $-17/32$, $67/32$, and $-9/32$ times of row 5 to rows 1, 2, 3, and 4, respectively, leading to the following feasible reduced tableau:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 246/5$
				1	$67/640$	$-147/640$		$-3/80$	$13/128$	$339/160$
			1		$-127/640$	$-113/640$		$-17/80$	$-17/128$	$1137/160$
					$-403/640$	$-797/640^*$	1	$-13/80$	$67/128$	$4845/1877$
				1	$-27/128$	$-21/128$		$-5/16$	$-9/128$	$4605/2041$
					$3/20$	$-3/20$		$1/5$	$1/4$	-4
u	6	9	12	11	5	14	7	10		
$\bar{x}$	$\frac{4605}{2041}$	$\frac{1137}{160}$	$\frac{339}{160}$	-4	-7	14	$\frac{4845}{1877}$	-6		
l	-7	-3	-5	-4	-7	-8	-7	-6		

Call Algorithm 18.1.2.

Iteration 5:

1. $\alpha = \min\{(12 - 339/160)/(13/128), (-3 - 1137/160)/(-17/128),$
 $(7 - 4845/1877)/(67/128), (-7 - 4605/2041)/(-9/128),$
 $(11 - (-4))/(1/4)\}$
 $= \min\{6324/65, 6468/85, 3191/378, 5924/45, 60\} = 3191/378, p =$
3.
3. $\bar{x}_B = (997/335, 401/67, 14, 1252/753, -2857/1512)^T, \bar{f} = 246/5 -$
 $3191/378 = 3709/91.$
4. $J = \{6\} \neq \emptyset.$
5. $\max\{|-797/640|\} = 797/640, q = 6.$
6. Multiply row 3 by $-640/797$, and then add $147/640, 113/640, 21/128$, and
 $3/20$ times of row 3 to rows 1, 2, 4, and 5, respectively

(without touching $\bar{x}_B$ column, only its 3rd component is replaced by $\bar{x}_6 = 14$):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = 3709/91$
			1		176/797		-147/797	-6/797	4/797	997/335
		1			-87/797		-113/797	-151/797*	-165/797	401/67
					403/797	1	-640/797	104/797	-335/797	14
	1				-102/797		-105/797	-232/797	-111/797	1252/753
				1	180/797		-96/797	175/797	149/797	-2857/1512
u	6	9	12	11	5	14	7	10		
$\bar{x}$	$\frac{1252}{753}$	$\frac{401}{67}$	$\frac{997}{335}$	$-\frac{2857}{1512}$	-7	14	7	-6		
l	-7	-3	-5	-4	-7	-8	-7	-6		

Iteration 6:

1. $\alpha = \min\{(12 - 997/335)/(4/797), (-3 - 401/67)/(-165/797),$
 $(-8 - 14)/(-335/7797), (-7 - 1252/753)/(-111/797),$
 $(11 - (-2857/1512))/(149/97)\}$
 $= \min\{219357/7122, 13758/317, 7537/144, 33028/531, 11514/167\}$
 $= 13758/317, p = 2.$
3. $\bar{x}_B = (527/165, 6, -140/33, -241/55, 1027/165)^T, \bar{f} = 3709/91 -$
 $13758/317 = -3628/1373.$
4. $J = \{5, 8\} \neq \emptyset.$
5. $\max\{|-87/797|, |-151/797|\} = 151/797, q = 8.$

6. Multiply row 2 by $-797/151$, and then add $6/797$, $-104/797$, $232/797$, and $-175/797$ times of row 1 to rows 1, 3, 4, and 5, respectively (without touching $\bar{x}_B$ column, only its second component is replaced by $\bar{x}_8 = -6$):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -3628/1373$
,		$-6/151$	1		$34/151$		$-27/151$		$2/151$	$527/165$
		$-797/151$			$87/151$		$113/151$	1	$165/151$	-6
		$104/151$			$65/151$	1	$-136/151$		$-85/151$	$-140/33$
	1	$-232/151$			$6/151$		$13/151$		$27/151$	$-241/55$
		$175/151$		1	$15/151$		$-43/151$		$-8/151$	$1027/165$
u	6	9	12	11	5	14	7	10		
$\bar{x}$	$-\frac{241}{55}$	-3	$\frac{527}{165}$	$\frac{1027}{165}$	-7	$-\frac{140}{33}$	7	-6		
l	-7	-3	-5	-4	-7	-8	-7	-6		

Iteration 7:

- $\alpha = \min\{(12 - 527/165)/(2/151), (10 - (-6))/(165/151), (-8 - (-140/33))/(-85/151), (6 - (-241/55))/(27/151), (-4 - 1027/165)/(-8/151)\}$
 $= \min\{4654/7, 2416/165, 5981/896, 2845/49, 11000/57\}$
 $= 5981/896, p = 3.$
- $\bar{x}_B = (279/85, 22/17, -8, -271/85, 499/85)^T, \bar{f} = -3628/1373 - 5981/896 = -2963/318.$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{f} = -2963/318$
		$-6/151$	1		$34/151$		$-27/151$		$2/151$	$279/85$
		$-797/151$			$87/151$		$113/151$	1	$165/151$	$22/17$
		$104/151$			$65/151$	1	$-136/151$		$-85/151$	-8
	1	$-232/151$			$6/151$		$13/151$		$27/151$	$-271/85$
		$175/151$		1	$15/151$		$-43/151$		$-8/151$	$499/85$
u	6	9	12	11	5	14	7	10		
$\bar{x}$	$-\frac{271}{85}$	-3	$\frac{279}{85}$	$\frac{499}{85}$	-7	-8	7	$\frac{22}{17}$		
l	-7	-3	-5	-4	-7	-8	-7	-6		

4. $J = \emptyset$. The optimal basic solution and optimal value are

$$x^* = (-271/85, -3, 279/85, 499/85, -7, -8, 7, 22/17)^T,$$

$$f^* = -2963/318.$$

18.2 Generalized Dual Reduced Simplex Algorithm

The generalized reduced Algorithm 18.1.2 pursues the dual feasibility while maintaining primal feasibility. Assume that reduced simplex tableau is dual feasible, i.e., for some row index p , it holds that

$$J = \{j \in \Gamma \mid \bar{a}_{p,n+1}\bar{a}_{pj} > 0\} \cup \{j \in \Pi \mid \bar{a}_{p,n+1}\bar{a}_{pj} < 0\} = \emptyset \quad (18.20)$$

and $\bar{x}_{j_p} = l_{j_p}$ if $\bar{a}_{p,n+1} < 0$, while $\bar{x}_{j_p} = u_{j_p}$ if $\bar{a}_{p,n+1} > 0$.

Introduce the following *bound violations*:

$$\rho_i = \begin{cases} l_{j_i} - \bar{x}_{j_i}, & \text{if } \bar{x}_{j_i} < l_{j_i}, \\ u_{j_i} - \bar{x}_{j_i}, & \text{if } \bar{x}_{j_i} > u_{j_i}, \\ 0, & \text{if } l_{j_i} \leq \bar{x}_{j_i} \leq u_{j_i}, \end{cases} \quad i = 1, \dots, m+1. \quad (18.21)$$

Then, seeking the primal feasibility while maintaining dual feasibility, we derive the following dual algorithm.

Algorithm 18.2.1 (Generalized Dual Reduced Simplex Algorithm: Tableau Form) *Initial: dual feasible reduced tableau of form (18.19). This algorithm solves the bounded-variable problem (18.1).*

1. Select row index $r \in \arg \max\{|\rho_i| \mid i = 1, \dots, m+1, i \neq p\}$, where ρ_i is defined by (18.21).
2. Stop if $\rho_r = 0$ (optimality achieved).
3. Set $p = r$, and go to step 7 if

$$J = \{j \in \Gamma \mid \text{sign}(\rho_r)\bar{a}_{rj} < 0\} \cup \{j \in \Pi \mid \text{sign}(\rho_r)\bar{a}_{rj} > 0\} = \emptyset.$$

4. Determine index q such that $\beta = |\bar{a}_{pq}/\bar{a}_{rj}| = \min_{j \in J} |\bar{a}_{pj}/\bar{a}_{rj}|$.
5. Convert $\bar{a}_{rj}$ to 1, and eliminate the other nonzeros of the column by elementary transformations (without touching the $\bar{x}_B$ column, except for its r th component replaced by the value of $\bar{x}_q$).
6. Update $\bar{x}_B = \bar{x}_B - \rho_r \bar{a}_{j_r}$, $\bar{x}_{j_r} = \bar{x}_{j_r} + \rho_r$.
7. Compute $\alpha = -\rho_p/\bar{a}_{p,n+1}$.
8. Stop if $\alpha < 0$ (dual unbounded or infeasible problem).
9. If $\alpha \neq 0$, set $\bar{x}_B - \alpha \bar{a}_{n+1}$ and $\bar{f} = \bar{f} + \alpha$.
10. Go to step 1.

Practically, the bound flipping tactic should be incorporated into Algorithm 18.2.1, with its revised version formed therewith.

Example 18.2.1 Solve the following problem by Algorithm 18.2.1:

$$\begin{aligned}
 \min f &= x_1 - 2x_2 + 5x_3 + x_4 \\
 \text{s.t. } 3x_1 - 2x_2 + x_3 - 4x_4 + x_5 &= -3, \\
 6x_1 + 3x_2 - 3x_3 + 6x_4 + x_6 &= 4, \\
 -5x_1 - 4x_2 - 6x_3 + 2x_4 + x_7 &= -7 \\
 0 \leq x_1 \leq 10, \quad 1 \leq x_2 \leq 10, \quad -20 \leq x_3 \leq 5, \\
 -4 \leq x_4 \leq 7, \quad 0 \leq x_5, x_6, x_7 \leq 15.
 \end{aligned}$$

Answer $B = \{5, 6, 7\}$, $N = \{1, 2, 3, 4\}$. Set $\bar{x}_N = (0_{(-)}, 10^{(+)}, -20_{(-)}, -4_{(-)})^T$, then

$$\bar{x}_B = b - Nx_N = \begin{pmatrix} -3 \\ 4 \\ -7 \end{pmatrix} - \begin{pmatrix} 3 & -2 & 1 & -4 \\ 6 & 3 & -3 & 6 \\ -5 & -4 & -6 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 10 \\ -20 \\ -4 \end{pmatrix} = \begin{pmatrix} 21 \\ -62 \\ -79 \end{pmatrix},$$

$$\bar{f} = (1, -2, 5, 1)(0, 10, -20, -4)^T = -124.$$

Initial tableau:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{f} = -124$
	3	-2	1	-4	1				21
	6	3	-3	6		1			-62
	-5	-4	-6	2			1		-79
	1*	-2	5	1				-1	124
u	10	10	5	7	15	15	15	$+\infty$	
$\bar{x}$	0	10	-20	-4	21	-62	-79	-124	
l	0	1	-20	-4	0	0	0	$-\infty$	

Iteration 1: Turn the preceding to the reduced tableau by taking $\bar{a}_{4,1}$ at the pivot. Add -3 , -6 , and 5 times of row 4 to rows 1, 2, and 3, respectively (without touching the $\bar{x}_B$ column, but its 4th component is replaced by the value, 0, of $\bar{x}_1$).

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{f} = -124$
		4	-14	-7	1			3	21
		15	-33	0		1		6	-62
		-14	19	7			1	-5	-79
	1	-2	5	1				-1	0
u	10	10	5	7	15	15	15	$+\infty$	
$\bar{x}$	0	10	-20	-4	21	-62	-79	-124	
l	0	1	-20	-4	0	0	0	$-\infty$	

which is a dual feasible reduced tableau with $r = 4$ and $p = 4$.

Iteration 2:

- $\rho_1 = 15 - 21 = -6$, $\rho_2 = 0 - (-62) = 62$, $\rho_3 = 0 - (-79) = 79$.
 $\max\{|-6|, |62|, |79|\} = 79$, $r = 3$.
- $J = \emptyset$, $p = 3$.
- $\alpha = -79/(-5) > 0$.
- $\bar{x}_B = (21, -62, -79, 0)^T - (79/5)(3, 6, -5, -1)^T = (-132/5, -784/5, 0, 79/5)^T$
 $\bar{f} = -124 + 79/5 = -541/5$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{f} = -541/5$
		4	-14	-7	1			3	-132/5
		15	-33*	0		1		6	-784/5
		-14	19	7			1	-5	0
	1	-2	5	1				-1	79/5
u	10	10	5	7	15	15	15	$+\infty$	
$\bar{x}$	79/5	10	-20	-4	-132/5	-784/5	0	-541/5	
l	0	1	-20	-4	0	0	0	$-\infty$	

Iteration 3:

- $\rho_1 = 0 - (-132/5) = 132/5$, $\rho_2 = 0 - (-784/5) = 784/5$, $\rho_4 = 10 - 79/5 = -29/5$.
 $\max\{|132/5|, |784/5|, |-29/5|\} = 784/5$, $r = 2$.
- $J = \{2, 3\}$.
- $\min\{|(-14)/15|, |19/(-33)|\} = 19/33$, $q = 3$.
- Multiply row 2 by $-1/33$, and add 14, -19 , -5 times of row 2 to rows 1, 3, 4, respectively (without touching the $\bar{x}_B$ column, but its 2nd component is replaced by the value, -20 , of $\bar{x}_3$).
- $\bar{x}_B = (-132/5, -20, 0, 79/5)^T - (784/5)(-14/33, -1/33, 19/33, 5/33)^T$
 $= (1324/33, -2516/165, -14896/165, -1313/165)^T$, $\bar{x}_6 = 0$.
- $\alpha = -(14896/165)/(-17/11) = 5199/89 > 0$.

$$\begin{aligned}
 9. \bar{x}_B &= (1324/33, -2516/165, -14896/165, -1313/165)^T - (5199/89)(5/11, \\
 &\quad -2/11, -17/11, -1/11)^T \\
 &= (502/37, -1828/395, 0, -1771/669)^T, \bar{f} = -541/5 + 5199/89 = \\
 &\quad -846/17.
 \end{aligned}$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{f} = -846/17$
		-26/11		-7	1	-14/33		5/11	$\frac{502}{37}$
		-5/11	1	0		-1/33		-2/11	$-\frac{1828}{395}$
		-59/11		7		19/33	1	-17/11	0
	1	3/11*		1		5/33		-1/11	$-\frac{1771}{669}$
u	10	10	5	7	15	15	15	$+\infty$	
$\bar{x}$	$-\frac{1771}{669}$	10	$-\frac{1828}{395}$	-4	$\frac{502}{37}$	0	0	$-\frac{846}{17}$	
l	0	1	-20	-4	0	0	0	$-\infty$	

Iteration 4:

- $\rho_1, \rho_2 = 0, \rho_4 = 0 - (-1771/669) = 1771/669$,
 $\max\{|-1771/669|\}, r = 4$.
- $J = \{2\}$.
- $\min\{|(3/11)/(-59/11)|\}, q = 2$.
- Multiply row 4 by 11/3, and add 26/11, 5/11, 59/11 times of row 4 to rows 1,2,3, respectively
 (without touching the $\bar{x}_B$ column, but its 4th component is replaced by the value, 10, of $\bar{x}_2$).
- $\bar{x}_B = (502/37, -1828/395, 0, 10)^T - (1771/669)(26/3, 5/3, 59/3, 11/3)^T$
 $= (-8822/941, -3851/426, -15046/289, 589/2007)^T, \bar{x}_1 = 0$.
- $\alpha = -(38992/749)/(-10/3) = 19319/1237 > 0$.
- $\bar{x}_B = (-8822/941, -3851/426, -15046/289, 589/2007)^T - (19319/1237)$
 $(-1/3, -1/3, -10/3, -1/3)^T$
 $= (-15504/3721, -2166/565, 0, 11/2)^T, \bar{f} = -846/17 + 19319/1237 =$
 $-205/6$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{f} = -205/6$
	26/3			5/3	1	8/9		-1/3	$-\frac{97125}{25337}$
	5/3		1	5/3		2/9		-1/3	$-\frac{2166}{565}$
	59/3			80/3		32/9	1	-10/3	0
	11/3	1		11/3		5/9		-1/3	$\frac{11}{2}$
u	10	10	5	7	15	15	15	$+\infty$	
$\bar{x}$	0	$\frac{11}{2}$	$-\frac{2166}{565}$	-4	$-\frac{97125}{25337}$	0	0	$-\frac{205}{6}$	
l	0	1	-20	-4	0	0	0	$-\infty$	

Iteration 5:

1. $\rho_1 = 97125/25337$, $\rho_2, \rho_4 = 0$. $\max\{|97125/25337|\}$, $r = 1$.
3. $J = \emptyset$, $p = 1$.
7. $\alpha = -(97125/25337)/(-1/3) = 325662/26053 > 0$.
9. $\bar{x}_B = (-15504/3721, -2166/565, 0, 11/2)^T - (325662/26053)(-1/3, -1/3, -10/3, -1/3)^T$
 $= (0, 1/3, 125/3, 29/3)^T$, $\bar{f} = -205/6 + 325662/26053 = -65/3$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{f} = -65/3$
	26/3			5/3	1	8/9		-1/3	0
	5/3		1	5/3		2/9		-1/3	1/3
	59/3			80/3*		32/9	1	-10/3	125/3
	11/3	1		11/3		5/9		-1/3	29/3
u	10	10	5	7	15	15	15	$+\infty$	
$\bar{x}$	0	29/3	1/3	-4	0	0	125/3	-65/3	
l	0	1	-20	-4	0	0	0	$-\infty$	

Iteration 6:

1. $\rho_2, \rho_2, \rho_4 = 0$, $\rho_3 = 15 - 125/3 = -80/3$, $r = 3$.
3. $J\{1, 4, 6\} \neq \emptyset$.
4. $\min\{|(26/3)/(59/3)|, |(5/3)/(80/3)|, |(8/9)/(32/9)|\} = (5/3)/(80/3)$, $q = 4$.
5. Multiply row 3 by $3/80$, and add $-5/3, -5/3, -11/3$ times of row 3 to rows 1, 2, 4, respectively (without touching the $\bar{x}_B$ column, but its 3rd component is replaced by the value, -4 , of $\bar{x}_4$).
6. $\bar{x}_B = (0, 1/3, -4, 29/3)^T - (-80/3)(-1/16, -1/16, 3/80, -11/80)^T = (-5/3, -4/3, -3, 6)^T$, $\bar{x}_7 = 125/3 - 80/3 = 15$.
7. $\alpha = -(5/3)/(-1/8) = 40/3 > 0$.
9. $\bar{x}_B = (-5/3, -4/3, -3, 6)^T - (40/3)(-1/8, -1/8, -1/8, 1/8)^T = (0, 1/3, -4/3, 13/3)^T$,
 $\bar{f} = -65/3 + 40/3 = -25/3$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{f} = -25/3$
	119/16				1	2/3	-1/16	-1/8	0
	7/16		1			0	-1/16	-1/8	1/3
	59/80			1		2/15	3/80	-1/8	-4/3
	77/80	1				1/15	-11/80	1/8	13/3
u	10	10	5	7	15	15	15	$+\infty$	
$\bar{x}$	0	13/3	1/3	-4/3	0	0	15	-25/3	
l	0	1	-20	-4	0	0	0	$-\infty$	

Iteration 7:

1. $\rho_2, \rho_3, \rho_4 = 0$.
2. Terminated with optimal solution and objective value:

$$x^* = (0, 13/3, 1/3, -4/3, 0, 0, 15)^T, \quad f^* = -25/3.$$

18.2.1 Generalized Dual Reduced Phase-I

When the components of l and u are all finite, it is, in principle, possible to assign appropriate values to nonbasic variables such that the tableau is dual feasible. Numerically, however, too large a bound value is unacceptable, let alone the magnitude of some components of bounds may be infinite. Therefore, it is still required to provide an initial dual feasible tableau or solution for the generalized dual reduced algorithm.

A choice is to directly modify the reduced simplex Algorithm 18.1.2, ignoring the requirement of the initial reduced tableau being feasible. The overall steps are as follows.

Algorithm 18.2.2 (Generalized Dual Reduced Phase-I: Tableau Form) *Initial: reduced tableau of form (18.19) and solution $\bar{x}$. This algorithm finds a dual feasible tableau to the bounded-variable problem (18.1).*

1. Determine stepsize α and row index p by (18.6) and (18.7).
2. If $\alpha \neq 0$, set $\bar{x}_B = \bar{x}_B + \alpha \bar{a}_{n+1}$ and $\bar{f} = \bar{f} - \alpha$.
3. Stop if $J = \{j \in \Gamma \mid \bar{a}_{p, n+1} \bar{a}_{pj} > 0\} \cup \{j \in \Pi \mid \bar{a}_{p, n+1} \bar{a}_{pj} < 0\} = \emptyset$
(dual feasibility achieved).
4. Determine column index $q \in \arg \max_{j \in J} |\bar{a}_{pj}|$.
5. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
6. Go to step 1.

Example 18.2.2 Find a dual feasible solution to the following problem by Algorithm 18.2.2:

$$\begin{aligned} \min \quad & f = 4x_1 - 5x_2 + 2x_3 + x_4, \\ \text{s.t.} \quad & 3x_1 - 4x_2 + x_3 - 5x_4 + x_5 = -3, \\ & -2x_1 - 6x_2 - 3x_3 + 6x_4 + x_6 = 26, \\ & -6x_1 + 2x_2 - 5x_3 + 3x_4 + x_7 = 21, \\ & -\infty \leq x_1 \leq 9, \quad -7 \leq x_2 \leq +\infty, \quad -1 \leq x_3 \leq 12, \\ & 0 \leq x_4 \leq 8, \quad 0 \leq x_5 \leq 10, \quad -5 \leq x_6 \leq 6, \\ & 0 \leq x_7 \leq +\infty. \end{aligned}$$

Answer Set $\bar{x}_N = (9^{(+)}, -7_{(-)}, -1_{(-)}, 0_{(-)})^T$ ($N = \{1, 2, 3, 4\}$). $\bar{x}_B = b - N\bar{x}_N = (-3, 26, 21)^T - (54, 27, -63, 69)^T = (-57, -1, 84)^T$, $\bar{f} = c_B^T \bar{x}_B = (4, -5, 2, 1)^T (9, -7, -1, 0) = 69$, $B = \{5, 6, 7\}$.

Initial: reduced tableau (where the RHS column is listed to be easier to go to Phase-II):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	RHS	$\bar{f} = 69$
	3	-4	1	-5	1				-3	-57
	-2	-6	-3	6		1			26	-1
	-6	2	-5	3			1		21	84
	4	-5*	2	1				-1		
u	9	∞	12	8	10	6	∞			
$\bar{x}$	9	-7	-1	0	-57	-1	84			
l	$-\infty$	-7	-1	0	0	-5	0			

Iteration 1: Turn the preceding to a reduced tableau.

Take $p = 4$, $J = \{1, 2\} \neq \emptyset$. $\max\{|4|, |-5|\} = 5$, $q = 2$.

Multiply row 4 by $-1/5$, and then add 4, 6, and -2 times of row 4 to rows 1, 2, and 3, respectively

(without touching $\bar{x}_B$ column, only its 4th component is replaced by $\bar{x}_2 = 7$).

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	RHS	$\bar{f} = 69$
	$-1/5$		$-3/5$	$-29/5$	1			$4/5$	-3	-57
	$-34/5^*$		$-27/5$	$24/5$		1		$6/5$	26	-1
	$-22/5$		$-21/5$	$17/5$			1	$-2/5$	21	84
	$-4/5$	1	$-2/5$	$-1/5$				$1/5$		-7
u	9	∞	12	8	10	6	∞			
$\bar{x}$	9	-7	-1	0	-57	-1	84			
l	$-\infty$	-7	-1	0	0	-5	0			

Call Algorithm 18.2.2.

Iteration 2:

- $\alpha = \min\{(10 - (-57))/(4/5), (6 - (-1))/(6/5), (0 - 84)/(-2/5) (\infty - (-7))/(1/5)\} = 35/6$, $p = 2$.
- $\bar{x}_B = (-157/3, 9, 245/3, -35/6)^T$, $\bar{f} = 69 - 35/6 = 379/6$.
- $J = \{1, 4\} \neq \emptyset$.
- $\max\{|-34/5|, |24/5|\} = 34/5$, $q = 1$.

5. Multiply row 2 by $-5/34$, and then add $1/5$, $22/5$, and $4/5$ times of row 2 to rows 1, 3, and 4, respectively (do not touch $\bar{x}_B$ column, except for the second component being replaced by $\bar{x}_1 = 9$):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	RHS	$\bar{f} = 379/6$
			$-15/34$	$-101/17$	1	$-1/34$		$13/17$	$-64/17$	$-157/3$
	1		$27/34$	$-12/17$		$-5/34$		$-3/17$	$-65/17$	9
			$-12/17^*$	$5/17$		$-11/17$	1	$-20/17$	$71/17$	$245/3$
		1	$4/17$	$-13/17$		$-2/17$		$1/17$	$-52/17$	$-35/6$
u	9	∞	12	8	10	6	∞			
$\bar{x}$	9	$-35/6$	-1	0	$-157/3$	6	$245/3$			
l	$-\infty$	-7	-1	0	0	-5	0			

Iteration 3.

- $\alpha = \min\{(10 - (-157/3))/(13/17), (-\infty - 9)/(-3/17), (0 - 245/3)/(-20/17), (\infty - (-35/6))/(1/17)\} = 833/12, p = 3.$
- $\bar{x}_B = (3/4, -13/4, -1, -7/4)^T$ column, $\bar{f} = 379/6 - 833/12 = -25/4.$
- $J = \{3\} \neq \emptyset.$
- $\max\{|-12/17|\}, q = 3.$
- Multiply row 3 by $-17/12$, and then add $15/34$, $-27/34$, and $-4/17$ times of row 3 to rows 1, 2, and 4, respectively (without touching $\bar{x}_B$ column, only the third component is replaced by $\bar{x}_3 = -1$).

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	RHS	$\bar{f} = -25/4$
				$-49/8$	1	$3/8$	$-5/8$	$3/2^*$	$-51/8$	$3/4$
	1			$-3/8$		$-7/8$	$9/8$	$-3/2$	$7/8$	$-13/4$
			1	$-5/12$		$11/12$	$-17/12$	$5/3$	$-71/12$	-1
		1		$-2/3$		$-1/3$	$1/3$	$-1/3$	$-5/3$	$-7/4$
u	9	∞	12	8	10	6	∞			
$\bar{x}$	$-13/4$	$-7/4$	-1	0	$3/4$	6	0			
l	$-\infty$	-7	-1	0	0	-5	0			

Iteration 4:

- $\alpha = \min\{(10 - 3/4)/(3/2), (-\infty - (-13/4))/(-3/2), (12 - (-1))/(5/3), (-7 - (-7/4))/(-1/3)\} = 37/6, p = 1.$
- $\bar{x}_B = (-25/2, 167/18, -137/36, 10)^T, \bar{f} = -25/4 - 37/6 = -149/12.$
- $J = \emptyset.$

Dual feasibility is achieved. Convert the end tableau to a conventional dual feasible one: delete the $\bar{x}_B$ column, and take the entry at the first row and f column

as the pivot to carry out elementary transformations (such that the pivot becomes -1). See the resulting tableau below (whose fourth row is now the objective row, which was the previous first row):

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	RHS	$\bar{f} = -149/12$
	1			$-13/2$	1	$-1/2$	$1/2$		$-11/2$	$-25/2$
			1	$115/18$	$-10/9$	$1/2$	$-13/18$		$7/6$	$167/18$
		1		$-73/36$	$2/9$	$-1/4$	$7/36$		$-37/12$	$-137/36$
				$49/12$	$-2/3$	$-1/4$	$5/12$	-1	$17/4$	10
u	9	∞	12	8	10	6	∞			
$\bar{x}$	$-25/2$	$-137/36$	$167/18$	0	10	6	0			
l	$-\infty$	-7	-1	0	0	-5	0			

$\rho_1, \rho_2, \rho_3 = 0$. The optimal basic solution and optimal value are

$$x^* = (-25/2, -137/36, 167/18, 0, 10, 6, 0)^T,$$

$$f^* = 4(-25/2) - 5(-137/36) + 2(167/18) = -149/12.$$

18.3 Generalized Dual D-Reduced Simplex Algorithm

Consider the D-reduced bounded-variable problem below:

$$\begin{aligned} \min f &= c^T x, \\ \text{s.t. } Ax &= e_m, \quad l \leq x \leq u. \end{aligned}$$

In this section, the dual D-reduced simplex algorithm (Sect. 17.2) is generalized to solve this problem. A tableau version will be derived first, and then its revised version will be formulated.

The key point is to update the objective row based on the datum row as far as possible.

To trace solution $\bar{x}$, three additional rows are listed at the bottom of tableau (17.3), i.e.,

	x_B^T	x_N^T	f	RHS
	I	$\bar{N}_R$		
		$\bar{w}_N^T$		1
		$\bar{z}_N^T$	-1	$-f$
u	l_B^T	u_N^T		
$\bar{x}$	$\bar{x}_B^T$	$\bar{x}_N^T$		
l	l_B^T	l_N^T		

where the associated sets B , N , and R are defined by

$$B = \{j_1, \dots, j_{m-1}\}, \quad N = A \setminus B, \quad R = \{1, \dots, m\} \setminus \{r\} = \{i_1, \dots, i_{m-1}\}, \quad (18.22)$$

$r \notin R$; q column is the datum column, satisfying

$$\bar{\omega}_q \neq 0, \quad \bar{z}_q = 0. \quad (18.23)$$

Let $\bar{x}$ be the basic solution, defined by (17.7), such that

$$\bar{x}_j = l_j \text{ or } u_j, \quad j \in N, \quad j \neq q \quad (18.24)$$

$$\bar{x}_B + \bar{N}_R \bar{x}_N = 0, \quad \sum_{j \in N} \bar{\omega}_j \bar{x}_j = 1. \quad (18.25)$$

If x_q enters the basis, $\bar{x}$ is clearly just the basic solution, associated with the new basis in the conventional sense, with objective value

$$\bar{f} = \bar{z}_N^T \bar{x}_N - \bar{z}_{n+1}. \quad (18.26)$$

Introduce the following column index sets:

$$\Gamma = \{j \in N, j \neq q \mid \bar{x}_j = l_j\}, \quad \Pi = \{j \in N, j \neq q \mid \bar{x}_j = u_j\}. \quad (18.27)$$

It is clear that $\Gamma \cup \Pi = N \setminus \{q\}$, $\Gamma \cap \Pi = \emptyset$.

The D-reduced tableau, satisfying the following conditions:

$$\bar{z}_\Gamma \geq 0, \quad \bar{z}_\Pi \leq 0, \quad (18.28)$$

is called dual feasible D-reduced tableau. If components of l and u are all finite, it is always possible to set the nonbasic variables to relevant bounds such that any D-reduced tableau is dual feasible.

Proposition 18.3.1 *The objective value $\bar{f}$ of the dual feasible D-reduced tableau is a lower bound of all feasible values.*

Proof Let x' be an arbitrary feasible solution. Then, it is known from (18.27) that

$$\bar{x}_\Gamma = l_\Gamma \leq x'_\Gamma, \quad x'_\Pi \leq u_\Pi = \bar{x}_\Pi,$$

which together with (18.28) gives

$$\bar{z}_\Gamma^T \bar{x}_\Gamma \leq \bar{z}_\Gamma^T x'_\Gamma, \quad \bar{z}_\Pi^T x'_\Pi \geq \bar{z}_\Pi^T \bar{x}_\Pi, \quad \bar{z}_{n+1} + \bar{z}_N^T x'_N \geq \bar{z}_{n+1} + \bar{z}_N^T \bar{x}_N = \bar{f}. \quad (18.29)$$

This indicates that the $\bar{f}$ is a lower bound on all feasible values if any. $\square$

Provided that $\bar{x}$ is dual feasible, if, in addition, it satisfies

$$l_q \leq \bar{x}_q \leq u_q, \quad l_B \leq \bar{x}_B \leq u_B, \quad (18.30)$$

then it is clearly a basic optimal solution. In the other case, a sequence of dual feasible tableaus can be generated such that the related objective value increases monotonically, until reaching a basic optimal solution or detecting primal infeasibility.

Introduce the “boundary violations” as follows:

$$\rho_j = \begin{cases} l_j - \bar{x}_j, & \text{If } \bar{x}_j < l_j, \\ u_j - \bar{x}_j, & \text{If } \bar{x}_j > u_j, \\ 0, & \text{If } l_j \leq \bar{x}_j \leq u_j, \end{cases} \quad j \in \{1, \dots, n\}. \quad (18.31)$$

There will be the following two cases to be dealt with:

(i) $\bar{x}_q$ violates boundary, i.e., $\rho_q \neq 0$.

Define the index set

$$N_1 = \{j \in \Gamma \mid \text{sign}(\rho_q \bar{\omega}_q) \bar{\omega}_j < 0\} \cup \{j \in \Pi \mid \text{sign}(\rho_q \bar{\omega}_q) \bar{\omega}_j > 0\}, \quad (18.32)$$

where $\text{sign}(t)$ is the sign function.

Lemma 18.3.1 Assume $\rho_q \neq 0$. If N_1 is empty, there is no feasible solution.

Proof In view of (18.27), the second expression of (18.25) can be expressed as

$$\bar{\omega}_q \bar{x}_q = 1 - \left(\sum_{j \in \Gamma} \bar{\omega}_j l_j + \sum_{j \in \Pi} \bar{\omega}_j u_j \right). \quad (18.33)$$

Assume that x' is a feasible solution, and hence it holds that

$$l_j \leq x'_j \leq u_j \quad (18.34)$$

and that

$$\bar{\omega}_q x'_q = 1 - \left(\sum_{j \in \Gamma} \bar{\omega}_j x'_j + \sum_{j \in \Pi} \bar{\omega}_j x'_j \right). \quad (18.35)$$

Consider the case of $\rho_q \bar{\omega}_q > 0$. In this case, $N_1 = \emptyset$ implies

$$\bar{\omega}_j \geq 0, \quad \forall j \in \Gamma, \quad \bar{\omega}_j \leq 0, \quad \forall j \in \Pi,$$

which along with (18.34) leads to

$$\sum_{j \in \Gamma} \bar{\omega}_j x'_j + \sum_{j \in \Pi} \bar{\omega}_j x'_j \geq \sum_{j \in \Gamma} \bar{\omega}_j l_j + \sum_{j \in \Pi} \bar{\omega}_j u_j.$$

Then, from the preceding expression, (18.33) and (18.35), it follows that

$$\begin{aligned} & \bar{\omega}_q x'_q - \left(\sum_{j \in \Gamma} \bar{\omega}_j x'_j + \sum_{j \in \Pi} \bar{\omega}_j x'_j \right) \\ & \leq - \left(\sum_{j \in \Gamma} \bar{\omega}_j l_j + \sum_{j \in \Pi} \bar{\omega}_j u_j \right) \\ & = -\bar{\omega}_q \bar{x}_q. \end{aligned}$$

If $\bar{\omega}_q > 0$, $\rho_q > 0$, the preceding expression gives $x'_q \leq \bar{x}_q < l_q$, as contradicts that x' is a feasible solution. If $\bar{\omega}_q < 0$, $\rho_q < 0$, on the other hand, it gives $x'_q \geq \bar{x}_q > u_q$, also contradicting that x' is a feasible solution.

In the case of $\rho_q \bar{\omega}_q < 0$, a contradiction can be derived similarly. Therefore, there is no feasible solution. $\square$

Now assume that $N_1 \neq \emptyset$. Determine β_1 and column index q' such that

$$\beta_1 = |\bar{z}_{q'}/\bar{\omega}'_q| = \min_{j \in N_1} |\bar{z}_j/\bar{\omega}_j|, \quad \theta_1 = \text{sign}(\rho_q \bar{\omega}_q) \beta_1. \quad (18.36)$$

If $\theta_1 \neq 0$, the associated D-reduced tableau is said to be *dual nondegenerate*.

Add θ_1 times of the datum row to the bottom row. We might as well denote the resulting tableau again by (18.3). Then it is clear that

$$\bar{z}_q = \theta_1 \bar{\omega}_q \quad (18.37)$$

and that $\bar{\omega}_{q'} \neq 0$, $\bar{z}_{q'} = 0$, where the q' column is taken as the datum column.

Let us determine the corresponding solution $\hat{x}$ and the associated objective value $\hat{f}$. Since $(\bar{x}, \bar{f})$ and $(\hat{x}, \hat{f})$ both are solutions to the system, presented by the new tableau, it holds that

$$\bar{x}_B = -\bar{N}_R \bar{x}_N, \quad \hat{x}_B = -\bar{N}_R \hat{x}_N, \quad (18.38)$$

$$\sum_{j \in N} \bar{\omega}_j \bar{x}_j = 1, \quad \sum_{j \in N} \bar{\omega}_j \hat{x}_j = 1, \quad (18.39)$$

$$\bar{f} = \bar{z}_N^T \bar{x}_N - \bar{z}_{n+1}, \quad \hat{f} = \bar{z}_N^T \hat{x}_N - \bar{z}_{n+1}. \quad (18.40)$$

Denote by $\bar{a}_j(R)$ the j th column of $\bar{N}_R$ for any $j \in N$.

Set $\hat{x}_q$ to the violated bound, i.e.,

$$\hat{x}_q = \bar{x}_q + \rho_q. \quad (18.41)$$

Besides $\hat{x}_q$ and $\hat{x}_{q'}$, let the other nonbasic components of $\hat{x}$ be equal to those of $\bar{x}$ correspondingly, i.e.,

$$\hat{x}_j = \bar{x}_j, \quad j \in N, \quad j \neq q', \quad j \neq q. \quad (18.42)$$

From (18.42) and subtracting the two equalities of (18.39), it follows that

$$\bar{\omega}_q(\hat{x}_q - \bar{x}_q) + \bar{\omega}_{q'}(\hat{x}_{q'} - \bar{x}_{q'}) = 0,$$

combining which and (18.41) gives

$$\hat{x}_{q'} = \bar{x}_{q'} - (\bar{\omega}_q / \bar{\omega}_{q'}) \rho_q. \quad (18.43)$$

From subtracting the two equalities of (18.38) together with (18.41) and (18.42), the updating formula follows, i.e.,

$$\hat{x}_B = \bar{x}_B - \rho_q \bar{a}_q(R) - (\hat{x}_{q'} - \bar{x}_{q'}) \bar{a}_{q'}(R). \quad (18.44)$$

It is easily verified that such a resulting $\hat{x}$ is a dual feasible solution, associated with the new tableau.

Noting $\bar{z}_{q'} = 0$, on the other hand, from (18.42), (18.41), and (18.36) together with subtracting the two equalities of (18.40), it follows that

$$\hat{f} = \bar{f} + \rho_q \bar{z}_q = \bar{f} + |\rho_q \bar{\omega}_q| \beta_1 \geq \bar{f}. \quad (18.45)$$

Therefore, the objective value strictly increases when the D-reduced tableau is dual nondegenerate, i.e., $\beta_1 > 0$.

Set $q = q'$, and repeat the preceding steps. Under the assumption of dual nondegeneracy, the procedure terminates in a finite number of iterations, finally satisfying $l_q \leq \bar{x}_q \leq u_q$, or detecting the nonexistence of a feasible solution. Note that such an iteration involves no basis change but only updates the objective row while maintaining the other part unchanged.

(2) $\bar{x}_q$ is within the boundary, i.e., $\rho_q = 0$.

Determine the row index

$$s \in \arg \max \{|\rho_{j_t}| \mid t = 1, \dots, m-1\}. \quad (18.46)$$

If $\rho_{j_s} = 0$, (18.30) clearly holds, and hence the optimality is achieved.

Now assume that $\rho_{j_s} \neq 0$. Note that $\rho_{j_s} > 0$ implies that $\bar{x}_{j_s}$ violates the lower bound, and $\rho_{j_s} < 0$ implies that it violates the upper bound. Only the following two cases arise:

(i) $\bar{a}_{i_s, q} \neq 0$.

Taking $\bar{a}_{i_s, q}$ as the pivot, convert it to 1, and eliminate the other nonzeros in the column by elementary transformations. Thus, j_s column leaves from and q column enters to the basis. Note that the bottom row remains unchanged since $\bar{z}_q = 0$. The bottom component of j_s th column is zero. Since the old $\bar{\omega}_q$ is nonzero, the new $\bar{\omega}_{j_s}$ must be nonzero, and therefore j_s th column becomes the new datum column, so set $q = j_s$. As the solution $\bar{x}$ is unchanged, it is clear that the new tableau maintains dual feasibility, though now the component, associated with the new datum column, violates the boundary. Consequently, it becomes case (1) via such an iteration.

(ii) $\bar{a}_{i_s, q} = 0$.

Using the set

$$N_2 = \{j \in \Gamma \mid \text{sign}(\rho_{j_s}) \bar{a}_{i_s, j} < 0\} \cup \{j \in \Pi \mid \text{sign}(\rho_{j_s}) \bar{a}_{i_s, j} > 0\}, \quad (18.47)$$

determine column index q' and β_2 such that

$$\beta_2 = |\bar{z}_{q'}/\bar{a}_{i_s, q'}| = \min_{j \in N_2} |\bar{z}_j/\bar{a}_{i_s, j}|, \quad \theta_2 = \text{sign}(\rho_{j_s})\beta_2. \quad (18.48)$$

Then, convert $\bar{a}_{i_s, q'}$ to 1, and eliminate the other nonzeros in the column by elementary transformations. The basis and nonbasis, associated with the resulting tableau, are

$$B = \{j_1, \dots, j_{s-1}, q', j_{s+1}, \dots, j_{m-1}\}, \quad N = N \setminus q' \cup \{j_s\}. \quad (18.49)$$

In this case, the associated elementary transformations do not change the q th column, which remains to be the datum column.

Similar to case (1), the $\hat{x}$, associated with the resulting tableau, can be obtained from (18.38) and (18.40) as follows. Set the nonbasic component $\hat{x}_{j_s}$ to the violated bound, i.e.,

$$\hat{x}_{j_s} = \bar{x}_{j_s} + \rho_{j_s}. \quad (18.50)$$

Let all nonbasic components of $\hat{x}$ be the same as those of $\bar{x}$, except for the q and j_s -indexed ones. Thus, from (18.39), it follows that

$$\bar{\omega}_q(\hat{x}_q - \bar{x}_q) + \bar{\omega}_{j_s}(\hat{x}_{j_s} - \bar{x}_{j_s}) = 0.$$

Thus, we have the following updating formula:

$$\hat{x}_q = \bar{x}_q - (\bar{\omega}_{j_s}/\bar{\omega}_q)\rho_{j_s}. \quad (18.51)$$

Then, by (18.38), it holds that

$$\hat{x}_B = \bar{x}_B - \rho_{j_s}(\bar{a}_{j_s}(R) - (\hat{x}_q - \bar{x}_q)\bar{a}_q(R)). \quad (18.52)$$

The new tableau is dual feasible for $\hat{x}$.

From that $\bar{z}_{j_s} = \theta_2$ holds for the new tableau, (18.40) and (18.48), it follows that

$$\hat{f} = \bar{f} + \rho_{j_s}\bar{z}_{j_s} = \bar{f} + \rho_{j_s}\theta_2 = \bar{f} + |\rho_{j_s}|\beta_2 \geq \bar{f}, \quad (18.53)$$

which implies that the objective value does not decrease and strictly increases when the D-reduced tableau is dual nondegenerate.

The overall steps can be summarized to the following algorithm, where the objective value is not updated by (18.53) iteration by iteration but is computed at the end of the algorithm instead.

Algorithm 18.3.1 (Generalized Dual D-Reduced Simplex Algorithm: Tableau Form) *Initial: dual feasible D-reduced tableau of form (18.3), q th column is the datum column, associated with solution $\bar{x}$. This algorithm solves the D-reduced bounded-variable problem (18.3).*

1. Go to step 7 if ρ_q defined by (18.31) is 0.
2. Stop if N_1 defined by (18.32) is empty.
3. Determine β_1 and column index q' by (18.36).
4. If $\beta_1 \neq 0$, add β_1 times of the datum row to the bottom row.
5. Update $\bar{x}_q$, $\bar{x}_{q'}$, and $\bar{x}_B$ by (18.41), (18.43), and (18.44).
6. Set $q = q'$, and go to step 1.
7. Determine s by (18.46).
8. Stop with $\bar{f} = c^T \bar{x}$ if $\rho_{j_s} = 0$.
9. Go to step 12 if $\bar{a}_{i_s, q} = 0$.
10. Convert $\bar{a}_{i_s, q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
11. Set $q = j_s$, and go to step 2.
12. Determine column index q' by (18.48).
13. Convert $\bar{a}_{i_s, q'}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
14. Update $\bar{x}_{j_s}$, $\bar{x}_{q'}$, and $\bar{x}_B$ by (18.50), (18.51), and (18.52).
15. Go to step 1.

Theorem 18.3.2 Algorithm 18.3.1 creates a sequence of basic feasible solutions. Under the dual nondegeneracy assumption, it terminates either at

- (i) Step 2, detecting infeasibility of the problem, or at
- (ii) Step 8, giving a basic optimal solution

Proof The validity comes from Lemma 18.3.1 and the analysis preceding Algorithm 18.3.1. □

Example 18.3.1 Solve the following problem by Algorithm 18.3.1:

$$\begin{aligned} \min \quad & f = 2x_1 - 4x_2 - 7x_3 + 5x_4, \\ \text{s.t.} \quad & -3x_1 + x_2 + 3x_3 + 5x_4 + x_5 = 5, \\ & -6x_1 + 2x_2 - 4x_3 - 4x_4 + x_6 = -20, \\ & 4x_1 - x_2 - 2x_3 + 3x_4 + x_7 = -8, \\ & -10 \leq x_1 \leq 10, \quad -8 \leq x_2 \leq 20, \quad 1 \leq x_3 \leq 15, \\ & -10 \leq x_4 \leq 17, \quad 0 \leq x_5 \leq 15, \quad -1 \leq x_6 \leq 12, \\ & -8 \leq x_7 \leq 15. \end{aligned}$$

Answer Initial tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
-3	1	3	5	1			5
-6	2	-4	-4		1		-20*
4	-1	-2	3			1	-8
2	-4	-7	5				

Iteration 1: Convert the preceding to a D-reduced tableau.

Take $\bar{x}_N = (-10_{(-)}, 20^{(+)}, 15^{(+)}, -10_{(-)})^T$ ($N = \{1, 2, 3, 4\}$).

$\bar{x}_B = b - N\bar{x}_N = (-40, -100, 112)^T$ ($B = \{5, 6, 7\}$).

$\rho_5 = 0 - (-40) = 40, \rho_6 = (-1) - (-100) = 99,$

$\rho_7 = 15 - 112 = -97. \max\{|-40|, |-99|, |97|\} = 99, \quad p = 2.$

Enter the right-hand side to the basis: multiply row 2 by 1/20, and then add 5 and -8 times of row 2 to rows 1 and 3, respectively:

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	-9/2	3/2	2	4	1	1/4		
	3/10	-1/10	1/5	1/5		-1/20		1
	32/5	-9/5	-2/5	23/5		-2/5	1	
	2	-4	-7	5				
u	10	20	15	17	15	12	15	
$\bar{x}$	-10	20	15	-10	-40	-100	112	
l	-10	-8	1	-10	0	-1	-8	

The second row is the datum row, and x_6 column is the datum column.

$$\bar{x}_B = (-40, 112)^T (B = \{5, 7\}).$$

$$\bar{x}_N = (-10_{(-)}, 20^{(+)}, 15^{(+)}, -10_{(-)}, -100)^T (N = \{1, 2, 3, 4, 6\}). \bar{f} = -255.$$

The resulting tableau is then dual feasible.

Call Algorithm 18.3.1: $r = 2$, $q = 6$.

Iteration 2:

1. $\rho_6 = -1 - (-100) = 99 > 0$.
2. $\text{sign}(\rho_6 \bar{a}_{2,6}) = -1$, $N_1 = \{1, 2, 4\} \neq \emptyset$.
3. $\beta_1 = \min\{|2/(3/10)|, |-4/(-1/10)|, |5/(1/5)|\} = 20/3$,
 $q' = 1$, $\theta_1 = -1(20/3) = -20/3$.
4. Add $-20/3$ times of row 2 to row 4.
5. $\bar{x}_6 = -100 + 99 = -1$. $\bar{x}_1 = -10 - ((-1/20)/(3/10))99 = 13/2$.
 $\bar{x}_B = (-40, 112)^T - 99(1/4, -2/5)^T - (13/2 - (-10))(-9/2, 32/5)^T$
 $= (19/2, 46)^T$, $B = \{5, 7\}$.
6. $q = 1$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	$-9/2$	$3/2$	2	4	1	$1/4$		
	$3/10$	$-1/10$	$1/5$	$1/5$		$-1/20$		1
	$32/5^*$	$-9/5$	$-2/5$	$23/5$		$-2/5$	1	
		$-10/3$	$-25/3$	$11/3$		$1/3$		$-20/3$
u	10	20	15	17	15	12	15	
$\bar{x}$	$13/2$	20	15	-10	$19/2$	-1	46	
l	-10	-8	1	-10	0	-1	-8	

Iteration 3:

1. $\rho_1 = 0$.
7. $\rho_5 = 0$, $\rho_7 = 15 - (46) = -31$. $i_3 = 3$, $j_3 = 7$.
9. $\bar{a}_{3,1} = 32/5 \neq 0$.
10. Multiply row 3 by $5/32$, and then add $9/2$ and $-3/10$ times of row 3 to rows 1 and 2, respectively.
11. $q = 7$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		15/64	55/32	463/64	1	-1/32	45/64	1
		-1/64	7/32	-1/64		-1/32	-3/64	
	1	-9/32	-1/16	23/32		-1/16	5/32	
		-10/3	-25/3	11/3		1/3		-20/3
u	10	20	15	17	15	12	15	
$\bar{x}$	13/2	20	15	-10	19/2	-1	46	
l	-10	-8	1	-10	0	-1	-8	

Iteration 4:

1. $\rho_7 = 15 - 46 = -30 < 0$.
2. $\text{sign}(\rho_7 \bar{a}_{2,7}) = 1$, $N_1 = \{3, 4, 6\} \neq \emptyset$.
3. $\beta_1 = \min\{|(-25/3)/(7/32)|, |(11/3)/(-1/64)|, |(1/3)/(-1/32)|\} = 32/3$,
 $q' = 6$, $\theta_1 = -(-1)(32/2) = 32/3$.
4. Add $32/3$ times of row 2 to row 4.
5. $\bar{x}_7 = 46 + (-31) = 15$; $\bar{x}_6 = -1 - ((-3/64)/(-1/32))(-31) = 91/2$;
 $\bar{x}_B = (19/2, 13/2)^T - (-31)(45/64, 5/32)^T - (91/2 - (-1))(-1/32, -1/16)^T$
 $= (131/4, 57/4)^T$, $B = \{5, 1\}$.
6. $q = 6$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		15/64	55/32	463/64	1	-1/32	45/64	1
		-1/64	7/32	-1/64		-1/32	-3/64	
	1	-9/32	-1/16	23/32		-1/16	5/32	
		-7/2	-6	7/2			-1/2	4
u	10	20	15	17	15	12	15	
$\bar{x}$	57/4	20	15	-10	131/4	91/2	15	
l	-10	-8	1	-10	0	-1	-8	

Iteration 5:

1. $\rho_6 = 12 - 91/2 = -67/2 < 0$.
2. $\text{sign}(\rho_6 \bar{a}_{2,6}) = 1$, $N_1 = \{3, 4\} \neq \emptyset$.
3. $\beta_1 = \min\{|(-6)/(7/32)|, |(7/2)/(-1/64)|\} = 192/7$, $q' = 3$, $\theta_1 = 192/7$.
4. Add $192/7$ times of row 2 to row 4.
5. $\bar{x}_6 = 91/2 + (-67/2) = 12$;
 $\bar{x}_3 = 15 - ((-1/32)/(7/32))(-67/2) = 143/14$;
 $\bar{x}_B = (131/4, 57/4)^T - (-67/2)(-1/32, -1/16)^T - (143/14 - (15))(55/32, -1/16)^T$
 $= (559/14, 83/7)^T$, $B = \{5, 1\}$.
6. $q = 3$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		15/64	55/32*	463/64	1	-1/32	45/64	
		-1/64	7/32	-1/64		-1/32	-3/64	1
	1	-9/32	-1/16	23/32		-1/16	5/32	
		-55/14		43/14		-6/7	-25/14	220/7
u	10	20	15	17	15	12	15	
$\bar{x}$	83/7	20	143/14	-10	559/14	12	15	
l	-10	-8	1	-10	0	-1	-8	

Iteration 6:

1. $\rho_3 = 0$.
7. $\rho_5 = 15 - 559/14 = -349/14$, $\rho_1 = 10 - 83/7 = -13/7$. $i_1 = 1$, $j_1 = 5$.
9. $\bar{a}_{1,3} = 55/32 \neq 0$.
10. Add $-7/32$, $1/16$ times of row 1 to rows 2,3, respectively:
11. $q = 5$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		3/22	1	463/110	32/55	-1/55	9/22	
		-1/22		-103/110*	-7/55	-3/110	-3/22	1
	1	-3/11		54/55	2/55	-7/110	2/11	
		-55/14		43/14		-6/7	-25/14	220/7
u	10	20	15	17	15	12	15	
$\bar{x}$	83/7	20	143/14	-10	559/14	12	15	
l	-10	-8	1	-10	0	-1	-8	

Iteration 7:

1. $\rho_5 = 15 - 559/14 < 0$.
2. $\text{sign}(\rho_5 \bar{a}_{2,5}) = 1$, $N_1 = \{4\} \neq \emptyset$.
3. $\beta_1 = \min\{|(43/14)/(-103/110)|\} = 2365/721$, $q' = 4$, $\theta_1 = 2365/721$.
4. Add $2365/721$ times of row 2 to row 4.
5. $\bar{x}_5 = 559/14 + (-349/14) = 15$;
 $\bar{x}_4 = -10 - ((-7/55)/(-103/110))(-349/14) = -681/103$;
 $\bar{x}_B = (143/14, 83/7)^T - (-349/14)(32/55, 2/55)^T$
 $\quad - (-681/103 - (-10))(463/110, 54/55)^T$
 $\quad = (1077/103, 972/103)^T$, $B = \{3, 1\}$.
6. $q = 4$.

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		3/22	1	463/110	32/55	-1/55	9/22	
		-1/22		-103/110	-7/55	-3/110	-3/22	1
	1	-3/11		54/55	2/55	-7/110	2/11	
		-420/103			-43/103	-195/206	-230/103	3575/103
u	10	20	15	17	15	12	15	
$\bar{x}$	972/103	20	1077/103	-681/103	15	12	15	
l	-10	-8	1	-10	0	-1	-8	

Iteration 8:

1. $\rho_4 = 0$.

7. $\rho_3 = \rho_1 = 0$. The optimal solution and optimal value are

$$x^* = (972/103, 20, 1077/103, -681/103, 15, 12, 15)^T,$$

$$f^* = (2, -4, -7, 5)(972/103, 20, 1077/103, -681/103)^T = -17240/103.$$

Based on the equivalence between the D-reduced tableau (17.3) and the revised tableau (17.19), the revised version of Algorithm 18.3.1 can be stated as follows.

Algorithm 18.3.2 (Dual D-Reduced Simplex Algorithm) *Initial:* B , R , N , B_R^{-1} , $\bar{z}_N = c_N - N_R^T B_R^{-T} c_B$, $\bar{x}$, and $\bar{f}$, satisfying (18.24) and (18.28). This algorithm solves the D-reduced bounded-variable problem (18.3).

1. Go to step 8 if ρ_q , determined by (18.31), is equal to 0.
2. Compute $\bar{\omega}_N = N^T e_m - N_R^T (B_R^{-T} B e_m)$.
3. Stop if $N_1 = \{j \in \Gamma \mid \text{sign}(\rho_q \bar{\omega}_q) \bar{\omega}_j < 0\} \cup \{j \in \Pi \mid \text{sign}(\rho_q \bar{\omega}_q) \bar{\omega}_j > 0\} = \emptyset$ (infeasible problem).
4. Determine θ_1 and column index q' such that $\beta_1 = |\bar{z}_{q'}/\bar{\omega}_{q'}| = \min_{j \in N_1} |\bar{z}_j/\bar{\omega}_j|$, $\theta_1 = \text{sign}(\rho_q \bar{\omega}_q) \beta$.
5. If $\theta_1 \neq 0$, update $\bar{z}_N = \bar{z}_N + \theta_1 \bar{\omega}_N$.
6. Update $\bar{x}_q$, $\bar{x}_{q'}$, and $\bar{x}_B$ by (18.41), (18.43), and (18.44).
7. Set $q = q'$, and go to step 1.
8. Determine s by (18.46).
9. If $\rho_{j_s} = 0$, compute $\bar{f} = c^T \bar{x}$, and stop (optimality achieved).
10. Compute $\sigma_N = N_R^T B_R^{-T} e_s$.
11. Go to step 15 if $\sigma_q = 0$.
12. Compute $\bar{a}_q(R) = B_R^{-1} a_q(R)$.
13. Update B_R^{-1} by (17.20).
14. Set $q = j_s$, and go to step 2.
15. Determine column index q' such that $|\bar{z}_{q'}/\sigma_{q'}| = \min_{j \in N_2} |\bar{z}_j/\sigma_j|$.
16. Compute $\bar{a}_{q'}(R) = B_R^{-1} a_{q'}(R)$.
17. Update B_R^{-1} by (17.20).

18. Update $\bar{x}_{j_s}$, $\bar{x}_{q'}$, and $\bar{x}_B$ by (18.50), (18.51), and (18.52).
19. Update (B, N) by exchanging j_s and q' .
20. Go to step 1.

Chapter 19

Deficient-Basis Method



As was mentioned, the assumption of nondegeneracy is entirely contrary to reality. When the simplex method is used to solve large-scale sparse problems, high degeneracy occurs frequently. In practice, the real problem caused by degeneracy is not cycling but stalling, in which case, the algorithm can be so time-consuming that an unacceptably large number of iterations stay at a degenerate vertex before exiting it.

In this chapter, the basis is generalized to allow deficiency to overcome this shortcoming. This brings up the possibility to take advantage of high degeneracy to reduce computational work significantly. The new algorithms cannot be classified to the simplex category, because each associated iteration no longer necessarily corresponds to a simplex.

In the next section, the concept of the deficient basis is introduced first. In the subsequent two sections, the deficient-basis algorithms in tableau form and revised form are derived in turn, together with the associated computational results reported. In the section after, we address some special implementation issues, such as the initial basis, the LU updating, and the Phase-I procedure using the single-artificial-variable scheme. In the final section, we incorporate the deficient basis into the reduced simplex algorithm together with the Phase-I procedure using the most-obtuse-angle rule.

Consider the standard LP problem but assume that the system $Ax = b$ is consistent, or in other words, the column space of A includes b , i.e.,

$$b \in \text{range}(B).$$

19.1 Concept of Deficient Basis

The *basis* has been the center of the simplex method for many years. Since it is a submatrix consisting of m linearly independent columns of A , however, it is unavoidable for some basic components of the solution to vanish, whenever the right-hand side belongs to a proper subset of $\text{range}(B)$. To cope with this difficulty, we use the following generalized basis instead.

Definition 19.1.1 Deficient basis is a nonsingular square submatrix of A such that the subspace spanned by the associated columns of A includes b .

It is clear that there exists such a basis under the assumption of $b \in \text{range}(A)$, and that a deficient basis is a normal basis if its order equals m .

The columns/rows associated with a deficient basis are said to be *basic*, and the other columns/rows are *nonbasic*. Variables (components) are said to be *basic* and *nonbasic*, respectively.

Let k be the order of a basis. Denote by B and R the index sets of basic columns and rows, respectively by

$$B = \{j_1, \dots, j_k\}, \quad \text{and} \quad R = \{i_1, \dots, i_k\}. \quad (19.1)$$

Denote the associated nonbasic column and row index sets, respectively by

$$N = A \setminus B, \quad \text{and} \quad R' = \{1, \dots, m\} \setminus R. \quad (19.2)$$

Once B, R are given, so are N, R' . Note that R' is empty when $k = m$ whereas N is always nonempty. Although $1 \leq k \leq m$ is usually assumed, it is sometimes useful and easy to include the case $k = 0$, in which B and R are both empty.

Without confusion, the preceding set notation will also be utilized to denote the submatrices or vectors consisting of the associated columns and rows; e.g., the submatrix consisting of basic columns is still denoted by B , as well as the following:

$B_R \in \mathcal{R}^{k \times k}$: basis, consisting of entries in basic columns and rows.

$B_{R'} \in \mathcal{R}^{(m-k) \times k}$: submatrix consisting of entries in basic columns and nonbasic rows.

$N_R \in \mathcal{R}^{k \times (n-k)}$: submatrix consisting of entries in nonbasic columns and basic rows.

$N_{R'} \in \mathcal{R}^{(m-k) \times (n-k)}$: submatrix consisting of entries in nonbasic columns and nonbasic rows.

$b_R \in \mathcal{R}^k$: subvector consisting of basic components (in basic rows) of the right-hand side.

$b_{R'} \in \mathcal{R}^{m-k}$: subvector consisting of nonbasic components (in nonbasic rows) of the right-hand side.

$a_q(R) \in \mathcal{R}^k$: the q -indexed column of N_R .

$a_q(R') \in \mathcal{R}^{m-k}$: the q -indexed column of $N_{R'}$.

Since $b \in \text{range}(B)$, an initial tableau of the standard LP problem, say

x_B	x_N	f	RHS
B_R	N_R		b_R
$B_{R'}$	$N_{R'}$		$b_{R'}$
c_B^T	c_N^T	-1	

(19.3)

can be converted to an equivalent form by elementary transformations with row and columns exchanges, i.e.,

x_B	x_N	f	RHS
I	$\bar{N}_R$		$\bar{b}_R$
	$\bar{N}_{R'}$		
	$\bar{z}_N^T$	-1	$-\bar{f}$

(19.4)

The preceding is referred to as *deficient-basis tableau*, where the unit matrix I corresponds to basis B_R , featured by basic columns and rows.

$\bar{z}_N^T$ is the *reduced cost(s)*. The solution to $Ax = b$, given by

$$\bar{x}_B = \bar{b}_R, \quad \bar{x}_N = 0,$$

is called *basic solution*. It is always possible to expand a deficient basis to a full one, associated with the same solution, by entering some nonbasic columns. Note that the j_t -indexed basic variable x_{j_t} is in the i_t -indexed row of the tableau. Therefore, it holds that

$$\bar{x}_{j_t} = \bar{b}_{i_t}, \quad t = 1, \dots, k.$$

The solution is feasible if $\bar{x}_B \geq 0$, and optimal if, in addition, the associated objective value attains the minimum value over the feasible region. The tableau is said to be feasible or optimal, respectively.

There are results similar to those in the conventional simplex context; for instance, we have the following.

Lemma 19.1.1 *Assume that the deficient-basis tableau (19.4) is feasible. If its reduced costs are all nonnegative, then it gives a pair of primal and dual basic optimal solutions.*

Proof It is clear that the tableau gives a pair of complementary primal and dual solutions, which are primal and dual feasible respectively, satisfying the optimality conditions. □

19.2 Deficient-Basis Algorithm: Tableau Form

Let us now discuss how to transform a feasible deficient-basis tableau to a new one while maintaining feasibility, and therefore derive a tableau version of the so-called *deficient-basis algorithm*.

Assume that the current deficient-basis tableau of form (19.4) is feasible ($\bar{b}_R \geq 0$) but $\bar{z}_N \not\geq 0$. We select a pivot column index q such that $\bar{z}_q < 0$, e.g., by the counterpart of Dantzig conventional rule below:

$$q \in \arg \min \{\bar{z}_j \mid j \in N\}. \quad (19.5)$$

Thus, it holds that $\bar{z}_q < 0$.

Denote the q -indexed columns of $\bar{N}_R$ and $\bar{N}_{R'}$ by $\bar{a}_q(R)$ and $\bar{a}_q(R')$, respectively.

The determination of a pivot row index depends on whether column $\bar{a}_q(R')$ vanishes, i.e., whether a_q belongs to the column space of B . There are the following two cases arising:

(i) Rank-increasing iteration: $R' \neq \emptyset$ and $\bar{a}_q(R') \neq 0$.

In this case, it is not allowed for $\bar{x}_q$ to be taken on any nonzero value while keeping the other nonbasic components of the current solution unchanged, as such doing violates some of the equality constraints, associated with nonbasic rows. To overcome the difficulty, we determine the row index

$$p \in \arg \max_{i \in R'} |\bar{a}_{i,q}|. \quad (19.6)$$

Then take $\bar{a}_{p,q}$ as a pivot to carry out elementary transformations, so that the q -indexed nonbasic column and p -index nonbasic row both become basic while the basic columns and rows remain basic. Thus, the rank of the basis increases by 1, and the iterations are complete. It is noted that the right-hand side remains unchanged.

(ii) Rank-remaining iteration: $R' = \emptyset$ or $\bar{a}_q(R') = 0$.

Increase $\bar{x}_q$ from zero to decrease the objective value while keeping the other nonbasic components of the current solution feasible. In the following case, such a decrease will be unlimited.

Lemma 19.2.1 *Assume that the deficient-basis tableau is feasible and that $R' = \emptyset$ or $\bar{a}_q(R') = 0$. If $\bar{z}_q < 0$ and $\bar{a}_q(R) \leq 0$, the original problem is lower unbounded.*

Proof The proof is an analogue to that of Lemma 3.1.2. $\square$

If, otherwise, the expression $\bar{a}_q(R) \not\leq 0$ is not satisfied, then there is a blocking variable, associated with α and s such that

$$\alpha = \bar{b}_{i_s} / \bar{a}_{i_s,q} = \min \{\bar{b}_{i_t} / \bar{a}_{i_t,q} \mid \bar{a}_{i_t,q} > 0, t = 1, \dots, k\} \geq 0, \quad (19.7)$$

where stepsize α will be used to update the feasible basic solution. Taking $\bar{a}_{i_s,q}$ as the pivot, carry out according elementary transformations. Consequently, the j_s -index

column leaves and q -index column enters the basis, while the basic rows remain basic. So, the rank of the basis remains, and an iteration is complete. It is noted such doing does not touch nonbasic rows at all (including entries in the right-hand side).

As $\bar{z}_q$ is eliminated, both cases change components of the dual solution, located at the bottom row. Since $\bar{b}_{i_s} = 0$, a rank-increasing iteration does not change tableau's right-hand side, hence the solution and objective value. On the other hand, a rank-remaining iteration changes these quantities, in general; in fact, it is known that the objective value becomes

$$\hat{f} = \bar{f} + \alpha \bar{z}_q \leq \bar{f},$$

where the right-most inequality results from (19.7) and $\bar{z}_q < 0$. Therefore, the objective value does not increase, and in case of nondegeneracy ($\bar{b}_R > 0$), α is positive, hence the objective value strictly decreases.

The overall procedure can be summarized into the following algorithm.

Algorithm 19.2.1 (Deficient-Basis Algorithm: Tableau Form) *Initial: feasible tableau of form (19.4) with $B, N, R, R', 1 \leq k \leq m$. This algorithm solves the standard LP problem.*

1. Determine column index $q \in \arg \min_{j \in N} \bar{z}_j$.
2. Stop if $\bar{z}_q \geq 0$ (optimality achieved).
3. Go to step 7 if $R' = \emptyset$ or $\bar{a}_q(R') = 0$.
4. Determine $p \in \arg \max_{i \in R'} |\bar{a}_{i,q}|$.
5. Convert $\bar{a}_{p,q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
6. Set $k = k + 1$, bring q from N to B and p from R' to R , and go to step 1.
7. Stop if $\bar{a}_q(R) \leq 0$.
8. Determine s by (19.7).
9. Convert $\bar{a}_{i_s,q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
10. Update (B, N) by exchanging j_s and q .
11. Go to step 1.

Theorem 19.2.2 *Assume that rank-remaining iterations are all nondegenerate. Algorithm 19.2.1 terminates either at*

- (i) *step 2, giving an optimal basic solution; or at*
- (ii) *step 7, detecting lower unboundedness.*

Proof The proof on termination of Algorithm 19.2.1 is similar to that in the conventional case. The meanings of its exits come from Lemmas 19.1.1 and 19.2.1 together with the related discussions preceding Algorithm 19.2.1. $\square$

Example 19.2.1 Solve the following problem by Algorithm 19.2.1:

$$\begin{array}{llllllllll}
 \min & f = & -2x_1 & +3x_2 & -4x_3 & +x_4 & +2x_7 & -3x_8, & & \\
 \text{s.t.} & & -x_1 & +x_2 & +x_3 & -x_4 & +x_5 & & +x_7 & -2x_8 = 2, \\
 & & 2x_1 & -x_2 & +x_3 & +x_4 & & +x_6 & -x_7 & +x_8 = 1, \\
 & & x_1 & -2x_2 & & -x_4 & & & & -3x_8 = 0, \\
 & & -x_1 & +x_2 & & +x_4 & & & & +x_8 = 0, \\
 & & 2x_1 & -x_2 & & +x_4 & & & & -x_8 = 0, \\
 & & x_j \geq 0, & j = 1, \dots, 8.
 \end{array}$$

Answer $k = 2$, $B = \{5, 6\}$, $R = \{1, 2\}$, $N = \{1, 2, 3, 4, 7, 8\}$, $R' = \{3, 4, 5\}$.
The initial feasible deficient-basis tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
-1	1	1	-1	1		1	-2	2
2	-1	1*	1		1	-1	1	1
1	-2		-1				-3	
-1	1		1				1	
2	-1		1				-1	
-2	3	-4	1			2	-3	

Iteration 1:

1. $\min\{-2, 3, -4, 1, 2, -3\} = -4 < 0$, $q = 3$.
3. $\bar{a}_q(R') = 0$.
7. $\bar{a}_q(R) \not\leq 0$.
8. $\min\{2/1, 1/1\} = 1$, $s = 2$, $i_s = 2$.
9. add $-1, 4$ times of row 2 to rows 1, 6, respectively.
10. $B = \{5, 3\}$, $N = \{1, 2, 4, 6, 7, 8\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
-3	2		-2	1	-1	2*	-3	1
2	-1	1	1		1	-1	1	1
1	-2		-1				-3	
-1	1		1				1	
2	-1		-2				-1	
6	-1		5		4	-2	1	4

Iteration 2:

1. $\min\{6, -1, 5, 4, -2, 1\} = -2 < 0$, $q = 7$.
3. $\bar{a}_q(R') = 0$.
7. $\bar{a}_q(R) \not\leq 0$.
8. $\min\{1/2\} = 1/2$, $s = 1$, $i_s = 1$.
9. Multiply row 2 by $1/2$, then add 1, 2 times of row 2 to rows 2, 6, respectively.
10. $B = \{7, 3\}$, $N = \{1, 2, 4, 5, 6, 8\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
$-3/2$	1		-1	$1/2$	$-1/2$	1	$-3/2$	$1/2$
$1/2$		1		$1/2$	$1/2$		$-1/2$	$3/2$
1	-2		-1				-3^*	
-1	1		1				1	
2	-1		-2				-1	
3	1		3	1	3		-2	5

Iteration 3:

1. $\min\{3, 1, 3, 1, 3, -2\} = -2 < 0$, $q = 8$.
3. $\bar{a}_q(R') \neq 0$;
4. $\max\{|-3|, |1|, |-1|\} = 3$, $p = 3$.
5. Multiply row 3 by $-1/3$, then add $-1, 1, 2$ times of row 3 to rows 4, 5, 6, respectively.
6. $k = 3$, $B = \{7, 3, 8\}$, $R = \{1, 2, 3\}$, $N = \{1, 2, 4, 5, 6\}$, $R' = \{4, 5\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
-2	2		$-1/2$	$1/2$	$-1/2$	1		$1/2$
$1/3$	$1/3$	1	$1/6$	$1/2$	$1/2$			$3/2$
$-1/3$	$2/3$		$1/3$				1	
$-2/3$	$1/3$		$2/3$					
$5/3$	$-1/3$		$-5/3$					
$7/3$	$7/3$		$11/3$	1	3			5

Iteration 4:

1. $\min\{7/3, 7/3, 11/3, 1, 3\} \geq 0$.
2. The optimal basic solution and optimal value are:

$$x^* = (0, 0, 3/2, 0, 0, 0, 1/2, 0)^T, \quad f^* = -5.$$

19.3 Deficient-Basis Algorithm

In this section, we derive the revised version of Algorithm 19.2.1.

The deficient basis characterizes an iteration, just as the normal basis in the conventional simplex context. All entries in a deficient-basis tableau can be expressed using the inverse of the deficient-basis matrix.

Let B_R be the basis matrix associated with sets B , R , and let B_R^{-1} be its inverse. Then, it is known that

$$\bar{b}_{R'} = b_{R'} - B_{R'} B_R^{-1} b_R = 0.$$

Thus, the revised tableau, corresponding to the deficient-basis tableau (19.4), is

$$\begin{array}{c|cc|c} x_B & x_N & f & \text{RHS} \\ \hline I & B_R^{-1} N_R & & B_R^{-1} b_R \\ & N_{R'} - B_{R'} B_R^{-1} N_R & & \\ \hline & c_N^T - c_B^T B_R^{-1} N_R & -1 & -c_B^T B_R^{-1} b_R \end{array} \quad (19.8)$$

Let $q \in N$ be the pivot column index selected by (19.5). The following tableau offers correspondence between wanted quantities in the preceding tableau and tableau (19.4) (see Table 19.1):

As for updating the inverse B_R^{-1} of the deficient basis, the following two types of iterations should be treated separately:

(i) Rank-increasing iteration: $R' \neq \emptyset$ and $\bar{a}_q(R') = a_q(R') - B_{R'} B_R^{-1} a_q(R) \neq 0$.

Assume that pivot row index $p \in R'$ was determined by (19.6). The new basis matrix (increasing rank by 1) is of the form

$$\widehat{B}_{\hat{R}} = \left(\begin{array}{c|c} B_R & a_q(R) \\ \hline e_p^T B & a_{i_s q} \end{array} \right) \begin{array}{c} k \\ 1 \end{array} \quad (19.9)$$

$k \quad 1$

where $e_p^T B$ denotes the p th row of B , a_{pq} denotes the p th component of a_q , and $a_q(R)$ corresponds to basic components of a_q . It is easy to verify that the inverse is

Table 19.1 Equivalence between the associated quantities

Quantity	Tableau	Relation	Revised tableau
Objective row	$\bar{z}_N$	=	$c_N^T - c_B^T B_R^{-1} N_R$
Basic components of pivot column	$\bar{a}_q(R)$	=	$B_R^{-1} a_q(R)$
Nonbasic components of pivot column	$\bar{a}_q(R')$	=	$a_q(R') - B_{R'} B_R^{-1} a_q(R)$
Basic components of the right-hand side	$\bar{b}_R$	=	$B_R^{-1} b_R$
Nonbasic entries of the p th basic row	$e_p^T \bar{N}_R$	=	$e_p^T B_R^{-1} N_R$

of the following form:

$$\widehat{B}_R^{-1} = \begin{pmatrix} U & v \\ d^T & \tau \end{pmatrix} \begin{matrix} k \\ 1 \end{matrix} \quad (19.10)$$

where

$$\begin{aligned} h^T &= e_p^T B, \\ \tau &= 1/(a_{pq} - h^T \bar{a}_q(R)), \\ v &= -\tau \bar{a}_q(R), \\ d^T &= -\tau h^T B_R^{-1}, \\ U &= B_R^{-1} - v h^T B_R^{-1}. \end{aligned}$$

(ii) Rank-remaining iteration: $R' = \emptyset$ or $\bar{a}_q(R') = a_q(R') - B_{R'} B_R^{-1} a_q(R) = 0$.

Essentially, the updating, in this case, is the same as in the conventional simplex context, only B_R^{-1} is now of order less than m , in general. Assume that $s \in \{1, \dots, k\}$ was determined by (19.7). The updating formula is

$$\widehat{B}_R^{-1} = E_s B_R^{-1}, \quad (19.11)$$

where

$$E_s = \begin{pmatrix} 1 & -(\bar{a}_q(R))_1/(\bar{a}_q(R))_s & & \\ & \ddots & \vdots & \\ & & -(\bar{a}_q(R))_{s-1}/(\bar{a}_q(R))_s & \\ & & 1/(\bar{a}_q(R))_s & \\ & & -(\bar{a}_q(R))_{s+1}/(\bar{a}_q(R))_s & \\ & & \vdots & \ddots \\ & & -(\bar{a}_q(R))_k/(\bar{a}_q(R))_s & 1 \end{pmatrix}$$

Based on Table 19.1, Algorithm 19.2.1 can be revised as follows.

Algorithm 19.3.1 (Deficient-Basis Algorithm) *Initial:* B , R , $1 \leq k \leq m$, B_R^{-1} ; $\bar{x}_B = B_R^{-1} b_R \geq 0$. This algorithm solves the standard LP problem.

1. Compute $\bar{z}_N = c_N - N_R^T \bar{y}$, where $\bar{y} = B_R^{-T} c_B$.
2. Determine pivot column index $q \in \arg \min_{j \in N} \bar{z}_j$.
3. If $\bar{z}_q \geq 0$, compute $\bar{f} = c_B^T \bar{x}_B$, and stop (optimality achieved).
4. Compute $\bar{a}_q(R) = B_R^{-1} a_q(R)$.
5. Go to step 9 if $R' = \emptyset$ or $\bar{a}_q(R') = a_q(R') - B_{R'} \bar{a}_q(R) = 0$.

6. Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{i,q}|$.
7. update B_R^{-1} by (19.10).
8. Set $k = k + 1$, bring q from N to B and p from R' to R , and go to step 1.
9. Stop if $\bar{a}_q(R) \leq 0$ (lower unbounded problem).
10. Determine s and α such that $\alpha = \bar{x}_{i_s}/\bar{a}_{i_s,q} = \min\{\bar{x}_{i_t}/\bar{a}_{i_t,q} \mid \bar{a}_{i_t,q} > 0, t = 1, \dots, k\}$.
11. If $\alpha \neq 0$, set $\bar{x}_q = \alpha$ and update $\bar{x}_B = \bar{x}_B - \alpha \bar{a}_q(R)$.
12. Update B_R^{-1} by (19.11).
13. Update (B, N) by exchanging j_s and q .
14. Go to step 1.

Example 19.3.1 Solve Example 19.2.1 by Algorithm 19.3.1.

Answer Initial: $k = 2$, $B = \{5, 6\}$, $R = \{1, 2\}$. $N = \{1, 2, 3, 4, 7, 8\}$, $R' = \{3, 4, 5\}$. $B_R = I$. $\bar{x}_B = (2, 1)^T \geq 0$.

Iteration 1:

1. $\bar{z}_N = (-2, 3, -4, 1, 2, -3)^T$.
2. $\min\{-2, 3, -4, 1, 2, -3\} = -4 < 0$, $q = 3$.
4. $\bar{a}_q(R) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
5. $\bar{a}_q(R') = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \neq 0$.
9. $\bar{a}_q(R) \not\leq 0$.
10. $\alpha = \min\{2/1, 1/1\} = 1$, $s = 2$, $i_s = 2$.
11. $\bar{x}_3 = 1$, $\bar{x}_B = (2, 1)^T - 1(1, 1)^T = (1, 0)^T$.
12. $B_R^{-1} = \begin{pmatrix} 1 & -1 \\ & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ & 1 \end{pmatrix}$.
13. $B = \{5, 3\}$, $N = \{1, 2, 4, 6, 7, 8\}$.

Iteration 2:

1. $\bar{y} = (0, -4)^T$,
 $\bar{z}_N = (-2, 3, 1, 0, 2, -3)^T - (-8, 4, -4, -4, 4, -4)^T = (6, -1, 5, 4, -2, 1)^T$.
2. $\min\{6, -1, 5, 4, -2, 1\} = -2 < 0$, $q = 7$.
4. $\bar{a}_q(R) = \begin{pmatrix} 1 & -1 \\ & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$.
5. $\bar{a}_q(R') = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} \neq 0$.
9. $\bar{a}_q(R) \not\leq 0$.
10. $\alpha = \min\{1/2\} = 1/2$, $s = 1$, $i_s = 1$.
11. $\bar{x}_3 = 1/2$, $\bar{x}_B = (1, 1)^T - (1/2)(2, -1)^T = (0, 3/2)^T$.

$$12. B_R^{-1} = \begin{pmatrix} 1/2 & 0 \\ 1/2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

$$13. B = \{7, 3\}, N = \{1, 2, 4, 5, 6, 8\}.$$

Iteration 3:

$$1. \bar{y} = (-1, -3)^T, \\ \bar{z}_N = (-2, 3, 1, 0, 0, -3)^T - (-5, 2, -2, -1, -3, -1)^T = (3, 1, 3, 1, 3, -2)^T.$$

$$2. \min\{3, 1, 3, 1, 3, -2\} = -2 < 0, q = 8.$$

$$4. \bar{a}_q(R) = \begin{pmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -3/2 \\ -1/2 \end{pmatrix}.$$

$$5. \bar{a}_q(R') = \begin{pmatrix} -3 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} -3/2 \\ -1/2 \end{pmatrix} = \begin{pmatrix} -3 \\ 1 \\ -1 \end{pmatrix} \neq 0.$$

$$6. \max\{|-3|, |1|, |-1|\} = 3, s = 3, i_s = 3.$$

$$7. \tau = (-3 - (0, 0)(-3/2, -1/2)^T)^{-1} = -1/3.$$

$$v = -(-1/3)(-3/2, -1/2)^T = (-1/2, -1/6)^T, d^T = (0, 0).$$

$$U = \begin{pmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{pmatrix} - \begin{pmatrix} -1/2 \\ -1/6 \end{pmatrix} (0, 0) = \begin{pmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

$$\hat{B}_{\hat{R}}^{-1} = \begin{pmatrix} 1/2 & -1/2 & -1/2 \\ 1/2 & 1/2 & -1/6 \\ 0 & 0 & -1/3 \end{pmatrix}.$$

$$8. k = 3, B = \{7, 3, 8\}, R = \{1, 2, 3\}, B_R^{-1} = \hat{B}_{\hat{R}}^{-1}, \bar{x}_8 = 0.$$

Iteration 4:

$$1. \bar{y} = (-1, -3, 2/3)^T,$$

$$\bar{z}_N = (-2, 3, 1, 0, 0)^T - (-13/3, 2/3, -8/3, -1, -3)^T = (7/3, 7/3, 11/3, 1, 3)^T \geq 0.$$

3. The optimal basic solution and optimal value:

$$x^* = (0, 0, 3/2, 0, 0, 0, 1/2, 0)^T, \quad f^* = (-4, 2)(3/2, 1/2)^T = -5,$$

which coincides with the result in Example 19.2.1.

It is seen that the deficient-basis algorithm solves two linear systems in each iteration, just as the simplex algorithm; but the systems are smaller, especially for large-scale and highly degenerate problems. Also, the deficient basis used appears better conditioned. Its advantages are conformed by computational experiments, as reported in the following subsection.

Note In the original work (Pan, 1998–2008), the basis is generalized as a rectangle submatrix B , whose column space includes b . It is then modified to a square submatrix. Unlike the simplex algorithm, therefore, the solution process of the deficient-basis algorithm cannot be interpreted as a movement from simplex to

simplex. Since the A is not of full row rank in general, there is no correspondence between the deficient bases and vertices.

As for the bounded-variable problem (25.1), the concept of basis can be generalized as follows.

Definition 19.3.1 Let B_R be the nonsingular square submatrix consisting of entries indexed by B and R , and let $\bar{x}_N$ be some fixed values of the nonbasic variables indexed by N . If the range space of B includes $b - N\bar{x}_N$, the submatrix B_R is a basis.

Based on the preceding definition, it is not difficult to generalize the deficient-basis algorithm for solving the bounded-variable problems.

19.3.1 Computational Results

We report computational results to demonstrate the performance of the deficient-basis simplex algorithm Pan (2008).

The experiments were conducted on a Pentium II 550E PC, with Windows 98 operating system, 256MB RAM, and about 16 decimal point precision. Visual Fortran 5.0 compiler was used.

There were two codes involved in the comparison:

1. MINOS 5.3.
2. PDBSA 1.0: is based on an algorithm close to Algorithm 19.3.1

The second code is implemented within MINOS 5.3; the changes are limited on its core part.

Total 65 standard test problems without bounds and ranges sections were tested, which are classified into two groups:

The first group includes 50 Netlib test problems (Appendix C: Table C.4: AFIRO-WOODW), classified into 3 sets according to increasing $m+n$: Small, Medium, and Large. The Small includes 20 problems, Medium and Large include 15 each.

The second group contains 15 larger problems, including 8 Kennington problems (Table C.5), 4 BPMPD (Table C.6), and the 3 largest Netlib problems (Table C.4: QAP12, MAROS-R7, STOCFOR3).

Iteration and time ratios of MINOS 5.3 to PDBSA 1.0 are listed in Table 19.2 where the end column gives degenerate iterations percentage ratio. The largest 3 Netlib problems are not included, as MINOS 5.3 failed to solve them.

At first, it is seen from Table 19.2 that the new code outperformed MINOS 5.3 overall:

The time ratios for the first and the second groups are 1.41 and 2.06, respectively.

It seems that the new code has greater advantages in dealing with large-scale problems.

Table 19.2 Iteration and time ratios of MINOS 5.3 to PDBSA 1.0

Problem	Iterations	Time	% Degen
Small (20)	1.09	1.50	1.06
Medium (15)	0.96	1.30	0.85
Large (15)	1.08	1.41	0.87
Average (50)	1.07	1.41	0.93
Kennington(8)	1.51	2.07	0.61
BPMPD(4)	1.64	2.05	0.63
Average (12)	1.52	2.06	0.63

The new code appeared to be more stable and reliable, as it solved all the 65 test problems, including NSCT2, MAROS-R7, and STOCFOR3, which MINOS 5.3 failed to solve.

It is noticeable that the time ratios are significantly larger than iteration ratios. This is not surprising since the systems solved in each iteration by the new code are small, compare with those solved by MINOS 5.3.

Finally, it is seen from the last column that except for the smallest 20 problems, degenerate iterations ratios are less than 1, that is, the proportion of degenerate iterations yielded by the new code is larger than that by MINOS 5.3. Even so, the new code was still an unambiguous winner. Forrest and Goldfarb (1992) are right that the efficiency of an LP solver does not depend on the proportion of degenerate iterations.

19.4 On Implementation

As a variant of the classical simplex algorithm, the implementation issues of the deficient-basis algorithm, such as the LU factorization of the basis matrix, L^{-1} stored in factors form, LU update in the rank-remaining iteration, etc., are analogous to those, presented in Part I. Besides all, one should pay close attention to taking advantage of basis deficiency to reduce computational work. In this section, only the following three themes are addressed.

19.4.1 Initial Basis

It is noticeable that

$$\text{rank}(A) = \text{rank}(A \mid b) \leq m.$$

The LU factorization of an initial basis matrix can be created by a crash procedure like that presented in Part I. In addition to balancing sparsity and stability, however,

the resulting initial basis should be of order as low as possible. To this end, some variant of the Markowitz Rule would be utilized.

Initially, set $B = \emptyset$, $R = \emptyset$. Suppose, for some k , Gauss elimination has been carried out on the matrix $(A \mid b)$ such that some k columns (excluding b) of the latter are upper triangular; in other words, k basic columns and rows have been determined, and the according $\bar{B}_R$ is an upper triangular k -order submatrix of $(\bar{A} \mid \bar{b})$.

If $k = m$, or if $k < m$ but the following row index set

$$I' = \{i \in R' \mid |\bar{b}_i| > \epsilon_0\} \quad (19.12)$$

is empty, where $\epsilon_0 > 0$ is a threshold for numerical 0, then the LU factorization of an k -order basis matrix has already been obtained. Otherwise, the following rule is utilized to determine the $(k + 1)$ th basic column and row.

Rule 19.4.1 *Given a constant σ with $0 < \sigma \leq 1$. Select the column index q and row index p such that*

$$(r_p - 1) \times (c_q - 1) = \min\{(r_i - 1) \times (c_j - 1) \mid |\bar{a}_{ij}| \geq \sigma\theta, i \in I', j \in N\}, \quad (19.13)$$

where r_i denotes the number of the nonzeros in the i th row and c_j the number of the nonzeros in the j th column, and

$$\theta = \max\{|\bar{a}_{ij}| \mid i \in I', j \in N\}. \quad (19.14)$$

Note that the preceding rule fails if

$$\{(i, j) \in I' \times N \mid |\bar{a}_{ij}| \geq \sigma\theta\} = \emptyset.$$

In this case, $Ax = b$ may be regarded numerically inconsistent.

19.4.2 LU Updating in Rank-Increasing Iteration

Updating LU factors in a rank-remaining iteration is essentially the same as that in the conventional simplex context. In contrast, updating LU factors in a rank-increasing iteration is simpler.

Let $L^{-1}B_R = U$ be the LU factorization of B_R . From (19.9) it follows that

$$\left(\begin{array}{c|c} L^{-1} & \\ \hline & 1 \end{array} \right) \hat{B}_{\hat{R}} = \left(\begin{array}{c|c} U & L^{-1}a_q(R) \\ \hline e_{i_s}^T B & a_{i_s q} \end{array} \right),$$

which is upper triangular, except for the bottom row. Premultiply a series of Gauss matrices $G_1, \dots, G_k$ to eliminate entries of the bottom row (except for the diagonal), it becomes an upper triangular matrix $\tilde{U}$. The LU factorization of the new

basis matrix is then

$$\tilde{L}^{-1} \widehat{B}_{\hat{R}} = \tilde{U}, \quad \tilde{L}^{-1} = G_k \dots G_1 \left(\frac{L^{-1}}{1} \right).$$

19.4.3 Phase-I: Single-Artificial-Variable

Standard Phase-I methods presented in Chap. 14 are applicable (or after slightly modified) in the deficient-basis context. It is also possible to design new Phase-I methods by taking advantage of the deficient basis, like in the Phase-I algorithm introduced in this section (Pans 2001).

Introducing an artificial variable f , associated with column the same as the right-hand side. The resulting auxiliary problem is then

$$\begin{aligned} \min \quad & f \\ \text{s.t.} \quad & Ax + bx_{n+1} = b, \quad x, f \geq 0 : \end{aligned} \tag{19.15}$$

which has the initial tableau below:

x^T	x_{n+1}	f	RHS
A	b		b
	1	-1	

Take the x_{n+1} column as the first basic column. Assuming $b_r \neq 0$, take the r th row as the first basic row, i.e.,

$$B = \{n+1\}, \quad R = \{r\}.$$

Thus,

$$B = \{n+1\}, \quad N = A, \quad R = \{r\}, \quad R' = \{1, \dots, m\} \setminus R.$$

Carry out elementary transformations to turn the r -th component of the x_{n+1} column to 1, and eliminate the other nonzeros in the column. By row and column exchanges, the resulting deficient-basis tableau is of the form

x_B^T	x_N^T	f	RHS
1	$\tilde{N}_R$		1
	$\tilde{N}_{R'}$		
	$-\tilde{N}_R$	-1	-1

(19.16)

Iteration 1:

Multiply row 1 by $-1/15$, then add 11 ; -2 ; -1 times of row 2 to rows 2, 3, 7, respectively:

The preceding auxiliary tableau is a feasible deficient-basis tableau:

$$s = 1, \quad B = \{9\}, \quad R = \{1\}, \quad N = \{1, 2, 3, 4, 5, 6, 7, 8\}, \quad R' = \{2, 3, 4, 5\}.$$

Phase-I: call Algorithm 19.2.1.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS(x_9)
1/3	-1/15	-2/15	-1/15	-1/15		1/15	1/5	1
11/3*	-56/15	-7/15	34/15	-26/15	-2	41/15	16/5	
-2/3	2/15	-41/15	77/15	-43/15	1	-47/15	8/5	
		1		1			-1	
		-1		-1			2	
-3	2	-1		3		2	-6	
-1/3	1/15	2/15	1/15	1/15		1/15	-1/5	

Iteration 2:

1. $\min\{-1/3, 1/15, 2/15, 1/15, 1/15, 1/15, -1/5\} = -1/3 < 0$, $q = 1$.
3. $\bar{a}_q(R') = (11/3, -2/3, 0, 0)^T \neq 0$.
4. $\max\{|11/3|, |-2/3|\} = 11/3$, $i_s = 2$.
5. Multiply row 2 by $3/11$, then add $-1/3$; $2/3$; 3 ; $1/3$ times of row 2 to rows 1, 3, 6, 7, respectively.
6. $k = 2$, $B = \{9, 1\}$, $R = \{1, 2\}$, $N = \{2, 3, 4, 5, 6, 7, 8\}$, $R' = \{3, 4, 5\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS(x_9)
	3/11	-1/11	-3/11	1/11	2/11	-2/11	-1/11	1
1	-56/55	-7/55	34/55	-26/55	-6/11	41/55	48/55	
	-6/11*	-31/11	61/11	-35/11	7/11	-29/11	24/11	
		1		1			-1	
		-1		-1			2	
	-58/55	-76/55	102/55	87/55	-18/11	233/55	-186/55	
	-3/11	1/11	3/11	-1/11	-2/11	2/11	1/11	

Iteration 3:

1. $\min\{-3/11, 1/11, 3/11, -1/11, -2/11, 2/11, 1/11\} = -3/11 < 0$, $q = 2$.
3. $\bar{a}_q(R') = (-6/11, 0, 0)^T \neq 0$.
4. $\max\{|-6/11|, 0, 0\} = 6/11$, $i_s = 3$.

5. Multiply row 3 by $-11/6$, then add $-3/11$; $56/55$; $58/55$; $3/11$ times of row 3 to rows 1, 2, 6, 7, respectively.
6. $k = 3$, $B = \{9, 1, 2\}$, $R = \{1, 2, 3\}$, $N = \{3, 4, 5, 6, 7, 8\}$, $R' = \{4, 5\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS(x_9)
		$-3/2$	$5/2^*$	$-3/2$	$1/2$	$-3/2$	1	1
1		$77/15$	$-146/15$	$82/15$	$-26/15$	$17/3$	$-16/5$	
	1	$31/6$	$-61/6$	$35/6$	$-7/6$	$29/6$	-4	
		1		1			-1	
		-1		-1			2	
		$61/15$	$-133/15$	$116/15$	$-43/15$	$28/3$	$-38/5$	
		$3/2$	$-5/2$	$3/2$	$-1/2$	$3/2$	-1	

Iteration 4:

1. $\min\{3/2, -5/2, 3/2, -1/2, 3/2, -1\} = -5/2 < 0$, $q = 4$.
3. $\bar{a}_q(R') = (0, 0)^T$.
7. $\bar{a}_q(R) \not\leq 0$.
8. $\alpha = \min\{1/(5/2)\} = 2/5$, $s = 1$, $i_s = 1$. x_9 leaves the basis.
9. Multiply row 1 by $2/5$, then add $146/15$; $61/6$; $133/15$; $5/2$ times of row 1 to rows 2, 3, 6, 7, respectively.
10. $B = \{4, 1, 2\}$, $N = \{3, 5, 6, 7, 8, 9\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS(x_9)
		$-3/5$	1	$-3/5$	$1/5$	$-3/5$	$2/5$	$2/5$
1		$-53/75$		$-28/75$	$16/75$	$-13/75$	$52/75$	$292/75$
	1	$-14/15$		$-4/15$	$13/15$	$-19/15$	$1/15$	$61/15$
		1		1			-1	
		-1		-1			2^*	
		$-94/75$		$181/75$	$-82/75$	$301/75$	$-304/75$	$266/75$
								1

As the artificial x_9 column has already become nonbasic, the preceding tableau is feasible. Delete the x_9 column and the bottom row, then turn to Phase-II.

Phase-II: call Algorithm 19.2.1.

Iteration 5:

1. $\min\{-94/75, 181/75, -82/75, 301/75, -304/75\} = -304/75 < 0$, $q = 8$.
3. $\bar{a}_q(R') = (-1, 2)^T \neq 0$.
4. $\max\{|-1|, |2|\} = 2$, $i_s = 5$.
5. Multiply row 5 by $1/2$, then add $-2/5$, $-52/75$, $-1/15$, 1 , $304/75$ times of row 5 to rows 1, 2, 3, 4, 6, respectively.

6. $k = 4$, $B = \{4, 1, 2, 8\}$, $R = \{1, 2, 3, 5\}$, $N = \{3, 5, 6, 7\}$, $R' = \{4\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
		$-2/5$	1	$-2/5$	$1/5$	$-3/5$		$2/5$
1		$-9/25$		$-2/75$	$16/75$	$-13/75$		$292/75$
	1	$-9/10$		$-7/30$	$13/15$	$-19/15$		$61/15$
		$1/2^*$		$1/2$				
		$-1/2$		$-1/2$			1	
		$-82/25$		$29/75$	$-82/75$	$301/75$		$266/75$

Iteration 6:

1. $\min\{-82/25, 29/75, -82/75, 301/75\} = -82/25 < 0$, $q = 3$.
3. $\bar{a}_q(R') \neq 0$.
4. $i_s = 4$.
5. Multiply row 4 by 2, then add $2/5, 9/25, 9/10, 1/2, 82/25$ times of row 4 to rows 1, 2, 3, 5, 6, respectively.
6. $k = 5$, $B = \{4, 1, 2, 8, 3\}$, $R = \{1, 2, 3, 5, 4\}$, $N = \{5, 6, 7\}$, $R' = \emptyset$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
			1		$1/5^*$	$-3/5$		$2/5$
1				$1/3$	$16/75$	$-13/75$		$292/75$
	1			$2/3$	$13/15$	$-19/15$		$61/15$
		1		1				
							1	
				$11/3$	$-82/75$	$301/75$		$266/75$

Iteration 7:

1. $\min\{11/3, -82/75, 301/75\} = -82/75 < 0$, $q = 6$.
3. $R' = \emptyset$.
7. $\bar{a}_q(R) \not\leq 0$.
8. $\alpha = \min\{1/(20/87)\} = 87/20$, $s = 1$, $i_s = 1$.
9. Multiply row 1 by 5, then add $-16/75, -13/15, 82/75$ times of row 1 to rows 2, 3, 6, respectively.
10. $B = \{6, 1, 2, 8, 3\}$, $N = \{4, 5, 7\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
			5		1	-3		2
1			-16/15	1/3		7/15		52/15
	1		-13/3	2/3		4/3		7/3
		1		1				
							1	
			82/15	11/3		11/15		86/15

Iteration 8:

1. $\min\{82/15, 11/3, 11/15\} \geq 0$.
2. The optimal basic solution and optimal value:

$$x^* = (52/15, 7/3, 0, 0, 0, 2, 0, 0)^T, \quad f^* = -86/15.$$

19.5 Deficient-Basis Reduced Algorithm

In this section, we incorporate the deficient basis into the reduced simplex algorithm to cope with large-scale and highly degenerate problems.

Consider the reduced problem (16.1), i.e.,

$$\begin{aligned} & \min f, \\ & \text{s.t. } [A \vdots -e_{m+1}] \begin{pmatrix} x \\ f \end{pmatrix} = b, \quad x \geq 0. \end{aligned} \quad (19.17)$$

Assume that B, R are basic columns and rows, denoted by (19.1) (The row index $m+1$ is the r th element of R); and N, R' are nonbasic columns and rows, denoted by (19.2). Its initial tableau is then

$$\begin{array}{cc|c|c} x_B & x_N & f & \text{RHS} \\ \hline B_R & N_R & -e_r & b_R \\ B_{R'} & N_{R'} & & b_{R'} \end{array} \quad (19.18)$$

where B_R is the deficient basis with the row index $r \in \{1, \dots, k\}$, and $-e_r \in \mathcal{R}^k$ denotes the basic row part of $a_{n+1} = -e_{m+1}$. Note that the f column is put alone as a special nonbasic column.

Let $\bar{f}$ be given. Setting $f = \bar{f}$ and $\bar{x}_N = 0$, it can be determined from the first k rows of (19.18) that

$$\bar{x}_B = B_R^{-1} b_R + \bar{f} B_R^{-1} e_r, \quad (19.19)$$

which determines the solution $\bar{x}$ to $Ax = b$ with the objective value $\bar{f}$. Then, if

$$B_{R'} B_R^{-1} e_r = 0, \quad b_{R'} - B_{R'} B_R^{-1} b_R = 0,$$

the tableau (19.18) can be transformed into the so-called *deficient-basis reduced tableau* below:

x_B	x_N	f	Solution
I	$\bar{N}_R$	$\bar{w}$	$\bar{x}_B$
	$\bar{N}_{R'}$		

(19.20)

If $\bar{x}_B \geq 0$, then (19.20) is a feasible tableau, and hence ready to carry out a rank-increasing or rank-remaining iteration (see Sect. 19.2). Note that the rank-increasing iteration is independent of both the f column and the $\bar{x}_B$ column.

Based on preceding discussions, integrating steps of Algorithms 16.1.1 and 19.2.1 gives the following algorithm.

Algorithm 19.5.1 (Deficient-Basis Reduced Algorithm: Tableau Form) *Initial: feasible deficient-basis reduced tableau of form (19.20) with B , R , N , R' , $1 \leq k \leq m + 1$. $\bar{x}_B$, $\bar{f}$. This algorithm solves the reduced problem (19.17).*

1. Stop if $\bar{w} \geq 0$ (unbounded problem).
2. Determine α and s such that $\alpha = -\bar{x}_{j_s}/\bar{w}_s = \min\{-\bar{x}_{j_t}/\bar{w}_t \mid \bar{w}_t < 0, t = 1, \dots, k\}$.
3. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha \bar{w}$, $\bar{f} = \bar{f} - \alpha$.
4. Determine column index $q \in \arg \min_{j \in N} \bar{a}_{i_s, j}$.
5. Stop if $\bar{a}_{i_s, q} \geq 0$ (optimality achieved).
6. Go to step 10 if $R' = \emptyset$ or $\bar{a}_q(R') = 0$.
7. Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{p, q}|$.
8. Convert $\bar{a}_{p, q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
9. Set $k = k + 1$, bring q from N to B and p from R' to R , and go to step 4.
10. Convert $\bar{a}_{i_s, q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
11. Update (B, N) by exchanging j_s and q .
12. Go to step 1.

Note $\bar{x}_B$ column is not changed in steps 8 and 10.

Example 19.5.1 Solve the following problem by Algorithm 19.5.1:

$$\begin{aligned}
 \min \quad & f = 3x_2 + 2x_4 + x_5 - 4x_6 - 2x_7, \\
 \text{s.t.} \quad & x_1 - 2x_2 \quad \quad + 3x_4 + x_5 - x_6 - 2x_7 = 4, \\
 & \quad \quad 4x_2 + x_3 - 5x_4 \quad \quad + 2x_6 + 3x_7 = 12, \\
 & \quad \quad 6x_2 \quad \quad \quad - 2x_5 \quad \quad \quad = 0, \\
 & \quad -5x_2 \quad \quad \quad \quad \quad - x_7 = 0, \\
 & \quad \quad x_2 \quad \quad \quad - 3x_5 \quad \quad + 4x_7 = 0, \\
 & x_j \geq 0, \quad j = 1, \dots, 7.
 \end{aligned}$$

Answer Initial: $k = 2$, $B = \{1, 3\}$, $R = \{1, 2\}$, and the feasible deficient-basis tableau is

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	RHS
1	-2		3	1	-1	-2		4
	4	1	-5		2	-3		12
	6			-2				
	-5					-1		
	1			-3		4		
	3		2	1	-4*	-2	-1	

$\bar{z}_6 = -4 < 0$, $\bar{a}_6(R') = (0, 0, 0)^T$, Take $q = 6$. Multiply row 6 by $-1/4$, then add 1, -2 times of row 6 to rows 1, 2, respectively, obtaining the feasible deficient-basis reduced tableau below:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{x}_B$
1	-11/4		5/2	3/4		-3/2	1/4	4
	11/2	1	-4*	1/2		-4	-1/2	12
	6			-1/2				
	-5					-1		
	1			-3		4		
	-3/4		-1/2	-1/4	1	1/2	1/4	

Call Algorithm 19.5.1: $k = 3$, $B = \{1, 3, 6\}$, $R = \{1, 2, 6\}$. $N = \{2, 4, 5, 7\}$, $R' = \{3, 4, 5\}$.

Iteration 1: $\bar{x}_B = (4, 12, 0)^T$, $\bar{f} = 0$

1. $\bar{w} = (14, -1/2, 1/4)^T \not\geq 0$.
2. $\alpha = \min\{-(12)/(-1/2)\} = 24$, $s = 2$.
3. Add 24 time of f column to $\bar{x}_B$ column; $\bar{f} = -24$.
4. $\min\{11/2, -4, 1/2, -4\} = -4 < 0$, $q = 4$.
6. $\bar{a}_q(R') = 0$.

10. Multiply row 2 by $-1/4$, then add $-5/2, 1/2$ times of row 2 to rows 1,6, respectively.
11. $B = \{1, 4, 6\}$, $R = \{1, 2, 6\}$, $N = \{2, 3, 5, 7\}$, $R' = \{3, 4, 5\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{x}_B$
1	11/16	5/8		17/16		-4	-1/16	10
	-11/8	-1/4	1	-1/8		1	1/8	
	6			-1/2				
	-5					-1		
	1			-3		4*		
	-23/16	-1/8		-5/16	1	1	5/16	6

Iteration 2:

- $\bar{w} = (-1/16, 1/8, 5/16)^T \not\geq 0$.
- $\alpha = \min\{-(10)/(-1/16)\} = 160$, $s, i_s, j_s = 1$.
- Add 160 times of f column to $\bar{x}_B$ column; $\bar{f} = -24 - 160 = -184$.
- $\min\{11/16, 5/8, 17/16, -4\} = -4 < 0$, $q = 7$.
- $\bar{a}_q(R') = (0, -1, 4)^T \neq 0$.
- $\max\{0, |-1|, |4|\} = 4$, $p = 5$.
- Multiply row 5 by $1/4$, then add $4, -1, 1, -1$ times of row 5 to rows 1, 2, 4, 6, respectively.
- $k = 4$, $B = \{1, 6, 4, 7\}$, $R = \{1, 2, 6, 5\}$, $N = \{2, 3, 5\}$, $R' = \{3, 4\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{x}_B$
1	27/16	5/8		-31/16			-1/16	
	-13/8	-1/4	1	5/8			1/8	20
	6			-1/2*				
	-19/4			-3/4				
	1/4			-3/4		1		
	-27/16	-1/8		7/16	1		5/16	56

Iteration 3:

- $\min\{27/16, 5/8, -31/16\} = -31/16 < 0$, $q = 5$.
- $\bar{a}_q(R') = (-2, 3/4)^T \neq 0$.
- $\max\{|-1/2|, |-3/4|\} = 3/4$, $p = 4$.
- Multiply row 4 by $-4/3$, then add $31/16, -5/8, 1/2, 3/4, -7/16$ times of row 4 to rows 1,2,3,5,6, respectively.
- $k = 5$, $B = \{1, 6, 4, 7, 5\}$, $R = \{1, 2, 4, 5, 6\}$, $N = \{2, 3\}$, $R' = \{3\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	$\bar{x}_B$
1	335/24	5/8					-1/16	
	-67/12	-1/4	1				1/8	20
	55/6							
	19/3			1				
	5					1		
	-107/24	-1/8			1		5/16	56

Iteration 4:

4. $\min\{335/24, 5/8\} \geq 0$.

5. The optimal basic solution and optimal value:

$$x^* = (0, 0, 0, 20, 0, 56, 0)^T, \quad f^* = -184.$$

It is obvious that the revised version of Algorithm 19.5.1 has features of both deficient-basis and reduced simplex methods. Assume that B_R is the deficient basis, then the revised deficient-basis reduced tableau is

x_B	x_N	f	Solution
I	$B_R^{-1}N_R$	$-B_R^{-1}e_r$	$\bar{x}_B$
	$N_{R'} - B_{R'}B_R^{-1}N_R$		

(19.21)

Based on equivalence between (19.20) and (19.21), it is straightforward to obtain:

Algorithm 19.5.2 (Deficient-Basis Reduced Algorithm) *Initial:* $1 \leq k \leq m + 1$, B , R , B_R^{-1} , $\bar{f}$; $\bar{w} = -B_R^{-1}e_r$, $\bar{x}_B = B_R^{-1}b_R - \bar{f}\bar{w} \geq 0$. This algorithm solves the reduced problem.

1. Stop if $\bar{w} \geq 0$ (unbounded problem).
2. Determine α and s such that $\alpha = -\bar{x}_{j_s}/\bar{w}_s = \min\{-\bar{x}_{j_t}/\bar{w}_t \mid \bar{w}_t < 0, t = 1, \dots, k\}$.
3. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha\bar{w}$, $\bar{f} = \bar{f} - \alpha$.
4. Compute $\sigma_N = N_R^T B_R^{-T} e_s$.
5. Determine column index $q \in \arg \min_{j \in N} \sigma_j$.
6. Stop if $\sigma_q \geq 0$ (optimality achieved).
7. Compute $\bar{a}_q(R) = B_R^{-1}a_q(R)$.
8. Go to step 13 if $R' = \emptyset$ or $\bar{a}_q(R') = a_q(R') - B_{R'}\bar{a}_q(R) = 0$.
9. Determine $p \in \arg \max_{i \in R'} |\bar{a}_{i,q}|$.
10. Update B_R^{-1} by (19.10).
11. Set $k = k + 1$, $\bar{x}_q = 0$, $\bar{a}_{k,n+1} = 0$, and bring q from N to B and p from R' to R .
12. Go to step 4.
13. Compute $\bar{\alpha} = \bar{a}_{i_s, n+1}/\bar{a}_{i_s, q}$.

14. If $\alpha \neq 0$, set $\bar{w}_s = \bar{\alpha}$, and update $\bar{w} = \bar{w} - \bar{\alpha}\bar{a}_q(R)$.

15. Update B_R^{-1} by (19.11).

16. Update (B, N) by exchanging j_s and q .

17. Go to step 1.

Note Phase-I procedure is usually needed to get the preceding algorithms started. To this end, construct an auxiliary problem by introducing artificial variable f with added column b in the coefficient matrix; then starting from $(x = 0, f = 1)$, minimize objective f to zero.

Example 19.5.2 Solve the following problem by the preceding algorithm:

$$\begin{aligned} \min f &= -3x_1 + x_4 + 4x_5 - 2x_6 + 2x_7, \\ \text{s.t. } &-3x_1 + x_3 - x_4 - 2x_5 + 4x_6 - 2x_7 = 9, \\ &\quad +x_2 + 4x_4 + 3x_5 - 3x_6 - 5x_7 = 2, \\ &6x_1 - 6x_4 - 2x_5 = 0, \\ &\quad +4x_7 = 0, \\ &\quad x_1 - x_4 - x_7 = 0, \\ &x_j \geq 0, \quad j = 1, \dots, 7. \end{aligned}$$

Answer The reduced problem is associated with the following tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	f	RHS
-3		1	-1	-2	4	-2		9
	1		4	3	-3	-5		2
6			-6	-2				
						4		
1			-1			-1		
-3			1	4	-2	2	-1	

Initial: The preceding is a feasible deficient-basis tableau. A feasible reduced (deficient) basis is yielded by transforming the bottom row to a basic row and x_6 column to a basic column, i.e., $k = 3$, $B = \{3, 2, 6\}$, $R = \{1, 2, 6\}$, $N = \{1, 4, 5, 7\}$, $R' = \{3, 4, 5\}$.

$$B_R = \begin{pmatrix} 1 & 4 \\ 1 & -3 \\ -2 \end{pmatrix}, \quad B_R^{-1} = \begin{pmatrix} 1 & 2 \\ 1 & -3/2 \\ -1/2 \end{pmatrix},$$

$$\bar{a}_{n+1} = -B_R^{-1}e_3 = \begin{pmatrix} -2 \\ 3/2 \\ 1/2 \end{pmatrix}. \quad \bar{f} = 0, \quad \bar{x}_B = B_R^{-1}b_R = \begin{pmatrix} 9 \\ 2 \\ 0 \end{pmatrix} \geq 0.$$

Iteration 1:

1. $\bar{w} = (-2, 3/2, 1/2)^T \not\geq 0$.
2. $\alpha = \min\{-9/(-2)\} = 9/2, s = 1$.
3. $\bar{x}_B = (9, 2, 0)^T + (9/2)(-2, 3/2, 1/2)^T = (0, 35/4, 9/4)^T, \bar{f} = -9/2$.
4. $\sigma_N = \begin{pmatrix} -3 & -1 & -2 & -2 \\ & 4 & 3 & -5 \\ -3 & 1 & 4 & 2 \end{pmatrix}^T \begin{pmatrix} 1 & 2 \\ & 1 & -3/2 \\ & & -1/2 \end{pmatrix}^T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -9 \\ 1 \\ 6 \\ 2 \end{pmatrix}$.
5. $\min\{-9, 1, 6, 2\} = -9 < 0, q = 1$.
7. $\bar{a}_q(R) = \begin{pmatrix} 1 & 2 \\ & 1 & -3/2 \\ & & -1/2 \end{pmatrix} \begin{pmatrix} -3 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} -9 \\ 9/2 \\ 3/2 \end{pmatrix}$.
8. $\bar{a}_q(R') = \begin{pmatrix} 6 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 6 \\ -3 \\ -2 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \\ 1 \end{pmatrix} \neq 0$.
9. $\max\{|6|, 0, |1|\} = 6, p = 3$.
10. $\tau = (6 - (0, 0, 0)(-9, 9/2, 3/2)^T)^{-1} = 1/6$.
 $v = -(1/6)(-9, 9/2, 3/2)^T = (3/2, -3/4, -1/4)^T$.
 $d^T = -(1/6)(0, 0, 0) \begin{pmatrix} 1 & 2 \\ & 1 & -3/2 \\ & & -1/2 \end{pmatrix} = (0, 0, 0)$.
 $U = \begin{pmatrix} 1 & 2 \\ & 1 & -3/2 \\ & & -1/2 \end{pmatrix} - \begin{pmatrix} 3/2 \\ -3/4 \\ -1/4 \end{pmatrix} (0 \ 0 \ 0) = \begin{pmatrix} 1 & 2 \\ & 1 & -3/2 \\ & & -1/2 \end{pmatrix}$.
 $\widehat{B}_{\hat{R}}^{-1} = \begin{pmatrix} 1 & 2 & 3/2 \\ & 1 & -3/2 & -3/4 \\ & & -1/2 & -1/4 \\ & & & 1/6 \end{pmatrix}$.
11. $k = 4, B = \{3, 2, 6, 1\}, R = \{1, 2, 6, 3\}, N = \{4, 5, 7\},$
 $R' = \{4, 5\}, B_R^{-1} = \widehat{B}_{\hat{R}}^{-1}, \bar{x}_1 = 0, \bar{a}_{4, n+1} = 0$.

Iteration 2:

4. $\sigma_N = \begin{pmatrix} -1 & -2 & -2 \\ & 4 & 3 & -5 \\ & 1 & 4 & 2 \\ & -6 & -2 \end{pmatrix}^T \begin{pmatrix} 1 & 2 & 3/2 \\ & 1 & -3/2 & -3/4 \\ & & -1/2 & -1/4 \\ & & & 1/6 \end{pmatrix}^T \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -8 \\ 3 \\ 2 \end{pmatrix}$.
5. $\min\{-8, 3, 2\} = -8 < 0, q = 4$.
7. $\bar{a}_q(R) = \begin{pmatrix} 1 & 2 & 3/2 \\ & 1 & -3/2 & -3/4 \\ & & -1/2 & -1/4 \\ & & & 1/6 \end{pmatrix} \begin{pmatrix} -1 \\ 4 \\ 1 \\ -6 \end{pmatrix} = \begin{pmatrix} -8 \\ 7 \\ 1 \\ -1 \end{pmatrix}$.

$$8. \bar{a}_q(R') = \begin{pmatrix} 0 \\ -1 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -8 \\ 7 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

$$13. \bar{\alpha} = -2/-8 = 1/4.$$

$$14. \bar{w} = (-2, 3/2, 1/2, 0)^T - (1/4)(-8, 7, 1, -1)^T = (0, -1/4, 1/4, 1/4)^T, \\ \bar{w}_1 = 1/4.$$

$$15. \widehat{B}_{\hat{R}}^{-1} = \begin{pmatrix} -1/8 \\ 7/8 \\ 1/8 \\ -1/8 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3/2 \\ 1-3/2-3/4 \\ -1/2-1/4 \\ 1/6 \end{pmatrix} = \begin{pmatrix} -1/8-1/4-3/16 \\ 7/8 & 1/4 & 9/16 \\ 1/8-1/4-1/16 \\ -1/8-1/4-1/48 \end{pmatrix}.$$

$$16. B = \{4, 2, 6, 1\}, N = \{3, 5, 7\}, B_R^{-1} = \widehat{B}_{\hat{R}}^{-1}.$$

Iteration 3:

$$1. \bar{a}_{n+1} = (1/4, -1/4, 1/4, 1/4)^T \not\geq 0.$$

$$2. \alpha = \min\{-(35/4)/(-1/4)\} = 35, s = 2.$$

$$3. \bar{x}_B = (0, 35/4, 9/4, 0)^T + 35(1/4, -1/4, 1/4, 1/4)^T = (35/4, 0, 11, 35/4)^T, \\ \bar{f} = -9/2 - 35 = -79/2.$$

$$4. \sigma_N = \begin{pmatrix} 1 & -2 & -2 \\ 3 & -5 \\ 4 & 2 \\ -2 \end{pmatrix}^T \begin{pmatrix} -1/8 & -1/4 & -3/16 \\ 7/8 & 1 & 1/4 & 9/16 \\ 1/8 & -1/4 & -1/16 \\ -1/8 & -1/4 & -1/48 \end{pmatrix}^T \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 7/8 \\ 9/8 \\ -25/4 \end{pmatrix}.$$

$$5. \min\{7/8, 9/8, -25/4\} = -25/4 < 0, q = 7.$$

$$7. \bar{a}_q(R) = \begin{pmatrix} -1/8 & -1/4 & -3/16 \\ 7/8 & 1 & 1/4 & 9/16 \\ 1/8 & -1/4 & -1/16 \\ -1/8 & -1/4 & -1/48 \end{pmatrix}^T \begin{pmatrix} -2 \\ -5 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/4 \\ -25/4 \\ -3/4 \\ -1/4 \end{pmatrix}.$$

$$8. \bar{a}_q(R') = \begin{pmatrix} 4 \\ -1 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1/4 \\ -25/4 \\ -3/4 \\ -1/4 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}.$$

$$9. \max\{|4|, |-1|\} = 4, p = 4.$$

$$10. \tau = (4 - 0)^{-1} = 1/4,$$

$$v = -(1/4)(-1/4, -25/4, -3/4, -1/4)^T = (1/16, 25/16, 3/16, 1/16)^T.$$

$$d^T = -(1/4)(0, 0, 0, 0)B_R^{-1} = (0, 0, 0, 0).$$

$$U = B_R^{-1} - (1/16, 25/16, 3/16, 1/16)^T(0, 0, 0, 0) = B_R^{-1}.$$

$$\widehat{B}_{\hat{R}}^{-1} = \begin{pmatrix} -1/8 & -1/4 & -3/16 & 1/16 \\ 7/8 & 1 & 1/4 & 9/16 & 25/16 \\ 1/8 & -1/4 & -1/16 & 3/16 \\ -1/8 & -1/4 & -1/48 & 1/16 \\ & & & & 1/4 \end{pmatrix}.$$

11. $k = 5$, $B = \{4, 2, 6, 1, 7\}$, $R = \{1, 2, 6, 3, 4\}$, $N = \{3, 5\}$, $R' = \{5\}$.
 $B_R^{-1} = \hat{B}_{\hat{R}}^{-1}$, $\bar{x}_7 = 0$, $\bar{a}_{5,n+1} = 0$.

Iteration 4:

$$4. \sigma_N = \begin{pmatrix} 1 & -2 \\ 3 \\ 4 \\ -2 \end{pmatrix}^T \begin{pmatrix} -1/8 & -1/4 & -3/16 & 1/16 \\ 7/8 & 1 & 1/4 & 9/16 & 25/16 \\ 1/8 & -1/4 & -1/16 & 3/16 \\ -1/8 & -1/4 & -1/48 & 1/16 \\ & & & 1/4 \end{pmatrix}^T \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 7/8 \\ 9/8 \end{pmatrix}.$$

5. $\min\{7/8, 9/8\} \geq 0$.

6. The optimal basic solution and optimal value:

$$x^* = (35/4, 0, 0, 35/4, 0, 11, 0)^T, \quad f^* = -79/2.$$

Note In general, a Phase-I procedure is needed to get the preceding algorithms started. To this end, an auxiliary problem can be formed by introducing artificial variable f with added column b in the coefficient matrix. It is clear that $x = 0$, $f = 1$ is feasible to the problem. Thus, what we to do next is to minimize objective f to reach zero.

19.5.1 Phase-I: Most-Obtuse-Angle Rule

As was seen in Sect. 19.4.3, a feasible deficient-basis tableau can be created by solving the auxiliary program (19.15) using the deficient-basis algorithm. Since the program is itself in reduced form; it seem to be attractive to use the deficient-basis reduced algorithm instead.

To begin with, put the initial tableau, say (19.15), into the following form:

x	f	RHS
A	b	b

Take $\bar{x} = 0$, $\bar{f} = 1$ as the initial solution. Assume that the current deficient-basis tableau is

x_B	x_N	f	RHS
I	$\bar{N}_R$	$\bar{b}_R$	$\bar{b}_R$
	$\bar{N}_{R'}$		

where $B = \{j_1, \dots, j_k\}$, $R = \{i_1, \dots, i_k\}$. It represents the equalities

$$x_B + \bar{N}_R x_N = -\bar{b}_R f + \bar{b}_R, \quad \bar{N}_{R'} x_N = 0.$$

If the set

$$S = \{t \mid \bar{b}_{it} < 0, t = 1, \dots, k\}$$

is empty, then $\bar{f}$ can be decreased from 1 until reaching 0, giving a feasible solution to the original problem; i.e.,

$$\bar{x}_B = \bar{b}_R \geq 0, \quad \bar{x}_N = 0.$$

If S is nonempty, the deficient-basis reduced algorithm determines a row index

$$s \in \arg \min_{t \in S} -\bar{x}_{j_t} / \bar{b}_{it}.$$

In our case, there is usually a tie in selection of s , since $\bar{x}_{j_t} = 0, \forall t \in S$. As the associated stepsize vanishes, no solution updating should happen. For such a highly degenerate case, we use the following rule instead:

$$s \in \arg \min_{t \in S} \bar{b}_{it},$$

which is not only stable but also advantageous in the sense of the most-obtuse-angle heuristic.

Therefore, the preceding tableau is simplified to

x_B^T	x_N^T	RHS
I	$\bar{N}_R$	$\bar{b}_R$
	$\bar{N}_{R'}$	

(19.22)

The subsequent steps are similar to Algorithm 19.5.1, as is formulated as follows.

Algorithm 19.5.1 (Deficient-Basis Phase-I: The Most-Obtuse-Angle Rule) Initial: deficient-basis tableau of form (19.22) with $B, R, N, R', 1 \leq k \leq m$. This algorithm finds a feasible deficient-basis tableau.

The same as Algorithm 19.5.1 except for its steps 1–5 replaced by

1. Determine $s \in \arg \min\{\bar{b}_{it} \mid t = 1, \dots, k\}$.
2. Stop if $\bar{b}_{is} \geq 0$ (feasibility achieved).
3. Do nothing.
4. Determine $q \in \arg \min_{j \in N} \bar{a}_{is, j}$.
5. Stop if $\bar{a}_{is, q} \geq 0$ (infeasible problem).

To get the preceding Algorithm started, the procedure described in Sect. 19.4.1 can be used to gain an initial deficient basis. For matching the Algorithm, it seems to be a good choice to utilize the following.

Algorithm 19.5.2 (Producing of Initial Basis) Initial: $(\bar{A} = A \mid \bar{b} = b)$, $B = \emptyset$, $R = \emptyset$, $N = A$, $R' = \{1, \dots, m\}$, $k = 0$. This algorithm finds a deficient-basis tableau.

1. Determine $p \in \arg \max_{i \in R'} |\bar{b}_i|$.
2. Stop if $|\bar{b}_p| = 0$ (deficient-basis attained).
3. Stop if $J = \{j \mid \bar{a}_{pj} \bar{b}_p > 0, j \in N\} = \emptyset$ (infeasible problem).
4. Determine $q \in \arg \max_{j \in J} |\bar{a}_{pj}|$.
5. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
6. Set $k = k + 1$, bring q from N to B and p from R' to R .
7. Go to step 1 if $R' \neq \emptyset$.
8. Stop (full basis attained).

Example 19.5.3 Solve the following problem :

$$\begin{aligned}
 \min \quad & f = 7x_1 + 2x_2 + 3x_3 - 2x_4 + 9x_7 - 2x_8 \\
 \text{s.t.} \quad & -5x_1 - x_2 - 5x_3 + 2x_4 + x_5 + x_6 - 3x_7 - 3x_8 = -9, \\
 & 2x_1 + 2x_2 - 2x_3 + x_4 + 3x_5 - x_6 + x_7 - 2x_8 = -9, \\
 & 5x_1 - 3x_2 \quad \quad - 3x_4 + 5x_5 - 3x_6 + 2x_7 + x_8 = 2, \\
 & \quad \quad 3x_2 \quad \quad + x_4 \quad \quad \quad + 3x_7 = 0, \\
 & \quad \quad \quad \quad \quad \quad \quad \quad -4x_7 = 0,
 \end{aligned}$$

Answer(1) Call Algorithm 19.5.2 to determine an initial basis. The equality constraints can be represented by the following tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
-5	-1	-5	2	1	1	-3	-3	-9
2	2	-2	1	3	-1	1	-2	-9
5	-3		-3	5	-3	2	1	2
	3		1			3		
						-4		

corresponding to $B = \emptyset$, $N = \{1, \dots, 8\}$, $R = \emptyset$, $R' = \{1, \dots, 5\}$, $k = 0$.

Iteration 1:

1. $\max \{|-9|, |-9|, |2|\} = 9$, $p = 1$.
4. $\max \{|-5|, |-1|, |-5|, |-3|, |-3|\} = 5$, $q = 1$.
5. Multiply row 1 by $-1/5$, then add $-2, -5$ times of row 1 to rows 2,3, respectively.
6. $k = 1$, $B = \{1\}$, $N = \{2, \dots, 8\}$, $R = \{1\}$, $R' = \{2, \dots, 5\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1	1/5	1	-2/5	-1/5	-1/5	3/5	3/5	9/5
	8/5	-4	9/5	17/5	-3/5	-1/5	-16/5	-63/5
	-4	-5	-1	6	-2	-1	-2	-7
	3		1			3		
						-4		

Iteration 2:

- 1. $\max \{|-63/5|, |-7|\} = 63/5, p = 2.$
- 4. $\max \{|-4|, |-3/5|, |-1/5|, |-16/5|\} = 4, q = 3.$
- 5. Multiply row 2 by $-1/4$, then add $-1, 5$ times of row 2 to rows 1 and 3, respectively.
- 6. $k = 2, B = \{1, 3\}, N = \{2, 4, \dots, 8\}, R = \{1, 2\}, R' = \{3, 4, 5\}.$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1	3/5		1/20	13/20	-7/20	11/20	-1/5	-27/20
	-2/5	1	-9/20	-17/20	3/20	1/20	4/5	63/20
	-6		-13/4	7/4	-5/4	-3/4	2	35/4
	3		1			3		
						-4		

Iteration 3:

- 1. $\max \{|35/4|\}, p = 3.$
- 4. $\max \{|7/4|, |2|\} = 2, q = 8.$
- 5. Multiply row 3 by $1/2$, then add $1/5, -4/5$ times of row 3 to rows 1 and 2, respectively.
- 6. $k = 3, B = \{1, 3, 8\}, N = \{2, 4, 5, 6, 7\}, R = \{1, 2, 3\}, R' = \{4, 5\}.$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1			-11/40	33/40	-19/40	19/40		-19/40
	2	1	17/20	-31/20	13/20	7/20		-7/20
	-3		-13/8	7/8	-5/8	-3/8	1	35/8
	3		1			3		
						-4		

A deficient basis attained

(2) Call Algorithm 19.5.1 to achieve feasibility.

Iteration 4:

1. $\min \{-19/40, -7/20, 35/8\} = -19/40, s = 1.$
3. $\min \{-11/40, -19/40\} = -19/40, q = 6.$
6. $\bar{a}_q(R') = 0.$
10. Multiply row 1 by $-40/19$, then add $-13/20, 5/8$ times of row 1 to rows 2,3, respectively.
11. $k = 3, B = \{6, 3, 8\}, N = \{1, 2, 4, 5, 7\}, R = \{1, 2, 3\}, R' = \{4, 5\}.$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
$-40/19$			$11/19$	$-33/19$	1	-1		1
$26/19$	2	1	$9/19$	$-8/19$		1		-1
$-25/19$	-3		$-24/19$	$-4/19$		-1	1	5
	3		1			3		
						-4		

Iteration 5:

1. $\min \{1, -1, 5\} = -1, s = 2.$
3. $\min \{-8/19\}, q = 5.$
6. $\bar{a}_q(R') = 0.$
10. Multiply row 2 by $-19/8$, then add $33/19, 4/19$ times of row 2 to rows 1 and 3, respectively.
11. $k = 3, B = \{6, 5, 8\}, N = \{1, 2, 3, 4, 7\}, R = \{1, 2, 3\}, R' = \{4, 5\},$

Feasibility is attained: $\bar{x}_B = (41/8, 19/8, 11/2)^T, \bar{x}_N = 0.$ Add the objective row at the bottom and $f = f$ column, and add 2 times row 3 to the bottom line to transform it to a feasible deficient-basis tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	RHS
$-31/4$	$-33/4$	$-33/8$	$-11/8$		1	$-41/8$			$41/8$
$-13/4$	$-19/4$	$-19/8$	$-9/8$	1		$-19/8$			$19/8$
-2	-4	$-1/2$	$-3/2$			$-3/2$	1		$11/2$
	3		1			3			
						-4			
3	-6	2	-5			6		-1	11

with solution

$$\bar{x} = (0, 0, 0, 0, 19/8, 41/8, 0, 11/2, 0)^T, \quad \bar{f} = -11. \quad (19.23)$$

(3) Convert the preceding to a reduced tableau.

Iteration 6:

$J = \{2, 4\}$, $\min\{-6, -5\} = -6$, $q = 2$, $\bar{a}_q(R') \neq 0$.

Multiply row 4 by $1/3$, then add $33/4$, $19/4$, 4 , 6 times of row 6 to rows 1, 2, 3, 6, respectively.

$k = 4$, $B = \{6, 5, 8, 2\}$, $N = \{1, 3, 4, 7\}$, $R = \{1, 2, 3, 4\}$, $R' = \{5\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	RHS
$-31/4$		$-33/8$	$11/8$		1	$25/8$			$41/8$
$-13/4$		$-19/8$	$11/24$	1		$19/8$			$19/8$
-2		$-1/2$	$-1/6$			$5/2$	1		$11/2$
0	1	0	$1/3$			1			
0		0				-4			
3		2	-3			12		-1	11

Iteration 7:

Put $\bar{x}_B$ column in place of RHS column (by zeroing the value 11 at the south-east corner).

$J = \{4\}$, $q = 4$, $\bar{a}_q R' = 0$.

Multiply row 6 by $-1/3$, then add $-11/8$, $-11/24$, $1/6$, $-1/3$ times of row 6 to rows 1, 2, 3, 4, respectively.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	f	$\bar{x}_B$
$-51/8$		$-77/24$			1	$69/8$		$-11/24$	$41/8$
$-67/24$		$-149/72$		1		$101/24$		$-11/72$	$19/8$
$-13/6$		$-11/18$				$11/6$	1	$1/18$	$11/2$
$1/3$	1	$2/9$				$7/3$		$-1/9$	
0		0				-4		0	
-1		$-2/3$	1			-4		$1/3$	

$k = 5$, $B = \{6, 5, 8, 2, 4\}$, $N = \{1, 3, 7\}$, $R = \{1, 2, 3, 4, 6\}$, $R' = \{5\}$.

(4) Call Algorithm 19.5.1:

Iteration 7:

1. $\bar{w} = (-11/24, -11/72, 1/18, -1/9, 1/3)^T \not\geq 0$.

2. $\alpha = 0$, $s = 4$.

4. $\min\{1/3, 2/9, 7/3\} \geq 0$.

5. Optimality achieved by solution (19.23).

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Chapter 20

Dual Deficient-Basis Method



This chapter attacks the standard LP problem from the dual side using the deficient basis. To achieve optimality, the method presented in the previous chapter seeks dual feasibility while maintaining primal feasibility. Seeking the primal feasibility while maintaining dual feasibility, in this chapter, we derive the dual deficient-basis algorithm in tableau form and its revised version. Combining the deficient-basis and D-reduced algorithms, we also develop the dual deficient-basis D-reduced algorithm in tableau form and its revised version and the deficient-basis D-reduced gradient algorithm in tableau form and its revised version.

20.1 Dual Deficient-Basis Algorithm: Tableau Form

Assume that the deficient-basis tableau (19.4) is dual feasible, satisfying $\bar{z}_j \geq 0$. If $\bar{b} \geq 0$, optimality is already achieved; else, determine row index i_s such that

$$i_s \in \arg \min_{i_t \in R} \bar{b}_{i_t} < 0. \quad (20.1)$$

This will let the basic infeasible variable x_{j_p} leave the basis, becoming a feasible variable.

Lemma 20.1.1 Assume that $\bar{z}_N \geq 0$ and $\bar{b}_{i_s} < 0$. If the index set

$$J = \{j \in N \mid \bar{a}_{i_s j} < 0\}$$

is empty, then problem is infeasible.

Proof It is seen that since $\bar{b}_{i_s} < 0$, the i_s th row of the tableau actually gives an upper-hill direction, with respect to the dual objective function. If the entries of this row are all nonnegative, then the stepsize taken along this direction is allowed

arbitrarily great while maintaining dual feasibility; that is to say that the dual problem is unbounded, and hence there is no feasible solution to the standard LP problem. $\square$

Now assume that $J \neq \emptyset$. The maximum possible stepsize that can be taken is

$$\beta = -\bar{z}_q / \bar{a}_{i_s q} = \min\{-\bar{z}_j / \bar{a}_{i_s j} \mid \bar{a}_{i_s j} < 0, j \in J\} \geq 0, \quad (20.2)$$

where column index q corresponds to a dual constraint, broking increasing of the dual objective value. Adding $\beta \geq 0$ times of the i_s th row to the bottom (objective) row updates the dual solution. Consequently, $\bar{z}_q$ vanishes, and the southeast entry of the tableau becomes

$$-\hat{f} = -\bar{f} + \beta \bar{b}_{i_s} \leq -\bar{f},$$

which means that the objective value $\hat{f}$ does not decrease, and it strictly increases under the assumption of dual nondegeneracy ($\beta > 0$).

There will be the following two cases to be handled differently:

- (i) Rank-remaining iteration: $R' = \emptyset$ or $\bar{a}_q(R') = 0$.

Convert $\bar{a}_{i_s, q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations. Consequently, the j_s -indexed basic column becomes nonbasic, and the q -indexed nonbasic column becomes basic, while the basic rows remain basic. So, the rank of the basis remains. This turns $\bar{b}_{i_s}$ to positive but does not touch nonbasic rows at all (including entries in the right-hand side).

- (ii) Rank-increasing iteration: $R' \neq \emptyset$ and $\bar{a}_q(R') \neq 0$.

Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{i q}|$. Convert $\bar{a}_{p, q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations. Then, while all basic columns and rows remain basic, the q -indexed nonbasic column and p -indexed nonbasic row become basic. Hence the rank of the basis increases by 1. Since the right-hand side remains unchanged, the same row index i_s will be selected in the next iteration, until encountering a rank-remaining iteration that makes $\bar{b}_{i_s}$ positive.

The associated steps can be summarized into the following algorithm.

Algorithm 20.1.1 (Dual Deficient-Basis Algorithm: Tableau Form) Initial: a dual feasible deficient-basis tableau of form (19.4) with $B, N, R, R', 1 \leq k \leq m$. This algorithm solves the standard LP problem.

1. Determine s such that $i_s \in \arg \min_{i \in R} \bar{b}_{i_t}$.
2. Stop if $\bar{b}_{i_s} \geq 0$.
3. Stop if $J = \{j \in N \mid \bar{a}_{i_s j} < 0\}$.
4. Determine column index q and stepsize β by (20.2).
5. If $\beta \neq 0$, add β times of the i_s th row to the bottom row.
6. Go to step 10 if $R' = \emptyset$ or $\bar{a}_q(R') = 0$.
7. Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{i q}|$.

8. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
9. Set $k = k + 1$, bring q from N to B and p from R' to R , and go to step 3.
10. Convert $\bar{a}_{isq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
11. Update (B, N) by exchanging j_s and q .
12. Go to step 1.

Theorem 20.1.1 *Under the dual nondegenerate assumption, Algorithm 20.1.1 terminates either at*

- (i) *Step 2, reaching a basic optimal solution, or at*
- (ii) *Step 3, detecting the infeasibility of the problem*

Proof The proof of finiteness of the algorithm is similar to that in the conventional simplex context. The meanings of its exits come from Lemmas 19.1.1 and 20.1.1, as well as discussions preceding the algorithm. $\square$

Example 20.1.1 Solve the following problem by Algorithm 20.1.1:

$$\begin{aligned}
 \min \quad & f = 4x_3 + x_4 + 3x_5 + x_6 + 3x_7 + 2x_8, \\
 \text{s.t.} \quad & x_1 + 2x_3 - x_4 + x_5 - x_6 - 3x_7 - 3x_8 = -10, \\
 & +x_2 + x_3 + 2x_4 - x_5 + x_6 - 2x_7 + x_8 = -7, \\
 & -3x_3 - x_4 - 3x_5 - x_6 + 2x_8 = 0, \\
 & +x_3 + x_4 + x_5 + 3x_6 - x_8 = 0, \\
 & -x_3 + x_4 - x_5 - 2x_6 + 2x_8 = 0, \\
 & x_j \geq 0, \quad j = 1, \dots, 8.
 \end{aligned}$$

Answer $s = 2$, $B = \{1, 2\}$, $R = \{1, 2\}$. $N = \{3, 4, 5, 6, 7, 8\}$, $R' = \{3, 4, 5\}$. The initial feasible dual deficient-basis tableau is

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1		2	-1	1	-1	-3	-3	-10
	1	1	2	-1	1	-2	1	-7
		-3	-1	-3	-1		2*	
		1	1	1	3		-1	
		-1	1	-1	-2		2	
		4	1	3	1	3	2	

Iteration 1:

1. $\min\{-10, -7\} = -10 < 0$, $s = 1$, $i_s = 1$.
3. $J = \{4, 6, 7, 8\} \neq \emptyset$.
4. $\beta = \min\{-1/(-1), -1/(-1), -3/(-3), -2/(-3)\} = 2/3$, $q = 8$.
5. Add $2/3$ times of the first row to the bottom row.

6. $\bar{a}_q(R') = (2, -1, 2)^T \neq 0$.
7. $\max\{|2|, |-1|, |2|\} = 2$, $p = 3$.
8. Multiply row 3 by $1/2$, and then add 3 , -1 , 1 , and -2 times of row 3 to rows 1, 2, 4, and 5, respectively.
9. $k = 3$, $B = \{1, 2, 8\}$, $r = \{1, 2, 3\}$, $N = \{3, 4, 5, 6, 7\}$, $R' = \{4, 5\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1		$-5/2$	$-5/2$	$-7/2$	$-5/2$	-3		-10
	1	$5/2$	$5/2$	$1/2$	$3/2$	-2		-7
		$-3/2$	$-1/2$	$-3/2$	$-1/2$		1	
		$-1/2$	$1/2$	$-1/2$	$5/2$			
		2	2^*	2	-1			
$2/3$		$16/3$	$1/3$	$11/3$	$1/3$	1		$-20/3$

Iteration 2:

3. $J = \{3, 4, 5, 6, 7\} \neq \emptyset$.
4. $\beta = \min\{-(16/3)/(-5/2), -(1/3)/(-5/2), -(11/3)/(-7/2), -(1/3)/(-5/2), -1/-3\} = 2/15$, $q = 4$.
5. Add $2/15$ times of the first row to the bottom row.
6. $\bar{a}_q(R') = (-1/2, 2)^T \neq 0$.
7. $\max\{|1/2|, |2|\} = 2$, $p = 5$.
8. Multiply row 5 by $1/2$, and then add $5/2$, $-5/2$, $1/2$, and $-1/2$ times of row 5 to rows 1, 2, 3, and 4, respectively.
9. $k = 4$, $B = \{1, 2, 8, 4\}$, $R = \{1, 2, 3, 5\}$, $N = \{3, 5, 6, 7\}$, $R' = \{4\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1				-1	$-15/4$	-3		-10
	1			-2	$11/4$	-2		-7
		-1		-1	$-3/4$		1	
		-1		-1	$11/4^*$			
		1	1	1	$-1/2$			
$4/5$		5		$16/5$		$3/5$		-8

Iteration 3:

3. $J = \{5, 6, 7\} \neq \emptyset$.
4. $\beta = \min\{-(16/5)/(-5/2), 0/(-15/4), -(3/5)/(-3)\} = 0$, $q = 6$.
6. $\bar{a}_q(R') = (11/4) \neq 0$.
7. $\max\{|11/4|\} = 11/4$, $p = 4$.
8. Multiply row 4 by $4/11$, and then add $15/4$, $-11/4$, $3/4$, and $1/2$ times of row 4 to rows 1, 2, 3, and 5 respectively.

9. $k = 5$, $B = \{1, 2, 8, 4, 6\}$, $R = \{1, 2, 3, 5, 4\}$, $N = \{3, 5, 7\}$, $R' = \emptyset$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1		-15/11		-26/11		-3*		-10
	1	1		-1		-2		-7
		-14/11		-14/11			1	
		-4/11		-4/11	1			
		9/11	1	9/11				
4/5		5		16/5		3/5		-8

Iteration 4:

3. $J = \{3, 5, 7\} \neq \emptyset$.
4. $\beta = \min\{-5/(-15/11), -(16/5)/(-26/11), -(3/5)/-3\} = 1/5$, $q = 7$.
5. Add $1/5$ times of the first row to the bottom row.
6. $R' = \emptyset$.
10. Multiply row 1 by $-1/3$, and then add 2 and $-3/5$ times of row 1 to rows 2 and 6, respectively.
11. $B = \{7, 2, 8, 4, 6\}$, $N = \{1, 3, 5\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
-1/3		5/11		26/33		1		10/3
-2/3*	1	21/11		19/33				-1/3
		-14/11		-14/11			1	
		-4/11		-4/11	1			
		9/11	1	9/11				
1		52/11		30/11				-10

Iteration 5:

1. $\min\{10/3, -1/3\} = -1/3 < 0$, $s = 2$, $i_s = 2$.
3. $J = \{1\} \neq \emptyset$.
4. $\beta = \min\{-1/(-2/3)\} = 3/2$, $q = 1$.
5. Add $3/2$ times of the second row to the bottom row.
6. $R' = \emptyset$.
10. Multiply row 2 by $-3/2$, and then add $1/3$ and -1 times of row 2 to rows 1 and 6, respectively.
11. $B = \{7, 1, 8, 4, 6\}$, $N = \{2, 3, 5\}$.

Iteration 6:

1. $\min\{7/2, 1/2\} \geq 0$.
2. The optimal basic solution and optimal value are

$$x^* = (1/2, 0, 0, 0, 0, 0, 7/2, 0)^T, \quad f^* = 21/2.$$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	-1/2	-1/2		1/2		1		7/2
1	-3/2	-63/22		-19/22				1/2
		-14/11		-14/11			1	
		-4/11		-4/11	1			
		9/11	1	9/11				
	3/2	167/22		79/22				-21/2

20.2 Dual Deficient-Basis Algorithm

Since all quantities involved in Algorithm 20.1.1 can be found in Table 19.1, it is straightforward to obtain its revised version.

Algorithm 20.2.1 [Dual Deficient-Basis Algorithm] Initial: B , N , R , R' , $1 \leq k \leq m$. B_R^{-1} ; $\bar{z}_N = c_N - N_R^T B_R^{-T} c_B \geq 0$; $\bar{z}_B = 0$; $\bar{x}_B = B_R^{-1} b_R$. This algorithm solves the standard LP problem.

1. Determine s such that $\bar{x}_{j_s} = \min\{\bar{x}_{j_t} \mid t = 1, \dots, k\}$.
2. Stop with $\bar{f} = c_B^T \bar{x}_B$ if $\bar{x}_{j_s} \geq 0$ (optimality achieved).
3. Compute $\sigma_N = N_R^T B_R^{-T} e_s$.
4. Stop if $J = \{j \in N \mid \sigma_j < 0\} = \emptyset$ (infeasible problem).
5. Determine β and column index q such that $\beta = -\bar{z}_q / \sigma_q = \min_{j \in J} -\bar{z}_j / \sigma_j$.
6. If $\beta \neq 0$, update $\bar{z}_N = \bar{z}_N + \beta \sigma_N$, $\bar{z}_{j_s} = \bar{z}_{j_s} + \beta$.
7. Compute $\bar{a}_q(R) = B_R^{-1} a_q(R)$.
8. Go to step 12 if $R' = \emptyset$ or $\bar{a}_q(R') = a_q(R') - B_{R'} \bar{a}_q(R) = 0$.
9. Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{i,q}|$.
10. Update B_R^{-1} by (19.10).
11. Update $k = k + 1$, $\bar{x}_q = 0$, bring q from N to B and p from R' to R , and go to step 3.
12. Compute $\alpha = \bar{x}_{j_s} / \bar{a}_{i_s, q}$.
13. If $\alpha \neq 0$, set $\bar{x}_q = \alpha$, and update $\bar{x}_B = \bar{x}_B - \alpha \bar{a}_q(R)$.
14. Update B_R^{-1} by (19.11).
15. Update (B, N) by exchanging j_s and q .
16. Go to step 1.

The preceding algorithm needs a dual feasible tableau (solution) to get itself started. The (conventional) dual simplex Phase-I methods (Chap. 15) are applicable to this purpose, almost without any modification.

Originally, the deficient basis is not defined as a square but a rectangular matrix, whose column space includes the right-hand side (Pan 2005). As the associated algorithm still reflects the behavior of Algorithm 20.2.1, we quote the computational results in the article as follows:

The computational experiments were carried out on a Pentium III 550E PC with Windows 98 operating system, 256MB inner storage, and about 16 decimal precision. Visual Fortran 5.0 compiler was used.

Table 20.1

Problem	Iteration counts	Time	% Degen
Small (20)	1.02	1.56	1.08
Medium (15)	1.10	1.06	0.89
Large (15)	1.44	1.25	1.07
Average (50)	1.37	1.24	1.02
Kennington (8)	5.68	2.83	0.30
BPMPD (5)	3.08	1.41	0.92
Netlib (2)	0.94	1.01	1.80
Average (15)	4.58	2.14	0.47

The following two codes were tested and compared:

1. MINOS 5.51
2. RDPPA 1.10

The second code was developed by modifying MINOS 5.5.1 with minor changes. The test set of 65 standard LP problems was the same as that used in Sect. 19.3.1, classified to the same sets.

Table 20.1 lists iterations and time ratios (MINOS 5.51/RDPPA 1.10), in which the last column gives degenerate iterations percentage ratios.

It is seen from the preceding table that the new code significantly outperformed MINOS 5.51 overall. For the first group of problems, the time ratio is 1.24, and for the second group, the time ratio reaches 2.14. Thus, the superiority margin is higher with large-scale sparse problems. Comparing these results with those listed in Sect. 19.3.1, we conclude that the dual deficient-basis algorithm appears to be better than its primal counterpart.

From the last column of the preceding table, it is seen that the degenerate iteration ratio for the first group is 1.02, very close to 1, whereas that for the second group of larger problems is significantly less than 1. Overall, the proportion of degenerate iterations is high with the new algorithm. Therefore, it is not true that degeneracy itself degrades the algorithm's efficiency.

Example 20.2.1 Solve the following problem by Algorithm 20.2.1:

$$\begin{aligned}
 \min f &= 4x_3 + x_4 + 3x_5 + x_6 + 5x_7 + 2x_8, \\
 \text{s.t. } x_1 &+ 2x_3 - x_4 + x_5 - x_6 - 3x_7 - 3x_8 = -9, \\
 &+ x_2 + x_3 + 2x_4 - x_5 + x_6 - 2x_7 + x_8 = -4, \\
 &- 3x_3 - 3x_5 + 2x_8 = 0, \\
 &+ x_3 + x_5 - x_8 = 0, \\
 &- x_3 - x_5 + 2x_8 = 0, \\
 x_j &\geq 0, \quad j = 1, \dots, 8.
 \end{aligned}$$

Answer Initial: $s = 2$, $B = \{1, 2\}$, $R = \{1, 2\}$. $N = \{3, 4, 5, 6, 7, 8\}$, $R' = \{3, 4, 5\}$. $B_R = I$. $\bar{z}_N = (4, 1, 3, 1, 5, 2)^T$. $\bar{z}_B = 0$, $\bar{x}_B = (-9, -4)^T$.

Iteration 1:

1. $\min\{-9, -4\} = -9 < 0, s = 1.$
3. $\sigma = (2, -1, 1, -1, -3, -3)^T.$
4. $J = \{4, 6, 7, 8\} \neq \emptyset.$
5. $\beta = \min\{-1/(-1), -1/(-1), -5/(-3), -2/(-3)\} = 2/3, q = 8.$
6. $\bar{z}_N = (4, 1, 3, 1, 5, 2)^T + (2/3)(2, -1, 1, -1, -3, -3)^T = (16/3, 1/3, 11/3, 1/3, 3, 0)^T.$
 $N = \{3, 4, 5, 6, 7, 8\}, \bar{z}_1 = 0 + 2/3 = 2/3.$
7. $\bar{a}_q(R) = (-3, 1)^T.$
8. $\bar{a}_q(R') = (2, -1, 2)^T \neq 0.$
9. $\max\{|2|, |-1|, |2|\} = 2, p = 3.$
10. $p = 3, \tau = (2 - (0, 0)(-3, 1)^T)^{-1} = 1/2.$
 $v = -(1/2)(-3, 1)^T = (3/2, -1/2)^T, d^T = (0, 0).$
 $U = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} - \begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix} (0 \ 0) = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}.$
 $\hat{B}_{\hat{R}}^{-1} = \begin{pmatrix} 1 & 3/2 \\ & 1 & -1/2 \\ & & 1/2 \end{pmatrix}.$
11. $k = 3, B = \{1, 2, 8\}, R = \{1, 2, 3\}, \bar{x}_8 = 0, N = \{3, 4, 5, 6, 7\}, R' = \{4, 5\}.$

Iteration 2:

3. $s = 1, \sigma_N = \begin{pmatrix} 2 & -1 & 1 & -1 & -3 \\ 1 & 2 & -1 & 1 & -2 \\ -3 & & -3 & 0 & \end{pmatrix}^T \begin{pmatrix} 1 \\ 0 \\ 3/2 \end{pmatrix} = (-5/2, -1, -7/2, -1, -3)^T.$
4. $J = \{3, 4, 5, 6, 7\} \neq \emptyset.$
5. $\beta = \min\{-(16/3)/(-5/2), -(1/3)/(-1), -(11/3)/(-7/2), -(1/3)/(-1), -(3)/(-3)\} = 1/3, q = 4.$
6. $\bar{z}_N = (16/3, 1/3, 11/3, 1/3, 3)^T + (1/3)(-5/2, -1, -7/2, -1, -3)^T$
 $= (9/2, 0, 5/2, 0, 2)^T, N = \{3, 4, 5, 6, 7\}, \bar{z}_1 = 2/3 + 1/3 = 1.$
7. $\bar{a}_q(R) = \begin{pmatrix} 1 & 3/2 \\ & 1 & -1/2 \\ & & 1/2 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix},$
8. $\bar{a}_q(R') = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$
12. $\alpha = (-9)/(-1) = 9.$
13. $\bar{x}_4 = 9, \bar{x}_B = (-9, -4, 0)^T - 9(-1, 2, 0)^T = (0, -22, 0)^T.$
14. $B_R^{-1} = \begin{pmatrix} -1 & \\ & 2 & 1 \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & 3/2 \\ & 1 & -1/2 \\ & & 1/2 \end{pmatrix} = \begin{pmatrix} -1 & -3/2 \\ & 2 & 1 & 5/2 \\ & & & 1/2 \end{pmatrix}.$
15. $B = \{4, 2, 8\}, N = \{1, 3, 5, 6, 7\}.$

Iteration 3:

1. $\min\{9, -22, 0\} = -22 < 0, s = 2.$
3. $\sigma_N = \begin{pmatrix} 1 & 2 & 1 & -1 & -3 \\ & 1 & -1 & 1 & -2 \\ & -3 & -3 & & \end{pmatrix}^T \begin{pmatrix} 2 \\ 1 \\ 5/2 \end{pmatrix} = (2, -5/2, -13/2, -1, -8)^T.$
4. $J = \{3, 5, 6, 7\} \neq \emptyset.$
5. $\beta = \min\{-(9/2)/(-5/2), -(5/2)/(-13/2), 0/(-1), -2/(-8)\} = 0, q = 6.$
7. $\bar{a}_q(R) = \begin{pmatrix} -1 & -3/2 \\ 2 & 1 & 5/2 \\ & & 1/2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \bar{a}_q(R') = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$
8. $\bar{a}_q(R') = 0.$
12. $\alpha = (-22)/(-1) = 22.$
13. $\bar{x}_B = (9, -22, 0)^T - 22(1, -1, 0)^T = (-13, 0, 0)^T, \bar{x}_6 = 22.$
14. $B_R^{-1} = \begin{pmatrix} 1 & 1 \\ & -1 \\ & & 1 \end{pmatrix} \begin{pmatrix} -1 & -3/2 \\ 2 & 1 & 5/2 \\ & & 1/2 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ -2 & -1 & -5/2 \\ & & 1/2 \end{pmatrix}.$
15. $B = \{4, 6, 8\}, N = \{1, 2, 3, 5, 7\}.$

Iteration 4:

1. $\min\{-13, 22, 0\} = -13 < 0, s = 1.$
3. $\sigma_N = \begin{pmatrix} 1 & 2 & 1 & -3 \\ & 1 & 1 & -1 & -2 \\ & & -3 & -3 & \end{pmatrix}^T \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = (1, 1, 0, -3, -5)^T.$
4. $J = \{5, 7\} \neq \emptyset.$
5. $\beta = \min\{-(5/2)/(-3), -2/(-5)\} = 2/5, q = 7.$
6. $\bar{z}_N = ((1, 0, 9/2, 5/2, 2)^T + (2/5)(1, 1, 0, -3, -5)^T) = (7/5, 2/5, 9/2, 13/10, 0)^T, N = \{1, 2, 3, 5, 7\}, \bar{z}_4 = 0 + 2/5 = 2/5.$
7. $\bar{a}_q(R) = \begin{pmatrix} 1 & 1 & 1 \\ -2 & -1 & -5/2 \\ & & 1/2 \end{pmatrix} \begin{pmatrix} -3 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ 8 \\ 0 \end{pmatrix}, \bar{a}_q(R') = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$
8. $\bar{a}_q(R') = 0.$
12. $\alpha = (-13)/(-5) = 13/5.$
13. $\bar{x}_B = (-13, 22, 0)^T - (13/5)(-5, 8, 0)^T = (0, 6/5, 0)^T, \bar{x}_7 = 13/5.$
14. $B_R^{-1} = \begin{pmatrix} -1/3 \\ -2/3 & 1 \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -2 & -1 & -5/2 \\ & & 1/2 \end{pmatrix} = \begin{pmatrix} -1/3 & -1/3 & -1/3 \\ -8/3 & -5/3 & -19/6 \\ & & 1/2 \end{pmatrix}.$
15. $B = \{7, 6, 8\}, N = \{1, 2, 3, 4, 5\}.$

Iteration 5:

1. $\min\{13/5, 6/5, 0\} \geq 0$.
2. The optimal basic solution and optimal value are

$$x^* = (0, 0, 0, 0, 0, 6/5, 13/5, 0)^T, \quad f^* = (1, 5)(6/5, 13/5)^T = 71/5.$$

20.3 Dual Deficient-Basis D-Reduced Algorithm: Tableau Form

In this section, the deficient basis is incorporated into the dual D-reduced simplex algorithm (Sect. 17.2). The new algorithm should be stable, compared with the dual deficient-basis algorithm, presented in the previous section, and the related search direction corresponds to the entire dual objective gradient.

Consider the D-reduced problem (17.1). Assume that the order of the deficient basis equals $1 \leq k \leq m-1$ and that the r -indexed row is the datum row ($1 \leq r \leq m$). The sets B , R , N , and R' are defined by

$$B = \{j_1, \dots, j_k\}, \quad \{r\} \notin R = \{i_1, \dots, i_k\}, \quad N = A \setminus B, \quad R' = \{1, \dots, m\} \setminus (R \cup \{r\}). \quad (20.3)$$

The initial tableau of (17.1) is

x_B	x_N	f	RHS
B_R	N_R		
$B_{R'}$	$N_{R'}$		
ω_B^T	ω_N^T		1
c_B^T	c_N^T	-1	

(20.4)

which can be converted to the following form by a series of elementary transformations:

x_B	x_N	f	RHS
I	$\bar{N}_R$		
	$\bar{N}_{R'}$		
	$\bar{\omega}_N^T$		1
	$\bar{z}_N^T$	-1	

(20.5)

The preceding is called *deficient-basis D-reduced tableau*, where the m th row is the datum row, indexed by $r \notin R, R'$.

Assume that the current tableau is dual feasible, i.e., $\bar{z}_N \geq 0$. If entries of the datum row are no more than zero, then there is no feasible solution to the original problem (Lemma 17.1.1). If this is not the case, determine β and column index q

such that

$$\beta = \bar{z}_q / \bar{\omega}_q = \min\{\bar{z}_j / \bar{\omega}_j \mid \bar{\omega}_j > 0, j \in N\} \leq 0.$$

Add $-\beta$ times of the datum row to the bottom row to turn $\bar{z}_q$ to 0. Then the southeast entry of the tableau becomes $-\beta$, and hence the maximum possible increment of the objective value just equals $\beta \geq 0$. Under the assumption of nondegeneracy, $\beta > 0$, and hence the objective value strictly increases.

The determination of the pivot row index depends on whether $\bar{a}_q(R')$ vanishes.

- (i) Rank-remaining iteration: $R' = \emptyset$ or $\bar{a}_q(R') = 0$.

Determine $s \in \arg \max\{\bar{a}_{i_t, q} \mid t = 1, \dots, k\}$. If $\bar{a}_{i_s, q} \leq 0$, it is clear by $\bar{\omega}_q > 0$ that the tableau corresponds to a basic feasible solution, and hence optimality is achieved (taking the q -indexed column as the datum column). If $\bar{a}_{i_s, q} > 0$, take it as the pivot to make the basis change. In such an iteration, the rank of the basis remains unchanged.

- (ii) Rank-increasing iteration: $R' \neq \emptyset$ and $\bar{a}_q(R') \neq 0$.

Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{i, q}|$. Take $\bar{a}_{p, q}$ as the pivot to make a basis change. Then the rank of the basis increases to $k + 1$.

In either iteration, the elementary transformations do not touch the datum row, the bottom row, and the right-hand side, so the resulting tableau is again a dual feasible D-reduced tableau. Repeat these steps until the optimality is achieved or the infeasibility of the problem is detected.

The overall steps are summarized into the following algorithm.

Algorithm 20.3.1 (Dual Deficient-Basis D-Reduced Algorithm 1: Tableau Form) Initial: dual feasible deficient-basis D-reduced tableau of form (20.5) with $B, R, N, R', 1 \leq k \leq m-1$. This algorithm solves the D-reduced problem 17.1.

1. Stop if $J = \{j \in N \mid \bar{\omega}_j > 0\} = \emptyset$ (infeasible problem).
2. Determine β and column index q such that $\beta = \bar{z}_q / \bar{\omega}_q = \min_{j \in J} \bar{z}_j / \bar{\omega}_j$.
3. If $\beta \neq 0$, add $-\beta$ times of the datum row to the bottom row.
4. Go to step 8 if $R' = \emptyset$ or $\bar{a}_q(R') = 0$.
5. Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{i, q}|$.
6. Convert $\bar{a}_{p, q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
7. Set $k = k + 1$, bring q from N to B and p from R' to R , and go to step 1.
8. Determine $s \in \arg \max\{\bar{a}_{i_t, q} \mid t = 1, \dots, k\}$.
9. If $\bar{a}_{i_s, q} \leq 0$, compute $\bar{x}_q = 1 / \bar{\omega}_q$, $\bar{x}_B = -\bar{x}_q \bar{a}_q(R)$, and stop (optimality achieved).
10. Convert $\bar{a}_{i_s, q}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
11. Update (B, N) by exchanging j_s and q .
12. Go to step 1.

Example 20.3.1 Solve the following problem by Algorithm 20.3.1:

$$\begin{aligned}
 \min f &= 3x_4 + 2x_5 + x_6 + 5x_7 + 6x_8 \\
 \text{s.t. } x_1 &- 6x_4 + x_5 + 4x_6 + 6x_7 = 9, \\
 &+ x_2 + 2x_4 - 7x_6 - 8x_7 + x_8 = -6, \\
 &+ x_3 - 4x_5 + 2x_8 = 0, \\
 &+ 3x_4 + 2x_5 + 3x_7 = 3, \\
 &-x_4 + 2x_5 - 4x_6 - 6x_7 + 3x_8 = 2, \\
 x_j &\geq 0, \quad j = 1, \dots, 8.
 \end{aligned}$$

Answer Initial tableau is

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1			-6	1	4	6		9
	1		2		-7	-8	1	-6
		1		-4			2	0
			3	2		3	0	3
			-1	2	-4	-6	3	2*
			3	2	1	5	6	0

Multiply row 5 by $1/2$, and then add -9 , 6 , and -3 times of row 5 to rows 1, 2, and 4, respectively.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1			$-3/2$	-8	22	33	$-27/2$	
	1		-1	6	-19	-26	10	
		1		-4			2	
			$9/2$	-1^*	6	12	$-9/2$	
			$-1/2$	1	-2	-3	$3/2$	1
			3	2	1	5	6	

Iteration 1: $k = 3$, $B = \{1, 2, 3\}$, $N = A \setminus B$, $R = \{1, 2, 3\}$, $R' = \{4\}$.

1. $J = \{5, 8\}$.
2. $\beta = \min\{2/1, 6/(3/2)\} = 2/1$, $q = 5$.
3. Add -2 times of row 5 to the bottom row.
4. $\bar{a}_q(R') \neq 0$.
5. $\max\{|-1|\}$, $p = 4$.
6. Multiply row 4 by -1 , and then add 8 , -6 , 4 , and -1 times of row 4 to rows 1, 2, 3, and 5, respectively.
7. $k = 4$, $B = \{1, 2, 3, 5\}$, $N = A \setminus B$, $R = \{1, 2, 3, 4\}$, $R' = \emptyset$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1			$-75/2$		-26	-63	$45/2$	
	1		26^*		17	46	-17	
		1	-18		-24	-48	20	
			$-9/2$	1	-6	-12	$9/2$	
			4		4	9	-3	1
			4		5	11	3	-2

Iteration 2:

1. $J = \{4, 6, 7\}$.
2. $\beta = \min\{4/4, 5/4, 11/9\} = 1, q = 4$.
3. Add -1 times of row 5 to the bottom row.
4. $R' = \emptyset$.
8. $\max\{-75/2, 26, -18, -9/2\} = 26 > 0, s = 2$.
10. Multiply row 2 by $1/26$, and then add $75/2, 18, 9/2$, and -4 times of row 2 to rows 1, 3, 4, and 5, respectively.
11. $k = 4, B = \{1, 4, 3, 5\}, N = A \setminus B, R = \{1, 2, 3, 4\}, R' = \emptyset$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1	$75/52$				$-77/52$	$87/26$	$-105/52$	
	$1/26$		1		$17/26^*$	$23/13$	$-17/26$	
	$9/13$	1			$-159/13$	$-210/13$	$107/13$	
	$9/52$			1	$-159/52$	$-105/26$	$81/52$	
	$-2/13$				$18/13$	$25/13$	$-5/13$	1
	$1/9$					$11/18$	$113/18$	$-67/18$

Iteration 3:

1. $J = \{6, 7\}$.
2. $\beta = 0, q = 6$.
4. $R' = \emptyset$.
8. $\max\{-77/52, 17/26, -159/13, -159/52\} = 17/26 > 0, s = 2$.
10. Multiply row 2 by $26/17$, and then add $77/52, 159/13, 159/52$, and $-18/13$ times of row 2 to rows 1, 3, 4, and 5, respectively:
11. $k = 4, B = \{1, 4, 3, 5\}, N = A \setminus B, R = \{1, 2, 3, 4\}, R' = \emptyset$.

Iteration 4:

1. $J = \{8\}$.
2. $\beta = 113/18, q = 8$.
4. $R' = \emptyset$.
8. $\max\{-7/2, -1, -4, -3.2\} = -1 \leq 0, s = 2$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1	26/17		77/34			125/17	-7/2	
	1/17		26/17		1	46/17	-1	
	24/17	1	318/17			288/17	-4	
	6/17		159/34	1		72/17	-3/2	
	-4/17		-36/17			-31/17	1	1
	1/9					11/18	113/18	-67/18

9. $\bar{x}_8 = 1$, $\bar{x}_B = (7/2, 1, 4, 3/2)^T$. The optimal basic solution and optima value are

$$x^* = (7/2, 0, 4, 0, 3/2, 1, 0, 1)^T, \quad f^* = 10.$$

It might be accepted that if the same approach is used, the search direction determined within a subspace of high dimension has better quality than that within a subspace of lower dimension. Therefore, it should be desirable to get Algorithm 20.3.1 started from $k = 0$. The associated sets are

$$B, R = \emptyset, \quad N = A, \quad R' = \{1, \dots, r-1, r+1, \dots, m\}, \quad (20.6)$$

and the basis matrix B_R may be regarded as of zero order. Subsequent steps are the same as those in the usual case. The corresponding tableau is called *zero-order D-reduced tableau*; if it is still denoted by tableau (20.5), I and $\bar{N}_R$ should be viewed as empty sets and $\bar{N}_{R'}$ the submatrix of A after the r -indexed (datum) row deleted. If the bottom row satisfies $\bar{z}_N \geq 0$, it is called zero-order dual feasible D-reduced tableau. Initially, therefore, a series of rank-increasing iterations are usually performed under the condition $\bar{a}_q(R') \neq 0$.

Example 20.3.2 Solve the following problem by Algorithm 20.3.1, starting from the zero-order D-reduced tableau:

$$\begin{aligned}
 \min \quad & f = 2x_1 + x_2 + 2x_3 + x_4 + 4x_5 + 5x_6 + 2x_7, \\
 \text{s.t.} \quad & +2x_2 \qquad \qquad -3x_5 -4x_6 +6x_7 = 0, \\
 & -2x_1 \qquad \quad -4x_3 \qquad \qquad +x_6 -x_7 = -1, \\
 & \qquad +3x_2 \qquad \quad -2x_4 +4x_5 \qquad \qquad = 8, \\
 & -3x_1 \qquad \quad +2x_3 -4x_4 \qquad \quad -2x_6 \qquad \quad = 0, \\
 & \qquad \quad 2x_2 -x_3 \qquad \quad +2x_5 \qquad \quad -5x_7 = -1, \\
 & x_j \geq 0, \quad j = 1, \dots, 7.
 \end{aligned}$$

Answer Initial tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	2			-3	-4	6	
-2		-4			1	-1	-1
	3		-2	4			8
-3		2	-4		-2		
	2	-1		2		-5	-1
2	1	2	1	4	5	2	

Convert the preceding to the zero-order D-reduced tableau with $\max\{|-1|, |8|, |-1|\} = 8$, $r = 3$. Take row 3 as the datum row (which involves the minimum number of nonzeros).

Multiply the datum row by $1/8$, and then add it to row 2 as well as row 5:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	2			-3	-4	6	
-2	3/8	-4	-1/4	1/2	1	-1	
	3/8		-1/4	1/2			1
-3		2	-4		-2		
	19/8*	-1	-1/4	5/2		-5	
2	1	2	1	4	5	2	

where $k = 0$, $B = \emptyset$, $R = \emptyset$, $N = \{1, \dots, 7\}$, $R' = \{1, 2, 4, 5\}$, datum row $r = 3$.

Iteration 1:

1. $J = \{2, 5\} \neq \emptyset$.
2. $\beta = \min\{1/(3/8), 4/(1/2)\} = 8/3$, $q = 2$.
3. Add $-8/3$ times of row 3 to the bottom row.
4. $\bar{a}_q(R') \neq 0$.
5. $\max\{|2|, |3/8|, |3/8|, |19/8|\} = 19/8$, $p = 5$.
6. Multiply row 5 by $8/19$, and then add -2 , $-3/8$, $-3/8$, and -1 times of row 5 to rows 1, 2, 3, and 6, respectively.
7. $k = 1$, $B = \{2\}$, $R = R \cup \{5\} = \{5\}$, $N = \{1, 3, 4, 5, 6, 7\}$, $R' = \{1, 2, 4\}$.

Iteration 2:

1. $J = \{3, 5, 7\} \neq \emptyset$.
2. $\beta = \min\{2/(3/19), (8/3)/(2/19), 2/(15/19)\} = 38/15$, $q = 7$.
3. Add $-38/15$ times of row 3 to the bottom row.
4. $\bar{a}_q(R') \neq 0$.
5. $\max\{|194/19|, |-4/19|\} = 194/19$, $p = 1$.
6. Multiply row 1 by $19/194$, and then add $4/19$, $-15/19$, $40/19$, and -2 times of row 1 to rows 2, 3, 4, and 6, respectively.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		16/19	4/19	-97/19	-4	194/19*	1
-2		-73/19	-4/19	2/19	1	-4/19	
		3/19	-4/19	2/19		15/19	
-3		2	-4		-2		
	1	-8/19	-2/19	20/19		-40/19	
2		2	5/3	8/3	5	2	-8/3

7. $k = 2$, $B = \{2, 7\}$, $R = \{5, 1\}$, $N = \{1, 3, 4, 5, 6\}$, $R' = \{2, 4\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		8/97	2/97	-1/2	-38/97	1	1
-2		-371/97	-20/97		89/97		
		9/97	-22/97	1/2	30/97		
-3		2	-4		-2		
	1	-24/97	-6/97		-80/97		
2		8/5	11/5	12/5	5		-26/5

Iteration 3:

1. $J = \{3, 5, 6\} \neq \emptyset$.
2. $\beta = \min\{(8/5)/(9/97), (12/5)/(1/2), 5/(30/97)\} = 24/5$, $q = 5$.
3. Add $-24/5$ times of row 3 to the bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
		8/97	2/97	-1/2	-38/97	1	1
-2		-371/97	-20/97		89/97		
		9/97	-22/97	1/2	30/97		
-3		2	-4		-2		
	1	-24/97	-6/97		-80/97		
2		112/97	319/97		341/97		-10

4. $\bar{a}_q(R') = 0$.
8. $\max\{-1/2, 0\} = 0 \leq 0$.
9. $\bar{x}_5 = 2$, $\bar{x}_B = (0, 1)^T$. The optimal basic solution and optimal value are

$$x^* = (0, 0, 0, 0, 2, 0, 1)^T, \quad f^* = 10.$$

After pivot column index q is determined, there may be choices for taking either $p \in R'$ or $i_s \in R$ as the pivot row index. In doing so, Algorithm 20.3.1 gives the

priority to R' . An alternative is to focus on numerical stability as follows. Define

$$\zeta = \max\{\zeta_1, \zeta_2\}, \quad \zeta_1 = |\bar{a}_{pq}| = \max_{i \in R'} |\bar{a}_{iq}| \quad \zeta_2 = \bar{a}_{is,q} = \max\{\bar{a}_{it,q} \mid t = 1, \dots, k\}$$

(where ζ_1 or ζ_2 is set to zero if R' or R is empty). Then, if $\zeta = \zeta_1$, take $p \in R'$, and if $\zeta = \zeta_2$, take $i_s \in R$.

Nevertheless, it seems to be preferable to keep the rank of the basis as low as possible. To this end, the following variant gives the priority to R rather than R' .

Algorithm 20.3.2 (Dual Deficient-Basis D-Reduced Algorithm 2: Tableau Form) Initial: dual feasible deficient-basis D-reduced tableau of form (20.5) with B, R, N, R' $1 \leq k \leq m-1$. This algorithm solves the D-reduced problem 17.1.

1. Stop if $J = \{j \in N \mid \bar{\omega}_j > 0\} = \emptyset$ (infeasible problem).
2. Determine column index q and β such that $\beta = \bar{z}_q / \bar{\omega}_q = \min_{j \in J} \bar{z}_j / \bar{\omega}_j$.
3. If $\beta \neq 0$, add $-\beta$ times of the datum row to the bottom row.
4. Determine $s \in \arg \max\{\bar{a}_{it,q} \mid t = 1, \dots, k\}$.
5. Go to step 8 if $\bar{a}_{is,q} \leq 0$.
6. Convert $\bar{a}_{is,q}$ into 1, and eliminate other nonzeros in the column by elementary transformations.
7. Update (B, N) by exchanging j_s and q , and go to step 1.
8. If $R' = \emptyset$ or $\bar{a}_q(R') = 0$, compute $\bar{x}_q = 1/\bar{\omega}_q$, $\bar{x}_B = -\bar{x}_q \bar{a}_q(R)$, and stop (optimality achieved).
9. Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{iq}|$.
10. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
11. Set $k = k + 1$, bring q from N to B and p from R' to R .
12. Go to step 1.

Example 20.3.3 Solve the problem in Example 20.3.1 by Algorithm 20.3.2:

$$\begin{aligned} \min f &= 2x_1 + x_2 - 2x_3 + x_4 - 3x_5 + x_6 + 2x_7 - 2x_8 \\ \text{s.t. } 2x_1 - 3x_2 + x_3 - 6x_4 + x_5 + 4x_6 + 6x_7 &= 0, \\ 5x_1 + 6x_3 - 2x_4 + 10x_6 + 8x_7 - x_8 &= 0, \\ -x_1 + 8x_2 - 4x_5 + 2x_8 &= 0, \\ 4x_1 - 2x_3 + 5x_4 + 8x_5 - 4x_6 + 3x_7 &= 1, \\ x_j &\geq 0, \quad j = 1, \dots, 8. \end{aligned}$$

Answer Begin with the second tableau in the Answer, i.e.,

Iteration 1: $k = 3$, $B = \{1, 2, 3\}$, $N = A \setminus B$, $R = \{1, 2, 3\}$, $R' = \{4\}$.

1. $J = \{5, 8\}$.
2. $\beta = \min\{2/1, 6/(3/2)\} = 2/1$, $q = 5$.
3. Add -2 times of row 5 to the bottom row.
4. $\max\{-8, 6, -4\} = 6$, $s = 2$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1			$-3/2$	-8	22	33	$-27/2$	
	1		-1	6	-19	-26	10	
		1	0	-4	0	0	2	
			$9/2$	-1	6	12	$-9/2$	
			$-1/2$	1	-2	-3	$3/2$	1
			3	2	1	5	6	

5. $\bar{a}_{i_s, q} > 0$

6. Multiply row 2 by $1/6$, and then add $8, 4, 1$, and -1 times of row 2 to rows 1, 3, 4, and 5, respectively.

7. $B = \{1, 5, 3\}$, $N = A \setminus B$, $R = \{1, 2, 3\}$, $R' = \{4\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1	$4/3$		$-17/6$		$-10/3$	$-5/3$	$-1/6$	
	$1/6$		$-1/6$	1	$-19/6$	$-13/3$	$5/3$	
	$2/3$	1	$-2/3$		$-38/3$	$-52/3$	$26/3$	
	$1/6$		$13/3$		$17/6$	$23/3$	$-17/6$	
	$-1/6$		$-1/3$		$7/6$	$4/3$	$-1/6$	1
	0		4		5	11	3	-2

Iteration 2:

1. $J = \{6, 7\}$.
2. $\beta = \min\{5/(7/6), 11/(4/3)\} = 30/7$, $q = 6$.
3. Add $-30/7$ times of row 5 to the bottom row.
4. $\max\{-10/3, -19/3, -38/3\} \leq 0$.
8. $\bar{a}_q(R') \neq 0$.
9. $\max\{|17/6|\}$, $p = 4$.
10. Multiply row 4 by $6/17$, and then add $10/3, 19/6, 38/3$, and $-7/6$ times of row 4 to rows 1, 2, 3, and 5, respectively.
11. $k = 4$, $B = \{1, 5, 3, 6\}$, $N = A \setminus B$, $R = \{1, 2, 3, 4\}$, $R' = \emptyset$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1	$26/17$		$77/34$			$125/17$	$-7/2$	
	$6/17$		$159/34$	1		$72/17$	$-3/2$	
	$24/17$	1	$318/17$			$288/17$	-4	
	$1/17$		$26/17$		1	$46/17$	-1	
	$-4/17$		$-36/17$			$-31/17$	1	1
	$5/7$		$38/7$			$37/7$	$26/7$	$-44/7$

Iteration 3:

1. $J = \{8\}$.
2. $\beta = 26/7$, $q = 8$.
3. Add $-26/7$ times of row 5 to the bottom row.
4. $\max\{-7/2, -3/2, -4, -1\} \leq 0$.
8. $R' = \emptyset$. $\bar{x}_8 = 1$, $\bar{x}_B = (7/2, 3/2, 4, 1)^T$.

The optimal basic solution and objective value are

$$x^* = (7/2, 0, 4, 0, 3/2, 1, 0, 1)^T, \quad f^* = 44/7 + 26/7 = 10.$$

20.4 Dual Deficient-Basis D-Reduced Algorithm

Assume that the r -indexed row is the datum row. Denote by B_R the basis matrix, associated with basic column and row index sets B and R , and so on. It is not difficult to show that the deficient-basis D-reduced tableau (20.5) is equivalent to the following revised tableau:

x_B	x_N	f	RHS
I	$B_R^{-1} N_R$		
	$N_{R'} - B_{R'} B_R^{-1} N_R$		
	$e_m^T N - e_m^T B B_R^{-1} N_R$		1
	$c_N^T - c_B^T B_R^{-1} N_R$	-1	

(20.7)

Based on such an equivalence, Algorithms 20.3.1 and 20.3.2 can be revised. But only the former's revision is formulated.

Algorithm 20.4.1 (Dual Deficient-basis D-Reduced Algorithm 1) Initial: B , R , N , R' , $1 \leq k \leq m - 1$. B_R^{-1} , $\bar{z}_N \geq 0$. This algorithm solves the D-reduced problem 17.1.

1. Compute $\bar{\omega}_N = N^T e_m - N_R^T B_R^{-T} B^T e_m$.
2. Stop if $J = \{j \in N \mid \bar{\omega}_j > 0\} = \emptyset$ (infeasible problem).
3. Determine column index q and β such that $\beta = \bar{z}_q / \bar{\omega}_q = \min_{j \in J} \bar{z}_j / \bar{\omega}_j$.
4. If $\beta \neq 0$, update $\bar{z}_N = \bar{z}_N - \beta \bar{\omega}_N$.
5. Compute $\bar{a}_q(R) = B_R^{-1} a_q(R)$.
6. Go to step 10 if $R' = \emptyset$ or $\bar{a}_q(R') = a_q(R') - B_{R'} \bar{a}_q(R) = 0$.
7. Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{i,q}|$.
8. Update B_R^{-1} by (19.10).
9. Set $k = k + 1$, bring q from N to B and p from R' to R , and go to step 1.
10. Determine $s \in \arg \max\{\bar{a}_{i_t,q} \mid t = 1, \dots, k\}$.

11. If $\bar{a}_{i_s, q} \leq 0$, compute $\bar{x}_q = 1/\bar{\omega}_q$, $\bar{x}_B = -\bar{x}_q \bar{a}_q(R)$, $\bar{f} = c_q \bar{x}_q + c_B^T \bar{x}_B$, and stop (optimality achieved).
12. Update B_R^{-1} by (17.20).
13. Update (B, N) by exchanging j_s and q .
14. Go to step 1.

The algorithm can get itself started if $\bar{z}_N = c_N - N_R^T B_R^{-T} c_B \geq 0$. In the other case, methods, presented in Sect. 17.2.1 or Chap. 15, can be utilized to provide an initial dual feasible solution, if slightly modified.

We point out that the so-called *reduced dual elimination* (Part I) is a convenient tool to convert the dual problem to the D-reduced form. The resulting problem can be solved by the dual algorithm presented in this section by generating a sequence of primal feasible solutions.

Example 20.4.1 Solve the following problem by Algorithm 20.4.1:

$$\begin{aligned}
 \min \quad & f = x_1 + 2x_3 + 3x_4 + 5x_6 + 2x_8, \\
 \text{s.t.} \quad & x_1 \qquad \qquad \qquad x_5 - 2x_6 \qquad + 3x_8 = 1, \\
 & -2x_1 \qquad -4x_3 + 3x_4 \qquad -3x_6 + x_7 - x_8 = 3, \\
 & \qquad \qquad -x_3 \qquad \qquad \qquad + 4x_8 = 0, \\
 & -2x_1 + x_2 + 2x_3 - 4x_4 \qquad + 4x_6 \qquad \qquad = -2, \\
 & 3x_1 \qquad \qquad \qquad + 2x_6 \qquad - 2x_8 = 0, \\
 & x_j \geq 0, \quad j = 1, \dots, 8.
 \end{aligned}$$

Answer The initial tableau is

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
1				1	-2		3	1
-2		-4	3		-3	1	-1	3
		-1					4	
-2	1	2	-4		-4			-2*
3					2		-2	
1		2	3		5		2	

which is a dual feasible deficient-basis tableau with $B = \{5, 7, 2\}$ and $R = \{1, 2, 4\}$.

Convert the preceding to a D-reduced tableau: we might as well take row 3 is the datum row. Multiply row 3 by $-1/3$, and then add -4 and 6 times of row 3 to rows 1 and 2, respectively.

For this tableau, $k = 2$, $B = \{5, 7\}$, $R = \{1, 2\}$, $N = \{1, 2, 3, 4, 6, 8\}$, $R' = \{3, 5\}$, the datum row: $r = 4$. $B_R = I$. $\bar{z}_N = (1, 0, 2, 3, 5, 2)^T \geq 0$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
	1/2	1	-2	1			3	
-5	3/2	-1	-3		3	1	-1	
		-1					4	
1	-1/2	-1	2		-2			1
3					2		-2	
1		2	3		5		2	

Iteration 1:

1. $\bar{\omega}_N = (1, -1/2, -1, 2, -2, 0)^T$.
2. $J = \{1, 4\}$.
3. $\beta = \min\{1/1, 3/2\} = 1, q = 1$.
4. $\bar{z}_N = (1, 0, 2, 3, 5, 2)^T - (1, -1/2, -1, 2, -2, 0)^T = (0, 1/2, 3, 1, 7, 2)^T$.
5. $\bar{a}_q(R) = (0, -5)^T$.
6. $\bar{a}_q(R') = (0, 3)^T \neq 0$.
7. $\max\{|3|\} = 3, p = 5$.
8. $\tau = 1/3, v = -(1/3)(0, -5)^T = (0, 5/3)^T, d^T = (0, 0), U = I$.

$$\widehat{B}_{\hat{R}}^{-1} = \begin{pmatrix} 1 & & \\ & 1 & 5/3 \\ & & 1/3 \end{pmatrix}.$$

9. $k = 3, B = \{5, 7, 1\}, R = \{1, 2, 5\}, N = \{2, 3, 4, 6, 8\}, R' = \{3\}$.

Iteration 2:

$$1. \bar{\omega}_N = (-1/2, -1, 2, -2, 0)^T - \begin{pmatrix} 1/2 & 1 & -2 & 3 \\ 3/2 & -1 & -3 & 3 & -1 \\ & & 2 & -2 \end{pmatrix}^T \begin{pmatrix} 1 & & \\ & 1 & 5/3 \\ & & 1/3 \end{pmatrix}^T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$= (-1/2, -1, 2, -8/3, 2/3)^T.$$

$$2. J = \{4, 8\}.$$

$$3. \beta = \min\{1/2, 2/(2/3)\} = 1/2, q = 4.$$

$$4. \bar{z}_N = (1/2, 3, 1, 7, 2)^T - (1/2)(-1/2, -1, 2, -8/3, 2/3)^T$$

$$= (3/4, 7/2, 0, 25/3, 5/3)^T.$$

$$5. \bar{a}_q(R) = \begin{pmatrix} 1 & & \\ & 1 & 5/3 \\ & & 1/3 \end{pmatrix} \begin{pmatrix} -2 \\ -3 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \\ 0 \end{pmatrix}.$$

$$6. \bar{a}_q(R') = 0.$$

$$10. \max\{-2, -3, 0\} \leq 0.$$

11. $\bar{x}_4 = 1/2, \bar{x}_B = -(1/2)(-2, -3, 0)^T = (1, 3/2, 0)^T$. The basic optimal solution and optimal value:

$$x^* = (0, 0, 0, 1/2, 1, 0, 3/2, 0)^T, \quad f^* = 3/2.$$

20.5 Deficient-Basis D-Reduced Gradient Algorithm: Tableau Form

It is noted that the search direction $-\bar{\omega}_N$, used in Algorithm 20.3.1 or 20.4.1, is independent of the current dual iterate. It is imaginable that if the current iterate is close to the boundary, the stepsize would become very small, and hence the associated change in the dual objective value is negligible. Exploiting the current iterate, in this section we modify the search direction so that it is not only ascent with respect to the dual objective but also has a “centering” tendency to leave the nearby boundary. This is a trick used in the interior-point method.

Let B, R, N , and R' be the sets defined by (19.1) and (19.2). The associated D-reduced tableau (20.5) represents the following D-reduced problem:

$$\begin{aligned} & \max y_r, \\ & \text{s.t.} \quad \begin{pmatrix} I & 0 & 0 \\ \bar{N}_R^T & \bar{N}_{R'}^T & \bar{\omega}_N \end{pmatrix} \begin{pmatrix} y_R \\ y_{R'} \\ y_r \end{pmatrix} + \begin{pmatrix} z_B \\ z_N \end{pmatrix} = \begin{pmatrix} 0 \\ \bar{z}_N \end{pmatrix}, \quad z_B, z_N \geq 0. \end{aligned} \quad (20.8)$$

Let $(\bar{z}_N \geq 0, \bar{z}_B = 0)$ be the current dual feasible solution. Setting $0 < \delta \ll 1$, introduce

$$\tilde{z}_N^{-1} = \tilde{Z}_N^{-1} e, \quad \tilde{z}_j = \begin{cases} \bar{z}_j, & \bar{z}_j \geq \delta, \\ \delta, & \bar{z}_j < \delta, \end{cases} \quad j \in N, \quad (20.9)$$

where $\tilde{Z}_N$ is the diagonal matrix, whose diagonal entries are components of $\tilde{z}_N$.

In order to create a “centering” direction, we maximize the following auxiliary function:

$$\max \left(\tilde{z}_N^{-1} \right)^T z_N,$$

subject to constraints of (20.8). From the equality constraints, it follows that

$$\left(\tilde{z}_N^{-1} \right)^T z_N = (n - k) - \left(\bar{\omega}_N^T \tilde{z}_N^{-1} \right) y_r - \left(\bar{N}_R \tilde{z}_N^{-1} \right)^T y_R - \left(\bar{N}_{R'} \tilde{z}_N^{-1} \right)^T y_{R'},$$

where the first term is constant. But, we do not want that the objective variable y_r be involved in the maximization of the auxiliary function. By constraints of (20.8), it is known that $y_R + z_B = 0$, and hence if the value of z_B does not change, the related value of y_R does not either. Thus, by neglecting the first three terms of the auxiliary function, the problem comes down to the maximization of the following objective:

$$\max \quad -(\bar{N}_{R'} \tilde{z}_j^{-1})^T y_{R'},$$

subject to constraints of (20.8). The reduced gradient of the preceding program is

$$\Delta y_{R'} = -\bar{N}_{R'} \bar{z}_N^{-1}. \quad (20.10)$$

If Δy_R , $\Delta y_r = 0$ is taken, the associated direction in z -space is then

$$\begin{aligned} \Delta z_N &= -\bar{N}_{R'}^T \Delta y_{R'} = \bar{N}_{R'}^T \bar{N}_{R'} \bar{z}_N^{-1}, \\ \Delta z_B &= 0. \end{aligned}$$

Now it is logical to take

$$-\tilde{\omega}_N = -\bar{\omega}_N + \mu \bar{N}_{R'}^T \bar{N}_{R'} \bar{z}_N^{-1}, \quad (20.11)$$

$$-\tilde{\omega}_B = 0 \quad (20.12)$$

rather than $-\bar{\omega}_N$ as the search direction, where the parameter $\mu \geq 0$ is a weight of the modification.

Denote by $\bar{a}_j(R')$, $j \in N$, the R' indexed subvector of $\bar{a}_j$. The vector $-\tilde{\omega}_N$ given by (20.11) with (20.9) has the following property.

Lemma 20.5.1 *Let $\mu > 0$ be given. If, for some $k \in N$, it holds that*

$$\bar{z}_k = 0, \quad \bar{a}_k(R') \neq 0, \quad \text{and} \quad \bar{z}_j \neq 0, \quad \forall j \in N, j \neq k,$$

then $-\tilde{\omega}_k > 0$ holds if the δ is small enough.

Proof From (20.9) and (20.11), it follows that

$$-\tilde{\omega}_k = -\bar{\omega}_k + \mu (\bar{a}_k(R'))^T \bar{N}_{R'} \bar{z}_N^{-1} = -\bar{\omega}_k + \mu \bar{a}_k(R')^T \bar{a}_k(R') / \delta + \mu \sum_{j \in N, j \neq k} \bar{a}_k(R')^T \bar{a}_j(R') / \bar{z}_j. \quad (20.13)$$

By the preceding, it is known that the sign of $-\tilde{\omega}_k$ is determined by

$$\mu \|\bar{a}_k(R')\|^2 / \delta > 0,$$

if the δ is small enough. Therefore, the claim is valid. $\square$

In addition, the following result can be shown similarly.

Lemma 20.5.2 *Let $\mu > 0$ and $N' \in N$ be given. If it holds that*

$$\bar{z}_j = 0, \quad 0 \leq \bar{a}_j(R') \neq 0, \quad \forall j \in N' \quad \text{and} \quad \bar{z}_j \neq 0, \quad \forall j \in N \setminus N',$$

then $-\tilde{\omega}_k > 0$ holds if the δ is small enough.

Based on the preceding lemmas, a sufficiently small δ ensures that the components, corresponding to zero-valued dual variables ($\bar{z}_j = 0$), of the new search direction $-\tilde{\omega}$ are strictly positive. Therefore, the effects of dual degeneracy can be avoided.

It is seen from (20.11) and (20.12) that this search direction yields from multiplying rows of R' of the D-reduced tableau by respective components of $\mu \Delta y_{R'}$ and adding the resulting rows to the datum row. Since the associated components of the right-hand side are zero, the whole right-hand side remains unchanged. Besides, the bottom row is not touched by these manipulations. Starting from such a D-reduced tableau, we proceed by a so-called “outer iteration,” consisting of a series of “inner” D-reduced rank-increasing iterations and terminating when $R' = \emptyset$ or $\bar{a}_q(R') = 0$; if the optimality conditions are not satisfied yet, i.e., $\bar{z}_q = 0$, $\bar{\omega}_q > 0$ but $\bar{a}_q(R) \not\leq 0$, the next outer iteration is carried out.

An outer iteration may start from any dual feasible deficient-basis D-reduced tableau, including the zero-order tableau, although it seems favorable to take the D-reduced tableau associated with the sets

$$B = B \setminus \{j_t \mid t \in T\}, \quad R = R \setminus \{i_t \mid t \in T\}, \quad T = \{t = 1, \dots, k \mid \bar{a}_{i_t, q} > 0\}. \quad (20.14)$$

This will separate the rows and columns corresponding to the negative components of the current primal feasible solution from the basis. Another scheme is to let a part of them leave the basis, e.g., to separate the rows and columns corresponding to the negative components of larger magnitude. In fact, under the assumption of Lemma 20.5.2, $-\bar{\omega}_{j_t} > 0$ holds for all $t \in T$ if δ is small enough. Therefore, the zero $\bar{z}_{j_t}$, $t \in T$, have nothing to do with the determination of the stepsize and the related index. However, in case $\bar{z}_N$ is degenerate, a positive stepsize is still not guaranteed if $\bar{a}_q(R) \not\leq 0$ and it holds for some $j \in N$ that $\bar{z}_j = 0$, $\bar{\omega}_j > 0$.

The respective algorithm can be obtained by modifying Algorithm 20.3.1.

Algorithm 20.5.1 (Deficient-Basis D-Reduced Gradient Algorithm: Tableau Form) Given $\mu > 0$, $\delta > 0$. Initial: dual feasible deficient-basis D-reduced tableau of form (20.5), associated with B , R , N , R' , $1 \leq k \leq m-1$. This algorithm solves the D-reduced problem 17.1.

1. Cover nonbasic components of the datum row by

$$\bar{\omega}_N^T = \bar{\omega}_N^T - \mu(\bar{z}_N^{-1})^T \bar{N}_{R'}^T \bar{N}_{R'}.$$
2. Stop if $J = \{j \in N \mid \bar{\omega}_j > 0\} = \emptyset$ (infeasible problem).
3. Determine column index q and β such that $\beta = \bar{z}_q / \bar{\omega}_q = \min_{j \in J} \bar{z}_j / \bar{\omega}_j$.
4. If $\beta \neq 0$, add $-\beta$ times of the datum row to the bottom row.
5. Go to step 9 if $R' = \emptyset$ or $\bar{a}_q(R') = 0$.
6. Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{i, q}|$.
7. Convert $\bar{a}_{pq}$ to 1, and eliminate the other nonzeros in the column by elementary transformations.
8. Set $k = k + 1$, bring q from N to B and p from R' to R , and go to step 2.

9. Determine $s \in \arg \max\{\bar{a}_{it,q} \mid t = 1, \dots, k\}$.
10. If $\bar{a}_{is,q} > 0$, update B, R by (20.14), and go to step 1.
11. Compute $\bar{x}_q = 1/\bar{\omega}_q$, $\bar{x}_{j_t} = -\bar{x}_q \bar{a}_{it,q}$, $t = 1, \dots, k$,
12. Stop (optimality achieved).

Note There are two layers of iterations in the preceding algorithm. The inner one consists of steps 2 to 7.

Example 20.5.1 Solve the following problem by Algorithm 20.5.1:

$$\begin{aligned}
 \min \quad & f = x_1 + 2x_2 + x_3 + 7x_4 + 3x_5 + x_6, \\
 \text{s.t.} \quad & -3x_1 \quad \quad -x_3 \quad \quad -3x_5 + 2x_6 = -1, \\
 & -2x_1 \quad -x_2 \quad +x_3 \quad -6x_4 + 6x_5 \quad -x_6 = -2, \\
 & \quad \quad +3x_2 \quad -2x_3 \quad \quad +8x_5 + 4x_6 = 0, \\
 & -6x_1 \quad \quad \quad +9x_4 - 3x_5 \quad \quad = 3, \\
 & x_j \geq 0, \quad j = 1, \dots, 6.
 \end{aligned}$$

Answer $\mu = 1$ and $\delta = 1/10$ are taken. Initial tableau:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-3		-1		-3	2	-1
-2	-1	1	-6	6	-1	-2
	3	-2		8	4	
-6			9	-3		3*
1	2	1	7	3	1	

Convert the preceding to a D-reduced tableau: $\max\{|-1|, |-2|, |0|, |3|\} = 3$, $r = 4$, and take row 4 as the datum row.

Multiply row 4 by $1/3$, and then add 1 and 2 times of row 4 to rows 1 and 2, respectively.

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-5		-1	3	-4	2	
-6	-1	1		4	-1	
	3	-2		8	4	
-2			3	-1		1
1	2	1	7	3	1	

Iteration 1:

$$k = 0. N = \{1, \dots, 6\}, R' = \{1, 2, 3\}.$$

$$1. \bar{N}_{R'} = \begin{pmatrix} -3 & -1 & -3 & -5 & 2 \\ -2 & -1 & 1 & -6 & 6 & -2 & -1 \\ 3 & -2 & & 8 & 3 & 4 \end{pmatrix}.$$

Set $\bar{\omega}_N^T = (-2, 0, 0, 3, -1, 0) - (1, 1/2, 1, 1/7, 1/3, 1)\bar{N}_{R'}^T\bar{N}_{R'}$
 $= (-1208/21, -71/3, 529/42, 124/7, -345/7, -841/42)$
 to as the left part of row 4:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-5		-1	3	-4	2	
-6	-1	1*		4	-1	
	3	-2		8	4	
-1208/21	-71/3	529/42	124/7	-345/7	-841/42	1
1	2	1	7	3	1	

2. $J = \{3, 4\} \neq \emptyset$.
3. $\beta = \min\{1/(529/42), 7/(124/7)\} = 1/(529/42)$, $q = 3$.
4. Add $-\beta$ times of the datum row to the bottom row.
5. $\bar{a}_q(R') \neq 0$.
6. $\max\{|-1|, |1|, |-2|\} = 2$, $p = 2$.
7. Add 1, 2, $-529/42$, -1 times of row 2 to rows 1,3,4,5, respectively.
8. $B = \{3\}$, $R = \{2\}$, $N = \{1, 2, 4, 5, 6\}$, $R' = \{1, 3\}$.

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-11	-1		3		1	
-6	-1	1		4	-1	
-12*	1			16	2	
379/21	-155/14		124/7	-299/3	-52/7	1
2945/529	2052/529		2959/529	159/23	1370/529	-42/529

Iteration 2:

2. $J = \{1, 4\} \neq \emptyset$.
3. $\beta = \min\{(2945/529)/(379/21), (2959/529)/(124/7)\}$
 $= (2945/529)/(379/21)$, $q = 1$.
4. Add $-\beta$ times of the datum row to the bottom row.
5. $\bar{a}_q(R') \neq 0$.
6. $\max\{|-11|, |-12|\} = 12$, $p = 3$.
7. Multiply row 3 by $-1/12$, and add 11, 6, $-379/21$, $-2945/529$ times of row 3 to rows 1, 2, 4, 5, respectively.
8. $B = \{3, 1\}$, $R = \{2, 3\}$, $N = \{2, 4, 5, 6\}$, $R' = \{1\}$.

Iteration 3:

2. $J = \{4\} \neq \emptyset$.
3. $\beta = (49/379)/(124/7)$, $q = 4$.
4. Add $-\beta$ times of the datum row to the bottom row.

x_1	x_2	x_3	x_4	x_5	x_6	RHS
	-23/12		3*	-44/3	-5/6	
	-3/2	1		-4	-2	
1	-1/12			-4/3	-1/6	
	-2411/252		124/7	-4763/63	-557/126	1
	5529/758		49/379	6477/172	1850/379	-147/379

5. $\bar{a}_q(R') \neq 0$.
6. $\max\{|3|\}, p = 1$.
7. Multiply row 3 by 1/3, and then add $-124/7$ and $-49/379$ times of rows 4 and 5, respectively.
8. $B = \{3, 1, 4\}$, $R = \{2, 3, 1\}$, $N = \{2, 5, 6\}$, $R' = \emptyset$.

x_1	x_2	x_3	x_4	x_5	x_6	RHS
	-23/36		1	-44/9	-5/18	
	-3/2	1		-4	-2	
1	-1/12			-4/3	-1/6	
	7/4			11	1/2	1
	1760/239			3477/91	1307/266	-49/124

Iteration 4:

2. $J = \{2, 5, 6\} \neq \emptyset$.
3. $\beta = \min\{(1760/239)/(7/4), (3477/91)/11, (1307/266)/(1/2)\}$
 $= (3477/91)/11, q = 5$.
4. Add $-\beta$ times of the datum row to the bottom row.

x_1	x_2	x_3	x_4	x_5	x_6	RHS
	-23/36		1	-44/9	-5/18	
	-3/2	1		-4	-2	
1	-1/12			-4/3	-1/6	
	7/4			11	1/2	1
	2027/1577				629/198	-383/99

5. $\bar{a}_q(R') = 0$.
9. $\max\{-44/9, -4, -4/3\} \leq 0$.
11. $\bar{x}_5 = 1/11$, $\bar{x}_B = (1/11)(4, 4/3, 44/9)^T = (4/11, 4/33, 4/9)^T$.
12. The optimal basic solution and optimal value are

$$x^* = (4/33, 0, 4/11, 4/9, 1/11, 0)^T, \quad f^* = 383/99.$$

Example 20.5.2 Solve the following D-reduced problem by Algorithm 20.5.1:

$$\begin{aligned}
 \min \quad & f = 3x_4 + x_6 + 2x_7, \\
 \text{s.t.} \quad & x_1 \quad \quad -2x_4 + 5x_5 - 5x_6 + 2x_7 = 0, \\
 & \quad +x_2 \quad \quad -3x_4 + 3x_5 - 2x_6 + x_7 = 0, \\
 & \quad \quad +x_3 + x_4 - 4x_5 + 6x_6 - 3x_7 = 0, \\
 & \quad \quad \quad -6x_4 \quad \quad + 3x_6 + x_7 = 0, \\
 & \quad \quad \quad +x_4 + 4x_5 + 2x_6 - x_7 = 1, \\
 & x_j \geq 0, \quad j = 1, \dots, 7.
 \end{aligned}$$

Answer $\mu = 1$ and $\delta = 1/10$ are taken. Initial tableau:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1			-2	5	-5	2	
	1		-3	3	-2	1	
		1	1	-4	6	-3	
			-6	0	3	1	
			1	4	2	-1	1
			3	0	1	2	

Row 5 is the datum row: $r = 5$. $B = \{1, 2, 3\}$, $R = \{1, 2, 3\}$, $N = \{4, 5, 6, 7\}$, $R' = \{4\}$.

Column 5 is the datum: $\bar{a}_{1,5} = 5 < 0$, $\bar{a}_{2,5} = 3 > 0$, $T = \{1, 2\}$.

Reset $B = \{3\}$, $R = \{3\}$, $N = \{1, 2, 4, 5, 6, 7\}$, $R' = \{1, 2, 4\}$.

Iteration 1:

$$1. \bar{N}_{R'} = \begin{pmatrix} 1 & -2 & 5 & -5 & 2 \\ & 1 & -3 & 3 & -2 & 1 \\ & & -6 & 3 & 1 \end{pmatrix}.$$

Cover nonbasic components of row 5 by

$$\begin{aligned}
 \bar{\omega}_N^T &= (0, 0, 1, 4, 2, -1) - (10, 10, 1/3, 10, 1, 1/2) \bar{N}_{R'}^T \bar{N}_{R'} \\
 &= (-166/3, -75/2, 1399/6, -2311/6, 2095/6, -452/3):
 \end{aligned}$$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
1			-2	5	-5*	2	
	1		-3	3	-2	1	
		1	1	-4	6	-3	
			-6	0	3	1	
-166/3	-75/2		1399/6	-2311/6	2095/6	-452/3	1
			3	0	1	2	

2. $J = \{4, 6\} \neq \emptyset$.

3. $\beta = \min\{3/(1399/6), 1/(2095/6)\} = 6/2095$, $q = 6$.

4. Add $-\beta$ times of the datum row to the bottom row.
5. $\bar{a}_q(R') \neq 0$.
6. $\max\{|-5|, |-2|, |3|\} = 5$, $p = 1$.
7. Multiply row 1 by $-1/5$, and then add 2, -6 , -3 , $-2095/6$, and -1 times of row 1 to rows 2, 3, 4, 5, and 6, respectively.
8. $B = \{3, 6\}$, $R = \{3, 1\}$, $N = \{1, 2, 4, 5, 7\}$, $R' = \{2, 4\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-1/5$			$2/5$	-1	1	$-2/5$	
$-2/5$	1		$-11/5$	1		$1/5$	
$6/5$		1	$-7/5$	2		$-3/5$	
$3/5^*$			$-36/5$	3		$11/5$	
$29/2$	$-75/2$		$187/2$	-36		-11	1
$332/2095$	$45/419$		$1397/599$	$1102/999$		$5094/2095$	$-6/2095$

Iteration 2:

2. $J = \{1, 4\} \neq \emptyset$.
3. $\beta = \min\{(332/2095)/(29/2), (1397/599)/(187/2)\}$
 $= (332/2095)/(29/2)$, $q = 1$.
4. Add $-\beta$ times of the datum row to the bottom row.
5. $\bar{a}_q(R') \neq 0$.
6. $\max\{|-2/5|, |3/5|\} = 3/5$, $p = 4$.
7. Multiply row 4 by $5/3$, and then add $1/5$, $2/5$, $-6/5$, $-29/2$, and $-332/2095$ times of row 4 to rows 1, 2, 3, 5, and 6, respectively.
8. $B = \{3, 6, 1\}$, $R = \{3, 1, 4\}$, $N = \{2, 4, 5, 7\}$, $R' = \{2\}$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
			-2		1	$1/3$	
	1		-7^*	3		$5/3$	
		1	13	-4		-5	
1			-12	5		$11/3$	
	$-75/2$		$535/2$	$-217/2$		$-385/6$	1
	$15/29$		$38/29$	$217/145$		$74/29$	$-2/145$

Iteration 3:

2. $J = \{4\} \neq \emptyset$.
3. $\beta = (38/29)/(535/2)$, $q = 4$.
4. Add $-\beta$ times of the datum row to the bottom row.
5. $\bar{a}_q(R') \neq 0$.
6. $\max\{|-7|\}$, $p = 2$.
7. Multiply row 2 by $-1/7$, and then add 2, -13 , 12 , $-535/2$, and $-38/29$ times of row 2 to rows 1, 3, 4, 5, and 6, respectively.

$$8. B = \{3, 6, 1, 4\}, R = \{3, 1, 4, 2\}, N = \{2, 5, 7\}, R' = \emptyset.$$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	-2/7			-6/7	1	-1/7	
	-1/7		1	-3/7		-5/21	
	13/7	1		11/7		-40/21	
1	-12/7			-1/7		17/21	
	5/7			43/7		-10/21	1
	75/107			217/107		920/321	-2/107

Iteration 4:

$$2. J = \{2, 5\} \neq \emptyset.$$

$$3. \beta = \min\{(75/107)/(5/7), (217/107)/(43/7)\} = (217/107)/(43/7), q = 5.$$

4. Add $-\beta$ times of the datum row to the bottom row:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	-2/7			-6/7	1	-1/7	
	-1/7		1	-3/7		-5/21	
	13/7	1		11/7		-40/21	
1	-12/7			-1/7		17/21	
	5/7			43/7		-10/21	1
	20/43					130/43	-15/43

$$5. \bar{a}_q(R') = 0.$$

$$9. \max\{11/7, -6/7, -1/7, -3/7\} = 11/7 > 0, s = 1.$$

$$10. T = \{1\}, B = \{6, 1, 4\}, R = \{1, 4, 2\}, N = \{2, 3, 5, 7\}, R' = \{3\}.$$

Iteration 5:

$$1. \bar{N}_{R'} = (13/7, 1, 11/7, -40/21). \text{ Cover nonbasic components of row 5 by } \\ \bar{\omega}_N^T = (5/7, 0, 43/7, -10/21) - (43/20, 10, 10, 43/130)\bar{N}_{R'}^T\bar{N}_{R'} \\ = (-22540/423, -12067/415, -7554/191, 15649/285):$$

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	-2/7			-6/7	1	-1/7	
	-1/7		1	-3/7		-5/21	
	13/7	1		11/7		-40/21*	
1	-12/7			-1/7		17/21	
	-22540/423	-12067/415		-7554/191		15649/285	1
	20/43					130/43	-15/43

$$2. J = \{7\} \neq \emptyset.$$

$$3. \beta = (130/43)/(15649/285), q = 7.$$

4. Add $-\beta$ times of the datum row to the bottom row.
5. $\bar{a}_q(R') \neq 0$.
6. $\max\{|-40/21|\}$, $p = 3$.
7. Multiply row 3 by $-21/40$, and then add $1/7$, $5/21$, $-17/21$, $-15649/285$, and $-130/43$ times of row 3 to rows 1, 2, 4, 5, and 6, respectively.
8. $B = \{6, 1, 4, 7\}$, $R = \{1, 4, 2, 3\}$, $N = \{2, 3, 5\}$, $R' = \emptyset$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	$-17/40$	$-3/40$		$-39/40$	1		
	$-3/8$	$-1/8$	1	$-5/8$			
	$-39/40$	$-21/40$		$-33/40$		1	
1	$-37/40$	$17/40$		$21/40$			
	$100463/401851$	$-19713/78851$		$23/4$			1
	$1397/411$	$658/411$		$2624/1205$			$-767/1899$

Iteration 6:

2. $J = \{2, 5\} \neq \emptyset$.
3. $\beta = \min\{(1397/411)/(100463/401851), (2624/1205)/(23/4)\}$
 $= (2624/1205)/(23/4)$, $q = 5$.
4. Add $-\beta$ times of the datum row to the bottom row.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	$-17/40$	$-3/40$		$-39/40$	1		
	$-3/8$	$-1/8$	1	$-5/8$			
	$-39/40$	$-21/40$		$-33/40$		1	
1	$-37/40$	$17/40$		$21/40$			
	$100463/401851$	$-19713/78851$		$23/4$			1
	$76/23$	$39/23$					$-18/23$

5. $\bar{a}_q(R') = 0$.
9. $\max\{-39/40, 21/40, -5/8, -33/40\} = 21/40 > 0$, $s = 2$.
10. $T = \{2\}$, $B = \{6, 4, 7\}$, $R = \{1, 2, 3\}$, $N = \{1, 2, 3, 5\}$, $R' = \{4\}$.

Iteration 7:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
	$-17/40$	$-3/40$		$-39/40$	1		
	$-3/8$	$-1/8$	1	$-5/8$			
	$-39/40$	$-21/40$		$-33/40$		1	
1	$-37/40^*$	$17/40$		$21/40$			
$-3379/222$	$4571/319$	$-3823/569$		$-3498/1561$			1
	$76/23$	$39/23$					$-18/23$

2. $J = \{2\} \neq \emptyset$.
3. $\beta = (76/23)/(4571/319)$, $q = 2$.
4. Add $-\beta$ times of the datum row to the bottom row.
5. $\bar{a}_q(R') \neq 0$.
6. $\max\{|-37/40|\}$, $p = 4$.
7. Multiply row 4 by $-40/37$, and then add $17/40$, $3/8$, $39/40$, $-4571/319$, and $-76/23$ times of row 4 to rows 1, 2, 3, 5, and 6, respectively.
8. $B = \{6, 4, 7, 2\}$, $R = \{1, 2, 3, 4\}$, $N = \{1, 3, 5\}$, $R' = \emptyset$.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-17/37$		$-10/37$		$-45/37$	1		
$-15/37$		$-11/37$	1	$-31/37$			
$-39/37$		$-36/37$		$-51/37$		1	
$-40/37$	1	$-17/37$		$-21/37$			
$3957/14641$		$-793/5868$		$218/37$			1
$2471/704$		$6039/1861$		$910/1761$			$-997/984$

Iteration 8:

2. $J = \{1, 5\} \neq \emptyset$.
3. $\beta = \max\{(2471/704)/(3957/14641), (910/1761)/(218/37)\}$
 $= (910/1761)/(218/37)$, $q = 5$.
4. Add $-\beta$ times of the datum row to the bottom row.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
$-17/37$		$-10/37$		$-45/37$	1		
$-15/37$		$-11/37$	1	$-31/37$			
$-39/37$		$-36/37$		$-51/37$		1	
$-40/37$	1	$-17/37$		$-21/37$			
$3957/14641$		$-793/5868$		$218/37$			1
$380/109$		$355/109$					$-120/109$

5. $\bar{a}_q(R') = 0$.
9. $\max\{-45/37, -31/37, -51/37, -21/37\} \leq 0$.
11. $\bar{x}_5 = 37/218$, $\bar{x}_B = (37/218)(45/37, 31/37, 51/37, 21/37)^T$
 $= (45/218, 31/218, 51/218, 21/218)^T$.
12. The basic optimal solution and optimal value:

$$\bar{x} = (0, 21/218, 0, 31/218, 37/218, 45/218, 51/218)^T, \quad \bar{f} = 120/109.$$

20.6 Deficient-Basis D-Reduced Gradient Algorithm

It is not easy to convert the tableau Algorithm 20.5.1 to the revised version because of the difficulty to update B_R^{-1} in step 10. The tableau Algorithm itself seems to be of practical value. However, the situation will be different if every outer iteration starts from the zero-order tableau instead. This idea is realized in the following algorithm.

Algorithm 20.6.1 (Deficient-Basis D-Reduced Gradient Algorithm) Given $\mu > 0, \delta > 0$. Initial: $\bar{z} \geq 0$. This algorithm solves the D-reduced problem 17.1.

1. Set $k = 0$, and give B, R, N, R' by (20.6).
2. Compute $\omega_N = N^T e_m - \mu N_{R'}^T N_{R'}^{-1} \bar{z}_N^{-1}$, and set $\bar{\omega}_N = \omega_N$.
3. Stop if $J = \{j \in N \mid \bar{\omega}_j > 0\} = \emptyset$ (infeasible problem).
4. Determine column index q and θ such that $\theta = \bar{z}_q / \bar{\omega}_q = \min_{j \in J} \bar{z}_j / \bar{\omega}_j$.
5. If $\theta \neq 0$, update $\bar{z}_N = \bar{z}_N - \theta \bar{\omega}_N$.
6. Compute $\bar{a}_q(R) = B_R^{-1} a_q(R)$ and $\bar{a}_q(R') = a_q(R') - B_{R'} \bar{a}_q(R)$.
7. Go to step 12 if $R' = \emptyset$ or $\bar{a}_q(R') = 0$.
8. Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{i,q}|$.
9. Update B_R^{-1} by (19.10).
10. Set $k = k + 1$, bring q from N to B and p from R' to R .
11. Compute $\bar{\omega}_N = \omega_N - N_R^T (B_R^{-T} \omega_B)$, and go to step 3.
12. Determine $s \in \arg \max \{\bar{a}_{i_s, q} \mid i_s = 1, \dots, k\}$.
13. Go to step 1 if $\bar{a}_{i_s, q} > 0$.
14. Compute $\bar{x}_q = 1 / \bar{\omega}_q$, $\bar{x}_B = -\bar{x}_q \bar{a}_q(R)$, $\bar{f} = c_q \bar{x}_q + c_B^T \bar{x}_B$.
15. Stop (the optimality achieved).

An alternative way is to utilize, to a larger extent, current information to form the datum row r . Only the first iteration starts from

$$\omega = A^T e_m - \mu A_{R'}^T \bar{A}_{R'}^{-1} \bar{z}^{-1}, \quad (20.15)$$

where $A_{R'}$ denotes the matrix, resulting from A by deleting row r . Assume that at the end of some subsequent inner iterations, the basic column and row index sets are B and R , not satisfying the optimality condition, the datum row $\bar{\omega}$ is

$$\bar{\omega}_B = 0, \quad \bar{\omega}_N = \omega_N - N_R^T (B_R^{-T} \omega_B),$$

and the basic row matrix $\bar{A}_R$ is

$$\bar{B}_R = I, \quad \bar{N}_R = B_R^{-1} N_R.$$

Then the datum row $\hat{\omega}$ for the next outer iteration is formed by combining the preceding two items as follows:

$$\begin{aligned}\hat{\omega}_B &= \bar{\omega}_B - \mu \bar{B}_R^T \bar{A}_R \bar{z}^{-1} = -\mu \left(\bar{z}_B^{-1} + B_R^{-1} N_R \bar{z}_N^{-1} \right), \\ \hat{\omega}_N &= \bar{\omega}_N - \mu \bar{N}_R^T \bar{A}_R \bar{z}^{-1} \\ &= \omega_N - N_R^T B_R^{-T} \omega_B - \mu \bar{N}_R^T \left(\bar{z}_B^{-1} + B_R^{-1} N_R \bar{z}_N^{-1} \right) \\ &= \omega_N - N_R^T B_R^{-T} (\omega_B - \hat{\omega}_B).\end{aligned}$$

Therefore, we have the following variant of Algorithm 20.6.1.

Algorithm 20.6.2 (Variant of the Dual Reduced Gradient Algorithm) *Same as Algorithm 20.6.1 except steps 1 and 2 replaced by*

1. Compute ω by (20.15), and set $\bar{\omega} = \omega$.
2. Set $\bar{\omega} = \omega$, $k = 0$; with sets B , R , N , and R' are determined by (20.6), and steps 13–15 replaced by
13. If $\bar{a}_{p,q} \leq 0$, compute $\bar{x}_q = 1/\bar{\omega}_q$, $\bar{x}_B = -\bar{x}_q \bar{a}_q(R)$, $\bar{f} = c_q \bar{x}_q + c_B^T \bar{x}_B$, and stop (optimality achieved).
14. Compute $v = -\bar{z}_B^{-1} - B_R^{-1} N_R \bar{z}_N^{-1}$.
15. Compute $\omega_N = \omega_N - N_R^T B_R^{-T} (\omega_B - v)$, $\omega_B = v$.
16. Go to step 1.

It is noted that the preceding variant solves two extra systems in each iteration.

Reference

Pan P-Q (2005) A revised dual projective pivot algorithm for linear programming. SIAM J Optim 16:49–68

Chapter 21

Face Method with Cholesky Factorization



The deficient-basis method performs well, but the only drawback is that the number of columns of the basis increases in the solution process, weakening its benefits, which lets us believe that it is impossible to overcome degeneracy by moving vertex by vertex.

Based on the Cholesky factorization, the face method presented in this chapter will no longer move vertex by vertex, but face by face. Therefore, it would bring about the hope of a complete escape from the degeneracy “nightmare.” In the subsequent sections, we address such themes as the steepest descent direction, update of the solution, face contraction, optimality test, face expansion, face algorithm supported by the Phase-I based on the single-artificial-variable approach, and related computational results. Before making our comments on the face method in the final section, we spend two sections on its variants: the affine face algorithm and generalized face algorithm.

Consider the reduced problem (16.1), i.e.,

$$\begin{aligned} \min \quad & f, \\ \text{s.t.} \quad & Ax = b, \quad x_j \geq 0, \quad j = 1, \dots, n, \end{aligned} \quad (21.1)$$

where $A \in \mathcal{R}^{(m+1) \times (n+1)}$, $b \in \mathcal{R}^{m+1}$, $\text{rank } A = m + 1$, $m < n$, $a_{n+1} = Ae_{n+1} = -e_{m+1}$. The assumption on the rank of A will be dropped later.

Assume that $B \in \mathcal{R}^{(m+1) \times k}$, $N \in \mathcal{R}^{(m+1) \times (n-k+1)}$ is a partition of A , where

$$\text{rank } B = m + 1, \quad m + 1 \leq k \leq n + 1.$$

The matrices B and N are called *face matrix* and *nonface matrix*, and the associated variables are *face variables* and *nonface variables*. Without confusion, denote the

associated index sets by B and N , respectively, i.e.,

$$B = \{1, \dots, k-1, n+1\}, \quad N = \{1, \dots, n+1\} \setminus B.$$

Note that $n+1$ is the last index of B .

Thereafter, it is assumed that the k th (last) column of the face matrix equals the $(n+1)$ th (last) column of A , i.e.,

$$Be_k = -e_{m+1}. \quad (21.2)$$

The face, associated with the face matrix B , is defined by

$$P_B = \{x \in A \mid Bx_B = b; x_s \geq 0, s = 1, \dots, k-1, x_N = 0\}. \quad (21.3)$$

A point/solution on the face is called *face point/solution*. Thus, the zero-dimensional face has the unique face point—vertex.

Definition 21.0.1 A face is a level face if the objective value is constant over it. A level face is an *optimal face* if the related objective value equals the optimal value.

Assume that $\bar{x}$ is the current face point on P_B , i.e., $\bar{x} \in P_B$. As $\bar{x}$ is uniquely determined by $\bar{x}_B$, the two will not be strictly distinguished hereafter; for simplicity, $\bar{x}_B \in P_B$ is often used in place of $\bar{x} \in P_B$.

21.1 Steepest Descent Direction

Firstly, we determine the search direction. Consider the following subprogram:

$$\begin{aligned} &\min f, \\ &\text{s.t. } Bx_B = b, \\ &\quad x_B \geq 0, \quad n+1 \notin B, \end{aligned} \quad (21.4)$$

which actually minimizes the objective over the face P_B .

Introduce $k \times k$ orthogonal projection matrix, i.e.,

$$P = I - B^T(BB^T)^{-1}B. \quad (21.5)$$

Then, the orthogonal projection of the objective gradient, e_k , of the subprogram onto the null space of B is

$$\Delta_B = Pe_k = e_k - B^T\bar{y}, \quad BB^T\bar{y} = -e_{m+1}, \quad (21.6)$$

and $-\Delta_B$ is a desirable search direction. In fact, we have the following result.

Proposition 21.1.1 *The vector Δ_B satisfies $B\Delta_B = 0$.*

Proof By (21.6), it is easy to verify validity of the equality. $\square$

Proposition 21.1.2 *The following statements are equivalent:*

$$(i) \Delta_B \neq 0; \quad (ii) \Delta_k > 0; \quad (iii) e_k \notin \text{range}(B^T).$$

Proof Premultiply the first expression of (21.6) by Δ_B^T . Then combining the result and Proposition 21.1.1 gives

$$\Delta_B^T \Delta_B = \Delta_k.$$

Therefore, statements (i) and (ii) are equivalent. If $\Delta_B = 0$, it is known by the first expression of (21.6) that $e_k = B^T \bar{y}$, i.e., $e_k \in \text{range}(B^T)$, and hence (iii) results in (i). Conversely, if $e_k \in \text{range}(B^T)$, then there is some vector u such that

$$e_k = B^T u. \quad (21.7)$$

Premultiplying the preceding by B together with (21.2) gives

$$-e_{m+1} = BB^T u.$$

Consequently, $u = \bar{y}$ follows from the preceding, the second expression of (21.6), and the full row rank of B . Furthermore, it holds by (21.7) that

$$e_k = B^T \bar{y}.$$

Thus substituting the preceding into the first expression of (21.6) gives $\Delta_B = 0$. Therefore, (i) and (iii) are equivalent. $\square$

Proposition 21.1.3 *If $\Delta_B \neq 0$, then it holds that*

$$e_k^T \Delta_B = \Delta_k > 0, \quad (21.8)$$

and

$$-e_k^T \Delta_B / \|\Delta_B\| \leq e_k^T v / \|v\|, \quad \forall 0 \neq v \in \text{Null}(B). \quad (21.9)$$

Proof The expression (21.8) can be derived directly from Proposition 21.1.2.

For P defined by (21.5), it is verified that

$$Pv = v \neq 0, \quad \forall 0 \neq v \in \text{Null}(B). \quad (21.10)$$

Furthermore, it follows from the Cauchy inequality that

$$(-Pe_k)^T(Pv) \leq \|Pe_k\| \|Pv\|.$$

Multiplying the preceding by $1/\|Pv\|$ gives

$$(-Pe_k)^T(Pv)/\|Pv\| \leq \|Pe_k\|,$$

i.e.,

$$(-Pe_k)^T(Pv)/\|Pv\| \leq (Pe_k)^T(Pe_k)/\|Pe_k\|.$$

Consequently, the preceding together with $P^2 = P$, $P^T = P$, (21.10), and $\Delta_B = Pe_k$ gives (21.9). $\square$

According to Proposition 21.1.3, vector $-\Delta_B \neq 0$ forms the most-obtuse-angle in the null space of B with the objective gradient e_k of the subprogram and is hence the steepest descent one. In this sense, taking the vector as a search direction is the “best” choice.

It is noted that the computation of Δ_B mainly lies on solving the $(m+1) \times (m+1)$ system

$$BB^T y = -e_{m+1}. \quad (21.11)$$

Since B is of full row rank, there exists the Cholesky factorization (Golub and Van Loan 1989)

$$BB^T = LL^T, \quad (21.12)$$

where L is nonsingular lower triangular. Therefore, system (21.11) becomes

$$LL^T y = -e_{m+1}, \quad (21.13)$$

which can be put into the following two triangular systems:

$$Lv = -e_{m+1}, \quad L^T y = v.$$

The solution to the first system is readily available, i.e., $v = -(1/v)e_{m+1}$, where v is the $(m+1)$ th diagonal of L . Consequently, there is only one triangular system left to be solved, i.e.,

$$L^T y = -(1/v)e_{m+1}. \quad (21.14)$$

21.2 Updating of Face Solution

The new iterate is defined by the following line search scheme:

$$\hat{x}_B = \bar{x}_B - \alpha \Delta_B, \quad (21.15)$$

where α is a stepsize to be determined. If the index set

$$J = \{j \in B \mid \Delta_j > 0, j \neq n+1\}$$

is nonempty, determine α and p such that

$$\alpha = \bar{x}_p / \Delta_p = \min_{j \in J} \bar{x}_j / \Delta_j \geq 0, \quad (21.16)$$

which is the maximum possible stepsize for the new iterate remaining within the feasible region. If $\bar{x}_j = 0$ for some $j \in B$, $j \neq n+1$, then face point $\bar{x}$ is said to be *degenerate*. In this case, the stepsize α could vanish, and the resulting iterate is actually not “new” but the same as the old one. Note that whether $\bar{x}$ is degenerate only depends upon the first $k-1$ components of $\bar{x}_B$.

Lemma 21.2.1 *Assume that $\bar{x}_B \in P_B$ and $\Delta_B \neq 0$.*

- (i) *If $J \neq \emptyset$, then $\hat{x}_B \in P_B$ is a boundary point. The objective value does not increase and strictly decreases if $\bar{x}_B$ is nondegenerate.*
- (ii) *If $J = \emptyset$, the original problem is lower unbound.*

Proof

- (i) $\Delta_B \neq 0$ and $J \neq \emptyset$ together ensure that (21.16) is well-defined. As $\bar{x}_B \in P_B$, it is known from Proposition 21.1.1 that $\hat{x}_B \in P_B$; furthermore, it is clear that $\hat{x}_p = 0$, and hence $\hat{x}$ is a boundary point, associated with the new objective value

$$\hat{f} = \bar{f} - \alpha \Delta_k. \quad (21.17)$$

It is known from (21.16) that $\alpha \geq 0$. Besides, from $\Delta_B \neq 0$ and Proposition 21.1.2, it follows that $\Delta_k > 0$, and hence $\hat{f} \leq \bar{f}$. If $\bar{x}_B$ is nondegenerate, then $\alpha > 0$, leading to $\hat{f} < \bar{f}$.

- (ii) When $J = \emptyset$, it is clear that $\hat{x}_B \in P_B$ holds for any $\alpha > 0$. It is then known by (21.17) that

$$\hat{f} \rightarrow -\infty \quad \text{as} \quad \alpha \rightarrow \infty.$$

Therefore the objective value is lower unbounded over P_B , hence over the feasible region.

□

21.3 Face Contraction

Assume that $\Delta_B \not\geq 0$ and that the new iterate $\hat{x}_B$ was determined by (21.15) with $\hat{x}_p = 0$. Updating B and N accordingly gives the face and nonface index sets, associated with the new iterate, i.e.,

$$\check{B} = B \setminus \{p\}, \quad \check{N} = N \cup \{p\}.$$

The associated face matrix $\check{B} \in \mathcal{R}^{(m+1) \times (k-1)}$ results from the old B by dropping the p -indexed column. Then we have the following result.

Proposition 21.3.1 *If rank $B = m + 1$, then rank $\check{B} = m + 1$.*

Proof It is clear that rank $\check{B} \leq m + 1$. It is known by (21.16) that $\Delta_p \neq 0$ and by Proposition 21.1.1 that

$$\sum_{j \in B} \Delta_j a_j = B \Delta_B = 0,$$

and thus

$$a_p = -(1/\Delta_p) \sum_{j \in B, j \neq p} \Delta_j a_j.$$

The above equality implies that $a_p \in \text{range } \check{B}$. Therefore, rank $\check{B} < m + 1$ leads to rank $B < m + 1$, as contradicts the assumption. Therefore rank $\check{B} = m + 1$. $\square$

The preceding proposition implies that the new face matrix $\check{B}$ is again of full row rank, and hence there exists the Cholesky factorization

$$\check{B} \check{B}^T = \check{L} \check{L}^T.$$

Saunders (1972) offers a procedure to obtain the new factor from its predecessor, as is put into the following algorithm.

Subalgorithm 21.3.1 (Subalgorithm: contraction) Initial: the Cholesky factor L of $B B^T$. This algorithm finds the Cholesky factor $\check{L}$ of $\check{B} \check{B}^T$.

- (1) Solve lower triangular system $Lu = a_p$.
- (2) Compute $\gamma = \sqrt{1 - \|u\|^2}$.
- (3) Determine Givens rotations $J_{m+1}, \dots, N_1$, where J_i is in the $(m+2, i)$ -plane, such that

$$N_1 \dots J_{m+1} \begin{pmatrix} u \\ \gamma \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

- (4) Compute $N_1 \dots J_{m+1} \begin{pmatrix} L^T \\ 0 \end{pmatrix} = \begin{pmatrix} \check{L}^T \\ a_p^T \end{pmatrix}$, to obtain $\check{L} \in \mathcal{R}^{(m+1) \times (m+1)}$.
- (5) Return.

21.4 Optimality Test

In case of $\Delta_B = 0$, the search direction is not well-defined by (21.6). In particular, it is the case when $k = m + 1$, as the projection matrix is a zero one.

Lemma 21.4.1 *Assume that Δ_B is defined by (21.6). If Δ_B vanishes, then P_B is a level face, and vice versa if $0 < \bar{x}_B \in P_B$.*

Proof Assume that $\bar{y}$ satisfies the second expression of (21.6). From $\Delta_B = 0$ and the first expression of (21.6), it follows that

$$e_k = B^T \bar{y}.$$

Transpose the two sides of the preceding equality, and postmultiply the result by any $x_B \in P_B$ to obtain

$$e_k^T x_B = \bar{y}^T B x_B,$$

which, combined with and $B x_B = b$, gives

$$f = \bar{y}^T b.$$

This implies that the objective value is constant over P_B , and hence P_B is a level face.

Now assume that P_B is a level face and that $0 < \bar{x}_B \in P_B$ with $\Delta_B \neq 0$. In the case when $\Delta_B \leq 0$, it is known by (ii) of Lemma 21.2.1 that the objective value is lower unbounded over P_B , which contradicts that P_B is a level face: when $\Delta_B \not\leq 0$, since x_B is nondegenerate, it is known from (i) of Lemma 21.2.1 that the objective value, associated with $\hat{x}_B \in P_B$ given by (21.15) and (21.16), strictly decreases, as also contradicts that P_B is a level face. Therefore, Δ_B vanishes. $\square$

Now consider the dual problem of (21.1)

$$\begin{aligned} & \max (b^T, 0)^T y, \\ & \text{s.t.} \quad \begin{pmatrix} B^T \\ N^T \end{pmatrix} y + \begin{pmatrix} z_B \\ z_N \end{pmatrix} = \begin{pmatrix} e_k \\ 0 \end{pmatrix}, \quad z_B, z_N \geq 0, \quad z_{n+1} = 0. \end{aligned} \quad (21.18)$$

Lemma 21.4.2 Assume that $\bar{x}_B \in P_B$ and $\Delta_B = 0$. If $\bar{z}_N = -N^T \bar{y} \geq 0$, then $(\bar{x}_B, \bar{x}_N)$ and $(\bar{z}_B, \bar{z}_N, \bar{y})$ are a pair of primal and dual optimal solutions, where $\bar{x}_N = 0$, $\bar{z}_B = 0$, and P_B is an optimal face to (21.1).

Proof $(\bar{x}_B, \bar{x}_N)$ ($\bar{x}_N = 0$) is clearly a feasible solution to (21.1). On the other hand, combining (21.6) and $\Delta_B = 0$ gives $B^T \bar{y} = e_k$. Therefore, it is easy to verify that $(\bar{z}_B, \bar{z}_N, \bar{y})$ is a feasible solution to (21.18), satisfying the complementarity condition, and hence is a pair of primal and dual optimal solutions. By Lemma 21.4.1, $\Delta_B = 0$ implies that P_B is a level face. It includes the optimal solution and hence is an optimal face. $\square$

According to Proposition 21.1.2, $\Delta_B = 0$ if and only if $\Delta_k = 0$. Therefore condition $\Delta_B = 0$ may be replaced by $\Delta_k = 0$ simply.

21.5 Face Expansion

Assume $\Delta_B = 0$ and $\bar{z}_N = -N^T \bar{y} \not\geq 0$. In this case, optimality of the level face P_B cannot be asserted, but a column index q can be determined such that

$$q \in \arg \min_{j \in N} \bar{z}_j.$$

Accordingly, update the face and nonface index sets by

$$\widehat{B} = B \cup \{q\}, \quad \widehat{N} = N \setminus \{q\}. \quad (21.19)$$

As a result, the number of columns of the face matrix will increase by 1.

Thus, the q -indexed column is added to the face matrix. Denote the resulting new face matrix by $\widehat{B} \in \mathcal{R}^{(m+1) \times (k+1)}$, which is still of full row rank. The corresponding updating of the Cholesky factor is simple (Golub 1965), stated as follows.

Subalgorithm 21.5.1 (Subalgorithm: expansion) Initial: the Cholesky factor L of BB^T . This algorithm finds the Cholesky factor $\widehat{L}$ of $\widehat{B}\widehat{B}^T$.

Determine Givens rotations $N_1, \dots, J_{m+1}$, where J_i is in the $(i, m+2)$ -plane, such that

$$J_{m+1} \dots N_1 \begin{pmatrix} L^T \\ a_q^T \end{pmatrix} = \begin{pmatrix} \widehat{L}^T \\ 0 \end{pmatrix},$$

which gives $\widehat{L} \in \mathcal{R}^{(m+1) \times (m+1)}$.

The validity of the preceding subalgorithm is verified by

$$\widehat{L}\widehat{L}^T = LL^T + a_q a_q^T = BB^T + a_q a_q^T = \widehat{B}\widehat{B}^T.$$

Another scheme, seeming to be attractive, is to use

$$\widehat{B} = B \cup \{j \in N \mid \bar{z}_j < 0\}, \quad \widehat{N} = A \setminus \widehat{B}$$

rather than (21.19), though adding two columns to the face matrix will complicate updating of the Cholesky factor.

21.6 Face Algorithm

Any submatrix $B \in \mathcal{R}^{(m+1) \times k}$ with $m+1 \leq k \leq n+1$ and $\text{rank}(B) = m+1$ can be taken as an initial face matrix although we prefer the maximum one, i.e., $k = n+1$, $B = A$.

Now drop the assumption that A is of full row rank. Assume that an orthogonal matrix $[Q_1, Q_2] \in \mathcal{R}^{(n+1) \times (n+1)}$ and a permutation $\Pi \in \mathcal{R}^{(m+1) \times (m+1)}$ have been found such that

$$A^T \Pi = [Q_1, Q_2] \begin{pmatrix} R & T \\ 0 & 0 \end{pmatrix},$$

where $R \in \mathcal{R}^{r \times r}$ is a nonsingular upper triangular. Then $L = R^T$ is the initial Cholesky factor. In fact, $\text{rank } A = r$, and the last $m - r + 1$ equations of $\Pi^T A x = \Pi^T b$ are redundant: if the original notation is still used to denote the problem, resulting from dropping these redundant rows, then $A^T = Q_1 R$, and the wanted Cholesky factorization is

$$A A^T = R^T Q_1^T Q_1 R = L L^T. \quad (21.20)$$

The overall steps can be put into the following algorithm.

Algorithm 21.6.1 [Face algorithm with Cholesky factorization] Initial: (B, N) , $m+1 \leq k \leq n+1$, feasible solution $\bar{x}$, and the Cholesky factor L of $B B^T$. This algorithm solves the reduced problem (21.1).

1. Solve upper triangular system $L^T \bar{y} = -(1/\nu) e_{m+1}$, where ν is the $(m+1)$ th diagonal of L .
2. If $k > m+1$, compute $\Delta_B = e_k - B^T \bar{y}$.
3. Go to step 10 if $k = m+1$ or $\Delta_B = 0$.
4. Stop if $J = \{j \in B \mid \Delta_j > 0, j \neq n+1\} = \emptyset$.
5. Determine stepsize α and index p such that $\alpha = \bar{x}_p / \Delta_p = \min_{j \in J} \bar{x}_j / \Delta_j$.
6. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B - \alpha \Delta_B$.
7. Update (B, N) by braining p from B to N .
8. Update L by Subalgorithm 21.3.1.
9. Set $k = k - 1$, and go to step 1.

10. Compute $\bar{z}_N = -N^T \bar{y}$.
11. Stop if $\bar{z}_N \geq 0$.
12. Determine index $q \in \arg \min_{j \in N} \bar{z}_j$.
13. Update L by Subalgorithm 21.5.1.
14. Bring q from N to B .
15. Set $k = k + 1$.
16. Go to step 1.

Theorem 21.6.1 *Under the assumption of nondegeneracy, Algorithm 21.6.1 terminates either at*

- (i) *Step 4, detecting lower unboundedness of (21.1), or at*
- (ii) *Step 11, giving an optimal face together with a pair of primal and dual optimal solutions*

Proof Under the assumption of nondegeneracy, the proof of the finiteness of Algorithm 21.6.1 is the same as for the simplex algorithm. The meanings of the exits come from Lemmas 21.2.1 and 21.4.2. $\square$

In a contraction iteration, the face algorithm solves two triangular systems (including one involved in updating L by the subalgorithm 21.3.1), whereas it solves only one in an expansion iteration, in contrast to the simplex algorithm that solves four. Therefore, the former involves less computational effort than the latter per iteration. In addition, there is no need at all for storing the Givens rotations, created to update the Cholesky factor.

The dimension of the end optimal face produced by the algorithm should be more than 0, in general, though it would not be necessarily the optimal set.

21.6.1 Face Phase-I: Single-Artificial-Variable

Any Phase-I algorithm can be utilized to provide a feasible solution to get the face algorithm started. This section introduces a Phase-I algorithm involving only a single artificial variable.

The auxiliary problem used is the same as (14.11), i.e.,

$$\begin{aligned} &\min f, \\ &\text{s.t. } Ax + a_{n+1}f = b, \quad x, f \geq 0, \end{aligned}$$

where $a_{n+1} = (b - Bh) / \|b - Bh\|$. Assume that $b - Bh \neq 0$ (see Sect. 14.2).

At the beginning of Phase-I, set

$$B = \{1, \dots, n, n+1\}, \quad N = \emptyset. \quad (21.21)$$

Thus, the components of the initial auxiliary feasible solution are all equal to 1, and the auxiliary problem can be solved by a slight modification of Algorithm 21.6.1.

In step 1, two triangular systems are solved to obtain $\bar{y}$. In step 5, $n + 1$ is taken as a candidate for selection of row index p . Once it leaves B (while $\bar{f}$ becomes 0), a feasible solution is obtained. In addition, step 4 should be dropped because the auxiliary problem is certainly lower bounded.

The steps are put in the following algorithm.

Algorithm 21.6.2 [Face Phase-I: single-artificial-variable] Initial: $B = \{1, \dots, n, n + 1\}$, $N = \emptyset$, $k = n + 1$; $\bar{x} = h$, $\bar{f} = 1$; $BB^T = LL^T$. This algorithm finds a feasible solution to (21.1).

Same as Algorithm 21.6.1, except for its steps 1–6 replaced by

1. Solve $LL^T\bar{y} = a_{n+1}$.
2. If $k > m + 1$, compute $\Delta_B = e_k - B^T\bar{y}$.
3. Go to step 10 if $k = m + 1$ or $\Delta_B = 0$.
4. Determine α and index p such that $\alpha = \bar{x}_p/\Delta_p = \min\{\bar{x}_j/\Delta_j \mid \Delta_j > 0, j \in B\}$.
5. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B - \alpha\Delta_B$.
6. Stop if $\bar{f} = 0$ (feasibility achieved).

Example 21.6.1 Solve the following problem by the two-phase face algorithm:

$$\begin{aligned} \min f &= x_1 + 2x_3 + 3x_4, \\ \text{s.t. } 2x_1 - x_2 - 2x_3 - x_4 &= -1, \\ &\quad +2x_2 - 2x_4 + x_5 = 1, \\ &\quad x_1 - 2x_2 - x_5 = -2, \\ x_j &\geq 0, \quad j = 1, \dots, 5. \end{aligned}$$

Answer Phase-I: call Algorithm 21.6.2.

Initial: $k = 6$, $B = \{1, 2, 3, 4, 5, 6\}$, $N = \emptyset$, $\bar{x} = e$, $a_6 = (1, 0, 0)^T$. B and the Cholesky factor of BB^T are

$$B = \begin{pmatrix} 2 & -1 & -2 & -1 & 1 \\ & 2 & & -2 & 1 \\ & & 1 & -2 & \\ & & & -1 & \\ & & & & 1 \end{pmatrix}, \quad L = \begin{pmatrix} -1257/379 & & & & \\ & -3 & & & \\ & & -2394/1985 & 5/3 & -2804/2109 \end{pmatrix}.$$

Iteration 1:

1. $\bar{y} = (29/175, -4/35, -36/175)^T$.
2. $\Delta_B = (-22/175, -3/175, 58/175, -11/175, -16/175, 146/175)^T \neq 0$.
4. $\alpha = \min\{1/(58/175), 1/(146/175)\} = 175/146$, $p = 6$.
- 5.

$$\begin{aligned} \bar{x}_B &= (1, 1, 1, 1, 1, 1)^T - (175/146)(-22/175, -3/175, 58/175, \\ &\quad -11/175, -16/175, 146/175)^T \\ &= (84/73, 149/146, 44/73, 157/146, 81/73, 0)^T. \end{aligned}$$

6. $\bar{x}_6 = 0$, return with

$$\bar{f} = (1, 0, 2, 3, 0)(84/73, 149/146, 44/73, 157/146, 81/73)^T = 815/146.$$

Phase-II: call Algorithm 21.6.1.

Initial: $k = 6$, $B = \{1, 2, 3, 4, 5, 6\}$, $N = \emptyset$, $\bar{x} = (84/73, 149/146, 44/73, 157/146, 81/73, 815/146)^T$. B and the Cholesky factor of BB^T are

$$B = \begin{pmatrix} 2 & -1 & -2 & -1 & & \\ & 2 & & -2 & 1 & \\ & 1 & -2 & & & -1 \\ & 1 & & 2 & 3 & 0 & -1 \end{pmatrix}, \quad L = \begin{pmatrix} -721/228 & & & & & \\ & -3 & & & & \\ -721/570 & 5/3 & -740/581 & & & \\ 721/456 & 2 & 905/3458 & -392/135 & & \end{pmatrix}.$$

Iteration 2:

1. $\bar{y} = (-61/1231, -114/1231, -30/1231, -146/1231)^T$.
2. $\Delta_B = (298/1231, 107/1231, 170/1231, 149/1231, 84/1231, 1085/1231)^T$.
4. $J \neq \emptyset$.
- 5.

$$\begin{aligned} \alpha &= \min\{(84/73)/(298/1231), (149/146)/(107/1231)(44/73)/(170/1231), \\ &\quad (157/146)/(149/1231), (81/73)/(84/1231)(815/146)/(1085/1231)\} \\ &= 1305/299, p = 3. \end{aligned}$$

6.

$$\begin{aligned} \bar{x}_B &= (84/73, 149/146, 44/73, 157/146, 81/73, 815/146)^T \\ &\quad - (1305/299)(298/1231, 107/1231, 170/1231, 149/1231, \\ &\quad 84/1231, 1085/1231)^T \\ &= (1211/12867, 109/170, 0, 93/170, 69/85, 15366/8855)^T. \end{aligned}$$

7. $B = \{1, 2, 4, 5, 6\}$, $N = \{3\}$.

$$8. L = \begin{pmatrix} -2158/881 & & & & \\ & -3 & & & \\ -1762/1079 & 5/3 & 963/1292 & & \\ & 881/2158 & 2 & -2889/1292 & 524/387 \end{pmatrix}.$$

9. $k = 5$.

Iteration 3:

1. $\bar{y} = (1, -14/11, -18/11, -6/11)^T$.
2. $\Delta_B = (2/11, 3/11, 1/11, -4/11, 83204/183049)^T$.
4. $J \neq \emptyset$.

5.

$$\begin{aligned}\alpha &= \min\{(1211/12867)/(2/11), (109/170)/(3/11), \\ &\quad (93/170)/(1/11)(15366/8855)/(83204/183049)\} \\ &= 1203/2324, \quad p = 1.\end{aligned}$$

6.

$$\begin{aligned}\bar{x}_B &= (1211/12867, 109/170, 93/170, 69/85, 15366/8855)^T \\ &\quad - (1203/2324)(2/11, 3/11, 1/11, -4/11, 83204/183049)^T \\ &= (0, 1/2, 1/2, 1, 3/2)^T.\end{aligned}$$

7. $B = \{2, 4, 5, 6\}$, $N = \{1, 3\}$.

$$8. L = \begin{pmatrix} 1393/985 & & & & \\ & -3 & & & \\ 1393/985 & 5/3 & -1121/2378 & & \\ -2378/1121 & 2 & 985/1393 & -1 & \end{pmatrix}.$$

9. $k = 4$.

Iteration 4:

$$1. \bar{y} = (0, -3/2, -3/2, -1)^T.$$

$$3. k = 4.$$

$$10. \bar{z}_N = N^T \bar{y} = (5/2, 2)^T \geq 0.$$

11. A zero-dimensional optimal face is obtained. The optimal basic solution and optimal value are

$$x^* = (0, 1/2, 0, 1/2, 1, 3/2)^T, \quad x_6^* = 3/2.$$

21.6.2 Computational Results

Preliminary computational experiments were conducted. Compiled using the Visual FORTRAN 5.0, test codes were run under a Windows XP system Home Edition Version 2002 on an IBM PC with an Intel(R) Pentium(R) processor 1.00 GB of 1.86 GHz memory and about 16 digits of precision.

The following two dense codes in FORTRAN 77 are compared:

1. RSA: The revised simplex algorithm, supported by the classical Phase-I (see Notation).
2. FALP: Face Algorithm 21.6.1, supported by Algorithm 21.6.2 being as the Phase-I.

The test set of problems are the 25 smallest Netlib problems listed in Appendix C (Table C.4: AFIRO-SCAGR25). From Appendix E (Table E.4), it is seen that average ratios of RSA to FALP are as follows:

Total iterations ratio:1.51, total time ratio:4.91.

It is indeed impressive that the face algorithm outperformed the simplex algorithm with a time ratio near 5. Related computational experiments are reported in detail in Appendix E.

21.7 Affine Face Algorithm

It is seen from (21.15) and (21.16) that in case $\bar{x}_B$ is degenerate, α would vanish and hence leading to a zero stepsize. It is more than that. It is known from (21.17) that the difference between the new and old objective values is

$$\hat{f} - \bar{f} = -\alpha \Delta_k.$$

Therefore, even though the first $k - 1$ components of $\bar{x}_B$ are positive in principle, the stepsize would still be so small that the decrement of the objective value is negligible if some components are too close to zero. Actually, this depends on the relative magnitudes of components. To avoid too small stepsizes, the affine interior-point algorithm (Part I) exploits affine transformation to attain a relative equilibrium between components. In this section, the affine transformation is employed to modify the search direction, used in the face algorithm, so that it is not only descent but also points to the interior of the feasible region.

Assume that $\bar{x}_B$ is nondegenerate. Denote by $\bar{X}_B$ the $k \times k$ diagonal matrix, whose diagonals are components of vector $\bar{x}_B$, except for the k th diagonal 1, i.e.,

$$\bar{X}_B = \text{diag}(\bar{x}_1, \dots, \bar{x}_{k-1}, 1). \quad (21.22)$$

Consider the affine transformation

$$x_B = \bar{X}_B x'_B. \quad (21.23)$$

This transformation is invertible and determines a 1-to-1 correspondence between $\bar{x}_B$ and

$$\bar{x}'_B = (\underbrace{1, \dots, 1}_{k-1}, \bar{f})^T.$$

Despite the fact that some of the first $k - 1$ components of $\bar{x}_B$ may be close to zero, the corresponding components of $\bar{x}'_B$ are all equal to 1, so that a line search along any direction in $\bar{x}'_B$ -space will not lead to a too small stepsize.

By carrying out transformation (21.23), the subprogram (21.4) is converted to

$$\begin{aligned} \min \quad & f, \\ \text{s.t.} \quad & (B\bar{X}_B)x'_B = b, \\ & x'_B \geq 0, \quad n+1 \notin B. \end{aligned}$$

The projection matrix onto the null space of $B\bar{X}_B$ is then

$$P = I - \bar{X}_B B^T (B\bar{X}_B^2 B^T)^{-1} B\bar{X}_B. \quad (21.24)$$

Therefore, the orthogonal projection of the negative objective gradient onto the null space is $-Pe_k$, corresponding to vector

$$\Delta_B = -\bar{X}_B Pe_k$$

in x_B -space. Then, combining

$$\bar{X}_B e_k = e_k, \quad B e_k = -e_{m+1},$$

and (21.24) gives that search direction below:

$$\Delta_B = -e_k + \bar{X}_B^2 B^T \bar{y}, \quad (B\bar{X}_B^2 B^T) \bar{y} = -e_{m+1}. \quad (21.25)$$

If the Cholesky factorization $B\bar{X}_B^2 B^T = LL^T$ is available, the $\bar{y}$ can be obtained by solving an upper triangular system only, i.e.,

$$L^T \bar{y} = -(1/v)e_{m+1},$$

where v is the $(m+1)$ th diagonal of L .

If $\bar{x}$ is degenerate, nevertheless, transformation (21.23) is not invertible and becomes useless. This difficulty may be overcome as follows. Let δ be a given small positive number. Define

$$\hat{x}_j = \begin{cases} \bar{x}_j, & \text{If } \bar{x}_j \geq \delta, \\ 1, & \text{If } \bar{x}_j < \delta, \end{cases} \quad j \in B, \quad j \neq n+1. \quad (21.26)$$

Then construct an affine transformation with $\hat{x}_B$ in place of $\bar{x}_B$, i.e., by using

$$\bar{X}_B = \text{diag}(\hat{x}_1, \dots, \hat{x}_{k-1}, 1), \quad (21.27)$$

instead of (21.22).

The difference between the resulting algorithm and the face algorithm lies in their search directions only. The overall steps are put into the following algorithm.

Algorithm 21.7.1 [Affine face algorithm] Given $0 < \delta \ll 1$. Initial: feasible solution $\bar{x}$. This algorithm solves the reduced problem (21.1).

1. Set $k = n + 1$, $B = A$, $N = \emptyset$, and compute the Cholesky factorization $B\bar{X}_B^2 B^T = LL^T$, where $\bar{X}_B$ is defined by (21.27).
2. Solve $L^T \bar{y} = -(1/\nu)e_{m+1}$, where ν is the $(m + 1)$ th diagonal of L .
3. Compute $\Delta_B = -e_k + \bar{X}_B^2 B^T \bar{y}$.
4. Go to step 11 if $\Delta_B = 0$.
5. Stop if $J = \{j \in B \mid \Delta_j < 0, j \neq n + 1\} = \emptyset$ (lower unbounded problem).
6. Determine stepsize α and index p such that $\alpha = -\bar{x}_p / \Delta_p = \min_{j \in J} -\bar{x}_j / \Delta_j$.
7. If $\alpha \neq 0$, update $\bar{x}_B = \bar{x}_B + \alpha \Delta_B$.
8. Update (B, N) by bringing p from B to N .
9. Update L by Algorithm 21.3.1.
10. Set $k = k - 1$; if $k > m + 1$, go to step 2.
11. Compute $\bar{z}_N = -N^T \bar{y}$.
12. Stop if $\bar{z}_N \geq 0$ (optimality achieved).
13. Go to step 1.

Note This algorithm has two layers of iterations. The inner layer consists of steps 2–10.

To reduce the effect of degeneracy in practice, it seems to be favorable to replace (21.26) alternatively by

$$\hat{x}_j = \begin{cases} \bar{x}_j, & \text{if } \bar{x}_j \geq \delta_j, \\ \delta_j, & \text{if } \bar{x}_j < \delta_j, \end{cases} \quad j \in B, j \neq n + 1,$$

where $\delta_j > 0$ and $n + 1 \neq j \in B$ are different small numbers conformable with magnitude of the first $k - 1$ components of $\bar{x}_B$.

Example 21.7.1 Solve the following problem by Algorithm 21.7.1, starting from feasible solution $(0, 0, 3, 2, 4)^T$:

$$\begin{aligned} \min \quad & x_1 + 5x_2 - 2x_4. \\ \text{s.t.} \quad & +3x_2 + x_3 - x_5 = -1, \\ & -3x_1 - x_2 + 3x_4 - 2x_5 = -2, \\ & x_1 - 2x_2 + 2x_3 + x_5 = 10, \\ & x_j \geq 0, \quad j = 1, \dots, 5. \end{aligned}$$

Answer Take $\delta = 10^{-6}$. Initial: feasible solution $\bar{x} = (0, 0, 3, 2, 4, -4)^T$.

Iteration 1:

1. $k = 6$, $B = \{1, 2, 3, 4, 5, 6\}$, $N = \emptyset$. $\bar{X}_B = \text{diag}(1, 1, 3, 2, 4, 1)^T$. Face matrix B and the Cholesky factor of $B\bar{X}_B^2 B^T$ are, respectively,

$$B = \begin{pmatrix} 3 & 1 & -1 \\ -3 & -1 & 3 & -2 \\ 1 & -2 & 2 & 1 \\ 1 & 5 & -2 & -1 \end{pmatrix},$$

$$L = \begin{pmatrix} -2449/420 \\ -5809/1168 & 3869/419 \\ 769/1121 & -4313/1346 & -3394/499 \\ -1402/545 & -14233/2934 & 2753/822 & -1941/1519 \end{pmatrix}.$$

2. $\bar{y} = (827/1382, -1205/2826, -269/892, -2737/4469)^T$.
3. $\Delta_B = (451/1235, -487/2051, -35/822, -635/2923, -576/763, -1732/4469)^T \neq 0$.
5. $J = \{2, 3, 4, 5\} \neq \emptyset$.
6. $\alpha = 0$, $p = 2$.
8. $B = \{1, 3, 4, 5, 6\}$, $N = \{2\}$.

$$9. L = \begin{pmatrix} -5 \\ -32/5 & 3019/366 \\ -2/5 & -2682/589 & -4040/713 \\ -1653/505 & 3573/1456 & 2273/2022 \end{pmatrix}.$$

10. $k = 5$.

Iteration 2:

2. $\bar{y} = (1977/2944, -1095/2176, -73/213, -1517/1917)^T$.
3. $\Delta_B = (80/213, -80/639, 560/1917, -80/639, -400/1917)^T \neq 0$.
5. $J = \{3, 5\} \neq \emptyset$.
6. $\alpha = \min\{-3/(-80/639), -4/(-80/639)\} = 1917/80$, $p = 3$.
- 7.

$$\bar{x}_B = (0, 3, 2, 4, -4)^T + (1917/80)(80/213, -80/639, 560/1917, -80/639, -400/1917)^T = (9, 0, 9, 1, -9)^T.$$

8. $B = \{1, 4, 5, 6\}$, $N = \{2, 3\}$.

$$9. L = \begin{pmatrix} -4 \\ -8 & -2207/329 \\ 4 & 1292/2889 & 2584/2889 \\ 1292/321 & -2584/2889 & -1 \end{pmatrix}.$$

10. $k = 4 = m + 1$.
11. $\bar{z}_N = (1097/1456, 80/5751)^T \geq 0$.

12. The zero-dimensional optimal face (optimal basic solution) and optimal value are

$$\bar{x} = (9, 0, 0, 9, 1, -9)^T, \quad \bar{x}_6 = -9.$$

21.8 Generalized Face Algorithm

In this section, we generalize Algorithm 21.6.1 to solve the bounded-variable reduced problem; i.e.,

$$\begin{aligned} \min \quad & f, \\ \text{s.t.} \quad & Ax = b, \\ & l \leq x \leq u, \end{aligned} \quad (21.28)$$

where A and b are the same as in (21.1). The formula for computing the search direction Δ_B and the iteration scheme are also applicable here. As bound constraints are added, however, (21.16) for determining the stepsize α and index p has to be replaced by

$$\alpha = \alpha_p = \min\{\alpha_j \mid j \in B, j \neq n+1\}, \quad (21.29)$$

where

$$\alpha_j = \begin{cases} (\bar{x}_j - u_j)/\Delta_j, & \text{If } \Delta_j < 0, \\ (\bar{x}_j - l_j)/\Delta_j, & \text{If } \Delta_j > 0, \\ \infty, & \text{If } \Delta_j = 0, \end{cases} \quad j \in B, j \neq n+1. \quad (21.30)$$

In addition, the optimality should be tested by examining whether the following set is empty:

$$J' = \{j \in \Gamma \mid \bar{z}_j < 0\} \cup \{j \in \Pi \mid \bar{z}_j > 0\}, \quad (21.31)$$

where Γ and Π are defined by (8.9).

The algorithm can be formulated as follows.

Algorithm 21.8.1 [Generalized face algorithm] Initial: (B, N) , $m+1 \leq k \leq n+1$, feasible solution $\bar{x}$, and the Cholesky factor L of BB^T . This algorithm solves the bounded-variable reduced problem (21.28).

Same as Algorithm 21.6.1, except for steps 4, 5, 11, and 12 replaced, respectively, by

4. Compute α_j by (21.30).
5. Determine stepsize α and index p by (21.29).

11. Stop if J' , defined by (21.31), is empty (optimality achieved).
12. Determine index $q \in \arg \max_{j \in J'} |\bar{z}_j|$.

Example 21.8.1 Solve the following problem by Algorithm 21.8.1:

$$\begin{aligned}
 \min f &= x_1 - x_3 + 2x_4, \\
 \text{s.t. } &+3x_1 + x_2 - 5x_3 + x_4 = 3, \\
 &-2x_1 + 6x_3 + x_5 = 5, \\
 &-6x_1 - 3x_2 + 3x_3 - 2x_5 = -11, \\
 &1 \leq x_1 \leq 9, \quad -2 \leq x_2 \leq 8, \\
 &0 \leq x_3 \leq 12, \quad 0 \leq x_4 \leq 10, \quad 0 \leq x_5 \leq 15.
 \end{aligned}$$

Answer There is a feasible solution $(1, 2, 1, 3, 1)^T$ with objective value 6.

Initial:

$$k = 6, \quad B = \{1, 2, 3, 4, 5, 6\}, \quad N = \emptyset,$$

feasible solution $\bar{x} = (1, 2, 1, 3, 1, 6)^T$. Face matrix B and the Cholesky factor of BB^T are

$$B = \begin{pmatrix} 3 & 1 & -5 & 1 \\ -2 & 6 & 1 \\ -6 & -3 & 3 & -2 \\ 1 & -1 & 2 & -1 \end{pmatrix},$$

$$L = \begin{pmatrix} -6 \\ 6 & -2889/1292 \\ 6 & 2279/637 & 549/181 \\ -5/3 & -2584/2889 & 1267/915 & -1001/816 \end{pmatrix}.$$

Iteration 1:

1. $v = -1001/816$, $\bar{y} = (772/623, 468/623, 27/89, -414/623)^T$.
2. $\Delta_B = (24/89, -205/623, 71/623, 8/89, -90/623, 209/623)^T \neq 0$.
- 4.

$$\begin{aligned}
 \alpha_{1,2,3,4,5} &= ((1 - 1)/(24/89), (8 - 2)/(-205/623), (1 - 0)/(71/623), \\
 &\quad (3 - 0)/(8/89), (1 - 10)/(-90/623)).
 \end{aligned}$$

5. $\alpha = 0$, $p = 1$.
6. $\bar{x}_B = (1, 2, 1, 3, 1, 6)^T$.
7. $B = \{2, 3, 4, 5, 6\}$, $N = \{1\}$.

$$8. L = \begin{pmatrix} -1351/260 & & & \\ 1351/234 & -1057/552 & & \\ 1351/390 & 1057/506 & -330/139 & \\ -260/193 & -843/908 & -2128/1401 & -251/249 \end{pmatrix}.$$

9. $k = 5$.

Iteration 2:

1. $v = -251/249, \bar{y} = (124/63, 220/189, 17/27, -62/63)^T$.
2. $\Delta_B = (-5/63, -1/63, 0, 2/21, 1/63)^T \neq 0$.
4. $\alpha_{2,3,5} = ((2-8)/(-5/63), (1-12)/(-1/63), (1-0)/(2/21))$.
5. $\alpha = 21/2, p = 5$.
- 6.

$$\begin{aligned} \bar{x}_B &= (1, 2, 1, 3, 1, 6)^T + (21/2)(5/63, 1/63, 0, -2/21, -1/63)^T \\ &= (17/6, 7/6, 3, 0, 35/6)^T. \end{aligned}$$

7. $B = \{2, 3, 4, 6\}, N = \{1, 5\}$.

$$8. L = \begin{pmatrix} -1351/260 & & & \\ 1351/234 & -1762/1079 & & \\ 1351/390 & 1079/881 & 2378/1121 & \\ -260/193 & -749/688 & 1393/985 & -1 \end{pmatrix}.$$

9. $k = 4$.

Iteration 3:

1. $v = -1, \bar{y} = (124/63, 220/189, 17/27, -62/63)^T$.
2. $k = 4$.
10. $\bar{z}_N = (4/3, 1/6)^T$.
11. $J' = \emptyset$. The optimal basic solution and optimal value are

$$\bar{x} = (1, 17/6, 7/6, 3, 0, 35/6)^T, \quad \bar{x}_6 = 35/6.$$

21.9 Notes on Face Method

The quality of an initial partition (B, N) is important to the face algorithm. Although initially setting $B = A$ would be a convenient choice, a small B should be favorable, ideally containing the “right” (unknown optimal indices) face columns. Therefore, it might be advisable to determine an initial B , whose cardinality is no more than $2m$, especially for large-scale problems with $n \gg m$, by using some crash procedure based on the most-obtuse-angle heuristic, balanced with sparsity considerations.

The related computational results with the face algorithm are very well although preliminary. This outcome is not surprising because it uses the steepest descent

search direction and involves less computational effort per iteration mainly due to fewer linear systems involved than the simplex algorithm.

Based on the *dual elimination* (Part I), it is possible to design another type of algorithm in the following way. By a series of elementary transformations with row exchanges, the standard dual problem can be deduced to the following form:

$$\begin{aligned} \min \quad & -g = -\bar{g} + \bar{x}^T z, \\ \text{s.t.} \quad & G_2 z = d_2, \quad z \geq 0, \end{aligned} \tag{21.32}$$

together with $Uy + G_1 z = d_1$. After the preceding standard problem is converted to the reduced form, face Algorithm 21.6.1 or affine face Algorithm 21.7.1 applies.

Gill and Murray (1973) use the Cholesky factorization $BB^T = LL^T$, where B is the normal basis, to improve the numerical stability of the simplex method. Based on the fact that BB^T and hence L is independent of the ordering of the columns of B , Saunders (1972) proposes an approach to keeping L sparse throughout the solution process. It initially carries out row and column permutations to turn the coefficient matrix A to the lower triangular form as much as possible and then uses the resulting row permutation in subsequent iterations. This approach is clearly applicable in the context of the face algorithm.

It is worth mentioning an interesting result just published by Zhang et al. (2013). They present a variant based on updating $(BB^T)^{-1}$ by the Sherman–Morrison formula to avoid the orthogonal transformations, but it turns out to be unstable (Pan, 2015).

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Chapter 22

Dual Face Method with Cholesky Factorization



The same idea of the previous face method applies to the dual problem. The resulting so-called *dual face method* turns out to be even more efficient. In the next sections, we put forward the steepest ascent direction, updating of the dual solution, dual face contraction, optimality test, and dual face expansion, and then formulate the dual face algorithm together with a dual face Phase-I and related computational results. The final section introduces the dual face algorithm without the Cholesky factorization.

To begin with, we use notation

$$A := \begin{pmatrix} A & -b \\ 0 & 1 \end{pmatrix}, \quad y := \begin{pmatrix} y \\ y_{m+1} \end{pmatrix}, \quad z := \begin{pmatrix} z \\ z_{n+1} \end{pmatrix}, \quad c := \begin{pmatrix} c \\ 0 \end{pmatrix},$$

to transform the standard dual problem into

$$\begin{aligned} & \max y_{m+1}, \\ & \text{s.t. } A^T y + z = c, \quad z \geq 0, \quad z_{n+1} = 0, \end{aligned} \quad (22.1)$$

where $A \in \mathcal{R}^{(m+1) \times (n+1)}$, $c \in \mathcal{R}^{(n+1)}$, $\text{rank } A = m + 1$, $m < n$, and

$$A e_{n+1} = (-b^T, 1)^T, \quad e_{m+1}^T A = e_n^T. \quad (22.2)$$

Assume that this problem is feasible.

Let (B, N) be a partition of matrix A , where $B \in \mathcal{R}^{(m+1) \times k}$ and

$$\text{rank } B = k, \quad 1 \leq k \leq m + 1.$$

So, B is of full column rank. The related variables are, respectively, termed *dual face* and *dual nonface* variables and matrices termed *dual face* and *dual nonface* matrices.

In the solution process, the k th (last) column of the face matrix is always the same as the $(n + 1)$ th (last) column of A . Thus,

$$Be_k = (-b^T, 1)^T, \quad e_{m+1}^T B = e_k^T, \quad e_{m+1}^T N = 0. \quad (22.3)$$

The partition (B, N) corresponds to dual face

$$D_N = \{(y, z) \in \mathcal{R}^m \times \mathcal{R}^n \mid A^T y + z = c, \ z_B = 0, \ z_N \geq 0\}. \quad (22.4)$$

A point on a dual face is termed *dual face point*. The zero-dimensional face ($k = m + 1$) has a unique dual face point—vertex. A face is said to be a *level dual face* if the dual objective value is constant over it and said to be *dual optimal* if the constant is equal to the optimal value.

For simplicity, thereafter $(y, z_N) \in D_N$ will often be used in place of $(y, z) \in D_N$.

22.1 Steepest Ascent Direction

We might as well assume that $B = \{1, \dots, k - 1, n + 1\}$. Consider the following subprogram:

$$\begin{aligned} & \max y_{m+1}, \\ & \text{s.t. } B^T y = c_B, \\ & \quad N^T y + z_N = c_N, \quad z_N \geq 0, \end{aligned} \quad (22.5)$$

whose feasible region is face D_N .

Using $(m + 1) \times (m + 1)$ projection

$$P = I - B(B^T B)^{-1} B^T, \quad (22.6)$$

the objective gradient e_{m+1} of the subprogram (22.5) is projected onto the null of B^T , yielding the following search direction:

$$\Delta y = P e_{m+1} = e_{m+1} - B \bar{x}_B, \quad B^T B \bar{x}_B = e_k, \quad (22.7)$$

$$\Delta z_N = -N^T \Delta y. \quad (22.8)$$

Proposition 22.1.1 For Δy defined by (22.7), it holds that $B^T \Delta y = 0$.

Proof By (22.7), it is easy to verify this equation is valid. $\square$

Proposition 22.1.2 *The following statements are equivalent:*

$$(i) \Delta y \neq 0; \quad (ii) (\Delta y)_{m+1} > 0; \quad (iii) e_{m+1} \notin \text{range}(B).$$

Proof Premultiplying the first expression of (22.7) by Δy^T , and then from Proposition 22.1.1 it follows that

$$0 \leq \Delta y^T \Delta y = (\Delta y)_{m+1},$$

which means the equivalence of (i) and (ii). If $\Delta y = 0$, then it is known by the first expression of (22.7) that $e_{m+1} = B\bar{x}_B$, hence $e_{m+1} \in \text{range}(B)$, so (iii) implies (i). Conversely, if $e_{m+1} \in \text{range}(B)$, then there is some vector u such that

$$e_{m+1} = Bu. \quad (22.9)$$

Premultiplying the preceding equality by B^T , and then from (22.3) it follows that

$$e_k = B^T Bu,$$

combining which and the second expression of (22.7) and B of full column rank gives $u = \bar{x}_B$. Thus, (22.9) implies

$$e_{m+1} = B\bar{x}_B.$$

Substituting the preceding equality to the first expression of (22.7) leads to $\Delta y = 0$. So (i) implies (iii). Furthermore, it can be asserted that (i) and (iii) are equivalent. $\square$

Proposition 22.1.3 *If $\Delta y \neq 0$, then*

$$e_{m+1}^T \Delta y = (\Delta y)_{m+1} > 0, \quad (22.10)$$

$$e_{m+1}^T \Delta y / \|\Delta y\| \geq e_{m+1}^T u / \|u\|, \quad \forall 0 \neq u \in \text{Null}(B^T). \quad (22.11)$$

Proof Equality (22.10) directly follows from Proposition 22.1.2. For P defined by (22.6), it holds that

$$Pu = u \neq 0, \quad \forall 0 \neq u \in \text{Null}(B^T). \quad (22.12)$$

It is known from the Cauchy inequality that

$$\|Pe_{m+1}\| \|Pu\| \geq (Pe_{m+1})^T (Pu).$$

Multiplying the preceding inequality by $1/\|Pu\|$ renders

$$\|Pe_{m+1}\| \geq (Pe_{m+1})^T(Pu)/\|Pu\|,$$

i.e.,

$$(Pe_{m+1})^T(Pe_{m+1})/\|Pe_{m+1}\| \geq (Pe_{m+1})^T(Pu)/\|Pu\|,$$

combining which, $P^2 = P$, $P^T = P$, (22.12), and $\Delta y = Pe_{m+1}$ leads to (22.11). $\square$

The preceding proposition implies that $\Delta y \neq 0$ forms the most acute angle with objective gradient e_{m+1} of the subprogram, in the null of B^T . It is therefore termed *the steepest ascent*. In this sense, taking it as a search direction in y -space is the “best” choice.

To compute search direction $(\Delta y, \Delta z_N)$, defined by (22.7) and (22.8), just solve the $k \times k$ system

$$B^T B x_B = e_k. \quad (22.13)$$

Assume that the QR factorization of B is

$$B = QR_1 = Q \begin{pmatrix} L^T \\ 0 \end{pmatrix}, \quad (22.14)$$

where $Q \in \mathcal{R}^{(m+1) \times (m+1)}$ is orthogonal, and $L \in \mathcal{R}^{k \times k}$ is nonsingular lower triangular. Then it is verified that

$$B^T B = LL^T.$$

Thus, the L is just the Cholesky factor of $B^T B$. Therefore, the system (22.13) becomes

$$LL^T x_B = e_k,$$

whose solution can be obtained by solving the following two triangular systems:

$$Lu = e_k, \quad L^T x_B = u.$$

As the solution to the first system is $v = (1/\mu)e_k$, where μ is the k th diagonal of L , consequently only the following lower triangular system needs to be solved:

$$L^T x_B = (1/\mu)e_k. \quad (22.15)$$

22.2 Updating of Dual Face Solution

Assume that $(\bar{y}, \bar{z}_N) \in D_N$ and $\Delta y \neq 0$. The new dual solution is determined by the following line search scheme:

$$\hat{y} = \bar{y} + \beta \Delta y, \quad \hat{z}_N = \bar{z}_N - \beta N^T \Delta y, \quad (22.16)$$

where the maximum possible stepsize, maintaining $\bar{z}_N \geq 0$, is

$$\beta = \bar{z}_q / (a_q^T \Delta y) = \min \left\{ \bar{z}_j / (a_j^T \Delta y) \mid a_j^T \Delta y > 0, j \in N \right\} \geq 0. \quad (22.17)$$

If some components of $\bar{z}_N$ vanish, then dual face point $(\bar{y}, \bar{z}_N)$ is said to be *dual degenerate*, as is a case where stepsize β vanishes, yielding a point the same as the old.

Lemma 22.2.1 Assume that $(\bar{y}, \bar{z}_N) \in D_N$ with $\Delta y \neq 0$.

- (i) If $N^T \Delta y \not\leq 0$, then $\hat{z}_N$ is a boundary point, belonging to D_N . The objective value does not decrease and strictly increases if $(\bar{y}, \bar{z}_N)$ is nondegenerate.
- (ii) If $N^T \Delta y \leq 0$, problem (22.1) is upper unbounded in D_N .

Proof

- (i) In this case, the new point $(\hat{y}, \hat{z}_N)$ is well-defined. By $(\bar{y}, \bar{z}_N) \in D_N$, (22.17), and Proposition 22.1.1, it is known that $(\hat{y}, \hat{z}_N) \in D_N$. In addition, it is clear that $\hat{z}_q = 0$, and hence the new iterate is on the boundary, associated with the objective value

$$\hat{y}_{m+1} = \bar{y}_{m+1} + \beta (\Delta y)_{m+1}. \quad (22.18)$$

Furthermore, from $\beta \geq 0$ and Proposition 22.1.2, it is known that $(\Delta y)_{m+1} > 0$, hence $\hat{y}_{m+1} \geq \bar{y}_{m+1}$. If $(\bar{y}, \bar{z}_N)$ is nondegenerate, then $\beta > 0$, hence $\hat{y}_{m+1} > \bar{y}_{m+1}$, as indicates that the objective value increases strictly.

- (ii) In this case, it is clear that $(\hat{y}, \hat{z}_N) \in D_N \forall \beta > 0$. It is then known by (22.18) and Proposition 22.1.2 that the objective value tends to ∞ , as β tends to ∞ .

□

22.3 Dual Face Contraction

Assume that $\hat{z}_q = 0$ and $N^T \Delta y \not\leq 0$. Update (B, N) by bring q from N to B . Denoting the new face matrix by $\check{B} \in \mathcal{R}^{(m+1) \times (k+1)}$, we have the following result.

Proposition 22.3.1 If rank $B = k$, then rank $\check{B} = k + 1$.

Proof It is clear that $k \leq \text{rank } \check{B} \leq k + 1$. Assume $\text{rank } \check{B} \neq k + 1$. Then, there is a vector $u \neq 0$ such that $a_q = Bu$. Thus, it is known by Proposition 22.1.1 that $B^T \Delta y = 0$, hence

$$a_q^T \Delta y = u^T B^T \Delta y = 0.$$

Furthermore, it follows from (22.17) that $a_q^T \Delta y > 0$, as is a contradiction. Therefore $\text{rank } \check{B} = k + 1$. $\square$

According to the preceding proposition, there exists the Cholesky factorization

$$\check{B}^T \check{B} = \check{L} \check{L}^T.$$

For the sake of simplicity, it is advisable to insert a_q as the second last column of $\check{B} \in \mathcal{R}^{(m+1) \times (k+1)}$ (followed by the last column $(-b^T, 1)^T$). Then, the new Cholesky factor can be computed by the following algorithm.

Algorithm 22.3.1 Assume that the QR factorization of B is given by (22.14). This algorithm finds the Cholesky factor $\check{L}$ of $\check{B}^T \check{B}$.

1. Compute $v = Q^T a_q$.
2. Insert v before the last $(k + 1)$ th column of R_1 , yielding matrix H .
3. Determine Givens rotation $J_{m+1}, \dots, J_{k+1}$, where J_i is in $(k, \cdot)$ -plane, such that

$$J_{k+1} \dots J_{m+1} H = \begin{pmatrix} \check{L}^T \\ 0 \end{pmatrix},$$

where $\check{L} \in \mathcal{R}^{(k+1) \times (k+1)}$ is lower triangular.

The validity of Algorithm 22.3.1 can be verified as follows:

It is clear that

$$\begin{aligned} J_{k+1} \dots J_{m+1} Q^T \check{B} &= J_{k+1} \dots J_{m+1} Q^T [a_1, \dots, a_{k-1}, a_q, a_{n+1}] \\ &= J_{k+1} \dots J_{m+1} H = \begin{pmatrix} \check{L}^T \\ 0 \end{pmatrix}. \end{aligned}$$

By Proposition 22.3.1, it is known that $\check{L} \in \mathcal{R}^{(k+1) \times (k+1)}$ is nonsingular. Thus, from the preceding expression, it follows that

$$\check{B}^T \check{B} = \check{B}^T Q J_{m+1}^T \dots J_{k+1}^T J_{k+1} \dots J_{m+1} Q^T \check{B} = (\check{L}, 0) \begin{pmatrix} \check{L}^T \\ 0 \end{pmatrix} = \check{L} \check{L}^T.$$

It is seen from step 1 of Algorithm 22.3.1 that it is now required to store the Givens rotations in product form, in contrast to the updating of the Cholesky factor in the face algorithm. For large-scale problems, therefore, the dual face algorithm should be restarted from scratch periodically.

22.4 Optimality Test

Assume that Δy vanishes and cannot be used as a search direction, e.g., when $k = m + 1$.

Lemma 22.4.1 *Assume that Δy is defined by (22.7). If $\Delta y = 0$, then D_N is a dual level face, and vice versa if there is $(\bar{y}, \bar{z}_N) \in D_N$ such that $\bar{z}_N > 0$.*

Proof Assume that $\bar{x}_B$ is the solution to $B^T B x_B = e_k$. From $\Delta y = 0$ and the first expression of (22.7), it follows that

$$e_{m+1} = B \bar{x}_B.$$

Transposing the two sides of the preceding equality and postmultiplying the result by y (for any $(y, z_N) \in D_N$) gives

$$e_{m+1}^T y = \bar{x}_B^T B^T y,$$

combining which and $B^T y = c_B$ leads to

$$y_{m+1} = \bar{x}_B^T c_B.$$

So the objective value is constant over D_N . Therefore, D_N is a level face.

Now assume that D_N is a level face but $\Delta y \neq 0$. If $N^T \Delta y \leq 0$, it is known by Lemma 22.2.1(ii) that the objective value is upper unbounded over D_N , as contradicts that it is a level face; if, otherwise, $N^T \Delta y \not\leq 0$, then since $(\bar{y}, \bar{z}_N)$ is nondegenerate, it is known from Lemma 22.2.1(i) that $(\hat{y}, \hat{z}_N)$, defined by (22.16) and (22.17), belongs to D_N , with a strictly increased objective value, as contradicts that D_N is a level face. Therefore $\Delta y = 0$. $\square$

Now consider the primal problem of (22.1), i.e.,

$$\begin{aligned} \min \quad & c^T x, \\ \text{s.t.} \quad & Ax = e_{m+1}, \quad x \geq 0. \end{aligned} \tag{22.19}$$

Theorem 22.4.1 *Assume that $(\bar{y}, \bar{z}_N) \in D_N$ and that $\Delta y = 0$ and $\bar{x}_B$ are determined by (22.7). If $\bar{x}_B \geq 0$, then $\bar{x}$ and $(\bar{y}, \bar{z})$ are a pair of primal and dual optimal solutions, and D_N is a dual optimal face to (22.1), with $\bar{x}_N = 0$ and $\bar{z}_B = 0$.*

Proof From (22.7) and $\Delta y = 0$, it follows that $e_{m+1} = B\bar{x}_B$. Also, it is easy to verify that $\bar{x}$ ($\bar{x}_N = 0$) is a feasible solution to (22.19). On the other hand, it is clear that $(\bar{y}, \bar{z})$ ($\bar{z}_B = 0$) is a feasible solution to (22.1). The two solutions exhibit slackness complementarity. Therefore, they are a pair of primal and dual optimal solutions. By Lemma 22.4.1, D_N is a (dual) level face. In fact, it is a (dual) optimal face since including an optimal solution. $\square$

22.5 Dual Face Expansion

If $\Delta y = 0$ but $\bar{x}_B \not\geq 0$, then the level dual face D_N cannot be declared optimal. In this case, determine the index

$$p \in \arg \min_{j \in B} \bar{x}_j, \quad (22.20)$$

and update (B, N) by

$$\widehat{B} = B \setminus \{p\}, \quad \widehat{N} = N \cup \{p\}. \quad (22.21)$$

Proposition 22.5.1 *The index p determined by (22.20) is different from $n + 1$.*

Proof It is known by (22.15) that $\bar{f} = (1/\mu^2) > 0$. As $\min_{j \in B} \bar{x}_j < 0$, it is not possible to select $n + 1$ to be as p . $\square$

Assume that $\widehat{B} \in \mathcal{R}^{(m+1) \times (k-1)}$ is the matrix resulting from dropping the p th column of B . Clearly, it is still of full column rank. The associated Cholesky factor can be obtained by the following steps.

Algorithm 22.5.1 Initial: the Cholesky factorization $B^T B = L L^T$. This algorithm generates the Cholesky factor $\widehat{L}$ of $\widehat{B} \widehat{B}^T$.

1. Drop the p th column from L^T , yielding the Hessenberg matrix H .
2. Determine Givens rotations $J_p, \dots, J_{k-1}$, where J_i is in $(i, i+1)$ -plane, such that

$$J_{k-1} \dots J_p H = \begin{pmatrix} \widehat{L}^T \\ 0 \end{pmatrix},$$

where $\widehat{L} \in \mathcal{R}^{(k-1) \times (k-1)}$ is lower triangular.

The validity of Algorithm 22.5.1 can be verified simply: since

$$\begin{aligned} J_{k-1} \dots J_p Q^T \widehat{B} &= J_{k-1} \dots J_p Q^T [a_1, \dots, a_{p-1}, a_{p+1}, \dots, a_{n+1}] \\ &= J_{k-1} \dots J_p H = \begin{pmatrix} \widehat{L}^T \\ 0 \end{pmatrix}, \end{aligned}$$

it holds that

$$\widehat{B}^T \widehat{B} = \widehat{B}^T Q J_p^T \dots J_{k-1}^T J_{k-1} \dots J_p Q^T \widehat{B} = (\widehat{L}, 0) \begin{pmatrix} \widehat{L}^T \\ 0 \end{pmatrix} = \widehat{L} \widehat{L}^T.$$

Seemingly attractive, an alternative way to expand face is to replace (22.21) by

$$\widehat{B} = B \setminus \{j \in B \mid \bar{x}_j < 0\}, \quad \widehat{N} = A \setminus \widehat{B}.$$

Unfortunately, updating the associated Cholesky factor becomes cumbersome.

22.6 Dual Face Algorithm

To be an initial dual face matrix, a submatrix $B \in \mathcal{R}^{(m+1) \times k}$, $1 \leq k \leq m+1$ with small k seem to be preferable, although any of full column rank is eligible.

The overall steps can be summarized into the following algorithm.

Algorithm 22.6.1 [Dual face algorithm with Cholesky factorization] Initial: B , N , $1 \leq k \leq m+1$; dual feasible solution $(\bar{y}, \bar{z}_N)$ and Cholesky factor L of $B^T B$. This algorithm solves the pair of reduced problems (22.1) and (22.19).

1. Solve $L^T \bar{x}_B = -(1/\mu)e_k$ for $\bar{x}_B$, where μ is the $(m+1)$ th diagonal of L .
2. If $k < m+1$, compute $\Delta y = e_{m+1} - B\bar{x}_B$.
3. Go to step 10 if $k = m+1$ or $\Delta y = 0$.
4. Stop if $N^T \Delta y \leq 0$.
5. Determine index q and stepsize β such that

$$\beta = \bar{z}_q / \left(a_q^T \Delta y \right) = \min \left\{ \bar{z}_j / \left(a_j^T \Delta y \right) \mid a_j^T \Delta y > 0, j \in N \right\}.$$

6. If $\beta \neq 0$, update $\bar{y} = \bar{y} + \beta \Delta y$, $\bar{z}_N = \bar{z}_N - \beta N^T \Delta y$.
7. Call Algorithm 22.3.1 to update L .
8. Update (B, N) by moving p from N to B .
9. Set $k = k+1$, and go to step 1.
10. Stop if $\bar{x}_B \geq 0$.
11. Determine $p \in \arg \min_{j \in B} \bar{x}_j$.
12. Call Algorithm 22.5.1 to update L .

13. Update $B = B \setminus \{p\}$, $N = N \cup \{p\}$.
14. Set $k = k - 1$.
15. Go to step 1.

Note Although $1 \leq k \leq m + 1$ is assumed in the preceding Algorithm, the $k = 0$ seems to be favorable. In this case $B = \emptyset$, $N = A$, the projection P is the unit matrix, and (22.7), which is not well-defined, is replaced by $\Delta y = e_{m+1}$. Then, the subsequent steps of the algorithm will go smoothly.

Theorem 22.6.1 *Assume dual nondegeneracy throughout the solution process. Algorithm 22.6.1 terminates either at*

- (i) *Step 4, detecting upper unboundedness of (22.1), or at*
- (ii) *Step 10, generating a (dual) optimal face together with a pair of primal and dual optimal solutions*

Proof Under the assumption of nondegeneracy, the proof of finiteness of the algorithm is the same as that for the simplex algorithm. The meanings of its exits are derived from Lemma 22.2.1 and Theorem 22.4.1. $\square$

22.6.1 Dual Face Phase-I

To provide a dual feasible solution to get Algorithm 22.6.1 started, any dual Phase-I methods presented in Chap. 15 apply, especially the single-artificial-variable algorithm described in Sect. 15.2 fit. In this section, an auxiliary problem, slightly simpler than (15.9), is solved by the dual face algorithm.

Let the pair $(\bar{y}, \bar{z})$ be given with $\bar{z} \geq 0$. If $A^T \bar{y} + \bar{z} = c$, then $(\bar{y}, \bar{z})$ is already a dual feasible solution. Now assume it is not the case.

Define

$$h = (\bar{z} - c + A^T \bar{y}) / \|\bar{z} - c + A^T \bar{y}\|,$$

and introduce variable y_{m+1} . Construct auxiliary problem

$$\begin{aligned} \max \quad & y_{m+1}, \\ \text{s.t.} \quad & A^T y + h y_{m+1} + z = c, \quad z \geq 0, \quad y_{m+1} \leq 0. \end{aligned} \tag{22.22}$$

It is clear that this problem has a feasible solution $(\bar{y}, \bar{y}_{m+1}, \bar{z})$, where

$$\bar{y}_{m+1} = -\|\bar{z} - c + A^T \bar{y}\|,$$

and is upper bounded, and hence there is an optimal solution. If its optimal value is equal to 0, then the part (y, z) of the optimal solution is a feasible solution to the original dual problem; else the dual problem is infeasible.

In order to decrease the possibility of zero stepsize, it is inadvisable to have many zero components of $\bar{z}$. A possible way is to set $\bar{y} = 0$, $\bar{z} = e$, thus

$$h = (e - c)/\|e - c\|, \quad (22.23)$$

and there is a feasible solution

$$\bar{y} = 0, \quad \bar{y}_{m+1} = -\|e - c\|, \quad \bar{z} = e, \quad (22.24)$$

to the auxiliary problem. Then by setting $A^T := [A^T : h]$, the auxiliary problem is converted to a form, similar to (22.1). Consequently, the following Phase-I algorithm yields by modifying Algorithm 22.6.1 slightly.

Algorithm 22.6.2 [Dual face Phase-I algorithm] Initial: $B = \emptyset$, $N = A$, $k = 0$, $\bar{y} = \|e - c\|e_{m+1}$, $\bar{z}_N = e$. This algorithm finds a feasible solution to (22.1).

1. Set $\Delta y = e_{m+1}$, and go to step 5.
2. Solve $LL^T \bar{x}_B = B^T e_{m+1}$, where μ is the $(m+1)$ th diagonal of L .
3. Compute $\Delta y = e_{m+1} - B\bar{x}_B$.
4. Go to step 10 if $\Delta y = 0$.
5. If $N^T \Delta y \not\leq 0$, then

(1) Determine q and β_1 such that

$$\beta = \bar{z}_q / \left(a_q^T \Delta y \right) = \min \left\{ \bar{z}_j / \left(a_j^T \Delta y \right) \mid a_j^T \Delta y > 0, j \in N \right\};$$

(2) Compute $\beta = \min\{-\bar{y}_m + 1/\Delta y_{m+1}, \beta_1\}$;
else

(3) Compute $\beta = -\bar{y}_{m+1}/\Delta y_{m+1}$.

6. Update: $\bar{y} = \bar{y} + \beta \Delta y$, $\bar{z}_N = \bar{z}_N - \beta N^T \Delta y$.
7. Stop if $\bar{y}_{m+1} = 0$ (dual feasibility achieved).
8. Update L by Algorithm 22.3.1.
9. Update (B, N) by bringing q from N to B .
10. Set $k = k + 1$, and go to step 2.
11. Stop if $\bar{x}_B \geq 0$ (dual infeasible problem).
12. Determine $p \in \arg \min_{j \in B} \bar{x}_j$.
13. Update L by Algorithm 22.5.1.
14. Update (B, N) by bringing p from B to N .
15. Set $k = k - 1$.
16. Go to step 2.

Example 22.6.1 Solve the following problem by the two-phase dual face algorithm:

$$\begin{aligned}
 \min \quad & 4x_1 - x_2 + 2x_3, \\
 \text{s.t.} \quad & 2x_1 + x_2 + x_3 + x_4 = 2, \\
 & x_1 - x_2 - x_3 - x_5 = -1, \\
 & 2x_1 - x_3 + x_5 = 0, \\
 & x_j \geq 0, \quad j = 1, \dots, 5.
 \end{aligned}$$

Answer The dual problem is

$$\begin{aligned}
 \max \quad & 2y_1 - y_2, \\
 \text{s.t.} \quad & 2y_1 + y_2 + 2y_3 + z_1 = 4, \\
 & y_1 - y_2 + z_2 = -1, \\
 & y_1 - y_2 - y_3 + z_3 = 2, \\
 & y_1 + z_4 = 0, \\
 & -y_2 + y_3 + z_5 = 0, \\
 & z_j \geq 0, \quad j = 1, \dots, 5.
 \end{aligned}$$

Construct the following auxiliary problem ($\|c - e\| = 4$):

$$\begin{aligned}
 \max \quad & y_4, \\
 \text{s.t.} \quad & 2y_1 + y_2 + 2y_3 - (3/4)y_4 + z_1 = 4, \\
 & y_1 - y_2 + (1/2)y_4 + z_2 = -1, \\
 & y_1 - y_2 - y_3 - (1/4)y_4 + z_3 = 2, \\
 & y_1 + (1/4)y_4 + z_4 = 0, \\
 & -y_2 + y_3 + (1/4)y_4 + z_5 = 0, \\
 & z_j \geq 0, \quad j = 1, \dots, 5.
 \end{aligned}$$

Phase-I: call Algorithm 22.6.2.

Initial: $k = 0$, $B = \emptyset$, $N = \{1, 2, 3, 4, 5\}$, $\bar{y} = (0, 0, 0, -4)^T$, $\bar{z} = (1, 1, 1, 1, 1)^T$.

Iteration 1:

1. $\Delta y = (0, 0, 0, 1)^T$.
4. $\Delta y \neq 0$.
5. $N^T \Delta y = (-3/4, 1/2, -1/4, 1/4, 1/4)^T \not\leq 0$.
 - (1) $\beta_1 = \min\{1/(1/2), 1/(1/4), 1/(1/4)\} = 2$, $q = 2$.
 - (2) $\beta = \min\{4/1, 2\} = 2$.
6. $\bar{y} = (0, 0, 0, -4)^T + 2(0, 0, 0, 1)^T = (0, 0, 0, -2)^T$.
 $\bar{z}_N = (1, 1, 1, 1, 1)^T - 2(-3/4, 1/2, -1/4, 1/4, 1/4)^T = (5/2, 0, 3/2, 1/2, 1/2)^T$.
7. $\bar{y}_4 = -2 \neq 0$.

8. $L = [3/2]$.
9. $B = \{2\}$, $N = \{1, 3, 4, 5\}$.
10. $k = 1$.

Iteration 2:

2. $\bar{x}_B = (2/9)$.
3. $\Delta y = (0, 0, 0, 1)^T - (1, -1, 0, 1/2)^T(2/9) = (-2/9, 2/9, 0, 8/9)^T$.
4. $\Delta y \neq 0$.
5. $N^T \Delta y = (-8/9, -2/3, 0, 0)^T \leq 0$.

$$(3) \quad \beta = -(-2)/(8/9) = 9/4.$$

6.

$$\begin{aligned} \bar{y} &= (0, 0, 0, -2)^T + (9/4)(-2/9, 2/9, 0, 8/9)^T = (-1/2, 1/2, 0, 0)^T. \\ \bar{z}_N &= (5/2, 3/2, 1/2, 1/2)^T - (9/4)(-8/9, -2/3, 0, 0)^T \\ &= (9/2, 3, 1/2, 1/2)^T. \end{aligned}$$

7. $\bar{y}_4 = 0$ achieves the dual feasibility.

Convert the dual problem to a form of (22.1):

$$\begin{aligned} \max \quad & y_4, \\ \text{s.t.} \quad & 2y_1 + y_2 + 2y_3 + z_1 = 4, \\ & y_1 - y_2 + z_2 = -1, \\ & y_1 - y_2 - y_3 + z_3 = 2, \\ & y_1 + z_4 = 0, \\ & -y_2 + y_3 + z_5 = 0, \\ & -2y_1 + y_2 + y_4 + z_6 = 0, \\ & z_j \geq 0, \quad j = 1, \dots, 6. \end{aligned} \tag{22.25}$$

Phase-II: call Algorithm 22.6.1.

Initial: $k = 2$, $B = \{2, 6\}$, $N = \{1, 3, 4, 5\}$ $\bar{y} = (-1/2, 1/2, 0, -3/2)^T$,

$$\bar{z}_N = (9/2, 3, 1/2, 1/2)^T, L = \begin{pmatrix} \sqrt{2} & 0 \\ -(3\sqrt{2})/2 & \sqrt{3/2} \end{pmatrix}.$$

Iteration 3:

1. $\bar{x}_B = (1, 2/3)^T$.
2. $\Delta y = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 & -2 \\ -1 & 1 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2/3 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/3 \\ 0 \\ 1/3 \end{pmatrix}$.
3. $\Delta y \neq 0$.

$$4. N^T \Delta y = \begin{pmatrix} 2 & 1 & 2 & 0 \\ 1 & -1 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1/3 \\ 1/3 \\ 0 \\ 1/3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1/3 \\ -1/3 \end{pmatrix}.$$

$$5. \beta = \min\{(9/2)/1, (1/2)/(1/3)\} = 3/2, \quad q = 4.$$

$$6. \bar{y} = (-1/2, 1/2, 0, -3/2)^T + (3/2)(1/3, 1/3, 0, 1/3)^T = (0, 1, 0, -1)^T.$$

$$7. L = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ -3/\sqrt{2} & -\sqrt{2}/2 & 1 \end{pmatrix}.$$

$$8. B = \{2, 4, 6\}, \quad N\{1, 3, 5\}.$$

$$9. k = 3.$$

Iteration 4:

$$1. \bar{x}_B = (1, 1, 1)^T.$$

$$2. \Delta y = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 & -2 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

$$3. \Delta y = 0.$$

10. $\bar{x}_B \geq 0$. The optimal solution and optimal value to (22.25) are

$$\bar{y} = (0, 1, 0, -1)^T, \quad \bar{z} = (3, 0, 3, 0, 1, 0)^T,$$

with the related optimal face

$$y_1 = 0, \quad y_2 = 1, \quad -3 \leq y_3 \leq 1, \quad y_4 = -1.$$

Finally, the optimal solution and optimal value to the original problem are

$$\bar{x} = (0, 1, 0, 1, 0)^T, \quad \bar{f} = -1,$$

and dual optimal solution is

$$\bar{y} = (0, 1, 0)^T, \quad \bar{z} = (3, 0, 3, 0, 1)^T.$$

22.6.2 Computational Results

Computational experiments were carried out in the same software and hardware environment, as described in Sect. 21.6.2.

The following two dense codes in FORTRAN 77 are compared:

1. RSA: the revised simplex algorithm, supported by the classical Phase-I (see Notation).
2. DFA: dual face Algorithm 22.6.1, supported by Algorithm 22.6.2 being as the Phase-I.

The test set of problems are the 25 smallest Netlib problems listed in Appendix C (Table C.4: AFIRO-SCAGR25). From Appendix E (Table E.5), it is seen that average ratios of RSA to DFA are as follows:

Total iteration ratio :1.19, Total time ratio: 10.04.

It is interesting to compare the performance of FALP (face algorithm) and DFA against RSA. It is seen that the iteration ratios of the two codes are 1.51 and 1.19, respectively, and time ratios are much higher: 4.91 and 10.04. Impressively, both methods outperformed the simplex algorithm with large margins.

As for a comparison between the face and dual face methods, the latter outperformed the former with a time ratio of 2.04. It is not surprising because the latter involves systems of order k , generally much less than the order, $m + 1$, of the systems involved in the former.

The detailed computational experiments are reported in Appendix E.

22.7 Dual Face Algorithm via Updating $(B^T B)^{-1}$

This section offers a variant of the dual face algorithm by updating $(B^T B)^{-1}$ rather than Cholesky factors. It is originally motivated by the idea of being amenable to large-scale sparse problems by getting rid of orthogonal transformations. Surprisingly, the variant turns out to be so simple that there is no need for solving any linear system although it might be less stable than the dual face algorithm.

Firstly, we consider how to update the inverse in a contracting iteration.

Lemma 22.7.1 *Let $B \in \mathcal{R}^{(m+1) \times k}$ with $1 \leq k < m + 1$ be a face matrix. If $0 \neq a_q \in \mathcal{R}^{m+1}$ satisfies $a_q \notin \text{range } B$, then it is well-defined that*

$$\mu = \left(a_q^T a_q - a_q^T B (B^T B)^{-1} B^T a_q \right)^{-1}. \quad (22.26)$$

Proof Since B is of full column rank k , $(B^T B)^{-1} \in \mathcal{R}^{k \times k}$ is well-defined. The condition $a_q \notin \text{range } B$ implies that the orthogonal projection of a_q onto the orthogonal complement space of range B does not vanish, i.e.,

$$a_q - B(B^T B)^{-1} B^T a_q = (I - B(B^T B)^{-1} B^T) a_q \neq 0,$$

and hence it holds that

$$a_q^T a_q - a_q^T B(B^T B)^{-1} B^T a_q = a_q^T (I - B(B^T B)^{-1} B^T) a_q \neq 0.$$

Thus, (22.26) is well-defined. $\square$

Theorem 22.7.1 Let $B \in \mathcal{R}^{(m+1) \times k}$ with $1 \leq k < m + 1$ be a face matrix, and let $(B^T B)^{-1}$ be available. If $\check{B} = (B \vdots a_q)$, where a_q is determined by step 5 of Algorithm 23.1.3, then it holds that

$$(\check{B}^T \check{B})^{-1} = \begin{pmatrix} (B^T B)^{-1} + \mu(B^T B)^{-1} B^T a_q a_q^T B(B^T B)^{-1} - \mu(B^T B)^{-1} B^T a_q & \\ -\mu a_q^T B(B^T B)^{-1} & \mu \end{pmatrix}, \quad (22.27)$$

where μ is defined by (22.26).

Proof Proposition 23.1.4 implies that $a_q \notin \text{range } B$. According to Lemma 22.7.1, therefore, μ (22.26) is well-defined. Then, since

$$\check{B}^T \check{B} = \begin{pmatrix} B^T \\ a_q^T \end{pmatrix} (B \ a_q) = \begin{pmatrix} B^T B & B^T a_q \\ a_q^T B & a_q^T a_q \end{pmatrix},$$

the validity of (22.27) is easily verified. $\square$

From the preceding theorem, the formula for updating $(B^T B)^{-1}$ follows, i.e.,

$$(\check{B}^T \check{B})^{-1} = \begin{pmatrix} D & v \\ v^T & \mu \end{pmatrix}, \quad (22.28)$$

where

$$u = B^T a_q, \quad w = (B^T B)^{-1} u, \quad \mu = (a_q^T a_q - u^T w)^{-1}, \quad v = -\mu w, \\ D = (B^T B)^{-1} - v w^T. \quad (22.29)$$

Nevertheless, the column a_q should be inserted before the last column of B (see step 8 of Algorithm 23.1.3). So, the true new face matrix may be obtained from $\check{B}$ by interchanging its last two columns. Denote the associated column permutation matrix by P^T . Then, it follows that

$$(\check{B}^T \check{B})^{-1} = ((\hat{B} P^T)^T (\hat{B} P^T))^{-1} = P(\hat{B}^T \hat{B})^{-1} P^T.$$

Therefore, the wanted inverse matrix can be obtained by interchanging the last two columns and rows of the matrix (22.27), which is referred to as *adding permutation*.

As for how to update the inverse in an expanding iteration, we have the following result.

Theorem 22.7.2 Let $\check{B} \in \mathcal{R}^{(m+1) \times k}$, $1 < k \leq m+1$, be a face matrix, and let $(\check{B}^T \check{B})^{-1}$ be of form (22.28). If B is created from $\check{B}$ by dropping its last column, then it holds that

$$(B^T B)^{-1} = D - \mu^{-1} v v^T. \quad (22.30)$$

Proof Assume that $\check{B} = (B \vdots a_q)$, and a_q is dropped. Then, (22.27) holds. Comparing (22.27) and (22.28) gives

$$v = -\mu(B^T B)^{-1} B^T a_q, \quad D = B^T B)^{-1} + \mu(B^T B)^{-1} B^T a_q a_q^T B(B^T B)^{-1},$$

from which (22.27) follows. $\square$

Therefore, it is quite simple to update the inverse in an expanding iteration: there are only $(k-1)^2/2 + k-1$ flops involved. Actually, the column selected to leave a face matrix, say $\hat{B}$, is always among the first $k-1$ columns of $\check{B}$. In this case, we first move the column to the end position to form a matrix, say $\check{B}$, and then drop it. Denote the corresponding column permutation by Q^T . Then, it holds that

$$(\check{B}^T \check{B})^{-1} = ((\hat{B} Q^T)^T (\hat{B} Q^T))^{-1} = Q(\hat{B}^T \hat{B})^{-1} Q^T.$$

Thus, $(\check{B}^T \check{B})^{-1}$ can be obtained by interchanging the last two columns and rows of the current inverse $(\hat{B}^T \hat{B})^{-1}$, which is referred to as *dropping permutation*.

In practice, an integer vector is enough to record positions of columns to avoid moving them all around. The overall steps can be put into the following algorithm.

Algorithm 22.7.1 [Dual face algorithm: updating $(B^T B)^{-1}$] Initial: B , N , $1 \leq k \leq m+1$, $(B^T B)^{-1}$, dual feasible solution $(\bar{y}, \bar{z}_N)$. This algorithm solves the pair of reduced problems (23.1) and (23.19).

1. Compute $\bar{x}_B = (B^T B)^{-1} e_k$.
2. If $k < m+1$, compute $\Delta y = e_{m+1} - B \bar{x}_B$.
3. Go to step 11 if $k = m+1$ or $\Delta y = 0$.
4. Stop if $N^T \Delta y \leq 0$ (problem (23.1) is upper unbounded).
5. Determine index q and stepsize β such that

$$\beta = \bar{z}_q / (a_q^T \Delta y) = \min \left\{ \bar{z}_j / (a_j^T \Delta y) \mid a_j^T \Delta y > 0, j \in N \right\}.$$

6. If $\beta \neq 0$, update $\bar{y} = \bar{y} + \beta \Delta y$, $\bar{z}_N = \bar{z}_N - \beta N^T \Delta y$.
7. Update $(B^T B)^{-1}$ by (22.28) with adding permutation.
8. Bring q from N , and insert before the last column of B .

9. Set $k = k + 1$, and go to step 1.
10. Stop if $\bar{x}_B \geq 0$ (optimality achieved).
11. Determine $p \in \arg \min_{j \in B} \bar{x}_j$.
12. Update $(B^T B)^{-1}$ by (22.30) with dropping permutation
13. Update $B = B \setminus \{p\}$, $N = N \cup \{p\}$.
14. Set $k = k - 1$.
15. Go to step 1.

It is seen that no linear system has to be solved in the solution process if $k = 1$ and $B = (a_{n+1})$ are set initially.

Example 22.7.1 Solve the following problem by Algorithm 22.7.1:

$$\begin{aligned}
 \min \quad & f = 5x_1 + x_2 + 3x_4 + 6x_6 + 2x_7 \\
 \text{s.t.} \quad & 3x_1 - 7x_3 + 5x_4 - 4x_5 - 6x_6 - 5x_7 = -18 \\
 & 1/3x_1 + 2x_2 + 1/3x_3 - 2/3x_5 + 2x_7 = 0 \\
 & -5x_2 + 11x_3 - 7x_4 - 3x_5 + 4x_6 = 9 \\
 & x_j \geq 0, \quad j = 1, \dots, 7.
 \end{aligned}$$

Answer

The problem is transformed to the following one in tableau form:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	RHS
3		-7	5	-4	-6	-5	18	
1/3	2	1/3		-2/3		2		
	-5	11	-7	-3	4		-9	
							1	1
5	1		3		6	2		

$$\begin{aligned}
 k = 1, \quad B = a_8 = (18, 0, -9, 1)^T, \quad N = A \setminus B; \quad (B^T B)^{-1} = (1/406), \\
 \bar{y} = (0, 0, 0, 0)^T, \quad \bar{z} = (5, 1, 0, 3, 0, 6, 2, 0)^T.
 \end{aligned}$$

Iteration 1:

1. $\bar{x}_B = (1/406)$.
2. $\Delta y = (-9/203, 0, 9/406, 405/406)^T$.
4. $N^T \Delta y = (-27/203, -45/406, 225/406, -153/406, 45/406, 72/203, 45/203)^T \not\leq 0$.
5. $\beta = \min\{0/(225/406), 0/(45/406), 6/(72/203), 2/(45/203)\} = 0, \quad q = 3$.
 $a_q = (-7, 1/3, 11, 0)^T$.
7. $(B^T B)^{-1} = \begin{pmatrix} 179/8130 & 91/7458 \\ 91/7458 & 517/56043 \end{pmatrix}$.

$$8. B = \begin{pmatrix} -7 & 18 \\ 1/3 & \\ 11 & -9 \\ & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 3 & 5 & -4 & -6 & -5 \\ 1/3 & 2 & & -2/3 & 2 \\ & -5 & -7 & -3 & 4 \end{pmatrix}.$$

9. $k = 2$.

Iteration 2:

$$1. x_B = (91/7458, 517/56043)^T.$$

$$2. \Delta y = (-227/2815, -53/13031, -294/5743, 537/542)^T \not\leq 0.$$

$$4. N^T \Delta y = (-642/2639, 314/1267, -84/1873, 781/1631, 1541/5522, 4977/12598)^T.$$

$$5. \beta = \min\{1/(314/1267), 0/(781/1631), 6/(1541/5522), 2/(4977/12598)\} = 0, \quad q = 5.$$

$$a_q = (-4, -2/3, -3, 0)^T.$$

$$7. (B^T B)^{-1} = \begin{pmatrix} 1027/977 & 420/271 & 2223/2947 \\ 420/271 & 1447/620 & 827/740 \\ 2223/2947 & 827/740 & 1049/1927 \end{pmatrix}.$$

$$8. B = \begin{pmatrix} -7 & -4 & 18 \\ 1/3 & -2/3 & \\ 11 & -3 & -9 \\ & & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 3 & 5 & -6 & -5 \\ 1/3 & 2 & & 2 \\ & -5 & -7 & 4 \end{pmatrix}.$$

9. $k = 3$.

Iteration 3:

$$1. x_B = (2223/2947, 827/740, 1049/1927)^T.$$

$$2. \Delta y = (-125/2599, 733/1485, -134/2941, 878/1927)^T \not\leq 0.$$

$$4. N^T \Delta y = (325/16051, 1893/1558, 51/650, 37/348, 1224/997)^T.$$

$$5. \beta = \min\{5/(325/16051), 1/(1893/1558), 3/(51/650), 6/(37/348), 2/(1224/997)\}$$

$$= 1/(1893/1558), \quad q = 2. \quad a_q = (0, 2, -5, 0)^T.$$

$$6. \bar{y} = (-217/5482, 13765/33883, -1165/31067, 33364/88971)^T,$$

$$\bar{z}_N = (299/60, 0, 1591/542, 11689/1977, 1709/1727)^T.$$

$$7. (B^T B)^{-1} = \begin{pmatrix} 2271/1000 & 1559/559 & 684/1115 & 2371/1581 \\ 1559/559 & 2407/670 & 1217/1953 & 778/415 \\ 684/1115 & 1217/1953 & 1106/3585 & 298/795 \\ 2371/1581 & 778/415 & 298/795 & 5281/5282 \end{pmatrix}.$$

$$8. B = \begin{pmatrix} -7 & -4 & 18 \\ 1/3 & -2/3 & 2 \\ 11 & -3 & -5 & -9 \\ & & & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 3 & 5 & -6 & -5 \\ 1/3 & & & 2 \\ & -7 & 4 & \end{pmatrix}.$$

9. $k = 4$.

Iteration 4:

11. The optimal solution and value are

$$\bar{x} = (0, 0.3748, 0, 1.4997, 0, 1.8747, 0)^T \geq 0, \quad \bar{f} = c_B^T \bar{x}_B = 0.3748.$$

The optimal solution obtained is

$$x^* = (0, 0.3750, 0, 1.5000, 0, 1.8750, 0), \quad f^* = 0.3750.$$

Chapter 23

Face Method with LU Factorization



The face and dual face methods, introduced in the previous two chapters, involve orthogonal transformations. But from a practical point of view, many researchers prefer LU factorization. In this chapter, we derive the face method based on LU factorization. In the following sections, we derive the descent search direction first and then address such topics as the update of the face solution, pivot operation, and optimality test. Then we spend the other sections to formulate the face algorithm in the tableau form and in the revised form, respectively, before making final comments.

Consider the standard LP problem; i.e.,

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b, \quad x \geq 0, \end{aligned} \tag{23.1}$$

where $A \in \mathcal{R}^{m \times n}$ ($m < n$), $b \in \mathcal{R}^m$, $c \in \mathcal{R}^n$; $\text{rank}(A) = m$.

23.1 Decent Search Direction

The quality of the search direction used is crucial to any LP solver. In this section, we derive a fresh search direction first.

The trick is to partition further by

$$\underline{B} = B \cup B', \quad \text{rank}(B) = m, \tag{23.2}$$

where $B \in \mathcal{R}^{m \times m}$ is called *basis (matrix)* and B' called *active matrix*. Associated items, such as indices, variables, etc., are said similarly.

Viewing the objective value f as a variable, we put subprogram (1.10) into the system of equations, expressed as

$$\begin{array}{c|ccc|c}
 x_B & x_{B'} & \bar{x}_N & f & RHS \\
 \hline
 B & B' & N & & b \\
 \hline
 c_B^T & c_{B'}^T & c_N^T & -1 &
 \end{array} \quad (23.3)$$

where entries under $\bar{x}_N$ correspond to the inactive (fixed) part.

Execute Gauss–Jordan elimination to convert the preceding tableau into the following form:

$$\begin{array}{c|ccc|c}
 x_B & x_{B'} & \bar{x}_N & f & RHS \\
 \hline
 I_m & \bar{B}' & \bar{N} & & \bar{b} \\
 \hline
 & \bar{c}_{B'}^T & \bar{c}_N^T & -1 & \mu
 \end{array} \quad (23.4)$$

which is termed (*canonical*) *face tableau*.

Consider the following direction:

$$\Delta x_{B \cup B'} = \begin{pmatrix} \Delta x_B \\ \Delta x_{B'} \end{pmatrix} = \begin{pmatrix} -\bar{B}' \bar{c}_{B'} \\ \bar{c}_{B'} \end{pmatrix}. \quad (23.5)$$

Theorem 23.1.1 *Let $\Delta x_{B \cup B'}$ be defined by (23.5). If it is nonzero, the following holds:*

$$(I_m, \bar{B}') \Delta x_{B \cup B'} = 0, \quad (23.6)$$

$$(0, \bar{c}_{B'}^T) \Delta x_{B \cup B'} > 0. \quad (23.7)$$

Proof The first expression of (23.7) is easily verified by (23.6) and (23.5).

In addition, it holds by (23.5) that

$$\left(0, \bar{c}_{B'}^T\right) \Delta x_{B \cup B'} = \bar{c}_{B'}^T \bar{c}_{B'} > 0,$$

which exhibits the validity of the second expression of (23.7). $\square$

The preceding implies that nonzero $-\Delta x_{B \cup B'}$ is descent in the null space of $(I_m, \bar{B}')$ and hence is eligible to be taken as search direction. It is more than that. In the next section, we will show that it is actually the best, in a certain sense.

23.2 Updating of Face Solution

Assume that $\bar{x}_{B \cup B'}$ is the current face solution (to subprogram (1.10)) and that a nonzero search direction $\Delta x_{B \cup B'}$ defined by (23.5) is available.

Define the index set

$$J = \{j \in B \cup B' \mid \Delta x_j > 0\}. \quad (23.8)$$

Assume now that J is nonempty, and hence the following line search scheme is well-defined:

$$\hat{x}_{B \cup B'} = \bar{x}_{B \cup B'} - \alpha \Delta x_{B \cup B'}, \quad (23.9)$$

where α is the stepsize such that

$$\alpha = \min_{j \in J} \bar{x}_j / \Delta x_j \geq 0. \quad (23.10)$$

For the new solution $\hat{x}_{B \cup B'}$, the following is claimed.

Theorem 23.2.1 *Assume that $\Delta x_{B \cup B'}$ is defined by (23.5), and J defined by (23.8) is nonempty. If $\bar{x}_{B \cup B'}$ is a feasible solution to subprogram (1.10), then so is $\hat{x}_{B \cup B'}$ given by (23.9) with (23.10).*

Proof Under the assumption, it holds that

$$(B \cup B')\bar{x}_{B \cup B'} = b, \quad \bar{x}_{B \cup B'} \geq 0. \quad (23.11)$$

Combining (23.9), (23.6), and the first expression in (23.11) gives

$$(B \cup B')\hat{x}_{B \cup B'} = (B \cup B')\bar{x}_{B \cup B'} - \alpha(B \cup B')\Delta x_{B \cup B'} = b. \quad (23.12)$$

Thus, $\hat{x}_{B \cup B'}$ satisfies the equality constraints.

Note that $\bar{x}_j \geq 0 \forall j \in B \cup B'$. Without loss of generality, assume that $(B \cup B') \setminus J \neq \emptyset$.

It is clear by (23.11) that

$$\Delta x_j \leq 0, \quad j \in (B \cup B') \setminus J,$$

from which and (23.9), for any $\alpha \geq 0$, it follows that

$$\hat{x}_j = \bar{x}_j - \alpha \Delta x_j \geq 0, \quad j \in (B \cup B') \setminus J. \quad (23.13)$$

On the other hand, it holds by (23.10) that

$$\bar{x}_j / \Delta x_j \geq \alpha \geq 0, \quad j \in J.$$

Multiplying the preceding by $\Delta x_j > 0$ leads to

$$\bar{x}_j \geq \alpha \Delta x_j \geq 0, \quad j \in J,$$

which together with (23.13) gives

$$\hat{x}_j = \bar{x}_j - \alpha \Delta x_j \geq 0, \quad j \in J. \quad (23.14)$$

Consequently, (23.12), (23.13), and (23.14) together imply that $\hat{x}_{B \cup B'}$ is a feasible solution to (23.14). $\square$

From (23.13) and (23.14), it is seen that only the requirement $\hat{x}_j \geq 0, \forall j \in J$, limits the increase of the stepsize, so that α determined by (23.14) is the maximum possible stepsize that can be taken along the search direction. Indices in J and corresponding variables are often said to be *blocking* ones.

Note that k -vector $x_{B \cup B'}$ is a feasible solution to (23.14) if and only if the corresponding n -vector

$$x = \begin{bmatrix} x_{B \cup B'} \\ x_N \end{bmatrix} = \begin{bmatrix} x_{B \cup B'} \\ 0 \end{bmatrix}$$

is a feasible solution to the original problem (23.1).

Now, let us explore the effect of the search direction further.

Assuming $\bar{c}_{B'} \neq 0$, we introduce the following simple result first.

Lemma 23.2.1 For $k \times (k - m)$ matrix

$$Z = \begin{pmatrix} -\bar{B}' \\ I_{k-m} \end{pmatrix}, \quad (23.15)$$

it holds that

$$(I_m, \bar{B}')Z = 0, \quad \text{and} \quad \text{rank}(Z) = k - m. \quad (23.16)$$

Proof The validity can be easily verified. $\square$

Z is known as a *null space matrix* of $(I_m, \bar{B}')$, which is the coefficient matrix of the equality constraints of face tableau (23.4). Thus, the set $\{-Zu \mid u \in R^{k-m}\}$ includes all possible search direction candidates. If $u = \bar{c}_{B'}$, in particular, it is just search direction $\Delta x_{B \cup B'}$ defined by (23.5).

For any given $u \in R^{k-m}$, the associated update of $\bar{x}_{B \cup B'}$ is

$$\hat{x}_{B \cup B'}(u) = \bar{x}_{B \cup B'} - \alpha Zu,$$

where α is stepsize. Premultiplying the preceding by $(0, \bar{c}_{B'}^T)$ and moving terms, we obtain the corresponding objective decrement, i.e.,

$$\gamma(u) = \left(0, \bar{c}_{B'}^T\right) \bar{x}_{B \cup B'} - \left(0, \bar{c}_{B'}^T\right) \hat{x}_{B \cup B'}(u) = \alpha \left(0, \bar{c}_{B'}^T\right) Zu, \quad (23.17)$$

which not only depends on the direction of u but also depends on the norm of it. It is therefore reasonable to restrict u by

$$\|u\| = \|\bar{c}_{B'}\|. \quad (23.18)$$

Using (23.17) and (23.18), we then turn to the following problem:

$$\begin{aligned} \max \gamma(u) &= \alpha(0, \bar{c}_{B'}^T)Zu \\ \text{s.t. } \|u\| &= \|\bar{c}_{B'}\|, \quad u \in \mathcal{R}^{k-m}. \end{aligned} \quad (23.19)$$

Theorem 23.2.2 *Problem (23.19) has unique optimal solution $u^* = \bar{c}_{B'}$, corresponding to maximum objective decrement $\gamma(u^*) = \alpha\|\bar{c}_{B'}\|^2$.*

Proof From (23.15), (23.17), and the constraint of (23.19), it follows that

$$\begin{aligned} \gamma(u) &= \alpha \left(0, \bar{c}_{B'}^T\right) \begin{pmatrix} -\bar{B}' \\ I_{k-m} \end{pmatrix} u \\ &= \alpha \bar{c}_{B'}^T u \\ &= \alpha \|\bar{c}_{B'}\| \|u\| \cos \angle \bar{c}_{B'}, u > \\ &= \alpha \|\bar{c}_{B'}\|^2 \cos \angle \bar{c}_{B'}, u > . \end{aligned} \quad (23.20)$$

It is therefore clear that $u^* = \bar{c}_{B'}$ is the unique optimal solution to problem (23.19), corresponding to $\gamma(u^*) = \alpha\|\bar{c}_{B'}\|^2$. $\square$

Theorem 23.2.3 ensures that $\gamma(u^*)$, being proportional to the square of $\|\bar{c}_{B'}\|$, is the maximum possible objective decrement that can be achieved. Therefore, the search direction $-\Delta x_{B \cup B'} = -Z\bar{c}_{B'}$ appears to be the “best.”

In the case when J is empty, we are then done, as the following says.

Theorem 23.2.3 *Assume that $\bar{x}_{B \cup B'}$ is a feasible solution and $\Delta x_{B \cup B'}$ defined by (23.5) is nonzero. If J defined by (23.8) is empty, problem (23.1) is lower unbounded.*

Proof The emptiness of J implies $\Delta x_{B \cup B'} \leq 0$, which along with $\bar{x}_{B \cup B'} \geq 0$ leads to

$$\hat{x}_{B \cup B'} = \bar{x}_{B \cup B'} - \alpha \Delta x_{B \cup B'} \geq 0, \quad \forall \alpha \geq 0. \quad (23.21)$$

From the preceding, (23.6), and $B \cup B' \bar{x}_B = b$, it follows that

$$B\hat{x}_B = B\bar{x}_B - \alpha B\Delta x_B = b, \quad \forall \alpha \geq 0.$$

Therefore, $\hat{x}_B$ is a feasible solution to (1.10) for all $\alpha \geq 0$.

Moreover, by (23.21) and (23.7), it holds that

$$c_B^T \bar{x}_B - c_B^T \hat{x}_B = \alpha c_B^T \Delta x_B > 0, \quad \forall \alpha > 0.$$

Consequently, the decrement in objective value goes to infinity, as so does α . $\square$

23.3 Pivoting Operation

Assume $k > m$, or $B' \neq \emptyset$. Define the following subset:

$$S = \{j \in B' \mid \bar{x}_j = 0, \bar{c}_j \geq 0\}, \quad (23.22)$$

which consists of all indices of B' exhibiting complementarity slackness. It is obvious that if S is empty, any zero stepsize will never yield from B' . This can be ensured at the beginning of each iteration by executing the following tactic.

Tactic 23.3.1 If S is nonempty, adjust (B', N, k) by

$$B' = B' \setminus S, \quad N = N \cup S, \quad k = k - |S|, \quad (23.23)$$

where $|S|$ denotes the cardinality of S (the same below).

Now assumed that a new face solution $\hat{x}_B$ was computed by (23.9) with α given by (23.10).

Introduce the following index subset:

$$J_\alpha = \{j \in J \mid \bar{x}_j / \Delta x_j = \alpha\}, \quad (23.24)$$

where J is defined by (23.8). It is clear that

$$\hat{x}_j = 0, \quad \forall j \in J_\alpha. \quad (23.25)$$

If $J_\alpha \subset B'$ (or $B \cap J_\alpha = \emptyset$), what to do is just going to the next iteration, where J_α will be moved from B' to N .

Now consider the other case: $J_\alpha \not\subset B'$ (or $B \cap J_\alpha \neq \emptyset$).

Designate by j_i the index of the i -th column, e_i , of $\bar{B} \triangleq I_m$. Thus, variable x_{j_i} corresponds to the i -th row of tableau (23.4).

At first, select a pivot row by the following rule.

Rule 23.3.1 [Row Rule] *Select the row index p such that*

$$j_p \in \arg \max_{j_i \in J_\alpha} \Delta x_{j_i}. \quad (23.26)$$

Thus, the index j_p is determined to leave B . The reason for choosing such a one is as follows.

It is evident that the cosine of the angle between the search direction $-\Delta x_B$ and the gradient of a constraint $x_j \geq 0$ equals

$$\alpha_j = -\Delta x_j / \|\Delta x_B\|, \quad \forall j \in B. \quad (23.27)$$

By the so-called “heuristic characterization of optimal solution” (pp. 56-60, Pan (2014)), for all $j_i \in B$, in particular, the constraint $x_{j_i} \geq 0$ corresponding to the smallest α_{j_i} , or the maximum Δx_{j_i} , tends to be active at the optimal solution. This indicates that the corresponding index should be selected to enter N , and the associated row be taken as the pivot row.

It is noticeable that $\Delta x_{j_p} > 0$ since $j_p \in J_\alpha \in J$. Extract the p -th row of $\bar{B}'$ from tableau (23.4), i.e.,

$$v_{B'}^T = e_p^T \bar{B}'. \quad (23.28)$$

Then, from (23.5), it follows that

$$\Delta x_{j_p} = -e_p^T \bar{B}' \bar{c}_{B'} = - \sum_{j \in B'} v_j \bar{c}_j > 0, \quad (23.29)$$

which implies that there exists index $j \in B'$ such that $v_j \bar{c}_j < 0$. Thus, we are able to bring up the following plausible column rule in accordance with Rule 23.3.1.

Rule 23.3.2 (Column Rule) *Select the column index q to enter B such that*

$$q \in \arg \min \{v_j \bar{c}_j \mid j \in B'\}. \quad (23.30)$$

If there is a tie, select one corresponding to the maximum $\hat{x}_q$.

Assume now that row index p and column index q have been determined. Update the face tableau (23.4) by premultiplying the p -th row by $1/\bar{a}_{p,q}$ and adding $-\bar{a}_{i,q}$ times of the row to the i -th row, for $i = \{1, \dots, m\}$, $i \neq p$. Then, adjust sets accordingly by the following tactic.

Tactic 23.3.2 Adjust (B, B', N, k) by

$$\hat{B} = (B \setminus \{j_p\}) \cup \{q\}, \quad \hat{B}' = B' \setminus \{q\}, \quad \hat{N} = N \cup \{j_p\}, \quad k = k - 1. \quad (23.31)$$

23.4 Optimality Test

It is noticeable that (23.4) is actually a simplex tableau of the original problem (23.1) if one takes into account of both the face and inactive sections.

Assume that search direction $\Delta x_{B'}$ vanishes, and hence line search scheme (23.9) becomes meaningless. It is clear that this case occurs if and only if $k = m$, or else $\bar{c}_{B'} = 0$.

Lemma 23.4.1 *If active reduced costs $\bar{c}_{B'}$ vanish, the face P_B defined by (21.3) is a level face.*

Proof By $\bar{c}_{B'} = 0$, it is clear that the equation corresponding to the objective row of (23.4) becomes

$$0x_B + 0x_{B'} + \bar{c}_N^T x_N - f = -\bar{f}. \quad (23.32)$$

Since all $\bar{x} \in P_B$ satisfy $\bar{x}_N = 0$, substituting them to (23.32) results in $f = \bar{f}$, which means that P_B is a level face. $\square$

Since the objective value is constant over the current face, in this case, it is not possible to make further progress from the associated feasible solution $\bar{x}$, listed in the bottom row of tableau (23.4). It is time to test for optimality.

Theorem 23.4.1 *Assume that the active reduced cost $\bar{c}_{B'}$ vanishes. Then P_B is an optimal face if the following index set is empty:*

$$T = \{j \in N \mid \bar{c}_j < 0\}. \quad (23.33)$$

Proof Note that $\bar{c}_N$ is nonnegative under the assumption.

By Lemma 23.4.1, $\bar{c}_{B'} = 0$ implies that P_B is a level face. Since $\bar{x}$ belongs to P_B , therefore, just examine whether $\bar{x}$ is optimal. In fact, it is so since $\bar{c}_{B'} = 0$ and $\bar{c}_N \geq 0$ together ensure that the simplex tableau (23.4) is optimal. Therefore, P_B is an optimal face. $\square$

Corollary 23.4.2 *Under assumptions of Theorem 23.4.1, the associated $\bar{x}$ and $\bar{c}$ are a pair of primal and dual optimal solutions to problem (23.1).*

In case when $\bar{c}_{B'}$ vanishes but T is nonempty, however, optimality cannot be claimed. Then perform the following tactic to expand the face, and carry out a new round of iterations.

Tactic 23.4.1 To expand the face, adjust (B', N, k) by

$$\widehat{B'} = B' \cup T, \quad \widehat{N} = N \setminus T, \quad k = k + |T|. \quad (23.34)$$

23.5 Face Algorithm: Tableau Form

In each iteration, a face tableau in form (23.4) is handled with $\bar{x}_{B \cup B'}$ updated and (B, B', N, k) adjusted so that k decreases by one, at least. Such a series of iterations referred to as *faces contraction* are performed until reaching a level face to test for optimality (unless detecting lower unboundedness). If optimality cannot be declared, we execute Tactic 23.4.1 to expand the face and carry out another face contraction.

We suggest some kind of Phase-1 using “the most-obtuse angle rule” (Sect. 24.3) although any simplex Phase-1 applies. However, we will not go into any details here.

Let B be the initial basis. Alternatively, it might be convenient to take the following initial partition:

$$B' = A \setminus B, \quad N = \emptyset, \quad k = m, \quad (23.35)$$

with feasible solution

$$\bar{x} \triangleq \bar{x}_{B \cup B'} = \{\bar{x}_B, \bar{x}_{B'}\}, \quad \bar{x}_{B'} = 0. \quad (23.36)$$

Overall steps are summarized as follows.

Algorithm 23.5.1 [Face Algorithm: Tableau Form] *Initial: basis B with partition (23.35); canonical face tableau in form (23.4); feasible solution (23.36). This algorithm solves problem (23.1).*

1. If S defined by (23.22) is nonempty, adjust (B', N, k) by (23.23).
2. Go to step 13 if $\bar{c}_{B'} = 0$.
3. Compute $\Delta x_{B \cup B'}$ by (23.5).
4. Stop if J defined by (23.8) is empty.
5. Determine α by (23.10).
6. If $\alpha > 0$, update $\bar{x}_{B \cup B'}$ by (23.9).
7. Go to step 1 if $J_\alpha \subset B'$, where J_α is defined by (23.24).
8. Determine row index p by Rule 23.3.1.
9. Determine column index q by Rule 23.3.2.
10. Multiply the p -th row by $1/\bar{a}_{p,q}$, and add $-\bar{a}_{i,q}$ times of the p -th row to the i -th row for all $i = 1, \dots, m+1$; $i \neq p$.
11. Adjust (B, B', N, k) by (23.31).
12. If $k > m$, go to step 1.
13. Stop if T defined by (23.33) is empty.
14. Adjust (B', N, k) by (23.34), and go to step 3.

Theorem 23.5.1 Assume nondegeneracy in a single iteration for each contraction process. Algorithm 23.5.1 terminates either at

- (i) Step 4, declaring lower unboundedness, or at
- (ii) Step 13, giving an optimal face together with a pair of primal and dual optimal solutions

Proof It is evident that Algorithm 23.5.1 reaches a level face at the end of every contraction process, within finitely many iterations, unless detecting lower unboundedness.

Assume that Algorithm 23.5.1 does not terminate. Then, it must carry out infinitely many contraction processes, each of which includes an iteration with nonzero stepsize. Therefore, by Theorem 23.2.3, the traversed level faces correspond to strictly reduced objective values. This contradicts that the number of level faces is finite. Therefore, the algorithm terminates.

By Theorem 23.2.1, exit step 4 indicates lower unboundedness. By Theorem 23.4.1 and Corollary 23.4.2, exiting from step 13 gives an optimal face together with a pair of primal and dual solutions. $\square$

We point out that degeneracy remains a problem, at this stage. But, the nondegeneracy assumption in Algorithm 23.5.1 seems to be quite relaxed. Indeed, it is unimaginable that a contraction process does not include any iteration involving nonzero stepsize.

It is noted that if stepsize is nonzero, components of $\hat{x}_{B'}$ are all strictly positive, and hence, any chosen entering $\hat{x}_q$ will never vanish. It is also evident that leaving index j_p will never reenter B until another faces contraction conducted.

Example 23.5.1 Solve Beale's cycling problem by Algorithm 23.5.1:

$$\begin{aligned} \min f &= -3/4x_4 + 20x_5 - 1/2x_6 + 6x_7, \\ \text{s.t. } x_1 &+ 1/4x_4 - 8x_5 - x_6 + 9x_7 = 0, \\ x_2 &+ 1/2x_4 - 12x_5 - 1/2x_6 + 3x_7 = 0, \\ x_3 &+ x_6 &= 1, \\ x_j &\geq 0, \quad j = 1, \dots, 7. \end{aligned}$$

Initial: $\bar{x} = [0, 0, 1, 0, 0, 0, 0]^T$, $B = \{1, 2, 3\}$, $B' = \{4, 5, 6, 7\}$, $N = \emptyset$.
Face tableau

x_1	x_2	x_3	x_4	x_5	x_6	x_7
1	0	0	1/4	-8	-1	9
0	1	0	1/2	-12	-1/2	3
0	0	1	0	0	1	0
0	0	0	-3/4	20	-1/2	6
0	0	1	0	*	0	*

Iteration 1:

1. $S = \{5, 7\} \neq \emptyset$, $B' = \{4, 6\}$, $N = \{5, 7\}$; $\bar{x}_{B \cup B'} = [1, 2, 3, 4, 6]^T$, $k = 5$.
2. $\bar{c}_{B'} = [-3/4, -1/2]^T \neq 0$.
3. $B' = \begin{bmatrix} 1/4 & -1 \\ 1/2 & -1/2 \\ 0 & 1 \end{bmatrix}$, $\Delta x_{B \cup B'} = [-5/16, 1/8, 1/2, -3/4, -1/2]^T \neq 0$.
4. $J = \{2, 3\} \neq \emptyset$.
5. $\alpha = \min\{0, 2\} = 0$.
7. $\{2\} = J_\alpha \not\subset B'$.
8. $j_p = 2$, $p = 2$.
9. $\min\{(1/2)(-3/4), (-1/2)(-1/2)\} = -3/8$, $q = 4$.
10. Multiply row 2 by $1/(1/2)$, and add $-1/4$ and $3/4$ times of row 2 to rows 1 and 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7
1	$-1/2$	0	0	-2	$-3/4$	$15/2$
0	2	0	1	-24	-1	6
0	0	1	0	0	1	0
0	$3/2$	0	0	2	$-5/4$	$21/2$
0	*	1	0	*	0	*

11. $B = \{1, 4, 3\}$, $B' = \{6\}$, $N = \{2, 5, 7\}$, $k = 4$.
12. $k > m$.

Iteration 2:

1. $S = \emptyset$.
2. $\bar{c}_{B'} = [-3/4, -1/2]^T \neq 0$.
3. $B' = [-3/4, -1, 1]^T$, $\Delta x_{B \cup B'} = [-15/16, -5/4, 5/4, -5/4]^T \neq 0$.
4. $J = \{3\} \neq \emptyset$.
5. $\alpha = \min\{4/5\}$.
6. $\bar{x}_{B \cup B'} = [0, 0, 1, 0]^T - (4/5)[-15/16, -5/4, 5/4, -5/4]^T = [3/4, 1, 0, 1]^T$.
7. $\{3\} = J_\alpha \not\subset B'$.
8. $j_p = 3$, $p = 3$.
9. $\min\{-5/4\}$, $q = 6$.
10. Add $3/4$, 1, and $5/4$ times of row 3 to rows 1, 2, and 4, respectively:

x_1	x_2	x_3	x_4	x_5	x_6	x_7
1	-1/2	3/4	0	-2	0	15/2
0	2	1	1	-24	0	6
0	0	1	0	0	1	0
0	3/2	5/4	0	2	0	21/2
3/4	*	*	1	*	1	*

11. $B = \{1, 4, 6\}$, $B' = \emptyset$, $N = \{2, 3, 5, 7\}$, $k = 3$.
12. $k = m$.
13. $T = \emptyset$. Terminated with primal and dual optimal solutions and optimal value:

$$x^* = [3/4, 0, 0, 1, 0, 1, 0]^T, \quad z^* = [0, 3/2, 5/4, 0, 2, 0, 21/2]^T, \quad f^* = -5/4.$$

23.6 Face Algorithm

Algorithm 23.5.1 can be transformed into a practicable version, using original data instead of the canonical face tableau as a whole. It involves some key quantities, which can be calculated based on (23.10) and (23.11).

We favor the LU factorization. Assume that $B = LU$ is available.

- (i) Search direction $\Delta x_{B \cup B'} \triangleq \{\Delta x_B, \Delta x_{B'}\}$ can be calculated as follows.
At first, solve the following two $m \times m$ triangular systems successively for $\bar{y} = B^{-T} c_B$:

$$U^T w = c_B, \quad L^T y = w. \quad (23.37)$$

Then compute active reduced costs by

$$\bar{c}_{B'} = c_{B'} - B'^T \bar{y}. \quad (23.38)$$

Therefore, $\Delta x_{B'} = \bar{c}_{B'}$ is available.

To gain $\Delta x_B = -B^{-1} B' \bar{c}_{B'}$, solve the following two $m \times m$ triangular systems successively:

$$Lw = -B' \bar{c}_{B'}, \quad U \Delta x_B = w. \quad (23.39)$$

- (ii) Assume that pivot row index p is determined. Then the corresponding row of $\bar{B}'$, i.e., $v_{B'}^T = e_p^T B_1^{-1} B'$, can be obtained as follows:
Firstly, solve the following two $m \times m$ triangular systems successively for $u = B^{-T} e_p$:

$$U^T w = e_p, \quad L^T u = w. \quad (23.40)$$

Then, compute

$$v_{B'} = B'^T u. \quad (23.41)$$

- (iii) Let $v_{B'}$ be available. Assume that $j_p \in B$ was moved to N and $q \in B'$ moved to B .

Then, active reduced costs can be updated for the next iteration by the following formula:

$$\hat{c}_{B'} = \bar{c}_{B'} + \beta v_{B'}, \quad \beta = -\bar{c}_q / v_q. \quad (23.42)$$

Overall steps are summarized in the following algorithm.

Algorithm 23.6.1 [Face algorithm: revised form] *Initial: Given basis B and partition (23.35); factorization $B = LU$; feasible solution (23.36). This algorithm solves problem (23.1).*

1. Compute $\bar{c}_{B'}$ by (23.37) and (23.38).
2. If S defined by (23.22) is nonempty, adjust (B', N, k) by (23.23).
3. Go to step 15 if $\bar{c}_{B'} = 0$.
4. Set $\Delta x_{B'} = \bar{c}_{B'}$, and compute Δx_B by (23.39).
5. Stop if $J = \{j \in B \cup B' \mid \Delta x_j > 0\}$ is empty (lower unbounded).
6. Determine α by (23.10).
7. If $\alpha > 0$, update $\bar{x}_{B \cup B'}$ by (23.9).
8. Go to step 2 if $J_\alpha \subset B'$, where J_α is defined by (23.24).
9. Determine row index p by Rule 23.3.1.
10. Compute $v_{B'}$ by (23.40) and (23.41).
11. Determine column index q by Rule 23.3.2.
12. Update L and U .
13. Adjust (B, B', N, k) by (23.31).
14. If $k > m$, update $\bar{c}_{B'}$ by (23.42), and go to step 2.
15. Compute $\bar{c}_N = c_N - N^T \bar{y}$, where $\bar{y}$ is defined by (23.37).
16. Stop if T defined by (23.33) is empty (optimality achieved).
17. Adjust (B', N, k) by (23.34), and go to step 4.

Example 23.6.1 Solve the following problem by Algorithm 23.6.1 in two phases:

$$\begin{aligned} \min \quad & x_{10} = -6x_1 + 5x_2 - 3x_3 - 4x_5 + 9x_6 - 2x_7, \\ \text{s.t.} \quad & 3x_1 - 2x_2 + 5x_3 + 4x_4 + 3x_6 + 5x_7 = 15, \\ & -x_2 + 6x_3 + 2x_4 + 8x_5 - 5x_6 + 4x_7 = 18, \\ & 5x_1 + 3x_2 - 2x_3 - 8x_4 - 4x_5 + x_6 - 3x_7 = 9, \\ & x_j \geq 0, \quad j = 1, \dots, 7. \end{aligned}$$

Any conventional Phase-I approach is applicable to provide a basis and a feasible solution to get Algorithm 23.6.1 started. Here, we form the auxiliary problem using artificial variables.

Phase-I:

Introduce artificial variables x_8, x_9 , and x_{10} to the three equalities, respectively, and eliminate them from auxiliary objective function $f' = -x_8 - x_9 - x_{10}$ by using the equalities. The resulting Phase-I problem is as follows:

$$\begin{aligned} \min f' &= -8x_1 - 9x_3 + 2x_4 - 4x_5 + 7x_6 - 6x_7, \\ \text{s.t. } 3x_1 - 2x_2 + 5x_3 + 4x_4 &+ 3x_6 + 5x_7 + x_8 &= 15, \\ &-x_2 + 6x_3 + 2x_4 + 8x_5 - 5x_6 + 4x_7 + x_9 &= 18, \\ 5x_1 + 3x_2 - 2x_3 - 8x_4 - 4x_5 &+ x_6 - 3x_7 + x_{10} &= 9, \\ x_j &\geq 0, \quad j = 1, \dots, 10. \end{aligned}$$

The objective gradient and the feasible solution are, respectively,

$$c = [-8, 0, -9, 2, -4, 7, -6, 0, 0, 0]^T \quad \text{and} \quad \bar{x} = [0, 0, 0, 0, 0, 0, 0, 15, 18, 9]^T.$$

Initially, we take

$$B = \{8, 9, 10\}, \quad B' = \{1, 2, 3, 4, 5, 6, 7\}, \quad N = \emptyset,$$

so that B and its LU factors are all unit matrices, and $\bar{x}_{B'} = 0$.

Iteration 1:

1. $\bar{y} = 0, \bar{c}_{B'} = c_{B'} = [-8, 0, -9, 2, -4, 7, -6]^T$.
2. $S = \{2, 4, 6\} \neq \emptyset, B' = \{1, 3, 5, 7\}, N = \{2, 4, 6\}, k = 7$.
3. $\bar{c}_{B_2} \neq 0$.
4. $L = U = I_3, w = -B'\bar{c}_{B'} = -\begin{bmatrix} 3 & 5 & 0 & 5 \\ 0 & 6 & 8 & 4 \\ 5 & -2 & -4 & -3 \end{bmatrix}[-8, -9, -4, -6]^T$
 $= [99, 110, -12]^T, \Delta x_{B \cup B'} = [99, 110, -12, -8, -9, -4, -6]^T \neq 0$.
5. $J = \{8, 9\} \neq \emptyset$.
6. $\alpha = \min\{5/33, 9/55\} = 5/33$.
7. $\bar{x}_B = [15, 18, 9, 0, 0, 0, 0]^T - (5/33)[99, 110, -12, -8, -9, -4, -6]^T$
 $= [0, 4/3, 119/11, 40/33, 15/11, 20/33, 10/11]^T$.
8. $\{8\} = J_\alpha \not\subset B'$.
9. $j_p = 8, p = 1$.
10. $u = w = e_1, v_{B'} = [3, 5, 0, 5]^T$.
11. $\min\{3(-8), 5(-9), 5(-6)\} = 5(-9), q = 3$.
12. Permuted $L = \begin{bmatrix} 5/6 & 1 & 0 \\ 1 & 0 & 0 \\ -1/3 & -2/5 & 1 \end{bmatrix}, U = \begin{bmatrix} 6 & 1 & 0 \\ 0 & -5/6 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

$$13. B = \{3, 9, 10\}, B' = \{1, 5, 7\}, N = \{2, 4, 6, 8\}, k = 6.$$

$$14. k = 6 > m, \bar{c}_{B'} = [-8, -4, -6]^T + (-(-9)/5)[3, 0, 5]^T = [-13/5, -4, 3]^T.$$

Iteration 2:

$$2. S = \emptyset.$$

$$3. \bar{c}_{B_2} \neq 0.$$

$$4. -B'\bar{c}_{B'} = - \begin{bmatrix} 3 & 0 & 5 \\ 0 & 8 & 4 \\ 5 & -4 & -3 \end{bmatrix} \begin{bmatrix} -13/5 \\ -4 \\ 3 \end{bmatrix} = \begin{bmatrix} -36/5 \\ 20 \\ 6 \end{bmatrix}, w = \begin{bmatrix} 20 \\ -358/15 \\ 78/25 \end{bmatrix}$$

$$\Delta x_{B \cup B'} = [-36/25, 716/25, 78/25, -13/5, -4, 3]^T \neq 0.$$

$$5. J = \{9, 10, 7\} \neq \emptyset.$$

$$6. \alpha = \min\{25/537, 1328/383, 10/33\} = 25/537.$$

$$7. \bar{x}_B = [15/11, 4/3, 119/11, 40/33, 20/33, 10/11]^T - (25/537)[-36/25, 716/25, 78/25, -13/5, -4, 3]^T \\ = [1166/815, 0, 2839/266, 2625/1969, 1560/1969, 911/1184]^T.$$

$$8. \{9\} = J_\alpha \not\subset B'.$$

$$9. j_p = 9, p = 2.$$

$$10. e_2 = [0, 1, 0]^T, w = [0, -6/5, 0]^T, u = [-6/5, 1, 0]^T, v_{B'} = [-18/5, 8, -2]^T.$$

$$11. \min\{-13/5, -4, 3\} = -4, J_1 = \{5\}, q = 5.$$

$$12. \text{Permuted } L = \begin{bmatrix} 5/6 & 1 & 0 \\ 1 & 0 & 0 \\ -1/3 & 1/5 & 1 \end{bmatrix}, U = \begin{bmatrix} 6 & 8 & 0 \\ 0 & -20/3 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$13. B = \{3, 5, 10\}, B' = \{1, 7\}, N = \{2, 4, 6, 8, 9\}, k = 5.$$

$$14. k > m, \bar{c}_{B'} = [-13/5, 3]^T + (-(-4)/8)[-18/5, -2]^T = [-22/5, 2]^T.$$

Iteration 3:

$$2. S = \emptyset.$$

$$3. \bar{c}_{B_2} \neq 0.$$

$$4. -B'\bar{c}_{B'} = - \begin{bmatrix} 3 & 5 \\ 0 & 4 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} -22/5 \\ 2 \end{bmatrix} = \begin{bmatrix} 16/5 \\ -8 \\ 28 \end{bmatrix}, w = \begin{bmatrix} -8 \\ 148/15 \\ 584/25 \end{bmatrix},$$

$$\Delta x_{B \cup B'} = [16/25, -37/25, 584/25, -22/5, 2]^T \neq 0.$$

$$5. J = \{3, 10, 7\} \neq \emptyset.$$

$$6. \alpha = \min\{959/429, 514/1125, 911/2368\} = 911/2368, J_\alpha = \{7\}.$$

$$7. \bar{x}_B = [1166/815, 1560/1969, 2839/266, 2625/1969, 911/1184]^T \\ - (911/2368)[16/25, -37/25, 584/25, -22/5, 2]^T \\ = [1509/1274, 625/459, 3249/1927, 2687/888, 0]^T.$$

$$8. J_\alpha \subset B'.$$

Iteration 4:

2. $S = \{7\}, B = \{3, 5, 10\}; B_2 = \{1\}, N = \{2, 4, 6, 7, 8, 9\}, k = 4 > m$.
3. $\bar{c}_{B'} \neq 0$.
4. $-B'\bar{c}_{B'} = -[3, 0, 5]^T[-22/5] = [66/5, 0, 22]^T, w = [0, 66/5, 484/25].$
 $\Delta x_{B \cup B'} = [66/25, -99/50, 484/25, -22/5]^T \neq 0$.
5. $J = \{3, 10\} \neq \emptyset$.
6. $\alpha = \min\{485/1081, 143/1642\} = 143/1642, J_\alpha = \{10\}$.
7. $\bar{x}_B = [1509/1274, 625/459, 3249/1927, 2687/888,]^T - (143/1642)[66/25,$
 $-99/50, 484/25, -22/5]^T$
 $= [24191/25343, 135/88, 0, 75/22]^T$.
8. $\{10\} = J_\alpha \not\subset B'$.
9. $j_p = 10, p = 3$.
10. $e_3 = [0, 0, 1]^T, w = e_3, u = [-1/5, 1/2, 1]^T, v = [22/5]$.
11. $\min\{-22/5\}, J_1 = \{1\}, q = 1$.
12. Permuted $L = \begin{bmatrix} 5/6 & 1 & 0 \\ 1 & 0 & 0 \\ -1/3 & 1/5 & 1 \end{bmatrix}, U = \begin{bmatrix} 6 & 8 & 0 \\ 0 & -20/3 & 3 \\ 0 & 0 & 22/5 \end{bmatrix}$.
13. $B = \{3, 5, 1\}, B' = \emptyset, N = \{2, 4, 6, 7, 8, 9, 10\}, k = 3$.
14. $k = m$. Phase-I ended while all artificial variables left the basis.

Phase-II:

The preceding Phase-1 produces an initial feasible solution

$$\bar{x} = [75/22, 0, 24191/25343, 0, 135/88, 0, 0]^T,$$

and the related basis $B = \{3, 5, 1\}$ with (permuted) LU factors.

Referring to the original objective gradient $c = [-6, 5, -3, 0, -4, 9, -2]^T$ and setting $B' = A \setminus B = \{2, 4, 6, 7\}, N = \emptyset$, we get Algorithm 23.6.1 started as Phase-II as follows:

Iteration 5:

1. $c_B = [-3, -4, -6]^T, w = [-1/2, 0, -15/11]^T, \bar{y} = [3/11, -13/11,$
 $-15/11]^T;$
 $\bar{c}_{B'} = \begin{bmatrix} 5 \\ 0 \\ 9 \\ -2 \end{bmatrix} - \begin{bmatrix} -2 & 4 & -3 & 5 \\ -1 & 2 & -5 & 4 \\ 3 & -8 & 1 & -3 \end{bmatrix}^T \begin{bmatrix} 3/11 \\ -13/11 \\ -15/11 \end{bmatrix} = \begin{bmatrix} 93/11 \\ -106/11 \\ 58/11 \\ -30/11 \end{bmatrix}.$
2. $S = \{2, 6\} \neq \emptyset, B' = \{4, 7\}, N = \{2, 6\}$.
3. $\bar{c}_{B'} \neq 0$.
4. $-B'\bar{c}_{B'} = -\begin{bmatrix} 4 & 5 \\ 2 & 4 \\ -8 & -3 \end{bmatrix} \begin{bmatrix} -106/11 \\ -30/11 \end{bmatrix} = \begin{bmatrix} 574/11 \\ 332/11 \\ -938/11 \end{bmatrix}, w = \begin{bmatrix} 332/11 \\ 892/33 \\ -4434/55 \end{bmatrix}.$
 $\Delta x_{B \cup B'} = [2593/121, -3038/247, -2217/121, -106/11, -30/11]^T \neq 0$.

5. $J = \{3\} \neq \emptyset$.
6. $\alpha = \min\{191/4288\}$.
7. $\bar{x}_B = [24191/25343, 135/88, 75/22, 0, 0]^T - (191/4288)$
 $[2593/121, -3038/247, -2217/121, -106/11, -30/11]^T$
 $= [0, 2007/964, 7204/1705, 743/1731, 233/1918]^T$.
8. $\{3\} = J_\alpha \not\subset B'$.
9. $j_p = 3, p = 1$.
10. $w = [1/6, 1/5, -3/22]^T, u = [5/22, -3/44, -3/22]^T, v_{B'} = [41/22,$
 $14/11]^T$.
11. $\min\{-106/11, -30/11\} = -106/11, J_1 = \{4\}, q = 4$.
12. Permuted $L = \begin{bmatrix} -1/2 & -2/7 & 1 \\ -1/4 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, U = \begin{bmatrix} -8 & -4 & 5 \\ 0 & 7 & 5/4 \\ 0 & 0 & 41/7 \end{bmatrix}$.
13. $B = \{4, 5, 1\}, B' = \{7\}, N = \{2, 3, 6\}, k = 4$.
14. $k > m, \bar{c}_{B'} = [-30/11]^T + (-(-106/11)/(41/22))[14/11]^T = [158/41]$.

Iteration 6:

2. $S = \emptyset$.
3. $\bar{c}_{B_2} \neq 0$.
4. $-B'\bar{c}_{B'} = -[5, 4, -3]^T[158/41] = [-790/41, -632/41, 474/41]^T$,
 $w = [474/41, -1027/82, -4898/287]$,
 $\Delta x_B = [-729/277, -2133/1681, -2905/997, 158/41]^T \neq 0$.
5. $J = \{7\} \neq \emptyset$.
6. $\alpha = \min\{335/10627\}$.
7. $\bar{x}_B = [743/1731, 2007/964, 7204/1705, 233/1918]^T - (335/10627)$
 $[-729/277, -2133/1681, -2905/997, 158/41]^T = [12244/23905,$
 $87/41, 177/41, 0]^T$.
8. $\{7\} = J_\alpha \subset B'$.

Iteration 7:

2. $S = \{7\}, B = \{4, 5, 1\}; B_2 = \emptyset, N = \{2, 3, 6, 7\}, k = 3 = m$.
 3. $\bar{c}_{B'} = 0$.
 15. $c_B = [0, -4, -6]^T, w = [0, -4/7, -37/41]^T, \bar{y} = [-37/41, -34/41,$
 $-27/41]^T$.
- $$\bar{c}_N = \begin{bmatrix} 5 \\ -3 \\ 9 \\ -2 \end{bmatrix} - \begin{bmatrix} -2 & 5 & -3 & 5 \\ -1 & 6 & -5 & 4 \\ 3 & -2 & 1 & -3 \end{bmatrix}^T \begin{bmatrix} -37/41 \\ -34/41 \\ -27/41 \end{bmatrix} = \begin{bmatrix} 178/41 \\ 212/41 \\ 115/41 \\ 158/41 \end{bmatrix} \geq 0.$$

16. $T = \emptyset$. Terminated with primal and dual optimal solutions:

$$x^* = [177/41, 0, 0, 12244/23905, 87/41, 0, 0]^T,$$

$$y^* = [-37/41, -34/41, -27/41]^T, \quad z^* = [0, 178/41, 212/41, 0, 0, 115/41, 158/41]^T.$$

Optimal value: $f^* = -1410/41$.

23.7 Notes on Face Method with LU Factorization

The face algorithm has the following features:

At first, unlike the simplex algorithm, which uses the search direction along some descent edge of the underlying polyhedron, the new algorithm uses a combination of all edges corresponding to nonzero active reduced costs.

Second, as a result, the new algorithm no longer moves from vertex to vertex, but from face to face, with a sequence of corresponding face points, until reaching an optimal face together with an optimal point. Therefore, it would have more opportunities to bypass degenerate vertices.

Third, the new algorithm is stable, compared with simplex algorithms, since its pivot is selectable, to some extent, while the latter can be arbitrarily close to zero. As we all know, the latter sometimes fails to maintain a usable basis and has to restart from scratch.

As for the computational cost in a single iteration, the new algorithm is about the same as the simplex algorithm. It solves four triangular systems, just as the latter. Using LU factorization, it can be implemented as in the traditional simplex context.

Finally, the unique and promising feature of the new algorithm might be the maximum objective decrement $\alpha \|\bar{c}_{B'}\|^2$ achieved per iteration. We expect that the number of iterations required could be reduced significantly. At this stage, however, we cannot talk much about its actual performance. Among the search directions (3.30) and (24.37), e.g., it is unknown which one is better, although we used the former. Alternatively, instead of the whole $\bar{c}_{B'}$, it is possible to use a part of it to formulate the search direction, e.g., such that

$$|\bar{c}_j| \geq \gamma, \quad j \in B', \quad (23.43)$$

where γ is a threshold decreasing in the solution process. All of these need to be considered. The implementation of the new algorithm and computational experiments are expected.

Reference

Pan PQ (2014) Linear programming computation. Springer, Berlin Heidelberg, Germany

Chapter 24

Dual Face Method with LU Factorization



It is fascinating to develop the dual face algorithm based on LU factorization. We handle this topic in this chapter.

In the next section, we first put forward the key to this method, which is important for understanding the subsequent derivation. In the following sections, we discuss topics, such as the ascent search direction, update of the dual face solution, pivot operation, and optimality test. We spend the subsequent two sections to formulate the dual face algorithm in the tableau form and revised form, respectively, and make remarks on the algorithm in the final section.

First of all, we introduce the so-called *pseudo-basis*, defined as the submatrix consisting of k independent columns of A (Pan 2005). If the k is less than m and the column space includes b , it is a deficient basis, and if $k = m$, it is a normal basis.

Now turn to the dual problem of (23.1), i.e.,

$$\begin{aligned} \max \quad & b^T y \\ \text{s.t.} \quad & A^T y + z = c, \quad z \geq 0. \end{aligned} \quad (24.1)$$

Partition A to (B, N) , where B is $m \times k$ pseudo-basis with $0 \leq k \leq m$.

Definition 24.0.1 A dual face is a dual feasible point/solution set, defined as

$$D_N = \{(y, z_N) \in \mathcal{R}^m \times \mathcal{R}^{n-k} \mid B^T y = c_B, N^T y + z_N = c_N, z_B = 0, z_N \geq 0\}. \quad (24.2)$$

The N and B are termed *dual face matrix* and *dual nonface matrix*, respectively. As $z_B = 0$ is fixed, the B is termed *inactive*.

The dual face D_N is an $(m - k)$ -dimensional set. In particular, if $k = m$, D_N is a zero-dimensional dual vertex, and if $k = 0$, it is the m -dimensional dual feasible region itself. Usually, $0 < k < m$ is assumed. Any solution in D_N is called *dual face point* or *solution*.

Definition 24.0.2 The dual face is a dual level face if the dual objective value is constant over it. A dual level face is *optimal* if the constant equals the dual optimal value.

24.1 Key of Method

Consider the following least squares problem:

$$\min_{x_B \in \mathcal{R}^k} \|b - Bx_B\|. \quad (24.3)$$

The unique solution to the preceding is known as

$$\bar{x}_B = (B^T B)^{-1} B^T b. \quad (24.4)$$

The residual of (24.3) at solution $\bar{x}_B$ is a key quantity, that is,

$$\Delta y = b - B\bar{x}_B. \quad (24.5)$$

It is clear that the Δy vanishes if and only if B is a deficient basis. Otherwise, it can be verified that

$$B^T \Delta y = 0, \quad b^T \Delta y > 0, \quad (24.6)$$

which means that the nonzero Δy is in the null of B^T , and is an ascent direction of the dual objective function. In fact, Δy is just the orthogonal projection of b onto the null of B^T . As the closest to the dual objective gradient b in the null of B^T , it is a favorable search direction in the y -space.

Without loss of generality, assume that B consists of the first k columns and R of the first k rows such that the $k \times k$ submatrix B_R has full rank.

Using

$$N = A \setminus B, \quad R_c = \{1, \dots, m\} \setminus R,$$

we put problem (23.1) into the following tableau:

x_B	x_N	f	RHS
B_R	N_R		b_R
B_{R_c}	N_{R_c}		b_{R_c}
c_B^T	c_N^T	-1	

(24.7)

Carrying out the Gauss–Jordan elimination to the preceding, we convert B_R into unit matrix, and eliminate all entries below it, yielding the following tableau:

x_B	x_N	f	RHS
I_k	$\bar{N}_R$		$\bar{b}_R$
	$\bar{N}_{R_c}$		$\bar{b}_{R_c}$
	$\bar{c}_N^T$	-1	$-\bar{f}$

(24.8)

where $\bar{c}_N$ is assumed nonnegative.

Tableau (24.8) corresponds to the following dual subproblem:

$$\begin{aligned}
 & \max \bar{b}_R^T y_R + \bar{b}_{R_c}^T y_{R_c} \\
 & \text{s.t.} \quad \begin{pmatrix} I_k & \bar{N}_R \\ \bar{N}_R^T & \bar{N}_{R_c}^T \end{pmatrix} \begin{pmatrix} y_R \\ y_{R_c} \end{pmatrix} + \begin{pmatrix} z_B \\ z_N \end{pmatrix} = \begin{pmatrix} 0 \\ \bar{c}_N \end{pmatrix}, \quad z_B = 0, z_N \geq 0.
 \end{aligned}
 \tag{24.9}$$

For the preceding, the counterpart of least squares problem (24.3) is much simpler, that is,

$$\min_{x_B \in \mathcal{R}^k} \left\| \begin{pmatrix} \bar{b}_R \\ \bar{b}_{R_c} \end{pmatrix} - \begin{pmatrix} I_k \\ 0 \end{pmatrix} x_B \right\|^2, \tag{24.10}$$

which exhibits its unique solution $\bar{x}_B = \bar{b}_R$ with residual

$$\Delta y = \begin{pmatrix} 0 \\ \bar{b}_{R_c} \end{pmatrix}. \tag{24.11}$$

If $\bar{b}_{R_c} \neq 0$, Δy is eligible to be a search direction in y -space. The matching search direction in z -space is then

$$\Delta z_B = 0, \quad \Delta z_N = -\bar{N}_{R_c}^T \bar{b}_{R_c}. \tag{24.12}$$

Tableau (24.7) can also be handled by block Gauss–Jordan elimination. Premultiplying the first row by B_R^{-1} , adding $-B_{R_c}$ times of the first row to the second row, and adding $-c_B^T$ times of the first row to the bottom row, we convert (24.7) to the following tableau:

x_B	x_N	f	RHS
I_k	$B_R^{-1} N_R$		$B_R^{-1} b_R$
	$N_{R_c} - B_{R_c} B_R^{-1} N_R$		$b_{R_c} - B_{R_c} B_R^{-1} b_R$
	$c_N^T - c_B^T B_R^{-1} N_R$	-1	$-c_B^T B_R^{-1} b_R$

(24.13)

Since the preceding is equivalent to (24.8), their corresponding entries are equal, e.g.,

$$\bar{c}_N = c_N - N_R^T B_R^{-T} c_B, \quad \bar{N}_{R_c} = N_{R_c} - B_{R_c} B_R^{-1} N_R. \quad (24.14)$$

From the equality constraint of (24.9), it is seen that $y_R = 0$ is actually fixed since so is $z_B = 0$. Dropping such an inactive part, subproblem (24.9) becomes

$$\begin{aligned} & \max \bar{b}_{R_c}^T y_{R_c} \\ & \text{s.t. } \bar{N}_{R_c}^T y_{R_c} + z_N = \bar{c}_N, \quad z_N \geq 0. \end{aligned} \quad (24.15)$$

Designating the feasible region of the preceding by D'_N , we have the following result.

Lemma 24.1.1 *If $\{y_{R_c}, z_N \geq 0\}$ is in D'_N , then $\{z_B = 0, y_R = B_R^{-T}(c_B - B_{R_c}^T y_{R_c}), y_{R_c}, z_N \geq 0\}$ is in D_N , defined by (24.2), and vice versa.*

Proof

$$\bar{c}_N = c_N - N_R^T B_R^{-T} c_B, \quad \bar{N}_{R_c} = N_{R_c} - B_{R_c} B_R^{-1} N_R. \quad (24.16)$$

Assume that the former point is in D'_N . From definition of D'_N and (24.16), it follows that

$$\begin{aligned} z_N &= \bar{c}_N - \bar{N}_{R_c}^T y_{R_c} \\ &= c_N - N_R^T B_R^{-T} c_B - (N_{R_c}^T - N_R^T B_R^{-T} B_{R_c}^T) y_{R_c} \\ &= c_N - N_{R_c}^T y_{R_c} - N_R^T B_R^{-T} (c_B - B_{R_c}^T y_{R_c}). \end{aligned} \quad (24.17)$$

Substituting

$$y_R = B_R^{-T} (c_B - B_{R_c}^T y_{R_c}) \quad (24.18)$$

into the preceding leads to

$$N_R^T y_R + N_{R_c}^T y_{R_c} + z_N = c_N,$$

which implies that $\{y = (y_R^T, y_{R_c}^T)^T, z_N\}$ satisfies

$$N^T y + z_N = c_N.$$

In addition, (24.18) also implies that

$$B^T y = \begin{pmatrix} B_R^T & B_{R_c}^T \end{pmatrix} \begin{pmatrix} y_R^T & y_{R_c}^T \end{pmatrix}^T = c_B.$$

Therefore, the latter point is in D_N .

Conversely, assume that the latter point is in D_N . From the definition of N_N and (24.18), it follows that

$$\begin{aligned} z_N &= c_N - N^T y \\ &= c_N - N_R^T B_R^{-T} \left(c_B - B_{R_c}^T y_{R_c} \right) - N_{R_c}^T y_{R_c} \\ &= c_N - N_R^T B_R^{-T} c_B - \left(N_{R_c}^T - N_R^T B_R^{-T} B_{R_c}^T \right) y_{R_c}, \end{aligned}$$

which along with (24.16) leads to

$$\bar{N}_{R_c}^T y_{R_c} + z_N = \bar{c}_N.$$

Therefore, the former point is in D'_N , and the proof is complete. $\square$

Lemma 24.1.1 establishes a 1-to-1 correspondence between points of D'_N and D_N . In particular, it is obvious that $\{y_{R_c} = 0, z_N = \bar{c}_N \geq 0\} \in D'_N$ corresponds to

$$\left\{ y_R = B_R^{-T} c_B, y_{R_c} = 0, z_B = 0, z_N = \bar{c}_N \geq 0 \right\} \in D_N. \quad (24.19)$$

Thus, solving subproblem (24.15) amounts to maximizing the objective of (1.7) over D_N .

Theorem 24.1.1 *If B is a deficient basis, then D_N is a dual level face, as well as D'_N .*

Proof Without loss of generality, assume that $k < m$.

Under the assumption, the residual Δy given by (24.11) is zero, and hence so is $\bar{b}_{R_c}$. Thus, D'_N is a dual level face, over which the objective of (24.15) equals zero.

By Lemma 24.1.1, any $\{y_{R_c}, z_N \geq 0\} \in D'_N$ corresponds to

$$\left\{ z_B = 0, y_R = B_R^{-T} \left(c_B - B_{R_c}^T y_{R_c} \right), y_{R_c}, z_N \geq 0 \right\} \in D_N.$$

Therefore, the objective value of the original dual problem (24.1) at the latter solution equals

$$\begin{aligned} b_R^T y_R + b_{R_c}^T y_{R_c} &= b_R^T B_R^{-T} \left(c_B - B_{R_c}^T y_{R_c} \right) + b_{R_c}^T y_{R_c} \\ &= b_R^T B_R^{-T} c_B - y_{R_c}^T B_{R_c} B_R^{-1} b_R + y_{R_c}^T b_{R_c} \\ &= b_R^T B_R^{-T} c_B + y_{R_c}^T \left(b_{R_c} - B_{R_c} B_R^{-1} b_R \right). \end{aligned} \quad (24.20)$$

On the other hand, a comparison between the right-hand sides of (24.8) and (24.13) gives that

$$0 = \bar{b}_{R_c} = b_{R_c} - B_{R_c} B_R^{-1} b_R.$$

Combining the preceding and (24.20) leads to

$$b_R^T y_R + b_{R_c}^T y_{R_c} = b_R^T B_R^{-T} c_B,$$

which is constant over D_N , and the proof is complete. $\square$

Corollary 24.1.2 *If $\bar{b}_{R_c}$ in tableau (24.8) vanishes, then the corresponding D_N is a dual level face.*

Nevertheless, it turned out that this approach coincides with one already used in Pan (2004, 2005). So, it appears to be blocked. Fortunately, the barrier can be broken by further partitions.

24.2 Ascent Search Direction

In this section, we derive a new dual search direction through a so-called *dual face subprogram*.

The trick is to partition further. Let N' be subset of N and R_1 subset of R_c , and set

$$N = N \setminus N', \quad R_c = R_c \setminus R_1, \quad |N'| = |R_1| = k'. \quad (24.21)$$

In particular, assume $\bar{N}'_R = 0$, $\bar{N}'_{R_1} = I_{k'}$, and $\bar{N}'_{R_c} = 0$, so that tableau (24.8) can be written

x_B	$x_{N'}$	x_N	f	RHS
I_k	$\bar{N}_R$			$\bar{b}_R$
	$I_{k'}$	$\bar{N}_{R_1}$		$\bar{b}_{R_1}$
		$\bar{N}_{R_c}$		$\bar{b}_{R_c}$
$\bar{c}_N^T$	$\bar{c}_{N'}^T$	$\bar{c}_N^T$	-1	

(24.22)

which corresponds to dual face with $z_B = 0$ fixed.

Handled from the dual side, the preceding is termed (*canonical*) *dual face tableau*, although it still represents the problem (23.1) from the primal side.

Using the active part of (24.22), construct the so-called *dual face subprogram* below:

$$\begin{aligned} & \max \bar{b}_{R_1}^T y_{R_1} + \bar{b}_{R_c}^T y_{R_c} \\ \text{s.t. } & \begin{pmatrix} I_{k'} & \\ \bar{N}_{R_1}^T & \bar{N}_{R_c}^T \end{pmatrix} \begin{pmatrix} y_{R_1} \\ y_{R_c} \end{pmatrix} + \begin{pmatrix} z_{N'} \\ z_N \end{pmatrix} = \begin{pmatrix} \bar{c}_{N'} \\ \bar{c}_N \end{pmatrix}, \quad z_{N'}, z_N \geq 0. \end{aligned} \quad (24.23)$$

We will use the preceding to create a search direction to take one step forward.

Consider the following direction:

$$\Delta y_{R_1 \cup R_c} = \bar{b}_{R_1 \cup R_c}, \quad (24.24)$$

$$\Delta z_{N' \cup N} = \{-\bar{b}_{R_1}, -\bar{N}_{R_1}^T \bar{b}_{R_1} - \bar{N}_{R_c}^T \bar{b}_{R_c}\}. \quad (24.25)$$

Then, the following can be claimed.

Lemma 24.2.1 *If $\Delta y_{R_1 \cup R_c}$ is nonzero, the direction defined by (24.24)–(24.25) is ascent in the null of the coefficient matrix of constraint equations, with respect to subprogram (24.23).*

Proof It is ascent, since $\bar{b}_{R_1 \cup R_c}$ is the gradient of the objective function. In addition, the following equation is easily verified:

$$\begin{pmatrix} I_{k'} & \\ \bar{N}_{R_1}^T & \bar{N}_{R_c}^T \end{pmatrix} \begin{pmatrix} \Delta y_{R_1} \\ \Delta y_{R_c} \end{pmatrix} + \begin{pmatrix} \Delta z_{N'} \\ \Delta z_N \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (24.26)$$

Therefore, the claim is valid. $\square$

Lemma 24.2.1 ensures that the direction defined by (24.24)–(24.25) is eligible to be a search direction. In fact, $\{\bar{b}_{R_1}, \bar{b}_{R_c}\}$ is the steepest ascent direction in $\{y_{R_1}, y_{R_c}\}$ -space.

In particular, we prefer matrix $B \cup N'$ being deficient basis, because $\bar{b}_{R_c}$ vanishes in this case, simplifying formulas (24.24)–(24.25) as

$$\Delta y_{R_1 \cup R_c} = \{\bar{b}_{R_1}, 0\}, \quad \Delta z_{N' \cup N} = \{-\bar{b}_{R_1}, -\bar{N}_{R_1}^T \bar{b}_{R_1}\}. \quad (24.27)$$

Furthermore, from a numerical point of view, it seems reasonable to use normalized $\bar{b}_{R_1}$ instead. This leads to the following search direction:

$$\Delta y_{R_1 \cup R_c} = \{\bar{b}_{R_1} / \|\bar{b}_{R_1}\|, 0\}, \quad \Delta z_{N' \cup N} = \{-\Delta y_{R_1}, -\bar{N}_{R_1}^T \Delta y_{R_1}\}. \quad (24.28)$$

24.3 Updating of Dual Face Solution

Assume that search direction $\Delta z_{N' \cup N}$ was determined by (24.25).

Introduce the index set

$$J = \{j \in N' \cup N \mid \Delta z_j < 0\}. \quad (24.29)$$

If J is nonempty, the current dual feasible solution to (24.23), i.e.,

$$y_{R_1 \cup R_c} = 0, \quad z_{N' \cup N} = \bar{c}_{N' \cup N} \geq 0, \quad (24.30)$$

can be updated by the following formula:

$$\hat{y}_{R_1 \cup R_c} = \beta \bar{b}_{R_1 \cup R_c}, \quad (24.31)$$

$$\hat{c}_{N' \cup N} = \bar{c}_{N' \cup N} + \beta \Delta z_{N' \cup N}, \quad (24.32)$$

where β is the maximum possible stepsize that can be achieved, i.e.,

$$\beta = \min_{j \in J} -\bar{c}_j / \Delta z_j \geq 0. \quad (24.33)$$

Dual feasible solution is said to be *degenerate* if some component of $\bar{c}_J$ is 0. In this case, dual stepsize β vanishes, and the “new” solution is not really new, but the same as the old.

Using the preceding notation, we claim the following result.

Theorem 24.3.1 *If J defined by (24.29) is nonempty, then (24.31)–(24.32) along with (24.33) give a new dual feasible solution to subprogram (24.23).*

Proof Note that β is well-defined by (24.33) under the nonempty assumption on J .

Since old solution (24.30) satisfies the equality constraint of (24.23) and direction (24.24)–(24.25) satisfies (24.26), the new solution satisfies the equality constraint.

From (24.32)–(24.33), it follows that

$$\beta \leq -\bar{c}_j / \Delta z_j, \quad j \in J. \quad (24.34)$$

Besides, (24.29) ensures that

$$\Delta z_j < 0, \quad j \in J.$$

Thus, multiplying (24.34) by Δz_j gives

$$\beta \Delta z_j \geq -\bar{c}_j, \quad j \in J,$$

which along with (24.32) leads to

$$\hat{c}_j = \bar{c}_j + \beta \Delta z_j \geq 0, \quad j \in J. \quad (24.35)$$

In case of $(N' \cup N) \setminus J \neq \emptyset$, on the other hand, (24.29) implies that

$$\Delta z_j \geq 0, \quad j \in (N' \cup N) \setminus J,$$

combining which, $\bar{c}_{N' \cup N} \geq 0$, and $\beta \geq 0$ results in

$$\hat{c}_j = \bar{c}_j + \beta \Delta z_j \geq 0, \quad j \in (N' \cup N) \setminus J. \quad (24.36)$$

Therefore, it holds that $\hat{c}_{N' \cup N} \geq 0$, and the proof is complete. $\square$

Theorem 24.3.2 *If search direction is determined by (24.24)–(24.25), the new solution (24.31)–(24.32) increases the dual objective value by*

$$\Delta f = \beta(\|\bar{b}_{R_1}\|^2 + \|\bar{b}_{R_c}\|^2), \quad (24.37)$$

with respect to dual subprogram (24.23).

Proof By (24.31), the old zero-valued solution in $y_{R_1 \cup R_c}$ space is updated to the new solution $\beta \bar{b}_{R_1 \cup R_c}$, and hence the corresponding objective value increases from zero up to $\beta \bar{b}_{R_1 \cup R_c}^T \bar{b}_{R_1 \cup R_c}$, which is equal to (24.37). $\square$

If the search direction (24.27) is used instead, the corresponding objective increment becomes $\beta \|\bar{b}_{R_1}\|^2$, whereas the increment becomes $\beta \|\bar{b}_{R_1}\|$ if the direction (24.28) is used.

In practice, it is not necessary to update the dual solution in y -space iteration by iteration, since the final one can be computed cheaply at the end of the solution process (this will be clear a little later). It is also possible to update $\bar{c}_{N' \cup N}$ by using the dual face tableau. It is noticeable, for instance, that using formulas (24.27) and (24.32) amounts to adding $-\beta \bar{b}_{R_1}$ times of rows corresponding to R_1 to the bottom row.

As for J being empty, a case when stepsize β is not well-defined, we are still done as the following says.

Theorem 24.3.3 *Assume that $\bar{b}_{R_1 \cup R_c}$ is nonzero. If $J = \emptyset$, the problem (23.1) is infeasible.*

Proof From the second half of the proof of Theorem 24.3.1, in case of $J = \emptyset$, new solutions (24.31)–(24.32) are dual feasible to (24.23) for all $\beta > 0$. Under the condition $\bar{b}_{R_1 \cup R_c} \neq 0$, the corresponding objective increment, given by Theorem 24.3.2, goes to ∞ , as so is β .

Therefore, dual subprogram (24.23) is upper unbounded, and hence there exists no feasible solution to the corresponding primal subprogram, as well as to problem (23.1). $\square$

24.4 Pivoting Operation

In this section, we discuss pivoting operations and associated index set adjustment.

Assume that new dual solution $\hat{c}_{N' \cup N}$ was determined by (24.32) along with (24.33). Introduce the index set

$$J' = \arg \min_{j \in J} -\bar{c}_j / \Delta z_j, \quad (24.38)$$

where J is defined by (24.29). It is clear that for each $j \in J'$, component $\hat{c}_j$ reaches the critical point zero. Related variables or indices are said to be *blocking* ones.

There are two types of iterations as follows:

(A) $\tilde{J} \triangleq N' \cap J'$ is nonempty (or $J' \not\subset N$).

There is then no need for pivoting, and the involved computational cost is quite cheap. Before going to the next iteration, what to do is only to execute the following tactic.

Tactic 24.4.1 Assume that row index set $\tilde{I}$ corresponds to $\tilde{J}$. Update $\{N', B, R_1, R, k\}$ by

$$\begin{aligned} N' &= N' \setminus \tilde{J}, \quad B = B \cup \tilde{J}, \\ R_1 &= R_1 \setminus \tilde{I}, \quad R = R \cup \tilde{I}, \quad k = k + |\tilde{J}|. \end{aligned} \quad (24.39)$$

(B) $\tilde{J} \triangleq N' \cap J'$ is empty (or $J' \subset N$).

There is a need for pivoting. To do so, execute the following rule first.

Rule 24.4.1 [Column rule] Select the column index q such that

$$q \in \arg \min_{j \in J'} \Delta z_j. \quad (24.40)$$

Then determine row index p such that

$$p \in \arg \max_{i \in R_c} |\bar{a}_{i q}|. \quad (24.41)$$

There will be the following two cases arising:

(i) $\bar{a}_{p q} \neq 0$.

We update the tableau by using $\bar{a}_{p q}$ as the pivot (see below) and then update the index sets conformably by the following tactic.

Tactic 24.4.2 Update $\{N, B, R_c, R, k\}$ by

$$\begin{aligned} N &= N \setminus q, \quad B = B \cup q, \\ R_c &= R_c \setminus p, \quad R = R \cup p, \quad k = k + 1. \end{aligned} \quad (24.42)$$

(ii) $\bar{a}_{pq} = 0$.

There are multiple choices for row pivoting. We bring up the following plausible one.

Rule 24.4.2 [Row rule] Assume that column index q was determined by (24.40). Reselect row index p such that

$$p \in \arg \min \{-\bar{b}_i \bar{a}_{i,q} \mid i \in R_1\}, \quad (24.43)$$

corresponding to $j_p \in N'$.

Using the preceding notation, we have the following result.

Proposition 24.4.1 Δz_q and $-\bar{b}_p \bar{a}_{p,q}$ have the same minus sign.

Proof It is evident that $\bar{a}_{i',q} = 0$ implies $\bar{N}_{R_c,q} = 0$, and hence

$$(\bar{N}_{R_c,q})^T \bar{b}_{R_c} = 0. \quad (24.44)$$

In addition, by (24.29) and $q \in J$, it is known that $\Delta z_q < 0$. Hence, combining (24.25) and (24.44) leads to

$$\Delta z_q = -\bar{N}_{R_1,q}^T \bar{b}_{R_1} = \sum_{i \in R_1} -\bar{b}_i \bar{a}_{i,q} < 0.$$

The preceding along with (24.43) gives $-\bar{b}_p \bar{a}_{p,q} < 0$, which has a minus sign as Δz_q . $\square$

The preceding ensures that the pivot $\bar{a}_{p,q}$ is nonzero. It is more than that. It also implies that Rule 24.4.2 conforms to the heuristic characteristic of the optimal solution, in the dual context.

Once the pivot is determined, what to do next is to update the tableau. To do so, modify the first m rows by premultiplying the p -th row by $1/\bar{a}_{p,q}$ and adding $-\bar{a}_{i,q}$ times of the row to the i -th row, for $i \in R_1 \cup R_c$, $i \neq p$.

Then, update the index sets by the following tactic respectively.

Tactic 24.4.3 Update $\{N, B, N', R_1, R, k\}$ by

$$\begin{aligned} N &= N \setminus q, \quad B = B \cup q, \quad N' = N' \setminus \{j_p\}, \\ R_1 &= R_1 \setminus p, \quad R = R \cup p, \quad N = N \cup \{j_p\}, \quad k = k + 1. \end{aligned} \quad (24.45)$$

We ignore related column and row permutations in conjunction with the tableau. After all, it is not necessary to move around entire column or row of entries in the implementation.

24.5 Optimality Test

Let (24.22) be current dual face tableau. It is noted that if $k = m$, none of formulas (24.25), (24.27) and (24.28) is well-defined, and if $k < m$ but $\bar{b}_{R_1 \cup R_c} = 0$, all these directions vanish.

Consider the following solution given by tableau (24.22):

$$\bar{x}_B = \bar{b}_R, \quad \bar{x}_{N' \cup N} = 0. \quad (24.46)$$

Clearly, if B is not a deficient basis, such a solution does not satisfy the equality constraint, let alone be a feasible solution to (23.1).

Lemma 24.5.1 *If B is a deficient basis, solution (24.46) satisfies the equality constraint of (23.1), corresponding to primal objective value $\bar{f}$.*

Proof Note that all resulting tableaus represent the same problem (23.1). It is obvious that B is deficient basis if and only if $k = m$ or $k < m$ but $\bar{b}_{R_1 \cup R_c} = 0$. Under the assumption, it is easily verified that solution (24.46) satisfies the equality constraint of (24.22), as well as that of (23.1), with the right-most entry of the bottom row giving the related objective value $\bar{f}$. $\square$

Denote by $D_{N' \cup N}$ the dual face defined by (1.10) with N replaced by $N' \cup N$.

Lemma 24.5.2 *If B is a deficient basis, $D_{N' \cup N}$ is a dual level face.*

Proof The validity of the statement is obvious through Corollary 24.1.2. $\square$

Theorem 24.5.1 *Assume that B is a deficient basis. If the following index set*

$$T = \{j_i \in B \mid \bar{b}_i < 0, i \in R\} \quad (24.47)$$

is empty, then $D_{N' \cup N}$ is a dual optimal face with a pair of primal and dual optimal solutions.

Proof Lemma 24.5.1 and emptiness of T together imply that $\bar{x}$ given by (24.46) is a primal feasible solution to (23.1). On the other hand, it is obvious that $\{y_{R_1 \cup R_c} = 0, z_{N' \cup N} = \bar{c}_{N' \cup N} \geq 0\}$ corresponds to

$$\{y_R = B_R^{-T} c_B, y_{R_1 \cup R_c} = 0, z_B = 0, z_{N' \cup N} = \bar{c}_{N' \cup N} \geq 0\} \in D_{N' \cup N}, \quad (24.48)$$

which is a dual feasible solution. The preceding and $\bar{x}$ exhibit complementarity slackness and are therefore a pair of primal and dual optimal solutions. According to Lemma 24.5.2, $D_{N' \cup N}$ is a dual level face. Including the dual optimal solution, it is a dual optimal face. $\square$

In the case of $T \neq \emptyset$ when optimality cannot be declared, execute the following tactic, and go on the next iteration.

Tactic 24.5.1 Using $T' = \{i \in R \mid \bar{b}_i < 0\}$, update $\{B, N', R, R_1, k\}$ by

$$\begin{aligned} B &= B \setminus T, \quad N' = N' \cup T, \\ R &= R \setminus T', \quad R_1 = R_1 \cup T', \quad k = k - |T|. \end{aligned} \quad (24.49)$$

Consequently, we are faced with an expanded new dual face. Note that the zero-valued new $\bar{c}_{N'}$ will never affect the determination of the next stepsize, since the new $\bar{b}_{R_1}$ is nonpositive, and hence $\Delta z_{N'} = -\bar{b}_{R_1}$ is nonnegative.

24.6 Dual Face Algorithm: Tableau Form

Based on the discussion made in previous sections, we formulate the dual face algorithm in tableau form in this section.

In each iteration, the dual face tableau is updated with $\bar{f}$ increasing. A series of iterations are performed until $k = m$ or $\bar{b}_{R_1 \cup R_c} = 0$ when a dual level face is reached (unless upper unboundedness detected). At the end of such a series, referred to as *dual face contraction*, optimality is tested: if set T is empty, optimality is achieved. In the other case, the dual level face is expanded, and the next dual face contraction is carried out.

Although the solution process can be initiated with any pseudo-basis, we prefer a deficient basis since this will make $\bar{b}_{R_c}$ vanish throughout the whole solution process. Therefore, the simpler search direction formula (24.27) will be used.

Let B be initial deficient basis. We take the following partition to get started:

$$\begin{aligned} N &= A \setminus B, \quad N' = \emptyset, \\ R &= R, \quad R_1 = \emptyset, \quad k \leq m. \end{aligned} \quad (24.50)$$

The overall steps are summarized into the following model.

Algorithm 24.6.1 [Dual face algorithm: tableau form] *Initial: dual face tableau in form (24.22) with partition (24.50), $\bar{b}_{R_c} = 0$, and $\bar{c}_{N' \cup N}$. This algorithm solves problem (23.1).*

1. If $k = m$ or $\bar{b}_{R_1} = 0$, then
 - (1) Stop if T defined by (24.47) is empty
 - (2) Execute Tactic 24.5.1
2. Compute $\Delta z_{N' \cup N}$ by (24.27).
3. Stop if J defined by (24.29) is empty.
4. Determine β by (24.33).
5. If $\beta > 0$, update $\bar{c}_{N' \cup N}$ by (24.32).
6. If $\tilde{J} = N' \cap J' \neq \emptyset$, where J' is defined by (24.38), execute Tactic 24.4.1, and go to step 1.

7. Determine column index q by (24.40) and row index p by (24.41).
8. If $\bar{a}_{p,q} = 0$, redetermine row index p by (24.43).
9. Premultiply the p -th row by $1/\bar{a}_{p,q}$, and then add $-\bar{a}_{i,q}$ times of the row to the i -th row, for $i \in R_1 \cup R_c$, $i \neq p$.
10. If $p \in R_c$, execute Tactic 24.4.2; else execute Tactic 24.4.3.
11. Go to step 1.

Theorem 24.6.1 *If each dual face contraction involves a nondegenerate dual feasible solution, Algorithm 23.5.1 terminates either at*

- (i) Step 1(1), giving a pair of primal and dual optimal solutions, or at
- (2) Step 3, declaring primal infeasibility

Proof It is seen that k increases by 1, at least, in each iteration. Therefore, within finitely many iterations, either k increases to m or $\bar{b}_{R_1 \cup R_c}$ vanishes so that each dual face contraction ends with a dual level face (Lemma 24.5.2) unless detecting primal infeasibility at step 3, by Theorem 24.3.3.

Assume that the algorithm does not terminate. Then, failing for optimality test, it performs infinitely many dual face contractions, each including a nonzero stepsize. Consequently, objective value $\bar{f}$, corresponding to the end dual level face, increases strictly (Theorem 24.3.2), as contradicts that the number of the dual faces is finite. Therefore, the algorithm terminates, with outlet step 1(1) giving a dual optimal face together with a pair of dual and primal optimal solutions, by Theorem 24.5.1. $\square$

24.7 Dual Face Algorithm

In this section, we convert Algorithm 24.6.1 into a revised form using LU factorization.

Introduce symbol $\widehat{B} \triangleq B \cup N'$. Let current $\bar{c}_{N' \cup N}$, $\bar{b}$, and LU factorization below be available:

$$\widehat{B}_{R \cup R_1} = LU. \quad (24.51)$$

The following quantity expressions required can be derived based on a comparison between (24.8) and (24.13), with B and R replaced, respectively, by $B \cup N'$ and $R \cup R_1$, e.g., giving

$$\bar{N}_{R \cup R_1} = \widehat{B}_{R \cup R_1}^{-1} N_{R \cup R_1}, \quad \bar{N}_{R_c} = N_{R_c} - \widehat{B}_{R_c} \widehat{B}_{R \cup R_1}^{-1} N_{R \cup R_1}, \quad (24.52)$$

which are useful for obtaining the search direction.

(A) Search direction $\Delta z_{N' \cup N} = \{-\bar{b}_{R_1}, -\bar{N}_{R_1}^T \bar{b}_{R_1} - \bar{N}_{R_c}^T \bar{b}_{R_c}\}$ (24.25).

$$(i) \ t_1^T \triangleq \bar{b}_{R_1}^T \bar{N}_{R_1} = (0_R^T, \bar{b}_{R_1}^T) \widehat{B}_{R \cup R_1}^{-1} N_{R \cup R_1}.$$

Solve following triangular systems for $v = \widehat{B}_{R \cup R_1}^{-T} (0_R^T, \bar{b}_{R_1}^T)^T$:

$$U^T u = \left(0_R^T, \bar{b}_{R_1}^T \right)^T, \quad L^T v = u, \quad (24.53)$$

and then compute

$$t_1 = N_{R \cup R_1}^T v. \quad (24.54)$$

(ii) $t_2 \triangleq \bar{N}_{R_c}^T \bar{b}_{R_c} = (N_{R_c} - \widehat{B}_{R_c} \widehat{B}_{R \cup R_1}^{-1} N_{R \cup R_1})^T \bar{b}_{R_c}$ (ignore if $\bar{b}_{R_c} = 0$).

Solve the following systems for $w = \widehat{B}_{R \cup R_1}^{-T} \widehat{B}_{R_c}^T \bar{b}_{R_c}$ first:

$$U^T u = \widehat{B}_{R_c}^T \bar{b}_{R_c}, \quad L^T w = u, \quad (24.55)$$

and then compute

$$t_2 = N_{R_c}^T \bar{b}_{R_c} - N_{R \cup R_1}^T w. \quad (24.56)$$

Finally, it follows that

$$\Delta z_N = \{-\bar{b}_{R_1}, -t_1 - t_2\}. \quad (24.57)$$

(B) Pivot subcolumns $\{\bar{a}_{R \cup R_1, q}, \bar{a}_{R_1, q}, \bar{a}_{R_c, q}\}$.

Solve the following triangular systems for $w = \bar{a}_{R \cup R_1, q} = \widehat{B}_{R \cup R_1}^{-1} a_{R \cup R_1, q}$:

$$L u = a_{R \cup R_1, q}, \quad U w = u. \quad (24.58)$$

Then it follows that

$$\bar{a}_{R_1, q} = w_{R_1}, \quad \bar{a}_{R_c, q} = a_{R_c, q} - \widehat{B}_{R_c} w. \quad (24.59)$$

(C) Update of right-hand side $\bar{b}$ (ignore if $p \in R_c$ and $\bar{b}_{R_c} = 0$).

Determine $\alpha = \bar{b}_p / \bar{a}_{p, q}$, and update

$$\hat{b} = \bar{b} - \alpha \bar{a}_q, \quad \hat{b}_p = \alpha. \quad (24.60)$$

As for updating LU factors of $\widehat{B}_{R \cup R_1}$, nothing has to do with Tactic 24.4.1 or 24.5.1, but the following two cases arise:

Case 1 Rank increasing with Tactic 24.4.2: rank k of $\widehat{B}_{R \cup R_1}$ increases by 1 (see Sect. 19.4.2).

Case 2 Rank remaining with Tactic 24.4.3: k remains unchanged. The related updating is the same as in conventional simplex context, in essence.

Now Algorithm 24.6.1 can be converted into the following model.

Algorithm 24.7.1 [Dual face algorithm with LU factorization] Initial: deficient-basis B , partition (24.50), $\bar{B}_{R \cup R_1} = LU$, $\bar{b}_{R \cup R_1}$, and $\bar{c}_{N' \cup N}$. This algorithm solves problem (23.1).

1. Go to step 4 if $k = m$.
2. Go to step 6 if $\Delta z_{N'} = -\bar{b}_{R_1} \neq 0$.
3. Stop if T defined by (24.47) is empty (optimality achieved).
4. Execute Tactic 24.5.1, and go to step 2.
5. Solve systems (24.53) for v , and compute $\Delta z_N = -N_{R \cup R_1}^T v$.
6. Stop if J defined by (24.29) is empty (infeasible problem).
7. Determine β by (24.33).
8. If $\beta > 0$, update $\bar{c}_{N' \cup N}$ by (24.32).
9. If $\tilde{J} = N' \cap J' \neq \emptyset$, where J' is defined by (24.38), execute Tactic 24.4.1, and go to step 1.
10. Determine column index q by (24.40).
11. Solve systems (24.58) for $w = \bar{a}_{R \cup R_1, q}$.
12. Compute $\bar{a}_{R_c, q}$ by the second expression of (24.59), and determine p by (24.41).
13. If $\bar{a}_{p, q} \neq 0$, then
 - (1) Update L , U by Case 1, and execute Tactic 24.4.2
 - else
 - (2) Redetermine p (with j_p) by (24.43)
 - (3) Update L , U by Case 2
 - (4) Update $\bar{b}_{R \cup R_1}$ by (24.60), where $\alpha = \bar{b}_p / \bar{a}_{p, q}$, and execute Tactic 24.4.3
14. Go to step 1.

24.8 Notes on Dual Face Method with LU Factorization

If the proposed algorithm gets started from a pseudo-basis rather than deficient basis, one should use the search direction (24.25) instead of (24.27), since the nonzero $\bar{b}_{R_c}$ then contributes extra growth $\beta \|\bar{b}_{R_c}\|^2$ to the dual objective increment, by Theorem 24.3.2. Accordingly, however, two more triangular systems have to be solved. It might be a good idea to get started with a pseudo-basis of small k , proceed with N' , $R_1 = \emptyset$, and search direction

$$\Delta z_N = -\bar{N}_{R_c}^T (\bar{b}_{R_c} / \|\bar{b}_{R_c}\|), \quad (24.61)$$

until $\bar{b}_{R_c}$ vanishes, and go over to Algorithm 23.6.1 with direction (24.27) instead.

Alternatively, instead of the whole $\bar{b}_{R_1}$, it is possible to use a part of it to form search direction, e.g., such that

$$|\bar{b}_i| \geq \rho, \quad i \in R_1, \quad (24.62)$$

where ρ is a threshold, decreasing in the solution process.

Surprisingly enough, it turns out that Algorithm 24.7.1 and Algorithm 23.6.1 are just a pigeon pair. It is interesting to compare the two algorithms. It seems to be natural to use the latter to achieve dual feasibility and then use the former to achieve optimality, or vice versa. It is noted that both algorithms can be implemented, the same as in the conventional simplex context.

An advantage of the new dual face algorithm is that it solves small systems, compared with those solved by the simplex algorithms. In a single iteration, therefore, the overall computational cost by the former is less than the latter.

Another advantage of the new algorithm is its good stability, compared with simplex algorithms. The former tends to select a large pivot in magnitude, while the latter's pivot can be arbitrarily close to zero. As we all know, the latter sometimes fails to maintain a usable basis and has to restart from scratch.

Finally, the most attractive feature of the proposed algorithm might be the dual objective increment $\beta \|\bar{b}_{R_1}\|^2$ achieved per iteration. It is expected that this would significantly reduce the number of iterations required. At this stage, however, there are no numerical results available. We do not even know which search direction is better in practice. The implementation of the new algorithm and computational experiments are expected.

References

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Chapter 25

Simplex Interior-Point Method



To cope with degeneracy, the face (or dual face) method introduced previously does not seek the optimal solution, vertex by vertex, but instead, face by face. However, it still faces degeneracy. It seems that the degeneracy is inevitable if the iterates are confined on the boundary of the feasible region. This leads us again to the interior-point method, approaching the optimal solution by moving across the interior of the feasible region. As it is known, nevertheless, the latter has been seriously restricted in applications because it cannot be *warmly started*. To obtain the advantages of both, therefore, it is very attractive to combine the two types of methods into one.

There is no insurmountable gap between them even if they look different. In this chapter, we show that the combination is achievable. In principle, any simplex algorithm can be transformed into an interior-point algorithm by appropriately modifying the pivot rule and the termination criterion and shortening the stepsize.

The so-called *simplex interior-point algorithm*, introduced in this chapter, is transformed from the conventional generalized simplex algorithm. Starting from an interior point, it generates a series of *interior points* to achieve the optimality. In the next section, we present the column rule and search direction first. In the subsequent section, we address the row rule and stepsize. In the remaining sections, we highlight the optimality criterion, formulate the algorithm, and report encouraging computational results obtained from our extensive experiments.

Consider the bounded-variable problem below:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b, \quad l \leq x \leq u, \end{aligned} \tag{25.1}$$

where $A \in R^{m \times n}$ ($m < n$); $\text{rank } A = m$. The values of components of l and u are assumed to be finite. For problems with infinite bounds, one may take numbers of large enough module instead.

Let B be the current (normal) basis and N the associated nonbasis, i.e.,

$$B = \{j_1, \dots, j_m\}, \quad N = A \setminus B.$$

Note that j_i is the index of the i th column of B or the basic index corresponding to the i th row of the associated simplex tableau.

25.1 Column Pivot Rule and Search Direction

Let $\bar{x}$ be the current interior solution. The reduced cost is

$$\bar{c}_B = 0, \quad \bar{c}_N = c_N - N^T \bar{y}, \quad B^T \bar{y} = c_B. \quad (25.2)$$

The normal generalized simplex algorithm selects an entering index based on $\bar{c}$ only. In contrast, we will not only consider $\bar{c}$ but also the possible ranges that the nonbasic components of the solution are allowed to change.

To begin with, introduce

$$\delta_j = \begin{cases} \bar{x}_j - u_j, & \text{if } \bar{c}_j < 0, \\ \bar{x}_j - l_j, & \text{if } \bar{c}_j > 0, \\ 0, & \text{if } \bar{c}_j = 0, \end{cases} \quad j \in N, \quad (25.3)$$

which is the distance from $\bar{x}_j$ to one of the associated bounds that is violated if $\bar{x}_j$ changes to reduce the objective value. If $\bar{c}_j \neq 0$, then $\delta_j \bar{c}_j > 0$ since $\bar{x}$ is an interior solution.

The following rule is the key to the proposed algorithm.

Rule 25.1.1 (Column Pivot Rule) *Select a nonbasic index q by*

$$q \in \arg \max_{j \in N} \bar{c}_j \delta_j. \quad (25.4)$$

Then, the following vector $\Delta x = \{\Delta_B, \Delta_N\}$ can be determined:

$$\Delta x_B = \text{sign}(\bar{c}_q) \bar{a}_q, \quad B \bar{a}_q = a_q. \quad (25.5)$$

$$\Delta x_j = \begin{cases} 0, & j \in N; j \neq q \\ -\text{sign}(\bar{c}_q), & j = q. \end{cases} \quad (25.6)$$

Using the preceding notation, the following lemma reveals the eligibility of Δx to be a search direction.

Lemma 25.1.1 Assume that q is determined by Rule 25.1.1. If $\bar{c}_q \delta_q > 0$, the vector Δx is a descent direction in the null of A with respect to the objective.

Proof Note that $\bar{c}_q \neq 0$.

From (25.5)–(25.6), it follows that

$$A \Delta x = \text{sign}(\bar{c}_q) B B^{-1} a_q - \text{sign}(\bar{c}_q) a_q = 0. \quad (25.7)$$

Thus, Δx is in the null of A .

Furthermore, it holds by (25.2) that

$$\bar{c}_q = c_q - c_B^T B^{-1} a_q,$$

which together with (25.5), (25.6), and (25.4) gives

$$\begin{aligned} c^T \Delta x &= -\text{sign}(\bar{c}_q) c_q + \text{sign}(\bar{c}_q) c_B^T B^{-1} a_q \\ &= -\text{sign}(\bar{c}_q) (c_q - c_B^T B^{-1} a_q) \\ &= -\text{sign}(\bar{c}_q) \bar{c}_q \\ &= -|\bar{c}_q| \\ &< 0. \end{aligned} \quad (25.8)$$

Therefore, Δx is a descent direction with respect to the objective function. □

25.2 Row Pivot Rule and Stepsize

Using Δx as search direction, we form the normal line search below:

$$\hat{x} = \bar{x} + \alpha \Delta x, \quad (25.9)$$

or equivalently,

$$\hat{x}_B = \bar{x}_B + \alpha \Delta x_B, \quad (25.10)$$

$$\hat{x}_q = \bar{x}_q - \text{sign}(\bar{c}_q) \alpha, \quad (25.11)$$

$$\hat{x}_j = \bar{x}_j, \quad j \in N; \quad j \neq q, \quad (25.12)$$

where α is the stepsize to be determined.

Introduce

$$\sigma_i = \begin{cases} (u_{j_i} - \bar{x}_{j_i})/\Delta x_{j_i} & \text{if } \Delta x_{j_i} > 0, \\ (l_{j_i} - \bar{x}_{j_i})/\Delta x_{j_i} & \text{if } \Delta x_{j_i} < 0, \end{cases} \quad i = 1, \dots, m, \quad (25.13)$$

which is the maximum range that $\bar{x}_q$ is allowed to change while keeping $\bar{x}_{j_i}$ within its bounds. Considering all rows, we are led to the following rule for selecting row index.

Rule 25.2.1 (Row Pivot Rule) *Determine row index p such that*

$$\sigma_p = \min \{\sigma_i \mid i = 1, \dots, m\}. \quad (25.14)$$

Therefore, the largest possible stepsize subject to $l \leq \hat{x} \leq u$ is

$$\bar{\alpha} = \min \{\delta_q, \sigma_p\} > 0, \quad (25.15)$$

where δ_j is defined by (25.3).

Then, there are two cases arising:

- (i) $\delta_q \leq \sigma_p$. There is no need for any basis change.
- (ii) $\delta_q > \sigma_p$. The basis change should be performed with leaving index j_p and entering index q .

For the new solution $\hat{x}$ to be interior, however, we do not use $\bar{\alpha}$ itself but a shortened one instead, i.e.,

$$\alpha = \lambda \bar{\alpha}, \quad 0 < \lambda < 1, \quad (25.16)$$

where λ is a predetermined parameter close to 1.

Then, the following result can be claimed.

Lemma 25.2.1 *Let $\bar{x}$ be an interior solution. If $\bar{c}_q \delta_q > 0$, the $\hat{x}$ defined by (25.10)–(25.12) with the α given by (25.16) together with (25.13)–(25.15) is an interior solution with the objective value reduced by $\lambda \bar{\alpha} \bar{c}_q > 0$.*

Proof Since $\bar{x}$ is an interior solution, it can be verified that $\hat{x}$ defined by (25.10)–(25.12) satisfies equality constraints $A\hat{x} = b$ for any real number α , in particular for the α determined by (25.16). Furthermore, from (25.10)–(25.16), it follows that $\hat{x}$ satisfies $l < \hat{x} < u$ and is hence an interior solution.

Moreover, from (25.9) with (25.8) and (25.16), it follows that

$$c^T \hat{x} - c^T \bar{x} = \alpha c^T \Delta x = -\lambda \bar{\alpha} |\bar{c}_q| < 0, \quad (25.17)$$

and the claim is therefore valid. $\square$

25.3 Optimality Condition and the Algorithm

The search direction Δx is useless if $\bar{c}_q \delta_q$ vanishes. In this case, the optimality condition is fulfilled, as the following says.

Theorem 25.3.1 *The current $\bar{x}$ is an optimal solution if and only if it holds that*

$$\bar{c}_q \delta_q = 0. \quad (25.18)$$

Proof Necessity. Assume that $\bar{x}$ is an optimal solution. If (25.18) does not hold, i.e., $\bar{c}_q \delta_q > 0$, then new solution $\hat{x}$ can be determined with strictly lower objective value (Lemma 25.2.1), leading to a contradiction.

Sufficiency. By (25.4), it is clear that condition (25.18) is equivalent to

$$\bar{c}_j \delta_j = 0, \quad \forall j \in N. \quad (25.19)$$

For each $j \in N$, the quantity $\bar{c}_j \delta_j$ is the upper bound of the amount by which the objective value can reduce as the value of $\bar{x}_j$ changes toward the associated bound. Therefore, (25.19) implies that the objective value cannot decrease any further, and the claim is therefore valid. $\square$

Note that the preceding theorem can also be derived from the optimal condition for the bounded-variable problem (Theorem 7.3.4 of Part I).

Instead of (25.18), however, we will use the following *approximate optimality condition*.

Definition 25.3.1 The current solution $\bar{x}$ is an ϵ -optimal solution if

$$|\bar{c}_j| \leq \epsilon_1 \quad \text{or} \quad |\delta_j| \leq \epsilon_2, \quad \forall j \in N; \quad (25.20)$$

where $\epsilon_1 > 0$ and $\epsilon_2 > 0$ are predetermined tolerances.

Correspondingly, δ_j given by (25.3) will be replaced by

$$\delta_j = \begin{cases} \bar{x}_j - u_j, & \text{if } \bar{c}_j < -\epsilon_1, \\ \bar{x}_j - l_j, & \text{if } \bar{c}_j > \epsilon_2, \\ 0, & \text{otherwise,} \end{cases} \quad j \in N. \quad (25.21)$$

The overall steps can be summarized into the following algorithm.

Algorithm 25.3.1 (Simplex Interior-Point Algorithm) *Given $0 < \lambda < 1$, $\epsilon_1, \epsilon_2 > 0$ and $M \gg 1$. Initial: basis B , associated with interior solution $\bar{x}$. This algorithm solves the bounded-variable problem.*

1. Solve $B^T \bar{y} = c_B$ for $\bar{y}$.
2. Compute $\bar{c}_N = c_N - N^T \bar{y}$.
3. Compute δ_j , $j \in N$ by (25.21).
4. Stop if (25.20) is satisfied.
5. Select column index q by $q \in \arg \max_{j \in N} \bar{c}_j \delta_j$.
6. Solve $B \bar{a}_q = a_q$ for $\bar{a}_q$.
7. Compute Δx by (25.5) and (25.6).
8. Determine row index p by (25.14) with (25.13).
9. Stop if $\sigma_p > M$.
10. Set $\alpha = \lambda \bar{\alpha}$, where $\bar{\alpha} = \min \{\delta_q, \sigma_p\}$
11. Update $\bar{x}$ by (25.10)–(25.12).
12. Go to step 1 if $\delta_q \leq \sigma_p$.
13. Update basis B by replacing its j_p -indexed column with q -indexed column.
14. Go to step 1.

Concerning meanings of the exits of the preceding, the following can be claimed.

Theorem 25.3.2 Assume the finiteness of Algorithm 25.3.1. It terminates either at

- (i) Step 4, producing an ϵ -optimal basic solution, or
- (ii) Step 9, declaring the lower unboundedness of the problem

Proof The meaning of the exit step 4 is based on Theorem 25.3.1 and Definition 25.3.1 and that of step 9 is obvious. $\square$

In some applications, where the required is a convex rather than an interior solution, the following simple purification tactic can be incorporated into Algorithm 25.3.1:

Assume that Algorithm 25.3.1 terminates at step 4 with an ϵ -optimal solution. We move nonbasic components of the solution onto their nearest bounds, respectively, maintaining basic components unchanged. If the resulting solution, say x^0 , satisfies $Ax^0 = b$ within some tolerance, it can be viewed as optimal convex solution. In the other case, a standard two-phase generalized simplex algorithm applies to attaining optimal convex solution, hopefully within few iterations.

Remarks

Algorithm 25.3.1 can get itself started from any *feasible* point, not just interior point. Moreover, if the first stepsize taken is positive, it generates a series of interior points subsequently.

The method of this type actually minimizes the objective function over a feasible region shrunken from the original feasible region but uses a special termination criterion. In each iteration, it still makes a basis change but uses a special pivot rule and takes a positive stepsize.

Table 25.1 Ratio

Problem	MINOS/FPSP		MINOS/FPS		FPSP/FPS	
	Itns	Time	Itns	Time	Itns	Time
Kennington(16)	6.6	3.2	9.0	3.5	1.4	1.1
BPMPD(15)	2.8	3.4	9.5	9.0	3.3	2.6
Average(31)	3.3	3.4	9.4	6.6	2.8	2.0

25.4 Computational Results

Extensive computational experiments have been performed to gain an insight into the behavior of Algorithm 25.3.1. A summary of the associated results is offered in this section.

Implemented and compared are the following three codes:

1. MINOS: MINOS 5.51 with full pricing
2. FPS: two-phase code based on Algorithm 25.3.1
3. FPSP: two-phase code based on Algorithm 25.3.1 with the purification

The first set of test problems included all 16 problems from Kennington and the second included all 17 problems from BPMPD that were more than 500KB in compressed form (Appendix C: Tables C.2–C.3).

In Table 25.1, a comparison between the three codes is made (see Appendix F for details).

From the preceding table, it is seen that FPSP and FPS outperformed MINOS remarkably, with average iteration ratios 6.6 and 9.0 and time ratios 3.2 and 3.5 for the 16 Kennington problems. They outperformed MINOS, by average iteration ratios 2.8 and 9.5 and time ratios 3.4 and 9.0 for BPMPD problems. For the entire set of the 31 test problems, FPSP and FPS defeated MINOS by average iterations ratio 3.3 and 9.4 and time ratios 3.4 and 6.6.

In summary, the simplex interior-point algorithm much overperforms the simplex algorithm.

Chapter 26

Facial Interior-Point Method



The simplex interior-point method, introduced in the previous chapter, is a combination of the simplex method and the normal interior-point method. In contrast, the so-called *facial interior-point method* proposed in this chapter is a combination of the face method and the normal interior-point method.

Any normal interior-point algorithm can be transformed into such a type, in principle. Starting from an interior point (in the n -dimensional face), the new algorithm moves along the same search direction to the nearest boundary and takes a scaled shortened stepsize, attaining an interior point in a lower-dimensional face. Such an iteration is termed *facial* iteration. Then, it performs another facial iteration by moving along the direction projected onto the face to the nearest boundary and takes a scaledly shorten stepsize, and so on, until approaching optimality (according to appropriate termination criteria) or performing another series of facial iterations.

Along this line, in this chapter, we spend three sections to introduce the facial affine face interior-point algorithm, facial D-reduced interior-point algorithm, and facial affine interior-point algorithm, respectively.

26.1 Facial Affine Face Interior-Point Algorithm

The search direction utilized in a normal interior-point algorithm is characterized by a combination of the steepest descent direction and a so-called centering direction towards the interior of the feasible region. The search direction in the affine face Algorithm 21.7.1 is just such a direction. In this section, therefore, we transform the algorithm into a normal interior-point algorithm first and then transform the latter further into the desired algorithm by merging the face algorithm.

Return to the reduced problem (21.1). Initially, the search direction is given by (21.25) with $k = n + 1$ or $B = A$, i.e.,

$$\Delta = -e_{n+1} + \bar{X}^2 A^T \bar{y}, \quad (A \bar{X}^2 A^T) \bar{y} = -e_{m+1}, \quad (26.1)$$

where

$$\bar{X} = \text{diag}(\bar{x}_1, \dots, \bar{x}_n, 1). \quad (26.2)$$

What needs to do is to start from an interior point and take λ times of the original stepsize, where $0 < \lambda < 1$ is close to 1 (e.g., 95%~99%).

Using the preceding notation, the overall steps can be put into the following algorithm.

Algorithm 26.1.1 (Affine Face Interior-Point Algorithm) Given $\epsilon \ll 1$ and $\lambda \in (0, 1)$ close to 1. Initial: an interior point $\bar{x} > 0$. This algorithm approximately solves the reduced problem (21.1).

1. Compute $A \bar{X}^2 A^T = L L^T$.
2. Solve $L^T \bar{y} = -(1/\nu) e_{m+1}$ for $\bar{y}$, where ν is the $(m + 1)$ th diagonal of L .
3. Compute $\Delta = -e_{n+1} + \bar{X}^2 A^T \bar{y}$.
4. Stop if $\|\Delta\| \leq \epsilon$. (optimality achieved)
5. Stop if $J = \{j \in A \mid \Delta_j < 0\} = \emptyset$. (lower unbounded)
6. Determine $\alpha = \lambda \min_{j \in J} -\bar{x}_j / \Delta_j$.
7. Update: $\bar{x} = \bar{x} + \alpha \Delta$.
8. Go to step 1.

The author was astonished that the preceding algorithm is the same as the Dikin's algorithm (Algorithm 26.3.1) in essence. The only difference is that the former solves the reduced problem while the latter solves the standard problem. Therefore, if Dikin had handled the reduced problem rather than the standard problem, he would have gotten Algorithm 26.1.1.

It should be indicated that Algorithm 26.1.1 is preferable from a computational point of view, since it solves one triangular system less in each iteration.

Further, Algorithm 26.1.1 can be transformed into a facial interior-point algorithm by merging the face algorithm.

Algorithm 26.1.2 (Facial Affine Interior-Point Algorithm) Given $\lambda \in (0, 1)$. Initial: interior point $\bar{x} > 0$. This algorithm solves the reduced problem (21.1).

1. Set $k = n + 1$, $B = A$, $N = \emptyset$ and compute Cholesky factorization $B \bar{X}_B^2 B^T = L L^T$, where $\bar{X}_B$ is defined by (21.27).
2. Solves $L^T \bar{y} = -(1/\nu) e_{m+1}$, where ν is the $(m + 1)$ th diagonal of L .
3. Compute $\Delta_B = -e_k + \bar{X}_B^2 B^T \bar{y}$.
4. Go to step 11 if $\Delta_B = 0$.
5. Stop if $J = \{j \in B \mid \Delta_j < 0, j \neq n + 1\} = \emptyset$ (lower unbounded problem).

6. Determine the stepsize α and index p such that $\alpha = -\bar{x}_p/\Delta_p = \min_{j \in J} -\bar{x}_j/\Delta_j$.
7. update $\bar{x}_B = \bar{x}_B + \lambda\alpha\Delta_B$.
8. Update (B, N) by bringing p from B to N .
9. Update L by Algorithm 21.3.1.
10. Set $k = k - 1$; if $k > m + 1$, go to step 2.
11. Compute $\bar{z}_N = -N^T\bar{y}$.
12. Stop if $\bar{z}_N \geq 0$ (optimality achieved).
13. Go to step 1.

Example 26.1.1 Solve the following problem by Algorithm 26.1.2:

$$\begin{aligned}
 \min \quad & f \\
 \text{s.t.} \quad & -4x_1 + 3x_2 + x_3 - 2x_4 + 2x_5 = 5, \\
 & 3x_1 - x_2 + 2x_3 - 3x_4 - 4x_5 = -8, \\
 & x_1 + x_2 + 2x_3 + x_4 + 3x_5 = 12, \\
 & -2x_1 - 3x_3 + 2x_4 + x_5 - f = 0, \\
 & x_j \geq 0, \quad j = 1, \dots, 6.
 \end{aligned}$$

Set $\lambda = 99/100$. Initial interior point $\bar{x} = (1, 2, 1, 1, 2, -1)^T$.

Answer Iteration 1:

1. $k = 6$, $B = \{1, 2, 3, 4, 5, 6\}$, $N = \emptyset$. $\bar{X}_B = \text{diag}(1, 2, 1, 1, 2, -1)^T$; face matrix B and the Cholesky factor of $B\bar{X}_B^2B^T$ are $B = \begin{pmatrix} -4 & 3 & 1 & -2 & 2 \\ 3 & -1 & 2 & -3 & -4 \\ 1 & 1 & 2 & 1 & 3 \\ -2 & -3 & 2 & 1 & -1 \end{pmatrix}$,

$$L = \begin{pmatrix} 1399/134 & & & & \\ -2362/411 & 7176/919 & & & \\ 1985/471 & -1327/373 & 916/207 & & \\ 1181/822 & -302/85 & -800/331 & 815/672 & \end{pmatrix}.$$

2. $\bar{y} = (-100/5011, -2485/5193, -215/579, -1219/1793)^T$.
3. $\Delta_B = (-583/1587, -2127/2168, 509/1591, -356/1393, 278/865, -1385/3339)^T \neq 0$.
5. $J = \{1, 2, 4\} \neq \emptyset$.
- 6.

$$\begin{aligned}
 \alpha &= (99/100) \min\{-1/(-583/1587), -2/(-2127/2168), -1/(-356/1393)\} \\
 &= (99/100)(1745/856) = 2333/1156, \quad p = 2.
 \end{aligned}$$

7.

$$\begin{aligned}\bar{x}_B &= (1, 2, 1, 1, 2, -1)^T \\ &\quad + (2333/1156)(-583/1587, -2127/2168, 509/1591, -356/1393, 278/865, \\ &\quad -1385/3339)^T \\ &= (323/1249, 750/37499, 1593/968, 261/539, 2005/757, -2019/1099)^T.\end{aligned}$$

8. $B = \{1, 3, 4, 5, 6\}$, $N = \{2\}$.

$$9. L = \begin{pmatrix} 882/145 & & & & \\ -290/49 & 307/43 & & & \\ 1450/441 & -959/279 & 3244/737 & & \\ & 145/98 & -7775/2199 & -1184/473 & 2500/2447 \end{pmatrix}.$$

10. $k = 5$.

Iteration 2:

$$2. \bar{y} = (-361/1907, -1405/1907, -1039/1907, -845/882)^T.$$

$$3. \Delta_B = (-156/1907, 232/1907, 244/1907, -184/1907, -277/6603)^T \neq 0.$$

$$5. J = \{1, 5\} \neq \emptyset.$$

6.

$$\begin{aligned}\alpha &= (99/100) \min\{-(323/1249)/(-156/1907), -(2005/757)/(-184/1907)\} \\ &= (99/100)(7761/2455) = 917/293, \quad p = 1.\end{aligned}$$

7.

$$\begin{aligned}\bar{x}_B &= (323/1249, 1593/968, 261/539, 2005/757, -2019/1099)^T \\ &\quad + (917/293)(-156/1907, 232/1907, 244/1907, -184/1907, -277/6603)^T \\ &= (109/42149, 1688/833, 583/659, 1117/476, -5360/2723)^T.\end{aligned}$$

8. $B = \{3, 4, 5, 6\}$, $N = \{1, 2\}$.

$$9. L = \begin{pmatrix} 3524/769 & & & & \\ -1435/274 & 1211/172 & & & \\ 1435/274 & -3433/1235 & 1187/491 & & \\ & 769/3524 & -2201/577 & -499/322 & 1 \end{pmatrix}.$$

10. $k = 4 = m + 1$.

$$11. \bar{z}_N = (156/1907, 717/1907)^T \geq 0.$$

12. The approximate basic optimal solution and optimal value are

$$\bar{x} = (0.0026, 0.0200, 2.0264, 0.8847, 2.3466)^T, \quad \bar{f} = -1.9684.$$

The outcome is close to the exact basic optimal solution and optimal value, i.e.,

$$x^* = (0, 0, 2.0256, 0.8718, 2.3590)^T, \quad f^* = -1.9744.$$

The maximum error in components of $\bar{x}_B$ is 0.02, and that in $\bar{f}$ is 0.006.

26.2 Facial D-Reduced Interior-Point Algorithm

The search direction, used in the D-reduced gradient algorithm (Sect. 20.5), is uphill for the dual objective, and points to the interior of the feasible region. Therefore, the direction is suitable for establishing a normal interior-point algorithm.

The overall steps are put in the following algorithm using the deficient basis.

Algorithm 26.2.1 (D-reduced Interior-Point Algorithm) Given $\mu > 0$, $\lambda \in (0, 1)$, $\epsilon > 0$. Initial: interior point $\bar{z} > 0$. This algorithm solves the D-reduced problem 17.1.

1. Compute ω by (20.15).
2. Stop if $J = \{j \in A \mid \omega_j > 0\} = \emptyset$ (infeasible problem).
3. Compute $\beta = \bar{z}_q / \bar{\omega}_q = \min_{j \in J} \bar{z}_j / \bar{\omega}_j$.
4. Stop if $\beta < \epsilon$ (approximate optimality achieved).
5. If $\beta \neq 0$, update $\bar{z} = \bar{z} - \lambda\beta\omega$.
6. Go to step 1.

As in the previous section, we transform the preceding algorithm into the following facial interior-point algorithm by merging the face algorithm.

Algorithm 26.2.2 (Facial D-reduced Interior-Point Algorithm) Given $\mu > 0$, $\lambda \in (0, 1)$. initial: $\bar{z}_N > 0$. This algorithm solves D-reduced problem 17.1.

1. Set $k = 0$ and give B, R, N, R' by (20.6).
2. Compute $\omega_N = N^T e_m - \mu N_{R'}^T N_{R'} \bar{z}_N^{-1}$, and set $\bar{\omega}_N = \omega_N$.
3. Stop if $J = \{j \in N \mid \bar{\omega}_j > 0\} = \emptyset$ (infeasible problem).
4. Determine β and q such that $\beta = \bar{z}_q / \bar{\omega}_q = \min_{j \in J} \bar{z}_j / \bar{\omega}_j$, and compute $\lambda\beta$.
5. If $\beta \neq 0$, update $\bar{z}_N = \bar{z}_N - \lambda\beta\bar{\omega}_N$.
6. Compute $\bar{a}_q(R) = B_R^{-1} a_q(R)$, $\bar{a}_q(R') = a_q(R') - B_{R'} \bar{a}_q(R)$.
7. Go to step 12 if $R' = \emptyset$ or $\bar{a}_q(R') = 0$.
8. Determine row index $p \in \arg \max_{i \in R'} |\bar{a}_{i,q}|$.
9. Update B_R^{-1} by (19.10).
10. Set $k = k + 1$, bring q from N to B and p from R' to R .
11. Compute $\bar{\omega}_N = \omega_N - N_R^T (B_R^{-T} \omega_B)$, and go to step 3.
12. Determine $s \in \arg \max \{\bar{a}_{i_s,q} \mid i_s = 1, \dots, k\}$.
13. Go to step 1 if $\bar{a}_{i_s,q} > 0$.
14. Compute $\bar{x}_q = 1/\bar{\omega}_q$, $\bar{x}_B = -\bar{x}_q \bar{a}_q(R)$, $\bar{f} = c_q \bar{x}_q + c_B^T \bar{x}_B$.
15. Stop (optimality achieved).

Example 26.2.1 Solve the following problem by Algorithm 26.2.2:

$$\begin{aligned}
 \min \quad & f = 2x_1 + x_2 + 3x_3 + x_4 + 2x_5 + 4x_6, \\
 \text{s.t.} \quad & x_1 + 3x_2 - 2x_3 + 3x_4 - 4x_5 + 2x_6 = 0, \\
 & -2x_2 \quad \quad -x_4 - 2x_5 + x_6 = 0, \\
 & -2x_1 + x_2 + 2x_3 - 4x_4 + 3x_5 - 3x_6 = 0, \\
 & \quad \quad + 2x_2 + 3x_3 \quad \quad - 2x_5 - x_6 = 1, \\
 & x_j \geq 0, \quad j = 1, \dots, 6.
 \end{aligned}$$

Answer For the convenience of comparison, the related tableau will be given for each iteration.

Initial tableau:

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	3	-2	3	-4	2	
	-2		-1	-2	1	
-2	1	2	-4	3	-3	
	2	3		-2	-1	1
2	1	3	1	2	4	

$$\lambda = 95/100, \mu = 1, r = 4.$$

$$N_{R'} = \begin{pmatrix} 1 & 3 & -2 & 3 & -4 & 2 \\ 0 & -2 & 0 & -1 & -2 & 1 \\ -2 & 1 & 2 & -4 & 3 & -3 \end{pmatrix}.$$

Iteration 1:

1. $k = 0$; $B, R = \emptyset$, $N = \{1, \dots, 6\}$, $R' = \{1, 2, 3\}$. $\bar{z}_N = (2, 1, 3, 1, 2, 4)^T$.
- 2.

$$\begin{aligned}
 \omega_N &= (0, 2, 3, 0, -2, -1)^T - N_{R'}^T N_{R'} (1/2, 1, 1/3, 1, 1/2, 1/4)^T \\
 &= (-19/2, -191/12, 101/6, -325/12, 187/12, -41/3)^T, \bar{\omega}_N = \omega_N.
 \end{aligned}$$

3. $J = \{3, 5\}$.
- 4.

$$\begin{aligned}
 \beta &= (95/100) \min\{3/(101/6), 2/(187/12)\} = (95/100)(24/187) \\
 &= 114/935, q = 5.
 \end{aligned}$$

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	3	-2	3	-4	2	
	-2		-1	-2	1	
-2	1	2	-4	3	-3	
-19/2	-191/12	101/6	-325/12	187/12	-41/3	1
2	1	3	1	2	4	

5.

$$\bar{z}_N = (2, 1, 3, 1, 2, 4)^T$$

$$- (114/935)(-19/2, -191/12, 101/6, -325/12, 187/12, -41/3)^T.$$

6. $\bar{a}_q(R') = (-4, -2, 3)^T.$

7. $\bar{a}_q(R') \neq 0.$

8. $p = 1, \tau = -1/4, \widehat{B}_R^{-1} = (-1/4).$

9. $k = 1, B = \{5\}, R = \{1\}. N = \{1, 2, 3, 4, 6\}, R' = \{2, 3\}.$

10.

$$\bar{\omega}_N = (-19/2, -191/12, 101/6, -325/12, -41/3)^T$$

$$- (1, 3, -2, 3, 2)^T(-1/4)(187/12)$$

$$= (-269/48, -203/48, 217/24, -739/48, -47/8)^T.$$

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-1/4	-3/4	1/2	-3/4	1	-1/2	
-1/2	-7/2	1	-5/2			
-5/4	13/4	1/2	-7/4		-3/2	
-269/48	-203/48	217/24	-739/48		-47/8	1
1696/537	644/219	886/935	1609/374	1/10	5298/935	-114/935

Iteration 2:

3. $J = \{3\}.$

4.

$$\beta = (95/100) \min\{(886/935)/(217/24)\} = (95/100)(875/8349)$$

$$= 205/2059, q = 3.$$

5.

$$\begin{aligned}\bar{z}_N &= (1696/537, 644/219, 886/935, 1609/374, 5298/935)^T \\ &\quad - (205/2059)(-269/48, -203/48, 217/24, -739/48, -47/8)^T \\ &= (1349/363, 9359/2784, 47/992, 4102/703, 2513/402)^T.\end{aligned}$$

$$6. \bar{a}_q(R) = (-1/4)(-2) = 1/2, \bar{a}_q(R') = (0, 2)^T - (-2, 3)^T(-2) = (-4, 8)^T.$$

$$7. \bar{a}_q(R') \neq 0.$$

8.

$$p = 3, \tau = (2 - 3(1/2))^{-1} = 2, v = -2(1/2) = -1, d^T = -2(3)(-1/4) = 3/2.$$

$$U = (-1/4) - (-1)3(-1/4) = -1, \hat{B}_{\hat{R}}^{-1} = \begin{pmatrix} -1 & -1 \\ 3/2 & 2 \end{pmatrix}.$$

$$9. k = 2, B = \{5, 3\}, R = \{1, 3\}. N = \{1, 2, 4, 6\}, R' = \{2\}.$$

10.

$$\begin{aligned}\bar{\omega}_N &= (-19/2, -191/12, -325/12, -41/3)^T \\ &\quad - \begin{pmatrix} 1 & 3 & 3 & 2 \\ -2 & 1 & -4 & -3 \end{pmatrix}^T \begin{pmatrix} -1 & -1 \\ 3/2 & 2 \end{pmatrix}^T (187/12, 101/6)^T \\ &= (17, -63, 65/4, 85/4)^T.\end{aligned}$$

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	-4		1	1	1	
2	-10		1		3	
-5/2	13/2	1	-7/2		-3	
17	-63		65/4		85/4	1
1349/363	9359/2784	47/992	4102/703	1/10	2513/402	-905/4086

Iteration 3:

$$3. J = \{1, 4, 6\}.$$

4.

$$\begin{aligned}\lambda\beta &= (95/100) \min\{(1696/537)/17, (4102/703)/(65/4), (2513/402)/(85/4)\} \\ &= (95/100)(473/2546) = 473/2680, q = 1.\end{aligned}$$

5.

$$\begin{aligned}\bar{z}_N &= (1349/363, 9359/2784, 4102/703, 2513/402)^T \\ &\quad - (473/2680)(17, -63, 65/4, 85/4)^T \\ &= (577/806, 9398/649, 1528/515, 1608/643)^T.\end{aligned}$$

6.

$$\begin{aligned}\bar{a}_q(R) &= \begin{pmatrix} -1 & -1 \\ 3/2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -5/2 \end{pmatrix}, \\ \bar{a}_q(R') &= 0 - (-2, 0)(1, -2)^T = 2.\end{aligned}$$

7. $\bar{a}_q(R') \neq 0$.

8.

$$\begin{aligned}p &= 2. \tau = (0 - (-2, 0)(1, -5/2)^T)^{-1} = 1/2. \\ v &= -(1/2)(1, -5/2)^T = (-1/2, 5/4)^T. \\ d^T &= -(1/2)((-2, 0)B_R^{-1}) = -(1/2)(2, 2) = (-1, -1). \\ U &= \begin{pmatrix} -1 & -1 \\ 3/2 & 2 \end{pmatrix} - (-1/2, 5/4)^T(2, 2) = \begin{pmatrix} 0 & 0 \\ -1 & -1/2 \end{pmatrix}, \\ \widehat{B}_R^{-1} &= \begin{pmatrix} 0 & 0 & -1/2 \\ -1 & -1/2 & 5/4 \\ -1 & -1 & 1/2 \end{pmatrix}.\end{aligned}$$

9. $k = 2$, $B = \{5, 3, 1\}$, $R = \{1, 3, 2\}$. $N = \{2, 4, 6\}$, $R' = \emptyset$.

10.

$$\begin{aligned}\bar{\omega}_N &= (-191/12, -325/12, -41/3)^T \\ &\quad - \begin{pmatrix} 3 & 3 & 2 \\ 1 & -4 & -3 \\ -2 & -1 & 1 \end{pmatrix}^T \begin{pmatrix} 0 & 0 & -1/2 \\ -1 & -1/2 & 5/4 \\ -1 & -1 & 1/2 \end{pmatrix}^T (187/12, 101/6, -19/2)^T \\ &= (22, 31/4, -17/4)^T.\end{aligned}$$

x_1	x_2	x_3	x_4	x_5	x_6	RHS
	1		1/2	1	-1/2	
1	-5		1/2		3/2	
	-6	1	-9/4		3/4	
	22		31/4		-17/4	1
577/806	9398/649	47/992	1528/515	1/10	1608/643	-1143/2872

Iteration 4:

3. $J = \{2, 4\}$.

4.

$$\begin{aligned}\lambda\beta &= (95/100) \min\{(8645/597)/22, (1528/515)/(31/4)\} \\ &= (95/100)(1557/4067) = 559/1537, \quad q = 4.\end{aligned}$$

5.

$$\begin{aligned}\bar{z}_N &= (9398/649, 1528/515, 1608/643)^T - (559/1537)(22, 31/4, -17/4)^T \\ &= (6460/997, 301/2029, 3395/839)^T.\end{aligned}$$

x_1	x_2	x_3	x_4	x_5	x_6	RHS
	1		1/2	1	-1/2	
1	-5		1/2		3/2	
	-6	1	-9/4		3/4	
	22		31/4		-17/4	1
577/806	6460/997	47/992	301/2029	1/10	3395/839	-636/835

6. $\bar{a}_q(R) = \begin{pmatrix} 0 & 0 & -1/2 \\ -1 & -1/2 & 5/4 \\ -1 & -1 & 1/2 \end{pmatrix} \begin{pmatrix} 3 \\ -4 \\ -1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -9/4 \\ 1/2 \end{pmatrix}.$

7. $\bar{a}_q(R') = 0.$

11. $\max\{1/2, -9/4, 1/2\} = 1/2 > 0, \quad s = 1, \quad p = 1.$

Iteration 5:

1. $k = 0$; $B, R = \emptyset$, $N = \{1, \dots, 6\}$, $R' = \{1, 2, 3\}$.
- 2.

$$\begin{aligned}\omega_N &= (0, 2, 3, 0, -2, -1)^T \\ &\quad - N_{R'}^T N_{R'} (806/577, 997/6460, 992/47, 2029/301, 10, 839/3395)^T \\ &= (9606/67, 1965/23, -22001/110, 22051/69, -23926/57, 6767/25)^T, \\ \bar{\omega}_N &= \omega_N.\end{aligned}$$

x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	3	-2	3	-4	2	
	-2		-1	-2	1	
-2	1	2	-4	3	-3	
9606/67	1965/23	-22001/110	22051/69	-23926/57	6767/25	1
577/806	6460/997	47/992	301/2029	1/10	3395/839	-636/835

3. $J = \{1, 2, 4, 6\}$.
- 4.

$$\begin{aligned}\lambda\beta &= (95/100) \min\{(577/806)/(9606/67), (6460/997)/(1965/23), \\ &\quad (301/2029)/(22051/69), (3395/839)/(6767/25)\} \\ &= (95/100)(37/79707) = 8/18141, \quad q = 4.\end{aligned}$$

5.

$$\begin{aligned}\bar{z}_N &= (577/806, 6460/997, 47/992, 301/2029, 1/10, 3395/839)^T \\ &\quad - (8/18141)(9606/67, 1965/23, -22001/110, 22051/69, \\ &\quad - 23926/57, 6767/25)^T \\ &= (295/452, 1604/249, 189/1394, 104/14021, 1074/3767, 1347/343)^T.\end{aligned}$$

6. $\bar{a}_q(R') = (3, -1, -4)^T$.
7. $\bar{a}_q(R') \neq 0$.
8. $p = 3$. $\tau = -1/4$. $\hat{B}_{\hat{R}}^{-1} = (-1/4)$.
9. $k = 1$, $B = \{4\}$, $R = \{3\}$. $N = \{1, 2, 3, 5, 6\}$, $R' = \{1, 2\}$.

10.

$$\begin{aligned}
\bar{\omega}_N &= (9606/67, 1965/23, -22001/110, -23926/57, 6767/25)^T \\
&\quad - (-2, 1, 2, 3, -3)^T (-1/4)(22051/69) \\
&= (-25331/1543, 15045/91, -10457/260, -7743/43, 6478/209)^T.
\end{aligned}$$

x_1	x_2	x_3	x_4	x_5	x_6	RHS
$-1/2$	$15/4$	$-1/2$		$-7/4$	$-1/4$	
$1/2$	$-9/4$	$-1/2$		$-11/4$	$7/4$	
$1/2$	$-1/4$	$-1/2$	1	$-3/4$	$3/4$	
$-25331/1543$	$15045/91$	$-10457/260$		$-7743/43$	$6478/209$	1
$295/452$	$1604/249$	$189/1394$	$104/14021$	$1074/3767$	$1347/343$	$-1022/1341$

Iteration 6:

3. $J = \{2, 6\}$.

4.

$$\begin{aligned}
\beta &= (95/100) \min\{(1604/249)/(15045/91), (1347/343)/(6478/209)\} \\
&= (95/100)(239/6134) = 311/8402, \quad q = 2.
\end{aligned}$$

5.

$$\begin{aligned}
\bar{z}_N &= (295/452, 1604/249, 189/1394, 1074/3767, 1347/343)^T \\
&\quad - (311/8402)(-25331/1543, 15045/91, -10457/260, \\
&\quad - 7743/43, 6478/209)^T \\
&= (1893/1502, 439/1363, 2019/1243, 1821/262, 1957/704)^T.
\end{aligned}$$

$$6. \bar{a}_q(R) = (-1/4)(1) = -1/4, \quad \bar{a}_q(R') = (3, -2)^T - (3, -1)^T(-1/4) = (15/4, -9/4)^T.$$

$$7. \bar{a}_q(R') \neq 0.$$

8.

$$\begin{aligned}
p &= 1, \quad \tau = (3 - 3(-1/4))^{-1} = 4/15, \quad v = -(4/15)(-1/4) = 1/15, \\
d^T &= -(4/15)(3)(-1/4) = 1/5.
\end{aligned}$$

$$U = (-1/4) - (1/15)3(-1/4) = -1/5, \quad \widehat{B}_{\hat{R}}^{-1} = \begin{pmatrix} -1/5 & 1/15 \\ 1/15 & 4/15 \end{pmatrix}.$$

9. $k = 2$, $B = \{4, 2\}$, $R = \{3, 1\}$. $N = \{1, 3, 5, 6\}$, $R' = \{2\}$.
 10.

$$\begin{aligned}\bar{\omega}_N &= (9606/67, -22001/110, -23926/57, 6767/25)^T \\ &\quad - \begin{pmatrix} -2 & 2 & 3 & -3 \\ 1 & -2 & -4 & 2 \end{pmatrix}^T \begin{pmatrix} -1/5 & 1/15 \\ 1/15 & 4/15 \end{pmatrix}^T (22051/69, 1965/23)^T \\ &= (1570/279, -1763/97, -9777/95, 14664/349)^T.\end{aligned}$$

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-2/15	1	-2/15		-7/15	-1/15	
1/5		-4/5		-19/5	8/5	
7/15		-8/15	1	-13/15	11/15	
1570/279		-1763/97		-9777/95	14664/349	1
1893/1502	439/1363	2019/1243	104/14021	1821/262	1957/704	-553/692

Iteration 7:

3. $J = \{1, 6\}$.
 4.

$$\begin{aligned}\lambda\beta &= (95/100) \min\{(1893/1502)/(1570/279), (1957/704)/(14664/349)\} \\ &= (95/100)(426/6439) = 436/6937, \quad q = 6.\end{aligned}$$

- 5.

$$\begin{aligned}\bar{z}_N &= (1893/1502, 2019/1243, 1821/262, 1957/704)^T \\ &\quad - (436/6937)(1570/279, -1763/97, -9777/95, 14664/349)^T \\ &= (437/482, 3118/1127, 5287/394, 113/813)^T.\end{aligned}$$

- 6.

$$\begin{aligned}\bar{a}_q(R) &= \begin{pmatrix} -1/5 & 1/15 \\ 1/15 & 4/15 \end{pmatrix} \begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 11/15 \\ -1/15 \end{pmatrix}, \\ \bar{a}_q(R') &= 1 - (-1, -2)(11/15, -1/15)^T = 8/5.\end{aligned}$$

7. $\bar{a}_q(R') \neq 0$.

8.

$$p = 2, \tau = (1 - (-1, -2)(11/15, -1/15)^T)^{-1} = 5/8.$$

$$v = -(5/8)(11/15, -1/15)^T = (-11/24, 1/24)^T.$$

$$d^T = -(5/8)((-1, -2)B_R^{-1}) = -(5/8)(-1/5, -3/5) = (1/8, 3/8).$$

$$U = \begin{pmatrix} -1/5 & 1/15 \\ 1/15 & 4/15 \end{pmatrix} - (-11/24, 1/24)^T(-1/5, -3/5) = \begin{pmatrix} -7/24 & -5/24 \\ 5/24 & 7/24 \end{pmatrix}.$$

$$\widehat{B}_R^{-1} = \begin{pmatrix} -7/24 & -5/24 & -11/24 \\ 5/24 & 7/24 & 1/24 \\ 1/8 & 3/8 & 5/8 \end{pmatrix}.$$

9. $k = 2$, $B = \{4, 2, 6\}$, $R = \{3, 1, 2\}$. $N = \{1, 3, 5\}$, $R' = \emptyset$.

10.

$$\begin{aligned} \bar{\omega}_N &= (9606/67, -22001/110, -23926/57)^T \\ &\quad - \begin{pmatrix} -2 & 2 & 3 \\ 1 & -2 & -4 \\ 0 & 0 & -2 \end{pmatrix}^T \begin{pmatrix} -7/24 & -5/24 & -11/24 \\ 5/24 & 7/24 & 1/24 \\ 1/8 & 3/8 & 5/8 \end{pmatrix}^T \\ &\quad \cdot (22051/69, 1965/23, 6767/25)^T \\ &= (515/1373, 25888/9137, -9428/3017)^T. \end{aligned}$$

x_1	x_2	x_3	x_4	x_5	x_6	RHS
-1/8	1	-1/6		-5/8		
1/8		-1/2		-19/8	1	
3/8		-1/6	1	7/8		
515/1373		25888/9137		-9428/3017		1
437/482	439/1363	3118/1127	104/14021	5287/394	113/813	-1755/2036

Iteration 8:

3. $J = \{1, 3\}$.

4.

$$\begin{aligned} \lambda\beta &= (95/100) \min\{(437/482)/(515/1373), (3118/1127)/(25888/9137)\} \\ &= (95/100)(2780/2847) = 500/539, \quad q = 3. \end{aligned}$$

5.

$$\begin{aligned}\bar{z}_N &= (437/482, 3118/1127, 5287/394)^T \\ &\quad - (500/539)(515/1373, 25888/9137, -9428/3017)^T \\ &= (733/1312, 214/1547, 9350/573)^T.\end{aligned}$$

x_1	x_2	x_3	x_4	x_5	x_6	RHS
$-1/8$	1	$-1/6$		$-5/8$		
$1/8$		$-1/2$		$-19/8$	1	
$3/8$		$-1/6$	1	$7/8$		
$515/1373$		$25888/9137$		$-9428/3017$		1
$733/1312$	$439/1363$	$214/1547$	$104/14021$	$9350/573$	$113/813$	$-2450/1369$

$$6. \bar{a}_q(R) = \begin{pmatrix} -7/24 & -5/24 & -11/24 \\ 5/24 & 7/24 & 1/24 \\ 1/8 & 3/8 & 5/8 \end{pmatrix} \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/6 \\ -1/6 \\ -1/2 \end{pmatrix}.$$

$$7. R' = \emptyset.$$

$$11. \max\{-1/6, -1/6, -1/2\} \leq 0.$$

$$13. \text{Setting } \bar{x}_3 = 9137/25888 \text{ gives}$$

$$\begin{aligned}\bar{x}_B &= (9137/25888)(1/6, 1/6, 1/2)^T \\ &= (9137/155328, 9137/155328, 9137/51776)^T.\end{aligned}$$

Thus, the approximate basic optimal solution and optimal value are

$$\begin{aligned}\bar{x} &\approx (0, 9137/155328, 9137/25888, 9137/155328, 0, 9137/51776)^T, \\ \bar{f} &= (1, 3, 1, 4)(9137/155328, 9137/25888, 9137/155328, 9137/51776)^T \\ &= 9137/4854.\end{aligned}$$

On the other hand, the exact basic optimal solution and optimal value are

$$x^* = (0, 1/17, 6/17, 1/17, 0, 3/17)^T, \quad f^* = 32/17.$$

The errors are about 10^{-5} , which are accumulated from the computation of $\bar{\omega}$.

26.3 Facial Affine Interior-Point Algorithm

In this section, a new interior-point algorithm is described first. The search direction, derived via the affine scaling, is just the same as that in Dikin's affine algorithm. However, the new framework allows for forming a better interior-point algorithm.

Let $\bar{x}$ be the current interior-point. Consider the dual problem below:

$$\begin{aligned} \max \quad & g = b^T y, \\ \text{s.t.} \quad & (\bar{X} A^T : I) \begin{pmatrix} y \\ z \end{pmatrix} = \bar{X} c, \quad z \geq 0. \end{aligned} \quad (26.3)$$

Note that there is a 1-to-1 correspondence between columns of the unit matrix I and indices of z .

To begin with, we perform the *dual elimination* by orthogonal transformations (Part I). Since A is of full row rank, hence $\bar{X} A^T$ is of full column rank, there exists the QR factorization

$$\bar{X} A^T = (Q_1, Q_3) \begin{pmatrix} R_1 \\ 0 \end{pmatrix} = Q_1 R_1, \quad (26.4)$$

where (Q_1, Q_3) is orthogonal, partitioned as $Q_1 \in \mathcal{R}^{n \times m}$ and $Q_3 \in \mathcal{R}^{n \times (n-m)}$, and $R_1 \in \mathcal{R}^{m \times m}$ is nonsingular upper triangular.

We cite from Sect. 10.2 of Part I the search direction below:

$$\Delta x' = P \bar{X} c = (I - \bar{X} A^T (A \bar{X}^2 A^T)^{-1} A \bar{X}) \bar{X} c. \quad (26.5)$$

The following reveals that the search direction in x' -space, used in Dikin's affine algorithm, can be obtained alternatively by using the matrix Q_3 .

Proposition 26.3.1 *The vector $\Delta x'$ defined by (26.5) is equal to $Q_3 Q_3^T \bar{X} c$.*

Proof Substituting (26.4) to (26.5) and noting $Q_1^T Q_1 = I$ and $Q_1 Q_1^T + Q_3 Q_3^T = I$ gives

$$\Delta x' = \left(I - Q_1 R_1 \left(R_1^T Q_1^T Q_1 R_1 \right)^{-1} R_1^T Q_1^T \right) \bar{X} c = \left(I - Q_1 Q_1^T \right) \bar{X} c = Q_3 \left(Q_3^T \bar{X} c \right). \quad (26.6)$$

□

Therefore, we are led to the following variant of the affine interior-point algorithm.

Algorithm 26.3.1 [Affine interior-point algorithm with QR factorization] Let $\lambda \in (0, 1)$, $\epsilon > 0$ be given. Initial: interior-point $\bar{x} > 0$.

This algorithm solves the standard LP problem.

1. Compute $\Delta x'$ by (26.6).
2. Stop if $\|\Delta x'\| < \epsilon$ (optimality achieved approximately).
3. Stop if $\Delta x' \leq 0$ (Unbounded below).
4. Update by $\bar{x} = \bar{x} - \lambda \bar{X} \Delta x' / \max(\Delta x')$.
5. Go to step 1.

Since Algorithm 26.3.1 and Dikin's algorithm are equivalent in theory, their computational effort involved depends on how to compute the same search direction, including the realization, the sparsity of A , the number $n - m$ compared with m , etc. We will not go into details here because what we are really interested in is not the preceding Algorithm but its facial variant, as derived as follows.

We premultiply the augmented matrix of the equality constraints of (26.3) by $Q^T = [Q_1, Q_3]^T$, leading to the so-called *triangular form*, i.e.,

$$Q^T(\bar{X}A^T : I \mid \bar{X}c) = \begin{pmatrix} R_1 & Q_1^T & Q_1^T \bar{X}c \\ 0 & Q_3^T & Q_3^T \bar{X}c \end{pmatrix}. \quad (26.7)$$

The preceding represents the linear system which is equivalent to the dual equality constraints. By Proposition 26.3.1, it is known that the south-east submatrix $(Q_3^T \mid Q_3^T \bar{X}c)$ gives the projection $\Delta x'$, defined by (26.6), i.e.,

$$\Delta x' = Q_3 \left(Q_3^T \bar{X}c \right). \quad (26.8)$$

Then, update $\bar{x}$ by

$$\hat{x} = \bar{x} - \lambda \bar{X} \Delta x' / \max(\Delta x'), \quad (26.9)$$

where $\lambda \in (0, 1)$ is the stepsize. If it goes to the next iteration at this point, the resulting is just the same as Algorithm 26.3.1.

On the contrary, we will carry out a series of facial iterations in x' -space, starting with

$$N = \emptyset, \quad B = A$$

and moves from face to lower-dimensional face.

Assume that $\Delta x'_B = \Delta x' \not\leq 0$. Update the $\bar{x}'_B$ by the following formula:

$$\hat{x}'_B = e - \lambda \Delta x'_B / \max(\Delta x'_B),$$

and determine an index q such that

$$q = \arg \max \left\{ \Delta x'_j \mid j = 1, \dots, n \right\}.$$

It is not difficult to show that $(\bar{X}A^T \vdots e_q)$ is of full column rank. If the QR factorization of it is available, then the orthogonal projection of the objective gradient $\bar{X}c$ onto the null space of

$$\begin{pmatrix} A\bar{X} \\ e_q^T \end{pmatrix}$$

can be computed similarly as before, and it is therefore able to update the solution once more in x' -space. Note that the q -indexed component of the projection equals 0, hence the q -indexed component of the solution remains unchanged.

Assume that after performing k ($< n - m$) such iterations, there are index sets

$$N = \{1, \dots, k\}, \quad B = \{k + 1, \dots, n\}.$$

Let the QR factorization $(\bar{X}A^T \vdots I_N) = QR$ be available. Premultiplying by Q^T the augmented matrix of equality constraints of (26.3) gives

$$\begin{aligned} & Q^T(\bar{X}A^T \vdots I_N \vdots I_B | \bar{X}c) \\ &= \begin{pmatrix} Q_1^T \bar{X}A^T & Q_1^T I_N & Q_1^T I_B & | & Q_1^T \bar{X}c \\ Q_2^T \bar{X}A^T & Q_2^T I_N & Q_2^T I_B & | & Q_2^T \bar{X}c \\ Q_3^T \bar{X}A^T & Q_3^T I_N & Q_3^T I_B & | & Q_3^T \bar{X}c \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & Q_1^T I_B & | & Q_1^T \bar{X}c \\ 0 & R_{22} & Q_2^T I_B & | & Q_2^T \bar{X}c \\ 0 & 0 & Q_3^T I_B & | & Q_3^T \bar{X}c \end{pmatrix} \end{aligned} \quad (26.10)$$

where the orthogonal matrix $Q = (Q_1, Q_2, Q_3)$ is partitioned as $Q_1 \in \mathcal{R}^{n \times m}$, $Q_2 \in \mathcal{R}^{n \times k}$ and $Q_3 \in \mathcal{R}^{n \times (n-m-k)}$, and $R_{11} \in \mathcal{R}^{(m+k) \times (m+k)}$ is nonsingular upper triangular. The preceding is the k th triangular form, whose south-east submatrix gives projection

$$\Delta x'_B = \left(Q_3^T I_B \right)^T \left(Q_3^T \bar{X}c \right). \quad (26.11)$$

Assume that $\Delta x'_B \not\leq 0$. Since $\Delta x'_N = (Q_3^T I_N)^T (Q_3^T \bar{X}c) = 0$ and $\bar{x}'_N$ remains unchanged, one just need to update

$$\hat{x}'_B = \bar{x}'_B - \lambda \alpha \Delta x'_B, \quad (26.12)$$

where

$$\alpha = \bar{x}'_q / \Delta x'_q = \min \left\{ \bar{x}'_j / \Delta x'_j \mid \Delta x'_j > 0, j \in B \right\}, \quad (26.13)$$

which is the stepsize from the current solution to the nearest boundary. It is clear that the new solution is again an interior point. Then move q from B to N , and go to the $(k + 1)$ th facial iteration.

The forgoing process terminates when $k = n - m$ or $Q_3^T \bar{X}c = 0$, hence the $\Delta x'_B$ defined by (26.11) vanishes. In this case, substituting $\bar{z}_B = 0$ to the dual equality constraints, represented by (26.10), leads to the upper triangular systems

$$\begin{pmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{pmatrix} \begin{pmatrix} y \\ z_N \end{pmatrix} = \begin{pmatrix} Q_1^T \bar{X}c \\ Q_2^T \bar{X}c \end{pmatrix}. \quad (26.14)$$

Assume that $(\bar{y}, \bar{z}_N)$ is the solution to this system.

If $\bar{z}_N \geq 0$, it is not difficult to show that $\bar{x}$ is the optimal solution to the following problem:

$$\begin{aligned} & \min c^T x, \\ & \text{s.t. } Ax = b, \\ & \quad x_B \geq 0, \quad x_N \geq \bar{x}_N. \end{aligned}$$

If $\lambda \in (0, 1)$ is sufficiently close to 1, then $\bar{x}_N$ can be arbitrarily close to 0. If λ is predetermined to be close to 1, therefore, $\bar{x}$ can be regarded as an approximately optimal solution to the original problem, and the solution process terminates.

In the other case when $\bar{z}_N \not\geq 0$, the facial iterations are finished. The related solution in the original space can be obtained by

$$\bar{x} = \bar{X} \bar{x}',$$

and a new $\bar{X}$ can be formed to be ready to carry out another series of facial iterations.

The trick of the algorithm is that the QR factors in each facial iteration can be obtained recursively. In fact, the last $n - m - k - 1$ components of the q -indexed column of (26.10) can be eliminated by Givens rotations.

Let $\tilde{Q} \in \mathcal{R}^{(n-m-k) \times (n-m-k)}$ be the product of Givens rotations such that

$$\tilde{Q} Q_3^T e_q = \eta e_1.$$

Then $\hat{Q}^T = [I : \tilde{Q}^T]^T Q^T$ is just the required factor for the $(k + 1)$ th facial iteration. It is seen that the k and $(k + 1)$ th triangular forms are the same, except for the submatrix, associated with Q_3 . On the other hand, at the beginning of each series of facial iterations, it is necessary to compute the QR factors from scratch since $\bar{X}$ changes.

The overall steps can be summarized to the following algorithm.

Algorithm 26.3.2 (Facial Affine Interior-Point Algorithm) Given $\lambda \in (0, 1)$. Initial: interior point $\bar{x} > 0$. This algorithm solves the standard LP problem.

1. Set $k = 0$, and compute triangular form (26.7).
2. Compute $\Delta x' = Q_3(Q_3^T \bar{X}c)$.
3. Stop if $\Delta x' \leq 0$ (lower unbounded).
4. Determine α and q such that $\alpha = \bar{x}'_q / \Delta x'_q = \min\{\bar{x}'_j / \Delta x'_j \mid \Delta x'_j > 0, j \in B\}$.
5. If $\alpha \neq 0$, update: $\bar{x}'_B = \bar{x}'_B - \lambda \alpha \Delta x'_B$.
6. Set $k = k + 1$, and update (B, N) by bringing q from B to N .
7. Go to step 9 if $k = n - m$ or $Q_3^T \bar{X}c = 0$.
8. Eliminate the $(m + k + 1)$ to n th components of the q -indexed column of the triangular form by Givens rotations.
9. Go to step 2.
10. Solve the upper triangular system (26.14).
11. Stop if $\bar{z}_N \geq 0$ (approximate optimality achieved).
12. Set $\bar{x} = \bar{X} \bar{x}'$.
13. Go to step 1.

The algorithm, developed by Pan (2013), is slightly different from the preceding algorithm, as it carries out the facial iteration using

$$\hat{x}'_B = \bar{x}'_B - \alpha \Delta x'_B$$

rather than (26.12). Therefore, the resulting iterate is not interior but boundary point. As Algorithm 26.3.2 is close to Pan's algorithm, we cite the latter's computational results to give a clue on its performance.

The associated computational experiments were carried out on a Pentium III 550E PC with Windows 98 operating system, 168 MB inner storage, and about 16 decimal precision. Visual Fortran 5.0 compiler was used. There were following three dense codes involved:

1. AIP: affine interior-point Algorithm 10.2.1 (Part I).
2. VAIP: Algorithm 26.3.1.
3. PAIP: Algorithm 26.3.2.

The preceding codes are tested on the 26 smallest (by $m + n$) Netlib standard problems. The first set involves 16 smaller problems, and the second set is the rest 10 problems (Appendix C: Table C.4, problems AFIRO-DEGEN2).

Table 26.1 below lists iterations and time ratios:

Table 26.1 Iteration and time ratios

Problem	AIP/VAIP		AIP/PAIP	
	Iterations	Time	Iterations	Time
Set 1 (16)	0.98	0.32	6.32	1.47
Set 2 (10)	0.92	0.25	3.55	1.52
Average (26)	0.95	0.26	4.56	1.52

From the bottom line of the preceding Table, it is seen that the total iteration and time ratios of AIP to VAIP are 0.95 and 0.26, respectively. So, the former outperforms the latter. However, PAIP outperforms AIP significantly: the total iteration and time ratios of AIP to PAIP are 4.56 and 1.52, respectively. Therefore, the introduction of the facial iterations is favorable.

On the other hand, the time ratio of AIP to PAIP is much less than their iteration ratio (1.52 vs 4.56), since each iteration of the latter is more time-consuming due to the orthogonal transformations.

Reference

Pan P-Q (2013) An affine-scaling pivot algorithm for linear programming. *Optimization* 62:431–445

Chapter 27

Decomposition Principle



It has long been a challenge to solve large-scale LP problems that require a huge number of iterations and storage. One option for solving such a problem is to use some decomposition methods, like the Dantzig–Wolfe or Benders decomposition method. Unfortunately, applications of these methods are limited to certain types of structured problems. Very large problems cannot be solved at present although the computer hardware is advanced.

On the reality side, however, decomposition arrangements have been put into effect successfully for many years. For instance, the football World Cup follows a “parallel decomposition” competition system: firstly preliminary selection contest is undertaken in each of the six continents; then the winners take part in the final round, and the champion stands out fi. On the other hand, an arena contest follows a “serial decomposition” competition system: the champion is produced through a series of games between a temporary winner and a challenger successively. Without a doubt, it is impossible to organize any big event without any decomposition scheme.

Solving very large-scale LP problems can be closely analogized to the above events. This is the theme of this chapter. Firstly, we introduce two decomposition algorithms in the next section and then address the decomposition principle—“*arena contest*,” which the new decomposition algorithms follow. In the subsequent sections, the decomposition method is illustrated with instances of optimization problems, such as the standard LP problem, bounded-variable LP Problem, and ILP Problem. Practical remarks are made in the final section.

27.1 New Decomposition Method

Firstly, we show that the standard problem can be solved by handling two smaller subprograms in succession.

Let B_0 be an initial feasible basis, associated with feasible basic solution $\bar{x}$. Partition nonbasis N to two sections (N_1, N_2). Denote the corresponding sections of c by $(c_{B_0}, c_{N_1}, c_{N_2})$. The first subprogram is of the following form:

$$\begin{aligned} \min \quad & c_{B_0}^T x_{B_0} + c_{N_1}^T x_{N_1}, \\ \text{s.t.} \quad & B_0 x_{B_0} + N_1 x_{N_1} = b, \quad x_j \geq 0, \quad j \in B_0 \cup N_1, \end{aligned} \quad (27.1)$$

whose optimal basis (basic solution or value) is termed *suboptimal* with respect to the original problem. Based on the resulting suboptimal basis, say B_1 , we construct a second subprogram as follows:

$$\begin{aligned} \min \quad & c_{B_1}^T x_{B_1} + c_{N_2}^T x_{N_2}, \\ \text{s.t.} \quad & B_1 x_{B_1} + N_2 x_{N_2} = b, \quad x_j \geq 0, \quad j \in B_1 \cup N_2. \end{aligned} \quad (27.2)$$

We might as well assume that the problem is not unbounded. Then each suboptimal solution corresponds to a feasible basic solution to the original problem; e.g., after the first subprogram is solved, the components, associated with $B_0 \cup N_1$, of the feasible solution $\bar{x}$ change to the new values of the suboptimal solution, while the other components remain unchanged. Note that all nonbasic components of the suboptimal solutions are zero, and so are those of the corresponding feasible solutions to the original problem.

It will be shown that optimality of the original problem is achieved if the suboptimal values of the two subprograms are equal. Otherwise, take the suboptimal basis, obtained by solving the second subprogram, as a new B_0 to repeat the previous steps.

The preceding can be generalized by partitioning the nonbasis to multiple sections. Let B_0 be the initial feasible basis, and let $(N_1, \dots, N_s)$, $s \geq 2$ be a partition of the nonbasis. The t th subprogram ($t = 1, \dots, s$) to be solved is of form

$$\begin{aligned} \min \quad & c_{B_{t-1}}^T x_{B_{t-1}} + c_{N_t}^T x_{N_t}, \\ \text{s.t.} \quad & B_{t-1} x_{B_{t-1}} + N_t x_{N_t} = b, \quad x_j \geq 0, \quad j \in B_{t-1} \cup N_t, \end{aligned} \quad (27.3)$$

where B_{t-1} is a suboptimal basis, obtained by solving the $(t-1)$ th subprogram ($t = 2, \dots, s$).

This scheme can be organized into the following algorithm model.

Algorithm 27.1.1 (Decomposition Algorithm 1) Let the integer ($s \geq 2$) be given. Initial: feasible basis B_0 . This algorithm solves the standard LP problem.

1. Partition $A \setminus B_0$ to $(N_1, \dots, N_s)$.
2. Set $t = 0$.
3. Set $t = t + 1$.
4. Solve subprogram (27.3) by the simplex algorithm (Part I):

- (1) Stop if lower unbounded.
- (2) If $t < s$, go to step 3 with the suboptimal basis B_t and objective value $\bar{f}_t$.

5. Stop if $\bar{f}_s = \bar{f}_1$.
6. Set $B_0 = B_s$.
7. Go to step 1.

Iterations associated with $t = 1 \dots s$ form a “circle” (solving the s subprograms). The solution process involves a series of circles, in general.

Lemma 27.1.1 *Algorithm 27.1.1 generates a sequence of feasible bases, with objective values decreasing (not necessarily strictly) monotonically.*

Proof We might as well assume that the algorithm terminates at the end of the second circle.

Let us take a look at the first circle. As the initial basis B_0 is feasible to the original problem, the suboptimal bases B_t are well-defined ($t = 1, \dots, s$). It is clear that each basis B_t and associated objective value $\bar{f}_t$ are not only optimal to the t th subprogram but also feasible to the original problem. Since each $\bar{f}_t$ is optimal to the t th subprogram and feasible to the $(t + 1)$ th subprogram, the objective values decrease (not necessarily strictly) monotonically. Likewise, it can be shown that the objective values decrease monotonically within the second circle; therefore, so do all the objective values since B_0 for the second circle is just the end suboptimal basis B_s of the first circle. $\square$

Theorem 27.1.2 *Under the assumption of nondegeneracy, Algorithm 27.1.1 terminates either at*

- (i) *Step 4(1), detecting lower unboundedness of the original problem (12.1), or at*
- (ii) *Step 5, generating an optimal basis to the original problem*

Proof Assume at the moment that the algorithm terminates. By Lemma 27.1.1, the condition $\bar{f}_s = \bar{f}_1$ implies that all the feasible values are equal.

It is clear that the termination at step 4(1) implies lower unboundedness of the original problem. Assume that the process terminates at step 5. We show (ii) by induction on the number s of sections.

Consider the case $s = 1$. Since B_1 is an optimal basis to the first subprogram, the corresponding optimal dual solution, say $(\tilde{y}, \tilde{z})$, satisfies

$$\tilde{z}_{B_0 \cup N_1} = c_{B_0 \cup N_1} - (B_0 \cup N_1)^T \tilde{y} \geq 0, \quad \tilde{y} = B_1^{-T} c_{B_1}. \quad (27.4)$$

As $A = \{B_0, N_1, N_2\}$, so we only need to show that

$$\tilde{z}_{N_2} = c_{N_2} - N_2^T \tilde{y} \geq 0. \quad (27.5)$$

Since $\bar{f}_2 = \bar{f}_1$, B_1 is also optimal to the second subprogram. If (27.5) does not hold, then it follows from $\bar{x}_{B_1} = B_1^{-1} b > 0$ (nondegeneracy assumption) that the B_1 is not optimal to the second subprogram, a contradiction. Therefore, (27.5) holds, which together with (27.4) implies together that B_1 is an optimal basis to the original problem.

Assuming that (ii) is valid for $1 \leq s \leq t$ with $t \geq 1$, we will show the validity for $s = t + 1$. Since $\bar{f}_{t+1} = \bar{f}_1$, it holds that $\bar{f}_t = \bar{f}_1$ (Lemma 27.1.1). According to the assumption of induction, B_1 is therefore an optimal basis to the problem

$$\begin{aligned} \min \quad & c_{B_0}^T x_{B_0} + c_{N'}^T x_{N'}, \\ \text{s.t.} \quad & B_0 x_{B_0} + N' x_{N'} = b, \quad x_j \geq 0, \quad j \in B_0 \cup N', \end{aligned} \quad (27.6)$$

where $N' = N_1 \cup N_2 \cup \dots \cup N_t$. Taking a look at the partition (B_0, N', N_{t+1}) of the original problem, and noting that the statement is valid for $s = 2$, it can be asserted that B_1 is an optimal basis to the original problem.

Let us turn to the termination of Algorithm 27.1.1. All associated subprograms are feasible, and hence the simplex algorithm utilized to solve them terminates under the assumption of nondegeneracy. Thus, it does not terminate only if there are infinitely many circles, and $\bar{f}_s < \bar{f}_1$ holds for each. This implies that there are infinitely many bases, as leads to a contradiction. Therefore, Algorithm 27.1.1 terminates. $\square$

The shortcoming of subprogram (27.1) (or (27.3)) is that the entering set N_1 (or N_t) is predetermined without considering the associated reduced costs at all. This may be remedied as follows.

Let ϵ be the optimality tolerance and let a real number ($\sigma > \epsilon > 0$) and an integer, ($m' \leq m$) be given. Denote B_0 the current feasible basis and by N the candidate indices from which to choose the entering set N_1 . Initially, just take the nonbasis as N , i.e., $N = A \setminus B_0$. Assume that $\bar{z}_N$ is the nonbasic reduced costs. Then, N_1 is determined by

$$N_1 = \{j \in N \mid \bar{z}_j \leq -\sigma\}, \quad (27.7)$$

subject to $|N_1| \leq m'$. So the number of indices in N_1 is no more than m . If $|N_1| > 0$, subprogram (27.1) is solved. Then we set $N = A \setminus (B_0 \cup N_1)$ for the next iteration. In the solution process, σ is decreased by a percentage τ to maintain $|N_1|$ reasonably large whenever $|N_1| < m'$. Assuming $|N_1| = 0$, optimality of the original problem is achieved if $\sigma = \epsilon$, or N_1 is determined by (27.7) again with a reduced σ if $\sigma > \epsilon$, and so on.

The preceding discussions lead to a variant of Algorithm 27.1.1, formulated as follows.

Algorithm 27.1.2 (Decomposition Algorithm 2) Give reals ϵ and τ such that $0 < \epsilon \ll 1$, $0 < \tau < 1$ and integer $m' \leq m$. Initial: $\sigma > \epsilon$; feasible basis B_0 and nonbasis N . This algorithm solves the standard LP problem.

1. Compute $\bar{z}_N = c_N - N^T \bar{y}$, $B_0^T \bar{y} = c_{B_0}$.
2. Determine set $N_1 = \{j \in N \mid \bar{z}_j \leq -\sigma\}$.
3. If $|N_1| < m'$, set $\sigma = \max\{\epsilon, \tau\sigma\}$.
4. Go to step 7 if $|N_1| > 0$.
5. Stop if $\sigma = \epsilon$ (optimality achieved).

6. Go to step 2.
7. Solve subprogram (27.1) by the simplex Algorithm 3.1.1 (Part I):
 - (1) Stop if lower unbounded.
 - (2) Set $N = A \setminus (B_0 \cup N_1)$, and then set the yielding suboptimal basis to B_0 .
8. Go to step 1.

27.2 Decomposition Principle: “Arena Contest”

The preceding decomposition algorithms work as if some “arena contest” is organized to generate the “team champion” via a series of “competitions.”

The number of players (columns) in the whole area (coefficient matrix) is too huge to match in the limited sports facilities (available hardware) to produce a team champion (optimal basis). Therefore, a system is formulated and executed to govern the competition (optimization), so that a series of relatively small games can go on smoothly in the limited facilities. The special idea behind each decomposition algorithm may be explained as follows.

(A) Algorithm 27.1.1:

At the beginning, an initial team (basis B_0) is selected from players (columns) of the area (coefficient matrix A) under some criterion. The other players (nonbasic columns N) are divided into s teams ($N_1, \dots, N_s$). The first match takes place in the facilities among the initial team and the first team (B_0, N_1), producing a first winner team (suboptimal basis B_1). Then, the second match takes place among the first winner and the second team (B_1, N_2), and so on. Once the s th match is finished, the last winner team (suboptimal basis B_s) is declared as the ultimate champion (“optimal” basis) if it has the same record (suboptimal value) as the first winner; otherwise, the next round of competitions is carried out by taking the last winner as a new initial team (setting $B_0 = B_s$).

(B) Algorithm 27.1.2:

After a temporary winner team (B_0) is selected, a challenging team (N_1) is determined in a tryout among all the other players (N) under a criterion, which will be relaxed whenever the member of the challenging team is less than some threshold. Then a match takes place between the temporary winner team and the challenging team (B_0, N_1) producing a new temporary winner team (setting the last suboptimal basis to B_0), and repeat. If no player wins out in the tryout under the criterion of the lowest standard ($\bar{z}_N \geq -\epsilon$), the last winner team; i.e., the end suboptimal basis, is declared as the ultimate team champion (the *optimal* basis).

27.3 Illustration on Standard LP Problem

In this subsection, an instance is offered for illustration of the decomposition principle on a standard problem.

Example 27.3.1 Solve the following LP problem by Algorithm 27.1.1 with $s = 2$:

$$\begin{aligned}
 \min \quad & f = -2x_1 + x_2 + 3x_3 - 5x_4 - 3x_6 - 7x_7 - 4x_8 - 2x_9 \\
 \text{s.t.} \quad & -3x_2 - 2x_3 + x_4 + 4x_7 + -2x_9 = 1 \\
 & 3x_1 + 4x_3 + x_5 + 2x_7 + x_8 + 5x_9 = 2 \\
 & -x_1 + x_2 + x_3 + x_6 + 8x_7 - 6x_8 + 3x_9 = 8 \\
 & x_j \geq 0, \quad j = 1, \dots, 9.
 \end{aligned}$$

Answer Initial feasible basis and nonbasis: $B_0 = \{4, 5, 6\}$, $N = \{1, 2, 3, 7, 8, 9\}$.

Outer iteration 1:

Subiteration 1:

1. $N_1 = \{1, 2, 3\}$, $N_2 = \{7, 8, 9\}$.
2. $t = 0$.
3. $t = 0 + 1 = 1$.
4. Call Algorithm 3.1.1 (Part I) to solve the first subprogram ($B_0 = \{4, 5, 6\}$, $N_1 = \{1, 2, 3\}$)

$$\begin{aligned}
 \min \quad & -2x_1 + x_2 + 3x_3 - 5x_4 - 3x_6 \\
 \text{s.t.} \quad & -3x_2 - 2x_3 + x_4 = 1 \\
 & 3x_1 + 4x_3 + x_5 = 2 \\
 & -x_1 + x_2 + x_3 + x_6 = 8 \\
 & x_j \geq 0, \quad j = 1, \dots, 6.
 \end{aligned}$$

6. The resulting optima basis: $B_0 = \{1, 2, 4\}$, associated with $\bar{x}_{B_0} = (2/3, 26/3, 27)^T$ with objective value $\bar{f}_1 = (-2, 1, -5)(2/3, 26/3, 27)^T = -383/3$.
- $t = 1 < s$.

Subiteration 2:

3. $t = 1 + 1 = 2$.
4. Call Algorithm 3.3.1 (Part I) to solve the second subprogram ($B_0 = \{1, 2, 4\}$, $N_2 = \{7, 8, 9\}$)

$$\begin{aligned}
 \min \quad & -2x_1 + x_2 - 5x_4 - 7x_7 - 4x_8 - 2x_9 \\
 \text{s.t.} \quad & -3x_2 + x_4 + 4x_7 - 2x_9 = 1 \\
 & 3x_1 + 2x_7 + x_8 + 5x_9 = 2 \\
 & -x_1 + x_2 + 8x_7 - 6x_8 + 3x_9 = 8 \\
 & x_j \geq 0, \quad j = 1, 2, 4, 7, 8, 9.
 \end{aligned}$$

6. The resulting optima basis: $B_1 = \{2, 4, 8\}$, associated with $\bar{x}_{B_1} = (20, 61, 2)^T$ with $\bar{f}_2 = (1, -5, -4)(20, 61, 2)^T = -293$.

$$t = s$$

7. $\bar{f}_2 \neq \bar{f}_1$.

8. $B_0 = \{2, 4, 8\}$, $N = \{1, 3, 5, 6, 7, 9\}$.

Outer iteration 2: Set $B_0 = B_1 = \{2, 4, 8\}$.

Subiteration 1:

1. $N_1 = \{1, 3, 5\}$, $N_2 = \{6, 7, 9\}$.

2. $t = 0$.

3. $t = 0 + 1 = 1$.

4. Call the simplex Algorithm 3.3.1 (Part I) to solve the first subprogram

$$\begin{aligned} \min \quad & -2x_1 + x_2 + 3x_3 - 5x_4 - 4x_8 \\ \text{s.t.} \quad & -3x_2 - 2x_3 + x_4 = 1 \\ & 3x_1 + 4x_3 + x_5 + x_8 = 2 \\ & -x_1 + x_2 + x_3 - 6x_8 = 8 \\ & x_j \geq 0, \quad j = 1, 2, 3, 4, 5, 8. \end{aligned}$$

6. The resulting suboptimal basis is $B_0 = \{2, 4, 8\}$, associated with $\bar{x}_{B_0} = (20, 61, 2)^T$ with $\bar{f}_1 = -293$.

$$t < s.$$

Subiteration 2:

3. $t = 1 + 1 = 2$.

4. Call Algorithm 3.3.1 (Part I) to solve the second subprogram ($B_0 = \{2, 4, 8\}$, $N_2 = \{6, 7, 9\}$)

$$\begin{aligned} \min \quad & x_2 - 5x_4 - 3x_6 - 7x_7 - 4x_8 - 2x_9 \\ \text{s.t.} \quad & -3x_2 + x_4 + 4x_7 - 2x_9 = 1 \\ & 2x_7 + x_8 + 5x_9 = 2 \\ & x_2 + x_6 + 8x_7 - 6x_8 + 3x_9 = 8 \\ & x_j \geq 0, \quad j = 2, 4, 6, 7, 8, 9. \end{aligned}$$

6. The resulting suboptimal basis is $B_1 = \{2, 4, 8\}$, associated with $\bar{x}_{B_1} = (20, 61, 2)^T$ with $\bar{f}_2 = (1, -5, -4)(20, 61, 2)^T = -293$.

$$t = s$$

7. $\bar{f}_2 = \bar{f}_1$. The optimal solution and optimal value are

$$\bar{x} = (0, 20, 0, 61, 0, 0, 0, 2, 0)^T, \quad \bar{f} = -293,$$

which can be verified by solving the original problem as a whole by the simplex algorithm.

27.4 Illustration on Bounded-Variable LP Problem

Algorithm 27.1.1 can be modified to solve the bounded-variable problem if the simplex algorithm called from step 4 is replaced by a bounded-variable problem solver.

Example 27.4.1 Solve the following bounded-variable problem by a modified algorithm, Algorithm 27.1.1, with $s = 2$, using Algorithm 8.1.1 (Part I) as the subalgorithm:

$$\begin{aligned}
 \min f &= 2x_1 - x_2 + 3x_3 - 6x_4 \\
 \text{s.t. } &-2x_1 + 3x_2 - 4x_3 + 2x_4 + x_5 &&= 14 \\
 &-3x_1 + 4x_2 - 5x_3 + 6x_4 &&+ x_6 &&= 16 \\
 &x_1 - 2x_2 + 2x_3 - 7x_4 &&+ x_7 &&= -15 \\
 &-15 \leq x_1 \leq 30, \quad -12 \leq x_2 \leq 20, \quad -17 \leq x_3 \leq 10, \\
 &-8 \leq x_4 \leq 15, \quad -200 \leq x_5 \leq 26, \quad -200 \leq x_6 \leq 34, \\
 &0 \leq x_7 \leq 200.
 \end{aligned}$$

Answer Initial feasible basis and nonbasis: $B_0 = \{5, 6, 7\}$, $N = \{1, 2, 3, 4\}$.

Outer iteration 1:

Subiteration 1:

1. $N_1 = \{1, 2\}$, $N_2 = \{3, 4\}$.
2. $t = 0$.
3. $t = 0 + 1 = 1$.
4. Construct the first subalgorithm with $B_0 = \{5, 6, 7\}$, $N_1 = \{1, 2\}$ by setting $\bar{x}_{N_2} = (-17, 15)^T$, where the right-hand side is changed to

$$b - N_2 \bar{x}_{N_2} = \begin{pmatrix} 14 \\ 16 \\ -15 \end{pmatrix} - \begin{pmatrix} -42 \\ -56 \\ 2 - 7 \end{pmatrix} = \begin{pmatrix} -17 \\ 15 \end{pmatrix} = \begin{pmatrix} -84 \\ -159 \\ 124 \end{pmatrix}, \text{ that is,}$$

$$\begin{aligned}
 \min & 2x_1 - x_2 \\
 \text{s.t. } &-2x_1 + 3x_2 + x_5 &&= -84 \\
 &-3x_1 + 4x_2 &&+ x_6 &&= -159 \\
 &x_1 - 2x_2 &&+ x_7 &&= 124 \\
 &-15 \leq x_1 \leq 30, \quad -12 \leq x_2 \leq 20, \quad -200 \leq x_5 \leq 26, \\
 &-200 \leq x_6 \leq 34, \quad 0 \leq x_7 \leq 200.
 \end{aligned}$$

Call Algorithm 3.1.1 (Part I) to solve the preceding.

6. The resulting suboptimal basis is $B_0 = \{2, 5, 7\}$, associated with the basic solution and objective value

$$\bar{x} = (-15, -1, -17, 15 - 111, -200, 137)^T, \quad \bar{f}_1 = (2, -1, 3, -6, 0, 0, 0)\bar{x} = -170.$$

$$t = 1 < s.$$

Subiteration 2:

$$3. \ t = 1 + 1 = 2.$$

4. Construct the second subalgorithm with $B_0 = \{2, 5, 7\}$, $N_2 = \{3, 4\}$ by setting $(\bar{x}_1, \bar{x}_6) = (-15, -200)$, where the right-hand side is changed to

$$b - N_2 \bar{x}_{N_2} = \begin{pmatrix} 14 \\ 16 \\ -15 \end{pmatrix} - \begin{pmatrix} -20 \\ -31 \\ 10 \end{pmatrix} = \begin{pmatrix} -15 \\ -200 \end{pmatrix} = \begin{pmatrix} -16 \\ 171 \\ 0 \end{pmatrix},$$

yielding

$$\begin{array}{ll} \min & -x_2 + 3x_3 - 6x_4 \\ \text{s.t.} & 3x_2 - 4x_3 + 2x_4 + x_5 = -16 \\ & 4x_2 - 5x_3 + 6x_4 = 171 \\ & -2x_2 + 2x_3 - 7x_4 + x_7 = 0 \\ & -12 \leq x_2 \leq 20, \quad -17 \leq x_3 \leq 10, \\ & -8 \leq x_4 \leq 15, \quad -200 \leq x_5 \leq 26, \quad 0 \leq x_7 \leq 200. \end{array}$$

Call Algorithm 3.1.1 (Part I) to solve the preceding.

6. The resulting suboptimal basis $B_1 = \{2, 5, 7\}$ is equal to B_0 .

$$t = s.$$

7. The optimal solution and optimal value are therefore

$$\bar{x} = (-15, -1, -17, 15-111, -200, 137)^T, \quad \bar{f}_2 = (2, -1, 3, -6, 0, 0, 0)\bar{x} = -170,$$

which can be verified by solving the original problem as a whole by Algorithm 8.1.1 (Part I).

27.5 Illustration on ILP Problem

It is possible to handle large-scale ILP problems by some variant of Algorithm 27.1.1 if there is an optimal solution to the initial ILP subprogram. In general, however, this approach may not produce an optimal solution to the ILP problem, but an approximate one only. Starting from such a point, however, methods presented in Chap. 11, in particular, the realization presented in the next section, are applicable for reaching an exact optimal solution. Besides, it is not difficult to extend the approach to solving the large-scale mixed-integer LP problems.

To start-, one would solve the associated relaxed LP problem first and then construct an initial subprogram, based on the resulting optimal basis, if any. But we will do differently with the following instance.

Example 27.5.1 Solve the following integer LP problem by a modified algorithm, Algorithm 27.1.1, with $s = 2$, supported by some solver:

$$\begin{aligned}
 \min \quad & f = x_1 - 3x_2 + 2x_3 - 5x_4 + x_5 \\
 \text{s.t.} \quad & 2x_1 + x_2 - x_3 + 3x_4 - 2x_5 + x_6 = 7 \\
 & 4x_1 - 3x_2 - 5x_4 - 4x_5 + x_7 = 2 \\
 & -3x_1 + 2x_2 + x_3 + 2x_4 + x_5 + x_8 = 8 \\
 & x_j \geq 0, \quad j = 1, \dots, 8 \text{ integers.}
 \end{aligned}$$

Answer Initial feasible basis and nonbasis: $B_0 = \{6, 7, 8\}$, $N = \{1, 2, 3, 4, 5\}$.

Outer iteration 1:

Subiteration 1:

1. $N_1 = \{1, 2\}$, $N_2 = \{3, 4, 5\}$.
2. $t = 0$.
3. $t = 0 + 1 = 1$.
4. Solve the first subalgorithm with $B_0 = \{6, 7, 8\}$, $N_1 = \{1, 2\}$, formed by setting $\bar{x}_{N_2} = (0, 0, 0)$:

$$\begin{aligned}
 \min \quad & x_1 - 3x_2 \\
 \text{s.t.} \quad & 2x_1 + x_2 + x_6 = 7 \\
 & 4x_1 - 3x_2 + x_7 = 2 \\
 & -3x_1 + 2x_2 + x_8 = 8 \\
 & x_j \geq 0, \quad j = 1, 2, 6, 7, 8 \text{ integers.}
 \end{aligned}$$

6. The resulting suboptimal basis is $B_0 = \{1, 2, 7\}$, associated with the solution and objective value

$$\bar{x}^{(1)} = (1, 5, 0, 0, 0, 0, 13, 1)^T, \quad \bar{f}_1 = (1, -3, 0, 0, 0, 0, 0, 0)\bar{x}^{(1)} = -14.$$

$$t = 1 < s.$$

Subiteration 2:

3. $t = 1 + 1 = 2$.
4. Construct the second subalgorithm with $B_0 = \{1, 2, 7\}$, $N_2 = \{3, 4, 5\}$ by setting $\bar{x}_{N_1} = (0, 1)^T$, where the right-hand side is changed to

$$b - N_2 \bar{x}_{N_2} = \begin{pmatrix} 7 \\ 2 \\ 8 \end{pmatrix} - \begin{pmatrix} 10 \\ 00 \\ 01 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \\ 7 \end{pmatrix},$$

yielding

$$\begin{aligned}
 \min \quad & x_1 - 3x_2 + 2x_3 - 5x_4 + x_5 \\
 \text{s.t.} \quad & 2x_1 + x_2 - x_3 + 3x_4 - 2x_5 = 7 \\
 & 4x_1 - 3x_2 - 5x_4 - 4x_5 + x_7 = 2 \\
 & -3x_1 + 2x_2 + x_3 + 2x_4 + x_5 = 7 \\
 & x_j \geq 0, \quad j = 1, 2, 3, 4, 5, 7 \text{ integers.}
 \end{aligned}$$

Solve the preceding by a code.

6. The resulting suboptimal basis is $B_1 = \{4, 5, 7\}$.

7. The optimal solution and optimal value of the subprogram are

$$\bar{x}^{(2)} = (0, 0, 0, 3, 1, 0, 21, 1)^T, \quad \bar{f}_2 = (1, -3, 2, -5, 1, 0, 0, 0)\bar{x}^{(2)} = -14.$$

7. $\bar{f}_2 = \bar{f}_1 = -14$. $\bar{x}^{(1)}$ or $\bar{x}^{(2)}$ would be an acceptable solution to the original problem.

The exact optimal solution to the original problem is

$$x^* = (3, 0, 0, 5, 7, 0, 43, 0)^T \quad f^* = -15.$$

27.6 Practical Remarks

Many large-scale LP problems involve a large number of variables, compared with the number of constraint equations ($m \ll n$). In this case, it is reasonable to form the subprograms involving columns fewer than m or less. Moreover, in the case when $n - m \ll m$, the problem can be converted by the *dual elimination* (Part I) to an $(n - m) \times n$ standard LP problem, so that a large-scale problem can be handled by solving small programs with the decomposition algorithm.

The partition $(N_1, \dots, N_s)$ involved in Algorithm 27.1.1 (or Algorithm 27.1.2) is not unique. Of course, we wish that an optimal basis is contained in the first few sections, e.g., in the N_1 section ideally. How to achieve this is worth further investigating. It seems promising to give the nonbasic indices an order by changing the pivoting-index (Part I) by taking sparsity into account, so that sparser columns are in front of the line to have priority to enter N_1 .

On the other hand, both Algorithms 27.1.1 and 27.1.2 require an initial feasible basis to get started, and hence a two-phase procedure is needed to solve general LP problems. Although any existing Phase-I approach applies for this purpose, it is preferable to deal with some Phase-I auxiliary problem by following the decomposition principle.

It should be pointed out that the feasibility of the initial basis is not necessary. It suffices to ensure that the initial subprogram is itself feasible.

The potential use of the decomposition principle would be of great importance. In fact, Algorithms 27.1.1 and 27.1.2 could be generalized along the following two lines:

- (A) Instead of the simplex algorithm, other basis-based algorithms, e.g., the dual simplex algorithm or the reduced simplex algorithm can be used as the subalgorithm, perhaps with slight modifications. Even the face algorithm, producing an optimal face, applies. But primal algorithms seem to be amenable, compared with dual algorithms.
- (B) Some hard optimization problems could be handled if they have linear constraints and separable variables; e.g., large integer or mixed-integer LP problems would be approximately solved by the algorithms, supported by the branch-bound or cutting-plane algorithm amenable to small such problems. Further investigation is expected in this aspect.

Algorithms 27.1.1 and 27.1.2 are designed for solving standard LP problems, where all nonbasic components of solutions are of default value 0, and hence may be omitted in the solution process. For other problems, such as the bounded-variable problem, a feasible basic solution should be available initially. In each iteration, the components, associated with the current subprogram, of the solution are updated, while the other components remain unchanged. For integer or mixed LP problems, an initial feasible integer solution is certainly preferable, but not necessary. A feasible solution to the associated relaxed LP problem can be taken as an initial solution to get some variant of Algorithm 27.1.1 or 27.1.2 started, though there is a need for the existence of an optimal solution to the initial integer subprogram.

Appendix A

On the Birth of LP and More

It would be interesting, inspiring, and beneficial to review the birth of LP.

LP can be traced back to as early as 1820s. French mathematician J.B.J. Fourier (well-known due to the mathematical series named after him) in 1823 and Belgian mathematician V. Poussin in 1911 each wrote a paper related to LP. Unfortunately, these isolated efforts went unnoticed. In 1939, the academician L.V. Kantorovich (1960) of former Soviet Union academy of sciences published a book entitled *Mathematical Method of Production Management and Planning*, concerning LP models and their solution, but it was unfortunately neglected by the authority and had not been known by the academic communities. In 1941, F.L. Hitchcock's original paper on the transportation problem once again did not draw any attention, until the late 1940s and early 1950s when it was finally rediscovered after the simplex method came around.

Indeed, human activities are the only source of all scientific theories and methodologies. Although the World War II led to devastating losses and casualties, the war efforts thrust science and technology forward and hastened emergence of many new branches of learning. In particular, Operations Research and Optimization are one area that has greatly benefited from the wartime demand.

After receiving his Ph.D., George B. Dantzig became a mathematical advisor to the US (Cottle et al. 2007) Air Force Controller in 1946. He was challenged to mechanize the planning process such that a time-staged deployment, training, and logistical supply program can be more rapidly computed using desk calculators. Using an active analysis approach, he formulated a model without an objective function and tried to solve it. Dantzig recalled the situation at that time at the XI-th International Symposium on Mathematical Programming (held at the University of Bonn, W. Germany, from August 23 to 27 in 1982), by giving the following example:

How to assign 70 men to 70 jobs ?

It is required that each job must be filled and each man must be assigned. Therefore, one needs to find a good “activity,” an 1-to-1 correspondence between the men and the jobs. There are $70! > 10^{100}$ activities. It is impracticable to determine the best among them by comparing all, because the number is too large even though being finite. Dantzig said (1982, Bonn):

Suppose we had an IBM 370-168 available at the time of the Big Bang 15 billion years ago. Would it have been able to look at all the $70!$ combinations by the year 1981? No! Suppose instead it could examine 1 billion assignments per second? The answer is still no. Even if the Earth were filled with such computers all working in parallel, the answer would still be no. If, however, there were 10^{44} suns all filled with nano-second speed computers all programmed in parallel from the time of the big bang until sun grows cold, then perhaps the answer is yes.

This example illustrates the ordeals faced by decision makers before 1947. Therefore, “experience,” “mature judgement,” and “rules-of-thumb” had to be relied on to come up with an “acceptable” scheme, which was often far from the optimal.

In the summer of 1947, Dantzig founded the simplex method, marking the birth of LP. The method’s prefix *simplex* came from a chat between G.B. Dantzig and T. Motzkin (Dantzig 1991) in the early days of LP. The latter indicated that the m columns of the basis matrix and the entering column just form a *simplex* in the m -dimensional space. Therefore, each iteration in Dantzig’s method can be viewed as a movement from a simplex to another. Dantzig accepted his suggestion by taking *simplex* as the prefix.

On October 3 of the year, Dantzig visited one of the greatest scientists of the twentieth century, J. von Neumann, and talked about his results. The latter immediately got the basic idea behind the method and indicated potential relationship between the method and game theory which he was working on.

In 1948, Dantzig attended a conference of the Econometric Society in Wisconsin held in Wisconsin, meeting well-known statisticians, mathematicians, and economists, such as H. Hotelling, T. Koopmans, J. von Neumann, and many others who became famous later but just had begun their career then. After young unknown Dantzig presented his simplex method and LP, the Chairman called for discussion. There was silence for a moment. Then, Hotelling said, “But we all know the world is nonlinear.” When Dantzig was frantically trying to properly reply, Von Neumann came to his rescue, “The speaker titled his talk ‘Linear Programming’ and he carefully stated his axioms. If you have an application that satisfies the axioms, use it. If it does not, then don’t.” Of course, what Hotelling said is right, the world is highly nonlinear; but, fortunately, the vast majority of nonlinear relations encountered in practice can be approximated by linear ones. As we all know, the subsequent development completely exceeded people’s imagination, even Dante’s own imagination.

The implementation of the simplex method began at National Bureau of Standards in the USA. Around 1952, a team led by A. Hoffman of National Bureau of Standards tested the simplex method and compared with T. Motzkin’s popular method at that time. Unambiguously, the former defeated the latter. In 1953–1954, W. Orchard-Hay (1954a) started his pioneer work. He worked out the first

simplex-method-based commercial software and performed on a primitive computer by today's standard. Subsequently, his implementation techniques were used and enhanced by many scholars. As a result, the simplex method became a powerful practical tool. If the simplex method is used, the assignment problem mentioned above can be solved in 1 s on IBM370-168, which is not even the fastest computer at that time when Dantzig told his story.

A number of Nobel Economic Prizes awarded in the past are related to LP. Russian-American economist Wassily Leontief took the Nobel Economic Prize in 1973 for developing the input–output model, representing the interdependencies between different branches of a national economy or different regional economies. Kantorovich and Koopmans shared Nobel Economic prize in 1975 for their contribution of resources allocation theory based on LP. The award was also given to Professors K. Arrow, P. Samuelson, H. Simon, and L. Herwicz et al., several decades later when they paid close attention to LP during the early days of their professional career. On the practice side, applications of LP and the simplex method have brought about enormous economic and social benefits in the past. The simplex method was selected as one of the “Top Ten Algorithms of the Century.” Highlighted in the January/February 2000 issue of *Computing in Science & Engineering*, a joint publication of the American Institute of Physics and the IEEE Computer Society, the ten algorithms were regarded as ones with the greatest influence on the development and practice of science and engineering in the twentieth century. Again and again, history confirms that it is practice that spawns theories and methods that are deep-rooted and exuberant.

Nevertheless, the simplex method has exponential time computational complexity (Klee and Minty 1972), while a “good” method is thought to be polynomial time. In fact, even finiteness of the simplex method is not guaranteed; in other words, the process may not terminate forever (Beale 1955; Hoffman 1953). On the other hand, a breakthrough was made by a former Soviet Union mathematician L.G. Khachiyan (1979) who proposed the first polynomial-time LP solver, the ellipsoid algorithm. Unfortunately, it performed poorly and is not competitive with the simplex method. The so-called polynomial-time complexity is only the worst-case complexity, and the case hardly happens in practice.

Soon, Indian mathematician N. Karmarkar (1984) published an interior-point algorithm for solving a special LP model. It is not only of lower order of polynomial-time complexity than the ellipsoid algorithm but also appeared promisingly fast in practice. The algorithm initiated an upsurge of interior-point method. Many researchers made contributions to this area. Some implementations of it are so efficient that many persons thought the interior-point method to be superior to the simplex method for solving large-scale sparse LP problems.

Unlike simplex methods, however, interior-point methods cannot be “warmly” started and hence not applicable for solving integer linear programming problems. Since the interior-point method cannot produce an optimal basic solution, moreover, there is a need for an additional purification process if a vertex solution is required.

Meanwhile, the art of the simplex method did not stand still, but rather, was developing and improving. For example, due to his outstanding work on software.

M. Saunders won the first Beale—Orchard-Hays Prize in 1985. After P.M.J. Harris (1973) tested some approximate steepest-edge pivot rules; J.J.H. Forrest and D. Goldfarb (1992) introduced several variants of the steepest-edge pivot rule and associated recurrence formulas and reported very favorable numerical results. Thus, the simplex method and interior-point method were neck-and-neck in competition.

In fact, the worst-case complexity is not appropriate from a practical point of view. The average time complexity might be more relevant. K.-H. Borgwardt (1982a,b) was able to show that on average the simplex method has a polynomial-time complexity for a certain probabilistic model, if using a certain pivot rule. S. Smale (1983a,b) offered a similar result. These coincide with the good performance of the simplex method.

Essentially, the evaluation of a method is a practical issue, and rigidly adhering to theory is not wise. The merit of a method, its efficiency, precision, and reliability and/or stability is ultimately determined by its performance in practice. In this respect, the method's finiteness or complexity is even unintentionally misleading. In fact, the finite/polynomial-time algorithms usually perform far worse than infinite/non-polynomial time ones overall, and the latter rather than the former plays a major role in applications. This is why this book draws its material from a practical point of view, focusing on effective theories, methods, and implementation techniques closely related to LP computations and applications.

Appendix B

MPS File

MPS (Mathematical Programming System) is a kind of data file format, used to represent LP problems and mixed-integer LP problems. It is widely applied in academic as well as commercial optimization codes.

There exist several expansions of MPS format. This appendix presents the MPS format based on that used in MINOS (Murtagh and Saunders 1998). Nevertheless, we are only concerned here with the part of linear programming only, though MINOS is also used to handle nonlinear programming problems.

In the MPS format, data of the problem to be solved have to appear in specific columns. A MPS file is sectioned under several items as follows:

```
NAME
ROWS
.
COLUMNS
.
RHS
.
RANGES (optional)
.
BOUNDS (optional)
.
ENDDATA
```

Each item must start from the first column. Symbols “.” under an item are of the following form:

Column	2–3	5–12	15–22	25–36	40–47	50–61
Content	Key	Name0	Name1	Value1	Name2	Value2

where note rows can be inserted, with symbol “*” at the first column, and any characters at the 2–22th columns.

1. NAME of the problem

Example B.0.1 NAME row

1.....4	15.....22
NAME	AFIRO

The name row is usually the first row of a MPS file.
Characters **NAME** are in the 1–4th columns. The name of the problem is filled in the 15–22th columns, which can be 1 to 8 any characters or blank spaces. The name may also be used to indicate outputs, at the first row of each output basis.

2. ROWS Section

Example B.0.2 ROWS

2	5.....12
ROWS	
E	ROW4
G	ROW1
L	ROW7
N	COST

This section lists given names of all (linear) constraints (e.g., ROW4,ROW1, ROW7, etc.), one row for each constraint, where a single character under **Key** gives the types of constraints, placed in columns 2–3. **Name0** gives a name of 8 characters, placed in columns 5–12. The types are

Key	Row type
E	=
G	≥
L	≤
N	Objective
N	Free

Types of row E,G, and L, respectively, indicate the constraints are 别标明相应约束条件为 “equal to,” “greater than or equal to,” and “less than or equal to,” and N indicates “free” or without restriction.

3. COLUMNS Section

Example B.0.3 COLUMNS section

1	5.....12	15....22	25.....36	40....47	50.....61
COLUMNS					
X1	ROW2	2.3	ROW3	−2.4	
X1	ROW1	−6.7	ROW6	5.22222	
X1	ROW7	15.88888			
X2	ROW2	1.0	ROW4	−4.1	
X2	ROW5	2.6666666			

For each variable x_j , COLUMNS section gives names and lists nonzeros a_{ij} of the corresponding columns of the coefficient matrix. Nonzero components of the first column must be listed before those of the second column, and so on. In case when there are multiple nonzeros in a column, these components are allowed to appear in any order.

Usually, **Key** is blank (except for comment line), **Name0** is the name of a column and **Name1**, **Value1** give the name of a row and the nonzero value of some component of the column. The name of another row of the same column and value can be listed in **Name2**, **Value2** of the same line, or put in the next row. **Name1** or **Name2** can be blank.

In the preceding example, the variable named by **X1** has 5 nonzero components located, respectively, at

rows **ROW2, ROW3, ROW1, ROW6, ROW7**.

4. RHS Section

Example B.0.4 RHS section

1	5.....12	15....22	25.....36	40....47	50.....61
RHS					
RHS1	ROW1	−3.5	ROW3	1.1111	
RHS1	ROW4	11.33333	ROW7	−2.4	
RHS1	ROW2	2.6			
RHS2	ROW4	17.3	ROW2	−5.6	
RHS2	ROW1	1.9			

This section lists nonzeros of the right-hand side of the constraint system, with the same format as the COLUMN section. If components of the right-hand side are all zero, then only line **RHS** appears in the section. This section may include multiple right-hand sides. But only the first will be used unless indicated otherwise in the SPECS file.

5. RANGES Section (optional)

Example B.0.5 RANGES section

```
1  5.....12  15....22  25.....36  40....47  50.....61
ROWS
E  ROW2
G  ROW4
L  ROW5
E  ROW1
.
COLUMNS
.
RHS
    RHS1      ROW5    2.0          ROW3    2.0
.
RANGES
    RANGE1  ROW1    3.0          ROW3    4.5
    RANGE2  ROW2    1.4          ROW5    6.7
```

RANGES section expresses constraints of form

$$l \leq a^T x \leq u,$$

where l and u both are finite values. The range of the constraint is $r = u - l$. This section only gives r , not l or u . The corresponding l or u will be determined by the row type of the constraint and the sign of r :

Type of row	Sign of r	Lower bound l	upper bound u
E	+	b	$b + r $
E	−	$b - r $	b
G	+ or −	b	$b + r $
L	+ or −	$b - r $	b

The format is the same as the COLUMN section, where **Name0** gives the name of a range.

6. BOUNDS Section (optional)

Example B.0.6 BOUNDS section

1	5.....12	15...22	25.....36
BOUNDS			
UP	BOUND1	X1	10.5
UP	BOUND1	X2	6.0
LO	BOUND1	X3	-5.0
UP	BOUND1	X3	4.5

Default bounds of all variables x_j (except for slack variables) are $0 \leq x_j \leq \infty$. If necessary, the default values 0 and ∞ can be modified to $l \leq x_j \leq u$ via **LOWER** and **UPPER** in the SPECS file.

In this section, **Key** gives the type of bounds (UP and LO, respectively, indicate upper and lower bounds). **Name0** gives the name of the bound set, and **Name1** and **Value1** list the name of column and the value of bound, respectively.

Example B.0.7 For the model of human arrangement problem

$$\min \quad f = 0.1x_2 + 0.2x_3 + 0.3x_4 + 0.8x_5 + 0.9x_6$$
$$\text{s.t.} \quad \begin{array}{rcl} x_1 & & + x_6 \geq 7 \\ x_1 + x_2 & & \geq 15 \\ x_2 + x_3 & & \geq 25 \\ x_3 + x_4 & & \geq 20 \\ x_4 + x_5 & & \geq 30 \\ x_5 + x_6 & & \geq 7 \\ x_j \geq 0, \quad j = 1, \dots, 6, \end{array}$$

The MPS file is as follows:

```
1 5....12 15....22 25....36 40....47 50....61
NAME ASSIGN.
ROWS
G TEAM1
G TEAM2
G TEAM3
G TEAM4
G TEAM5
G TEAM6
N HANDS
COLUMNS
X1 TEAM1 1.0 TEAM2 1.0
X2 TEAM2 1.0 TEAM3 1.0
X2 HANDS 0.1
X3 TEAM3 1.0 TEAM4 1.0
X3 HANDS 0.2
X4 TEAM4 1.0 TEAM5 1.0
X4 HANDS 0.3
X5 TEAM5 1.0 TEAM6 1.0
X5 HANDS 0.8
X6 TEAM1 1.0 TEAM6 1.0
X6 HANDS 0.9
RHS
RHS1 TEAM1 7.0 TEAM2 15.0
RHS1 TEAM3 25.0 TEAM4 20.0
RHS1 TEAM5 30.0 TEAM6 7.0
ENDDATA
```

Appendix C

Test LP Problems

In this appendix, test problems (see Gay 1985) are listed according to increasing $m + n$ order (Tables C.1, C.2, C.3, C.4).

Source:

NETLIB (<http://www.netlib.org/lp/data>)

Kennington (<http://www-fp.mcs.anl.gov/otc/Guide/TestProblems/LPtest/>)

BPMPD (<http://www.sztaki.hu/meszaros/bpmpd/>).

Notations in the tables:

m : the number of rows of the coefficient matrix.

n : the number of columns of the coefficient matrix.

Nonzeros: the number of nonzeros of the coefficient matrix.

BR: indicates whether there are BOUNDS and RANGES sections.

Optimal value: optimal values given by the file.

Table C.1 Netlib problems

No.	Name	$m + n$	m	n	Nonzeros	BR	Optimal value
1	AFIRO	60	28	32	88		$-4.6475314286\text{E}+02$
2	KB2	85	44	41	291	B	$-1.7499001299\text{E}+03$
3	SC50B	99	51	48	119		$-7.0000000000\text{E}+01$
4	SC50A	99	51	48	131		$-6.4575077059\text{E}+01$
5	ADLITTLE	154	57	97	465		$2.2549496316\text{E}+05$
6	BLEND	158	75	83	521		$-3.0812149846\text{E}+01$
7	SHARE2B	176	97	79	730		$-4.1573224074\text{E}+02$
8	SC105	209	106	103	281		$-5.2202061212\text{E}+01$
9	STOCFOR1	229	118	111	474		$-4.1131976219\text{E}+04$
10	SCAGR7	270	130	140	553		$-2.3313897524\text{E}+06$
11	RECIPE	272	92	180	752	B	$-2.6661600000\text{E}+02$
12	BOEING2	310	167	143	1339	BR	$-3.1501872802\text{E}+02$
13	ISRAEL	317	175	142	2358		$-8.9664482186\text{E}+05$
14	SHARE1B	343	118	225	1182		$-7.6589318579\text{E}+04$
15	VTP.BASE	402	199	203	914	B	$1.2983146246\text{E}+05$
16	SC205	409	206	203	552		$-5.2202061212\text{E}+01$
17	BEACONFD	436	174	262	3476		$3.3592485807\text{E}+04$
18	GROW7	442	141	301	2633	B	$-4.7787811815\text{E}+07$
19	LOTFI	462	154	308	1086		$-2.5264706062\text{E}+01$
20	BRANDY	470	221	249	2150		$1.5185098965\text{E}+03$
21	E226	506	224	282	2767		$-1.1638929066\text{E}+01$
22	BORE3D	549	234	315	1525	B	$1.3730803942\text{E}+03$
23	FORPLAN	583	162	421	4916	BR	$-6.6421896127\text{E}+02$
24	CAPRI	625	272	353	1786	B	$2.6900129138\text{E}+03$
25	AGG	652	489	163	2541		$-3.5991767287\text{E}+07$
26	BOEING1	736	352	384	3865	BR	$-3.3521356751\text{E}+02$
27	SCORPION	747	389	358	1708		$1.8781248227\text{E}+03$
28	BANDM	778	306	472	2659		$-1.5862801845\text{E}+02$
29	SCTAP1	781	301	480	2052		$1.4122500000\text{E}+03$
30	SCFXM1	788	331	457	2612		$1.8416759028\text{E}+04$
31	AGG3	819	517	302	4531		$1.0312115935\text{E}+07$
32	AGG2	819	517	302	4515		$-2.0239252356\text{E}+07$

(continued)

Table C.1 (continued)

No.	Name	$m + n$	m	n	Nonzeros	BR	Optimal value
33	STAIR	824	357	467	3857	B	-2.5126695119E+02
34	SCSD1	838	78	760	3148		8.6666666743E+00
35	TUFF	921	334	587	4523	B	2.9214775747E-01
36	GROW15	946	301	645	5665	B	-1.0687094129E+08
37	SCAGR25	972	472	500	2029		-1.4753433061E+07
38	DEGEN2	979	445	534	4449		-1.4351780000E+03
39	FIT1D	1051	25	1026	14430	B	-9.1463780924E+03
40	ETAMACRO	1089	401	688	2489	B	-7.5571521687E+02
41	FINNIS	1112	498	614	2714	B	1.7279096547E+05
42	FFFFF800	1379	525	854	6235		5.5567967533E+05
43	GROW22	1387	441	946	8318	B	-1.6083433648E+08
44	PILOT4	1411	411	1000	5145	B	-2.5809984373E+03
45	STANDATA	1435	360	1075	3038	B	1.2576995000E+03
46	SCSD6	1498	148	1350	5666		5.0500000078E+01
47	STANDMPS	1543	468	1075	3686	B	1.4060175000E+03
48	SEBA	1544	516	1028	4874	BR	1.5711600000E+04
49	STANDGUB	1546	362	1184	3147	B	1.2576995000E+03
50	SCFXM2	1575	661	914	5229		3.6660261565E+04
51	SCRS8	1660	491	1169	4029		9.0430601463E+02
52	GFRD-PNC	1709	617	1092	3467	B	6.9022359995E+06
53	BNL1	1819	644	1175	6129		1.9776292440E+03
54	SHIP04S	1861	403	1458	5810		1.7987147004E+06
55	PEROLD	2002	626	1376	6026	B	-9.3807477973E+03
56	MAROS	2290	847	1443	10006	B	-5.8063743701E+04
57	FIT1P	2305	628	1677	10894	B	9.1463780924E+03
58	MODSZK1	2308	688	1620	4158	B	3.2061972906E+02
59	SHELL	2312	537	1775	4900	B	1.2088253460E+09
60	SCFXM3	2362	991	1371	7846		5.4901254550E+04
61	25FV47	2393	822	1571	11127		5.5018458883E+03
62	SHIP04L	2521	403	2118	8450		1.7933245380E+06
63	QAP8	2545	913	1632	8304		2.0350000002E+02
64	WOOD1P	2839	245	2594	70216		1.4429024116E+00

(continued)

Table C.1 (continued)

No.	Name	$m + n$	m	n	Nonzeros	BR	Optimal value
65	PILOT.JA	2929	941	1988	43220	B	−6.1130535369E+03
66	SCTAP2	2971	1091	1880	8124		1.7248071429E+03
67	GANGES	2991	1310	1681	7021	B	−1.0958577038E+05
68	SCSD8	3148	398	2750	11334		9.0499999993E+02
69	PILOTNOV	3148	976	2172	13129	B	−4.4972761882E+03
70	SHIP08S	3166	779	2387	9501		1.9200982105E+06
71	SIERRA	3264	1228	2036	9252	B	1.5394392927E+07
72	DEGEN3	3322	1504	1818	26230		−9.8729400000E+02
73	PILOT.WE	3512	723	2789	9218	B	−2.7201037711E+06
74	NESM	3586	663	2923	13988	BR	1.4076065462E+07
75	SHIP12S	3915	1152	2763	10941		1.4892361344E+06
76	SCTAP3	3961	1481	2480	10734		1.4240000000E+03
77	STOCFOR2	4189	2158	2031	9492		−3.9024408538E+04
78	CZPROB	4453	930	3523	14173	B	2.1851966989E+06
79	CYCLE	4761	1904	2857	21322	B	−5.2263930249E+00
80	SHIP08L	5062	779	4283	17085		1.9090552114E+06
81	PILOT	5094	1442	3652	14706	B	−5.5728790853E+02
82	BNL2	5814	2325	3489	16124		1.8112404450E+03
83	SHIP12L	6579	1152	5427	21597		1.4701879193E+06
84	D6CUBE2	6600	416	6184	43888		3.1473177974E+02
85	D6CUBE	6600	416	6184	43888	B	3.1549166667E+02
86	PILOT87	6914	2031	4883	73804	B	3.0171261980E+02
87	D2Q06C	7339	2172	5167	35674		1.2278421653E+05
88	GREENBEA	7798	2393	5405	31499	B	−7.2461393630E+07
89	WOODW	9504	1099	8405	37478		1.3044763331E+00
90	FIT2D	10526	26	10500	138018	B	−6.8464293007E+04
91	QAP12	12049	3193	8856	44244		5.2289435056E+02
92	80BAU3B	12062	2263	9799	29063	B	9.8724313086E+05
93	MAROS-R7	12545	3137	9408	151120		1.4971851665E+06
94	FIT2P	16526	3001	13525	60784	B	6.8464293232E+04
95	DFL001	18302	6072	12230	41873	B	1.1266503030E+07
96	STOCFOR3	32371	16676	15695	74004		−3.9976256537E+04

Table C.2 Kennington problems

No.	Name	$m + n$	m	n	Nonzeros	BR	Optimal value
1	KEN-07	6029	2427	3602	11981	B	−6.7952044338E+08
2	CRE-C	6747	3069	3678	16922		2.5275116141E+07
3	CRE-A	7584	3517	4067	19054		2.3595410589E+07
4	PDS-02	10489	2954	7535	21252	B	2.8857862010E+10
5	OSA-07	25068	1119	23949	167643		5.3572251730E+05
6	KEN-11	36044	14695	21349	70354	B	−6.9723822625E+09
7	PDS-06	38537	9882	28655	82269	B	2.7761037600E+10
8	OSA-14	54798	2338	52460	367220		1.1064628448E+06
9	PDS-10	65322	16559	48763	140063	B	2.6727094976E+10
10	KEN-13	71292	28633	42659	139834	B	−1.0257394789E+10
11	CRE-D	78907	8927	69980	312626		2.4454969898E+07
12	CRE-B	82096	9649	72447	328542		2.3129640065E+07
13	OSA-30	104375	4351	100024	700160		2.1421398737E+06
14	PDS-20	139603	33875	105728	304153	B	2.3821658640E+10
15	OSA-60	243247	10281	232966	1630758		4.0440725060E+06
16	KEN-18	259827	105128	154699	512719	B	−5.2217025287E+10

Table C.3 BPMPD problems

No.	Name	$m + n$	m	n	Nonzeros	BR	Optimal value
1	RAT7A	12545	3137	9408	275180		2.0743714157E+06
2	NSCT1	37883	22902	14981	667499		−3.8922436000E+07
3	NSCT2	37985	23004	14981	686396		−3.7175082000E+07
4	ROUTING	44818	20895	23923	210025	B	5.9416502767E+03
5	DBIR1	46160	18805	27355	1067815		−8.1067070000E+06
6	DBIR2	46262	18907	27355	1148847		−6.1169165000E+06
7	T0331-4L	47580	665	46915	477897	B	2.9730033352E+04
8	NEMSEMM2	49077	6944	42133	212793	B	6.2161463095E+05
9	SOUTHERN	54160	18739	35421	148318	B	1.8189401971E+09
10	RADIO.PRIM	66919	58867	8052	265975	B	1.0000000000E+00
11	WORLD.MOD2	67393	35665	31728	220116	B	4.3648258595E+07
12	WORLD	68245	35511	32734	220748	B	6.9133457165E+07
13	RADIO.DUAL	74971	8053	66918	328891	B	−1.0000000000E+00
14	NEMSEMM1	75359	3946	71413	1120871	B	5.1442135978E+05
15	NW14	123483	74	123409	1028319	B	6.1844000000E+04
16	LPL1	164952	39952	125000	462127	B	6.7197548415E+10
17	DBIC1	226436	43201	183235	1217046	B	−9.7689730000E+06

Table C.4 Netlib problems without bounds and ranges

No.	Name	$m + n$	m	n	Nonzeros	Optimal value
1	AFIRO	60	28	32	88	-4.6475314286E+02
2	SC50B	99	51	48	119	-7.0000000000E+01
3	SC50A	99	51	48	131	-6.4575077059E+01
4	ADLITTLE	154	57	97	465	2.2549496316E+05
5	BLEND	158	75	83	521	-3.0812149846E+01
6	SHARE2B	176	97	79	730	-4.1573224074E+02
7	SC105	209	106	103	281	-5.2202061212E+01
8	STOCFOR1	229	118	111	474	-4.1131976219E+04
9	SCAGR7	270	130	140	553	-2.3313897524E+06
10	ISRAEL	317	175	142	2358	-8.9664482186E+05
11	SHARE1B	343	118	225	1182	-7.6589318579E+04
12	SC205	409	206	203	552	-5.2202061212E+01
13	BEACONFD	436	174	262	3476	3.3592485807E+04
14	LOTFI	462	154	308	1086	-2.5264706062E+01
15	BRANDY	470	221	249	2150	1.5185098965E+03
16	E226	506	224	282	2767	-1.1638929066E+01
17	AGG	652	489	163	2541	-3.5991767287E+07
18	SCORPION	747	389	358	1708	1.8781248227E+03
19	BANDM	778	306	472	2659	-1.5862801845E+02
20	SCTAP1	781	301	480	2052	1.4122500000E+03
21	SCFXM1	788	331	457	2612	1.8416759028E+04
22	AGG3	819	517	302	4531	1.0312115935E+07
23	AGG2	819	517	302	4515	-2.0239252356E+07
24	SCSD1	838	78	760	3148	8.6666666743E+00
25	SCAGR25	972	472	500	2029	-1.4753433061E+07
26	DEGEN2	979	445	534	4449	-1.4351780000E+03
27	FFFFF800	1379	525	854	6235	5.5567967533E+05
28	SCSD6	1498	148	1350	5666	5.0500000078E+01
29	SCFXM2	1575	661	914	5229	3.6660261565E+04
30	SCRS8	1660	491	1169	4029	9.0430601463E+02
31	BNL1	1819	644	1175	6129	1.9776292440E+03
32	SHIP04S	1861	403	1458	5810	1.7987147004E+06

(continued)

Table C.4 (continued)

No.	Name	$m + n$	m	n	Nonzeros	Optimal value
33	SCFXM3	2362	991	1371	7846	5.4901254550E+04
34	25FV47	2393	822	1571	11127	5.5018458883E+03
35	SHIP04L	2521	403	2118	8450	1.7933245380E+06
36	QAP8	2545	913	1632	8304	2.0350000002E+02
37	WOOD1P	2839	245	2594	70216	1.4429024116E+00
38	SCTAP2	2971	1091	1880	8124	1.7248071429E+03
39	SCSD8	3148	398	2750	11334	9.0499999993E+02
40	SHIP08S	3166	779	2387	9501	1.9200982105E+06
41	DEGEN3	3322	1504	1818	26230	−9.8729400000E+02
42	SHIP12S	3915	1152	2763	10941	1.4892361344E+06
43	SCTAP3	3961	1481	2480	10734	1.4240000000E+03
44	STOCFOR2	4189	2158	2031	9492	−3.9024408538E+04
45	SHIP08L	5062	779	4283	17085	1.9090552114E+06
46	BNL2	5814	2325	3489	16124	1.8112404450E+03
47	SHIP12L	6579	1152	5427	21597	1.4701879193E+06
48	D6CUBE2	6600	416	6184	43888	3.1473177974E+02
49	D2Q06C	7339	2172	5167	35674	1.2278421653E+05
50	WOODW	9504	1099	8405	37478	1.3044763331E+00
51	QAP12	12049	3193	8856	44244	5.2289435056E+02
52	MAROS-R7	12545	3137	9408	151120	1.4971851665E+06
53	STOCFOR3	32371	16676	15695	74004	−3.9976256537E+04

Table C.5 Kennington problems without bounds and ranges

No.	Name	$m + n$	m	n	Nonzeros	Optimal value
1	CRE-C	6747	3069	3678	16922	2.5275116141E+07
2	CRE-A	7584	3517	4067	19054	2.3595410589E+07
3	OSA-07	25068	1119	23949	167643	5.3572251730E+05
4	OSA-14	54798	2338	52460	367220	1.1064628448E+06
5	CRE-D	78907	8927	69980	312626	2.4454969898E+07
6	CRE-B	82096	9649	72447	328542	2.3129640065E+07
7	OSA-30	104375	4351	100024	700160	2.1421398737E+06
8	OSA-60	243247	10281	232966	1630758	4.0440725060E+06

Table C.6 BPMPD problems without bounds and ranges

No.	Name	$m + n$	m	n	Nonzeros	Optimal value
1	RAT7A	12545	3137	9408	275180	2.0743714157E+06
2	NSCT1	37883	22902	14981	667499	−3.8922436000E+07
3	NSCT2	37985	23004	14981	686396	−3.7175082000E+07
4	DBIR1	46160	18805	27355	1067815	−8.1067070000E+06
5	DBIR2	46262	18907	27355	1148847	−6.1169165000E+06

Appendix D

Empirical Evaluation for Nested Pivot Rules

This appendix offers detailed numerical results, obtained from our extensive computational experiments on selected pivot rules, associated with those presented in Chap. 5 of Part I.

Compiled using Visual Fortran 5.0, all codes were run under a Windows XP system Home Edition Version 2002 on an IBM PC with an Intel(R) Pentium(R) processor 1.00GB of 1.86GHz memory, and about 16 digits of precision.

The test sets included three types of problems from Netlib, Kennington, and BPMPD test sets (see below). These problems are ordered by their sizes, in terms of $m + n$, where m and n are the numbers of rows and columns of the constraint matrix, excluding slack variables.

In results listed below, all reported CPU times were measured in seconds with utility routine CPU_TIME, excluding the time spent on preprocessing and scaling. Final objective values reached are not listed. All runs were terminated within the primal and dual feasibility tolerance 10^{-6} .

MINOS was used as the testing benchmark and platform. To compare easy and fairly, each rule is implemented within MINOS 5.51, the latest version of MINOS at that time. Actually, all resulting codes are the same as MINOS 5.51, except for its pivot rule replaced by relevant ones. All codes used the default options, except for

```
Rows 200000;  
Columns 300000;  
Elements 5000000;  
Iterations 20000000;  
Scale yes;  
Solution no;  
Log frequency 0;  
Print level 0.
```

(1) We first report results associated with the following five codes and make comparisons between one another:

1. Dantzig: MINOS 5.51 with the full pricing option.
2. Devex: Devex rule.
3. P-Dantzig: MINOS 5.51 with the default *partial* pricing option.
4. N-Devex: Nested-Devex Rule.
5. N-Dantzig: Rule [12.5.1](#).

A. Results for 48 Netlib problems

The first set of test problems included the 48 largest Netlib problems. Associated results are listed in Table [D.1](#), where total iterations and time required for solving each problem are listed in columns labeled Iters and Time for the five codes. The totals listed in the second bottom line are for the 47 problems, excluding QAP15, as it is too time consuming to solve QAP15 with Dantzig and P-Dantzig, and results with the other three codes would heavily dominate the entire computations. Results associated with QAP15 are listed separately in the bottom row of the table.

To serve as a comparison between the codes, Table [D.2](#) lists iterations and run time ratios of the first four codes to N-Dantzig. From the second bottom line, it is seen that N-Dantzig and N-Devex outperformed all the standard codes, in terms of both iterations and run time. N-Devex performed slightly better than N-Dantzig with time ratio 0.95. On the other hand, code Dantzig performed worst on this set with largest iterations ratio 5.16 and time ratio 6.00. It is also noted that Devex performed better than P-Dantzig.

B. Results for 16 Kennington problems

The second set of problems includes all 16 problems in Kennington test set. These problems are larger than problems in sets 1, overall. Associated numerical results are listed in Table [D.3](#) and compared in Table [D.4](#). From the bottom line of the latter, it is seen that N-Dantzig and N-Devex again defeated all the standard codes in terms of both iterations and run time. Again, N-Devex performed a little better than N-Dantzig with time ratio 0.91. Code Dantzig performed worst on this set with largest iterations ratio 5.63 and time ratio 5.65. In contrast to the first set, however, Devex performed worse than P-Dantzig with this set.

C. Results for 17 BPMPD problems

The third set consists of the 17 largest BPMPD problems, in terms of more than 500KB in compressed form. Associated numerical results are listed in Table [D.5](#) and compared in Table [D.6](#). From the bottom line of Table [D.6](#), it is observed that the codes performed similarly to the first two sets. Even with larger margins, N-Dantzig and N-Devex defeated the standard codes in terms of both iterations and run time. In fact, N-Dantzig performed best with this set while code Dantzig did worst with iterations ratio 8.29 and time ratio 12.86!

Table D.1 Statistics for 48 Netlib problems

Problem	$m + n$	Dantzig		Devex		P-Dantzig		N-Devex		N-Dantzig	
		Iters	Time	Iters	Time	Iters	Time	Iters	Time	Iters	Time
SCR58	1660	742	0.2	386	0.1	737	0.2	526	0.1	579	0.2
GFRD-PNC	1709	668	0.2	608	0.2	668	0.2	755	0.2	713	0.2
BNL1	1819	1333	0.5	934	0.4	1236	0.4	1659	0.4	1509	0.4
SHIP04S	1861	154	0.1	150	0.1	169	0.1	233	0.1	228	0.1
PEROLD	2002	3032	1.4	2200	1.1	3816	1.4	4653	1.8	4751	1.8
MAROS	2290	1642	0.8	1442	0.7	2377	0.9	2455	0.9	3194	1.3
FIT1P	2305	890	0.5	554	0.3	839	0.4	919	0.4	945	0.5
MODSZK1	2308	768	0.4	644	0.2	1025	0.3	951	0.3	980	0.3
SHELL	2312	256	0.1	252	0.1	340	0.1	275	0.1	275	0.1
SCFXM3	2362	936	0.5	927	0.5	1100	0.5	1351	0.5	1408	0.6
25FV47	2393	7146	4.3	3728	2.3	7129	3.4	5319	2.4	6018	2.7
SHIP04L	2521	226	0.1	229	0.1	262	0.1	347	0.1	344	0.1
QAP8	2545	11274	11.6	5921	6.1	11055	10.4	9496	8.7	11161	10.7
WOOD1P	2839	681	0.9	610	0.9	816	0.6	818	0.6	1050	0.7
PILOT.JA	2929	6802	4.4	3379	2.3	6562	3.3	6944	3.7	6567	3.3
SCTAP2	2971	663	0.3	731	0.4	767	0.3	760	0.3	733	0.3
GANGES	2991	574	0.3	590	0.3	719	0.3	743	0.3	699	0.3
PILOTNOV	3148	1706	1.1	1310	0.9	2047	1.0	2795	1.5	2762	1.4
SCSD8	3148	1695	0.7	1900	0.8	2779	0.7	2277	0.5	2330	0.6
SHIP08S	3166	249	0.2	235	0.2	256	0.1	262	0.1	258	0.1
SIERRA	3264	1275	0.7	1160	0.6	1107	0.5	1231	0.5	1221	0.5
DEGEN3	3322	9035	9.3	3599	3.5	7301	6.3	3541	2.9	4064	3.3
PILOT.WE	3512	5635	2.9	2090	1.2	4215	1.6	6358	2.4	5953	2.2

NESM	3586	3110	1.5	3035	1.5	3237	0.9	8556	2.6	6917	2.1
SHIP12S	3915	379	0.3	376	0.3	414	0.2	422	0.2	418	0.2
SCTAP3	3961	750	0.5	835	0.6	915	0.5	899	0.4	907	0.5
STOCFOR2	4189	1490	1.2	1400	1.1	1922	1.4	3303	2.2	3651	2.5
CZPROB	4453	1492	0.8	1043	0.6	1604	0.6	1504	0.6	1420	0.5
CYCLE	4761	2462	2.4	2194	2.1	2711	2.1	3098	2.3	3288	2.4
SHIP08L	5062	449	0.3	446	0.3	438	0.2	595	0.3	594	0.3
PILOT	5094	13374	29.7	8884	21.2	16865	31.2	21692	33.9	22138	38.3
BNL2	5814	5454	6.2	3788	4.3	4682	4.1	5105	4.2	5356	4.5
SHIP12L	6579	792	0.7	818	0.7	831	0.4	906	0.5	875	0.5
D6CUBE	6600	77001	78.7	21235	23.6	106878	44.4	17705	7.2	13792	5.8
D6CUBE2	6600	115853	120.4	20824	23.2	136114	57.0	22273	9.2	25157	11.1
PILOT87	6914	19641	112.1	10028	55.9	20055	98.9	26289	86.7	25615	97.2
D2Q06C	7339	49302	77.5	11456	17.8	46638	53.3	22956	23.9	18784	20.5
GREENBEA	7798	27878	38.6	13201	18.3	24005	24.5	13992	13.8	13032	13.2
WOODW	9504	3968	4.4	3098	3.7	3710	2.4	5548	3.1	5331	2.9
TRUSS	9807	33065	42.1	9552	12.6	13253	8.5	14607	8.3	14259	8.5
FIT2D	10526	46627	83.3	10296	20.5	32150	8.1	22278	9.5	23077	8.5
QAPI2	12049	417689	3685.5	77356	696.7	254428	1966.8	169080	1336.3	170669	1454.1
80BAU3B	12062	14514	17.0	8069	9.6	10488	7.5	13472	8.6	13263	9.0
MAROS-R7	12545	2515	16.0	2720	18.0	2520	13.3	9026	39.3	7615	36.2
FIT2P	16526	13653	44.6	9171	25.6	16064	43.5	14395	33.6	14081	37.0
DFL001	18302	2177575	8999.0	453186	1682.6	2019465	6569.3	156242	429.7	136487	398.6
STOCFOR3	32371	11982	80.1	12220	78.3	14189	77.4	15634	76.4	15603	78.8
Total(47)	275734	3098397	13484.4	718810	2742.4	2790898	9049.6	624245	2161.6	600071	2264.9
QAPI5	28606	—	—	596500	20272.6	—	—	613641	21269.9	625953	23279.8

Table D.2 Ratio to N-Dantzig for 48 Netlib problems

Problem	Dantzig		Devex		P-Dantzig		N-Devex	
	Iters	Time	Iters	Time	Iters	Time	Iters	Time
SCRS8	1.28	1.00	0.67	0.50	1.27	1.00	0.91	0.50
GFRD-PNC	0.94	1.00	0.85	1.00	0.94	1.00	1.06	1.00
BNL1	0.88	1.25	0.62	1.00	0.82	1.00	1.10	1.00
SHIP04S	0.68	1.00	0.66	1.00	0.74	1.00	1.02	1.00
PEROLD	0.64	0.78	0.46	0.61	0.80	0.78	0.98	1.00
MAROS	0.51	0.62	0.45	0.54	0.74	0.69	0.77	0.69
FIT1P	0.94	1.00	0.59	0.60	0.89	0.80	0.97	0.80
MODSZK1	0.78	1.33	0.66	0.67	1.05	1.00	0.97	1.00
SHELL	0.93	1.00	0.92	1.00	1.24	1.00	1.00	1.00
SCFXM3	0.66	0.83	0.66	0.83	0.78	0.83	0.96	0.83
25FV47	1.19	1.59	0.62	0.85	1.18	1.26	0.88	0.89
SHIP04L	0.66	1.00	0.67	1.00	0.76	1.00	1.01	1.00
QAP8	1.01	1.08	0.53	0.57	0.99	0.97	0.85	0.81
WOOD1P	0.65	1.29	0.58	1.29	0.78	0.86	0.78	0.86
PILOT.JA	1.04	1.33	0.51	0.70	1.00	1.00	1.06	1.12
SCTAP2	0.90	1.00	1.00	1.33	1.05	1.00	1.04	1.00
GANGES	0.82	1.00	0.84	1.00	1.03	1.00	1.06	1.00
PILOTNOV	0.62	0.79	0.47	0.64	0.74	0.71	1.01	1.07
SCSD8	0.73	1.17	0.82	1.33	1.19	1.17	0.98	0.83
SHIP08S	0.97	2.00	0.91	2.00	0.99	1.00	1.02	1.00
SIERRA	1.04	1.40	0.95	1.20	0.91	1.00	1.01	1.00
DEGEN3	2.22	2.82	0.89	1.06	1.80	1.91	0.87	0.88
PILOT.WE	0.95	1.32	0.35	0.55	0.71	0.73	1.07	1.09
NESM	0.45	0.71	0.44	0.71	0.47	0.43	1.24	1.24
SHIP12S	0.91	1.50	0.90	1.50	0.99	1.00	1.01	1.00
SCTAP3	0.83	1.00	0.92	1.20	1.01	1.00	0.99	0.80
STOCFOR2	0.41	0.48	0.38	0.44	0.53	0.56	0.90	0.88
CZPROB	1.05	1.60	0.73	1.20	1.13	1.20	1.06	1.20
CYCLE	0.75	1.00	0.67	0.87	0.82	0.87	0.94	0.96
SHIP08L	0.76	1.00	0.75	1.00	0.74	0.67	1.00	1.00
PILOT	0.60	0.78	0.40	0.55	0.76	0.81	0.98	0.89
BNL2	1.02	1.38	0.71	0.96	0.87	0.91	0.95	0.93
SHIP12L	0.91	1.40	0.93	1.40	0.95	0.80	1.04	1.00
D6CUBE	5.58	13.57	1.54	4.07	7.75	7.66	1.28	1.24
D6CUBE2	4.61	10.85	0.83	2.09	5.41	5.14	0.89	0.83
PILOT87	0.77	1.15	0.39	0.58	0.78	1.02	1.03	0.89
D2Q06C	2.62	3.78	0.61	0.87	2.48	2.60	1.22	1.17
GREENBEA	2.14	2.92	1.01	1.39	1.84	1.86	1.07	1.05
WOODW	0.74	1.52	0.58	1.28	0.70	0.83	1.04	1.07
TRUSS	2.32	4.95	0.67	1.48	0.93	1.00	1.02	0.98

(continued)

Table D.2 (continued)

Problem	Dantzig		Devex		P-Dantzig		N-Devex	
	Iters	Time	Iters	Time	Iters	Time	Iters	Time
FIT2D	2.02	9.80	0.45	2.41	1.39	0.95	0.97	1.12
QAP12	2.45	2.53	0.45	0.48	1.49	1.35	0.99	0.92
80BAU3B	1.09	1.89	0.61	1.07	0.79	0.83	1.02	0.96
MAROS-R7	0.33	0.44	0.36	0.50	0.33	0.37	1.19	1.09
FIT2P	0.97	1.21	0.65	0.69	1.14	1.18	1.02	0.91
DFL001	15.95	22.58	3.32	4.22	14.80	16.48	1.14	1.08
STOCFOR3	0.77	1.02	0.78	0.99	0.91	0.98	1.00	0.97
Average	5.16	6.00	1.20	1.21	4.65	4.00	1.04	0.95
QAP15	—	—	0.95	0.87	—	—	0.98	0.91

Table D.3 Statistics for 16 Kennington problems

Problem	$m + n$	Dantzig		Devex		P-Dantzig		N-Devex		N-Dantzig	
		Iters	Time	Iters	Time	Iters	Time	Iters	Time	Iters	Time
KEN-07	6029	1724	1.7	1657	1.5	1822	1.4	1723	1.3	1718	1.3
CRE-C	6747	4068	5.2	3113	4.1	4252	4.3	4485	4.4	4345	4.4
CRE-A	7584	4066	5.9	2986	4.4	3642	4.2	4222	4.8	4228	4.9
PDS-02	10489	5110	6.6	3550	4.6	3118	2.8	2022	1.7	2099	1.8
OSA-07	25068	1767	5.5	1757	6.0	1919	1.9	1494	1.6	1504	1.5
KEN-11	36044	13327	82.8	12814	77.4	15328	72.5	13449	62.2	13361	61.7
PDS-06	38537	59521	306.6	32424	170.1	26127	81.6	13107	40.7	13100	40.6
OSA-14	54798	3821	24.9	3730	27.1	4115	7.4	3372	7.0	3212	6.0
PDS-10	65322	177976	1615.0	105469	988.1	78125	440.3	30149	176.7	24179	134.1
KEN-13	71292	35709	475.7	29196	355.6	47041	500.6	33056	314.3	33074	339.5
CRE-D	78907	377243	3428.1	415765	4141.5	295538	1051.9	70807	341.8	73528	312.1
CRE-B	82096	328160	3159.1	363018	3833.4	188706	726.4	63250	293.0	63703	277.1
OSA-30	104375	7536	91.6	7048	96.3	8141	26.2	6128	25.7	5963	20.5
PDS-20	139603	1557953	31598.4	1589107	33356.3	823110	10913.9	89139	1129.2	93045	1199.4
OSA-60	243247	16278	458.2	16188	505.9	17098	125.5	13992	126.5	13751	112.1
KEN-18	259827	157224	10037.9	127756	6818.1	198019	10017.3	137508	5712.3	137975	6565.8
Total(16)	1229965	2751483	51303.2	2715578	50390.4	1716101	23978.2	487903	8243.2	488785	9082.8

Table D.4 Ratio to N-Dantzig for 16 Kennington problems

Problem	Dantzig		Devex		P-Dantzig		N-Devex	
	Iters	Time	Iters	Time	Iters	Time	Iters	Time
KEN-07	1.00	1.31	0.96	1.15	1.06	1.08	1.00	1.00
CRE-C	0.94	1.18	0.72	0.93	0.98	0.98	1.03	1.00
CRE-A	0.96	1.20	0.71	0.90	0.86	0.86	1.00	0.98
PDS-02	2.43	3.67	1.69	2.56	1.49	1.56	0.96	0.94
OSA-07	1.17	3.67	1.17	4.00	1.28	1.27	0.99	1.07
KEN-11	1.00	1.34	0.96	1.25	1.15	1.18	1.01	1.01
PDS-06	4.54	7.55	2.48	4.19	1.99	2.01	1.00	1.00
OSA-14	1.19	4.15	1.16	4.52	1.28	1.23	1.05	1.17
PDS-10	7.36	12.04	4.36	7.37	3.23	3.28	1.25	1.32
KEN-13	1.08	1.40	0.88	1.05	1.42	1.47	1.00	0.93
CRE-D	5.13	10.98	5.65	13.27	4.02	3.37	0.96	1.10
CRE-B	5.15	11.40	5.70	13.83	2.96	2.62	0.99	1.06
OSA-30	1.26	4.47	1.18	4.70	1.37	1.28	1.03	1.25
PDS-20	16.74	26.35	17.08	27.81	8.85	9.10	0.96	0.94
OSA-60	1.18	4.09	1.18	4.51	1.24	1.12	1.02	1.13
KEN-18	1.14	1.53	0.93	1.04	1.44	1.53	1.00	0.87
Average	5.63	5.65	5.56	5.55	3.51	2.64	1.00	0.91

Table D.5 Statistics for 17 BPM/DP problems

Problem	m+n	Dantzig		Devex		P-Dantzig		N-Devex		N-Dantzig	
		Iters	Time	Iters	Time	Iters	Time	Iters	Time	Iters	Time
RAIT7A	12545	4073	558.6	5226	93.6	3801	135.8	18814	186.9	20074	296.8
NSCT1	37883	2190	31.8	1607	27.6	2151	19.1	5868	45.5	9090	65.1
NSCT2	37985	11555	169.2	6266	88.1	12298	103.1	6258	44.2	5908	42.8
ROUTING	44818	190627	1697.1	68504	606.5	74550	424.8	18742	105.5	22336	124.5
DBIR1	46160	2289	42.9	1848	45.8	1710	16.4	11858	88.0	12209	82.9
DBIR2	46262	42575	819.8	13031	248.8	53839	469.1	7704	54.7	8012	60.4
T0331-4L	47580	62333	638.0	24435	249.7	74755	277.4	45302	276.4	49932	282.1
NEMSEMM2	49077	23581	154.4	17117	118.1	23423	73.4	13899	43.6	10392	30.0
SOUTHERN	54160	29693	432.6	16774	234.6	23031	316.9	23937	315.8	23771	318.1
RADIO.PR	66919	2	1.4	2	1.4	5	1.5	2	1.4	2	1.4
WORLD.MD	67393	1405151	23856.3	427776	7403.1	1241489	18068.8	217502	2979.3	220329	3077.7
WORLD	68245	2627208	46257.3	501004	8747.2	2182226	31763.3	261210	3590.0	278284	3899.0
RADIO.DL	74971	3	0.9	3	1.0	7	1.0	3	0.9	3	0.9
NEMSEMM1	75359	13650	207.9	10574	168.8	13065	60.2	12787	41.7	9514	30.9
NW14	123483	407	9.1	394	9.5	953	5.7	778	5.9	736	5.2
LPL1	164952	3343112	89309.0	2337341	60073.4	1047339	17532.1	445616	7320.2	262076	4474.3
DBIC1	226436	624907	25384.9	437658	18252.6	342960	7404.3	102935	2822.4	78334	1949.6
Total(17)	1244228	8383356	189571.2	3869560	96369.8	5097602	76672.9	1193215	17922.4	1011002	14741.7
Total(80)	14232326	254358.8	7303948	149502.6	9604601	109700.7	2305363	28327.2	2099858	26089.4	

Table D.6 Ratio to N-Dantzig for 17 BPMPD problems

Problem	Dantzig		Devex		P-Dantzig		N-Devex	
	Iters	Time	Iters	Time	Iters	Time	Iters	Time
RAT7A	0.20	1.88	0.26	0.32	0.19	0.46	0.94	0.63
NSCT1	0.24	0.49	0.18	0.42	0.24	0.29	0.65	0.70
NSCT2	1.96	3.95	1.06	2.06	2.08	2.41	1.06	1.03
ROUTING	8.53	13.63	3.07	4.87	3.34	3.41	0.84	0.85
DBIR1	0.19	0.52	0.15	0.55	0.14	0.20	0.97	1.06
DBIR2	5.31	13.57	1.63	4.12	6.72	7.77	0.96	0.91
T0331-4L	1.25	2.26	0.49	0.89	1.50	0.98	0.91	0.98
NEMSEMM2	2.27	5.15	1.65	3.94	2.25	2.45	1.34	1.45
SOUTHERN	1.25	1.36	0.71	0.74	0.97	1.00	1.01	0.99
RADIO.PR	1.00	1.00	1.00	1.00	2.50	1.07	1.00	1.00
WORLD.MD	6.38	7.75	1.94	2.41	5.63	5.87	0.99	0.97
WORLD	9.44	11.86	1.80	2.24	7.84	8.15	0.94	0.92
RADIO.DL	1.00	1.00	1.00	1.11	2.33	1.11	1.00	1.00
NEMSEMM1	1.43	6.73	1.11	5.46	1.37	1.95	1.34	1.35
NW14	0.55	1.75	0.54	1.83	1.29	1.10	1.06	1.13
LPL1	12.76	19.96	8.92	13.43	4.00	3.92	1.70	1.64
DBIC1	7.98	13.02	5.59	9.36	4.38	3.80	1.31	1.45
Average	8.29	12.86	3.83	6.54	5.04	5.20	1.18	1.22

D. A summary

Table D.7 offers ratios of the first four codes to N-Dantzig for each test set and for all (excluding QAP15). It is seen that the codes performed consistently with the three sets, overall. Further, N-Dantzig and N-Devex defeated the standard codes in terms of both total iterations and run time. N-Dantzig even outperformed Devex rule with total iterations ratio 3.48 and average time ratio 5.73. It can be asserted that N-Dantzig performed best overall while code Dantzig did worst with iterations ratio 6.78 and time ratio 9.75.

The ratio of the number of iterations required by a code to the number of rows and columns should be an indicator for problem’s difficulty to the code. The smaller the ratio is, the easier the problem solved by the code. Following Forrest and Goldfarb (1992), we, therefore, list in Tables D.8 and D.9 normalized iterations required by the five codes for the Kennington and BPMPD test sets, respectively.

Table D.7 A ratio summary

Problem	Dantzig		Devex		P-Dantzig		N-Devex	
	Iters	Time	Iters	Time	Iters	Time	Iters	Time
Netlib(47)	5.16	5.95	1.20	1.21	4.65	4.00	1.04	0.95
Kennington(16)	5.63	5.65	5.56	5.55	3.51	2.64	1.00	0.91
BPMPD(17)	8.29	12.86	3.83	6.54	5.04	5.20	1.18	1.22
Average(80)	6.78	9.75	3.48	5.73	4.57	4.20	1.10	1.09

Table D.8 Normalized iterations for 16 Kennington problems

No	Problem	Dantzig	Devex	P-Dantzig	N-Devex	N-Dantzig
1	KEN-07	0.29	0.27	0.30	0.29	0.28
2	CRE-C	0.60	0.46	0.63	0.66	0.64
3	CRE-A	0.54	0.39	0.48	0.56	0.56
4	PDS-02	0.49	0.34	0.30	0.19	0.20
5	OSA-07	0.07	0.07	0.08	0.06	0.06
6	KEN-11	0.37	0.36	0.43	0.37	0.37
7	PDS-06	1.54	0.84	0.68	0.34	0.34
8	OSA-14	0.07	0.07	0.08	0.06	0.06
9	PDS-10	2.72	1.61	1.20	0.46	0.37
10	KEN-13	0.50	0.41	0.66	0.46	0.46
11	CRE-D	4.78	5.27	3.75	0.90	0.93
12	CRE-B	4.00	4.42	2.30	0.77	0.78
13	OSA-30	0.07	0.07	0.08	0.06	0.06
14	PDS-20	11.16	11.38	5.90	0.64	0.67
15	OSA-60	0.07	0.07	0.07	0.06	0.06
16	KEN-18	0.61	0.49	0.76	0.53	0.53
	Average	2.24	2.21	1.40	0.40	0.40

Table D.9 Normalized iterations for 17 BPMPD Problems

No	Problem	Dantzig	Devex	P-Dantzig	N-Devex	N-Dantzig
1	RAT7A	0.32	0.42	0.30	1.50	1.60
2	NSCT1	0.06	0.04	0.06	0.15	0.24
3	NSCT2	0.30	0.16	0.32	0.16	0.16
4	ROUTING	4.25	1.53	1.66	0.42	0.50
5	DBIR1	0.05	0.04	0.04	0.26	0.26
6	DBIR2	0.92	0.28	1.16	0.17	0.17
7	T0331-4L	1.31	0.51	1.57	0.95	1.05
8	NEMSEMM2	0.48	0.35	0.48	0.28	0.21
9	SOUTHERN	0.55	0.31	0.43	0.44	0.44
10	RADIO.PR	0.00	0.00	0.00	0.00	0.00
11	WORLD.MD	20.85	6.35	18.42	3.23	3.27
12	WORLD	38.50	7.34	31.98	3.83	4.08
13	RADIO.DL	0.00	0.00	0.00	0.00	0.00
14	NEMSEMM1	0.18	0.14	0.17	0.17	0.13
15	NW14	0.00	0.00	0.01	0.01	0.01
16	LPL1	20.27	14.17	6.35	2.70	1.59
17	DBIC1	2.76	1.93	1.51	0.45	0.35
	Average	6.74	3.11	4.10	0.96	0.81

From the bottom lines of the Tables, it is seen that, on average, normalized iterations required by N-Dantzig and N-Devex are less than one, whereas those required by their standard counterparts are greater than two. In particular, normalized iterations required by N-Dantzig and N-Devex are less than one for each Kennington problem. Overall, normalized iterations required by them are quite stable, while those required by their standard counterparts fluctuate.

There is no doubt that nested pricing rules are amenable to the test sets, as is the case can be expected in general.

(2) We also report our computational experiments with the following three codes:

- 1. Steepest-Edge(Stp): the Steepest-edge rule.
- 2. N-Steepest(N-Stp): Nested-steepest-edge rule.
- 3. N-Dantzig(N-D): Rule 12.5.1.

Computational results associated with Netlib, Kennington, and BPMPD problems are given in Tables D.10, D.11, D.12, D.13, D.14, D.15. The first two codes failed to solve QAP15 in Netlib, KEN-18 in Kennington, and LPL1 and DBIC1 in BPMPD within 48 hours. So, we report results, excluding the four problems.

Table D.10 Statistics for 47 Netlib problems

Problem	Stp		N-Stp		N-D	
	Iters	Time	Iters	Time	Iters	Time
SCRS8	6372	567.4	486	0.4	579	0.2
GFRD-PNC	626	0.3	722	0.4	713	0.2
BNL1	624	0.8	958	1.0	1509	0.4
SHIP04S	145	0.2	146	0.2	228	0.1
PEROLD	1638	2.0	2772	3.3	4751	1.8
MAROS	1115	1.8	1725	2.7	3194	1.3
FIT1P	396	0.6	804	1.1	945	0.5
MODSZK1	596	1.1	748	1.0	980	0.3
SHELL	232	0.2	244	0.2	275	0.1
SCFXM3	715	1.0	947	1.2	1408	0.6
25FV47	1577	3.0	3716	6.6	6018	2.7
SHIP04L	223	0.3	223	0.3	344	0.1
QAP8	3315	9.9	4351	13.9	11161	10.7
WOOD1P	763	3.2	775	2.9	1050	0.7
PILOTJ	1862	3.8	4114	8.0	6567	3.3
SCTAP2	612	1.2	845	1.5	733	0.3
GANGES	571	0.6	727	0.8	699	0.3
PILOTN	1108	2.6	2256	5.0	2762	1.4
SCSD8	751	1.1	1634	2.4	2330	0.6

(continued)

Table D.10 (continued)

Problem	Stp		N-Stp		N-D	
	Iters	Time	Iters	Time	Iters	Time
SHIP08S	237	0.6	267	0.6	258	0.1
SIERRA	1005	2.0	1018	2.0	1221	0.5
DEGEN3	1854	6.7	2887	10.1	4064	3.3
PILOT.W	1339	3.3	3736	8.2	5953	2.2
NESM	2200	4.0	5558	10.5	6917	2.1
SHIP12S	393	1.1	446	1.2	418	0.2
SCTAP3	771	2.2	960	2.7	907	0.5
STOCF.2	1429	3.9	2119	5.3	3651	2.5
CZPROB	900	2.5	1189	2.8	1420	0.5
CYCLE	1125	4.7	3004	12.3	3288	2.4
SHIP08L	408	1.8	538	1.8	594	0.3
PILOT	19344	251.7	15765	319.5	22138	38.3
BNL2	2032	11.3	3118	17.0	5356	4.5
SHIP12L	813	4.4	912	4.6	875	0.5
D6CUBE	7793	40.9	9387	45.0	13792	5.8
D6CUBE2	9585	48.8	12911	62.4	25157	11.1
PILOT87	9224	304.4	19541	759.1	25615	97.2
D2Q06C	5986	62.9	10970	117.3	18784	20.5
GREENBE	7210	63.6	11242	98.9	13032	13.2
WOODW	2524	21.5	3907	33.1	5331	2.9
TRUSS	8877	83.0	12743	118.7	14259	8.5
FIT2D	5970	33.0	11558	57.3	23077	8.5
QAP12	29494	3055.0	70841	7583.5	170669	1454.1
80BAU3B	4565	48.1	8006	95.1	13263	9.0
MAROSR7	6029	232.6	7598	409.5	7615	36.2
FIT2P	6645	197.5	12138	418.3	14081	37.0
DFL001	24704	1020.6	52010	2153.4	136487	398.6
STOCF.3	10628	1082.6	15703	1612.0	15603	78.8
Total(47)	196325	7195.9	328265	14015.0	600071	2264.7

Table D.11 Ratio for 47 Netlib problems

Problem	m	n	Stp/N-Stp		Stp/N-D		N-Stp/N-D	
			Iters	Time	Iters	Time	Iters	Time
SCRS8	491	1169	13.11	1383.93	11.01	3546.31	0.84	2.56
GFRDP	617	1092	0.87	0.89	0.88	2.00	1.01	2.24
BNL1	644	1175	0.65	0.74	0.41	1.70	0.63	2.32
SH04S	403	1458	0.99	1.00	0.64	2.13	0.64	2.13
PEROL	626	1376	0.59	0.61	0.34	1.14	0.58	1.86
MAROS	847	1443	0.65	0.66	0.35	1.39	0.54	2.10
FIT1P	628	1677	0.49	0.55	0.42	1.26	0.85	2.28
MODS1	688	1620	0.80	1.09	0.61	4.04	0.76	3.70
SHELL	537	1775	0.95	1.00	0.84	2.75	0.89	2.75
SCFX3	991	1371	0.76	0.79	0.51	1.76	0.67	2.24
25F47	822	1571	0.42	0.46	0.26	1.10	0.62	2.41
SH04L	403	2118	1.00	0.94	0.65	2.82	0.65	3.00
QAP8	913	1632	0.76	0.71	0.30	0.93	0.39	1.30
WOOD1	245	2594	0.98	1.10	0.73	4.40	0.74	4.00
PIL.J	941	1988	0.45	0.47	0.28	1.14	0.63	2.42
SCTA2	1091	1880	0.72	0.78	0.83	4.30	1.15	5.48
GANGE	1310	1681	0.79	0.81	0.82	2.03	1.04	2.52
PLTNO	976	2172	0.49	0.51	0.40	1.83	0.82	3.56
SCSD8	398	2750	0.46	0.44	0.32	1.89	0.70	4.32
SH08S	779	2387	0.89	1.04	0.92	4.46	1.03	4.31
SIERR	1228	2036	0.99	1.01	0.82	3.94	0.83	3.88
DEGE3	1504	1818	0.64	0.66	0.46	2.01	0.71	3.03
PIL.W	723	2789	0.36	0.41	0.22	1.53	0.63	3.77
NESM	663	2923	0.40	0.38	0.32	1.85	0.80	4.90
SH12S	1152	2763	0.88	0.88	0.94	4.91	1.07	5.59
SCTA3	1481	2480	0.80	0.82	0.85	4.74	1.06	5.79
STCF2	2158	2031	0.67	0.73	0.39	1.54	0.58	2.12
CZPRO	930	3523	0.76	0.90	0.63	4.77	0.84	5.28
CYCLE	1904	2857	0.37	0.39	0.34	1.95	0.91	5.06
SH08L	779	4283	0.76	1.00	0.69	7.24	0.91	7.24
PILOT	1442	3652	1.23	0.79	0.87	6.58	0.71	8.35
BNL2	2325	3489	0.65	0.67	0.38	2.51	0.58	3.77
SH12L	1152	5427	0.89	0.96	0.93	8.82	1.04	9.18
D6CUB	416	6184	0.83	0.91	0.57	7.06	0.68	7.76
D6CU2	416	6184	0.74	0.78	0.38	4.42	0.51	5.64
PIL87	2031	4883	0.47	0.40	0.36	3.13	0.76	7.81
D2Q06	2172	5167	0.55	0.54	0.32	3.07	0.58	5.73
GREEN	2393	5405	0.64	0.64	0.55	4.81	0.86	7.48

(continued)

Table D.11 (continued)

Problem	m	n	Stp/N-Stp		Stp/N-D		N-Stp/N-D	
			Iters	Time	Iters	Time	Iters	Time
WOODW	1099	8405	0.65	0.65	0.47	7.38	0.73	11.32
TRUSS	1001	8806	0.70	0.70	0.62	9.76	0.89	13.96
FIT2D	26	10500	0.52	0.58	0.26	3.88	0.50	6.74
QAP12	3193	8856	0.42	0.40	0.17	2.10	0.42	5.22
80BAU	2263	9799	0.57	0.51	0.34	5.36	0.60	10.59
MARR7	3137	9408	0.79	0.57	0.79	6.43	1.00	11.31
FIT2P	3001	13525	0.55	0.47	0.47	5.33	0.86	11.29
DFL00	6072	12230	0.47	0.47	0.18	2.56	0.38	5.40
STOO3	16676	15695	0.68	0.67	0.68	13.73	1.01	20.45
Ave.	–	–	0.60	0.51	0.33	3.18	0.55	6.19

Table D.12 Statistics for 15 Kennington problems

Problem	Stp		N-Stp		N-D	
	Iters	Time	Iters	Time	Iters	Time
KEN-07	1699	6.0	1723	6.1	1718	1.3
CRE-C	1391	13.7	2249	22.5	4345	4.4
CRE-A	1497	18.4	2325	29.2	4228	4.9
PDS-02	911	10.6	1133	12.8	2099	1.8
OSA-07	1507	39.3	1570	38.2	1504	1.5
KEN-11	11931	1068.0	12247	1090.8	13361	61.7
PDS-06	3394	467.2	4032	583.7	13100	40.6
OSA-14	3180	334.2	3227	336.8	3212	6.0
PDS-10	6562	2605.2	8377	3440.4	24179	134.1
KEN-13	23769	9181.5	25464	9861.8	33074	339.5
CRE-D	17782	8227.1	22075	10265.2	73528	312.1
CRE-B	16411	8534.1	24482	12764.0	63703	277.1
OSA-30	6254	2264.3	7727	2720.3	5963	20.5
PDS-20	24854	52549.0	28443	59942.4	93045	1199.4
OSA-60	12809	24627.3	16731	30928.0	13751	112.1
Total(15)	133951	109945.9	161805	132042.2	350810	2517.0

Table D.16 offers ratios for each test set and for all the 77 problems as a whole. It is seen that N-Dantzig outperformed Steepest-Edge with average time ratio as high as 25.22 despite it required more total iterations (with iterations ratio 0.34). It is also noted that Nested-Steepest-Edge was inferior to Steepest-Edge.

In summary, the nested pricing are unambiguously superior to standard one. As the best among nested ones, the nested Dantzig criteria should be used whenever possible.

Table D.13 Ratio for 15 Kennington problems

Problem	m	n	Stp/N-Stp		Stp/N-D		N-Stp/N-D	
			Iters	Time	Iters	Time	Iters	Time
CRE-C	3069	3678	0.62	0.61	0.32	3.10	0.52	5.10
CRE-A	3517	4067	0.64	0.63	0.35	3.73	0.55	5.91
PDS-02	2954	7535	0.80	0.83	0.43	5.80	0.54	7.02
OSA-07	1119	23949	0.96	1.03	1.00	27.11	1.04	26.34
KEN-11	14695	21349	0.97	0.98	0.89	17.32	0.92	17.69
PDS-06	9882	28655	0.84	0.80	0.26	11.50	0.31	14.37
OSA-14	2338	52460	0.99	0.99	0.99	55.69	1.00	56.13
PDS-10	16559	48763	0.78	0.76	0.27	19.42	0.35	25.65
KEN-13	28633	42659	0.93	0.93	0.72	27.04	0.77	29.05
CRE-D	8927	69980	0.81	0.80	0.24	26.36	0.30	32.89
CRE-B	9649	72447	0.67	0.67	0.26	30.80	0.38	46.07
OSA-30	4351	100024	0.81	0.83	1.05	110.72	1.30	133.02
PDS-20	33875	105728	0.87	0.88	0.27	43.81	0.31	49.98
OSA-60	10281	232966	0.77	0.80	0.93	219.63	1.22	275.82
Average	–	–	0.83	0.83	0.38	43.68	0.46	52.46

Table D.14 Statistics for 15 BPMPD problems

Problem	Stp		N-Stp		N-D	
	Iters	Time	Iters	Time	Iters	Time
RAT7A	6372	567.4	11779	1498.3	20074	296.8
NSCT1	1956	541.1	1981	573.0	9090	65.1
NSCT2	3867	1023.8	4211	1126.5	5908	42.8
ROUTING	7956	2350.6	8350	2459.9	22336	124.5
DBIR1	2544	1134.8	2443	1090.4	12209	82.9
DBIR2	1980	801.3	2085	839.6	8012	60.4
T0331-4L	6304	1445.1	22066	6031.8	49932	282.1
NEMSEMM2	11680	2485.7	9282	1913.5	10392	30.0
SOUTHERN	16472	4008.5	21608	5491.0	23771	318.1
RADIO.PR	2	29.7	2	29.7	2	1.4
WORLD.MD	74418	92591.3	111230	132327.3	220329	3077.7
WORLD	79915	103793.9	129969	163859.8	278284	3899.0
RADIO.DL	4	36.8	4	36.8	3	0.9
NEMSEMM1	7459	2374.7	7215	2275.8	9514	30.9
NW14	215	12.8	313	14.2	736	5.2
Total(15)	221144	213197.3	332538	319567.4	670592	8317.8
Total(77)	551420	330339.1	822608	465624.6	1621473	13099.5

Table D.15 Ratio for 15 BPMPD problems

Problem	m	n	Stp/N-Stp		Stp/N-D		N-Stp/N-D	
			Iters	Time	Iters	Time	Iters	Time
RAT7A	3137	9408	0.54	0.38	0.32	1.91	0.59	5.05
NSCT1	22902	14981	0.99	0.94	0.22	8.31	0.22	8.80
NSCT2	23004	14981	0.92	0.91	0.65	23.94	0.71	26.34
ROUTING	20895	23923	0.95	0.96	0.36	18.88	0.37	19.76
DBIR1	18805	27355	1.04	1.04	0.21	13.69	0.20	13.15
DBIR2	18907	27355	0.95	0.95	0.25	13.27	0.26	13.90
T0331-4L	665	46915	0.29	0.24	0.13	5.12	0.44	21.38
NEMSEMM2	6944	42133	1.26	1.30	1.12	82.99	0.89	63.89
SOUTHERN	18739	35421	0.76	0.73	0.69	12.60	0.91	17.26
RADIO.PR	58867	8052	1.00	1.00	1.00	21.81	1.00	21.82
WORLD.MD	35665	31728	0.67	0.70	0.34	30.08	0.50	43.00
WORLD	35511	32734	0.61	0.63	0.29	26.62	0.47	42.03
RADIO.DL	8053	66918	1.00	1.00	1.33	38.68	1.33	38.75
NEMSEMM1	3946	71413	1.03	1.04	0.78	76.80	0.76	73.60
NW14	74	123409	0.69	0.90	0.29	2.44	0.43	2.71
Average	–	–	0.67	0.67	0.33	25.63	0.50	38.42

Table D.16 A second ratio summary

Problem	Stp/N-Stp		Stp/N-D		N-Stp/N-D	
	Iters	Time	Iters	Time	Iters	Time
Netlib(47)	0.60	0.51	0.33	3.18	0.55	6.19
Kennington(15)	0.83	0.83	0.38	43.68	0.46	52.46
BPMPD(15)	0.67	0.67	0.33	25.63	0.50	38.42
Average(77)	0.67	0.71	0.34	25.22	0.51	35.55

Appendix E

Empirical Evaluation for Primal and Dual Face Methods with Cholesky Factorization

To give an insight into the behavior of face methods, the following three codes in FORTRAN 77 were tested and compared:

1. RSA: The revised simplex algorithm supported by the classical Phase-I (see Notation).
2. FALP: Face Algorithm 21.6.1, supported by Algorithm 21.6.2, with components of the initial $\hat{x}$ being all ones.
3. DFA: Dual face Algorithm 22.6.1, supported by Algorithm 22.6.2.

All the preceding were dense codes in the sense that they did not exploit sparsity structure. Harris' two-pass practical tactic was used. Number 10^{-6} was taken as primal and dual feasibility tolerance, and 10^{-4} as equality relative tolerance. In FALP, $\|\Delta_B\|_\infty < 10^{-8}$ was used in place of $\Delta_B = 0$, while in DFA, $\|\Delta y\|_\infty < 10^{-8}$ used in place of $\Delta y = 0$.

Compiled using the Visual FORTRAN 5.0, all codes were run under a Windows XP system Home Edition Version 2002 on an IBM PC with an Intel(R) Pentium(R) processor 1.00GB of 1.86GHz memory, and about 16 digits of precision. All reported CPU times were measured in seconds with utility routine CPU_TIME.

Tested was a set of 26 standard problems from NETLIB that do not have BOUNDS and RANGES sections in their MPS files. Specifically, these were the first 26 such problems in the order of increasing sum of numbers of rows and columns of the constraint matrix, before adding slack variables (see Table C.4 in Appendix C).

Test statistics obtained with RSA, FALP, and DFA are listed in Tables E.1, E.2, and E.3, respectively. Total iterations and time, required for solution of each problem, are displayed in columns labeled Itn and Time, respectively, and those associated with Phase-1 alone in columns labeled Itn1 and Time1. Percentages of total degenerate iterations are listed in columns labeled % Deg; those associated with Phase-1 are in the column labeled % Deg1. In the last column labeled k-m of Table E.2, listed are the dimensions of the optimal face finally reached, whereas the

Table E.1 Statistics for code RSA

Problem	Total			Phase 1		
	Itn	Time	%Deg	Itn1	Time1	%Deg1
AFIRO	29	0.03	69.0	28	0.03	71.4
SC50B	57	0.06	80.7	50	0.06	90.0
SC50A	55	0.09	74.5	51	0.09	78.4
ADLITTLE	134	0.13	12.7	69	0.13	24.6
BLEND	115	0.17	40.0	90	0.17	51.1
SHARE2B	196	0.28	57.1	149	0.27	67.8
SC105	123	0.36	73.2	110	0.36	77.3
STOCFOR1	174	0.50	68.4	150	0.48	79.3
SCAGR7	181	0.67	34.3	146	0.64	42.5
ISRAEL	507	1.41	0.8	188	0.97	1.1
SHARE1B	309	1.67	7.1	190	1.58	11.6
SC205	255	2.23	66.7	211	2.16	80.1
BEACONFD	213	2.61	51.2	159	2.53	57.9
LOTFI	354	3.13	16.4	190	2.91	22.6
BRANDY	356	3.86	33.7	238	3.63	49.2
E226	615	5.50	23.7	394	4.94	32.7
AGG	639	13.64	6.4	571	12.89	7.0
SCORPION	405	16.64	61.0	380	16.47	64.5
BANDM	666	20.42	21.8	467	19.39	30.0
SCTAP1	481	23.11	36.6	349	22.44	44.4
SCFXM1	587	26.98	35.1	429	26.03	45.5
AGG2	824	39.47	5.5	629	36.70	5.6
AGG3	837	52.14	5.5	627	49.17	6.1
SCSD1	186	52.31	81.7	80	52.23	97.5
SCAGR25	922	64.52	34.1	596	60.42	44.5
Total(25)	9220	331.93	39.9	6541	316.69	47.3
DEGEN2	10000	177.83	48.7	1990	87.84	45.7

column labeled m-k in Table E.3 are dimensions of dual optimal faces. Objective values attained are not listed—all runs were terminated within the feasibility tolerances except for problem DEGEN2, which RSA failed to solve within 10000 iterations, and the associated results were listed separately, in the bottom lines of the tables.

(1) A comparison between RSA and FALP.

Table E.4 gives iteration and time ratios of RSA to FALP. From the second bottom line, it is seen that the total iteration ratio with the 25 test problems is 1.51, while the total time ratio is 4.91! As the time ratio with SCSD1 is exceptionally high (83.03), the bottom line lists total ratios with the 24 problems, excluding SCSD1. It is seen that the associated iterations and time ratio are 1.69 and 4.17, respectively, as indicates that results with SCSD1 alone do not dominate too much.

Table E.2 Statistics for FALP

Problem	Total			Phase 1			k-m
	Itn	Time	%Deg	Itn1	Time1	%Deg1	
AFIRO	26	0.02	0.0	7	0.00	0.0	9
SC50B	30	0.02	0.0	1	0.00	0.0	20
SC50A	32	0.02	0.0	1	0.00	0.0	20
ADLITTLE	175	0.13	0.0	40	0.03	0.0	15
BLEND	49	0.08	0.0	11	0.03	0.0	44
SHARE2B	85	0.06	0.0	55	0.03	0.0	14
SC105	65	0.06	0.0	1	0.02	0.0	45
STOCFOR1	66	0.09	0.0	23	0.03	0.0	63
SCAGR7	134	0.23	0.0	91	0.13	0.0	84
ISRAEL	294	0.67	0.3	79	0.16	0.0	0
SHARE1B	243	0.34	0.0	140	0.16	0.0	89
SC205	139	0.53	0.0	1	0.08	0.0	91
BEACONFD	135	0.44	3.0	60	0.16	5.0	144
LOTFI	353	0.73	0.6	198	0.30	0.0	95
BRANDY	246	0.91	1.2	76	0.23	0.0	142
E226	486	1.81	0.2	192	0.53	0.0	37
AGG	160	4.45	0.0	94	1.81	0.0	49
SCORPION	119	2.36	3.4	23	0.61	17.4	249
BANDM	410	5.02	0.0	167	1.55	0.0	305
SCTAP1	449	3.55	2.9	112	1.02	0.0	120
SCFXM1	528	6.23	0.2	383	3.78	0.0	191
AGG2	282	7.97	0.0	67	2.14	0.0	76
AGG3	290	8.25	0.0	86	2.50	0.0	76
SCSD1	736	0.63	9.8	1	0.03	0.0	77
SCAGR25	556	23.06	0.0	361	13.47	0.0	300
Total(25)	6088	67.66	0.9	2270	28.80	0.9	94
DEGEN2	452	11.13	0.0	190	3.94	0.0	219

(2) A comparison between RSA and DFA.

Table E.5 gives iterations and time ratios of RSA to DFA. If is seen from the second bottom line that the total iteration ratio with the 25 test problems is 1.19, while the total time ratio is as high as 10.04! As the time ratio with SCSD1 is as exceptionally high (1046.20), the bottom line lists total ratios with the other 24 problems: the associated iteration and time ratios are 1.17 and 8.47, respectively.

Considering that the time ratio is the only indication for efficiency, we conclude that the primal and dual face codes outperformed RSA unambiguously.

Such outcome might be astonishing, but not surprising:

On the one hand, FALP solves a single triangular $(m + 1) \times (m + 1)$ -system in each iteration, compared with RSA solving four triangular $m \times m$ -systems.

Table E.3 Statistics for code DFA

Problem	Total			Phase 1			m-k
	Itn	Time	%Deg	Itn1	Time1	%Deg1	
AFIRO	30	0.00	20.0	22	0.00	0.0	3
SC50B	50	0.00	0.0	48	0.00	0.0	2
SC50A	53	0.00	0.0	47	0.00	0.0	1
ADLITTLE	164	0.03	15.2	29	0.02	0.0	0
BLEND	104	0.03	2.9	75	0.02	0.0	0
SHARE2B	142	0.03	11.3	69	0.02	1.4	0
SC105	110	0.03	0.0	102	0.02	0.0	1
STOCFOR1	152	0.08	0.0	127	0.05	0.0	0
SCAGR7	254	0.23	11.8	110	0.06	10.9	0
ISRAEL	321	0.36	12.8	140	0.11	0.0	0
SHARE1B	245	0.13	4.5	119	0.06	0.0	0
SC205	209	0.27	0.0	202	0.20	0.0	2
BEACONFD	136	0.06	27.9	1	0.00	0.0	50
LOTFI	264	0.27	8.3	137	0.13	0.0	0
BRANDY	556	0.89	36.7	1	0.00	0.0	16
E226	381	0.78	7.3	156	0.23	0.0	9
AGG	542	3.59	9.0	122	0.63	0.8	11
SCORPION	362	1.44	22.1	1	0.00	0.0	21
BANDM	558	2.75	0.7	294	1.22	0.3	0
SCTAP1	366	0.75	79.0	1	0.02	0.0	33
SCFXM1	713	3.30	5.6	216	0.70	0.5	1
AGG2	571	4.97	5.3	161	1.25	0.0	0
AGG3	588	5.14	5.1	151	1.14	0.0	0
SCSD1	78	0.05	35.9	1	0.00	0.0	0
SCAGR25	822	7.89	16.8	360	2.75	13.3	8
Total(25)	7771	33.07	13.5	2692	8.63	1.1	6
DEGEN2	1627	18.09	6.0	434	3.53	0.2	4

On the other hand, it is seen from the last column of Table E.2 that all optimal faces, finally reached by FALP, are not vertices, except for ISREAL, and that the average dimension of these faces is about 94. The ratio of average dimensions to average rows is as low as $94/213 = 0.44$. In general, a high-dimensional optimal face is easier to achieve than an optimal vertex.

As for a comparison between the two face codes, DFA is even superior to FALP, partially due to that the former solves a smaller triangular $k \times k$ system in each iteration, where k varies dynamically but never exceeds m . From the last column of Table E.3, in fact, it is seen that k did not reach m for 14 out of the 26 test problems.

Table E.4 Ratio of RSA to FALP

Problem	<i>m</i>	<i>n</i>	<i>m + n</i>	Total		Phase 1	
				It _n	Time	It _n	Time
AFIRO	28	32	60	1.12	1.50	4.00	300.00
SC50B	51	48	99	1.90	3.00	50.00	600.00
SC50A	51	48	99	1.72	4.50	51.00	900.00
ADLITTLE	57	97	154	0.77	1.00	1.73	4.33
BLEND	75	83	158	2.35	2.13	8.18	5.67
SHARE2B	97	79	176	2.31	4.67	2.71	9.00
SC105	106	103	209	1.89	6.00	110.00	18.00
STOCFOR1	118	111	229	2.64	5.56	6.52	16.00
SCAGR7	130	140	270	1.35	2.91	1.60	4.92
ISRAEL	175	142	317	1.72	2.10	2.38	6.06
SHARE1B	118	225	343	1.27	4.91	1.36	9.88
SC205	206	203	409	1.83	4.21	211.00	27.00
BEACONFD	174	262	436	1.58	5.93	2.65	15.81
LOTFI	154	308	462	1.00	4.29	0.96	9.70
BRANDY	194	249	443	1.45	4.24	3.13	15.78
E226	224	282	506	1.27	3.04	2.05	9.32
AGG	489	163	652	3.99	3.07	6.07	7.12
SCORPION	358	358	716	3.40	7.05	16.52	27.00
BANDM	306	472	778	1.62	4.07	2.80	12.51
SCTAP1	301	480	781	1.07	6.51	3.12	22.00
SCFXM1	331	457	788	1.11	4.33	1.12	6.89
AGG2	517	302	819	2.92	4.95	9.39	17.15
AGG3	517	302	819	2.89	6.32	7.29	19.67
SCSD1	78	760	838	0.25	83.03	80.00	1741.00
SCAGR25	472	500	972	1.66	2.80	1.65	4.49
Average(25)	213	248	461	1.51	4.91	2.88	11.00
Average(24)	218	226	445	1.69	4.17	2.85	9.19

Table E.5 Ratio of RSA to DFA

Problem	m	n	$m + n$	Total		Phase 1	
				Itn	Time	Itn	Time
AFIRO	28	32	60	0.97	-	1.27	-
SC50B	51	48	99	1.14	-	1.04	-
SC50A	51	48	99	1.04	-	1.09	-
ADLITTLE	57	97	154	0.82	4.33	2.38	6.50
BLEND	75	83	158	1.11	5.67	1.20	8.50
SHARE2B	97	79	176	1.38	9.33	2.16	13.50
SC105	106	103	209	1.12	12.00	1.08	18.00
STOCFOR1	118	111	229	1.14	6.25	1.18	9.60
SCAGR7	130	140	270	0.71	2.91	1.33	10.67
ISRAEL	175	142	317	1.58	3.92	1.34	8.82
SHARE1B	118	225	343	1.26	12.85	1.60	26.33
SC205	206	203	409	1.22	8.26	1.04	10.80
BEACONFD	174	262	436	1.57	43.50	159.00	-
LOTFI	154	308	462	1.34	11.59	1.39	22.38
BRANDY	194	249	443	0.64	4.34	238.00	-
E226	224	282	506	1.61	7.05	2.53	21.48
AGG	489	163	652	1.18	3.80	4.68	20.46
SCORPION	358	358	716	1.12	11.56	380.00	-
BANDM	306	472	778	1.19	7.43	1.59	15.89
SCTAP1	301	480	781	1.31	30.81	349.00	1122.00
SCFXM1	331	457	788	0.82	8.18	1.99	37.19
AGG2	517	302	819	1.44	7.94	3.91	29.36
AGG3	517	302	819	1.42	10.14	4.15	43.13
SCSD1	78	760	838	2.38	1046.20	80.00	-
SCAGR25	472	500	972	1.12	8.18	1.66	21.97
Average(25)	213	248	461	1.19	10.04	2.43	36.70
Average(24)	218	226	445	1.17	8.47	2.40	30.64

Appendix F

Empirical Evaluation for Simplex Interior-Point Algorithm

Computational experiments have been performed to gain an insight into the behavior of Algorithm 25.3.1.

Implemented and compared are the following three codes:

1. MINOS: MINOS 5.51 with full pricing.
2. FPS: Two-Phase code based on Algorithm 25.3.1.
3. FPSP: Two-Phase code based on Algorithm 25.3.1 with the purification.

In both FPS and FPSP, $\epsilon_1 = 10^{-8}$, $\epsilon_2 = 10^{-8}$, and $\lambda = 0.6$ were used for Phase-1, and $\epsilon_1 = 10^{-3}$, $\epsilon_2 = 10^{-6}$, and $\lambda = 0.99$, for Phase-2, and $M = 10^{11}$.

Codes FPS and FPSP were developed using MINOS 5.51 as a platform. Therefore, the three codes shared such features as preprocessing, scaling, LUSOL etc. Only the Mi50lp module was replaced by programs written by the author himself, with minor changes in few other modules. All codes used the default options.

Compiled using Visual Fortran 5.0, all codes were run under a Microsoft Windows XP Professional version 2002 on an ACER PC with an Intel(R) Pentium(R) 4 CPU 3.06GHz, 1.00GB memory, and about 16 digits of precision.

The first set of test problems included all 16 problems from Kennington and the second included all 17 problems from BPMPD that were more than 500KB in compressed form (Appendix C: Tables C.2, C.3).

In the tables, test problems are ordered by their sizes, in terms of $m+n$ as before, where m and n are the numbers of rows and columns of the constraint matrix, excluding slack variables. All reported CPU times were measured in seconds with utility routine CPU_TIME, excluding the time spent on preprocessing and scaling. FPS and FPSP used 10^{-6} as the equality tolerance (for holding of $Ax = b$), compared with MINOS, which used 10^{-4} , as usual.

Numerical results obtained with Kennington and BPMPD problems are displayed, respectively, in Tables F.1 and F.2, where the total iterations and time required for solving each problem are listed in the columns labeled Itns and Time

Table F.1 Statistics for 16 Kennington problems

Problem	MINOS		FPSP		FPS	
	<i>Itns</i>	Time	<i>Itns</i>	Time	<i>Itns</i>	Time
KEN-07	1724	1.97	1754	2.02	1750	2.02
CRE-C	4068	6.81	7487	11.89	2751	4.66
CRE-A	4066	7.67	8972	16.77	4069	8.09
PDS-02	5110	9.06	1171	1.84	1025	1.61
OSA-07	1767	7.16	1949	8.00	1204	5.02
KEN-11	13327	111.39	13549	113.67	13542	113.59
PDS-06	59521	433.67	8860	57.41	6317	40.36
OSA-14	3821	33.48	3581	32.16	1747	16.19
PDS-10	177976	2253.64	11191	122.13	7110	75.47
KEN-13	35709	637.92	35301	629.66	35162	627.05
CRE-D	377243	4796.42	28042	358.75	1144	13.23
CRE-B	328160	4406.19	32228	430.81	1422	17.91
OSA-30	7536	126.30	5886	101.59	2778	49.33
PDS-20	1557953	44803.02	88068	2258.83	64777	1643.14
OSA-60	16278	638.50	10976	449.45	4714	199.36
KEN-18	157224	19395.88	155355	19315.88	154801	19253.55
Total	2751483	77669.08	414370	23910.86	304313	22070.58

Table F.2 Statistics for 17 BPMPD problems

Problem	MINOS		FPSP		FPS	
	<i>Itns</i>	Time	<i>Itns</i>	Time	<i>Itns</i>	Time
RAT7A	4073	573.59	4102	52.06	4100	52.03
NSCT1	2190	35.83	13198	197.36	95	1.44
NSCT2	11555	196.63	11971	182.23	92	1.45
RIUTING	190627	2143.70	75759	813.08	72774	780.00
DBIR1	2289	47.44	12781	242.64	34	0.61
DBIR2	42575	966.77	12750	235.83	12745	235.72
T0331-4L	62333	789.83	108780	1330.63	108575	1328.14
NEMSEMM2	23581	204.59	18109	136.98	7327	58.00
SOUTHERN	29693	462.67	27550	314.17	27550	314.16
WORLD.MO	1405151	30953.42	822287	19193.38	94542	2929.00
WORLD	2627208	61340.78	1260355	27315.31	165045	4132.83
NEMSEMM1	13650	263.20	14575	255.17	14491	253.64
NW14	407	8.45	899	19.20	690	14.42
LPL1	3343112	127379.39	203967	7214.02	2414	104.41
DBIC1	624907	33335.58	377655	18746.17	375734	18679.69
Total	8383351	258701.88	2964738	76248.23	886208	28885.54
RADIO.PR	2	0.50	639702	14371.16	623810	14009.16
RADIO.DU	3	0.09	5055	56.11	5018	55.66

Table F.3 Rario for 16 Kennington problems

Problem	M	N	MINOS/FPSP		MINOS/FPS		FPSP/FPS	
			Itns	Time	Itns	Time	Itns	Time
KEN-07	2427	3602	1.0	1.0	1.0	1.0	1.0	1.0
CRE-C	3069	3678	0.5	0.6	1.5	1.5	2.7	2.6
CRE-A	3517	4067	0.5	0.5	1.0	0.9	2.2	2.1
PDS-02	2954	7535	4.4	4.9	5.0	5.6	1.1	1.1
OSA-07	1119	23949	0.9	0.9	1.5	1.4	1.6	1.6
KEN-11	14695	21349	1.0	1.0	1.0	1.0	1.0	1.0
PDS-06	9882	28655	6.7	7.6	9.4	10.7	1.4	1.4
OSA-14	2338	52460	1.1	1.0	2.2	2.1	2.0	2.0
PDS-10	16559	48763	15.9	18.5	25.0	29.9	1.6	1.6
KEN-13	28633	42659	1.0	1.0	1.0	1.0	1.0	1.0
CRE-D	8927	69980	13.5	13.4	329.8	362.5	24.5	27.1
CRE-B	9649	72447	10.2	10.2	230.8	246.0	22.7	24.1
OSA-30	4351	100024	1.3	1.2	2.7	2.6	2.1	2.1
PDS-20	33875	105728	17.7	19.8	24.1	27.3	1.4	1.4
OSA-60	10281	232966	1.5	1.4	3.5	3.2	2.3	2.3
KEN-18	105128	154699	1.0	1.0	1.0	1.0	1.0	1.0
Ave.(16)	–	–	6.6	3.2	9.0	3.5	1.4	1.1

under MINOS, FPSP, and FPS, separately. As MINOS solved problems RADIO.PR and RADIO.DU in exceptional few iterations. Results for the two problems are listed separately in the bottom lines and excluded from comparison with BPMPD set.

In Table F.3, a performance comparison between the three codes with Kennington set is made by giving iteration and time ratios of MINOS to FPSP, MINOS to FPS, and FPSP to FPS for each problem. It is seen from the bottom line that FPSP and FPS outperformed MINOS remarkably, with average iterations ratios 6.6 and 9.0, and time ratios 3.2 and 3.5! It is noted that the difference between FPSP and FPS is small (with iterations ratio 1.4 and time ratio 1.1).

In Table F.4, a comparison between the three codes with BPMPD set is made. It is seen from the second bottom line that FPSP and FPS outperformed MINOS, by average iterations ratios 2.8 and 9.5, and time ratios 3.4 and 9.0! The margin between FPSP and FPS is smaller (with iterations ratio 3.3 and time ratio 2.6).

In the bottom line of Table F.4, listed are average ratios for the entire set of the 31 test problems. FPSP and FPS again defeated MINOS: the average iterations ratios are 3.3 and 9.4, while the time ratios are 3.4 and 6.6.

In summary, the simplex interior-point algorithm is significantly superior to the standard simplex algorithm, at least with the tested problems.

Table F.4 Ratio for 15 BPMPD problems

Problem	M	N	MINOS/FPSP		MINOS/FPS		FPSP/FPS	
			Itns	Time	Itns	Time	Itns	Time
RAT7A	3137	9408	1.0	11.0	1.0	11.0	1.0	1.0
NSCT1	22902	14981	0.2	0.2	23.1	24.9	138.9	137.1
NSCT2	23004	14981	1.0	1.1	125.6	135.6	130.1	125.7
RIUTI	20895	23923	2.5	2.6	2.6	2.7	1.0	1.0
DBIR1	18805	27355	0.2	0.2	67.3	77.8	375.9	397.8
DBIR2	18907	27355	3.3	4.1	3.3	4.1	1.0	1.0
T0331	665	46915	0.6	0.6	0.6	0.6	1.0	1.0
NEM.2	6944	42133	1.3	1.5	3.2	3.5	2.5	2.4
SOUTH	18739	35421	1.1	1.5	1.1	1.5	1.0	1.0
WOR.M	35665	31728	1.7	1.6	14.9	10.6	8.7	6.6
WORLD	35511	32734	2.1	2.2	15.9	14.8	7.6	6.6
NEM.1	3946	71413	0.9	1.0	0.9	1.0	1.0	1.0
NW14	74	123409	0.5	0.4	0.6	0.6	1.3	1.3
LPL1	39952	125000	16.4	17.7	1384.9	1220.0	84.5	69.1
DBIC1	43201	183235	1.7	1.8	1.7	1.8	1.0	1.0
Ave.(15)	–	–	2.8	3.4	9.5	9.0	3.3	2.6
Ave.(31)	–	–	3.3	3.4	9.4	6.6	2.8	2.0

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